सत्यमेव जयते

MOFPI

MINISTRY OF FOOD PROCESSING INDUSTRIES
GOVERNMENT OF INDIA

STUDY TO DETERMINE POST-HARVEST LOSSES OF AGRI PRODUCES IN INDIA

NABCONS

NABARD CONSULTANCY SERVICES
(A wholly owned subsidiary of NABARD)










FOREWORD

Agriculture is the backbone of the Indian economy, a fact reaffirmed by the exceptional performance of this sector even during the outbreak of the COVID-19 pandemic. The strength of the sector lies in the fact that India is the leading producer in a number of commodities like milk, cereals, tea, sugarcane, fruits and vegetables etc. It has the potential to become a food basket for the world considering the scope and ever increasing demand for food processing.

On the other side, huge quantities of harvested produce get wasted due to fragmented supply chains and inadequacy of post-harvest infrastructure. A strong and dynamic food processing sector plays an important role in reducing the wastage of perishable products, enhancing shelf-life and ensuring value addition to agricultural produce. It also leads to diversification & commercialization of agriculture, generation of employment, enhancing income of farmers and creating surplus for the agri exports. With a view to making policy level decisions and for repositioning our focus in establishment of post-harvest infrastructure for value addition, it becomes essential that estimation of losses may be done at various levels in the post-harvest chain. The Ministry of Food Processing Industries (MoFPI), Government of India, accordingly commissioned this study to assess the post-harvest losses in agriculture & allied produce in India.

Earlier ICAR-CIPHET had taken up first study in 2005-07 to estimate the harvest and post-harvest losses of agriproduce on recommendations of the Parliamentary Standing Committee on Agriculture. In 2012, ICAR-CIPHET again took up the second study which was sponsored by the Ministry of Food Processing Industries, Govt of India to determine the levels of losses and to ascertain if there were changes in losses since the first study period.

For 3rd study, MoFPI awarded the work to NABARD Consultancy Services (NABCONS) to minimise the information gaps that exist in the sector. The present study was conceptualised to provide greater insights into the subject by giving focus to the perishable products like fruits and vegetables also.

I am happy to share that the results and findings of this study are now being brought out in a report form. The information presented in this report will provide insights in quantifying the post-harvest losses in various districts, states, agro climatic zones and national level. The study will help in identifying the areas of losses where policy guidance and support from government is required in terms of setting up of post-harvest infrastructure for reducing post-harvest losses. The data provided in this report will help policy makers, researchers and funding agencies in the sector.

I sincerely appreciate the inputs from Technical Committee (TC) members of MoFPI, inputs from Secretary, MoFPI during the review meetings, professionals, policy makers at the stage of finalising the report and financial support provided by MoFPI.

I congratulate Team NABCONS, Survey Agencies and Enumerators who carried out this enormous task and completed this study.

(Niraj Kumar Verma)

New Delhi

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New Delhi

Team NABCONS

LIST OF CONTRIBUTORS

Team Leader

Dr. B R Premi (Team Leader & Senior Vice President, NABCONS/ GM, NABARD)

Technical Experts

Dr. Pritam Chand Sharma (Horticulture Expert, NABCONS)
Dr. V Esakkimuthu (Animal Husbandry Expert, NABCONS/ DGM, NABARD)
Dr. P S Harikrishnaraj (Animal Husbandry Expert, NABCONS/ DGM, NABARD)
Mr. Raja Mithelesh Kumar (Fisheries Expert, NABCONS)
Dr. Anju Sharma (Statistician, NABCONS)
Dr. Girish Chandra (Statistician, NABCONS)

Study Team

Ms. Gunjan Khera (Consultant, NABCONS)
Ms. Anshumani Pandey (Consultant, NABCONS)
Mr. Kiran P Mali (Project Consultant, NABCONS)
Dr. Vaishali Wankhade (Project Consultant, NABCONS)
Ms. Monika Rautela (Associate Consultant, NABCONS)
Ms. Neeti Jain (Associate Consultant, NABCONS)
Ms. Neha Mahto (Associate Consultant, NABCONS)
Ms. Usha Rani (Associate Consultant, NABCONS)
Dr. Aaradhana Chilwal (Research Associate, NABCONS)

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CONCEPTS AND DEFINITIONS

Post-harvest losses – Post-harvest loss can be defined as the degradation in both quantity and quality of a food production from harvest to consumption. Quality losses are intrinsic in nature leading to internal and external biochemical changes leading to changes in colour, flavour, nutritive value of food, calorific value etc. without significant change in weight /volume. Quantitative loss is more serious in nature leading to value loss to the actors in the supply chain. It is mainly caused due to poor post-harvest management practices such as loss in weight due to moisture loss, physical damage, insect pest infestation, spoilage or rotting due to senescence and bacterial/ fungal infections etc.

Losses occurring during harvesting operation and thereafter were considered as post-harvest loss for assessment.

Cold chain - A cold chain is a climate-controlled supply chain, consisting of storage and distribution activities, which maintain product at a given temperature range from farm to fork.

Logistics chain- Activities related to packaging and transportation of produce from production locations to market.

Modern pack-house - It includes facilities like conveyor belt sorting, grading units, washing, drying and weighing, which enables the unit to undertake the first steps in organising the horticulture supply chain. It enables a small lot of sourcing of horticulture produce and should be built close to the farming area.

Pre-cooling unit - It refers to a specialised cooling room that rapidly removes field heat from fresh produce after harvest and thereby prepares the cargo for subsequent shipping. Precooling or post-harvest cooling is one of the most critical steps in preparing fruits and vegetables for the extended cold-chain.

Cold room - It is an insulated and refrigerated chamber, which is a necessary combination for pre-cooling unit and serves as transient storage, while allowing the pre-cooler to be utilised for the next batch load of incoming produce.

Cold storage units - It refers to temperature-controlled storages designed for post-harvest storage of horticulture produce. It can be designed to suit single or multiple product storage with multi-temperature requirements.

Reefer trucks - It refers to vehicle road transport with fixed insulated carriage body equipped with active refrigeration designed for the temperature-controlled carriage of horticulture produce, having GPS based location tracking system and installed with data logging temperature and humidity sensors.

Reefer container - A reefer container describes a multi-modal insulated container with integrated refrigeration equipment. Unlike fixed body trucks, reefer containers can be released from the trailer chassis and stationed on-site for localised use as a temporary temperature controlled store pending subsequent operations. This allows the prime motive and/or trailer to be utilised for other carriages.

Ripening chamber - It is one of the last mile facilities in the cold-chain, designed to function for controlled and hygienic ripening of fresh fruit. Modern ripening facilities are used extensively for ripening bananas but are also used to ripen other fruits like mangoes, avocadoes, kiwis, tomatoes, stone fruit, pears, etc.

Controlled atmosphere storage - Controlled atmosphere (CA) storage involves altering and maintaining an atmospheric composition that is different from air composition (about 78% N_2, 21% O_2, and 0.03% CO_2); generally, O_2 below 8 percent and CO2 above 1 percent are used. The atmospheric modification should be considered as a supplement to the maintenance of optimum ranges of temperature and RH for each commodity in preserving quality and safety of fresh fruits, ornamentals, vegetables, and their products throughout post-harvest handling.

LIST OF ABBREVIATIONS

ACZs	Agro-climatic Zones
ADDs	Agriculture Development Districts
AICMIP	All India Coordinated Millet Improvement Project
AICRPO	All India Coordinated Research Project on Oilseeds
ANOVA	Analysis of Variance
APEDA	Agricultural and Processed Food Products Export Development Authority
APHLIS	African Postharvest Losses Information System
APMC	Agriculture Produce Market Committee
CAS	Controlled Atmospheric Storage
CAP	Cover and Plinth
CAPI	Computer-Assisted Personal Interviews
CAS	Controlled Atmospheric Storage
CDB	Coconut Development Board
CFB	Corrugated Fibre Board
CFLD	Cluster Front Line Demonstrations
CIFA	Central Institute of Freshwater Aquaculture
CIFE	Central Inland Fisheries Research Institute
CIFT	Central Institute of Fisheries Technology
CIH	Central Institute of Horticulture
CIPHET	Central Institute of Postharvest Engineering & Technology
CLP	Critical Loss Point
CWC	Central Warehousing Corporation
DAC&FW	Department of Agriculture, Co-operation and Farmers' Welfare
DAHD	Department of Animal Husbandry and Dairying
DOGR	Directorate of Onion and Garlic Research
e-NAM	National Agriculture Market
EPC	Export Promotion Council
FAO	Food and Agriculture Organisation of the United Nations
FCI	Food Corporation of India
FFDA	Fish Farmers Development Agency
FICCI	Federation of Indian Chambers of Commerce & Industry
FLD	Front Line Demonstrations
FLI	Food Loss Index
FPC	Farmers Producer Company
FPO	Farmers Producer Organisation
FSI	Fishery Survey of India
FTL	Food Testing Laboratories
FWI	Food Waste Index
GDP	Gross Domestic Product
GIZ	Deutsche Gesellschaft für Internationale Zusammenarbeit
GoI	Government of India

GPS	Global Positioning System
HDPE	High Density Polyethylene
HMNEH	Horticulture Mission for North East & Himalayan States
HOPCOMS	Horticultural Producers' Co-operative Marketing and Processing Society Ltd.
ICAR	Indian Council of Agricultural Research
ICRISAT	International Crops Research Institute for the Semi-Arid Tropics
IFAD	International Fund for Agricultural Development
IFPRI	International Food Policy Research Institute
IIHR	Indian Institute of Horticultural Research
IIOR	Indian Institute of Oilseeds Research
IQF	Individual Quick Freezing
ISMA	Indian Sugar Mills Association
KCC	Kisan Credit Card
KVK	Krishi Vigyan Kendra
MIDH	Mission for Integrated Development of Horticulture
MIS	Market Intervention Scheme
MoFPI	Ministry of Food Processing Industries
MoU	Memorandum of Understanding
MSAMB	Maharashtra State Agricultural Marketing Board
MT	Metric Tonnes
NABARD	National Bank for Agriculture and Rural Development
NABCONS	NABARD Consultancy Services Private Limited
NARES	National Agricultural Research & Education System
NARI	Nimbkar Agricultural Research Institute
NCCD	National Centre for Cold-chain Development
NFDB	National Fisheries Development Board
NFSM	National Food Security Mission
NHB	National Horticulture Board
NHM	National Horticulture Mission
NIFPHATT	National Institute of Fisheries Post-Harvest Technology and Training
NLM	National Livestock Mission
NRC	National Research Centres
PHL	Post-Harvest Losses
PHM	Post-Harvest Management
PMKSY	Pradhan Mantri Kisan SAMPADA Yojana
PMMSY	Pradhan Mantri Matsya Sampada Yojana
PPS	Probability Proportion to Size Sampling
RBPD	Rural Backyard Poultry Development
SAU’s	State Agricultural Universities
SDGs	Sustainable Development Goals
SFAC	Small Farmers Agribusiness Consortium
SHGs	Self Help Groups
SIS	Sugarcane Information System

SMAM	Sub Mission on Agricultural Mechanisation
SNNP	Southern Nations, Nationalities, and Peoples' Region
SPSS	Statistical Package for Social Sciences
TMOC	Technology Mission on Coconut
TOR	Terms of Reference
UNDP	United Nation Development Programme
USDA	United States Department of Agriculture
WDRA	Warehousing Development and Regulatory Authority
WFLO	World Food Logistics Organization
WFP	World Food Programme
ZECC	Zero Energy Cool Chamber

EXECUTIVE SUMMARY

The main objective of the "S***tudy to determine post-harvest losses of agri produces in India",*** was a comprehensive evaluation of the post-harvest losses at various levels of the food supply chain, so as to finalise a national policy framework for the sector to reduce post-harvest losses of agriculture produce in India. The study was conceptualised to provide greater insights into the subject by giving more focus to the perishable products like fruits and vegetables. This study identified the gaps in infrastructure development, technology infusion, and skills requirements in the sector along with highlighting investment needs to fill the gaps. The study also assessed the effectiveness of post-harvest management schemes of Government of India (GoI) to develop appropriate policy and implementation directives. The study also highlighted region-specific and product-specific gaps, wherein intervention and support is required from GoI.

2. Broadly, the scope of the study was to assess post-harvest losses with respect to 54 crops/commodities across all 15 agro-climatic zones of India. All of the major producing districts of identified crops were covered under the study.

3. An effective post-harvest management is very important for planning effective utilisation of the agricultural and livestock produce. As post-harvest losses occur at each stage of the supply chain, therefore it is important to check the different channels through which crop and livestock produce reaches consumers. The present study was targeted to map post-harvest losses at each channel at farm and market level.

4. The data for estimating post-harvest losses were collected from 292 districts covering 15 agro-climatic zones. Stratified multistage random sampling method as described in chapter 3 of report was used to select the respondents. The data were collected through enquiry as well as, $1/5^{th}$ by observations, by visiting the farmer/ market player by trained manpower/ enumerators. Computer Aided Personal Interview (CAPI) based application was used for data collection from all the respondents. The collected data was cross checked, scrutinised and analysed for estimation of post-harvest losses at district level, state level, agro-climatic zone level and national level.

5. The salient findings of the study and range of post-harvest losses observed in this study are summarised below.

a) In the ***cereals*** category, the post-harvest losses for paddy, wheat, maize, Bajra and sorghum were estimated. The post-harvest losses ranged from 3.89 percent in maize to 5.92 percent in sorghum. Use of manual methods for operations such as sickle and sticks for harvesting and threshing majorly contributed to the post-harvest losses.

b) The estimated post-harvest losses in ***pulses*** (pigeon pea, chickpea, black gram and green gram) ranged from 5.65 percent in pigeon pea to 6.74 percent in chickpea. Shattering of grains during harvesting, spillage during various operations, and mishandling resulted in post-harvest losses.

c) In case of ***oilseeds,*** post-harvest losses of crops such as mustard, cottonseed, soybean, safflower, sunflower and groundnut were estimated. The post-harvest losses were ranging from 2.87 percent in cottonseed to 7.51 percent in soybean. Harvesting and threshing at farm operations and improper storage conditions at market channels resulted in post-harvest losses.

d) Among 11 ***fruits (***apple, banana, citrus, grapes, guava, mango, papaya, sapota, pineapple, pomegranate and muskmelon) selected for study, post-harvest losses were in the range of 6.02 percent in pineapple to 15.05 percent in guava. Harvesting and sorting/ grading in farm operations and improper handling and storage conditions majorly contributed towards post-harvest losses. This indicated that better harvesting tools, proper handling and effective cold chain infrastructure was essential for further reduction in post-harvest losses.

e) Among 14 ***vegetables*** (cabbage, cauliflower, green peas, mushroom, onion, potato, tomato, tapioca, bottle gourd, brinjal, beans, radish, capsicum and okra) the estimated post-harvest losses ranged from 4.82 percent in tapioca to 11.61 percent in tomato. Farm operations such as harvesting and sorting/grading resulted in post-harvest losses. Lack of proper storage and improper handling practices by various stakeholders in market channels were the major contributing factors towards post-harvest losses. The cold chain infrastructure for vegetables in the country was still at a very nascent stage and needed special focus for reduction in post-harvest losses as well as for retention of good quality as desirable by the consumers.

f) The post-harvest losses of ***plantation crops*** (areca nut, cashew nut, coconut and sugarcane) ranged from 3.72 percent in cashew nut to 7.33 percent in sugarcane. Operations such as harvesting and threshing majorly contributed towards post-harvest losses.

g) In the case of ***spices*** **(**black pepper, coriander, chilli and turmeric), the estimated post-harvest losses were in the range of 1.29 percent in black pepper to 6.11 percent in chilli. Shattering during harvesting and spillage at various stages contributed towards post-harvest losses.

i) The post-harvest loss of ***inland fish*** was 4.86 percent, whereas post-harvest loss of marine fish was 8.76 percent. Post-harvest loss in marine fish was majorly attributed by throwing the uneconomical fish after harvesting along with post-harvest losses at wholesaler and retailer levels, which can be overcome by strengthening of cold chain infrastructure including cold storages, ice plants, freezing plants.

j) The post-harvest loss of ***meat*** was 2.34 percent, whereas post-harvest loss of poultry meat was 5.63 percent. The overall improvement in farm level operations and significant shift of standalone slaughter houses to integrated facilities led to reduced post-harvest losses.

k) The post-harvest loss of ***milk*** was observed to be 0.87 percent. Improvement in handling of milk at farm level and in processing units may help in further reduction of post-harvest losses.

l) In comparison to post-harvest losses during 2014-15, the post-harvest losses during 2020-22 reduced significantly for 25 crops and non-significant reduction was observed in 17 crops. However, in 3 crops viz. black paper, turmeric and tapioca, estimated post-harvest losses were slightly higher than previous assessment in 2015.

6. For better post-harvest management, development of infrastructure at both farm and market level is required. At farm level, support for mechanical harvesters, threshers and packaging materials may be provided to the farmers. Creation of collection centres, pre-cooling units, storage structures and packing units near the production centres is essential for better handling of produce at farm level. At market channels, forward and backward linkages, adequate storage structures along with modern retail infrastructure are needed.

8. At both farm and market operations, a number of technologies and policy interventions could be deployed to potentially reduce the post-harvest losses. For instance, organised marketing through creation of FPOs/ FPCs, improvement in crop production and protection technologies and training and capacity building of farmers through extension activities to impart skills regarding better management of crops.

The study recommended various strategies to reduce post-harvest losses both at farm level and market level which are discussed in detail in **Chapter 8.**

CHAPTER 1 INTRODUCTION

1.1 Background

India is among the leading producers of many agriculture and allied produce in the world. It is the largest producer of milk and dairy products and the second-largest producer of cereals, fruits, vegetables and fish in the world. India ranks second in terms of total food production globally. Though, outbreak of pandemic during last two years affected every sector of the Indian economy yet the agriculture sector had shown resilience. It registered a positive growth of 3.6 percent in 2020-21, which improved to 3.9 percent in 2021-22, resulting in an increase in its share in Gross Domestic Product (GDP) to 19.9 percent in 2020-21 from 17.8 percent in 2019-20. However, disruptions in the supply chain adversely affected the flow of agricultural goods leading to high food inflation and detrimental impact on major agricultural exports.

The production for the years 2018-19, 2019-20 and 2020-21 are presented in the **Table 1.1**. In 2020-21, India produced 310.74 million MT of food grains, 302.93 million MT of horticulture produce, 209.96 million MT of milk, 14.50 million MT of fish, 8.80 million MT of meat and 122.05 billion numbers of eggs. It can be seen that the production of all categories has increased over the years. But at the same time, high post-harvest losses in the supply chain always remained a cause of concern to the planners and policy makers.

Table 1.1 Production of different commodities in India

S.No	Name of the crop/ commodity	Production (million MT)		
		2018-19	2019-20	2020-21
1	Cereals	263.14	274.48	285.28
2	Pulses	22.08	23.03	25.46
3	Oilseeds	31.52	33.22	35.95
4	Fruits	97.97	102.08	102.48
5	Vegetables	183.17	188.28	200.45
6	Plantation crops	16.35	16.12	16.63
7	Sugarcane	405.42	370.50	405.40
8	Spices	9.43	10.14	11.12
9	Milk	187.70	198.40	209.96
10	Meat	8.10	8.60	8.80
11	Eggs (billion numbers)	103.32	114.40	122.05
12	Fish	13.57	14.16	14.50

Source: Directorate of Economics and Statistics, Department of Agriculture and Farmers Welfare, MoA&FW, GoI

Better post-harvest supply chain management will minimize wastages in the sector, create surpluses, improve the level of real consumption and enhance income in the economy at large. Better management at every stage of supply chain will lead to an increase in the real income of the producers. In order to reduce post-harvest loss of agriculture and allied produce a number of schemes were launched by different Ministries of GoI. The Ministry of Food Processing Industries (MoFPI), GoI is operating a Pradhan Mantri Kisan SAMPADA Yojana (PMKSY) for creation of modern infrastructure with efficient supply chain management from farm gate to retail outlet. The Ministry of Agriculture, GoI launched a "Mission for Integrated Development of Horticulture (MIDH)" in 2014, for holistic growth of the horticulture sector. The Ministry of Commerce is operating a scheme to enhance exports, which includes necessary export oriented cold-chain infrastructure. National Livestock Mission, Department of Animal husbandry and Dairying is working for development of poultry and meat sector. The Department of Fisheries, Ministry of Fisheries, Animal Husbandry and Dairying, Government of India is implementing "Pradhan Mantri Matsya Sampada Yojana (PMMSY)" for the development of fisheries sector.

Development of new varieties, training programmes, and technological innovations by State Agricultural/Horticultural Universities (SAU's), National Institutes for different crops, livestock, animals, fisheries and technology development, National Research Centres (NRCs), Krishi Vigyan Kendra (KVKs) under Indian Council of Agricultural Research (ICAR) and National Agricultural Research and Education Systems (NARES) have contributed towards better post-harvest management of agricultural commodities.

Post-harvest loss adversely impacts food security, nutrition, and economic stability of several stakeholders such as farmers, exporters, traders, processors and consumers. The reduction in post-harvest losses is thus not only essential for narrowing the demand-supply gap in the food chain, but also in reducing economic value losses to the country. In order to monitor the quantum of post-harvest losses and reorientation of policy to contain it, the Ministry of Food Processing Industries (MoFPI) conducts studies at regular intervals. The previous studies sponsored by MoFPI were conducted by ICAR-CIPHET (2012) and ICAR-CIPHET (2015) and they indicated substantial post-harvest losses of major crops/ commodities and livestock produce in India due to improper post-harvest management practices.

ICAR-CIPHET (2015) reported post-harvest losses of 4.65-5.99 percent in cereals, 6.36-8.41 percent in pulses, 3.08-9.96 percent in oilseeds, 6.70-15.88 percent in fruits, 4.58-12.44 percent in vegetables, 1.18-7.89 percent in plantation crops and spices and 7.19 percent in egg, 5.23-10.52 percent in fish, 2.71-6.74 percent in meat and 0.92 percent in milk in India. They further estimated economic value losses to the tune of ₹92651 crore at that time. Since, a period of 7 years has elapsed for the previous study, there was an urgent need for reassessment of post-harvest losses of agricultural and allied produce in the country. In this backdrop, MoFPI felt the need to reassess the extent of post-harvest losses occurring across different agriculture value chains. Therefore, the Ministry of Food Processing Industries (MoFPI), Government of India, commissioned this study to assess the post-harvest losses in agriculture & allied produce in India in terms of award letter no. 19012/2/2019-Economic Division dated 21st July 2020 to NABARD Consultancy Services Pvt Ltd (NABCONS).

1.2 Study rationale and objectives

The present study was conceptualised to provide greater insights into the subject by giving more focus to the perishable products like fruits and vegetables. A MoU was signed between NABCONS and MoFPI on 19 August 2020 with the following objectives:

i. To assess the effectiveness of post-harvest management schemes of MoFPI to develop appropriate policy and implementation directives.
ii. To highlight region-specific and product-specific gaps and allow for target support from Government of India.
iii. To finalise a National Policy framework for the sector to reduce post-harvest losses of agriculture produce in India.
iv. To identify the gaps in infrastructure development, technology infusion, and skills requirements in the sector along with highlighting investment needs to fill the gaps.

1.3 Scope of the study

The broad scope of the study was to assess post-harvest losses with respect to 54 crops/commodities across all 15 agro-climatic zones of India. All of the major producing districts of identified crops were covered under the study. The specific scope of study is outlined as follows:

a) Assessment of post-harvest losses of agri produce along the supply chain at all India level, Agro Climatic Zone wise and agricultural and livestock produce wise.
b) Agro Climatic Zone wise post-harvest losses for each crop along with the reasons and interventions needed for reducing the post-harvest losses.
c) Holding level assessment of post-harvest losses of agricultural produce. The different stages to be covered for the assessment of the losses packaging, transportation in farm gate opeartions and storage channels such as farm, godown, wholesaler, retailer, processing units etc.
d) Understanding and documentation of the current business practices of the market players (collectors, wholesalers and traders). Category and sub-category wise cross country comparison of post-harvest losses

1.4 Reference year

The field survey was conducted between August 2020 to May 2022 for 54 crops and the final report was submitted in August 2022.

1.5 Limitations of study

- Outbreak of COVID 19 pandemic had disrupted the survey.
- Farmers' agitation, cyclone and other such natural factors affected the study resulting in delay in conduct of survey or change in selected districts.

CHAPTER 2: REVIEW OF LITERATURE

India has reached self-sufficiency in food grain production, however food security of rising population is still a major concern in the country. Increase in food availability by reducing post-harvest losses of various crops is one of the major solutions as a significant quantity of food is lost due to improper post-harvest management of crops. Therefore, reducing post-harvest losses continues to be an important issue for all concerned. Various studies were conducted in the past on assessment of post-harvest losses, methodologies followed for such assessments, identification of methods to contain losses and various channels affecting such losses. The majority of such studies were conducted at laboratory scale experiments and were limited to one or more crops/ commodities for a specific location. An attempt was made to scan through the findings of these studies and the same is presented in this chapter.

2.1 Data collection methodologies

The adoption of appropriate methodology for assessment of post-harvest losses of agricultural produce is of utmost importance. Efforts were made to study various approaches and methodologies followed by different researchers to assess post-harvest losses in their research so that most appropriate methodology & analytical tools could be adopted in this assessment. A brief description of the methodologies followed by various researchers are presented here under:

According to UNDP, the Sustainable Development Goals (SDGs), also known as the Global Goals, were adopted by the United Nations in 2015 as a universal call to action to end poverty, protect the planet, and ensure that by 2030 all people enjoy peace and prosperity. The second goal of zero hunger, aims to end all forms of hunger and malnutrition by 2030, making sure all people–especially children–have sufficient and nutritious food all year. This involves promoting sustainable agricultural, supporting small-scale farmers and equal access to land, technology and markets. It also requires international cooperation to ensure investment in infrastructure and technology to improve agricultural productivity.

According to Food and Agriculture Organisation (FAO), food loss refers to the decrease in edible food mass at the production, post-harvest and processing stages of the food chain and food waste refers to the discard of edible foods at the retail and consumer levels. According to the UNEP Food Waste Index report (2021), Food Loss Index (FLI) measures losses for key commodities in a country across the supply chain, up to but not including retail. Food Waste Index (FWI), measures food waste at retail and consumer level (households and food service).

According to FAO, the expression "post-harvest losses" means a measurable quantitative and qualitative loss in a given product. Such losses can occur during any of various phases of the post-harvest system. Food losses can be quantitative as measured by decreased weight or volume or can be qualitative, such as reduced nutrient value and unwanted changes in taste, colour, texture, or cosmetic features of food (Buzby and Hyman, 2012).

The Ministry of Food Processing Industry (MoFPI), GoI has been conducting studies to assess the post-harvest losses in the country with the help of reputed institutions. The first such study was conducted by MoFPI with the help of ICAR-CIPHET (Central Institute of Post-Harvest Engineering & Technology) on assessment of harvest and post-harvest losses of major crops and livestock produce in India (ICAR-CIPHET (2012)). They had developed methodologies and schedules to assess post-harvest losses by both enquiry and observation for 46 commodities. For selection of respondents, stratified multistage random sampling method was used. The agro-climatic zones were taken as strata. Districts in each stratum were taken as the first stage, blocks as the second stage, villages as the third stage, and farmers as the fourth stage units. Fourteen, out of 15 agro climatic zones of the country were considered for selection of representative districts. The zone pertaining to the island region was not included in the survey as the total contribution to Indian agricultural production from this zone was reported to be quite low. A total of 120 districts were randomly selected from 14 ACZs followed by selection of 2 blocks from each district and 5 villages from each block at random. From each village, 10 farmers were randomly drawn for data collection after enumeration. Again, MoFPI in 2015 got conducted a similar study from ICAR-CIPHET on assessment of quantitative harvest and post-harvest losses of 45 major crops and commodities in India based on the methodology adopted by ICAR-CIPHET (2012).

Musasa et al. (2013) conducted an exploratory study in Rusitu valley in which 80 sweet orange growing farmers were covered. The value chain of sweet orange was mapped and information on soil nutrient supplementation, water and fruit tree management, pest and disease management was collected during the study. Likert type questionnaire scale was used to elicit information and they found that 40 percent of the mature fruits from Rusitu citrus farmers were lost in the field during temporary storage.

Bantyehu et al. (2019) conducted a study on estimation of pre and post-harvest losses of tropical fruits in Ethiopia. The aim of the research was to quantify and identify major factors of fruit loss. The semi-structured questionnaires were used to collect data from 180 randomly selected fruit producers of six districts. In this study, the purposive sampling method was used to select fruit production areas, whereas respondent farmers were sampled randomly. From the six districts, 30 tropical fruit producing farmers were selected. A total of 180 sample fruit producers were selected from the six districts randomly. Data analysis was made using statistical software (SPSS). Descriptive statistics were used to estimate post-harvest losses of fruits, whereas multiple regression was used for identification of major factors.

Sarkar et al. (2020) conducted an assessment of pre- and post-harvest losses of vegetables in the Kanpur district of Uttar Pradesh. From the district, 2 blocks having maximum area under the vegetables were selected and out of each block, 5 villages were selected from the list of villages with a minimum 15 percent area under production of vegetables. From each village, 10 farmers were selected on the basis of the probability proportion of the area under the vegetable. A total sample size of 100 farmers and 30 traders were selected under the study. At the farm level, 2 types of losses during market preparation and transportation, and damage during handling and sorting at trader level were observed.

2.2 Post-harvest losses of cereals

The systematic assessment of post-harvest losses in food grains started as early as 1971, when GoI appointed "Panse Committee" to assess post-harvest losses (Government of India, 1971). There were very high losses ranging between 6 percent in Bajra to 11 percent in Paddy. Appiah et al. (2011) conducted a study on post-harvest losses of rice from harvesting to milling in Ghana. Total post-harvest losses of 4.60 percent to 17.88 percent were reported in the study. The harvesting losses ranged from 3.03 percent to 12.05 percent, threshing losses from 0.53 percent to 4.07 percent and drying losses from 1.57 percent to 1.76 percent. The losses at farm level were 3.91 percent and at storage level were 1.28 percent. GIZ (2014) in the study on post-harvest losses of rice in Nigeria reported a total loss of 24.9 percent, which included 11.39 percent losses during operations such as harvesting of paddy, threshing, winnowing, drying, storage and transportation, 4.4 percent for traditional milling, 5.2 percent for parboiled rice and 7.54 percent for milled rice. Nath et al. (2016) conducted a study on post-harvest loss assessment of rice at selected areas of Gazipur district in Bangladesh. The post-harvest losses of 10 percent (9.78% during Aus season, 9.77% during Aman season and 9.91% during Boro season) were indicated.

FAO (2018) conducted a case study on rice value chain in the republic of India covering East Godavari and Nellore districts of Andhra Pradesh. Study reported 6 percent loss during harvesting and threshing of paddy, 0.2 percent during drying of paddy, 0.5 percent during transportation of paddy, 0.3 percent during storage at mills, 0.2 percent during storage by CWC warehouses and 0.2 percent during storage by FCI warehouse. Grains shattering and left over grains were the major causes for losses during harvesting and threshing operations. Damage by birds and dispersal during drying and spillage during loading and unloading also contributed to the losses. In another similar study conducted by FAO (2018(h)) in the Democratic Republic of Timor-Leste, losses of 3.5 percent during harvesting, 5 percent during threshing, 5 percent during storage and 10 percent during milling were reported.

Ridolfi et al. (2018) in their review on post-harvest losses in Ghana reported a total loss of 13.5 percent for rice. The preliminary processing (5.9%) and on-farm storage (4.3%) were the key loss hotspots in Ghana. FAO, IFAD and WFP (2021) in the technical brief on grain supply chains in Burkina Faso, the Democratic Republic of the Congo and the Republic of Uganda reported losses up to 12 percent during harvesting, 4 percent during drying, 5 percent during on-farm storage, 11 percent during manual hulling and 22 percent during mechanised hulling.

Arun and Ghimire (2019) conducted a study on estimating post-harvest loss of wheat at the farm level to enhance food security in Nepal. The study reported that the post-harvest loss of wheat was 4.88 percent. The Pyongyang Agricultural Campus and Kim II Sung University, in collaboration with the FAO and the UNDP on the post-harvest losses of wheat and barley estimated that the loss in North Korea was 16.35 percent for wheat across the supply chain (FAO 2017). Bartholomeu et al. (2017) conducted a case study for Brazil on post-harvest losses of wheat logistics chain. The study reported that 11.8 percent of wheat was lost during logistic stages, 6 percent post-harvest losses on the farm level and 5 percent losses during the storage in cooperatives and 0.8 percent losses during transportation. Dessalegn et al. (2014) conducted a study on the post-harvest wheat losses in Africa. The case study reported that the overall loss in the wheat supply chain was 17.1 percent. The study revealed that humid conditions, insects and rodents in storage were found to be the major causes of post-harvest loss. Grover (2013) conducted a study to assess the pre and post-harvest losses in wheat in Punjab and reported losses of 2.62, 1.94, 2.37 and 2.26 percent at harvesting stage in stratum I, II, III & IV, respectively with an overall loss of 2.30 percent at the harvesting stage in the sample area.

FAO (2020) conducted a study in two agriculture development districts (ADDs) of Malawi, Salima and Lilongwe reported total 4.4 percent losses in harvesting, 2.9 percent in shelling, 1.7 percent at the time of winnowing, 1.6 percent during drying and 3.5 percent in storage of maize. Arun and Ghimire (2019) conducted a study on estimating post-harvest loss at the farm level to enhance the food security in Nepal. A mean harvest loss of 4.00 percent at farm level was reported for maize. The study reported average losses of 2.99 percent and 2.88 percent for winter maize and summer maize, respectively. In another case study by FAO (2018(a)) for analysis of food loss, its causes and solutions in the maize value chain in the republic of Malawi, Rome reported 3.5 percent post-harvest losses at stacking, 5 percent at the time of removing cobs from the stalks, 4.2 percent during shelling and 17.5 percent in storage. Study revealed that pre-harvest conditions and actions in the field indirectly led to post-harvest losses at later stages in the chain, as production and agronomic practices influenced quality at harvest, suitability for transport and shipping, storage stability and shelf-life after harvest. FAO (2018(b)) in the Federal Democratic Republic of Ethiopia, Rome reported 1 percent post-harvest losses during harvesting, 2.1 percent during stacking, 0.5 percent during shelling, 0.4 percent during farm level transportation, 6.9 percent in farm level storage, 1.75 percent at wholesaler level and 4 percent at retailer level.

FICCI-PWC (2018) in a study on maize stated that approximately 5 to 7 percent of maize produced in India was lost due to improper storage. The use of scientific warehousing for maize could reduce the post-harvest losses to 0.5-1 percent. Ghanashyam Bhandari et al. (2015) conducted a study on maize post-harvest losses and its management practices in the western hills of Nepal and reported that insects were the main problem in the maize field followed by weed and disease leading to 10-20 percent post-harvest losses in storage. Abass et al. (2013) estimated an average post-harvest loss of 13.45 percent in the semi-arid Savannah area of Tanzania.

A study conducted by Grant Thorton in 2017-18 for the state of Rajasthan reported that the post-harvest losses in Bajra were around 14-18 percent during harvesting, storage, transportation, processing, distribution and handling. The study also estimated that about 6 to 8 percent of Bajra was lost at farmers' level during harvesting, threshing, winnowing, transportation and storage. In the case of handling through local commission agents, a further 1 percent post-harvest loss was reported due to improper handling. In the APMC premises, 3-5 percent post-harvest loss was reported during the transportation, weighing, unpacking, repacking and auction handling process. At the processors' doorstep, 3-5 percent post-harvest loss was reported during processing due to wastage, pilferage and moisture loss. According to the African Post-harvest Losses Information System (APHLIS) post-harvest losses in 2012 for sorghum were at 11.6 percent.

In Indian context, ICAR-CIPHET (2012) and ICAR-CIPHET (2015) conducted important studies on harvest and post-harvest losses covering both farm operations as well as market channels. ICAR-CIPHET (2012) reported losses of 3.87-5.96 percent in cereals, while ICAR-CIPHET (2015) reported losses of 4.65-5.99 percent. The estimated crop wise losses are indicated in **Table 2.1** and **Table 2.2**.

The overall losses for cereals reported over the years by various authors in India and abroad have been summarised and given in **Table 2.3**. It can be seen that the losses varied from 8.78 percent in maize to 15.72 percent in wheat.

Table 2.1 Harvest and post-harvest losses (%) of cereals, pulses and oilseeds at national level (ICAR-CIPHET 2012)

Crop	Farm operations							Total loss in farm operations	Storage in channels					Total loss in storage	Overall total loss
	Harvest-ing	Collec-tion	Thresh -ing	Winnow-ing/ cleaning	Drying	Packag -ing	Trans -port		Farm	Godown	Whole-salers	Retailers	Processing unit		
Paddy	1.2 ±1.1	0.74 ±0.53	1.15 ±0.44	0.36 ±0.48	0.23± 0.23	-	0.13 ±0.16	3.91 ±0.58	0.64 ±0.29	0.03 ±0.02	0.22 ±0.04	0.02 ±0.02	0.39 ±0.11	1.29 ±0.31	5.19 ±0.49
Wheat	1.69 ±0.7	0.56 ±0.43	1.62 ±0.48	0.48 ±0.59	0.04 ±0.11	-	0.16 ±0.13	4.67 ±0.47	0.59 ±0.24	0.06 ±0.06	0.13 ±0.12	0.04 ±0.03	.048 ±0.14	1.28 ±0.31	5.96 ±0.18
Maize	0.5 ±0.42	0.19 ±0.17	1.57 ±0.36	0.21 ±0.36	0.17 ±.09	-	0.07 ±0.08	2.81 ±0.26	0.41 ±0.15	0.01 ±.02	0.66 ±0.26	0.11 ±0.09	0.11 ±0.07	1.29 ±0.31	4.10 ±0.29
Bajra	0.8 ±0.64	0.56 ±0.41	1.30 ±0.60	0.33 ±.072	0.10 ±0.25	-	0.17 ±0.16	3.82 ±0.52	0.50 ±0.39	0.01 ±0.01	0.24 ±0.12	0.12 ±0.09	0.07 ±0.03	0.98 ±0.42	4.80 ±0.47
Sorghum	0.6 ±0.78	0.38 ±0.39	0.67 ±0.82	0.39 ±0.49	0.34 ±0.20	-	0.13 ±0.09	2.76 ±0.52	0.36 ±0.11	0.03 ±0.01	0.66 ±0.12	0.04 ±0.03	0.01 ±0.01	1.12 ±0.17	3.87 ±0.38
Pigeon pea	0.48 ±1.58	0.44 ±1.24	0.69 ±0.63	1.04 ±1.23	0.47 ±0.54	0.19 ±0.52	0.11 ±0.33	0.43 ±1.03	1.36 ±0.59	0.01 ±0.01	0.22 ±0.4	0.08 ±0.07	0.30 ±0.05	1.96 ±0.60	5.39 ±0.78
Chick pea	0.72 ±1.14	0.56 ±0.83	0.82 ±0.88	0.38 ±0.81	0.56 ±0.83	0.22 ±0.29	0.16 ±0.14	3.41 ±0.80	0.31 ±0.24	0.02 ±0.01	0.37 ±0.13	0.08 ±0.09	0.12 ±0.21	0.86 ±0.28	4.28 ±0.56
Black gram	1.13 ±1.79	1.04 ±1.22	1.62 ±1.84	0.31 ±0.53	0.50 ±0.99	0.23 ±0.61	0.14 ±0.36	4.96 ±1.56	0.73 ±0.84	0.01 ±0.00	0.16 ±0.03	0.09 ±0.20	0.11 ±0.04	1.07 ±0.86	6.06 ±1.28
Green gram	0.86 ±1.28	0.62 ±1.12	1.63 ±0.89	0.32 ±1.02	0.31 ±1.09	0.14 ±0.21	0.21 ±0.27	4.09 ±0.97	0.68 ±0.68	0.01 ±0.00	0.47 ±0.06	0.80 ±0.12	0.04 ±0.01	1.42 ±0.69	5.51 ±0.84
Mustard	4.54 ±2.11	1.09 ±1.17	1.78 ±0.92	0.33 ±0.98	0.23 ±0.42	0.21 ±0.55	0.24 ±0.45	8.43 ±1.19	0.24 ±0.16	0.02 ±0.00	0.10 ±0.03	0.04 ±0.03	0.10 ±0.10	0.46 ±0.20	8.89 ±0.84
Cottonseed	0.59 ±1.85	0.24 ±1.12	-	6.86 ±1.13	0.38 ±0.91	0.06 ±0.19	0.06 ±0.14	2.19 ±1.31	0.01 ±0.02	-	0.52 ±0.32	-	0.06 ±0.01	0.56 ±0.32	2.76 ±1.08
Soybean	1.14 ±1.02	046 ±0.39	1.24 ±0.67	0.49 ±0.35	0.22 ±0.21	0.14 ±0.10	0.17 ±0.13	5.84 ±0.57	0.10 ±0.05	0.01 ±0.02	0.24 ±0.05	0.04 ±0.04	0.01 ±0.01	0.41 ±0.08	6.26 ±0.47
Safflower	0.47 ±0.52	0.73 ±0.69	1.06 ±1.82	0.52 ±0.45	0.33 ±0.31	0.08 ±0.15	0.07 ±0.10	3.26 ±0.71	0.03 ±0.04	0.00 ±0.00	0.08 ±.09	-	0.29 ±0.25	0.43 ±0.27	3.68 ±0.54
Sunflower	1.08 ±0.36	0.47 ±0.17	1.31 ±0.47	0.52 ±0.21	0.31 ±0.17	0.13 ±0.05	0.11 ±0.06	3.93 ±0.26	0.04 ±0.01	0.01 ±0.01	0.19 ±.04	0.01 ±0.01	0.35 ±0.12	0.62 ±0.13	4.55 ±0.24
Groundnut	4.79 ±3.75	0.68 ±2.26	2.72 ±1.81	0.34 ±0.78	0.28 ±0.54	0.20 ±0.26	0.10 ±0.26	9.11 ±1.99	0.16 ±0.11	0.00 ±0.00	0.50 ±0.13	0.07 ±0.04	0.27 ±0.08	0.96 ±0.19	10.47 ±1.41

Table 2.2 Harvest and post-harvest losses (%) of cereals, pulses and oilseeds at national level (ICAR-CIPHET 2015)

Crop	Farm operations							Total loss in farm operations	Storage in channels					Total loss in storage	Overall total loss
	Harves-ting	Collect-ion	Thresh-ing	Winnow-ing/ cleaning	Drying	Packaging	Trans-port		Farm	Godown	Whole -salers	Retailers	Processing unit		
Paddy	2.08 ±0.79	0.37 ±0.29	1.44 ±0.39	0.50 ±0.50	0.10 ±0.15	0.08 ±0.04	0.09 ±0.06	4.67 ±0.44	0.39 ±0.15	0.07 ±0.03	0.21 ±0.07	0.02 ±0.01	0.16 ±0.04	0.86 ±0.14	5.53 ±0.34
Wheat	1.43 ±0.47	0.56 ±0.22	1.43 ±0.41	0.40 ±0.19	0.07 ±0.09	0.10 ±0.07	0.08 ±0.04	4.07 ±0.29	0.53 ±0.14	0.03 ±0.02	0.10 ±0.07	0.02 ±0.01	0.17 ±0.04	0.86 ±0.13	4.93 ±0.20
Maize	1.42 ±0.31	0.42 ±0.16	1.20 ±0.40	0.40 ±0.65	0.18 ±0.24	0.16 ±0.06	0.13 ±0.09	3.90 ±0.33	0.21 ±0.20	0.04 ±0.03	0.30 ±0.17	0.12 ±0.07	0.08 ±0.06	0.75 ±0.20	4.65 ±0.29
Bajra	1.15 ±0.54	0.43 ±0.42	2.15 ±0.32	0.19 ±0.09	0.16 ±0.22	0.20 ±0.12	0.15 ±0.16	4.43 ±0.33	0.38 ±0.09	0.02 ±0.01	0.21 ±0.12	0.12 ±0.03	0.06 ±0.03	0.79 ±0.09	5.23 ±0.24
Sorghum	1.47 ±0.48	0.33 ±0.06	2.04 ±0.38	0.47 ±0.20	0.08 ±0.05	0.28 ±0.14	0.09 ±0.03	4.78 ±0.28	0.24 ±0.09	0.08 ±0.01	0.73 ±0.18	0.15 ±0.05	0.02 ±0.01	1.21 ±0.10	5.99 ±0.20
Pigeon pea	1.18 ±0.38	0.39 ±0.29	2.13 ±0.71	0.41 ±0.59	0.18 ±0.18	0.22 ±0.26	0.19 ±0.31	4.69 ±0.45	1.02 ±0.15	0.10 ±0.03	0.08 ±0.04	0.16 ±0.05	0.32 ±0.06	1.67 ±0.13	6.36 ±0.30
Chick pea	1.87 ±0.53	1.19 ±0.39	2.60 ±0.59	0.58 ±0.19	0.40 ±0.24	0.25 ±0.08	0.35 ±0.08	7.23 ±0.38	0.41 ±0.11	0.04 ±0.01	0.34 ±0.11	0.17 ±0.05	0.21 ±0.03	1.18 ±0.10	8.41 ±0.26
Black gram	1.82 ±0.42	1.01 ±0.55	1.94 ±1.04	0.48 ±0.31	0.26 ±0.26	0.23 ±0.04	0.15 ±0.04	5.89 ±0.53	0.62 ±0.20	0.04 ±0.02	0.20 ±0.05	0.19 ±0.05	0.13 ±0.01	1.18 ±0.15	7.07 ±0.39
Green gram	2.00 ±0.39	0.76 ±0.20	1.54 ±0.86	0.36 ±0.52	0.33 ±0.32	0.22 ±0.03	0.14 ±0.03	5.37 ±0.47	0.41 ±0.17	0.00 ±0.00	0.39 ±0.11	0.31 ±0.09	0.13 ±0.04	1.24 ±0.15	6.60 ±0.35
Mustard	1.85 ±0.88	0.54 ±0.29	1.78 ±0.52	0.64 ±0.30	0.19 ±0.05	0.18 ±0.12	0.14 ±0.08	5.32 ±0.45	0.11 ±0.06	0.02 ±0.02	0.06 ±0.03	0.03 ±0.03	0.01 ±0.01	0.22 ±0.06	5.54 ±0.31
Cottonseed	2.01 ±0.57	0.32 ±0.34	-	-	0.02 ±0.01	0.05 ±0.05	0.14 ±0.11	2.54 ±0.37	0.04 ±0.03	0.01 ±0.01	0.47 ±0.24	0.02 ±0.01	0.00 ±0.00	0.54 ±0.06	3.08 ±0.28
Soybean	5.45 ±0.52	1.17 ±0.62	1.45 ±0.31	0.52 ±0.37	0.07 ±0.06	0.16 ±0.11	0.14 ±0.10	8.95 ±0.38	0.12 ±0.06	0.14 ±0.04	0.34 ±0.19	0.15 ±0.04	0.25 ±0.01	1.00 ±0.07	9.96 ±0.30
Safflower	1.08 ±0.53	0.49 ±0.36	0.49 ±0.46	0.25 ±0.14	0.11 ±0.02	0.20 ±0.11	0.17 ±0.12	2.80 ±0.35	0.01 ±0.00	0.02 ±0.01	0.30 ±0.13	-	0.11 ±0.00	0.44 ±0.09	3.24 ±0.27
Sunflower	0.96 ±0.22	0.40 ±0.07	1.76 ±0.30	0.25 ±0.19	0.11 ±0.07	0.10 ±0.10	0.07 ±0.05	3.65 ±0.18	0.04 ±0.02	0.02 ±0.01	0.16 ±0.06	0.05 ±0.01	1.34 ±0.49	1.61 ±0.21	5.26 ±0.19
Groundnut	2.05 ±0.60	0.52 ±0.28	1.64 ±0.51	0.43 ±0.25	0.13 ±0.13	0.19 ±0.08	0.12 ±0.07	5.09 ±0.37	0.09 ±0.04	0.06 ±0.01	0.44 ±0.15	0.06 ±0.17	0.30 ±0.14	0.95 ±0.11	6.03 ±0.28

Table 2.3 Estimates of cereal grain losses in percent by various studies

S.No	Name of crop	Years of reference	No. of studies	Min	Max	Average	SD
1	Paddy	2011, 2012, 2014, 2015, 2016, 2018	6	5.19	13.50	9.45	3.38
2	Wheat	2012, 2014, 2015, 2017 (2)	5	4.93	26.50	15.72	6.78
3	Maize	2012, 2013, 2015 (2), 2018	5	4.10	13.45	8.78	5.74
4	Bajra	2012, 2015	2	4.80	5.23	5.02	-
5	Sorghum	2012, 2015	2	3.87	5.99	4.93	-

2.3 Post-harvest losses of pulses

Pulses play an important role in human diet as a source of protein. India imports considerable quantities of pulses to meet its demand. The reduction in post-harvest losses will help in increasing the availability of pulses in the country. Earlier studies have shown that considerable quantities of pulses are lost during the supply chain. Shah (2014) conducted an empirical assessment study on pre and post-harvest losses of pigeon pea in Maharashtra and reported that 14 to 18 percent of the total production of pigeon pea was lost during the supply chain. Abass et al. (2013) assessed post-harvest handling practices and food losses in a maize-based farming system in semi-arid areas of Central and Northern Tanzania. The study revealed an estimated loss of 10 percent in pigeon pea for manual methods of processing. The losses were in the form of mechanical damage such as breakage, puncture, compression, rupture, dent bruises, or spillages of the crops

Chichaybelu (2014) conducted a study on chickpea in four regions namely Tigray, Amhara, Oromia & SNNP of Ethiopia. They concluded that on an average 2 percent of chickpea grain was lost during harvesting and 0.01 percent during cleaning. In packaging & transportation, the losses were not significant. The commonly used packing material was polyethylene bags and they lost an average of 0.32 percent chickpea grain during transportation. Among various operations, the largest grain loss was estimated at storage i.e. 2 percent. Farmers from Oromia region reported the largest storage loss at 4 percent, while the least loss of 1.5 percent was reported by farmers from SNNP region. Average storage losses were 2.5 percent.

Mannan et al. (2011) conducted a study on the assessment of storage losses of different pulses at the farmers' level in Jamalpur region of Bangladesh. They found that the highest moisture content was 17.9 percent in case of pulses stored in a jute bag and the lowest (6.3%) in case of jute bag along with polythene. The percent infestation of black gram was the highest (13.7%), when it was stored in a jute bag and the lowest (7.7%), when it was stored in the jute bag along with polythene. Similarly, percent weight loss was maximum (26.3%) in case of jute bags followed by 23.1 percent loss in case of synthetic bags and the lowest (4.8 %) in case of jute bags along with polythene.

Vishwakarma et al. (2019) found that the extent of post-harvest losses in green gram during farm operations ranged between 2.28 percent in ACZ3 and 6.10 percent in ACZ7. Most of the post-harvest losses occurred due to threshing and harvesting operations. The post-harvest losses from threshing ranged from 0.58 percent in ACZ3 to 2.69 percent in ACZ4. The post-harvest loss during storage at the farm ranged from 0.73 percent in ACZ9 to 1.07 percent in ACZ7. The major reasons for storage loss were unscientific methods of storage and insect attacks. The maximum post-harvest losses at the wholesale level were found to be 1.09 percent, which was due to poor storage facilities in the markets. At the retailer level, the post-harvest loss was assessed to be highest in ACZ7 (0.54%), followed by ACZ13 (0.28%).

ICAR-CIPHET (2012) and ICAR-CIPHET (2015) in studies on harvest and post-harvest losses reported losses for pulses such as pigeon pea, chickpea, black gram and green gram at the national level. ICAR-CIPHET (2012) reported losses of 4.28-6.06 percent in pulses, while ICAR-CIPHET (2015) reported losses of 4.60-8.41 percent. The estimated crop wise losses are indicated in **Table 2.1** and **Table 2.2**.

The overall losses for pulse reported over the years by various authors in India and abroad have been summarised and given in **Table 2.4**. It can be seen that the losses varied from 6.06 percent in green gram to 9.25 percent in pigeon pea.

Table 2.4 Estimates of losses of pulses in percent by various studies

S.No	Name of crop	Years of reference	No. of studies	Min	Max	Average	SD
1	Pigeon pea	2013, 2014, 2015	3	5.39	16.00	9.25	5.87
2	Chick pea	2012, 2015	2	4.28	8.41	6.35	-
3	Black gram	2012, 2015	2	6.04	7.07	6.56	-
4	Green gram	2012, 2015	2	5.51	6.6	6.06	-

2.4 Post-harvest losses of oilseeds

India is one of the largest importers of edible oils as its domestic production is unable to meet its domestic demand. The country has to rely heavily on imports to meet the gap between demand and supply. Around 56-60 percent of the edible oils consumed in the country is met through imports. In this backdrop, not only domestic production of edible oil is important but the reduction in post-harvest losses in oilseed crops is equally important. Cardoen et al. (2015) in the study on agriculture biomass in India and post-harvest losses, reported that during transport, the losses in cottonseed were around 5 percent and the total post-harvest losses were 10 percent along the cotton-seed supply chain. Another study was conducted by Kamran et al. (2020) on harvesting and post-harvest management for improving seed quality and subsequent crop yield of cotton. The study reported that the drying of cottonseed with zeolite beads reduced seed moisture content to lower levels than could be attained by sun-drying. There was a linear increase in moisture throughout the storage period in cloth bags and in cold storage, however, when cotton seeds were placed in hermetic bags, moisture entry to seeds was efficiently resisted throughout storage. Searcy et al. (2010) in their report on seed cotton handling and storage indicated that use of mechanical harvester and module builder increased the effectiveness of harvesting and also allowed to store cotton in the farm for longer periods before sending to gins without exposing cotton to undesirable weather thereby reducing losses and improving the yield.

According to the study conducted by Kannan (2014), the total post-harvest losses of soybean for the states like Madhya Pradesh, Maharashtra and Rajasthan were 12.56 percent 3.7 percent and 3.4 percent, respectively. Shah (2013) conducted a study for assessment of pre and post-harvest losses in Tur and soybean crops in Maharashtra. The study reported total post-harvest loss for soybean as 3.66 percent, which covered losses during different operations as 1.13 percent in harvesting, 0.50 percent in threshing, 0.36 percent in winnowing, 0.46 percent in transportation, 0.63 percent in handling and 0.58 percent in storage. Dutta et al. (2013) reported that the total loss at the farm level for soybean crop was 3.41 percent covering harvesting, threshing, winnowing, transportation, handling and storage operations in Rajasthan. A study by Fine et al. (2015) on food losses and waste in the French oil crops sector, reported 6 percent in soybean and 2 percent in sunflower crops.

FAO (2019) in its study conducted in Uganda for food loss analysis indicated the losses for sunflower at various critical loss points (CLP) at various levels. The critical losses at farmer level were reported during harvesting (2.50%), threshing (2.40%), drying (3.00%) and storage (2.00%). At trader and cooperatives level, the losses were reported around 1.80 percent and at small scale millers, major losses were reported during milling i.e. 1.5 percent.

Sugri et al. (2021) in their study on post-harvest losses and mitigating technologies reported 8.9 percent post-harvest storage losses on-farm for groundnut. In a study conducted by FAO (2020) estimated harvest and post-harvest losses of groundnut in Malawi in two agriculture development districts. The losses reported were 7.2 percent at harvesting, 3.2 percent during threshing/shelling, 2.9 percent during winnowing/cleaning, 3.6 percent during drying and 4.9 percent at storage level. In another study conducted by Tsusaka et al. (2018) on on-farm assessment of post-harvest losses of groundnut in Malawi reported that the total harvest and post-harvest losses incurred at farmers' level were 13.7 percent of the harvest. Cardoen et al. (2015) in their review study on agriculture biomass in India and post-harvest losses, reported that during transport, the losses in groundnut were around 2 percent and the total post-harvest losses were 10 percent.

The estimates of harvest and post-harvest losses of oilseeds at various farm operations and market channels at the national level were reported by ICAR-CIPHET (2012) and ICAR-CIPHET (2015). ICAR-CIPHET (2012) indicated losses of 2.76 percent in cottonseed to 10.07 percent in groundnut. The post-harvest losses reported by ICAR-CIPHET (2015) ranged between 3.08 percent in cottonseed to 9.96 percent in soybean. The estimated losses are indicated in **Table 2.1** and **Table 2.2,** respectively.

The overall losses for oilseed crops reported over the years by various authors in India and abroad have been summarised and given in **Table 2.5.** It can be seen that the losses varied from 2.92 percent in cottonseed crop to 8.70 percent in groundnut.

Table 2.5 Estimates of losses of oilseeds in percent by various studies

S. No.	Name of crop	Years of reference	No. of studies	Min	Max	Average	SD
1	Mustard	2012, 2015	2	5.34	8.89	7.12	2.51
2	Cottonseed	2012, 2015	2	2.75	3.08	2.92	-
3	Soybean	2012, 2013, 2014, 2015	4	3.41	12.56	6.98	3.61
4	Safflower	2012, 2015	2	3.24	3.68	3.46	-
5	Sunflower	2012, 2015	2	4.55	5.26	4.91	-
6	Groundnut	2015 (2), 2012	3	6.03	10.07	8.70	2.31

2.5 Post-harvest losses of fruits

Fruits being rich sources of vitamins, minerals and micronutrients, plays an important role in human nutrition. Being highly perishable, they are prone to high post-harvest losses that require effective post-harvest management solutions. Rinchenz et al. (2019) conducted a study on post-harvest damage and loss of apples in eleven major apple-growing villages in the Thimphu and Paro districts of Bhutan. The study findings revealed 12.78 percent post-harvest loss, of which 7.60 percent loss was due to diseases, physiological disorders, and insect and bird damages and the remaining 5.18 percent damage was observed in the handling operations. They further reported that 0.36 percent loss was caused due to improper harvesting equipment & bruising; 0.02 percent loss caused due to improper packaging material; 2.70 percent loss caused due to field to depot transportation & 2.10 percent depot to market operation. Mohammad (2011) conducted a study on post-harvest losses in apples in the Nerkh District, Afghanistan and found that the overall loss was approximately 31 percent of the total produce. At disaggregate levels, 5.8 percent loss occurred due to improper harvesting; 11.2 percent loss due to improper packaging material; 14 percent loss caused at storage level for storing after 2.5 months due to pest & disease, decay, weight losses, off-flavour, and over-ripening.

Luiz et al. (2021) conducted a study on characterization and quantification of post-harvest losses of apple fruit stored under commercial conditions in Santa Catarina state, Brazil by observation method. Two experiments were conducted using Gala & Fuji apples. The first experiment was performed from 2007 to 2010 to characterise and quantify apple losses during long-term CA storage and shelf life periods by treating the apples with 1-methyl cyclopropane (1-MCP). The second experiment was conducted in 2011 to identify the main pathogenic fungi responsible for apple fruit decay. Depending on the year and 1-MCP treatment, total apple losses observed in "Gala" during storage varied from 3.9 percent to 12.1 percent, while in 'Fuji', it varied from 6.6 percent to 8.4 percent. During storage, deterioration caused by fungal decay was 60 percent and 80 percent of total losses for 'Gala' and 'Fuji', respectively. During shelf life, additional losses caused by fungal decay ranged from 8.4 percent to 17.6 percent for 'Gala' and 12.4 percent to 27.2 percent for 'Fuji', depending on the year. Senescent breakdown and superficial scald were the major physiological disorders.

FAO (2018(a)) in a study on managing quality, assuring safety and reducing post-harvest losses in fruit and vegetable supply chains in South Asian Countries reported 28.8 percent post-harvest loss in the traditional banana supply chain in Sri Lanka, wherein 9 percent losses were observed at the farmer level, 5.4 percent at wholesaler level and 14.41 percent at the retailer level. Ilyas et al. (2007) reported that the estimated losses in bananas transported from Nawabshah, Mirpur Khas and Hyderabad to Faisalabad market in the months of December, February and March amounted to 37, 39 and 43 percent, respectively, Kumar and Kumari (2018) in their study on assessment of post-harvest losses in banana in Vaishali district of Bihar found losses of 2.90 percent in banana due to disease, inefficient harvesting techniques, transportation and loading, packing and packaging, as well erratic post-harvest techniques leading to major losses physically and economically. Abewoy et al. (2021) conducted a study to assess the extent of post-harvest losses of bananas along the wholesaler and retailer supply chain at Jimma town, Ethiopia. The study reported 26.5 percent post-harvest losses due to mechanical damage, improper transport, improper storage, improper ripening and mechanical damage.

Rana et al. (2005) conducted a study on estimation of post-harvest losses in Kinnow in Punjab, Haryana and Himachal Pradesh. They concluded that the losses in Kinnow were due to rotting (4.9%), damage during plucking (1.6%) and other losses (0.1%). Losses at commission/forwarding agent level were reported as 5.1 percent, which were mainly due to pressing during packing and damage due to transportation. Gangwar et al. (2007) conducted a study on estimation of post-harvest losses in Kinnow mandarin in Punjab and reported losses of 10.63 percent at orchard level.

The damaged fruit segregated during sorting, grading and packing were accounted as post-harvest loss at orchard level. Kinnow from Punjab were mainly marketed to distant markets in Delhi (Azadpur) and Karnataka (Bangalore). The losses in Kinnow due to transportation and wholesale level storage at Azadpur were 2.30 percent and post-harvest losses due to transportation and wholesale level storage at Bangalore were found to be 5.70 percent. The post-harvest losses at retailers' level in Azadpur and Bangalore markets were estimated at 10.06 percent and 13.7 percent, respectively majorly due to press/ bumped and physical injury.

Musasa et al. (2013) conducted a study on post-harvest losses in oranges in Rusitu Valley, Zimbabwe. They found that 40 percent of the mature fruits from Rusitu citrus farmers were lost in the field during temporary storage. A report was compiled at IIHR, Bangalore by Narayana et al. (2014) from findings of various studies on post-harvest losses in selected fruits and vegetables in India. According to the report, the overall post-harvest losses in oranges were found to be 6.38 percent, of which 4.84 percent at farm level and 1.54 percent in storage. The total losses reported in mandarins were 3.00 percent in Sikkim and 6.15 percent in Assam from the farm to the retail level. IIHR, Bangalore, 2014 reported the post-harvest losses of 3.5 percent in Coorg mandarin. While, the post-harvest losses in Nagpur mandarin from Nagpur, Amravati and Wardha were reported to be 8.39 percent. In market packed fruit, losses ranged from 18.34 percent to 23.48 percent when transported by trucks and 21.95 percent to 24.59 percent by train up to the Delhi market. The losses in farmhouse packed fruit were 15.6 percent to 20.73 percent by trucks and 19.21 percent to 21.85 percent by train transport up to Delhi. The losses in acid limes from Nagpur and Akola were assessed and the losses at Nagpur were found to be 8.21 percent at the farm, 8.58 percent at the wholesaler and 20.50 percent at retailers' level, aggregating to 37.29 percent. Similarly, in the Akola region, the losses at the farm were found to be 5.47 percent, at the wholesale level it was 7.04 percent and at a retail level loss was 10.50 percent, thus aggregating to 23.41 percent. While the losses in acid lime in the Ahmednagar region were found to be 10 percent. The causes of post-harvest losses indicated in various studies conducted were pests, diseases, poor production practices, harvesting injury, rough handling, and retention of fruit after harvest, poor temporary storage facilities, and poor physical infrastructure.

A study by Ladaniya (2015) on post-harvest management of citrus fruit in South Asian countries indicated that the total losses from farm to retail level were 25 to 35 percent in South Asian countries like India, Pakistan, Nepal, Bangladesh, Bhutan and Sri Lanka. Kitinoja and Kader (2015) in their study reported that the average losses for oranges in Benin at the farm level were 10 percent, at the wholesale level losses were 11.6 percent and at the retail level, the losses were 10.9 percent. The study reported an overall post-harvest loss of 32.5 percent. Hossain et al. (2017) reported post-harvest losses of 9.4 percent at farm level and 10.60 percent at traders' level in oranges in Bangladesh. The total post-harvest losses were 20 percent. Insect attack, spoilage of fruit at farm level, poor storage facilities, damage during transportation and delayed selling of fruits at traders' level were reported as major reasons of losses. Kasso and Bekele (2018) found that the estimated losses in oranges were 35.58 percent and losses in mandarin were 34.25 percent in Dire Dawa Region, Ethiopia.

FAO (2018 (a)) reported post-harvest losses of 7.25 percent at wholesale level and 13.04 percent at retail level in mandarin in South Asian Countries. At the wholesale level, the losses were due to transport and packaging related damage (4.35%), poor quality (2.20%) and weight loss (0.70%) in fruit while at retail level, the losses were due to weight loss (6.84%) and deterioration (6.20%) of the fruit. Overall, post-harvest losses were 20.29 percent. Porat et al. (2018) in their study reported a grading loss of 3 percent, packaging loss of 0.5 percent and retail waste of 2-2.5 percent in citrus. The study indicated inadequate cold storage, inadequate maintenance of the cold chain and overstocking as the main causes of losses at retail level while lack of planning, over-purchasing and poor home storage at consumers' level.

Murthy et al. (2004) in their study on post-harvest losses in grape at various levels in the Bijapur district of Karnataka reported a loss of 7.31 percent at field level, 4.24 percent during transit & local wholesale market and 10.80 percent during transit and distant wholesale market, 2.85 percent during local retail market and 3.27 percent during distant retail marketing. The total aggregate loss in the district ranged from 14.4 percent in the local retail market to 21.3 percent in the distant market. Aujla et al. (2011) in their study on post-harvest losses and marketing of grapes in Pakistan reported an aggregate post-harvest losses ranging from 16-23 percent. Murthy et al. (2014) reported a loss of 3.40 percent in Thompson seedless variety of grape in Andhra Pradesh. At retailer level, the loss was 4.56 percent. For the export market, the losses were estimated as 7.82 percent at field level and 12.13 percent at cold storage.

Sharma et al. (2018) reported various studies conducted on post-harvest losses of grapes in India found that post-harvest losses in grapes were within the range of 8.23 to 16 percent in the country. FAO (2021) conducted a study on food loss analysis for grape value chains over two years viz. 2016 and 2017 in Egypt. In 2016, losses reported were 18.6 percent at farm level, 5.7 percent at wholesale market level and 6.7 percent at retail market level. The losses reported in 2017 were 10.30 percent at farm level, 16.41 percent at wholesale market level and 19.05 percent at retail level. The aggregate losses were 31 percent in 2016, while 45.76 percent in 2017. According to various studies, causes of losses at farm level were improper harvesting, rough handling, bruising, physical damage, crushing of berries, bad weather conditions, damage due to insects, and sunburn; while at market level, causes were delay in selling, rough handling by consumers, open air markets, improper/ bad handling at earlier stages and packaging.

Bantayehu et al. (2017) conducted a study on post-harvest losses assessment in North-Western Ethiopia and reported an average loss of 16 percent at the producer level in guava. Major contributing factors for post-harvest losses were dropping of fruits on the ground from tall varieties, high temperature during harvesting, harvesting stick injury, harvesting immature fruits, and sunburning. Another study on post-harvest loss and quality deterioration of horticultural crops in Ethiopia's Dire Dawa Region by Mohammed et al. (2018) reported an estimated loss of 23.10 percent due to skin scratch, bruising, rotting, softening & discoloration. Mendes et al. (2019) conducted a study on post-harvest loss in guava in the retail trade at Chapadinha, Brazil and estimated a loss of 21 percent in supermarkets & 17 percent in groceries stores, which was due to the physiological loss (accelerated ripening). Gajanana et al. (2015) conducted a study for post-harvest loss of guava in Karnataka. The study reported 9.17 percent of the loss in guava at the field level due to over-ripening & physical damage and 4.12 percent loss at the retail level. The total post-harvest loss was observed at 13.29 percent. Gajanana et al. (2019) reported a total post-harvest loss (PHL) in pink flesh guava at 17.06 percent consisting of field level loss of 11.47 percent and retail level loss of 5.59 percent in the Bangalore market. However, the total loss at the distant market in Kerala was 8.19 percent. The post-harvest loss at the wholesalers' level was observed to be at 3.6 percent, which was due to pressed and crushed fruits during transit. At the retail level, the loss was 4.59 percent, mainly due to storage for more than two days, resulting in decaying, rotting, yellowing, etc. Further, the total loss at a distant market in Tamil Nadu was at 10.71 percent. The post-harvest loss at the wholesalers' level was observed to be 4.62 percent due to pressed and crushed fruits during transit. The retail level loss was 6.09 percent, mainly caused due to the pressing of fruits during handling.

WFLO report (2010) on identification of appropriate post-harvest technologies for improving market access and income for small horticultural farmers in Sub-Saharan Africa and South Asia had reported losses in mango. The physical losses in mango in Ghana were estimated to be 6 percent at farm level and 10.4 percent at wholesale market. The average percent losses at farm level were 3.6 percent, at wholesale level 5.26 percent and at retail level 4.5 percent. FAO (2018(b)) in the study for assessing food loss analysis for mango value chain in the Republic of Guyana reported that the losses at harvest level were 17.5 percent and at retail level post-harvest losses were 4.3 percent. Anonymous (2017) in a study on value chain analysis of mango in Chittoor district, Andhra Pradesh reported a total post-harvest loss from harvesting to consumption stage in the range of 20 - 30 percent in mango. During harvesting, the study reported a loss of 5-10 percent, 10 percent during cleaning, grading and sorting, 10-15 percent was recorded in ripening through traditional practice & 3-6 percent in ripening through ripening chambers. FAO (2016) in Chittoor district of Andhra Pradesh reported that there was a loss of 15 percent at harvesting due to improper harvesting, 5 percent during cleaning, grading and sorting, 15 percent during transportation due to long distance transportation & improper sorting & 5-10 percent at retail level due to lack of demand & improper handling of mango. Overall, the aggregate losses in the mango supply chain worked out to be as high as 40-45 percent in Andhra Pradesh.

Muhie et al. (2013) conducted a study on the post-harvest losses in papaya in South Wollo, Ethiopia. They found that the losses at the farmers' level, transportation and storage levels were 1.5 percent, 1 percent and 3.3 percent, respectively. Another study was conducted by Gajanana et al. (2010) on assessment of post-harvest loss during handling and marketing of papaya var. Taiwan 786 grown in Andhra Pradesh and marketed in Bangalore Karnataka. The authors estimated total losses of 25.49 percent in papaya consisting of 1.66 percent at the field level, 4.12 percent during transit and 8.22 percent during ripening at the market level and 11.49 percent at the retail level. Mohammed & Afework (2016) conducted a study in Dire Dawa region, Ethiopia and revealed post-harvest losses of 30.31 percent in papaya. Major causes for losses were fruit damages such as scratch, flaccid, decay, bleach, spot, compression, lesion and crash. Bantayehu et al. (2017) conducted a study on post-harvest losses assessment of tropical fruits in North Western Ethiopia. The study reported an average loss of 19 percent at producer level in papaya. The major contributing factors to post-harvest losses were dropping of fruit on ground from tall varieties, high temperature, harvesting stick injury, harvesting immature fruit, sun burning harvesting container damage and harvesting immature fruit. Ferreira et al. (2020) in their study conducted in different microregions of Maranhao, Brazil, reported post-harvest losses of 17.90 percent in papaya in the main retail markets of seven cities in Maranhao.

Gajanana et al. (2006) undertook a study in the Kolar district of Karnataka to assess the post-harvest losses in handling and marketing of sapota and reported post-harvest loss (PHL) of 15.98 percent in the wholesale marketing channel (WMS) and about 14.07 percent in HOPCOMS channel. The usage of mechanical harvester, pre harvest management of fruit for protection against fruit borer and opening of procurement centres of HOPCOMS in the producing region were suggested in order to reduce the PHL and also for improving the efficiency of the marketing system. Siddiqui et al. (2014) conducted a concise review of the post-harvest biology and technology of sapota and reported that the post-harvest losses were high (25–30%) in tropical countries. These losses occurred due to lack of proper storage facilities, improper handling, rapid ripening, and microbial spoilage.

Hassan et al. (2010) estimated that the overall post-harvest losses along the pineapple supply chain were 43 percent in Bangladesh. The study found a loss of 10.40 percent at grower level, 11.55 percent at the Bepari level, 14.1 percent at wholesalers' level and 6.97 percent at retailers' level. Gajanana et al. (2011) reported a total post-harvest loss of 29.25 percent in pineapple. The losses were 2.19 percent during sorting and grading, 16.39 percent losses during handling during wholesale marketing and 10.67 percent at retail level. Ningombam et al. (2019) reported that the total post-harvest loss was estimated at 32.12 percent in pineapple in Manipur. At the disaggregated level, the report found that a 6.82 percent loss was estimated at the farm level, 3.09 percent at the wholesale market including transportation, 5.64 percent at the storage level. At the retail and processing levels, the losses were estimated around 5.44 percent and 4.31 percent, respectively. The loss of 6.82 percent was observed at consumer level. Troger et al. (2020) in their study on pineapple value chain in Uganda indicated that the two major hotspots for the occurrence of damage were transportation and storage. The chances of damage during transportation was higher as compared to storage. In case of transportation, there were higher chances of damage in case of matured fruit, while during storage, losses occurred when the fruit were stored for a longer period due to problems in selling of the fruit. Bad roads were also one of the causes of damage as bicycles and vehicles broke down and the fruit got damaged due to sun or rotting.

Gajanana et al. (2011) reported a total post-harvest loss of 35.38 percent for pomegranate in South India. The losses reported were 9.86 percent at sorting and grading level, 10.10 percent at wholesale marketing level and 15.48 percent at retail marketing level. In another study conducted by Sudharshan et al. (2013) on pomegranate in Karnataka, the aggregate post-harvest loss in Bangalore market was estimated at 25.48 percent comprising 6 percent post-harvest losses at farm level, 8.09 percent at transit & wholesale market level and 11.39 percent at retail level. In Mangalore market, the aggregate post-harvest loss of 38.44 percent was estimated as 8 percent post-harvest losses at farm level, 12.4 percent at transit & wholesale market level and 18.00 percent at retail level. Dhinesh and Ramasamy (2016) in their review found that the post-harvest losses in pomegranate in India were very high in the range of 20-40 percent. Opara et al. (2021) conducted a study to assess post-harvest losses of pomegranate at pack house in the Western Cape province of South Africa. They estimated total loss ranging from 6.74 percent to 7.69 percent among all 3 cultivars (Acco, Hershkawitz, Wonderful) selected under the study. The cause of loss was rough handling resulting in fruit bruises, superficial injuries, and rejection of fruit due to the high market standard.

Kitinoja and Kader (2015) prepared a white paper, wherein they reported about a study, which was conducted in Ethiopia on assessment of post-harvest losses of fruits and vegetables. The study estimated post-harvest losses in melons to be 16.7 percent. Meena et al. (2019) in their study on scientific approaches in muskmelon cultivation from field to market suggested room cooling, icing and forced air cooling for pre-cooling, hot water treatment at 52°C-55°C for 2-3 minutes to reduce the risk of fungal infection caused by *Fusarium a*nd fruit fly, coating with different agents like chitosan, aloe vera gel, natural bee wax, shellac wax etc. by dipping into coating solutions for 10-20 minutes on the basis of type and nature of waxes to reduce moisture loss and maintain its freshness and firmness and storage around 7-10 °C as best practices to reduce post-harvest losses in muskmelon.

ICAR-CIPHET (2012) and ICAR-CIPHET (2015) estimated the losses of fruits such as apple, banana, citrus, grapes, guava, mango, papaya and sapota at the national level. ICAR-CIPHET (2012) indicated losses ranged from 5.77 percent in sapota to 18.05 percent in guava. ICAR-CIPHET (2015) reported losses ranged from 6.70 percent in papaya to 15.88 percent in guava. The estimated operation-wise losses for fruits and vegetables are indicated in **Table 2.6** and **Table 2.7** respectively.

The overall losses for fruits reported over the years by various authors in India and abroad have been summarised and given in **Table 2.8.** It can be seen that the losses varied from 14.12 percent in sapota to 34.68 percent in pineapple.

Table 2.6 Harvest and post-harvest losses (%) of fruits and vegetables at national level (ICAR-CIPHET 2012)

Crop	Farm Operations					Total loss in farm operations	Storage in Channels					Total loss in storage	Overall total loss
	Harvesting	Collection	Sorting/ Grading	Packaging	Transport		Farm	Godown	Wholes-alers	Retailers	Processing unit		
Apple	4.56 ±1.41	0.42 ±1.03	0.79 ±1.51	0.10 ±0.08	1.19 ±0.22	11.06 ±1.08	0.04 ±0.2	0.12 ±0.10	0.52 ±0.32	0.23 ±0.10	0.29 ±0.73	1.20 ±0.81	12.26 ±1.05
Banana	1.33 ±8.59	0.36 ±0.54	0.93 ±0.35	0.44 ±0.46	1.14 ±1.29	4.18 ±0.08	0.04 ±0.03	0.16 ±0.01	1.85 ±0.82	0.36 ±0.23	0.01 ±0.00	2.42 ±0.85	6.60 ±3.43
Citrus	0.92 ±1.39	0.48 ±1.01	1.79 ±1.16	0.35 ±0.94	1.30 ±1.67	4.84 ±1.30	0.03 ±0.07	0.00 ±0.00	0.69 ±0.35	0.77 ±0.27	0.01 ±0.02	1.54 ±0.45	6.38 ±1.15
Grapes	0.94 ±1.94	0.24 ±1.10	0.21 ±1.26	0.26 ±0.98	1.93 ±1.13	6.57 ±1.39	0.02 ±0.00	-	0.54 ±0.28	0.84 ±0.29	0.30 ±0.12	1.73 ±0.42	8.30 ±1.00
Guava	4.36 ±1.53	1.20 ±1.05	4.64 ±1.18	0.94 ±0.69	2.77 ±0.73	13.92 ±1.17	0.41 ±0.26	-	1.83 ±0.34	1.80 ±0.75	0.06 ±0.01	4.13 ±0.86	18.05 ±1.12
Mango	4.11 ±2.37	0.68 ±0.71	2.80 ±1.54	0.51 ±0.54	2.53 ±2.49	10.64 ±1.89	0.06 ±0.03	-	0.92 ±0.42	0.93 ±0.59	0.19 ±.09	2.11 ±0.73	12.74 ±1.57
Papaya	1.45 ±1.56	0.28 ±0.26	1.97 ±0.92	0.23 ±0.20	1.13 ±1.35	5.06 ±1.08	0.08 ±0.12	0.00 ±0.00	1.02 ±0.66	1.20 ±0.64	0.00 ±0.00	2.28 ±0.92	7.36 ±1.04
Sapota	1.53 ±1.14	0.23 ±0.52	1.43 ±0.83	0.08 ±0.51	1.06 ±0.67	4.31 ±0.89	0.02 ±0.00	-	0.75 ±0.25	0.73 ±0.32	-	1.46 ±0.40	5.77 ±0.69
Cabbage	1.08 ±2.84	0.30 ±0.39	1.64 ±2.10	0.27 ±0.64	1.32 ±0.79	4.61 ±1.71	0.14 ±0.15	0.06 ±0.04	0.88 ±0.27	1.21 ±0.39	0.03 ±0.02	2.33 ±0.50	6.94 ±1.51
Cauliflower	0.84 ±1.76	0.27 ±0.45	1.66 ±2.30	0.18 ±0.35	1.61 ±1.13	4.85 ±1.38	0.08 ±0.07	0.04 ±0.01	1.00 ±0.78	0.89 ±0.63	-	2.03 ±1.01	6.88 ±1.32
Green pea	3.46 ±2.63	1.08 ±1.07	3.30 ±2.22	0.23 ±0.50	0.50 ±0.54	8.58 ±1.78	0.06 ±0.09	0.01 ±0.00	0.72 ±1.05	0.92 ±0.25	-	1.70 ±1.08	10.28 ±1.67
Mushroom	1.37 ±2.45	1.77 ±0.08	4.26 ±4.59	1.64 ±2.10	2.00 ±0.85	11.03 ±2.96	-	-	-	1.51 ±0.78	-	1.51 ±0.78	12.54 ±1.40
Onion	2.70 ±0.54	0.23 ±0.49	1.64 ±1.37	0.14 ±0.56	0.44 ±0.58	5.17 ±1.47	0.54 ±0.40	0.40 ±0.17	0.83 ±0.20	0.57 ±0.22	0.01 ±0.00	2.34 ±0.53	7.51 ±1.09
Potato	0.18 ±4.02	0.69 ±0.23	2.23 ±3.41	0.10 ±0.14	0.54 ±0.36	6.73 ±2.53	0.35 ±0.19	0.76 ±0.35	0.96 ±0.15	0.19 ±0.05	0.01 ±0.00	2.26 ±0.43	8.99 ±1.87
Tomato	1.73 ±1.26	1.06 ±0.49	3.24 ±0.98	0.77 ±0.53	3.14 ±1.20	9.94 ±1.03	1.22 ±0.72	0.01 ±0.00	1.06 ±0.29	0.58 ±0.30	0.17 ±0.14	3.04 ±0.84	12.98 ±1.00
Tapioca	3.61 ±2.75	0.51 ±0.62	1.54 ±1.32	0.53 ±1.08	1.28 ±0.78	7.47 ±1.79	4.13 ±4.10	0.17 ±0.17	-	0.68 ±0.35	0.74 ±0.46	0.13 ±0.04	9.19 ±1.52

Table 2.7 Harvest and post-harvest losses (%) of fruits and vegetables at national level (ICAR-CIPHET 2015)

Crop	Farm Operations					Total loss in farm operations	Storage in Channels					Total loss in storage	Overall total loss
	Harvesting	Collection	Sorting/ Grading	Packaging	Transport		Farm	Godown	Wholes-alers	Retailers	Processing unit		
Apple	4.33 ±0.35	0.29 ±0.22	3.94 ±0.44	0.11 ±0.11	0.42 ±0.19	9.08 ±0.30	0.02 ±0.00	0.13 ±0.02	0.57 ±0.08	0.34 ±0.29	0.25 ±0.11	1.31 ±0.11	10.39 ±0.24
Banana	1.62 ±0.35	0.26 ±0.14	2.06 ±0.37	0.19 ±0.31	1.91 ±0.40	6.04 ±0.33	0.03 ±0.01	0.08 ±0.04	1.16 ±0.35	0.45 ±0.07	0.00 ±0.00	1.72 ±0.17	7.76 ±0.29
Citrus	1.68 ±0.48	0.33 ±0.10	3.71 ±0.59	0.18 ±0.13	1.65 ±0.24	7.55 ±0.38	0.04 ±0.01	0.02 ±0.01	0.91 ±0.13	1.12 ±0.20	0.06 ±0.07	2.14 ±0.11	9.69 ±0.29
Grapes	1.77 ±0.2	0.30 ±0.05	3.36 ±0.42	0.10 ±0.06	0.98 ±0.18	6.52 ±0.26	0.01 ±0.00	-	0.78 ±0.14	1.24 ±0.17	0.09 ±0.03	2.11 ±0.15	8.63 ±0.22
Guava	5.33 ±1.78	0.31 ±0.28	4.95 ±1.86	0.09 ±0.20	1.21 ±0.64	11.90 ±1.24	0.23 ±0.06	-	1.62 ±0.52	2.08 ±0.73	0.04 ±0.03	3.98 ±0.55	15.88 ±1.11
Mango	2.09 ±0.71	0.30 ±0.44	3.26 ±0.77	0.23 ±0.07	1.04 ±0.14	6.92 ±0.54	0.11 ±0.01	0.01 ±0.01	0.69 ±0.39	1.18 ±0.42	0.25 ±0.09	2.24 ±0.28	9.16 ±0.50
Papaya	0.98 ±0.45	0.42 ±0.15	1.46 ±0.47	0.34 ±0.06	0.92 ±0.39	4.12 ±0.37	0.05 ±0.02	0.01 ±0.00	0.79 ±0.17	1.71 ±0.25	0.03 ±0.01	2.58 ±0.12	6.70 ±0.26
Sapota	2.53 ±1.18	0.35 ±0.07	2.55 ±0.56	0.28 ±0.17	1.70 ±0.48	7.41 ±0.67	0.01 ±0.01	0.25 ±0.05	0.89 ±0.20	1.13 ±0.13	0.03 ±0.02	2.31 ±0.09	9.73 ±0.46
Cabbage	1.74 ±0.55	0.38 ±0.16	3.32 ±0.46	0.36 ±0.06	1.02 ±0.49	6.81 ±0.42	0.16 ±0.06	0.08 ±0.02	0.89 ±0.15	1.42 ±0.35	0.02 ±0.01	2.56 ±0.21	9.37 ±0.36
Cauliflower	2.21 ±0.79	0.26 ±0.38	3.78 ±0.48	0.38 ±0.13	0.92 ±0.49	7.55 ±0.52	0.09 ±0.07	0.07 ±0.05	0.83 ±0.25	1.00 ±0.38	0.00 ±0.00	2.00 ±0.22	9.56 ±0.45
Green pea	2.25 ±0.46	0.32 ±0.12	2.41 ±0.47	0.13 ±0.09	0.61 ±0.08	5.72 ±0.32	0.05 ±0.02	0.00 ±0.00	1.09 ±0.27	0.55 ±0.34	0.03 ±0.03	1.73 ±0.21	7.45 ±0.28
Mushroom	0.99 ±0.14	0.04 ±0.00	5.34 ±0.33	0.18 ±0.06	0.77 ±0.06	7.32 ±0.19	0.66 ±1.22	-	-	1.52 ±1.66	-	2.19 ±1.45	9.51 ±0.65
Onion	2.62 ±0.25	0.44 ±0.32	2.35 ±0.72	0.12 ±0.27	0.51 ±0.15	6.05 ±0.40	0.35 ±0.19	0.30 ±0.08	0.77 ±0.16	0.72 ±0.14	0.01 ±0.01	2.16 ±0.17	8.20 ±0.28
Potato	2.58 ±0.84	0.25 ±0.30	2.93 ±0.99	0.06 ±0.07	0.72 ±0.18	6.54 ±0.63	0.15 ±0.06	0.17 ±0.18	0.34 ±0.08	0.11 ±0.05	0.02 ±0.01	0.78 ±0.06	7.32 ±0.44
Tomato	3.16 ±0.53	0.52 ±0.22	3.74 ±0.48	0.24 ±0.14	1.75 ±0.29	9.41 ±0.39	0.12 ±0.04	-	1.26 ±0.20	1.63 ±0.33	0.02 ±0.00	3.03 ±0.23	12.44 ±0.35
Tapioca	1.23 ±0.39	0.30 ±0.07	0.99 ±0.16	0.09 ±0.05	0.61 ±0.18	3.22 ±0.24	0.28 ±0.23	-	0.31 ±0.06	0.59 ±0.20	0.17 ±0.02	1.36 ±0.20	4.58 ±0.23

Table 2.8 Estimates of losses of fruits in percent by various studies

S.No	Name of crop	Years of reference	No. of studies	Min	Max	Average	SD
1	Apple	2010, 2011, 2015, 2019	4	10.39	31	16.61	9.65
2	Banana	2007, 2012, 2015, 2018, 2021	5	2.9	39.67	18.70	14.95
3	Citrus	2005, 2007, 2010, 2013, 2014, 2015 (3), 2017, 2018 (3)	12	5.75	40.00	20.68	12.11
4	Grapes	2004, 2011, 2012, 2014, 2015, 2018 (2)	7	7.96	45.76	18.90	11.92
5	Guava	2010, 2015(2),2018, 2019	5	13.29	23.10	16.92	3.50
6	Mango	2010, 2015, 2016, 2018	4	9.16	45	22.83	17.24
7	Papaya	2010, 2012, 2013, 2015, 2016, 2017, 2020	7	5.8	30.31	16.08	9.77
8	Pomegranate	2011, 2013, 2016, 2021	4	25.48	38.40	32.33	5.74
9	Pineapple	2010, 2011, 2019	3	29.25	42.66	34.68	7.06
10	Sapota	2006, 2012, 2014, 2015.	4	5.77	25	14.12	8.38

2.6 Post-harvest losses of vegetables

Koza et al. (2018) while conducting a study on economics of cabbage cultivation and post-harvest losses in Chizami, block of Phek district of Nagaland estimated post-harvest loss of 17.14 percent for medium farmers, 14.20 percent for small and 11.94 percent for marginal farmers. At market level, the average loss was 25.37 percent at the retailer and 15.86 percent at wholesaler levels. Gonzales et al. (2016) conducted a study on post-harvest losses of cabbage in Cebu, Philippines. The total post-harvest loss of cabbage was almost similar in both traditional (26.4%) and modern supply chains (26.5%). However, farmers in the modern supply chain incurred a higher loss of 13.2 percent than traditional chain farmers of about 10.1 percent loss. Further, in traditional chains, wholesalers incurred 6.8 percent loss and retailers incurred 9.6 percent loss. In the modern chain, wholesalers and retailers incurred an average loss of 6 percent and 7.2 percent, respectively. Kitinoja et al. (2010) conducted a study on post-harvest losses of fruits and vegetables in Ghana. The study reported that there were 20.1 percent, 6.5 percent and 28.1 percent losses at farm level, wholesale market level and retail market level, respectively for cabbage. Chonhenchob et al. (2009) conducted a study on the effect of truck vibration during transport on damage to fresh produce shipments in Thailand. The study measured the transportation environment in truck shipments from various pack houses to major retail distribution centres and then the subsequent distribution to regional stores in smaller trucks. Vibration levels measured in truck shipments showed the highest levels in vertical direction, followed by lateral and longitudinal orientations. The highest level of vibration occurred from the grower to pack houses, due to the poor road conditions. The highest level of damage occurred between the fruits and vegetables being transported to the pack houses.

Singh et al. (2019) conducted a study on assessment of post-harvest losses of cole crops in Jaunpur District of Uttar Pradesh. At field level, average losses in cauliflower were 25-35 percent, 0.5 percent, 0.6 percent, 0.5 percent and 1.2 percent due to disease and insect pest incidence, mechanical injury, rotting, packaging and transportation, respectively. At wholesale level, the losses were 4 percent, 0.5 percent, 1.25 percent, 0.50 percent and 2 percent during storage, packaging, transportation and sorting, respectively. At retail level, the post-harvest losses were 2 percent, 0.5 percent and 2.15 percent during sorting, transportation and mechanical injury, respectively. FAO (2018(a)) reported a total loss of 52 percent in the traditional supply chain of cauliflower in Dhanding district of Nepal, which comprised 11 percent losses during transportation and 41 percent during wholesale marketing. The loss in the improved supply chain was 18.3 percent, which consisted of 4.5 percent during transportation & wholesale marketing and 13.8 percent during retail marketing. Udas et al. (2005) in their study in Eastern hills of Nepal found 6 percent loss at farm level and 41 percent loss at retail level in cauliflower.

Kumar et al. (2015) examined the nature and extent of post-harvest losses in the vegetable supply chain in the Varanasi district of Uttar Pradesh and reported 7.78 percent post-harvest loss at producer level and 5.27 percent at market level in green peas. In another study by Sharma and Singh, (2011) post-harvest losses of 10.06 percent at farm level and 6.31 percent at market level were found in green peas in Kumaon Division of Uttarakhand. The losses during harvesting (4.87 %), grading and packing (1.05%), transportation (3.70%) and marketing (0.44%) contributed to total farm level losses. While losses during loading/ unloading (1.79%), transportation (0.80%), grading (0.73%) and selling (4.71%) made up the total losses at market level. FAO and Asian Productivity Organization (2006) conducted a study on post-harvest management of fruits and vegetables in the Asia-Pacific region. They found that most of the physical loss for selected vegetables was during transportation in the Philippines. The losses for green peas during refrigerated transport were divided into two parts, cold storage loss of 10 percent and the loss of 13.5 percent under ambient storage conditions.

Thakur et al. (2022) reported that the washing or pre-treatment, packaging, transport and marketing were some of the important standardised techniques for short-term storage of mushrooms. They specified that the value addition of mushrooms enhanced the quality and addressed the demand for ready-made or ready-to-make food products. In another study by Harun (2017) reported that mushroom after harvesting undergoes respiration and hence it becomes mature and then gets spoiled if stored for long duration. Das et al. (2010) studied the efficacy of washing (control and 5% hydrogen peroxide) and post-harvest treatments (control, hot water treatment, wrapping with non-perforated and perforated plastic bags, wrapping with brown paper bag and wrapping with aluminium foil) on shelf life and quality of oyster mushroom. The study reported that the post-harvest treatments caused significant effects to influence weight loss, dry matter and protein contents, disease incidence and shelf life of mushrooms. At the 7th day of storage, the highest weight loss (98.08%) was recorded in the untreated mushroom, whereas the lowest weight loss (33.62%) was observed in the mushroom wrapped in a perforated plastic bag. Unwashed mushrooms held at ambient temperature had the highest incidence of disease. The longest shelf life of 6 days was recorded in mushrooms wrapped in unperforated plastic bags, whereas the shortest shelf life (3 days) was found in the untreated mushroom.

Emana et al. (2017) in their study on the assessment of post-harvest losses and marketing of onion in Ethiopia, found that over 30 percent of post-harvest losses were registered due to poor cultural practices and disease attack during production, harvesting, poor handling practices, storage and transport. In another study by Calica and Cabanayan (2018) on the assessment of the post-harvest systems and losses of bulb onions in Nueva Ecija, Philippines, it was reported that the industry was incurring post-harvest physical losses of 31.49 percent and quality losses of 32.41 percent. Chattha et al. (2017) conducted a study on cultivators associating post-harvest losses of onion in Sindh's Mirpurkhas district of Pakistan. They found that most of the post-harvest loss in the region was due to the lack of information to farmers across the supply chain. Kitinoja and Kader (2015) in their study estimated that the post-harvest physical loss due to various reasons for onion in Niger, Pakistan, and Egypt were 15 percent, 9 percent and 19 percent, respectively. FAO (2010) estimated post-harvest loss of 14 percent and 13 percent in onion in the major and minor growing season in Ghana, respectively. Porat et al. (2018) conducted a study in the United

Kingdom and reported post-harvest loss of 3-5 percent at farm level, 9-20 percent during sorting-grading, 3-10 percent during storage and 2-3 percent during packaging in retail of onion.

Raghuvanshi et al. (2018) conducted a study on post-harvest losses in potatoes in Chhattisgarh. At farm level, losses were estimated to be 8.07 percent followed by grading and packaging being 11.65 percent. At market level, losses were 3.27 percent, 2.68 percent and 19.79 at wholesaler and retailer and storage level, respectively. Tadesse et al. (2018) conducted a study on assessment of post-harvest loss along potato value chain in Mhasa District of Southwest Ethiopia, Sheka Zone. The study found that the post-harvest losses in potato value chain were 21.72, 0.59, 0.655 and 1.92 percent at producer, local trader, and wholesaler and retailer level respectively. Delgado et al. (2017) conducted a study to measure food loss across food value chains in Ecuador and Peru. The study used four methods viz., self-reported method, category method, attribute method and price method. In Ecuador, the post-harvest losses at producer level were 8.11 percent, 12.82 percent, 12.17 percent and 11.84 percent. At middlemen level post-harvest losses were 1.70 percent, 0.91 percent, 1.77 percent and 1.52 percent. At wholesale level post-harvest losses were 2.54 percent, 2.54 percent, 2.45 percent and 2.45 percent. At the entire value chain level, the post-harvest losses were 11.50 percent, 16.18 percent, 16.39 percent and 15.80 percent. In Peru, the post-harvest losses were 9.38 percent, 15.99 percent, 19.62 percent and 19.84 percent at producer level. At middlemen level, the post-harvest losses were 1.22 percent, 1.60 percent, 3.72 percent and 2.05 percent. At wholesale level the post-harvest losses were 2.27 percent, 2.27 percent, 2.27 percent and 2.27 percent. At the entire value chain level 12.87 percent, 19.86 percent, 25.62 percent and 24.17 percent losses were estimated.

FAO (2018(c)) conducted a study for assessing food loss analysis for tomato value chain in the Republic of Trinidad and Tobago and reported a loss of 7 percent at harvesting, 8 percent during pack house operations, and 12 percent at the marketing level. Overall, the aggregate losses worked out to be as high as 27 percent in the Republic of Trinidad and Tobago. WFLO Report (2010) reported the physical losses of 25.1 percent at farm level, 21.5 percent at wholesale market, and 23 percent at retail market for tomato in Ghana. In Benin, the physical losses in tomato were estimated to be 23 percent at farm level, 31.2 percent at wholesale market, and 26.4 percent at retail market. In Rwanda, the physical losses were estimated to be 7.8 percent at farm level, 10.7 percent at wholesale market, and 14.7 percent at retail market. FAO (2018(d)) reported that the total post-harvest losses for tomatoes were measured at 34 percent in the Republic of Guyana. Physical damage leading to unmarketable fruit was estimated at 9.5 percent, physiological losses at 7.5 percent, pathological and entomological losses at 17 percent. Tiwari et al. (2020) conducted a study on post-harvest practices and loss assessment in tomato in Kathmandu, Nepal. The results show that 10 percent of total losses were found from harvesting to marketing. In the field, there was a 2 percent loss. However, 4 percent loss was found during the transportation of tomato and 2 percent of the loss during the storage of tomato. Sharma (2018) conducted a study on post-harvest losses of tomato value chain in select districts of Bangladesh. The study indicated that there was 9.25 percent post-harvest loss at farmers' level, 1.70 percent at traders' level, 1.98 percent at wholesalers' level, 3.99 percent at retailers, 5.35 percent at processors level, and 2.36 percent consumers level, respectively. Overall, the total loss after harvest was estimated at 22.93 percent along with the tomato value chain in the study area. Sarkar et al. (2019) conducted a study on assessment of pre- and post-harvest losses of vegetables in Kanpur district of Uttar Pradesh. The results show that there was a pre harvest loss at farm level of 5.13 percent, post-harvest loss at farm level of 6.18 percent and total loss at trader's level was 10.62 percent. The overall post-harvest loss was observed at 21.93 percent.

Parmar et al. (2018) conducted a study on post-harvest management and associated post food losses and by-products of cassava in southern Ethiopia. It was found that critical loss points in tapioca were sun-drying (4%) and stockpiling at farms and the marketplace (30-50%). FAO (2018(e)) in its study in the Republic of Trinidad and Tobago indicated 6.5 percent loss at harvesting, 2 percent at pack house, 14.5 percent at retailing, 1.5 percent at supermarkets and 0.5 percent at wholesale markets. Another study conducted by FAO (2018(f)) in Guyana indicated 6.5 percent loss at harvesting stage, 2 percent loss attributed to damage at packaging and transport and 14.5 percent loss at retailing for tapioca. FAO (2018(g)) in its study in Santa Lucia identified critical loss points as 10 to 12 percent at processing level.

Sarkar et al. (2020) conducted an assessment of pre-and post-harvest losses of vegetables in the Kanpur district of Uttar Pradesh and reported an overall loss of 13.20 percent for bottle gourd with 2.97 percent at the farm level and 10.23 percent at market level. At the farm level, losses during market preparation and transportation were observed, while at trader level, damage during handling, and sorted out thrown out weight was observed.

Mitrannavar and Yeledalli (2014) study found that at the farm level, losses in brinjal were 6.63 percent and at the commission agent-cum-wholesaler and retail level the losses were 7.42 percent and 7.56 percent, respectively in Karnataka. Therefore, the overall post-harvest losses were estimated to be 21.61 percent in brinjal. Kumar et al. (2015) in their study found that the overall post-harvest loss was 15.28 percent in brinjal with 9.57 percent loss at producer level and 5.71 percent at trader level. Kaysar et al. (2016) conducted a study on post-harvest loss assessment of brinjal in some selected areas of Bangladesh. They reported a total loss of 23.38 percent with the highest loss of 12.51 percent at the growers' level due to causes such as insect damage, rotten damage, cleaning, storage and transportation. A loss of 2.26 percent at Aratdar level, 2.65 percent at Bepari (wholesale) level and 5.96 percent at the retail level was observed in brinjal. Another study conducted by Khatun and Rahman (2019) in Bangladesh reported 13.90 percent post-harvest loss of brinjal at farm level. The causes of post-harvest losses were insect infestation, rotting, disease, bird attack, skinning, bruising, and absence of storage system. Sarkar et al. (2020) reported an overall loss of 14.80 percent in brinjal in the Kanpur district of Uttar Pradesh with 6.45 percent at the farm level, 8.35 percent at the trader level.

Mitrannavar and Yeledalli (2014) carried out a management appraisal study on post-harvest losses of major vegetables in Karnataka. A loss of 6.63 percent was observed at farm level for beans, while at commission agent-cum-wholesaler level, it was observed to be 8.76 percent. The losses assessed at the retail level were 6.97 percent in beans. The overall loss estimated by the study was 22.36 percent in beans in Karnataka. Another study conducted by Tibagonzeka et al. (2018) estimated a loss of 10.28 percent during harvesting level, 15.43 percent during storage level and 4.99 percent during transportation level. A total loss of 30.7 percent was observed across the crop value chain. FAO (2018(a)) indicated an overall post-harvest loss of 27.91 percent for snap beans in Sri Lanka. Post-harvest losses at wholesale and retail level were 7.3 percent and 20.61 percent, respectively.

Sharma and Singh (2011) in their study on economic analysis of post-harvest losses in marketing of vegetables in Uttarakhand reported an overall post-harvest loss of 6.52 percent in radish with 3.89 percent at grower level and 2.63 percent at retail level. Another study conducted by Ayub et al. (2013) to ascertain the post-harvest preservation of crimson giant radishes. The radish was stored in two different conditions viz. at room temperature (20 ± 2°C and 75 % relative humidity) with or without leaves, and at refrigeration temperature of 5°C and 90 percent relative humidity with leaves, without leaves or minimally processed. The study reported weight losses exceeding 5 percent after one day in case of radish stored at room temperature, while at the refrigeration temperature of 5°C, minimally processed radish bulbs could be stored for more than six days and non-processed not more than one day. Kumar et al. (2015) conducted a study on the extent of physical post-harvest losses of important vegetables of Varanasi in Uttar Pradesh and found that the overall post-harvest losses observed were 12.68 percent in radish with 7.43 percent at producer level and 5.25 percent at trader level. Chandra et al. (2018) conducted a study on the effects of packaging on shelf life and post-harvest qualities of radish roots during storage at low temperature for an extended period. The study estimated the effect of packaging during long-term storage on water loss of radish in South Korea. They found that 18 percent water loss during storage was observed in non-packaged radishes, whereas 10 percent water loss was observed in case of packaged radishes. The study also estimated that these losses can be reduced to about 2.2 percent and 3 percent if radishes are given a curing + High Density Polyethylene (HDPE) + Plastic Crates (PC) treatment during the storage period.

Kumar et al. (2015) examined the extent of post-harvest losses in the vegetable supply chain in the Varanasi district of Uttar Pradesh. Post-harvest losses of 9.09 percent at producer level and 5.86 percent at market level were found in capsicum in this study. Another study conducted by Sharma and Singh (2011) on nature and extent of post-harvest losses in vegetable supply chain in the Kumaon division of Uttarakhand reported post-harvest losses of 4.59 percent at farm level and 5.84 percent at market level for capsicum. Harvesting (1.93%), grading and packing (0.89 %) and transportation (1.78%) mainly contributed to total farm level losses. While losses during loading/ unloading (0.35%), transportation (0.37%), grading (1.15%) and selling (3.97%) made up the total losses at market level. Santos et al. (2020) in their study on post-harvest losses of fruits and vegetables in supply centres in Salvador, Brazil reported a total loss of 33.33 percent in capsicum, which was estimated as per the traders for bell peppers in the supply chain centres. Kitinoja and Kader (2015) in their study on measuring the post-harvest losses of fresh fruits and vegetables in developing countries found a loss of 12 percent at the farm level and around 15 percent loss during transit in Nigeria in capsicum. They estimated a total loss of 30 percent across the whole supply chain in Brazil. In another study conducted by Olayemi et al. (2010) assessed the post-harvest challenges of small scale farm holders of bell pepper (*Capsicum annuum*) and hot pepper (*Capsicum chinense*) in Kano State of Nigeria. The results obtained revealed that bell and hot pepper farmers experienced loss of 10-30 percent during harvesting and transportation stages.

Sarkar et al. (2020) conducted an assessment of pre-and post-harvest losses of vegetables in the Kanpur district of Uttar Pradesh. Study reported an overall loss of 10.35 percent in okra with 1.42 percent at the farm level, while at the trader level it was estimated to be 8.93 percent. Vala and Rathod (2019) conducted a study on post-harvest handling and assessment of losses of freshly harvested selected vegetables in Anand district of Gujarat. The study reported that at the farm level, there were 1-2 percent losses, at the APMC level losses were 3-5 percent and at the retailer level, there were 2-4 percent losses. Therefore, the total losses reported were 6-11 percent for okra. A review of post-harvest losses by Ridolfi et al. (2018) published by International Food Policy Research Institute (IFPRI), reported total losses covering the key loss hotspots in okra were 24.4 percent including losses during harvesting (16.6%) and losses at retail level (5.1%) in Ghana. Kumar et al. (2015) in their study found that the overall post-harvest loss observed was 17.80 percent in okra with 11.16 percent loss at producer level and 6.64 percent at trader level. Sharma and Singh (2011) conducted a study on economic analysis of post-harvest losses in marketing of vegetables in Uttarakhand in Kumaon division of the state. The study reported an overall post-harvest loss of 15.63 percent in okra with 8.54 percent at the farm level (maximum loss during harvesting - 3.83%) and 7.09 percent at retail level. Hassan et al. (2010) conducted a study on post-harvest loss assessment in Bangladesh. Losses at farmer's level were estimated to be 9.4 percent. Higher losses were estimated at market level including 9.8 percent at *Bepari* level, 4.9 percent at wholesaler level and 8.3 percent at retailer level. The total post-harvest losses in okra in the supply chain was found to be 32.3 percent. The higher post-harvest losses were reported at the intermediary levels due to the lack of proper transportation and storage facilities.

Earlier, harvest and post-harvest losses for vegetables were assessed by studies conducted by ICAR-CIPHET (2012) and ICAR-CIPHET (2015) at the national level. The losses varied between 6.94 percent (cabbage) to 12.98 percent (tomato) in the study conducted by ICAR-CIPHET (2012) and between 4.58 percent (tapioca) to 12.44 percent (tomato) in the study conducted by ICAR-CIPHET (2015). The estimated operation-wise losses for vegetables are indicated in **Table 2.6** and **Table 2.7** respectively.

The overall losses for vegetables reported over the years by various authors in India and abroad have been summarised and given in **Table 2.9.** It can be seen that the losses varied from 7.18 percent in tapioca to 29.44 percent in cauliflower.

The spoilage/loss of vegetables at the grower level resulted from lack of knowledge about proper post-harvest management, improper grading, packing, lack of storage and inadequate transportation facilities.

Table 2.9 Estimates of losses of vegetables in percent by various studies

S. No.	Name of crop	Years of reference	No. of studies	Min	Max	Average	SD
1	Cabbage	2010, 2012, 2015, 2016, 2018	5	6.90	26.40	16.65	8.73
2	Cauliflower	2005, 2012, 2015, 2018,2019	5	6.88	52.00	29.44	21.56
3	Green peas	2010, 2011, 2015 (2)	4	7.45	16.37	11.91	3.81
4	Mushroom	2012, 2015	2	9.51	12.54	11.03	-
5	Onion	2010, 2012, 2015 (4)	6	7.50	19.00	13.25	4.56
6	Potato	2012, 2015, 2018	3	7.32	24.88	16.10	9.69
7	Tomato	2010, 2015, 2018, 2020	4	10	34	19.89	9.59
8	Tapioca	2012, 2015	2	4.58	9.77	7.18	-
9	Radish	2011, 2015	2	6.52	12.68	9.60	4.36
10	Brinjal	2014, 2015, 2016,2020	4	14.80	23.38	18.77	4.37
11	Okra	2010, 2011, 2015, 2018, 2019, 2020	6	8.50	32.30	18.13	8.91
12	Beans	2014, 2018 (2)	3	22.36	30.7	26.99	4.25
13	Capsicum	2010, 2011, 2015(2)	4	10.43	27.00	18.71	8.46

2.7 Post-harvest losses of plantation crops

Plantation crops are traditionally grown as cash crops in India. Therefore, most of the farmers grow them as a livelihood enterprise. These crops required processing after their harvesting before final disposal in the market. A considerable quantity of these products get lost in the supply chain. Swain et al. (2016) conducted a study on post-harvest processing and value addition in coconut, areca nut and spices in the Islands. They found that lack of awareness and skills on post-harvest technologies caused significant losses in various harvest and post-harvest operations starting from harvesting, drying and storage etc. Khandekar (2017) studied coconut and areca nut production protocol under the aberrant climate of the coastal region. As per the study, the climatic vagaries experienced in coastal climate affect the growth and yield of these crops, which led to severe economic loss to the farmers. Priyashantha et al.

(2020) conducted a study on pre and post-harvest losses of cashewnut in Batticaloa district, Sri Lanka, to find out preliminary investigation of the causes. The study reported 15 percent of post-harvest losses in cashew due to animals and birds.

Punchihewa and Arancon (1999) conducted a study on post-harvest operations of coconut in collaboration with FAO and reported that lack of awareness and actual skills on coconut post-harvest technologies caused significant losses starting from the harvesting of the nuts, seasoning, drying and storage. Nuts were allowed to fall naturally, without harvesting or picking of the nuts from the tree. The losses as high as 10 percent were reported during harvesting especially for varieties that were early germinating. Losses due to pilferage and losses due to nuts that were hidden or covered by thick weeds or shrubs ranged from 5 to 10 percent of the total harvest. Studies reveal that after one year of storage, copra weight loss due to pests was 5 to 10 percent.

ICAR-Sugarcane Breeding Institute, Coimbatore stated that staling of sugarcane beyond 24 hours resulted in a considerable loss in cane weight due to moisture loss and reduction in juice sucrose content due to inversion. The reduction in the cane weight ranged between 7.4 and 17.0 percent, and sugar recovery by about 2.0 percent at different places in India due to staling of cane for 96 hours. Ali et al. (2018) conducted a study on post-harvest losses and control of unprocessed sugarcane at the department of farm structures, Sindh Agriculture University, Tandojam. They conducted an experiment on freshly harvested SPF-234 sugarcane variety collected and stored for 10 days under two different methods i.e. one kept open and other covered with sugarcane trash. A maximum loss of 27.8 and 15.7 percent after 10 days of storage was observed, respectively for sugarcane kept open and covered with trash. They suggested that local farmers should adopt sugarcane trash cover method, which provided a convenient and safe storage system.

Estimates of harvest and post-harvest losses of plantation crops at the national level were reported by ICAR-CIPHET (2012) and ICAR-CIPHET (2015). The overall total losses were the highest in areca nut (7.87%) and the lowest in cashew (1.12%). ICAR-CIPHET (2015) conducted a study on the similar methodology adopted by ICAR-CIPHET (2012). The overall total losses were the highest in sugarcane (7.89%) and the lowest in cashew (4.17%). The estimated losses for oilseeds are indicated in **Table 2.10** and **Table 2.11** respectively.

The overall losses for plantation crops reported over the years by various authors in India and abroad have been summarised and given in **Table 2.12.** It can be seen that the losses varied from 2.64 percent in cashew nuts to 8.27 percent in sugarcane.

Table 2.10 Harvest and post-harvest losses (%) of plantation crops and spices at national level (ICAR-CIPHET 2012)

Crop	Farm Operations								Total loss in farm operations	Storage in Channels					Total loss in storage	Overall total loss
	Harvest -ing	Collec -tion	Thresh-ing	Sorting/ Grading	Winnowing / cleaning	Drying	Packag -ing	Trans-port		Farm	Godown	Whole-salers	Retailers	Processing unit		
Areca nut	1.90 ±0.37	0.16 ±0.16	3.32 ±0.83	0.39 ±0.59	0.41 ±0.20	0.09 ±0.37	0.06 ±0.04	0.19 ±0.09	0.62 ±0.32	0.03 ±0.01	0.00 ±0.00	0.88 ±0.13	0.27 ±0.02	0.12 ±.02	1.26 ±0.13	7.87 ±0.54
Black pepper	0.71 ±0.33	0.16 ±0.11	0.46 ±0.70	-	0.83 ±0.49	0.92 ±0.58	0.24 ±0.19	0.21 ±0.41	3.60 ±0.49	0.01 ±0.02	0.00 ±0.00	0.19 ±0.02	0.10 ±0.02	0.00 ±0.01	0.27 ±0.03	3.86 ±0.41
Cashew	0.16 ±0.09	0.14 ±0.45	0.37 ±0.11	-	-	0.21 ±0.05	0.00 ±0.00	0.01 ±0.05	0.89 ±0.26	0.00 ±0.00	0.00 ±0.00	0.06 ±0.02	0.02 ±0.01	0.14 ±0.07	0.23 ±0.07	1.12 ±0.23
Chilli	1.64 ±1.05	0.71 ±0.58	-	0.67 ±0.85	0.56 ±0.29	0.12 ±0.49	0.12 ±0.22	0.15 ±0.47	3.95 ±0.70	0.03 ±0.04	0.03 ±0.01	1.19 ±0.53	0.36 ±0.15	0.09 ±0.06	1.66 ±0.55	5.60 ±0.66
Coconut	1.52 ±0.47	0.22 ±0.26	-	1.36 ±0.77	-	0.66 ±0.51	0.13 ±0.22	0.17 ±0.16	4.10 ±0.87	0.14 ±0.05	0.02 ±0.02	0.46 ±0.12	0.22 ±0.12	0.36 ±0.07	1.27 ±0.19	5.36 ±0.71
Coriander	2.19 ±1.36	0.68 ±0.68	3.10 ±1.66	-	0.21 ±0.57	0.37 ±0.30	0.16 ±0.23	0.20 ±0.19	6.81 ±1.03	0.09 ±0.03	0.00 ±0.00	0.31 ±0.08	0.10 ±0.07	0.00 ±0.00	0.51 ±0.11	7.31 ±0.66
Sugarcane	2.78 ±1.75	0.86 ±0.61	-	0.31 ±0.55	-	0.52 ±0.03	0.09 ±0.29	0.22 ±0.63	7.79 ±0.85	0.08 ±0.23	-	-	0.01 ±0.01	0.81 ±0.23	0.00 ±0.76	0.64 ±1.82
Turmeric	3.60 ±6.47	0.77 ±0.99	-	0.67 ±1.93	1.14 ±1.46	0.12 ±0.07	0.13 ±0.34	0.22 ±0.59	6.72 ±3.34	0.18 ±0.13	-	0.43 ±0.14	0.08 ±0.08	-	0.66 ±0.20	7.37 ±2.50

Table 2.11 Harvest and post-harvest losses (%) of plantation crops and spices at national level (ICAR-CIPHET 2015)

Crop	Farm Operations								Total loss in farm operations	Storage in Channels					Total loss in storage	Overall total loss
	Harvest -ing	Collec -tion	Thresh -ing	Sorting/ Grading	Winnowing / cleaning	Drying	Packaging	Trans -port		Farm	Godown	Whole -salers	Retailers	Proces sing unit		
Areca nut	1.24 ±0.35	0.39 ±0.10	-	0.71 ±0.27	1.19 ±0.35	0.19 ±0.15	0.05 ±0.01	0.17 ±0.06	3.94 ±0.23	0.02 ±0.00	-	0.48 ±0.31	0.10 ±0.06	0.36 ±0.5	0.97 ±0.15	4.91 ±0.22
Black pepper	0.47 ±0.19	0.21 ±0.10	-	0.23 ±0.11	0.02 ±0.10	0.04 ±0.04	0.01 ±0.00	0.00 ±0.00	0.99 ±0.11	0.01 ±0.00	-	0.00 ±0.00	0.18 ±0.14	-	0.20 ±0.07	1.18 ±0.11
Cashew	1.45 ±1.59	0.57 ±2.23	-	1.34 ±0.53	0.30 ±0.76	0.07 ±0.08	0.00 ±0.00	0.07 ±0.04	3.82 ±1.12	0.00 ±0.00	0.00 ±0.00	0.14 ±0.11	0.03 ±0.03	0.17 ±0.07	0.35 ±0.07	4.17 ±0.84
Chilli	1.60 ±0.34	0.84 ±0.09	2.18 ±0.59	-	-	0.02 ±0.07	0.15 ±0.05	0.30 ±0.36	5.11 ±0.36	0.03 ±0.00	-	0.99 ±0.24	0.31 ±0.12	0.06 ±0.02	1.40 ±0.14	6.51 ±0.28
Coconut	1.37 ±0.27	0.20 ±0.14	-	1.02 ±0.32	0.37 ±0.10	0.36 ±0.13	0.08 ±0.05	0.05 ±0.13	3.45 ±0.23	0.08 ±0.01	-	0.61 ±0.23	0.25 ±0.10	0.38 ±0.05	1.32 ±0.07	4.77 ±0.20
Coriander	2.48 ±0.14	0.92 ±0.05	-	1.07 ±0.21	0.45 ±0.05	0.01 ±0.01	0.09 ±0.01	0.31 ±0.02	5.33 ±0.10	0.03 ±0.00	-	0.27 ±0.07	0.26 ±0.06	-	0.55 ±0.05	5.87 ±0.09
Sugarcane	2.11 ±0.22	0.04 ±0.16	1.02 ±0.08	-	-	3.95 ±0.30	0.07 ±0.01	0.10 ±0.15	7.29 ±0.18	0.04 ±0.06	-	0.42 ±0.25	0.11 ±0.07	0.04 ±0.05	0.60 ±0.09	7.89 ±0.17
Turmeric	2.41 ±0.28	0.10 ±0.19	0.79 ±0.17	-	-	0.16 ±0.13	0.09 ±0.04	0.04 ±0.29	3.60 ±0.21	0.09 ±0.03	-	0.62 ±0.05	0.06 ±0.05	0.06 ±0.01	0.84 ±0.04	4.44 ±0.15

Table 2.12 Estimates of losses of plantation crops in percent in various studies

S. No.	Name of crop	Years of reference	No. of studies	Min	Max	Average	SD
1	Areca nut	2012, 2015	2	4.91	7.87	6.39	-
2	Cashew nut	2012, 2015	2	1.12	4.17	2.64	-
3	Coconut	2012, 2015	2	4.77	5.36	5.07	-
4	Sugarcane	2012, 2015	2	7.89	8.65	8.27	-

2.8 Post-harvest losses of spices

Spices play an important role in earning foreign exchange for India. Therefore, the post-harvest management of spices is a very vital step in its supply chain. Aneeshya et al. (2013) conducted a study on fabrication of three model harvesters for black pepper and evaluation of the model harvesters on the basis of efficiency in the cutting action and easiness in operation. Usually the pepper spikes were nipped by hand and collected in bags. The model where simultaneous cutting and collection of spikes was done was found to be the most efficient, user friendly and with least loss percent at harvesting. In another study conducted by Korikanthimath (2013) reported that the management strategies help in reduction of harvest and post-harvest losses in black pepper, cardamom and ginger. It was reported that in the case of black pepper, if the berries were allowed to over-ripe on the vines, there was a high amount of loss due to berries drop and bird damage, therefore, it was crucial to harvest green pepper at the right stage of maturity.

Sharma and Singh (2011) observed that total post-harvest losses in chilli were 16.75 percent at both the levels i.e. farmer and market level. The major causes of post-harvest losses were non-awareness about technical know-how, improper grading, packaging, inadequate transportation and storage facilities. Shil et al. (2018) conducted an experiment on the physiological loss in weight of chilli in 4 different genotypes under different storage conditions. The chilli was stored in a zero energy cool chamber (ZECC) and at ambient temperature for different time periods. The highest physiological losses (36.13%) in weight were observed under ambient temperature conditions and the lowest under ZECC (3.48%).

Estimates of harvest and post-harvest losses of spices were reported by ICAR-CIPHET (2012) and ICAR-CIPHET (2015). The overall total losses were the highest in turmeric (7.37%) and the lowest in black pepper (3.86%). ICAR-CIPHET (2015) conducted a study on the similar methodology adopted by ICAR-CIPHET (2012). The overall total losses were the highest in chilli (6.51%) and the lowest in black pepper (1.18%). The estimated losses for spices are indicated in **Table 2.10** and **Table 2.11** respectively.

The overall losses for spices reported over the years by various authors in India and abroad have been summarised and given in **Table 2.13**. It can be seen that the losses varied from 2.51 percent in black pepper to 6.59 percent in coriander.

Table 2.13 Estimates of losses of spices in percent in various studies

S. No.	Name of crop	Years of reference	No. of studies	Min	Max	Average	SD
1	Black pepper	2012, 2015	2	1.18	3.86	2.52	-
2	Chilli	2012, 2015	2	5.60	6.51	6.06	-
3	Coriander	2012, 2015	2	5.87	7.31	6.59	-
4	Turmeric	2012, 2015	2	4.44	7.37	5.91	-

2.9 Post-harvest losses of eggs

ICAR-CIPHET (2012) conducted a study on harvest and post-harvest losses of major crops and livestock production in India. The overall total post-harvest losses in eggs were 6.55 percent. In a similar study, ICAR-CIPHET (2015) reported overall post-harvest losses of 7.19 percent for eggs at the national level. The operation wise estimates of losses reported by ICAR-CIPHET (2012) and ICAR-CIPHET (2015) are indicated in **Table 2.14** and **Table 2.15,** respectively. Hegazy (2016) in his study on post-harvest losses in India reported data from various studies conducted on post-harvest losses in agri-produce in India and suggested that the losses in the supply chain of eggs were in the range of 10 to 25 percent. FAO (2010) in its study on post-harvest losses reduction reported 12 percent post-harvest losses in eggs in Ghana. Analysis of post-harvest handling of table eggs produced in San Jose, Batangas, Philippines conducted by Abigail and Julieta (2017) revealed that only 57 percent of farmers in Philippines cleaned dirty eggs by wiping them out with a piece of cloth in the case of small-scale farmers and using tap water for large-scale farmers. In the case of intermediary-respondents, those who picked up from the farms were the ones who cleaned dirty eggs.

Bannor et al. (2021) examined the influence of commercialization and post-harvest losses on the choice of marketing outlet among poultry farmers and reported that the supply chain of poultry eggs in Ghana was not that well developed and had significant post-harvest losses. The losses included direct physical losses along with quality losses, which minimised the economic value of the produce. The study reported that the losses were as high as 80 percent of total output in extreme cases. Inefficient technology, low expertise in handling produce as well as lack of logistical support were the prime reasons for high losses (about 50%) in case of underdeveloped countries. In another study on post-harvest food losses at the retailer and consumer levels in the United States, USDA reported a total loss of 7 percent at the retailer level. Muhammad et al. (2014) in their study on the effect of storage time on eggs quality and hatchability characteristics of Rhode Island Red (RIR) hens, reported that the weight of the eggs decreased during the storage period attributed to evaporative water loss through thousands of pores contained on eggshell surface, as around 80 percent of the weight of eggs constituted of water.

2.10 Post-harvest losses of fish

FAO (2010(a)) reported an overall loss of 21.93 percent for inland fish in Ghana. Inadequate infrastructure for landing and handling, inadequate pre-shipment storage, shortage of ice making facilities were the major reasons for post-harvest losses of inland fish. FAO (2011) reported losses of 9.4 percent at fishing, 0.5 percent at post-harvest handling and storage, 6 percent at processing and packaging and 9 percent at supermarket retail for Europe; 12 percent at fishing, 0.5 percent at post-harvest handling and storage, 6 percent at processing and packaging and 9 percent at supermarket retail for North America; 5.7 percent at fishing, 5 percent at post-harvest handling and storage, 9 percent at processing and packaging and 10 percent at distribution channel for Latin America; and 15 percent at fishing, 2 percent at post-harvest handling and storage, 6 percent at processing and packaging and 11 percent at distribution channel for Asia. FAO (2018) reported losses of 6.5 percent and 8.9 percent in the catch operation of commercial fishes in inland reservoir gillnet and hook-and-line fisheries of Odisha, India. The causes of losses were spoilage from the inflow of muddy water, too long gear soak time, and catch being damaged because of poor handling practices. Rashid and Sarkar (2020) reported post-harvest losses of 4.47 percent for culture fish and 7.01 percent for capture fish. Major causes of post-harvest losses were improper handling by the fishermen during the harvesting operation. Thomas et al. (2020) conducted a study in Gujarat, Kerala, Tamil Nadu, and Andhra Pradesh and reported physical losses of 8.0 percent and quality losses of 5.0 percent in gillnet inland fisheries.

FAO (2010(b)) reported an overall loss of 21.08 percent for marine fish in Ghana. Inadequate infrastructure for landing and handling, inadequate pre-shipment storage, shortage of ice making facilities were the major reasons for post-harvest losses of fishes. ICAR-CIPHET (2012) reported an overall total loss of 6.92 percent and 2.78 percent for inland fish and marine fish, respectively at national level. ICAR-CIPHET (2015) conducted a similar study based on methodology followed by ICAR-CIPHET and reported a loss of 5.23 percent for inland fish and 10.52 percent for marine fish. The operation wise losses are indicated in **Table 2.14** and **Table 2.15**. Buzby et al. (2014) reported post-harvest losses of 8 percent at retail level for fish in the United States. FAO (2014) conducted a study on food loss assessments in Kenya. The study reported post-harvest losses of 1-2 percent during drying operations for marine fish. Chattopadhyay et al. (2016) conducted a study in West Bengal on marine fisheries and found that at the fishermen/farm level, the post-harvest losses were in the range of 5-15 percent. Rashid and Sarkar (2020) conducted a study on post-harvest losses of culture, capture and marine fisheries of Bangladesh. The study reported post-harvest losses of 11.67 percent for marine fish species. Major causes of post-harvest losses were improper handling by the fishermen during the harvesting operation. Geethalakshmi (2020) conducted a study on marine fisheries at the Central Institute of Fisheries Technology and reported a maximum loss of 18.73 percent during harvesting, during multi-day fishing using mechanised trawlers, due to the capture of juveniles and their discards. The traditional fisheries sector reported minimal or no loss (1.14%) during the period. For fresh fish, losses were found to be 3.79 percent (at the wholesale market) and 3.12 percent (at the retail market), however, the losses were higher in the case of dry fish (7.56% at the wholesale market and 8.23% at the retail market). The pre-processing and processing level losses were around 1.19 percent during pre-processing and 0.38 percent during the processing of marine fish.

2.11 Post-harvest losses of meat

Hegazy (2016) in his review on post-harvest situation and losses in India reported data from various studies conducted on post-harvest losses in agri-produce in India and suggested that the losses in the supply chain of egg, meat and poultry meat were in the range of 10 to 25 percent. Buzby et al. (2014) in an USDA study on post-harvest food losses in the United States reported post-harvest loss of 4 percent in terms of quantity and leading to monetary loss of $161.6 billion at the retail level. Out of food groups studied, the highest loss was observed in case of meat, poultry, and fish (30%, amounting to $48 billion). Post-slaughter, the absence of infrastructure including adequate cold chains was considered to be the crucial factor in reducing post-harvest losses, after underutilization of by-products was taken into account (Tomlins et al. 2016). Gustavsson et al. (2011) suggested an overall 'production or waste' loss for meat products as varying between 20 and 28 percent. They reported that the losses were more in small-scale and extensive livestock systems (e.g. sub-Saharan Africa) due to mortality (mainly disease) on farms than in the systems that were intensive. Where high losses occurred at the level of the consumer, the accumulated economic cost along the chain was magnified (Hodges et al. 2011). In a report by FAO (2011) on global food losses and food waste, an estimated loss of 3.1 percent at the production level, 0.7 percent during post-harvest handling and storage, 5 percent during processing and packaging and 4 percent during distribution for Europe including Russia for meat including poultry meat. In Turkey a study of live weight shrinkage and mortality of broilers during transport found 5.9 percent average losses, of which 0.4 percent was mortality and the rest live weight loss. Similar studies in China (Liu 2014) and Jordan (Al-Sharafat et al. 2013) support similar levels of loss for chicken.

ICAR-CIPHET (2012) conducted a study on harvest and post-harvest losses of major crops and livestock produce in India. The total post-harvest losses in meat and poultry meat were 2.23 percent and 3.65 percent, respectively. ICAR-CIPHET (2015) conducted the study based on methodology adopted by ICAR-CIPHET (2012). Overall post-harvest losses reported for meat and poultry meat were 2.71 percent and 6.74 percent, respectively. The operation wise post-harvest losses are indicated in **Table 2.14** and **Table 2.15**. Tomlins et al. (2016), in a study on reducing post-harvest losses in the OIC member countries, estimated physical loss of 25-40 percent in the meat value chain in Oman. High losses reported during the transport to the market were attributed to a lack of feed for the animals and poor transportation conditions (animals dehydrated and losing weight). A review of the goat value chain in Kenya (Roba, 2013) identified key causes of post-harvest losses as: lack of market information, poor market coordination and insufficient infrastructure (e.g. roads, cold chain, butchery equipment). A quantitative study using willing-to-pay methods suggested a loss for small ruminants at 3 percent (Juma 2007). FAO (2010) in their study on post-harvest losses reduction reported 14 percent post-harvest losses in meat in Ghana. FAO (2011) estimated extent, causes and prevention of global food losses and food waste and reported post-harvest loss of 0.6-1.1 percent in processing and 5 percent during handling and storage of meat.

2.12 Post-harvest losses of milk

The supply chain of milk in India, at one side, is fairly well developed, whereas a considerable quantity of milk is sold in an unorganised way leading to losses. ICAR-CIPHET (2012) conducted a study in 11 districts for milk. The total post-harvest loss at farm operations in milk was 0.67 percent and post-harvest losses during storage at the market level was 0.10 percent. The overall total post-harvest losses in milk were 0.77 percent. ICAR-CIPHET (2015) conducted the study of milk in 14 districts from 14 agro-climatic zones. Overall post-harvest losses reported for milk were 0.92 percent at the national level. The post-harvest loss at farm level operations was found to be 0.71 percent, whereas, post-harvest losses during market level storage was 0.21 percent. The operation wise post-harvest losses are indicated in **Table 2.14** and **Table 2.15**. FAO (2017) in its study on milk value chain-food loss analysis reported 1.14 percent losses in Krishna district of Andhra Pradesh. The total quantitative loss in the milk supply chain was 5.8 percent. Milk loss at the producer level was mainly due to discarding of infected milk resulting from mastitis and other infections. It was observed that the major milk rejections occurred during summer due to increased microbial content in milk owing to high temperature, which was compounded by the time gap between milking and delivery at the chilling centre. Ndungu et al. (2019) conducted a study on the determinants and causes of post-harvest milk losses among milk producers in Nyandarua, Kenya. The most common method of preservation was cooling, whereby 85.4 percent of farmers cooled milk by dipping it in a bucket of cold water, 8.2 percent boiled the milk and 6.4 percent used other means of preservation like smoking of storage containers. Another study by Odongo et al. (2016) noted delay in cleaning milk-handling containers as a risk factor, because it offered a suitable environment for microorganisms to multiply and increased microbial load.

According to FAO (2014), Kenya loses approximately 95 million litres of milk annually accounting for US$22.4 million per year and the impact was mainly big at the farm level. Buzby et al. (2014) reported a total loss of 12 percent of milk at the retail level. In a report by FAO (2013) on global food losses and food waste, an estimated loss of 3.5 percent at the production level, 0.5 percent during post-harvest handling and storage, 1.2 percent during processing and packaging and 0.5 percent during distribution for Europe including Russia was observed for milk. Bart et al. (2019) in their study on post-harvest losses in rural-urban milk value chain in Ethiopia reported post-harvest loss of 9.4 percent in milking, 0.5 percent during handling and storage and 6 percent in processing and packaging. FAO (2011) estimated extent, causes and prevention of global food losses and food waste and reported post-harvest loss of 3.5 percent in milking, 0.5 percent during handling and storage and 1.2 percent in processing and packaging in Europe and North America. Higher post-harvest losses of 6 percent during handling and storage and 2 percent in processing and packaging were reported in Latin America.

The overall losses for various livestock products reported over the years by various authors in India and abroad have been summarised and given in **Table 2.16**. It can be seen that the losses varied from 5.20 percent in poultry meat to 13.12 percent in meat.

Table 2.14 Harvest and post-harvest losses (%) of livestock produce at national level (ICAR-CIPHET 2012)

Crop	Farm Operations					Total loss in farm operations	Storage in Channels					Total loss in storage	Overall total loss
	Harvesting	Collect-ion	Sorting/ Grading	Packaging	Trans-port		Farm	Godown	Wholesalers	Retailers	Processing-unit		
Egg	-	2.08 ±0.58	-	0.98 ±0.25	1.83 ±0.90	4.88 ±0.56	0.04 ±0.04	-	0.98 ±0.28	0.66 ±0.34	0.02 ±0.01	1.67 ±0.44	6.55 ±0.52
Inland Fish	2.62 ±2.58	0.14 ±0.25	1.63 ±2.16	0.46 ±0.89	0.32 ±0.33	5.18 ±1.66	0.04 ±0.09	-	0.82 ±0.26	0.84 ±0.73	-	1.75 ±0.78	6.92 ±1.49
Marine Fish	-	0.17 ±0.01	0.09 ±0.01	-	1.56 ±0.66	1.81 ±0.04	-	-	0.28 ±0.02	0.28 ±0.01	0.44 ±0.66	0.97 ±0.00	2.78 ±0.04
Meat	1.36 ±0.09	-	-	-	-	1.36 ±0.09	-	-	0.48 ±0.40	0.42 ±0.27	-	0.87 ±0.49	2.23 ±0.22
Poultry Meat	2.67 ±0.16	-	-	-	-	2.67 ±0.16	-	-	0.31 ±0.34	0.68 ±0.57	0.01 ±0.00	0.98 ±0.66	3.65 ±0.36
Milk	0.12 ±0.16	0.49 ±0.09	-	-	0.07 ±0.04	0.67 ±0.13	0.02 ±0.01	-	-	-	0.08 ±0.01	0.10 ±0.01	0.77 ±0.12

Table 2.15 Harvest and post-harvest losses (%) of livestock produce at national level (ICAR-CIPHET 2015)

Crop	Farm Operations					Total loss in farm operations	Storage in Channels					Total loss in storage	Overall total loss
	Harvesting	Collect-ion	Sorting/ Grading	Packaging	Trans-port		Farm	Godown	Wholesalers	Retailers	Processing-unit		
Egg	-	1.92 ±0.15	1.40 ±0.20	-	1.21 ±0.12	0.36 ±0.21	4.88 ±0.16	0.07 ±0.04	-	1.35 ±0.24	0.89 ±0.19	-	2.31 ±0.14
Inland Fish	1.74 ±0.33	0.37 ±0.00	1.72 ±0.43	-	0.18 ±0.00	0.17 ±0.00	4.18 ±0.27	0.09 ±0.00	-	0.24 ±0.15	0.72 ±0.32	-	1.05 ±0.24
Marine Fish	7.40 ±0.01	0.75 ±0.07	0.41 ±0.37	0.13 ±0.00	0.00 ±0.00	0.91 ±0.10	9.61 ±0.20	-	-	0.65 ±0.19	0.26 ±0.16	-	0.91 ±0.17
Meat	1.78 ±0.12	-	0.21 ±0.05	-	-	0.00 ±0.00	1.99 ±0.11	0.00 ±0.00	0.01 ±0.00	0.46 ±0.16	0.25 ±0.11	0.00 ±0.00	0.72 ±0.07
Poultry Meat	1.62 ±0.29	-	0.46 ±1.64	-	0.00 ±0.00	0.66 ±0.28	2.74 ±0.72	0.00±	-	3.02 ±0.65	0.97 ±0.25	-	4.00 ±0.16
Milk	0.21 ±0.17	0.18 ±0.03	-	-	0.30 ±0.00	0.02 ±0.00	0.71 ±0.11	0.00 ±0.00	-	-	-	0.21 ±0.20	0.21 ±0.16

Table 2.16 Estimates of losses of livestock produce in percent in various studies

S.No	Name of crop	Years of reference	No. of studies	Min	Max	Average	SD
1	Eggs	2012, 2015	2	6.55	7.19	6.87	0.45
2	Marine fish	2010, 2012, 2015, 2016, 2020	5	2.78	21.08	11.21	6.53
3	Inland fish	2010, 2012, 2015, 2018, 2020 (2)	6	5.23	21.93	9.64	6.26
4	Meat	2011, 2012, 2014, 2015	4	2.23	24	13.12	10.54
5	Poultry meat	2011, 2012, 2013, 2015	4	3.65	6.74	5.20	1.69
6	Milk	2013, 2014, 2015, 2017	4	1.7	12	6.85	5.09

CHAPTER 3 SURVEY CONSIDERATIONS AND SCHEDULES

Post-harvest losses may occur in any crop/ commodity at various levels in the supply chain such as farm level and market level. The post-harvest losses at farm level consists of losses during harvesting, collection, sorting & grading, threshing, winnowing, drying, packaging, farm level storage, transport and handling. At market level post-harvest losses take place during storage in godowns/ warehouses, cold storages, silos, etc., as well as storage and handling of these commodities by wholesalers, retailers, transporters and processing units/ pack houses etc. The present study was conducted to provide estimates of post-harvest losses for 54 crops/ commodities at the national level and to compare the results obtained with previous reported results. The present study also attempted to understand the change in extent of post-harvest losses with respect to previous studies and identify the need for intervention by the concerned stakeholders. Under the study, the surveys were carried out in all 15 Agro-Climatic Zones (ACZs) of the country by both enquiry and observation method. The current chapter deals with the assumptions and different considerations taken during the assessment, crops and commodities selected and sampling design and procedures followed for the study.

3.1 Assumptions

Following the methodology of ICAR-CIPHET (2015) for the ICAR-CIPHET study sponsored by MoFPI, post-harvest losses were estimated both by enquiring various stakeholders and conducting actual observations in the field for 1/5th of the sample stakeholders. The deviations, if any, from the methodology of the ICAR-CIPHET (2015) have been discussed in the report in respective sections.

Major assumptions and considerations taken for the study include the following:

1) Selection of districts, blocks, villages, respondents and stakeholders was done using standard statistical methods.
2) Data collection was done by trained manpower engaged for the sole purpose of this study.
3) Data of the post-harvest losses was collected for one full crop cycle (one crop year) of the selected commodities. The data was revalidated, wherever needed in the next crop cycle/ season. The starting point of data collection at farm level was harvesting operation.
4) For some crops grown more than once in a year, the data of farm operation were collected for each harvest season in that year.
5) For collection of data from selected farmers through observation, data from actual ongoing farm operations were recorded by research team itself.
6) The data collection for harvesting (catching) of marine fish was collected only by enquiry.
7) Wherever feasible actual experiments were carried out by our research team to record post-harvest losses.
8) In case of perishables, transport level post-harvest losses were estimated by recording data both at origin and destination.

3.2 Crops/ commodities, their channels and unit operations

As per the MoU with the Ministry of Food Processing Industries (MoFPI), 54 crops / commodities including livestock produce of India were taken up for assessment of post-harvest losses at both farm level and market level. The selected crops were categorised into 12 categories (**Table 3.1**) namely cereals, pulses, fruits, vegetables, oilseeds, spices, plantation, milk, egg, meat, inland fish and marine fish. The various farm operations and channels for the selected crops / commodities and the extent of the coverage in the study are summarised in **Table 3.2**.

Table 3.1 Categorisation of 54 crops in 12 categories

S.No	Category	Crops
1	Cereals	Bajra, paddy, wheat, maize, sorghum (5)
2	Pulses	Black gram, green gram, chickpea, pigeon pea (4)
3	Fruits	Grapes, pomegranate, citrus, pineapple, guava, apple, papaya, sapota, mango, muskmelon, banana (11)
4	Vegetables	Mushroom, green pea, onion, brinjal, radish, bottle gourd, beans, cabbage, potato, cauliflower, tomato, tapioca, capsicum, okra (14)
5	Oilseeds	Cottonseed, groundnut, safflower, sunflower, soybean, rapeseed/ mustard (6)
6	Spices	Dry chilli, coriander, black pepper, turmeric (4)
7	Plantation	Areca nut, sugarcane, coconut, cashew nut (4)
8	Milk	Milk (1)
9	Egg	Egg (1)
10	Meat	Meat and poultry meat (2)
11	Inland fish	Inland fish (1)
12	Marine fish	Marine fish (1)

Table 3.2 Farm level operations/ market level channels and the extent of the coverage for selected crops/ commodities

S.No	Operation/ channel	Extent of coverage of the operation	Crops Covered
1	**Harvesting**	Cutting of the standing crop	Bajra, paddy, wheat, maize, green gram, sorghum, black gram, chickpea, pigeon pea, soybean, safflower, mustard/ rapeseed, sugarcane, coriander
		Plucking of crops from trees/ plants/ bunches	Grapes, pomegranate, citrus, pineapple, guava, apple, papaya, banana, sapota, mango, muskmelon, cottonseed, brinjal, beans, bottle gourd, tomato, coconut, cashew nut, areca nut, chilli, okra, black pepper, cauliflower, cabbage, sunflower
		Uprooting/ digging of crops from the soil	Radish, onion, potato, tapioca, turmeric, groundnut
		Milking	Milk
		Slaughter	Meat, poultry meat
		Catching	Inland fish, marine fish
2	**Collection**	Stacking, piling, bundling and transportation to threshing floor	Bajra, paddy, wheat, maize, green gram, sorghum, black gram, chickpea, pigeon pea, soybean, safflower, sunflower, mustard/ rapeseed, sugarcane, chilli, turmeric, groundnut, black pepper, coriander
		Stacking, piling, filling in crates/ baskets/ bags of crops at a place	Grapes, pomegranate, citrus, pineapple, guava, apple, papaya, banana, sapota, mango, muskmelon, cottonseed, brinjal, beans, bottle gourd, radish, tomato, coconut, cashew nut, areca nut, okra, cauliflower, cabbage, potato, tapioca
		Collection of eggs	Egg
		Removal of leaves, stacking, bundling	Sugarcane
		Collection of fish from net, filling in containers/ baskets/tanks	Inland fish
		Unloading the fish from boat at landing centre	Marine fish
3	**Sorting/ Grading**	Separation of fruits not fit for consumption due to damage or injury. Grading may be done on the basis of colour, size and extent of damages.	Apple, mango, guava, grapes, tomato, pomegranate, citrus, pineapple, papaya, sapota, banana, muskmelon, chilli, turmeric, egg, mushroom, capsicum, green peas, onion, brinjal, beans, bottle gourd, cauliflower, cabbage, radish, potato, tapioca, sugarcane, okra, areca nut
		Separation of dead, spoiled, unwanted fish	Inland fish, marine fish
4	**Threshing/ dehusking**	Separation of grains/ seeds, pods,nuts and removal of husk	Pigeon pea, sorghum, soybean, maize, paddy, black gram, green gram, mustard, Bajra, safflower, sunflower, wheat, groundnut, black pepper, areca nut, coriander
		Separation of nuts from husk or shell	Coconut, cashew nut
5	**Winnowing/ cleaning**	Removal of husk, chaff, dust and other materials	Maize, paddy, black gram, green gram, sorghum, soybean, pigeon pea, Bajra, chickpea, mustard, safflower, sunflower, wheat, groundnut, black pepper, areca nut, coriander
		Transportation from field to crushing unit, before crushing starts	Sugarcane

S.No	Operation/ channel	Extent of coverage of the operation	Crops Covered
7	**Packaging**	Filling of crop/seed in crates, boxes, bags and other packaging material	Grapes, pomegranate, citrus, chilli, turmeric, egg, mushroom, capsicum, green peas, onion, brinjal, beans, cauliflower, cabbage, potato, tapioca, bottle gourd, radish, okra, sugarcane, maize, paddy, black gram, green gram, Bajra, papaya, banana, sapota, pigeon pea, soybean, safflower, sunflower, cottonseed, wheat, groundnut, black pepper, sorghum, areca nut, coconut, cashew nut, coriander, chickpea, mustard
		Packaging in ice box/ application of ice	Inland fish, marine fish
8	**Farm level transportation**	Loading of packaged crops from farm to the market.	Grapes, pomegranate, citrus, pineapple, chilli, turmeric, milk, egg, mushroom, capsicum, green peas, onion, brinjal, beans, bottle gourd, radish, okra, cauliflower, cabbage, potato, tapioca, sugarcane, maize, paddy, black gram, green gram, Bajra, safflower, sunflower, cottonseed, wheat, groundnut, black pepper, sorghum, inland fish, marine fish, soybean, areca nut, coconut, cashew nut, coriander, banana, papaya, sapota, pigeon pea, muskmelon
9	**Storage at farm/ household level**	Temporary storage at farm for few hours	Grapes, pomegranate, citrus, pineapple, chilli, turmeric, egg, mushroom, capsicum, green peas, okra, onion, brinjal, beans, bottle gourd, radish, cauliflower, cabbage, potato, tapioca, maize, paddy, black gram, green gram, Bajra, safflower, sunflower, cottonseed, wheat, groundnut, black pepper, inland fish, sorghum, soybean, pigeon pea, papaya, areca nut, coriander, banana, sapota
10	**Wholesalers**	Loading, during storage, unloading for further sale	Grapes, pomegranate, citrus, pineapple, chilli, turmeric, egg, meat, poultry meat, mushroom, capsicum, green peas, onion, okra, brinjal, beans, bottle gourd, radish, cauliflower, cabbage, potato, tapioca, maize, paddy, black gram, green gram, Bajra, safflower, sunflower, cottonseed, wheat, groundnut, black pepper, inland fish, marine fish, areca nut, coconut, cashew nut, coriander, sorghum, soybean, pigeon pea, banana, sapota, papaya, muskmelon,
11	**Retailers**	Loading, during storage, unloading for further sale	Grapes, pomegranate, citrus, pineapple, chilli, turmeric, milk, egg, meat, poultry meat, mushroom, capsicum, green peas, onion, brinjal, beans, bottle gourd, okra, radish, cauliflower, cabbage, potato, tapioca, maize, paddy, black gram, green gram, Bajra, safflower, sunflower, cottonseed, wheat, groundnut, black pepper, inland fish, marine fish, areca nut, coconut, cashew nut, coriander, sorghum, soybean, pigeon pea, banana, sapota, papaya, muskmelon
12	**Transporter**	Loading of packaged/ unpacked at production centre and unloading at consumption centres/ from one district to another district/ other state	Grapes, pomegranate, citrus, pineapple, chilli, turmeric, meat, poultry meat, milk, mushroom, capsicum, green peas, onion, okra, brinjal, beans, bottle gourd, radish, cauliflower, cabbage, potato, tapioca, sugarcane, maize, paddy, black gram, green gram, Bajra, safflower, sunflower, cottonseed, wheat, groundnut, black pepper, inland fish, marine fish, areca nut, coconut, cashew nut, coriander, sorghum, soybean, pigeon pea, banana, sapota, papaya, muskmelon
13	**Storage at godown/ warehouse/ cold stores**	Unloading, during storage, unloading for further sale	Grapes, pomegranate, citrus, pineapple, chilli, turmeric, meat, poultry meat, egg, capsicum, onion, cauliflower, cabbage, potato, tapioca, maize, paddy, black gram, green gram, Bajra, safflower, sunflower, cottonseed, wheat, groundnut, black pepper, marine fish, areca nut, coconut, cashew nut, coriander, sorghum, soybean, pigeon pea, banana, sapota, papaya, muskmelon
14	**Processing unit / pack house**	Unloading, sorting, grading, packaging storage, unloading for further sale	Grapes, pomegranate, citrus, milk, meat, poultry meat, onion, sugarcane, cauliflower, cabbage, potato, tapioca, maize, paddy, black gram, green gram, Bajra, safflower, sunflower, cottonseed, groundnut, marine fish, areca nut, coconut, cashew nut, coriander, soybean, pigeon pea, banana, sapota, black pepper

3.3 Sampling design and sampling procedure

Sampling is a procedure of selection of a sample from an individual or from a large group of population for a certain kind of research purpose. The study was conducted to estimate the post-harvest losses of the crops/ commodities across the country. Similar to the study conducted by ICAR-CIPHET (2015), stratification of the country approved by the then Planning Commission of India in the form of agro-climatic zones was selected for the study. The whole country was divided into 15 Agro-Climatic Zones (ACZs) and all 15 ACZs were selected under the study for data collection purposely. The survey was conducted at the farmers' field, markets, public and private agencies, godowns, cold storages, processing units/ pack houses, wholesalers, retailers, and transporters levels. Based on a standard statistical sampling procedure, selection of respondents for data collection was done by using random sampling procedure. Sampling units were selected using random sampling techniques for all crops/ commodities. The procedure followed for sampling in each stage are indicated in the following subsections.

3.3.1 District selection

The districts were selected as sampling units for further selection of respondents. To use the methods of sample survey, it is essential to cover at least 10 percent units of first stage sampling. Hence, out of 737 districts of the country, a total of 292 districts were selected from 15 agro-climatic zones across India.

Following the methodology of ICAR-CIPHET (2015), the **Stratified Multistage Random Sampling method** was used for selection of the respondents for data collection. All fifteen ACZs were considered as a strata and the districts, blocks, villages and farmers were taken as the first, second, third and fourth stage units, respectively in each stratum. The districts were selected as sampling units for further selection of respondents. To use the methods of sample survey, it is essential to cover at least 10 percent units of first stage sampling units. Further, based upon the production of districts and variation present among the productions, a total of 292 districts were selected from all the 15 agro-climatic zones of India. The districts in which the crops are not cultivated were excluded for the selection. The number of districts in stratum were proportionally allocated. Trained manpower/ enumerators across the country were deployed to conduct the survey in all 15 ACZs. The agro-climatic zone wise districts and crops/ commodities selected for the study are given in **Table 3.3**.

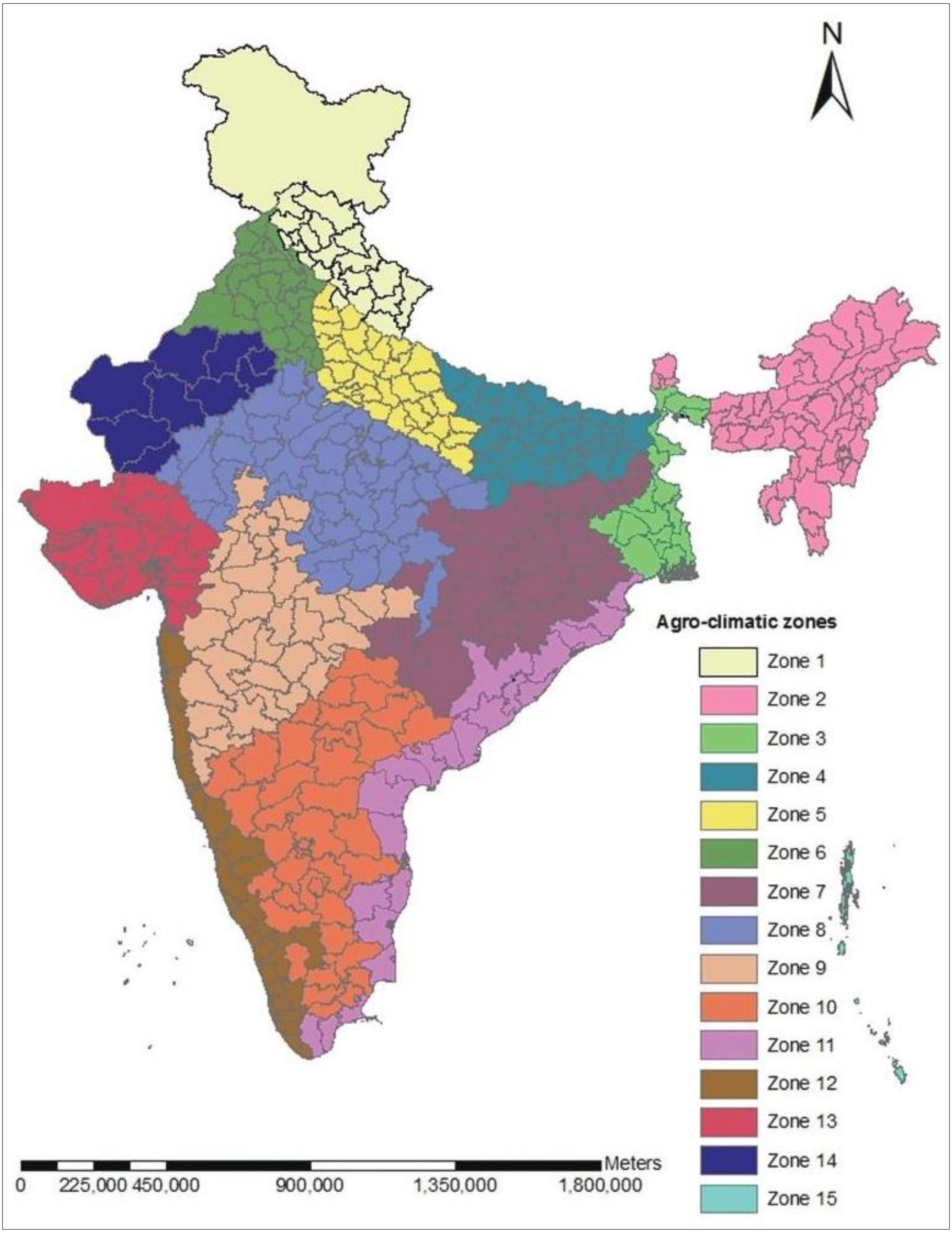
Table 3.3 List of districts and crops/ commodities selected for the study

S.No	ACZ	States	Districts selected	Crops/ commodities selected
1	ACZ1	Jammu & Kashmir	Anantnag, Baramulla, Rajouri, Jammu	Apple, mushroom, paddy, sugarcane, maize, wheat, capsicum, egg, green peas, milk
		Uttarakhand	Nainital, Udham Singh Nagar, Haridwar	
		Himachal Pradesh	Kinnaur, Shimla, Kangra, Solan	
2	ACZ 2	Nagaland	Mokokchung	Tapioca, turmeric, radish, mushroom, paddy, pineapple, wheat, cabbage, egg, meat, poultry meat, cashew nut
		Meghalaya	East Garo Hills, West Jaintia Hills, West Garo Hills, South Garo Hills	
		Assam	Kamrup, Barpeta, Golaghat, Nagaon, Sonitpur, Dima Hasao	
		West Bengal	Darjeeling, Cooch Behar, North Dinajpur	
3	ACZ 3	West Bengal	Bankura, Murshidabad, South 24 Parganas, Nadia, Barddhaman, Paschim Medinipur, Dinajpur Uttar, North 24 Parganas, West Bardhaman, Hooghly, Purba Medinipur, Jalpaiguri	Green gram, beans, brinjal, radish, paddy, pineapple, wheat, maize, cabbage, cauliflower, egg, meat, milk, poultry meat, potato, sunflower, groundnut, okra, inland fish, marine fish
4	ACZ 4	Uttar Pradesh	Firozabad, Azamgarh, Jaunpur, Ghazipur, Mirzapur, Basti	Green gram, muskmelon, bottle gourd, brinjal, radish, paddy, sugarcane, pineapple, wheat, maize,

S.No	ACZ	States	Districts selected	Crops/ commodities selected
		Bihar	Muzaffarpur, Vaishali, Nalanda, Aurangabad, Rohtas, West Champaran , Kishanganj, Bhagalpur, Begusarai, Katihar, Nalanda, Madhepura, Supaul, Darbhanga, Bhabhua (Kaimur)	cabbage, cauliflower, potato, Bajra, sunflower, okra, inland fish, egg
5	**ACZ 5**	Uttar Pradesh	Aligarh, Farrukhabad, Agra, Mainpuri, Budaun, Kaushambi, Bareilly, Barabanki, Shahjahanpur, Bijnor, Muzaffarnagar, Hardoi, Bulandshaher, Kasganj , Kanpur Nagar, Meerut, Unnao, Saharanpur, Kanpur, Hathras	Muskmelon, bottle gourd, black gram, mushroom, paddy, sugarcane, wheat, maize, green peas, egg, meat, milk, poultry meat, Bajra, inland fish
6	**ACZ 6**	Rajasthan	Hanumangarh, Sri Ganganagar	Green gram, muskmelon, citrus, radish, mushroom, paddy, sugarcane, wheat, maize, capsicum, green peas, egg, meat, milk, poultry meat, potato, Bajra, sunflower, cottonseed
		Punjab	Kapurthala, Muktsar, Fazilka, Hoshiarpur, Ludhiana, Sangrur, Gurdaspur, Rupnagar, Amritsar , Moga, Jalandhar	
		Haryana	Sonipat, Karnal, Rohtak, Yamunanagar, Sirsa, Jind , Mewat, Bhiwani, Mahendergarh, Kurukshetra	
7	**ACZ 7**	Rajasthan	Ajmer, Pali, Tonk, Bhilwara, Bundi, Jhansi, Lalitpur, Bundi,Udaipur, Jaipur , Alwar	Green gram, papaya, pigeon pea, soybean, sorghum, bottle gourd, brinjal, citrus, radish, black gram, mushroom, paddy, wheat, maize, cabbage, cauliflower, green peas, egg, onion, milk, coriander, Bajra, groundnut, okra, inland fish, cottonseed
		Jharkhand	Sahebganj	
		Madhya Pradesh	Chhindwara, Narsinghpur, Betul, Sagar, Indore, Khandwa, Guna, Morena, Jabalpur, Shivpuri, Mandla	
		Maharashtra	Chandrapur, Bhandara, Gondiya	
		Chhattisgarh	Durg, Kondagaon, Korba, Janjgir Champa,	
		Uttar Pradesh	Jhansi, Jalaun, Lalitpur	
		Odisha	Sundargarh, Kalahandi, Kendujhar, Bargarh	
8	**ACZ 8**	Rajasthan	Baran, Jhallawar, Chittorgarh, Dausa	Soybean, sorghum, pigeon pea, black gram, paddy, wheat, maize, green peas, coriander, Bajra, chickpea
		Uttar Pradesh	Mahoba	
		Madhya Pradesh	Singrauli, Alirajpur, Shahdol	
9	**ACZ 9**	Madhya Pradesh	Barwani, Khargone, Mandsaur, Ujjain, Dewas, Neemuch, Indore, Ujjain , Jhabua , Ratlam, Rajgarh	Chilli, green gram, muskmelon, soybean, sorghum, wheat, capsicum, green peas, potato, coriander
		Odisha	Mayurbhanj	
		Maharashtra	Jalgaon, Nanded, Latur, Amravati, Yavatmal, Akola, Solapur, Pune, Sangli, Osmanabad, Aurangabad, Jalna, Nagpur, Ahmednagar, Nashik, Washim, Kolhapur, Buldana, Nashik , Dhule, Satara, Hingoli, Parbhani	Banana, pigeon pea, sapota, soybean, sorghum, citrus, pomegranate, black gram, grapes, paddy, mushroom, sugarcane, green gram, maize, egg, meat, onion, poultry meat, Bajra, sunflower, groundnut, safflower, cottonseed
10	**ACZ10**	Andhra Pradesh	Prakasam, Guntur, Anantapur, Chittoor,	Chilli, chickpea, mango, tomato, green gram,

S.No	ACZ	States	Districts selected	Crops/ commodities selected
			Kadapa, Kurnool, Krishna, East Godavari, West Godavari, Guntur, Krishna, Vishakhapatnam, Srikakulam, Nellore	papaya, turmeric, banana, muskmelon, sorghum, citrus, sapota, pigeon pea, pomegranate, soybean, beans, coconut, black gram, grapes, paddy, mushroom, sugarcane, maize, capsicum, egg, meat, onion, areca nut, milk, poultry meat, cashew nut, Bajra, sunflower, groundnut, tapioca, safflower, inland fish, black pepper, marine fish, cottonseed
		Telangana	Warangal Rural, Khammam, Mahabubnagar, Nalgonda, Vikarabad, Siddipet,Rangareddy, Hyderabad ,	
		Tamil Nadu	Dindigul, Dharmapuri, Namakkal	
		Odisha	Cuttack, Puri, Ganjam	
		Karnataka	Chamarajanagara, Gulbarga, Bidar, Dharwad, Bagalkot, Bijapur, Belgaum, Kolar, Chitradurga, Hassan , Tumkur, Chikkaballapur , Bangalore rural ,Bellary, Gadag , Davangere, Koppal, Raichur	
11	**ACZ 11**	Tamil Nadu	Tuticorin, Tirupathur, Virudhunagar, Ramanathapuram, Vellore, Cuddalore, Thanjavur, Villupuram, Perambalur, Kancheepuram, Kanniyakumari, Nagapattinam	Banana, sapota, sorghum, beans, black gram, paddy, sugarcane, green gram, maize, cabbage, cauliflower, meat, egg, groundnut, okra, marine fish
		Odisha	Jagatsinghpur, Kandhamal, Baleshwar, Bhadrak	
12	**ACZ 12**	Kerala	Kollam, Idduki, Thiruanathapuram, Ernakulum, Kasargod, Kannur Kozhikode, Malappuram, Kodagu	Sapota, banana, coconut, paddy, pineapple, maize, meat, areca nut, poultry meat, cashew nut, black pepper, marine fish
		Maharashtra	Palghar, Raigarh, Thane, Mumbai, Ratnagiri	
		Karnataka	Shimoga, North-Kannada, South-Kannada, Chikmagalur	
		Tamil Nadu	Theni	
13	**ACZ 13**	Gujarat	Kutch, Anand, Bharuch, Vadodara, Panch Mahals, Junagadh, Navsari, Banas Kantha, Chhota Udaipur, Ahmedabad, Sabarkantha, Ahmadabad , Dohad , Rajkot, Bhavnagar, Mehsana , Banas Kantha, Kheda, Surat, Tapi, Amreli, Gir Somnath, Porbander	Green gram, papaya, banana, pigeon pea, sapota, beans, brinjal, black gram, paddy, wheat, maize, cabbage, cauliflower, onion, milk, potato, Bajra, groundnut, okra, marine fish, cottonseed
14	**ACZ 14**	Rajasthan	Nagaur, Sikar, Jhunjhunu, Rajsamand , Jalore, Bikaner, Jodhpur	Sorghum, wheat, maize, Bajra, groundnut, cottonseed
15	**ACZ 15**	Union Territory Andaman and Nicobar Islands	Nicobar	Coconut, areca nut

Fig. 3.1 Agro Climatic Zones of India



Source: Planning Commission, 1989

3.3.2 Selection of blocks in the district

All blocks in the selected districts were enlisted and two blocks were randomly selected from this list using a randomisation method.

3.3.3 Selection of villages

All villages producing the selected crop/ commodity in the selected block were enlisted. A sample size of 5 villages were randomly selected from the list using the random sampling method.

3.3.4 Selection of farmers

A sample of 10 farmers from each selected village was randomly drawn from the list of farmers growing that particular crop/ commodity. During the selection of villages, it was ensured that the number of farmers cultivating the selected crops/ commodities were sufficient from which samples would further be drawn. In case the number of farmers were not sufficient in a particular village or block, more numbers of villages and blocks were covered for data collection from adjoining areas.

Mushrooms: Cluster based approach was used for the selection of farmers. As the farmers were not sufficient in a specific village, farmers from more than 5 villages were enumerated and then 10 farmers from the prepared list were drawn randomly.

Milk: From each village, a sample of 10 milk producing farmers from the prepared list were were randomly selected.

Eggs: All farmers having backyard poultry farms were enumerated and 50 farmers were randomly selected. Thereafter, 6 commercial farms were also selected randomly.

Meat and poultry meat: From a sampled district, all farmers having butcher shops/slaughter houses were enumerated and 25 samples were randomly selected for meat, whereas 50 samples were selected for poultry meat.

Fish: A sample of 3 fishermen from each village totalling to 30 fishermen from each district were randomly drawn for the data collection by enquiry.

3.3.5 Selection of field and plot

For recording the data related to post-harvest losses through observation during farm operations, this selection was made for all the categories of cereals, pulses, oilseeds, fruits, vegetables, plantation crops, and spices. To estimate losses in the harvesting operation, 4-5 fruit bearing trees/ plants/ vines or a plot size of 5*5 metre or a plot size allowed by the farmer was taken.

For ***fisheries***, ponds/ landing centers were randomly selected from the list of all enumerated fishponds. For estimation of post-harvest losses in ***egg, meat and poultry meat***, 10 poultry farms and 5 slaughter houses were taken.

3.3.6 Selection of wholesalers and retailers

From a sampled district, 2 wholesalers and retailers, each were randomly selected for data collection by both enquiry and observation. For estimation of post-harvest losses by observation, three lots (boxes/crates/bags/piles) were selected. Both wholesalers and retailers were not covered for sugarcane while assessing post-harvest losses as sugarcane majorly transported to sugar mills directly from farms.

3.3.7 Selection of processing units and godowns

From a sampled district, 2 processing units and godowns each were randomly selected for data collection by both enquiry and observation. For estimation of post-harvest losses by observation, three lots (boxes/crates/bags/piles) were selected. In cases where processing units or godowns were not available in the selected district, they were covered in adjacent districts or consumption centres or in other sampled districts as per the availability. Pack houses (with pre-cooling facility) for fruits such as grapes and pomegranate, waxing units for citrus fruits, ripening chambers for banana and mango were also covered under the study.

3.3.8 Selection of transporters

From a district, 2 transporters were randomly selected for data collection by both enquiry and observation. To estimate the post-harvest losses at transportation for fruits and vegetables, the observations on post-harvest losses both at loading point and unloading point were taken by selecting three to five sample lots (boxes/crates/bags). In case of other crop categories, observations for post-harvest losses were taken at unloading point.

3.4 Sample size

Operation/ channel wise sample size are discussed in the following sections.

3.4.1. Farm operations

From a sampled district, two blocks were randomly selected. From each block, five villages were randomly selected from a list of villages totalling to 10 villages from a district. A sample of 10 farmers from each village was randomly drawn from the list of farmers growing that particular crop/ commodity in the selected village and thereby totalling to 100 farmers in a district for the data collection by enquiry. From the list of 10 farmers, 2 farmers were randomly selected for data collection by observation.

Fish: A sample of 3 fishermen from each village totalling to 30 fishermen from each district was randomly drawn for the data collection by enquiry

Eggs: Fifty backyard poultry farms and 6 commercial farms were selected randomly.

Meat and poultry meat: A sample of 25 and 50 butcher shops/slaughter houses were randomly selected for meat and for poultry meat, respectively.

Fish: A sample of 3 fishermen from each village totalling to 30 fishermen from each district were randomly drawn for the data collection by enquiry.

3.4.2 Storage at producer level

Samples of farmers selected for the data collection of farm operations were taken for data collection at storage also by both enquiry and observation.

3.4.3 Storage at market level

In a sampled district, 2 units of stakeholders such as wholesalers, retailers, transporters, storage units and processing units were selected randomly from the list prepared for the purpose using randomisation method. In case, any channel was not available in the selected district, they were covered in nearby/ adjoining areas. Data by both enquiry and observation was collected from all the selected stakeholders. The category wise sample size covered under the study is indicated in **Table 3.4.**

Table 3.4 Sample size covered under various categories

Categories	Farmers	Wholesaler	Retailer	Transporter	Storage	Processing unit	Total
Cereals	12960	216	216	216	216	204	14028
Eggs	1254	19	19	19	19	-	1330
Fish	1368	190	190	190	72	72	2082
Fruits	9000	150	150	150	132	136	9718
Meat	990	22	22	22	-	22	1078
Milk	1560	-	26	26	-	26	1638
Oilseeds	6660	128	128	128	128	128	7300
Plantation Crops	3840	38	38	64	36	60	4076
Pulses	8400	126	126	126	126	126	9030
Spices	2400	40	40	40	40	20	2580
Vegetables	14640	244	244	244	108	113	15593
Grand Total	**63072**	**1173**	**1199**	**1225**	**877**	**907**	**68453**

The total number of respondents for the assessment of both farm operations and market channels are indicated in ***Appendix 2*** *and* ***Appendix 3****, respectively. Also, crop-wise number of ACZ, states and districts covered are indicated in* ***Appendix 4****.*

3.5 Survey schedule development

The schedules/ questionnaires for data collection developed in the ICAR-CIPHET (2015) were adopted with the following modifications.

1. Computer Assisted Personal Interview (CAPI) mobile based application was used for capturing of data at the field. The recording of geotagged locations is one of the features of this application.
2. Infrastructure gaps across all channels of the supply chain were enquired from all the stakeholders.
3. Marketing details such as quantity consumed, quantity sold were captured at the farmer level.
4. The data regarding post-harvest losses during transport for perishable products like fruits were recorded at source and destination.
5. Data from transporters of various nodes of the supply chain for both fresh and frozen meat and rejection of meat during ante mortem and post mortem, edible and inedible offals were captured.
6. Information regarding assistance by Fish Farmers Development Agency (FFDA), Kisan Credit Card (KCC) and any other subsidy availed by fishermen was enquired. Losses by observation at storage and transport operations at fishermen level were also captured.
7. The questionnaires with respect to various holding levels at farm level and market level used during the study are given as **Appendix 1**.

3.6 Training for assessment of post-harvest losses

Training on data collection was given to all field investigators through video conferencing mode appointed for conduct of study. All the concerned were instructed to follow methodology for data collection properly and to report the actual data collected from the field.

In addition, regular monitoring of the field survey under the supervision of NABCONS team was carried out.

3.7 Webinar on post-harvest losses-assessment & prevention strategies

A webinar on “**Post-harvest Losses-Assessment & Prevention Strategies**” was held on 05th March 2021 through video conferencing mode under the Chairpersonship of Smt. Pushpa Subrahmanyam, IAS, the then Secretary, MoFPI, GoI. The webinar was attended by esteemed panellists and participants from various International and National organisations pertaining to the sector. Various recommendations and suggestions were received from the panellists for smooth conduct of study.

3.8 Seminar on post-harvest losses-assessment & prevention strategies

A seminar on “**Assessment of Post-Harvest Losses of Agriculture and Allied Produce India**” was conducted on 30th September 2022 at India Habitat Centre, New Delhi under the Chairmanship of Shri. Minhaj Alam, IAS, Additional Secretary, MoFPI, GoI. In the seminar, there were 4 technical sessions wherein study findings and views by domain experts were discussed.

The seminar was attended by delegates from various Ministries, government bodies, and academic institutions. Valuable inputs and suggestions received from the delegates during the seminar were incorporated in the study report.

The eminent speakers of the seminar were from various renowned institutes which includes ICAR-Central Institute of Post-Harvest Engineering and Technology (CIPHET), Ludhiana, ICAR-Indian Agricultural Research Institute (IARI), New Delhi, National Institute of Food Technology Entrepreneurship and Management (NIFTEM), Kundli, Sonipat, ICAR-Indian Institute of Pulses Research, Kanpur, ICAR-Indian Institute of Maize Research Centre, Hyderabad, ICAR-Indian Institute of Oilseeds Research, Hyderabad, National Research Centre for Meat (NRCM), Hyderabad, Gujarat Cooperative Milk Marketing Federation Limited (Amul), ICAR- Central Avian Research Institute (CARI), Bareilly, Assam Fisheries Development Corporation Ltd, ICAR-Central Institute of Fisheries Technology (CIFT), Cochin, National Fisheries Development Board (NFDB), Hyderabad.

CHAPTER 4 DATA COLLECTION AND DATA ENTRY

The farm level survey data was collected in the schedules by trained field investigators, who were provided training by subject matter specialists. The market level survey was undertaken by the experts in post-harvest management and related fields. The outcome of study depends upon the method of conducting the survey and also the quality of the questionnaire utilised for the same. The survey schedules are described below:

4.1 Data collection by enquiry

Category-wise schedules were made for fruits, vegetables, spices, plantation crops, milk, meat (including poultry meat), eggs, inland fish, marine fish, oilseeds, cereals and pulses for collection of data by both enquiry and observation method. These schedules were interpreted with a CAPI based application for online data collection. A separate schedule was made for sugarcane, which was completed through google forms. Based on the enumeration, farmers and respondents from the market were selected.

4.1.1 Farm level

(i) Complete enumeration of farmers - The schedules/ questionnaires used for collection of farm level data under the study contained basic information about the farmer and the crop related particulars such as agro-climatic zone, state, district, block, village, name of the farmer, area under the crop, major crops grown by farmers, variety of the crop grown, etc. This information was collected for all respondents.

Table 4.1 Category-wise number of schedules

Categories	No. of schedules	Type of schedule/ questionnaires
Fruits, vegetables	5	Farm level, wholesalers/ retailers, storage, pack house/primary processing units and transporters
Cereals, pulses, oilseeds, plantation	5	Farm level, wholesalers/ retailers, storage, processing units and transporters
Egg, spices	4	Farm level, wholesalers/ retailers, storage, and transporters
Milk	4	Farm level, retailers, processing units and transporters
Meat and poultry meat	6	Slaughter house (farm level), municipal slaughterhouse (farm level) wholesalers/ retailers, storage, processing units and transporters
Inland fish and marine fish	5	Farm level (inland fish), farm level (marine fish), wholesalers/ retailers, storage/ processing units and transporters

(ii) Losses during farm level operations - This data was collected during the harvesting period of all the crops/ commodities. Information such as season of crop, month of harvesting, total quantity produced, quantity consumed by farmers were recorded. Data related to the method of operation, equipment used and post-harvest losses for various farm operations such as harvesting, collection, threshing/ dehusking, sorting/ grading, winnowing/ cleaning, drying, packaging, storage and transportation were recorded. Causes/ reasons for the occurrence of the post-harvest losses and gaps in the existing infrastructure were also recorded.

For **eggs**, information such as the type of birds, system of poultry rearing, type of poultry house was recorded. For the estimation of post-harvest losses for **meat and poultry meat**, slaughter houses were covered. Type of animals, number of animals, method of milking and other related details were recorded for assessment of post-harvest losses in **milk**. For estimating the post-harvest losses in **inland fish** and **marine fish**, data related to all the operations performed by fishermen was recorded. The data for harvesting (catching) of marine fish was collected only by enquiry method.

4.1.2 Market level

(i) Complete enumeration of market players -The schedules/ questionnaires used for collection of market level data under the study contained basic information about the stakeholder and the crop related particulars such as agro-climatic zone, state, district, block, village, name of the market player, crops/ commodity handled, capacity of storage etc. This information was collected for all respondents.

(ii) Losses during market level operations - Data for post-harvest losses at the market level were collected in 5 schedules/ questionnaires, namely for wholesalers, retailers, transporters, storage channels as well as primary processing units. The schedule contained basic information regarding quantity stored, withdrawn, addition, total quantity stored, quantity lost during storage and handling were recorded. The causes/ reasons for occurrence of post-harvest losses and gaps in the existing infrastructure were also recorded. The schedule for transporters also contained information regarding the type of fleet, source and destination details, distance covered and time taken for reaching the destination.

4.2 Data collection by observation

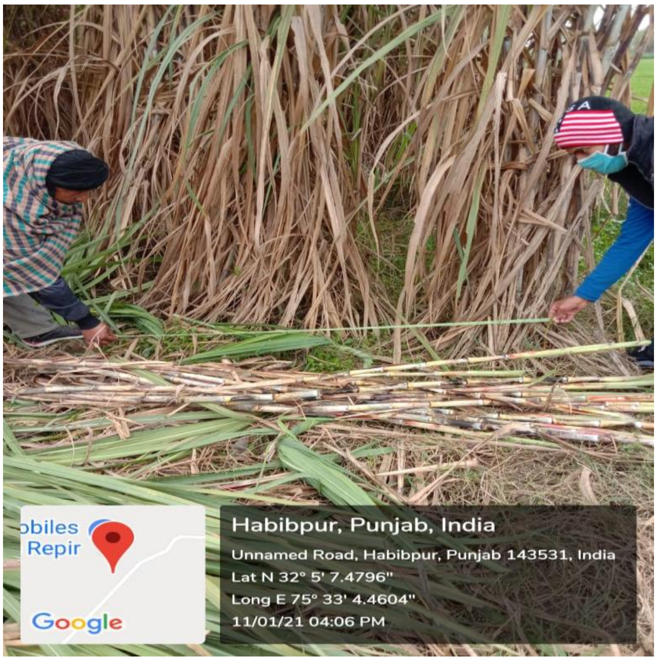
The schedules that were developed for eliciting information from respondents by enquiry consisted of 2 parts- enquiry and observation, therefore, the same schedules were used for collection of data by observation. The observations for different operations, which were going on at the time of survey were considered.

4.2.1 Farm level

(i) Losses at farm level in cereals, pulses, oilseeds and spices- The questionnaires prepared for inquiry of farmers growing cereals, pulses, oilseed and spices were used by the survey team for recording losses by observation also for the ongoing farm operation by the selected farmer. To estimate losses in the harvesting operation, a plot size of 5*5 metre or a plot size allowed by the farmer was taken during the observation method. Crop obtained from the harvesting of the selected plot size was taken as the harvesting lot. Crops damaged or fallen during the harvesting operation were observed by the field investigator. In case of harvesting the crops with a combine harvester, the production from demarcated fields was recorded after completion of the harvesting operation. After measuring the actual area of the selected field in which harvesting was carried out by combine harvester, the yield from 5m x 5m plot was estimated. The fallen grains from the demarcated field were collected and weighed to estimate harvesting losses in both the cases.

For the estimation of post-harvest losses during threshing and cleaning/ winnowing operations, the produce and straw obtained from the operations were weighed separately. From the straw obtained, a sample of 250/ 500 g was drawn and the grains in the straw were separated, weighed and accounted as loss. For operations such as storage and transportation, three lots (bags/piles) per farmer were selected randomly. In each of the lot, any crop damaged or fallen/ spilled during the operation were checked and were recorded as loss. Causes of post-harvest losses were also recorded.

(ii) Losses at farm level in fruits and vegetables - The questionnaires prepared for inquiry of farmers for fruits & vegetables were used by the survey team for recording losses by observation and also for the ongoing farm operation by the selected farmer. To estimate losses in the harvesting operation, fruits harvested from the selected trees were weighed and taken as the harvesting lot. In the case of vegetables, a plot of 5*5 metre or a plot size allowed by the farmer was taken and the harvested produce obtained was taken as the harvesting lot. The harvest produce was then analysed for any damages or injuries during the operation. The fallen produce during the operation was also collected. Produce unfit for human consumption and thrown during the operation was weighed and recorded as loss. For estimation of post-harvest losses at the sorting/ grading operation, three lots were drawn randomly and were sorted/graded. Damaged or discarded produce during the operation were recorded. For estimation of post-harvest losses at farm storage, three lots were randomly drawn and the damaged and spoiled produce was recorded. During transportation, 3-5 lots were randomly selected and were observed for post-harvest losses at loading and unloading points. The damaged and spoiled produce were separated, weighed and accounted for loss.



Measurement of plot size for observation

(iii) Losses at farm level in plantation crops- The questionnaires prepared for inquiry of farmers of plantation crops were used by the survey team for recording post-harvest losses by observation and also for the ongoing farm operation by the selected farmer. To estimate losses at harvesting, produce from selected trees was checked for any kind of damages or injuries due to the harvesting method in coconut, cashew nut and areca nut. Nuts not fit for human consumption were segregated from the rest of the lot and were accounted as post-harvest loss. Causes of post-harvest loss were also recorded for estimating losses during operations such as collection, sorting and grading and packaging, three lots (boxes/crates/bags/piles) per farmer for each operation were selected randomly. In each of the lot, damaged or injured produce during the operation were checked and were recorded as loss. To estimate loss at farm level transportation, sample lots (boxes/crates/bags) were selected and any damages to produce was checked and recorded. For estimation of post-harvest losses in sugarcane at farm level, 5m x 5m size plot was demarcated and farmers were allowed to harvest crop in that plot. The produce of the demarcated plot was collected separately and weighed to get the production data. After harvesting, the stubbles left in the demarcated plot were separated and collected. Weight of stubble pieces left in the field and stubbles cut while cleaning and detopping of sugarcane were considered as post-harvest losses at farm level operations. Collected bundles from the same plot were marked and loaded in trucks for observation of staling loss during transportation.

(iv) Losses at farm level in eggs - Data on post-harvest losses during collection, sorting and grading, packaging, storage and transportation were collected from 20 percent of farmers selected for enquiry. For the survey, all the eggs laid in the selected shed of the randomly selected farm were collected. Total number of eggs laid and damaged eggs were counted separately to record post-harvest losses during collection. For estimating post-harvest losses during operations such as packaging, storage and transportation, three lots (boxes/crates) per farmer for each operation were selected randomly. In each of the lot, any eggs damaged or injured during the operation were checked and were recorded as loss.

(v) Losses at farm level in inland fish- For the harvesting operation, catching of fish was covered. The schedule/ questionnaire recorded particulars such as method, fishing gear, causes of post-harvest losses for all the operations. After catching, fishes were unloaded from the net at the boat for collection and sorting and grading. Spoiled fish unfit for human consumption and the fish left behind in the boat after unloading were weighed for assessing post-harvest losses. Post-harvest losses during packaging and storage in ice boxes or in ice slabs were observed and recorded.

(vi) Losses at farm level in marine fish- The schedule/ questionnaire for marine fish covered data for all the applicable operations. Fishes left behind in the boat after unloading were weighed and accounted as loss during collection operation. Spoiled or damaged fish discarded by fishermen during sorting and grading was observed. Weight of fishes which were damaged/spoiled and were unfit for human consumption was computed as the post-harvest loss.

(vii)Losses at farm level in meat- To estimate post-harvest losses during slaughtering and dressing, 1-2 carcasses of goat, /buffalo or pigs were selected for meat, whereas 5-10 birds were selected for poultry meat. After slaughtering and dressing, the weight of the fresh carcass was taken. Part of the carcass not fit for human consumption due to disease, damage or injury was segregated from the rest of the lot, weighed and was accounted as loss.

(viii) Losses at farm level in milk- To estimate losses at milking, the milk that was collected during one session was observed and any leakages/ spillages/ spoilage was recorded. For estimating post-harvest losses during transportation, three lots per farmer were selected randomly and post-harvest losses during transportation were recorded.

Collection of milk for observation

4.2.2 Market level

Schedules used for collection of data on post-harvest losses at market level channels (wholesaler, retailer, godown, transporter and processing unit) through enquiry were followed for observation also. The respondents who were enquired about the losses were further observed to estimate the post-harvest losses.

(i) Losses at market level in cereals, pulses, oilseeds and spices- For observation of post-harvest losses at wholesaler and retailer level in cereals, pulses, oilseeds and spices, three lots were selected in which losses during storage and handling were recorded. At transporters level, total commodity weight transported by three vehicles at source and destination was recorded to estimate post-harvest losses as reduction in weight during the transport. At the storage level, a godown visit was done in which three bags were selected and weighed followed by a second visit during which the same bags were weighed again to record the losses.

(ii) Losses at market level in fruits and vegetables- For observation of post-harvest losses at wholesaler and retailer level in fruits and vegetables, sample lot was selected and spoiled or damaged fruit/ vegetable from the sample lot was weighed. At transporters level, at the source/ loading point, 5 crates/ boxes were randomly selected. A sample lot from each crate/ box was randomly drawn and observed for damage/ spoilage. The selected crates/ boxes were then marked, labelled and kept in the vehicles. At the destination/ unloading point, the same labelled crates/ boxes were opened and samples were drawn from all the 5 crates/ boxes. The damaged or spoiled fruits and vegetables were separated, weighed and computed as loss during the transportation operation. Similarly, at the storage level, godown/ cold store visit was made in which five sample lots were drawn from randomly selected crates/ boxes. After recording the reading for loss, the samples were packed, marked and labelled and were kept in the storage. A second visit was then conducted where the marked crates/ boxes were opened again. Samples were drawn for assessment of damaged or spoiled fruits and vegetables during the period of storage. The spoiled and damaged weight was then accounted as the loss during storage at market level. In the processing unit, three sample lots were randomly selected.

Selection of samples for observation at storage

(iii) Losses at market level in plantation crops- At market level, respondents such as wholesalers and retailers, transporters, godowns and processing units were covered in the survey of plantation crops viz. coconut, cashew nut and areca nut. For estimating losses, three lots (boxes/crates/bags/piles) per market respondents were selected at random. In each of the lot, nuts damaged during the operation were checked and recorded as loss. To estimate loss at market level transportation, sample lots (boxes/crates/bags) were selected and any damages to nuts/ reduction in weight during transportation was checked and recorded as loss. To estimate post-harvest losses at storage level, 2 visits to warehouses were made. During each visit post-harvest loss was recorded in identified lots during storage period. For estimating post-harvest losses of sugarcane at market level channels i.e. transportation and at sugar mill level was done by observation. Data of weight of quantity loaded and unloaded along with distance of mill from source was collected and loss in weight was considered as staling loss in sugarcane during transportation. Similarly, loss in weight of sugarcane after unloading at mill to before crushing in the sugar mill was considered as staling loss in sugarcane at sugar mill level.

(iv) Losses at market level in eggs For estimating post-harvest losses in market channels through observation, three to five lots (boxes/crates) per market respondent were selected at random and losses were recorded along with the causes. To estimate loss during transportation, sample lots (boxes/crates) were selected and any damages to eggs during loading and unloading was checked and recorded.

(v) Losses at market level in fish- For fisheries, damaged and spoiled produce was separated and weighed at the unloading points at various stakeholders and accounted as post-harvest losses. During storage, spoiled fish were sorted and weight was recorded for assessment of losses. Data was collected based on the operations performed by the fishermen. During the drying operation, a lot was randomly drawn and spoiled fish was observed and computed as loss.

Transportation of marine fish

(vi) Losses at market level in meat- For estimating post-harvest losses at market channels through observation, three to five lots (animal/ bird carcasses) per market respondent (wholesaler and retailer) were selected at random and losses were recorded. To estimate loss in transportation, sample lots were selected and any damage to carcasses during loading and unloading was checked and recorded. In processing units, any damage to the carcasses during storage was recorded.

(vii) Losses at market level in milk- For estimating post-harvest losses at retailer level through observation, three to five lots (pouches) per market respondent were selected at random and losses were recorded. In processing units, any loss during storage was accounted for. The causes of post-harvest losses were also recorded.

Selection of samples of apple at loading point in Anantnag, Jammu and Kashmir

Unloading points in Kolkata and Uttar Pradesh

CHAPTER 5 DATA ANALYSIS

Data analysis is the most crucial part of any research for drawing the inferences about the population under study. The number of respondents, data volume, types (nominal, ordinal etc.) and number of variables, methods, tools and techniques used play an important role in the analysis of data. In the present study, the main aim was to compute post-harvest losses from the data collected from fields at district levels and then pooling them at agro-climatic zone/state and national levels.

5.1 Analysis tools and techniques

The data of the selected crops from the selected districts/ blocks/villages/households collected by the designated survey agencies were entered in data entry software (Computer Assisted Personal Interviews (CAPI) based). In this software, the functionality of internal consistency checks of data at the time of data entry was inbuilt.

The web application inbuilt with geo-tagging feature was created to track progress of project and monitor the activities of survey team working pan India. Geo-tagging has already proved to be a powerful tool in effective monitoring of the project. Subsequently, the data entries were scrutinised for any omissions and errors incurred during data collection and/or entry. Wherever there were inconsistencies, possible corrections were made and revalidating of the same was done by resending the field investigators to the field. If it was not possible to rectify the errors/ discrepancies, data was discarded as outliers. The outliers were removed using the standard method of Box Plots.

The estimation of post-harvest losses was carried out at district level followed by pooling at agro-climatic zone (ACZ) level by assigning appropriate weights. National level estimations of post-harvest losses were obtained by pooling the estimates of ACZ levels. Different estimators (formulae) were employed to estimate the post-harvest losses at various levels as described in the Terms of Reference (ToR) and other project documents. In addition, the state wise estimates were also obtained by different estimators (formulae) used for ACZ level. The study team broadly followed ICAR-CIPHET 2015 study for estimation of post-harvest losses as mentioned in the Terms of Reference (TOR). The PPS based estimators were also obtained in this study for the further research and advancement in this area and the estimates have been presented in the **Appendix 9**. The sanitised/scrutinised data were analysed using the statistical software SPSS and advanced version of Microsoft Excel, using macros.

In this analysis, sampling weights were obtained for each record according to sampling design implemented for data collection at district level (i.e. weightage of the sample, no. of farmers, villages and blocks to their actual number). Same weights of ICAR-CIPHET (2015) were used at the district level estimation. For estimating losses at ACZ and state level, weightage was assigned based on the production of the specific crop/commodity in all the sampled districts. Similarly, post-harvest losses at the national level were estimated by assigning weightage based on the production of a specific crop/commodity in all the ACZs.

Wherever some crops were grown more than once a year, the data for post-harvest farm operations were collected for each harvest in that year and data were collected from each channel along the chain. Information on available stocks from the previous year, addition/withdrawal, the total quantity stored, and loss during the enquiry period were recorded. Type of storage and causes of loss were recorded carefully after cross-verification. More visits within a week or month in case of fruits, vegetables and plantation crops were undertaken depending upon storage periods.

The first few schedules of the questionnaire covered the complete enumeration of the selected farmers and market players. Based on the enumeration, farmers and respondents from the market were selected. Thereafter, sections for the collection of post-harvest loss data from all the channels along the supply chain, i) farm level (loading, transporting etc.), ii) storage and crop protection (packaging, ripening, cold storage, warehousing etc.) iii) market linkages (wholesale, retailer etc.,) entailing primary processing were covered.

Data analysis procedure are indicated in **Appendix 9.**

CHAPTER 6 RESULTS AND DISCUSSIONS

Mechanical harvesting of paddy using combine harvester

Data for the assessment of post-harvest losses for 54 crops and commodities was collected, scrutinised and analysed. This chapter deals in identification of post-harvest operations and the assessment of post-harvest losses in selected 54 crops and commodities across all 15 agro-climatic zones and at national level. Losses of crops and commodities in different districts, states and agro-climatic zones of India are indicated in **Appendix 5**, **Appendix 6** and **Appendix** 7.

6.1 Cereals

Cereals are an important component of human diet in India. The country has attained self-sufficiency in food grain production post green revolution with 289.11 million MT of cereals production in 2021-22 (Directorate of Economics and Statistics, 2021-22). Despite several initiatives by GoI and State Govt., a considerable quantity of cereals go waste due to lack of adequate post-harvest management infrastructure.

6.1.1 Farm level operations

Harvesting - Harvesting of cereal crops was mainly done manually using sickle. The machines viz. combine harvester, tractor mounted and reaper were also used for crops like wheat, paddy and Bajra. Wheat was cut above the root using sickle and kept in the field itself in bundles. Bajra was harvested either by cutting the ear head or cutting the entire plant using a sickle. Maize cobs were harvested when husk turned to yellow and grain became hard. Earheads of wheat and paddy getting detached while cutting were left in the field while harvesting. Maize cobs hidden in the stubble due to negligence of labour led to losses during harvesting. In case of harvesting with combine harvester losses were reported due to splitting and shattering of grains. The plants off side to the cutter area and at the corners of the field were not harvested. Also due to high moisture at harvesting stage mechanical damage was observed while harvesting with combine harvester.

Collection – The harvested cereal bundles were collected by labourers at one place for threshing/ drying by carrying them in bamboo baskets, plastic crates, basket and gunny bags on head/ tractor/ tempo. Being dry and delicate to handle breakage of spikes occurred, while collecting the harvested wheat. The maize plants were stacked in the field for some days followed by removal of sheath. There was practice of drying cobs on rooftops in the sun. In some places, the cobs were spread on the big tarpaulin plastic sheets for drying and making them ready for shelling. Some of the grains during this operation got detached from the cob contributing to post-harvest losses. Birds or rodents damage was also observed. The post-harvest losses in Bajra and sorghum at this operation were mainly due to improper handling, shattering of grains, damage by rain/bird due to delay in collection.

Threshing/ shelling - Threshing of manually harvested cereals was done by mechanical thresher. Physical damage, splitting and shattering occurred during threshing resulting in post-harvest losses. Light weight and immature grains were blown with straw while passing through the aspiration unit for separation of straw and dust causing post-harvest losses in wheat, sorghum, Bajra and paddy. Spillage was also observed during threshing operations around the threshing area. In the case of Bajra, a large number of farmers were observed to be using thresher machines, while a few resorted to traditional methods using sticks and hand pounding for threshing of produce. The manual threshing was performed by farmers retaining crops for self-consumption. The post-harvest losses observed in case of mechanical thresher were due to overheating and malfunctioning due to low voltage and jamming due to power cuts. In maize, shelling was done by hand, hand tools and with mechanical threshers during which incomplete shelling because of less efficient threshers resulted in higher post-harvest losses. Practice of re-passing of such cobs through thresher was followed by farmers at some places to overcome these post-harvest losses. Shattering and breakage of kernels was also observed during shelling, due to excessive drying of cobs and improper thresher adjustments by the operators.

Threshing of paddy using sticks

Cleaning/winnowing – The threshed cereal grains were cleaned removing extraneous matter by winnowing. Cleaning was done by winnowing kernels in front of the air stream, usually with the help of a table fan by farmers. The post-harvest losses during the operation were mainly due to spilling of grains.

Drying – Sun drying of cereal grains by spreading the grains and maize kernels on the waterproof tarpaulin sheets in the yard of farmers or in the field itself was performed after threshing to reduce its moisture content for safe storage over a long duration. During drying spoilage occurred due to birds and stray animals. Drying was not a prevalent operation in wheat, being usually harvested in the dry season.

Packaging - Packaging of cereals was done manually by farmers usually in gunny bags and plastic bags with the help of plastic tubs and then were sealed with the coconut coir/jute fibre thread for secure storage and transport. Spillage occurred during filling of bags causing post-harvest losses. In some areas drum was also observed as a packaging material for storage of small quantities of maize kernels for domestic use. Damaged/defective and old gunny bags were also found to leak the filled grains in some cases.

Transport - Bulk grains were transported from farm to farmer's home in trolleys directly getting loaded from combine harvester after harvesting. Bullock carts were also used by marginal farmers for transporting small quantities. Post-harvest losses occurred due to spillage caused by mishandling, while loading and unloading and defective packaging materials causing spillage.

Farm level storage – Storage was done at farmer level for a period ranging from few days to few months, before the sale to traders after proper packaging in gunny bags. The storage of grains at farm level was done in sacs or bins (wooden/ polyvinyl/metal) without any fumigation arrangement. The main cause of spoilage during storage was pest/insect attack and rodent attack. The pest transferred from earlier stored and infected grain lots stored in the same area caused spoilage in the stored grains. Some evidence of uses of dried Neem leaves and Aluminium phosphide tablets by farmers were reported to prevent the stored grains from spoilage in wheat.

6.1.2 Market level operations

Wholesalers and retailers - Cereals were handled and cleaned manually at the wholesalers and retailers level. For initial few days grains were kept in the open yard of Mandi by wholesalers and retailers and were transferred later to the godowns for long term storage. The major post-harvest losses were mainly due to spilling, physical damage/ crushing of grains and improper storage conditions.

Transporters - Cereal grains were transported through bullock carts, tractors, trucks, tempos and pickup vehicles. Improper packaging, improper loading and unloading practices and stacking practice resulted in spillage during transportation. Parkhhi/ bag spear was used to draw samples by wholesaler after unloading to test the sample before storage. The holes made for drawing samples, if not closed properly, caused spillage of grains leading to post-harvest losses.

Spillage of wheat at market level

Storage - Storage of cereal grains was done in godowns without any temperature control conditions at warehouses of State Warehouse Corporation, Central Warehouse Corporation, Food Corporation of India, private players etc. Cover and plinth storage was also used in some areas in the yard of APMC. Of late, bulk steel silo storage systems for rice and wheat are gaining momentum in states like Punjab, Haryana, Madhya Pradesh, Tamil Nadu etc. The post-harvest losses observed during storage of cereals were very low mainly due to spilling as a result of improper stacking and due to physical damage/ crushing. Spoilage due to insect infestation during storage was also observed.

Storage of grains in scientific godown

Processing - Milling units were observed for estimation of post-harvest losses during processing of cereal grains. The post-harvest losses observed at the milling units were mainly due to spilling of grains and improper handling. In case of maize, storage of maize kernels at processing level was assessed for post-harvest losses. Losses were caused due to spoilage by insects and pests from the infected lots. Also due to weather changes, moisture spots were observed at some storage areas.

Processing of greengram

6.1.3 Post-harvest losses at farm and market level

Assessment of harvest and post-harvest losses were carried out in 5 major cereals (paddy, wheat, maize, Bajra and sorghum). The extent of post-harvest losses at district level, state level and ACZ level, respectively are presented in **Appendix 5, Appendix 6** and **Appendix 7** and at national level are reported in **Table 6.1.**

The survey for estimating the post-harvest losses of ***paddy*** was conducted in 25 districts in 13 states covering 13 agro-climatic zones (ACZ1, ACZ2, ACZ3, ACZ4, ACZ5, ACZ6, ACZ7, ACZ8 ACZ9, ACZ10, ACZ11, ACZ12 and ACZ13). The overall post-harvest losses in paddy were the highest in Nagaon district (6.15%) and the lowest in Udham Singh Nagar district (2.91%). The overall post-harvest losses in paddy at state level were the highest in Assam (6.07%) and the lowest in Uttarakhand (2.87%). The overall post-harvest losses in paddy were the highest in ACZ2 (6.1%) and the lowest in ACZ1 (2.91%). The lower post-harvest losses in ACZ1 (Uttarakhand) & ACZ6 (Punjab) could be due to increased use of combine harvesters in these states. At national level, the post-harvest losses in farm operations were 4.16 percent and in market level operations it was 0.61 percent with an overall post-harvest loss of 4.77 percent in paddy. Among different operations, harvesting (1.42%) and threshing (0.97%) in farm operations and wholesaler (0.23%) level in market operations majorly contributed to the post-harvest losses in paddy. Earlier ICAR-CIPHET (2015) reported an overall loss of 5.53 percent for paddy, while the present study reported significantly lower post-harvest losses (4.77%) at 5 percent level of significance (**Fig. 6.1**). This decrease in overall post-harvest losses may be attributed to overall improvements such as increase in storage capacity.

Table 6.1 Post-harvest losses of cereals in percent at national level

Crop	Farm Operations									Market Level Assessment						Overall total loss
	Harves-ting	Collec-tion	Thresh-ing	Winnowing / cleaning	Drying	Packag-ing	Storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retailers	Processing unit	Trans-port	Total loss in market level	
Paddy	1.42 ±0.0131	0.42 ±0.0048	0.97 ±0.0081	0.4 ±0.0047	0.33 ±0.0037	0.13 ±0.002	0.40 ±0.007	0.1 ±0.0014	4.16 ±0.0187	0.06 ±0.0094	0.23 ±0.0381	0.03 ±0.0053	0.16 ±0.0193	0.12 ±0.0323	0.61 ±0.0546	4.77 ±0.0577 (5.53)
Wheat	1.44 ± 0.0567	0.33 ± 0.0164	1.05 ± 0.0483	0.34 ± 0.0263	0.05 ± 0.0045	0.01 ± 0.0016	0.29 ± 0.0186	0.1 ± 0.0029	3.61 ±0.083	0.02 ± 0.0062	0.13 ± 0.0315	0.05 ± 0.0203	0.28 ± 0.0977	0.08 ± 0.0738	0.56 ± 0.1282	4.17 ± 0.1527 (4.93)
Maize	0.9 ± 0.0323	0.37 ± 0.0274	1.2 ± 0.0406	0.19 ± 0.0081	0.16 ± 0.0127	0.11 ± 0.0085	0.17 ± 0.01	0.12 ±0.0108	3.22 ± 0.0629	0.003 ± 0.0005	0.29 ± 0.0974	0.14 ± 0.0625	0.09 ± 0.0297	0.14 ± 0.0446	0.67 ± 0.1275	3.89 ±0.1422 (4.65)
Bajra	0.97 ±0.0198	0.29 ±0.006	1.13 ±0.0238	0.16 ±0.0031	0.22 ±0.0043	0.25 ±0.0051	0.31 ±0.0067	0.26 ±0.0055	3.59 ±0.0335	0.13 ±0.006	0.12 ±0.0227	0.07 ±0.0093	0.13 ±0.0075	0.32 ±0.0283	0.78 ±0.0387	4.37 ±0.0512 (5.23)
Sorghum	1.55 ±0.0284	0.33 ±0.0064	1.8 ±0.0414	0.35 ±0.0094	0.1 ±0.0019	0.25 ±0.0064	0.23 ±0.0045	0.13 ±0.0025	4.75 ±0.0521	0.05 ±0.0122	0.76 ±0.1214	0.16 ±0.0221	0.02 ±0.005	0.19 ±0.0396	1.17 ±0.1303	5.92 ±0.1403 (5.99)

Note: *Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*
Figures in parentheses represent the losses of previous study by ICAR-CIPHET (2015)

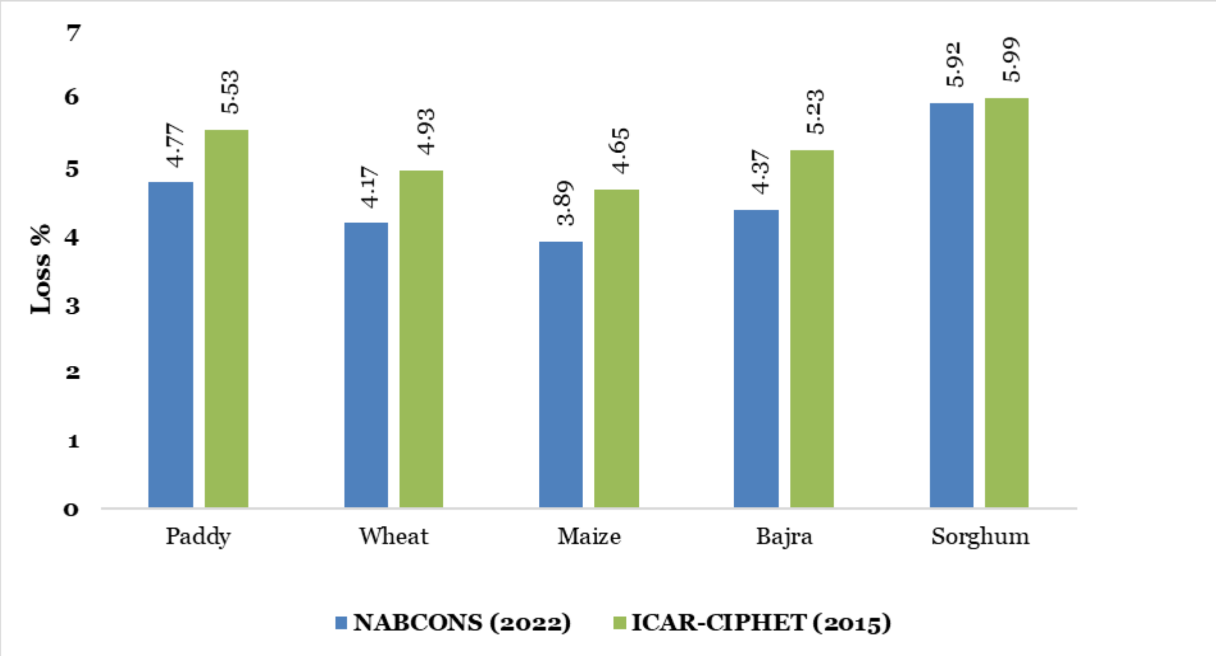
The survey for estimating the post-harvest losses of ***wheat*** was conducted in 19 districts across 9 states spread over 10 agro-climatic zones (ACZ1, ACZ3, ACZ4, ACZ5, ACZ6, ACZ7, ACZ8, ACZ9, ACZ13 and ACZ14). The highest overall post-harvest losses at district level were found in Jaunpur district (6.56%) and the lowest in Udham Singh Nagar district (1.39%) in wheat. The highest overall post- harvest losses in wheat at state level were observed in Uttar Pradesh (6.36%) and the lowest in Uttarakhand (2.51%) state. At ACZ level, the highest overall post- harvest losses in wheat were observed in ACZ14 (5.77%) and the lowest in ACZ1 (2.10%). The post-harvest losses in farm operations at national level in wheat were found to be 3.61 percent, while for market level, the post-harvest losses were 0.56 percent. The overall total post-harvest losses of wheat were 4.17 percent. The highest post-harvest losses were estimated during harvesting (1.44%) operation at farm level and during processing (0.28%) at market level assessment in wheat. Wheat harvested at low moisture content caused breakage and spillage during harvesting in both manual as well as mechanical methods, which led to higher post-harvest losses.

During storage at the processing unit, wheat was stored for a longer duration and impact of weather deviation resulted in moisture gain in stored wheat causing insect infestation. As indicated in **Fig. 6.1,** earlier study by ICAR-CIPHET (2015) reported overall level post-harvest loss of 4.93 percent for wheat, however the present study estimated significantly lower overall post-harvest losses (4.20%) at 5 percent level of significance. This decrease in overall post-harvest losses may be attributed to overall improvements in post-harvest infrastructure such as increase in use of combine harvesters, increase in storage capacity and modernisation of threshers etc.

The survey for estimating the post-harvest losses of ***maize*** was conducted in 23 districts in 14 states covering 13 agro-climatic zones (ACZ1, ACZ3, ACZ4, ACZ5, ACZ6, ACZ7, ACZ8, ACZ9, ACZ10, ACZ11, ACZ12, ACZ13 and ACZ14). The highest overall post-harvest losses in maize were found in Bhilwara district (5.34%) and the lowest in Kangra district (2.38%). At state level, the highest overall post- harvest losses in maize were observed in Punjab (6.91%) and the lowest in Himachal Pradesh (2.27%). At ACZ level, overall post-harvest losses in maize were the highest in ACZ6 (4.97%) and the lowest in ACZ1 (2.60%). At national level, the overall post-harvest losses for maize at the national level were observed as 3.89 percent comprising 3.22 percent in farm operations and 0.67 percent in market operations.

The highest post-harvest losses were estimated during shelling (1.2%) of maize at farm level, whereas the highest post-harvest losses were observed at wholesale level (0.29%) during market level assessment. During shelling of maize cobs, cobs were not being completely shelled and kernels were left on the surface of it after shelling. Also spillage occurred while shelling, immature and light weight kernels were blown with the aspirator in husk resulting in huge post-harvest losses. At wholesaler level huge post-harvest losses were observed as maize kernels were stored in open places causing spillage. Earlier ICAR-CIPHET (2015) reported overall level post-harvest loss of 4.65 percent for maize, however the present study estimated significantly lower (3.89%) post-harvest losses (**Fig. 6.1)** at 5 percent level of significance. This decrease in overall post-harvest losses may be attributed to overall improvements in post-harvest management practices, increase in storage capacity, technological interventions and promotion of new technologies (self-propelled maize combine harvester and maize dehusker and sheller by ICAR-Indian Institute of Maize Research).

Fig. 6.1 Comparison of post-harvest losses of cereals



The survey for estimating the post-harvest losses of ***Bajra*** was conducted in 18 districts from 8 states covering 9 agro-climatic zones (ACZ4, ACZ5, ACZ6, ACZ7, ACZ8, ACZ9, ACZ10, ACZ13 and ACZ14). The highest overall post-harvest losses in Bajra were reported in Prakasam district (5.11%), while the lowest losses were reported in Bhiwani district (2.93%). This could be due to non-awareness about modern farm machinery among farmers, use of improper harvesting techniques, shattering of grains during threshing, rough handling and delayed time in collection. The highest overall post-harvest losses in Bajra were reported in Andhra Pradesh (5.07%) and the lowest in Haryana (3.77%). At ACZ level, the highest overall post-harvest losses in Bajra were reported in ACZ10 (5.04%) and the lowest in ACZ6 (3.76%). The lowest post-harvest was observed in Haryana in ACZ6 due to use of mechanical threshers by some of the farmers and due to proper and careful operations carried out by farmers. It was observed during the study that the farmers of Haryana were progressive and used better farm management techniques thus minute losses were observed during harvesting. At national level, the post-harvest losses observed at farm level operations were 3.59 percent, whereas, for market level operations post-harvest losses were 0.78 percent with overall post-harvest losses of 4.37 percent. Among different operations, the highest post-harvest losses were observed at threshing (1.13%) in farm operations and at transport level (0.32%) in market channels. As indicated in **Fig 6.1,** the overall losses for Bajra reported by ICAR-CIPHET (2015) were 5.23 percent. The present study observed significantly lower post-harvest losses (4.37%) than ICAR-CIPHET (2015) at 5 percent level of significance. There have been continuous efforts in research and training conducted under the All India Coordinated Millet Improvement Project (AICMIP) on Bajra by ICAR majorly in the states of Rajasthan, Haryana, Gujarat, Tamil Nadu, Maharashtra, Karnataka, Madhya Pradesh and Andhra Pradesh.

The survey for estimating the post-harvest losses of ***sorghum*** was conducted in 24 districts in 6 states covering 6 agro-climatic zones (ACZ7, ACZ8, ACZ9, ACZ10, ACZ11 and ACZ14). The overall post-harvest losses at district level in sorghum were the highest in Neemuch district (7.42%) and the lowest in Guntur district (5.36%). The high post-harvest losses in Neemuch, Dewas and Khargone districts could be due to disease and pest infestation, spillage during harvesting and other farm operations, manual threshing by few farmers and improper storage conditions. The overall post-harvest losses in sorghum at state level were the highest in Madhya Pradesh (6.76%) and the lowest was reported in Andhra Pradesh (5.67%). At ACZ level, the overall post-harvest losses in sorghum were the highest in ACZ9 (6.29%) and the lowest in ACZ14 (5.82%). The post-harvest losses for sorghum in farm operations at national level were 4.75 percent and for market operations post-harvest losses were 1.17 percent with overall total losses of 5.92 percent in sorghum at national level. Among different operations, harvesting (1.55%) and threshing (1.80%) majorly contributed to the losses in farm level operations and at wholesaler (0.76%) level in market operations. ICAR-CIPHET (2015) reported an overall loss of 5.99 percent for sorghum, while the present study reported lower post-harvest losses (5.92%) and the difference was non-significant at 5 percent level of significance as depicted in **Fig 6.1**. This decrease in total post-harvest losses may be attributed to overall improvements such as enhanced usage of threshers for threshing, usage of hybrid varieties and increased use of covered spaces and godowns for storage.

Among cereals, shattering of grains during harvesting, manual methods of operations such as use of sticks for threshing, spillage of grains during various operations, damage by birds & rodents during drying were the major causes observed during farm level operations. Spillage during loading and unloading during transportation and improper storage conditions and mishandling at various market channels resulted in post-harvest losses during market operations. The estimated post-harvest losses in paddy, wheat, maize and Bajra obtained from the present study were significantly lower than the earlier conducted study (ICAR-CIPHET 2015), which can be attributed to the improved post-harvest infrastructure. As per WDRA, the storage capacity has increased up to 166.2 million MT (as on 31.03.2021) from 126.96 million MT (as on 2015). In order to boost farm mechanisation, the Government of India had introduced 'Sub Mission on Agricultural Mechanisation (SMAM)' in 2014-15. As of now, more than 13 lakh agricultural machines have been distributed and more than 27.5 thousand Custom Hiring Institutions have been established. Also, as per economic survey, the number of APMCs has been increased to 6946 in 2020-21 from 2477 in 2014-15 contributing towards reduction in post-harvest losses at market level.

Box 6.1 Good practices to be followed for cereals

- *Use of combine harvesters and threshers*
- *Direct loading into vehicles designed for the purpose and transport in bulk*
- *Storage in silos and scientific warehouses*
- *Negotiable warehouse receipt system by warehouse owners*
- *Availment of pledge finance from banks by the farmers*
- *Regular fumigation and pest control measures during storage*
- *Use of improved and hybrid varieties in crops such as sorghum and paddy.*
- *Use of e-commerce like e-NAM portal & Kisan Rath*
- *Branding and packaging of cereals*
- *Organized marketing of cereals through FPOs/ FPC.*
- *Use of drones for spraying chemicals over agricultural fields for efficient pest management of crops.*

6.2 Pulses

Pulses being an important source of nutrition is consumed abundantly in India and around the world. As per Directorate of Economics and Statistics 2021-22, the total pulses production in the country was 26.96 million MT. Although pulses production is increasing every year, a considerable quantity gets wasted due to lack of adequate post-harvest management infrastructure resulting in reduction in total availability of pulses for human consumption.

6.2.1 Farm level operations

Harvesting- Harvesting of pulses viz. black gram, green gram, chickpea and pigeon pea was done manually by using sickle. The post-harvest losses during harvesting operation were mainly due to shattering of pods and spilling due to mishandling by the farmers.

Mechanical harvesting of chickpea – combine harvester

Harvesting of green gram

Collection- Harvested pulse pods were collected by farmers using bullock cart, tractor and head load. The post-harvest losses observed during this operation were less and mainly due to improper handling and spilling of grains.

Threshing- Threshing of pulses was done manually by sticks as well as mechanically by threshers. The post-harvest losses observed during the operation were due to physical damage/ crushing and spilling of grains.

Threshing of pigeon pea

Winnowing/ cleaning- The threshed pulses were cleaned by winnowing removing extraneous matter. The operation was done manually by the farmers using fans or even by hand with natural wind. The post-harvest losses during the operation were mainly due to spilling of grains and swaying away of grains with stalks.

Winnowing/ cleaning of black gram

Drying- The pulses, after threshing and cleaning were sun dried before packaging. The post-harvest losses during drying were mainly due to spilling and shrinkage of grains and damage due to birds and rodents.

Drying of pigeon pea

Packaging- The pulses were packed in gunny bags and plastic bags. There was minute post-harvest during the operation and were mainly due to spilling during filling in the bags.

Transport- The pulses were transported through bullock cart, tractor and trolley from the farm to traders in the market. Losses during transportation were less as pulses were packaged properly before transporting. The losses observed were mainly due to spilling of grains due to improper loading and unloading practices.

6.2.2 Market level operations

Wholesalers and retailers- The pulses were handled and cleaned manually at the wholesalers and retailers level. The major post-harvest losses at the wholesalers and retailers were mainly due to spilling and physical damage/ crushing of grains and improper storage conditions.

Transporters- The pulses were transported through bullock cart, tractor and trolley. The post-harvest losses observed were mainly due to spilling of grains due to improper packaging and improper loading and unloading practices.

Wholesaler of chickpea

Storage- Storage of pulses was done in godowns without any temperature control conditions. The post-harvest losses observed during storage were low and were due to spilling as a result of improper stacking and due to physical damage/ crushing.

Processing- Dal milling units were observed for estimation of post-harvest losses during processing of pulses. The post-harvest losses observed at the processing units were mainly due to spilling of grain and improper handling.

Storage of pigeon pea

Storage of chickpea in processing unit

6.2.3 Post-harvest losses at farm and market level

Survey for assessment of harvest and post-harvest losses in pulses was conducted for 4 pulse crops (pigeon pea, chickpea, black gram and green gram). The extent of post-harvest losses at district level, state level and ACZ level are presented, respectively in **Appendix 5, Appendix 6** and **Appendix 7** and at national level are reported in **Table 6.2.**

The survey for estimating the post-harvest losses of ***pigeon pea*** was conducted in 14 districts in 6 states covering 5 agro-climatic zones (ACZ7, ACZ8, ACZ 9, ACZ10, and ACZ13). The overall post-harvest losses in pigeon pea were the highest in Yavatmal district (7.63%) and the lowest in Bharuch district (3.89%). The high post-harvest losses in Yavatmal and Latur districts were due to improper methods of harvesting, disease and pest infestation, use of sticks for manually threshing of grains resulting in grain shattering and mishandling resulting in spillage of grains. The overall post-harvest losses in pigeon pea at the state level were the highest in Maharashtra (6.9%) and the lowest was reported in Gujarat (4.27%). The overall post-harvest losses at ACZ level were the highest in ACZ8 (7.07%) and the lowest was reported in ACZ13 (4.2%). Improper harvesting method and mishandling resulted in high losses in ACZ 8 (Madhya Pradesh) and ACZ 9 (Maharashtra), while proper handling methods, better irrigation system and interventions by government departments for creating awareness amongst the farmers, may have resulted in the lowest post-harvest losses in ACZ13 (Gujarat). Post-harvest losses in farm operations at national level were found to be 4.67 percent in pigeon pea while for market level, the post-harvest losses were 0.99 percent. The overall total post-harvest losses of pigeon pea were 5.65 percent. Among different operations, harvesting (1.1%) and threshing (1.84%) majorly contributed to the losses in farm level operations, while processing (0.35%) and transport (0.27%) level in market operations. ICAR-CIPHET (2015) reported an overall loss of 6.36 percent for pigeon pea, while the present study reported lower post-harvest losses (5.65%). The estimated post-harvest losses obtained from the present study were significantly lower than the earlier conducted study (ICAR-CIPHET 2015) at 5 percent level of significance (**Fig. 6.2)**. This decrease in overall post-harvest losses may be attributed to overall improvements such as increase in the usage of threshers for threshing of pigeon pea, demonstrations on improved package of practices of pulses for cultivation of pigeon pea by ICAR, KVKs and SAUs and short duration of storage at the farm level.

Table 6.2 Post-harvest losses of pulses in percent at national level

Crop	Farm operations									Market operations						Overall total loss
	Harvesting	Collection	Threshing	Winnowing / cleaning	Drying	Packaging	Storage	Transport	Total loss in farm operations	Godown	Wholesalers	Retailers	Processing unit	Transport	Total loss in market level	
Pigeon Pea	1.1 ±0.0463	0.28 ±0.011	1.84 ±0.09	0.33 ±0.0088	0.2 ±0.0074	0.18 ±0.0071	0.52 ±0.028	0.21 ±0.0123	4.67 ±0.107	0.12 ±0.0306	0.1 ±0.0221	0.15 ±0.0382	0.35 ±0.0735	0.27 ±0.0619	0.99 ±0.1058	5.65 ±0.1505 (6.36)
Chickpea	1.49 ±0.0367	1.02 ±0.0153	2.00 ±0.0466	0.36 ±0.0064	0.35 ±0.0049	0.19 ±0	0.21 ±0	0.27 ±0.0091	5.89 ±0.0627	0.23 ±0.033	0.11 ±0.0154	0.12 ±0.021	0.09 ±0.0148	0.3 ±0.047	0.85 ±0.0647	6.74 ±0.0901 (8.41)
Green Gram	1.81 ±0.0669	0.69 ±0.0185	1.53 ±0.0532	0.32 ±0.0102	0.3 ±0.0081	0.19 ±0.0051	0.36 ±0.0135	0.11 ±0.0037	5.3 ±0.0897	0.02 ±0.0018	0.33 ±0.0522	0.29 ±0.0387	0.13 ±0.0167	0.12 ±0.0079	0.89 ±0.0676	6.19 ±0.1123 (6.60)
Black Gram	1.44 ±0.0294	0.77 ±0.0139	1.17 ±0.032	0.37 ±0.0071	0.35 ±0.0024	0.21 ±0.0061	0.51 ±0.0106	0.13 ±0.0056	4.95 ±0.0481	0.03 ±0.007	0.24 ±0.034	0.23 ±0.0422	0.14 ±0.0318	0.23 ±0.0512	0.88 ±0.0813	5.83 ±0.0945 (7.07)

Note: *Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*
Figures in parentheses represent the losses of previous study by ICAR-CIPHET (2015)

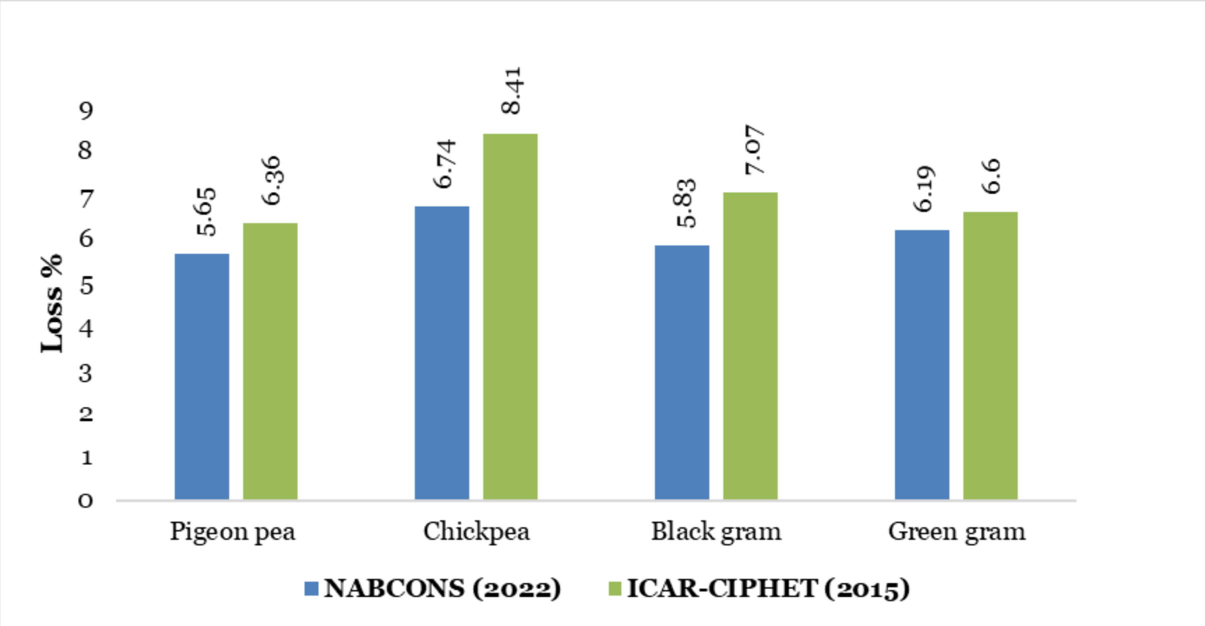
The survey for estimating the post-harvest losses of ***chickpea*** was conducted in 19 districts in 6 states covering 6 agro-climatic zones (ACZ6, ACZ7, ACZ8, ACZ9, ACZ10 and ACZ14). The overall post-harvest losses at district level were the highest in Alwar district (8.80%) and the lowest in Amravati and Krishna districts (5.63%). The reason for high post-harvest losses in Alwar and Baran districts were due to high losses observed during harvesting and threshing operations. The same may be due to improper method of harvesting and use of sticks and animals for threshing resulting in shattering and physical damage. The overall post-harvest losses at state level were the highest in Odisha (6.09%) and the lowest in Maharashtra (5.39%). The overall ACZ wise post-harvest losses were the highest in ACZ14 (6.69%), and the lowest in ACZ9 (5.29%). The major reason for high post-harvest losses in ACZ14 was due to manual harvesting, manual threshing and mishandling during transportation. Post-harvest losses in farm operations at national level were found to be 5.89 percent while for market level, the post-harvest losses were 0.85 percent. The overall total post-harvest losses of chickpea were 6.74 percent. Among different operations, harvesting (1.49%) and threshing (2.00%) majorly contributed to the losses in farm operations and transport (0.30%) in market channels.

ICAR-CIPHET (2015) reported an overall loss of 8.41 percent for chickpea while the present study reported lower post-harvest losses (6.74%). The estimated post-harvest losses obtained from the present study were significantly lower than the earlier conducted study (ICAR-CIPHET 2015) at 5 percent level of significance (**Fig. 6.2**). This decrease in overall post-harvest losses may be attributed to overall improvements such as improved varieties, increased use of threshers, and increased storage capacity during the recent years.

The survey for estimating the post-harvest losses of ***black gram*** was conducted in 12 districts in 7 states covering 7 agro-climatic zones (ACZ5, ACZ7, ACZ8, ACZ9, ACZ10, ACZ11 and ACZ13). The overall post-harvest losses at district level were the highest in Nanded district (7.80%), and the lowest in Chhota Udaipur district (4.87%) in black gram. The high post-harvest losses in black gram in Nanded and Washim districts may be due to improper method of harvesting resulting in grain shattering and mishandling resulting spillage of grain. The overall post-harvest losses at state level were the highest in Maharashtra (7.48%), and the lowest was reported in Gujarat (4.80%). The overall post-harvest losses in black gram at ACZ level were the highest in ACZ9 (7.60%) and the lowest was reported in ACZ13 (4.87%). Improper harvesting methods and mishandling resulted in high losses in ACZ9 (Maharashtra) in black gram, while proper handling methods and interventions by government departments such as conduct of seminars for farmer's awareness, may have resulted in the lowest losses in ACZ13 (Gujarat). Post-harvest losses in farm operations in black gram at national level were found to be 4.95 percent while for market level, the post-harvest losses were 0.88 percent. The overall total post-harvest losses of black gram were 5.83 percent. Among different operations, harvesting (1.44%) and threshing (1.17%) majorly contributed to the losses in farm level operations and at wholesaler (0.24%) and retailer (0.23%) level in market operations.

As depicted in **Fig. 6.2**, ICAR-CIPHET (2015) reported an overall loss of 7.07 percent for black gram, while the present study reported lower post-harvest losses (5.83%). The estimated post-harvest losses obtained from the present study were significantly lower than the earlier conducted study (ICAR-CIPHET 2015) at 5 percent level of significance. This decrease in overall post-harvest losses may be attributed to overall improvements such as increase in storage capacity established in the country since 2015 and increased use of threshers.

Fig. 6.2 Comparison of post-harvest losses of pulses



The survey for estimating the post-harvest losses of ***green gram*** was conducted in 18 districts in 8 states covering 7 agro-climatic zones (ACZ3, ACZ4, ACZ6, ACZ7, ACZ9, ACZ10 and ACZ13). The overall post-harvest losses at district level in green gram were the highest in South 24 Paraganas district (6.50%), and the lowest in Ajmer district (5.54%). The reason for high post-harvest losses in green gram in Bankura and South 24 Paraganas districts were due to high post-harvest losses observed during harvesting and threshing operations. The overall post-harvest losses at state level were the highest in West Bengal and Telangana (6.24%), and the lowest was reported in Gujarat (5.85%). The overall post-harvest losses at ACZ level were the highest in ACZ3 (6.24%) followed by ACZ7 (6.20%), and the lowest was reported in ACZ6 (5.75%). Improper harvesting method and mishandling resulted in high post-harvest losses in ACZ 3. Post-harvest losses in farm operations at national level in green gram were found to be 5.30 percent, while for market level, the post-harvest losses were 0.89 percent. The overall total post-harvest losses of green gram were 6.19 percent. Among different operations, harvesting (1.81%) and threshing (1.53%) majorly contributed to the post-harvest losses in farm level operations and wholesaler (0.33%) level in market operations.

Box 6.2 Good practices to be followed for pulses

- *Use of threshers instead of manual methods such as use of sticks and animals*
- *Storage in the scientific warehouses or temperature controlled warehouses*
- *Negotiable warehouse receipt system by warehouse owners*
- *Availment of pledge finance from banks by the farmers*
- *Regular fumigation and pest control measures during storage*
- *Use of improved varieties released in crops such as black gram.*
- *Use of e-commerce like e-NAM portal & Kisan Rath*
- *Organized marketing of pulses through FPOs/ FPC.*

ICAR-CIPHET (2015) reported an overall post-harvest loss of 6.60 percent for green gram while the present study reported lower post-harvest loss (6.19%). The estimated post-harvest losses obtained from the present study were insignificantly lower than the earlier conducted study (ICAR-CIPHET 2015) at 5 percent level of significance as depicted in **Fig. 6.2**.

Among pulses under the study, shattering of grains during harvesting, use of manual methods of harvesting and threshing, disease and pest infestation, spillage of grains during the various operations, damage by birds during collection and drying were some of the major causes observed during farm level operations. Spillage during loading and unloading was observed during the transportation and improper storage conditions and mishandling resulted in post-harvest losses at various market stakeholders. The estimated post-harvest losses in pigeon pea, chickpea, and black gram were significantly lower than the earlier conducted study (ICAR-CIPHET 2015) which can be attributed to the improved varieties, increased use of threshers, increase in the number of dal mills, and improvement in the storage capacity.

6.3 Oilseeds

As one of the major oilseeds grower and importer of edible oils, India has produced 37.15 million MT of oilseeds during 2021- 22 (Directorate of Economics and Statistics, 2021-22). Being an important source of income as an importer, establishment of efficient post-harvest infrastructure for oilseeds is essential.

6.3.1 Farm level operations

Harvesting- Harvesting of oilseeds was done manually using sickle or hand tools. Cotton bolls once started splitting open were plucked manually. The losses during harvesting in cotton were mainly due to improper harvesting and dropping of cotton bolls in the field due to mishandling by the pluckers. Also, the cotton bolls left on bracts in fields after harvesting contributed to the losses. In some areas combine harvesters for mustard and soybean reapers were used for harvesting of soybean. Groundnuts were harvested manually either by simply uprooting/ pulling the plant with hands form the soil or by digging or loosening the soil using hand hoe or plough or blade harrow. The pods left in the soil after uprooting of the plants were picked manually by the farmers or labourers employed by the farmers led to post-harvest losses in groundnut during harvesting. It was also observed that the fields were irrigated in some cases to soften the soil and to assist in easy pulling of the plant along with groundnut pods. The losses during harvesting operation in sunflowers were mainly due to the shattering of seed from the flower as a result of improper harvesting. The flower heads left unharvested in the field also added to the losses.

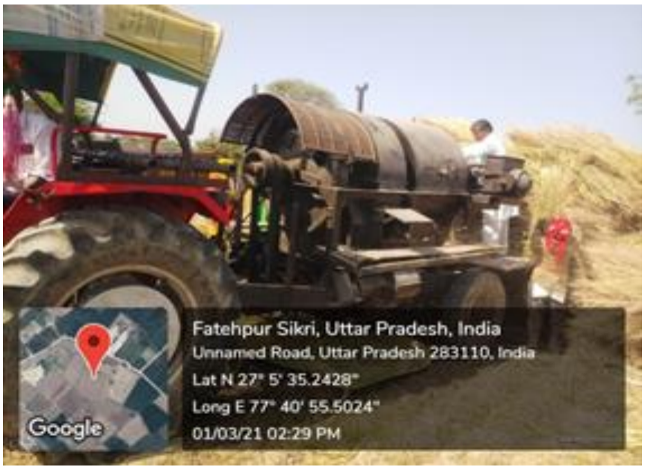
Manual harvesting of sunflower

Collection- Harvested sunflower heads and mustard stalks were collected manually by farmers at one place in the field on cloth or tarpaulin or in plastic bags and left for some time in the field. Plucked cotton bolls, safflower along with the base stalk, groundnut and soybean plant consisting of the pods kept in bundles at different points in the field were collected manually by farmers. The losses observed during this operation were mainly due to physical damage to the flower head and thereby leading to shattering of sunflower seed on the ground. The unfavourable weather conditions and damage by birds contributed to the post-harvest losses. Also, if the collected plants were heaped for a longer time, post-harvest losses due to mould growth were also observed in soybean.

Collection of sunflower at farm level

Threshing/ stripping - Both manual and mechanical threshing was being done by farmers depending upon the volume of production of selected oilseed crops. It was observed that more farmers were using mechanical threshers for threshing sunflower and soybean. Sunflower and safflower threshing by beating the brittle flower heads with sticks on ground was common amongst the small farmers. Both manual and mechanical stripping were observed in groundnut in the farm to remove pods from the plant. During manual threshing, the losses occurred due to spilling, physical damage to the seed due to beating with sticks or other tools and the seed remaining on the head because of improper threshing in oilseeds. In case of mechanical stripping of groundnut, the post-harvest losses were attributed to physical damage to pods by the machine, mishandling during loading the plants in the machine and collection of the stripped pods and spilling of pods in groundnut.

Winnowing/ cleaning- The threshed oilseed kernels were cleaned by picking of extraneous matter or by winnowing to separate the lighter chaff and dust particles. Sometimes, sieving was also done to clean the seed. Also, the shrivelled or infested seeds were separated during this operation. During the operation, the post-harvest losses mainly included swayed away seed or spilling losses and due to damaged seed.



Mechanical threshing of mustard

Drying- The threshed and cleaned oilseeds were sun dried to reduce the moisture content of the seed. The cleaned groundnut pods were dried before packaging and storage or before selling in the market. The drying operation was done by spreading the pods on a clean surface, tarpaulin or mats and were sun dried for two-three days in groundnut. The post-harvest during drying were mainly due to spilling and physical damage to pods and due to birds or rodents.

Drying of mustard

Packaging- Oilseeds were packed in gunny bags and plastic bags. Losses during this operation were low and were mainly due to spilling because of mishandling.

Packaging of groundnut at farm level

Transport- The oilseeds packaged in gunny bags and plastic bags were transported using tempo, tractor and trucks from the farmer level to traders in the market to directly transport to the oil expelling units. During transportation, the post-harvest losses were low as the seeds were packaged properly before transportation. The losses included spilling losses as result of defect in packaging or improper packaging. Also, during loading and unloading in the fleet, spilling losses were observed.

Storage- Oilseeds were stored at farm level after packaging in gunny bags/ plastic bags. The storage period was ranging from a few days to a few months before sale to oil expellers units or traders and ginning units in case of cotton. The post-harvest losses during storage of oilseeds included spilling losses due to improper packaging and stacking and losses due to fungal attack and insect infestation during longer periods of storage.

6.3.2 Market level operations

Wholesalers and retailers- Oilseeds were handled manually at the wholesalers and retailers level. The post-harvest losses were due to spilling, shattering, physical damage/ crushing of seeds and pods. In case of groundnut, pods stored for longer periods of time under humid conditions were damaged by fungal attack.

Wholesaler level of safflower

Transporters- Oilseeds were transported through bullock cart, vans, tempo and trucks within the same district, from one district to another and from one state to another state. The post-harvest losses observed during transportation were mainly due to spilling of seeds and pods as a result of improper/ defective packaging, physical damage or crushing of seeds and pods due to impact in transit and improper stacking and improper handling during loading and unloading from the vehicle.

Storage- Oilseeds were stored in godowns without any temperature or humidity control conditions. Post-harvest losses observed during storage included losses due to spilling of pods as a result of improper stacking and packaging and also due to physical damage/ crushing of seeds and pods. In groundnut, biological losses such as fungal attack in case of high humidity conditions or storage of improperly dried groundnut pods were also observed.

Processing- Post-harvest losses were estimated at the processing units during storage and unloading from the fleet at the unit. The post-harvest losses observed at the processing units were low and were mainly due to spilling, damage to the seeds and pods as a result of mishandling and fungal attack in cases of improper storage.

Storage of sunflower seeds and spilling losses during storage at godown

6.3.3 Post-harvest losses at farm and market level

The survey for assessment of post-harvest losses was carried out for 6 oilseed crops (mustard, cottonseed, soybean, safflower, sunflower and groundnut) in this study. The extent of post-harvest losses at district level, state level and ACZ level are presented respectively, in **Appendix 5, Appendix 6** and **Appendix 7** and national level are reported in **Table 6.3.**

The survey for estimating the post-harvest losses of ***mustard*** was conducted in 15 districts in 5 states covering 6 agro-climatic zones (ACZ3, ACZ5, ACZ6, ACZ7, ACZ9, and ACZ13). The overall post-harvest losses at district level in mustard were the highest in Nadia district (7.33%) and the lowest in Bharatpur district (4.71%). The reason for higher post-harvest losses in Nadia and Murshidabad districts was improper method of harvesting resulting in shattering of seed and mishandling resulting spillage of seed during manual threshing.

The overall post-harvest losses at state level in mustard were the highest in West Bengal (6.20%) and the lowest in Gujarat (4.38%) in mustard. At ACZ level, the overall post-harvest losses in mustard were the highest in ACZ3 (6.25%) and the lowest in ACZ14 (3.84%). The improper harvesting method, use of sticks and animals for threshing, lack of adequate storage units contributed to more losses in ACZ3.

Post-harvest losses in farm operations at national level in mustard were found to be 4.08 percent, while for market level, the post-harvest losses were 0.38 percent with overall total losses of 4.46 percent. Among different operations, harvesting (1.67%) and threshing (1.18%) majorly contributed to the losses in farm level operations of mustard. Earlier ICAR-CIPHET (2015) reported an overall loss of 5.54 percent for mustard, while the present study reported lower post-harvest losses (4.46%), which were significantly lower than the earlier conducted study at 5 percent level of significance **(Fig. 6.3)**.

This decrease in overall post-harvest losses may be attributed to overall improvements such as increased use of combine harvesters and threshers and increase in the oil milling units in recent years for this crop.

Transportation of green gram

Table 6.3 Post-harvest losses of oilseeds in percent at national level

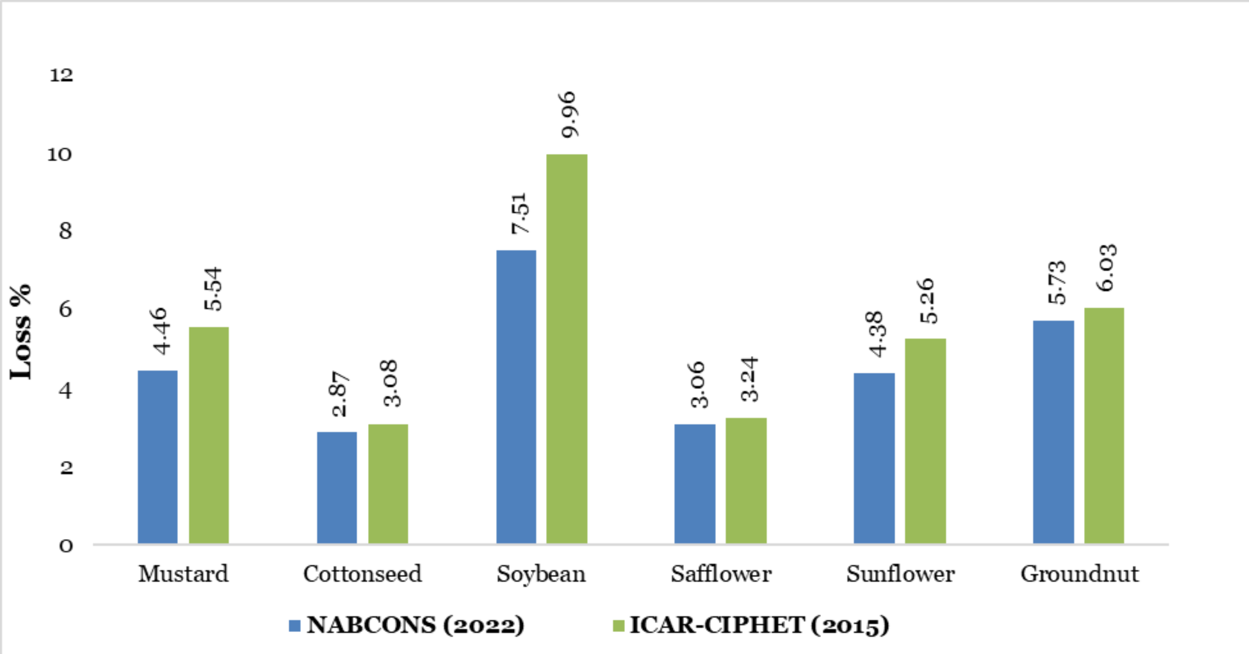
Crop	Farm operations									Market operations						Overall total loss
	Harve-sting	Collect-ion	Thresh-ing	Winnowi-ng/ cleaning	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retail-ers	Proces-sing unit	Trans-port	Total loss in market level	
Mustard	1.67 ±0.0298	0.41 ±0.0157	1.18 ±0.0207	0.35 ±0.0067	0.12 ±0.0041	0.09 ±0.0036	0.07 ±0.0034	0.19 ±0.0247	4.08 ±0.1086	0.05 ±0.0075	0.04 ±0.0059	0.03 ±0.0035	0.03 ±0.0039	0.23 ±0.0165	0.38 ±0.0183	4.46 ±0.1101 (5.54)
Cottonseed	1.81 ±0.0167	0.27 ±0.0025	-	-	0.08 ±0.0014	0.07 ±0.001	0.07 ±0	0.17 ±0.002	2.47 ±0.0171	0.01 ±0.0014	0.22 ±0.021	0.03 ±0.0033	0.03 ±0.0047	0.11 ±0.0135	0.4 ±0.0255	2.87 ±0.0308 (3.08)
Soybean	2.68 ± 0.1216	0.88 ± 0.0355	1.21 ± 0.0595	0.22 ± 0.0171	0.11 ± 0.0043	0.44 ± 0.0178	0.78 ± 0.0422	0.25 ± 0.0124	6.58 ± 0.1488	0.11 ± 0.0034	0.32 ± 0.1311	0.18 ± 0.0546	0.21 ± 0.0328	0.1 ± 0.0082	0.93 ± 0.146	7.51 ± 0.2085 (9.96)
Safflower	1.04 ±0.0296	0.4 ±0.0069	0.48 ±0.0086	0.25 ±0.005	0.13 ±0.0035	0.17 ±0.0026	0.02 ±0.0003	0.16 ±0.003	2.66 ±0.0324	0.02 ±0.0045	0.15 ±0.0471	0.04 ±0.0168	0.07 ±0.0212	0.12 ±0.0672	0.4 ±0.0864	3.06 ±0.0923 (3.24)
Sunflower	0.91 ±0.0367	0.38 ±0.0192	1.46 ±0.0688	0.2 ±0.0074	0.08 ±0.0027	0.12 ±0.0049	0.05 ±0.0027	0.1 ±0.0053	3.3 ±0.0809	0.02 ±0.0037	0.17 ±0.0341	0.04 ±0.0063	0.74 ±0.1023	0.11 ±0.0245	1.08 ±0.1107	4.38 ±0.1372 (5.26)
Groundnut	1.89 ±0.0176	0.49 ±0.0041	1.42 ±0.0116	0.4 ±0.0035	0.14 ±0.0012	0.14 ±0.0014	0.08 ±0.001	0.13 ±0.0012	4.7 ±0.022	0.05 ±0.0137	0.39 ±0.0617	0.08 ±0.0114	0.34 ±0.0574	0.17 ±0.0596	1.03 ±0.1039	5.73 ±0.1062 (6.03)

Note: *Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*
Figures in parentheses represent the losses of previous study by ICAR-CIPHET (2015)

The survey for estimating the post-harvest losses of ***cottonseed*** was conducted in 15 districts in 5 states covering 6 agro-climatic zones (ACZ6, ACZ7, ACZ9, ACZ10, ACZ13 and ACZ14). The overall post-harvest losses in cottonseed at district level were observed the highest in Jalgaon district (3.32%) and the lowest in Bhiwani district (2.14%). The lowest post-harvest losses in Bhiwani district of Haryana were mainly due to efficient manual harvesting and careful plucking of cotton bolls from the plant. The overall post-harvest losses in cottonseed at state level were observed the highest in Telangana (3.04%) and the lowest overall post-harvest losses were observed in Haryana (2.55%). The reason for lower losses in the state of Haryana were because of the careful plucking of cotton bolls by the farmers and timely collection on the field avoiding the spilling losses. At ACZ level, the overall post-harvest losses in cottonseed were observed the highest in ACZ10 (3.06%), while the lowest were observed in ACZ6 (2.34%). The reason for lower losses in the ACZ6 were efficient manual harvesting and careful plucking of cotton bolls by the farmers. The post-harvest losses at farm operations at national level in cottonseed were found to be 2.47 percent, while for market level, the post-harvest losses were 0.40 percent with overall post-harvest losses of 2.87 percent at the national level. Among different operations, the highest post-harvest losses in cottonseed were observed at harvesting operations (1.81%) at farm level and at wholesale level (0.22%) in market channels. The higher losses during harvesting operation were mainly due to spilling of cotton bolls and unplucked cotton bolls left on plant as a result of improper harvesting by farmers or labourers employed by farmers. The major causes of losses at market level were mainly due to spilling of cotton seeds and damage to the seeds. The overall losses for cottonseed reported by ICAR-CIPHET (2015) were 3.08 percent, while the present study estimated lower post-harvest losses (2.87%), however, the difference was non-significant at 5 percent level of significance (**Fig. 6.3)**. The reduction in post-harvest losses at farm level compared to previous study can be attributed to the careful harvesting of cotton bolls and better handling practices in the supply chain. Further, research efforts and various training programmes conducted by ICAR- Central Institute for Research on Cotton Technology (CIRCOT) have contributed to better utilisation of cotton by-products including cottonseed and better handling practices at farm.

The survey for estimating the post-harvest losses of ***soybean*** was conducted in 7 districts in 5 states covering 4 agro-climatic zones (ACZ7, ACZ8, ACZ9 and ACZ10). The highest overall post-harvest losses in soybean were observed in Latur district (8.84%) and the lowest in Baran district (6.19%). The lower losses in Baran district were due to use of mechanised harvesters by many farmers in the harvesting operation of soybean. However, most of the farmers in Latur district were manually harvesting soybean leading to losses. The highest overall post-harvest losses in soybean at state level were observed in Maharashtra (8.84%) and the lowest in Rajasthan (6.54%). At ACZ level, the highest overall post-harvest losses in soybean were observed in ACZ9 (7.72%) and the lowest in ACZ8 (6.54%). At national level, the post-harvest losses observed at farm level operations in soybean were 6.58 percent, whereas, for market level operations post-harvest losses were 0.93 percent with overall post-harvest losses of 7.51 percent for this crop at national level. Among different operations, the highest post-harvest losses were observed at harvesting (2.68%) in farm operations and at wholesale level (0.32%) in market channels. As depicted in **Fig. 6.3,** the overall losses for soybean reported by ICAR-CIPHET (2015) were 9.96 percent. The present study estimated significantly lower overall post-harvest losses (7.51%) in comparison to previous study conducted by ICAR-CIPHET (2015) at 5 percent level of significance. The reduction in post-harvest losses over a period of time may be attributed to the infrastructure and technology advancement in recent years.

Fig. 6.3 Comparison of post-harvest losses of oilseeds



The survey for estimating the post-harvest losses of ***safflower*** was conducted in 5 districts in 2 states covering 2 agro-climatic zones (ACZ9 and ACZ10). The highest overall post-harvest losses in safflower were observed in Gulbarga district (3.19%) and the lowest in Hingoli district (3.00%). The highest post-harvest losses in Gulbarga were mainly due to inefficient manual harvesting and threshing and delayed time in collection. The safflower plant stalks were left in the field for a longer time during collection leading to higher losses and damage by birds to the flowers bearing safflower seeds. The lowest losses were observed in Hingoli district of Maharashtra due to use of mechanical threshers by some of the farmers and due to proper and careful operations carried out by farmers. The highest overall post-harvest losses in safflower at state level were observed in Karnataka (3.1%) and the lowest overall post-harvest losses were observed in Maharashtra (3.02%). The main reason for the higher post-harvest losses in Karnataka were higher losses at the farm level due to delayed collection and manual threshing operation by beating the safflower stalks with sticks. At ACZ level, the highest overall post-harvest losses were observed in ACZ10 (3.1%) and the lowest overall post-harvest losses were observed in ACZ9 (3.02%). The higher losses in ACZ10 were due to manual threshing and delayed collection time. At national level**,** the post-harvest losses in safflower observed at farm level operations were 2.66 percent, whereas, for market level operations post-harvest losses were 0.40 percent. The overall post-harvest losses for safflower at the national level were observed as 3.06 percent. Among different operations the highest post-harvest losses were observed at harvesting (1.04%) in farm operations and at wholesale level (0.15%) in market channels. The overall losses for safflower reported by ICAR-CIPHET (2015) were 3.24 percent. The present study observed lower post-harvest losses (3.06%) in comparison to ICAR-CIPHET 2015, however, the difference was non-significant at 5 percent level of significance (**Fig. 6.3**). There have been continuous efforts in research and training conducted under the All India Coordinated Research Project on Oilseeds (AICRPO) on Safflower by ICAR majorly in the states of Maharashtra and Karnataka, ICAR- Indian Institute of Oilseeds Research (IIOR), Hyderabad and efforts by non-governmental organisations such as Nimbkar Agricultural Research Institute (NARI) in Maharashtra has led to development of various varieties of safflower and better farm practices inculcation in farmers. The use of non-spiny varieties launched by ICAR has contributed to the ease of harvesting and thereby lowered the losses during manual harvesting.

The survey for estimating the post-harvest losses of ***sunflower*** was conducted in 9 districts in 5 states covering 5 agro-climatic zones (ACZ3, ACZ4, ACZ6, ACZ9 and ACZ10). The overall post-harvest losses in sunflower were found the highest in South 24 Paraganas district (5.21%) and the lowest in Koppal district (4.2%). The highest post-harvest losses in South 24 Paragans district of West Bengal were mainly due to inefficient manual harvesting, prolonged period of collection and manual threshing leading to more spilling and un-separated seeds still attached to the flower heads. The lowest losses were observed in Koppal district of Karnataka as most of the farmers were using mechanical threshers and better handling practices. The highest overall post-harvest losses in sunflower were estimated in West Bengal (5.21%) and the lowest in Karnataka (4.22%). At ACZ level, the highest overall post-harvest losses in sunflowers were found in ACZ3 (5.21%) and the lowest in ACZ10 (4.22%). The lowest post-harvest losses in ACZ 10 covering the state of Karnataka were mainly due to careful collection of sunflower heads, use of mechanical threshers by most of the farmers and better storage practices in plastic bags. At national level, overall the post-harvest losses of 4.38 percent were observed, comprising 3.3 percent at farm level and 1.08 percent at market level. Among various operations, threshing (1.46%), and harvesting (0.91%) majorly contributed to post-harvest losses during farm operations, which was obviously due practice of manually harvesting and threshing by beating sunflower heads with sticks on the ground leading to splitting and physical damage to the seeds. As indicated in **Fig. 6.3,** the overall losses for sunflower reported by ICAR-CIPHET (2015) were 5.26 percent. The estimated post-harvest losses obtained from the present study (4.38%) were significantly lower than the earlier conducted study (ICAR-CIPHET 2015) at 5 percent level of significance. The production and development in the supply chain of oilseeds has been continuously promoted by the Government of India through the National Food Security Mission (Oilseeds) (NFSM- Oilseeds) in all districts of India. Further, technology transfer through Front Line Demonstrations (FLDs) and Cluster Front Line Demonstrations (CFLDs) are provided to the farmers through Indian Council of Agricultural Research (ICAR). The use of mechanical threshers for sunflower at the farm level has significantly increased. IT enabled awareness and support to farmers by mobile application developed by ICAR-Indian Institute of Oilseeds Research on Sunflower in 2017 has contributed to the better management of sunflower at farmer level. There is increase in processing units for sunflowers in recent years. After 2011, a total of 967 vegetable oil industries were registered under Vegetable Oil Products Production and Availability (Regulation) Order, 2011 as on June 2021.

The survey for estimating the post-harvest losses of ***groundnut*** was conducted in 13 districts in 5 states covering 7 agro-climatic zones (ACZ3, ACZ7, ACZ9, ACZ10, ACZ11, ACZ13 and ACZ14). The highest overall post-harvest losses in groundnut were observed in East Medinipur district (7.24%) and the lowest in Sabar Kantha district (4.84%). The highest post-harvest losses in East Medinipur district of West Bengal was due to the fact that majority of the farmers were manually stripping the groundnut pods and the use of mechanical stripper to separate the pods from the plant was limited. Also, losses during harvesting were high in the East Medinipur district due to poor handling practices during the operation. The highest overall post-harvest losses in groundnut at state level were observed in West Bengal (7.24%) and the lowest overall in Rajasthan (5.03%). At ACZ level, the highest overall post-harvest losses in groundnut were found in ACZ3 (7.24%) and the lowest overall in ACZ14 (5.08%). The national level post-harvest losses observed for groundnut at farm level operations were 4.70 percent, whereas, for market level operations post-harvest losses were 1.03 percent.

The overall post-harvest losses for groundnut at the national level were observed as 5.73 percent. Among different operations at farm level, harvesting (1.89%), and stripping/ majorly contributed towards post-harvest losses. The higher losses during harvesting were mainly due to manual uprooting of the plant or digging and pulling of plant leading to breakage of pegs and increased number of pods remaining in the soil. The major causes of losses in groundnut pods at the market level were spilling of the pods and physical damage/ crushing of pods and fungal attack due to improper storage conditions.

Box 6.3 Good practices to be followed for oilseeds crops

- *Use of mechanical groundnut diggers and mechanical strippers*
- *Use of combine harvester and mechanical thresher for crops like mustard and soybean*
- *Use of improved varieties e.g. non-spiny varieties for safflower, which are easy for harvesting.*
- *Use of cotton modulers for collection and packaging of cotton bundles*
- *Storage in dry warehouses with humidity control*
- *Organized marketing of oilseeds through FPOs/ FPC.*

As indicated in **Fig. 6.3,** the overall losses in groundnut reported in ICAR-CIPHET (2015) were 6.03 percent. The estimated post-harvest losses obtained from the present study (5.73%) were significantly lower than the earlier conducted study (ICAR-CIPHET 2015) at 5 percent level of significance. The reduction in the post-harvest losses can be attributed to the increased use of mechanical strippers for separating the pods and increasing use of mechanical groundnut diggers.

Among oilseeds under the study, shattering of grains, use of sticks and animals, damage by birds, spillage during loading and unloading, improper storage conditions, mishandling resulted in post-harvest losses. The estimated post-harvest losses in mustard, soybean, and sunflower were significantly lower than the earlier conducted study (ICAR-CIPHET 2015) which can be attributed to the increased use of threshers, and increase in oil mills - 967 vegetable oil mills registered under Vegetable Oil Products Production and Availability (Regulation) Order, 2011 (June 2021).

6.4 Fruits

India is the second largest producer of fruits in the world with a total production of 102.92 million MT in 2021-22 (DAC&FW, 2021-22). Due to the highly perishable nature, even after several initiatives taken by GoI and State Govt, a considerable quantity of fruits go waste due to lack of adequate post-harvest management infrastructure.

6.4.1 Farm level operations

Harvesting- Harvesting of fruits was done manually by the farmers. Some of the farmers were using fruit pickers and a stick or long pole picking net bag or a pole with a knife/hook was used for harvesting mango & sapota. Grape was harvested using scissors for harvesting. Banana bunches were cut using knives and carried by the farmers/labourers. The major causes of losses observed during harvesting were improper and rough handling, hand pressing, harvesting of immature and over mature fruits, and improper collecting material by labourers resulting in physical damage, bruising and cracking in fruits. Physical damage of fruit as well as dropping of fruit was also observed in grapes and apple. The major cause of losses observed during harvesting in mango was improper harvesting methods such as shaking of the branches, plucking mango without leaving the pedicle, which led to oozing out of milky/resinous sap spoiling mango appearance. Further, the fruit harvested with 8-10 mm long stalks appeared better on ripening as undesired spots on skin caused by sap burn were prevented. Such fruits were less prone to stem-end rot and other storage diseases.

Mango harvesting - Bamboo stick with a hook

Collection- Collection of fruit was done manually by farmers in plastic trays or jute bags or plastic Katta bags or bamboo baskets. The fruit in most of the cases were collected in the orchard without any shelter. In Maharashtra, collection of mandarin and sweet orange (Mausambi) was done either by using baskets and carrying them on head or by directly loading in the vehicles present between the trees. Kinnow in Punjab were collected by using a plastic sheet in the field while oranges in Madhya Pradesh were collected using crates. The causes of losses included either improper handling or unavoidable weather conditions during the operation.

Trimming, washing and desapping - This operation was done manually by the farmers in most of the states especially in mango and banana. The mango and banana fruits were washed with clean water. Pineapple heads were trimmed to facilitate packing.

Waxing, sorting and grading - This operation was one of the most important operations after harvesting and was done manually on the field. During this operation, cutting, washing and sorting of fruits were done and the fruits unfit for consumption were removed. Fruit discarded during the operation accounted for post-harvest losses. The mango fruits were arbitrarily graded by the farmers on the basis of their size and maturity. Sorting grading was done mechanically with the help of a sorting and grading machine at the farm level in apples to remove highly damaged, overripe, infested (apple scab) & small size fruits. A coating of edible wax was provided to the apple and citrus fruits after washing to reduce post-harvest losses.

Sorting, grading of citrus

Packaging- Packaging of fruits was done manually in crates as well as in Corrugated Fibre Board (CFB) cartons at the field level. During transportation, pomegranate and banana fruits were packed in CFB cartons along with shredded newspapers and banana leaves, respectively. Packaging of papaya was performed manually by wrapping the fruit using newspaper. For the distant market, the apples were packed in corrugated boxes in fruit trays which were found efficient in handling the thrust during transportation. The packaging of mango in Maharashtra and Andhra was majorly done in plastic perforated crates and fibreboard boxes with holes for ventilation. The citrus fruits in Maharashtra were majorly packed in crates and plastic perforated sacs. It was also observed that many farmers were not packaging citrus fruit and were transporting it in loose form from the field. Overfilling of fruits in boxes / overloading during transport were the major reasons for pressing/ abrasion of fruits leading to bruising and spoilage. Rough handling such as throwing of fruits in crates by the packers and mishandling during stacking led to damage to the fruits.

Storage- Storage of fruits at the field for a few hours before transportation to the respective destination was considered as farm level storage. Storage of papaya at the field occurred for a few hours or short period of time before the arrangement of the transportation facility. Also, many farmers did not store the fruit at the farm level. The storage of guava at the field level was considered for one to two days before transportation to the respective destination. Secondly, when the harvested fruits were immature, the farmers generally kept them in the field and covered them with leaves to ripen, which led to rotting and pigmentation in the fruits. Storage of papaya at the field occurred for a few hours or short period of time before the arrangement of the transportation facility. Also, many farmers did not store the fruit at the farm level. During the short period of storage at farm level, minute losses due to stacking of crates/containers and development of spots in fruit due to improper storage conditions were observed. In most of the states like Uttar Pradesh and Bihar, some small farmers stored mangoes in separate rooms for ripening. The mango fruits were arranged in several layers on the ground spread with paddy straw or the gunny bags and the top of the lot was also covered with straw and were kept for about four weeks for uniform ripening. Due to a short period of storage, minute/ negligible losses due to stacking of boxes/ crates and normal storage conditions (without any temperature controlled conditions) were observed.

Transport- Transportation of fruits was done by various types of fleet such as tempo, trucks, tractors and trolleys, Tata ACE trucks etc. to various destinations such as markets in the same district and different districts/ states. Losses were observed during the transportation owing to physical damage due to stacking, vibration/ impact/ bruising during the journey and subsequent rotting of the fruit.

6.4.2 Market level operations

Wholesalers and retailers- All operations were done manually at the wholesalers and retailers level. The losses during handling, marketing, transportation and storage were observed. Post-harvest losses during transportation due to physical damage, improper handling, over ripening, rotting and shrivelling of fruit during marketing and storage were observed. In grapes losses were observed due to pressing of fruits while in pomegranate cracking and breakage of fruits was observed.

Wholesalers of guava

Transporters- Various routes such as transportation within the same district, from one district to another and from one state to another state were observed. Types of vehicles used for transportation of fruit included truck, tractor, tempo, Tata AC trucks and pick up vans. Sapota was transported in gunny bags/ plastic crates. Banana was transported in loose bunches as well as in plastic crates. Physical damage due to improper stacking, impact during the journey and subsequent rotting of the fruit were the major causes observed during the transportation. Of late, fruits were transported in reefer vans, especially by organised players.

Transportation of fruits

Storage- Cold storage, godowns/ warehouses (without any temperature controlled conditions) as well as the storage under a shed were considered under the storage of fruits. Post-harvest losses in godowns/ warehouses were observed majorly due to ripening and rotting of fruit and improper/ rough handling. Losses at storage level due to rotting or fungal infections were observed mainly due to storage at ambient conditions in the majority of districts by the stakeholders instead of a cold storage. The losses at storage level for bananas were assessed at ripening chambers cum cold storages. During storage, the major causes for losses were spoilage due to temperature during traditional methods of ripening and physical damage of the fruits. The harvested apples were stored in the cold store and CA store for three to six months. The post-harvest losses during the storage were observed very low and were due to physiological loss in weight or chilling injury.

Processing unit- Both packhouse and processing units of pineapple, grapes, pomegranate, citrus, sapota, banana, mango, guava and apple were observed under this operation. During this operation, the losses were observed due to over ripening and rotting, immature fruit and physical injury due to bruising. In grapes, sapota and banana primary processing operations such as sorting and grading, packaging and precooling were observed. The fruits were first sorted and graded manually by the labourers, packed into the crates or boxes and then pre-cooled at pre-cooling chamber. In citrus, after waxing, packaging was done in both crates as well as Corrugated Fiber Board (CFB) boxes for further transit at their respective destinations.

Storage of fruits

6.4.3 Post-harvest losses at farm and market level

Eleven fruits (apple, banana, citrus, grapes, guava, mango, papaya, sapota, pineapple, pomegranate and muskmelon) were selected for survey of harvest and post-harvest losses in fruits. The extent of post-harvest losses at district level, state level and ACZ level, respectively are presented in **Appendix 5, Appendix 6** and **Appendix** 7 and national level are reported in **Table 6.4.**

The survey for estimating the post-harvest losses of ***apples*** was conducted in 5 districts in 3 states covering 1 agro-climatic zone (ACZ1). The overall post-harvest losses at district level in apple were the highest in Nainital district (11.82%) and the lowest in Shimla district (8.97%). The overall post-harvest losses at state level in apples were the highest in Uttarakhand (11.49%) and the lowest in Himachal Pradesh (9.20%). The lowest post-harvest losses in Shimla and Kinnaur districts of Himachal Pradesh district were mainly due to high density plantation being easy to harvest and availability of better post-harvest management infrastructure such as packhouse, CA/ cold storage etc. in these districts. At national level, post-harvest losses in farm operations for apples were found to be 7.87 percent, while at market level, the post-harvest losses were 1.64 percent. The overall total losses of apple were 9.51 percent. Harvesting operation (3.55%) and sorting and grading (3.61%) under farm operations, whereas transportation (0.91%) and retail (0.39%) under market operations majorly contributed to post-harvest losses. The major reason for post-harvest losses during harvesting was due to accidental falling of fruit on the ground during harvesting leading to bruising and pressing. Post-harvest losses due to mishandling and improper practices by various stakeholders and lack of availability of good road infrastructure were majorly observed under the present study. At the market level, losses were mainly observed due to injury of fruit during loading and unloading, pressing/ abrasion of fruits during transportation, and over-ripening (physiological loss).

Cold storage of apple in Shimla

As depicted in **Fig. 6.4(a),** ICAR-CIPHET (2015) reported an overall loss of 10.39 percent for apples, while the present study reported lower post-harvest losses (9.51%). The estimated post-harvest losses obtained from the present study were significantly lower than the earlier conducted study (ICAR-CIPHET 2015) at 5 percent level of significance. This decrease in overall post-harvest losses may be attributed to overall improvements in various farm operations such as careful harvesting, cleaning, packaging, and sorting grading of apples and also the considerable improvement in the post-harvest infrastructure. There were 72 cold/ controlled atmospheric storages in Jammu & Kashmir with 2,64,053 MT storage capacity, 80 cold/ CA storage with 1,66,192 MT capacity in Himachal Pradesh and 60 cold/ CA storage with 2,06,821 MT capacity in Uttarakhand as on 31.01.2022.

Some of the improvements in the use of advanced technologies, infrastructure and post-harvest management practices for apple are given as under:

- Use of plastic crates & ropeways for farm level collection and transportation in J&K and Himachal Pradesh.
- Availability of the low capacity sorting and grading machines at the farm level in Himachal Pradesh.
- The market intervention scheme (MIS) of Govt. of HP on procurement of low grade fruit at a fixed price has helped in farm level grading of fruits, thereby reducing post-harvest losses.
- Use and easy availability of the good packaging material such as the wooden box or Corrugated Fiber Board (CFB) packaging.
- Use of reefer vans in some states such as Himachal Pradesh for long-distance transportation.
- Availability of the cold storage, CA store, and processing units near production centres.
- Application of edible coatings (waxing) to increase the shelf life by reducing the moisture loss (physiological losses) and respiration process in the fruit.
- Development of good roads and highway infrastructure and good connectivity has reduced the transportation time of the produce from the farm gate to the storage and consumption centre.

The survey for estimating the post-harvest losses of ***banana*** was conducted in 7 districts in 4 states covering 5 agro-climatic zones (ACZ9, ACZ10, ACZ11, ACZ12 and ACZ13). The overall post-harvest losses in banana at district level were the highest in East Godavari district (9.33%) and the lowest in Jalgaon district (5.54%). This may be due to the presence of post-harvest infrastructure such as pack houses and cold storage in the districts such as Jalgaon resulting in reduced losses and increased shelf life of bananas. Also, loose banana bunches were transported from the farm level leading to higher losses in transportation in the east Godavari district. The overall post-harvest losses in bananas at state level were the highest in Andhra Pradesh (9.33%) and the lowest in Maharashtra (6.06%). The overall post-harvest losses in bananas at ACZ level were the highest in ACZ10 (9.33%) and the lowest in ACZ12 (5.84%). Inadequate post-harvest management infrastructure for bananas in the East Godavari district of Andhra Pradesh (ACZ10) resulted in comparatively higher losses as compared to Theni district of Tamil Nadu (ACZ12). Post-harvest losses in farm operations at national level in banana were found to be 5.17 percent while for market level, the post-harvest losses were 2.40 percent.

The overall total post-harvest losses of bananas were 7.57 percent. Among different operations, sorting and grading (1.80%) and harvesting (1.45%) majorly contributed to the losses in farm level operations. At market level, wholesaler (1.00%) and transport level (0.93%) mainly contributed to post-harvest losses in market operations. Pest and disease infected bunches, improper harvesting method, removal of physically damaged fruits during sorting and grading and transportation of loose bunches contributed to the losses at the farm level. At the market level, losses were mainly observed due to physical damage of bananas during ripening and improper handling while loading and unloading during transportation. Earlier ICAR-CIPHET (2015) reported an overall loss of 7.76 percent for bananas, while the present study reported lower post-harvest losses (7.57%), however the difference was non-significant at 5 percent level of significance **(Fig. 6.4(a)).** This decrease in overall post-harvest losses may be attributed to overall improvements in various farm level operations such as careful harvesting, cultivation of improved varieties having better shelf-life, improvement in management practice such as cleaning of fruits at the farm level by usage of chemicals like sodium hypochlorite and usage of crates for transportation. Export facility centre was also established by MSAMB (Maharashtra State Agricultural Marketing Board) in Jalgaon district, which included a mechanical handling system, pre-cooling, cold storage and ripening chamber for bananas. This facility helped the farmers by development of infrastructure at the market level. Also, there has been an increase in pack houses established since 2015.

Vacuum packaging of banana

The survey for estimating the post-harvest losses of ***citrus*** was conducted in 9 districts in 5 states covering 4 agro-climatic zones (ACZ6, ACZ7, ACZ9 and ACZ10). The overall post-harvest losses at district level in citrus were the highest in Amravati district (9.63%) for loose mandarin and the lowest in Fazilka district (5.03%) for Kinnow. The reason for lowest post-harvest losses in Muktsar and Fazilka district could be attributed to low temperature during the harvesting period of Kinnow. The overall post-harvest losses at state level in citrus were the highest in Maharashtra

(8.79%) and the lowest in Punjab (5.4%). The reason for high post-harvest losses of citrus in Maharashtra may be due to comparatively higher temperature at the time of harvesting in comparison to Punjab. Further, Kinnow grown in Punjab were more hardy than loose mandarins grown in Maharashtra. At ACZ level, the overall post-harvest losses in citrus were the highest in ACZ9 (8.79%), and the lowest in ACZ6 (5.40%). Lower temperature conditions as well as presence of a number of waxing plants for Kinnow in Punjab resulted in lower post-harvest losses in ACZ6 as compared to other ACZs. Post-harvest losses in farm operations at national level in citrus were found to be 5.53 percent, while for market level, the post-harvest losses were 2.18 percent. The overall total losses of citrus were 7.71 percent. The highest post-harvest losses were observed during sorting and grading (1.92%) in farm operations and at retailer (0.63%) level in market operations. Improper harvesting, handling of citrus at ambient conditions, injury of fruit during loading and unloading, skin puncture of Kinnow by pedicel leading to injection and pressing of fruit during the transportation were major causes of post-harvest losses in citrus. ICAR-CIPHET (2015) reported an overall loss of 9.69 percent for citrus while the present study reported significantly lower post-harvest losses (7.71%) at 5 percent level of significance **(Fig. 6.4(a))**. This decrease in overall post-harvest losses may be attributed to overall improvements in various farm operations such as careful harvesting and collection of fruit after 2015. APEDA has set up 160 pack houses in Maharashtra, which is a major citrus growing area. Further, about 30 cold chain projects in Maharashtra, 7 in Punjab and 3 in Madhya Pradesh under the Scheme of Cold Chain, Value Addition and Preservation Infrastructure of MoFPI were set up after 2015. Although a number of pack houses with waxing facilities were installed in the country after 2005 for primary processing operations such as cleaning/ washing, sorting and grading, waxing and packaging, more units were established during the recent years with semi and fully automatic systems. These facilities have been set up by private players and farmer producer companies (FPC). Kharisureran Farmer Producer Company Limited in Haryana is one such FPC having most advanced integrated packhouses for Kinnow **(Box 6.4).**

The survey for estimating the post-harvest losses of ***grapes*** was conducted in 5 districts in 2 states covering 2 agro-climatic zones (ACZ9 and ACZ10). The overall post-harvest losses at district level in grapes were the highest in Bijapur district (7.92%) and the lowest in Nashik district (6.69%). This may be due to the presence of post-harvest infrastructure such as pack houses and cold storage in the districts such as Nashik resulting in reduced losses and increased shelf life of grapes. The overall post-harvest losses at state level in grapes were the highest in Karnataka (8.02%) and the lowest in Maharashtra (6.73%). At ACZ level, the overall post-harvest losses in grapes were the highest in ACZ10 (8.02%) and the lowest in ACZ9 (6.73%). Inadequate post-harvest management infrastructure in Karnataka (ACZ10) resulted in comparatively higher losses as compared to Maharashtra (ACZ9), where a number of pack houses (160 pack houses by APEDA) were established for effective post-harvest management. Post-harvest losses in farm operations at national level were found to be 5.09 percent, while for market level, the post-harvest losses were 2.05 percent in grapes. The overall total post-harvest losses of grapes were 7.15 percent. Among different operations, harvesting (1.53%) and sorting and grading (2.43%) majorly contributed to the losses in farm level operations. At market level, post-harvest losses were the highest at retailer level (0.9%). Dropping of fruit due to mishandling and improper harvesting method was majorly observed during harvesting. The overall post-harvest losses in grapes (7.15%) in the present study were significantly lower at 5 percent level of significance than the post-harvest losses reported in the previous study (8.63%) by ICAR-CIPHET (2015) **(Fig. 6.4(a))**. This decrease in overall post-harvest losses may be attributed to overall improvements in various farm operations such as careful harvesting and collection of fruits. APEDA has set up 160 pack houses in Maharashtra, a major grape growing area. There has been an increase in pack houses established since 2015. Further, there was significant improvement in other supporting infrastructure such as the Food Testing Laboratories (FTL) created with financial assistance from the Ministry of Food Processing Industries (MoFPI), Govt. of India. As on 25 May 2022, 175 FTL labs were sanctioned by MoFPI. These pack houses and supporting FTLs have contributed to an increase in export of grapes from 96,277 MT in 2015 to 2.89 lakh MT in 2021 from India.

Box 6.4 Kharisureran Farmer Producer Company Limited

The FPC was incorporated on 22 February 2019 in the Sirsa district of Haryana. Kinnow is the major crop grown and marketed by the FPC. As per Small Farmers Agri-Business Consortium Haryana (SFACH), the FPC has one of the most advanced Integrated Packhouse for Kinnow in the state of Haryana. The packhouse is equipped with fully integrated automatic electronic optical grading and packing lines. It is equipped with optical sensors which provides grade wise data for further analysis. Domestic market of FPC includes states such as Chennai, Bangalore, Hyderabad, Mumbai, Chitrakoot, Delhi and Srinagar and international market of Dubai.

Table 6.4 Post-harvest losses of fruits in percent at national level

Crop	Farm operations							Market operations						Overall total loss
	Harvest-ing	Collec-tion	Sorting/ Grading	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retailers	Processing unit	Trans-port	Total loss in market level	
Apple	3.55 ±0.0885	0.01 ±0.0006	3.61 ±0.1126	0.02 ±0.0008	0.05 ±0.0033	0.62 ±0.0241	7.87 ±0.0933	0.01 ±0.007	0.39 ±0.3579	0.17 ±0.2178	0.16 ±0.2146	0.91 ±0.4094	1.64 ±0.624	9.51 ±0.6309 (10.39)
Banana	1.45 ±0.0235	0.3 ±0.0051	1.8 ±0.0276	0.17 ±0.003	0.04 ±0.0007	1.41 ±0.0246	5.17 ±0.0442	0.06 ±0.0166	1.00 ±0.2823	0.4 ±0.1274	0.01 ±0.0028	0.93 ±0.1997	2.40 ±0.3689	7.57 ±0.3716 (7.76)
Citrus	1.12 ±0.0159	0.23 ±0.003	1.92 ±0.0191	0.35 ±0.0029	0.98 ±0.0104	0.93 ±0.0317	5.53 ±0.0418	0.04 ±0.0075	0.58 ±0.1168	0.63 ±0.2427	0.02 ±0.0028	0.91 ±0.1395	2.18 ±0.3034	7.71 ±0.3063 (9.69)
Grapes	1.53 ±0.0417	0.15 ±0.006	2.43 ±0.0529	0.06 ±0.0022	0.5 ± 0.0119	0.42 ±0.0131	5.09 ±0.0699	0.02 ±0.0047	0.49 ±0.1562	0.9 ±0.1339	0.07 ±0.0292	0.57 ±0.1672	2.05 ±0.2667	7.15 ±0.2757 (8.63)
Guava	5.05 ±0.1527	0.45 ±0.0259	4.59 ±0.1396	0.31 ±0.0112	0.45 ±0.0216	0.73 ±0.0227	11.59 ±0.2112	-	1.69 ±0.6875	3.12 ±1.1304	1.31 ±0.3057	1.04 ±0.1896	3.46 ±1.3711	15.05 ±1.3872 (15.88)
Mango	1.8 ±0.0248	0.29 ±0.0059	2.74 ±0.0379	0.04 ±0.0012	0.31 ±0.0068	0.85 ±0.0106	6.03 ±0.0474	0.4 ±0.122	1.03 ±0.1001	0.08 ±0.0191	0.08 ±0.0012	0.98 ±0.1056	2.5 ±0.1908	8.53 ±0.1966 (9.16)
Papaya	1.31 ± 0.0224	0.45 ± 0.0103	0.66 ± 0.0093	0.3 ± 0.0065	0.21 ± 0.0017	0.84 ± 0.0156	3.77 ± 0.0313	0.02 ± 0.0044	0.54 ± 0.1048	1.58 ± 0.3663	0.05 ±0.0115	0.63 ± 0.107	2.82 ± 0.3959	6.59 ± 0.3971 (6.70)
Sapota	2.31 ±0.0375	0.32 ±0.0049	1.95 ±0.0353	0.31 ±0.0044	0.04 ±0.0005	1.28 ±0.019	6.22 ±0.0553	0.17 ±0.0317	0.83 ±0.1104	0.97 ±0.1207	0.04 ±0.0048	1.30 ±0.1176	3.32 ±0.204	9.53 ±0.2114 (9.73)
Pineapple	2.09 ±0.1035	0.08 ±0.0037	1.81 ±0.0086	0.03 ±0.0016	0.08 ±0.0035	0.42 ±0.0198	4.22 ± 0.1057	0.07 ±0.018	0.58 ±0.0606	0.52 ±0.1131	0.13 ±0.0299	0.5 ±0.1062	1.8 ±0.1702	6.02 ±0.2003
Pomegranate	1.39 ±0.0268	0.19 ±0.0088	1.72 ±0.0382	0.22 ±0.0088	0.3 ±0.0143	0.27 ±0.0104	4.09 ±0.0514	0.15 ±0.0274	0.67 ±0.0896	0.8 ±0.096	0.12 ±0.0206	1.00 ±0.163	2.73 ±0.2121	6.82 ±0.2182
Muskmelon	1.52 ±0.0015	0.29 ±0.0054	1.17 ±0.0011	0.33 ±0.0239	0.07 ±0.0002	0.74 ±0.0009	4.12 ±0.0246	0.02 ±0.0031	0.58 ±0.0713	1.29 ±0.1804	-	0.82 ±0.1067	2.71 ±0.2214	6.83 ±0.2228

Note: *Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*

Figures in parentheses represent the losses of previous study by ICAR-CIPHET (2015)

The survey for estimating the post-harvest losses of ***guava*** was conducted in 9 districts in 3 states covering 5 agro-climatic zones (ACZ3, ACZ4, ACZ5, ACZ7 and ACZ9). The overall post-harvest losses in guava were the highest in South 24 Paraganas district (16.61%) and the lowest in Korba district (13.62%). The reason for lower post-harvest losses in Korba district of Chhattisgarh could be attributed to its harvesting during the winter season. The overall post-harvest losses in guava at state level were the highest in West Bengal (15.86%) and the lowest in Chhattisgarh (13.79%). The reason for the high post-harvest losses of guava in West Bengal may be due to improper harvesting and lack of post-harvest infrastructure. Comparatively, lower post-harvest losses in Madhya Pradesh may be due to good post-harvest management practices like use of protection bags, precooling, use of better packaging materials and use of reefer vans for long distance transportation. The overall post-harvest losses in guava at ACZ level were the highest in ACZ3 (17.08%) and the lowest in ACZ9 (14.14%). Post-harvest losses in farm operations at national level were found to be 11.58 percent, while at market level, the post-harvest losses for guava were 3.46 percent with an overall post-harvest loss of 15.05 percent. Harvesting operation (5.049%) and sorting and grading (4.58%) operations at farm level and retail marketing (3.11%) in market channels majorly contributed to the post-harvest losses in guava.

As indicated in **Fig. 6.4(b),** ICAR-CIPHET (2015) reported an overall loss of 15.88 percent for guava, while the present study reported lower post-harvest losses (15.05%) which was non-significant at 5 percent level of significance. Some of the best practices followed observed during the study are given as under:

- There was an improvement in various farm operations such as harvesting i.e. plucking of guava along with leaves and use of the protection bag at the time of harvesting.
- The adoption of the improved variety of guava for cultivation, which has a longer shelf life and are large in size with thick pericarp.
- The use of good packaging material such as wooden box or CFB cartons along with wrapping material such as newspaper, sponge bag & paper have increased.
- There was increased use of reefer vans and pre-cooling units in some states for long-distance transportation of guava
- Application of edible coatings and chemical treatment was followed in some states, which helped in the increase of shelf life by reducing the moisture loss (physiological losses) and respiration process in the fruit.

The survey for estimating the post-harvest losses of ***mango*** was conducted in 6 districts in 4 states covering 4 agro-climatic zones (ACZ4, ACZ5, ACZ10 and ACZ12). The overall post-harvest losses at district level in mango were the highest in Darbhanga district (10.40%) and the lowest in Chittoor district (7.87%). The lower post-harvest losses in Krishna and Chittoor districts of Andhra Pradesh may be due to good harvesting practices and availability of the good post-harvest infrastructure such as pack houses, processing units and ripening chambers. The overall post-harvest losses at state level in mango were the highest in Bihar (10.10%) and the lowest in Andhra Pradesh (7.89%). The reason for the high post-harvest losses of mango in Bihar may be attributed to improper harvesting and lack of post-harvest infrastructure. The overall post-harvest losses in mango were the highest in ACZ4 (9.95%) and the lowest in ACZ10 (7.84%). The availability of packhouses, sorting and grading facilities, multi commodity cold storages, ripening chambers, processing units and good connectivity has contributed to lower loss in ACZ10 as compared to other ACZs. Further, the adoption of high density mango plantation enabling easy harvesting, use of fruit pickers for mango plucking, availability of the good packaging material were major reasons for lower loss. Post-harvest losses in farm operations in mango at national level were found to be 6.03 percent, while at market level, the post-harvest losses were 2.50 percent. The overall total losses of mango were 8.53 percent. Harvesting operation (1.80%), and sorting and grading (2.74%) in farm operations and retailer (1.03%) level in market operations majorly contributed towards post-harvest losses.

As depicted in **Fig. 6.4(b),** ICAR-CIPHET (2015) reported an overall loss of 9.16 percent for mango, while the present study reported significantly lower post-harvest losses (8.53%) at 5 percent level of significance. This decrease in overall post-harvest losses may be attributed to overall improvements in various farm operations such as careful harvesting, cleaning, packaging, and sorting grading of mangoes as also an increase in post-harvest infrastructure such as pack houses, ripening chambers, reefer vehicles etc.

Fig. 6.4(a) Comparison of post-harvest losses of fruits

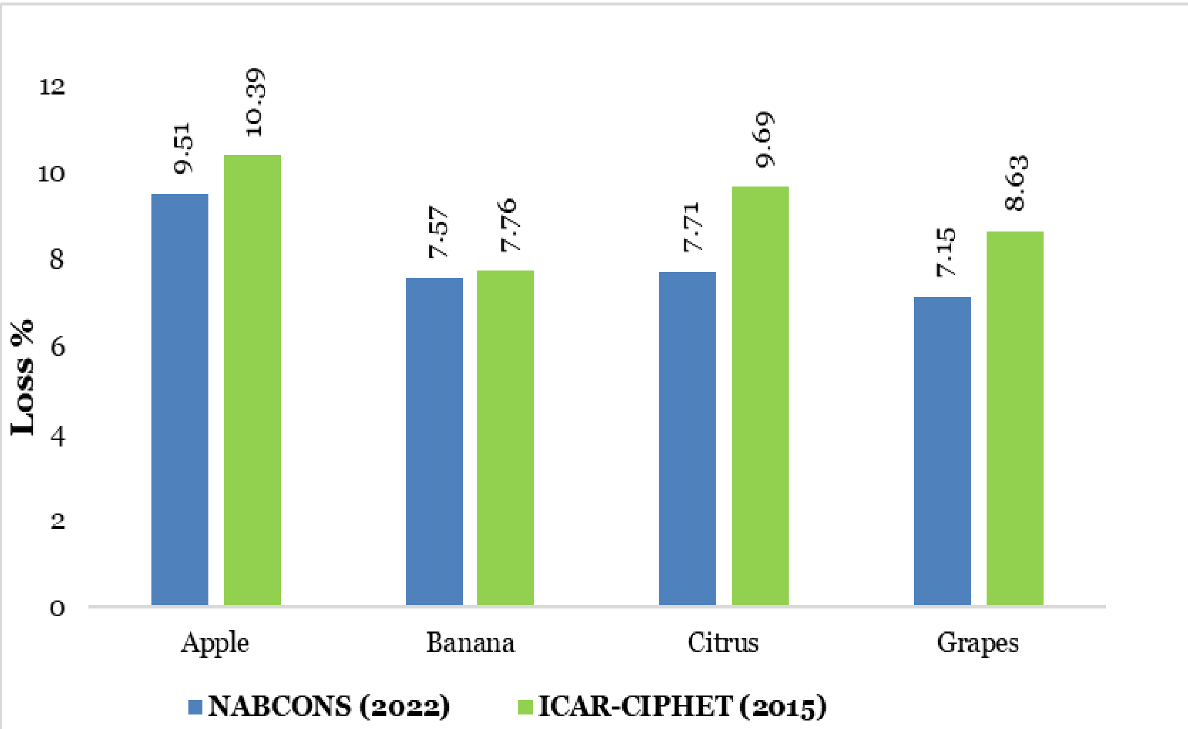
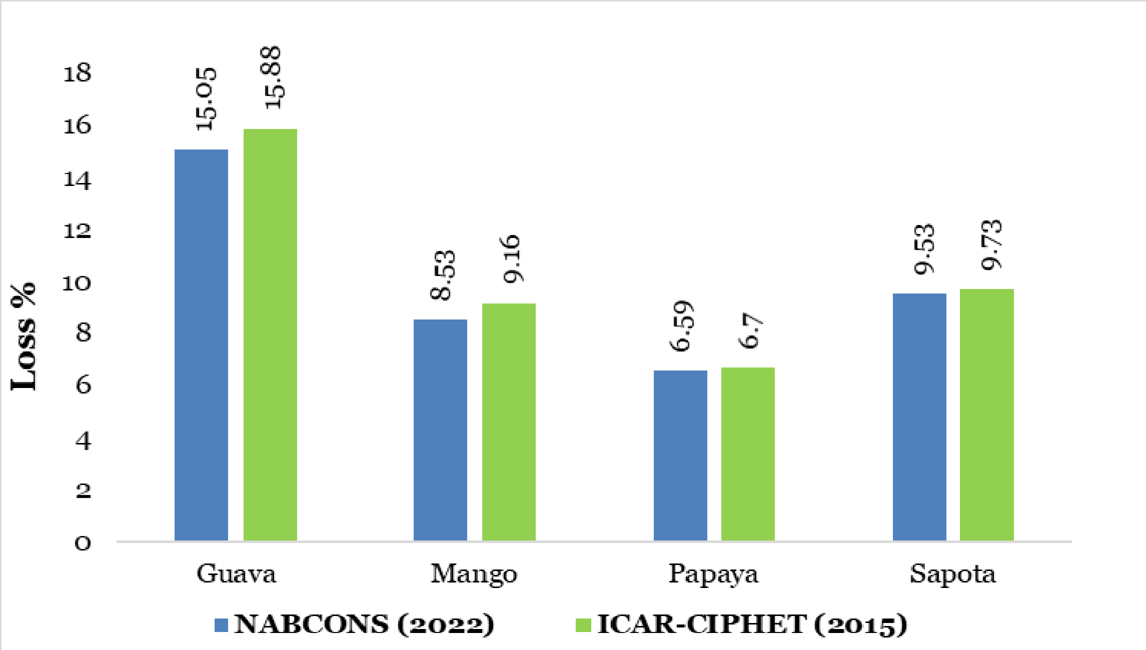


Fig. 6.4(b) Comparison of post-harvest losses of fruits



The survey for estimating the post-harvest losses of ***papaya*** was conducted in 5 districts in 3 states covering 3 agro-climatic zones (ACZ7, ACZ10 and ACZ13). The highest overall post-harvest losses in papaya were observed in Sahebganj district (6.73%) and the lowest in Kachchh district (5.81%). The farmers in the Kachchh district of Gujarat adopted cultivation of dwarf varieties of papaya in which harvesting operations were easily conducted by the farmers/labourers. Also, the deployment of trained labour for hand harvesting of papaya and usage of crates and carts for collecting the harvested papaya contributed to the lower losses during harvesting. The highest overall post-harvest losses at state level in papaya were observed in Jharkhand (6.73%) and the lowest in Gujarat (5.81%). At ACZ level, the highest overall post-harvest losses in papaya were observed in ACZ7 (6.73%) and the lowest in ACZ13 (5.81%). The post-harvest losses observed at farm operations at national level in papaya were found to be 3.77 percent, whereas, for market level operations post-harvest losses were 2.82 percent with an overall post-harvest loss of 6.59 percent. Among different operations, the highest post-harvest losses were observed at harvesting (1.31%) operation. Improper harvesting methods, fruits infected by diseases and pests, rough handling by farmers/labours and pressing/impact were the major causes of losses observed during harvesting, collection, sorting & grading and transportation in farm level operations. It was observed that in some of the cases, either traders directly procured the fruit from the farm or farmer collectives supplied the produce in the market thereby reducing the number of intermediaries resulting in less handling losses as well as reduction in duration of storing of fruit at the farm level. The overall losses for papaya reported by ICAR-CIPHET (2015) were 6.70 percent. The present study estimated lower post-harvest losses (6.59%) in comparison to previous study (ICAR-CIPHET 2015), which was non-significant at 5 percent level of significance **(Fig. 6.4(b))**. The decrease in the overall post-harvest losses may be attributed to introduction of dwarf varieties, use of better management practices viz. increased usage of paper, use of crates and cartons for packaging, usage of better harvesting tools and gloves during harvesting, etc.

The survey for estimating the post-harvest losses of ***sapota*** was conducted in 9 districts in 5 states covering 5 agro-climatic zones (ACZ9, ACZ10, ACZ11, ACZ12 and ACZ13). The overall post-harvest losses in sapota at district level were the highest in Tirupathur district (11.63%) and the lowest in Junagadh district (7.36%). The overall post-harvest losses in sapota at state level were the highest in Tamil Nadu (11.27%) and the lowest in Gujarat (7.47%). The overall post-harvest losses were the highest in ACZ11 (11.48%) and the lowest in ACZ13 (7.47%). Lack of proper harvesting equipment, physical damage due to dropping of the fruits, usage of gunny bags for transportation and inadequate processing units contributed to higher losses of sapota in ACZ11. Post-harvest losses in farm level operations at national level in sapota were found to be 6.22 percent while for market level, the post-harvest losses were 3.32 percent with an overall post-harvest loss of 9.53 percent. Among different operations, harvesting (2.31%) and sorting and grading (1.95%) majorly contributed to the losses in farm level operations and at transport (1.30%) in market operations. Earlier, ICAR-CIPHET (2015) reported an overall loss of 9.73 percent for sapota, while the present study reported lower post-harvest losses (9.53%), which were non-significantly lower than the earlier conducted study at 5 percent level of significance **(Fig. 6.4(b))**. The decrease in overall post-harvest losses may be attributed to overall improvements in various farm operations such as use of sapota harvesters, plastic bags and crates for transportation, cultivation of improved varieties, processing of sapota for production of value added products and export of sapota.

The survey for estimating the post-harvest losses of ***pineapple*** was conducted in 6 districts in 4 states covering 4 agro-climatic zones (ACZ2, ACZ3, ACZ4 and ACZ12). The overall post-harvest losses at district level in pineapple were the highest in Darjeeling district (8.01%) and the lowest in Shimoga district (3.34%). In the districts such as Shimoga and Uttara Kannada, harvesting of the majority of pineapple was done by hand resulting in less harvesting losses as compared to other districts, where knife and scythe were used resulting in damage (cuts and bruises) to fruit. Also, pineapples were stored in cold storage and were placed with crown portions downwards for cushioning resulting in less post-harvest losses in the districts of Shimoga and Uttara Kannada. The overall post-harvest losses at state level in pineapple were the highest in Assam (7.11%) and the lowest in Karnataka (3.73%), which could be due to the fact that pineapple was being stored at ambient conditions or even under a shed resulting in rotting of fruit. At ACZ level, the overall post-harvest losses in pineapple were the highest in ACZ2 (7.41%) and the lowest in ACZ12 (3.75%). The post-harvest handling practices such as proper harvesting, storage at right temperature and placement of pineapple in upside down position (for cushioning) during transportation resulted in less overall post-harvest losses in ACZ12 as compared to other ACZs, where such practices were not practised properly. Post-harvest losses in farm operations at national level in pineapple were found to be 4.22 percent, while for market level, the post-harvest losses were 1.80 percent. The overall post-harvest losses of pineapple were 6.02 percent. Harvesting operation (2.09%) and sorting and grading (1.81%) in farm operations and wholesaler (0.58%) and retailer (0.52%) level in market operations majorly contributed to the post-harvest losses. Improper harvesting methods such as damage of pineapple by using knife and scythe and rough handling by labourers were the major causes of losses observed during harvesting. At the market level, post-harvest losses observed were mainly due to handling of pineapple at ambient conditions resulting in ripening and rotting and due to rough handling, stacking and impact during the transportation.

The survey for estimating the post-harvest losses of ***pomegranate*** was conducted in 5 districts in 2 states covering 2 agro-climatic zones (ACZ9 and ACZ10). The overall post-harvest losses at district level were the highest in Anantapur district (8.69%) and the lowest in Ahmednagar district (5.47%). One of the reasons for lower post-harvest losses in districts such as Nashik, Solapur and Ahmednagar was due to the presence/establishment of better infrastructure such as pack houses (precooling and storage facilities) which resulted in proper handling of fruit. The overall post-harvest losses at state level in pomegranate were the highest in Andhra Pradesh (8.69%) and the lowest were observed in Maharashtra (6.02%). There was a lack of infrastructure for cold storage and pack houses in states such as Andhra Pradesh and Karnataka resulting in highest post-harvest losses. At ACZ level, the overall post-harvest losses observed were the highest in ACZ10 (8.58%) followed by ACZ9 (6.02%). The post-harvest losses in farm operations at national level in pomegranate were found to be 4.09 percent, while 2.73 percent at market level. The total overall post-harvest losses observed were 6.82 percent. Among different operations, harvesting (1.39%) and sorting & grading (1.72%) at farm level and retailer (0.8%) at market level majorly contributed to the post-harvest losses. At the market level, losses were observed due to handling of produce at ambient conditions resulting in rotting, black spots and over ripening (major cause) and physical damage due to improper handling (minor cause) of fruit by the labourers.

The survey for estimating the post-harvest losses of ***muskmelon*** was conducted in 7 districts in 4 states covering 5 agro-climatic zones (ACZ4, ACZ5, ACZ6, ACZ9 and ACZ10). The overall post-harvest losses at district level in muskmelon were the highest in Farrukhabad district (7.75%) and the lowest in Kapurthala district (5.50%). This may be due to the higher usage of hybrid seeds, early sowing practice and awareness amongst the farmers for improved cultivation practices in the districts, such as Kapurthala resulting in reduced losses of muskmelon. The overall post-harvest losses at state level in muskmelon were the highest in Andhra Pradesh (7.42%) and the lowest in Punjab (5.48%). The overall post-harvest losses in muskmelon at ACZ level were the highest in ACZ10 (7.42%) and the lowest in ACZ6 (5.50%). Disease and pest infestation, usage of local seeds and less awareness amongst the farmers resulted in comparatively higher losses in Andhra Pradesh state of ACZ10 and Uttar Pradesh state of ACZ5 as compared to Punjab state of ACZ6, where hybrid seeds, early sowing practices and greater awareness amongst the farmers for cultivation was observed. The post-harvest losses in farm operations at national level in muskmelon were found to be 4.12 percent, while for market level, the post-harvest losses were 2.71 percent. The overall total losses of muskmelon were 6.83 percent. Among different operations, harvesting (1.52%) and sorting and grading (1.17%) in farm level operations and at retailer (1.29%) level in market operations majorly contributed to the losses.

Among fruits covered under the study, pest and disease infected fruits, manual harvesting of fruits by the use of stick and knife, dropping of fruits, improper and rough handling of fruits by labourers leading to bruising, cracking & hand pressing, removal of physically damaged fruits during sorting & grading contributed to the losses at the farm level. At the market level, losses were mainly observed due to physical damage of fruits during handling and loading and unloading during transportation. The estimated post-harvest losses in apple, citrus, grapes, guava and mango were significantly lower than the earlier conducted study (ICAR-CIPHET 2015) which can be attributed to the overall improvements such as improved varieties, increased use of crates for handling & transport, and increase in the number of pack houses established.

Study / Reference	Number of pack house
Modern Pack Houses as per All India Cold-chain Infrastructure Capacity Assessment of Status & Gap (NCCD 2015)	249
Total Pack Houses (as on 2021) Integrated pack house created under MIDH (as on 2021)- 260 224 active pack houses as per APEDA (as on July 2021)	484

There has been an increase in pack houses established since 2015. As per study published by NCCD on all India cold-chain infrastructure capacity assessment of status & gap (2015), there were 249 pack houses. Under the Mission for Integrated Development of Horticulture (MIDH), about 260 integrated pack houses have been created (up to 2021). Also, as per APEDA, as of July 2021, there are 224 active pack houses in the country. Therefore, currently there are a total of

484 pack houses. As per the agriculture statistics data 2021, the total number of cold storage all India are 8354 with a total storage capacity of 3,80,88249 MT as on 31.01.2022. Out of around 4100 FPOs registered under the schemes promoted by NABARD, more than 1200 FPOs are working in fruits and vegetable sector. Also, various training courses conducted by ICAR on post-harvest management across the country has contributed towards reduction of post-harvest losses.

Box 6.5 Good practices to be followed for fruits

- *Application of edible coatings as is being done for Kinnow*
- *Use of ethylene scrubbers & SO2 generating pad in absorbent tissue paper in the box as is being done for grapes.*
- *Use of light/ sticky trap @ 1/ha to monitor the activity of adults (fruit fly) and insecticides like malathion 50 EC for pest management in guava.*
- *Dipping of mangoes in hot water to minimize fruit fly damage, anthracnose, and stem-end rot infection*
- *Cleaning of bananas using alum or sodium hypochlorite to protect against fungal attack.*
- *Ethylene based ripening for banana.*
- *Use of reefer vans for long-distance transportation, as is being done by modern retail.*
- *Organized marketing of fruits through FPOs/ FPC.*
- *Setting up of portable solar powered cold stores in rural areas where continuous electricity supply is not available for pre-cooling of fruits and vegetables on the lines of a cold storage facility, Ecofrost, developed by Ecozone Solutions.*

Box 6.5 Good practices to be followed for fruits

- *Use of digital products like Intello Track, Farmtrace for traceability across the value chain.*
- *Cultivation of improved varieties of sapota and guava and dwarf varieties of papaya.*
- *High density plantation of apples using dwarfing rootstocks.*
- *Use of fruit pickers for harvesting and plastic crates for collection at farm. Lining of crates using bubble sheet and polythene sheet as is being done for grapes*
- *Small capacity sorting/grading machines at the farm level to ensure uniform packaging on the lines of apple in Himachal Pradesh*
- *Installation of ropeways collection and transportation in hilly areas on the lines of Himachal Pradesh for apples.*
- *The use of individual shrink film wrapping being used for bananas and fruit foam nets being used in guava.*
- *Use of Corrugated Fiber Board (CFB) and fruit trays for packing fruits. The fruits should be packed in a single or minimum number of layers.*

6.5 Vegetables

With production of 199.88 million MT in 2021-22 (DAC&FW, 2021-22), India is one of the largest producers of vegetables in the world. Losses due to absence of adequate post-harvest infrastructure resulted in reduction in the availability of vegetables for human consumption.

6.5.1 Farm level operations

Harvesting- At the farm level, vegetables were harvested manually by plucking from the plant by hand or by cutting the stalk with a knife. Harvesting of green peas was done manually by plucking the mature pods by hand. While harvesting farmers carried plastic buckets, tubs or *kilta* with them to place the pea pods immediately after harvest. Harvesting of onion was done manually by gently pulling out the onion bulbs from the soil, followed by shaking off the adhering soil. Yellowing and falling over of the tops from the neck portion indicated the maturity of onion bulbs. Many farmers used forked tools to loosen the soil around bulbs for easier harvesting. Potatoes were harvested by Kudar, Lambi and potato diggers. The harvesting of tomato was generally done manually in the morning hours. As tomato is the climacteric fruit, maturity indices play an important role in post-harvest losses of tomato. In some states, the colour charts were distributed among the farmers to ensure the timely harvesting. The harvesting of the tomato crop at an appropriate stage of maturity was important for long term storage and transport to the distant market. Okra was harvested manually by plucking from plants or by cutting the okra using knives or pruners or scissors. The losses during harvesting of okra included splitting or cutting okra due to improper plucking or cutting and physical damage to okra due to improper manual harvesting. The main causes of post-harvest losses were improper harvesting, harvesting of under ripe or overripe vegetables resulting in damage to vegetables by the sickle during manual harvesting. Farmers harvested radish manually by digging out the vegetable. The main cause of post-harvest losses observed were due to physical injury by implements leading to splitting of radish. Some portions of radish were being left in the field due to the splitting of radish during manual harvesting. Harvesting operation of mushroom was carried out manually by plucking with clean hands and by lightly twisting the bottom of fully mature first flush in 3 to 5 days and the second flush after 5-7 days. Wet bubble disease infection was reported in some cases right after harvesting of mushroom which led to heavy post-harvest losses.

Harvesting of radish

Collection- Collection of vegetables was done manually using gunny bags, plastic baskets, wooden baskets and carpets by farmers. Improper handling resulted in damaged leaves during the operation. Minimal post-harvest losses were observed during this operation majorly attributed to improper handling while collection leading to dropping of capsicum. Sometimes losses due to unfavourable weather conditions during the operation were also observed. After harvesting, drying/ curing of onion bulbs was done for which bulbs were left in the field in a single layer for the stems to dry out, necks to wither and papery skin to tighten around the bulbs. Once the necks were completely tight and dry and the stems contained no moisture, the roots were cut off from the bottom of each bulb and tops were trimmed back to 1-2 inches using scissors. After this, the onion bulbs were either collected at one place in the field or brought to a farm shed. Farmers who planned to braid the onion bulbs together during storage avoided trimming off of the tops. Post-harvest losses at this level were observed due to poor cutting techniques leading to the cuts in stem near the top of the bulb, which made the bulbs susceptible to attacks from fungus-like black moulds, bacteria and other surface injuries during storage.

Collection of tomato in plastic crates

Precooling – This operation was not common in vegetables, but some of the farmers manually did this operation to reduce the field heat in tomato to extend its shelf life. The collected tomato was heaped in shade or dipped in cold water to remove the farm heat. In most of the states, this operation was generally not practised by the farmers.

Trimming- The trimming operation was mostly performed in the case of shitake mushroom to remove the woody stem and in button mushroom to cut the inedible root part. Trimming was performed with the help of a knife after the harvesting operation. The physical and mechanical loss was noted during trimming operation due to improper handling and inappropriate cutting.

Cleaning- A cleaning operation was performed to remove adhering dirt and compost from the surface of the mushroom. Harvested mushrooms were gently soaked in the sodium metabisulphite water in the plastic tub and then washed with water while cleaning. The cleaning was done manually. Being delicate to handle, the physical and mechanical loss was noticed during the cleaning of the mushroom.

Sorting and grading- Sorting and grading was performed manually in the field to remove the vegetables, which were unfit for consumption from the lot such as over ripe and under ripe pods, disease and pest infested pods and rotten pods. In mushroom, sorting and grading operation was mainly performed at the end of the seasons during the 3rd and 4th cycle of harvesting as inferior quality mushrooms were also being harvested at this stage. Sorting and grading of mushroom was done to segregate the mushroom into different grades based on the quality and market standards. Due to improper handling and bad weather conditions, the physical and mechanical loss was observed during sorting and grading. The thick necked, bolted, doubled, bruised and decayed onion bulbs were picked out during sorting grading. The outer dry scales were also rubbed off during the grading process to give the bulbs a better appearance for the market. Onion bulbs were sorted based upon the size as well.

Packaging- Gunny bags, plastic bags and large bamboo baskets were used for packaging of vegetables in the field level. The operation was performed manually by farmers. Tomato were packed in plastic crates, wooden crates, cartons, fibreboard boxes, bamboo baskets or plastic bags. The fruit were damaged due to pressing / compression due to excess load in the plastic crates / cartons or boxes. Cuts and bruises due to the sharp edges in the wooden or bamboo baskets were observed in tomatoes. Packaging operation at the farm level was performed manually and with the help of a hand-operated sealing machine. Transparent polythene pouches and plastic mushroom packaging trays of 200 grams capacity were used for packing. Few growers also used punnet to pack mushroom. Physical losses and mechanical injuries were reported in mushroom due to manual handling while packaging. Mostly jute gunny bags and leno bags were used for packaging of onions at the field level.

Improved packaging of mushroom

Storage- Storage of vegetables at farm for a few hours before transportation to the respective destinations was considered as farm level storage. The onions were bulk stored in special houses with thatched roofs and side walls made up with bamboo sticks or wire mesh for good air circulation. At some places, the sides were also covered with gunny cloth. Onions were stored in these sheds by spreading them on dry and damp proof floors or racks. Periodical turning of bulbs or removal of rotten, damaged and sprouted bulbs was also done. Storage by braiding the onion bulbs together and hanging them in a cool, dry, well-ventilated area was also followed by some farmers. Post-harvest losses in this operation were observed due to unfavourable temperature and humidity conditions leading to sprouting and rotting of bulbs.

Transport- Vegetables were transported to nearby markets. Motorcycles, bicycles and TATA Ace were used for transportation. The post-harvest losses were observed during the transportation owing to physical damage due to overfilling and tearing of plastic bags. Farmers were sending their mushrooms to markets in the roadways buses too, which was noted as one of the economical transport fleet at farm level. Collaborative transport with the vehicle chain was also noticed at some survey locations. Farm level aggregation and collective of farm produce by FPOs / FPC was observed in vegetable growing clusters. Improper packaging/ stacking, and non-availability of standard transport facilities led to post-harvest losses during transportation from farm to market.

6.5.2 Market level operations

Wholesalers and retailers- At wholesalers and retailers level, all operations of marketing, handling and storage were being done manually. Wholesalers received regular supply of green peas from farmers and supplied the same to retailers in smaller lots depending upon the market requirement. Sometimes wholesalers and retailers also arranged direct pick up of produce from the field. The post-harvest losses observed were due to rough handling, delay in sales and improper storage conditions. For vegetables like green peas, capsicum, okra, brinjal, bottle gourd etc. markets were lacking normal or cold storage facilities leading to losses. In states like Himachal Pradesh, traders distributed plastic crates and packing materials to farmers in advance to collect the vegetable produce.

Wholesale market of tomato

Godowns- The freshly received stock of vegetables and unsold produce were stored in godowns/sheds. Potato was purchased by the wholesalers from farmers, who in turn kept the purchased potato in cold storage from June to Dec. These cold storages were mostly privately owned and available within a range of 15-25 kms from the farm. Onions were stored either loose or in bags in traditional and unscientific storage structures, where losses in the form of weight loss and rotting due to unfavourable conditions of temperature and humidity, and poor ventilation were observed. The post-harvest losses observed in cold storage were due to moisture loss, rotting, storage of pest-infected potato and sprouting. Onion required a specialised cold storage with the facilities of drying after removal of the onion from the cold storage. In other vegetables, post-harvest losses observed at storage level were due to yellowing of produce, damage by rodents-rat, delay in sales and improper storage conditions, as there were no cold storages and closed space available in the market area. Storage spaces specifically dedicated for the storage of vegetables like cabbage, cauliflower, green peas, mushroom, brinjal etc. were rarely found.

Transport- Vegetables were transported to various locations within the district and outside the district. Tempo, mini canters, pick-up vans, bicycles and motorcycles were being used for transportation. The reefer vans were sparingly used for the transportation of green peas for the domestic market. Losses at this level were observed due to damage to packaging material during transport, mishandling of produce during loading and unloading, improper stacking and vibration during transportation as a result of undulating roads. Rotting of pods was also observed in case of long distance transport as no temperature and humidity control facilities were available in the vehicles.

Processing unit - Processing units were observed for cabbage, cauliflower, green peas, onion, potato, tomato and tapioca. During this operation, the losses were observed due to over ripening, spoilage, immature fruit, physical injury due to bruising, loading & unloading. Onion was getting processed into a wide variety of products like minimally processed ready to use or ready to cook fresh onions, onion paste, dehydrated onion flakes, onion powder, onion oil, onion vinegar, onion sauce, pickled onion, onion wine and beverage etc. Green peas processing units aimed at producing frozen peas targeting off-season supply of peas to consumers. Frozen peas were supplied to food industries as well for further use in producing different processed food products.

6.5.3 Post-harvest losses at farm and market level

Assessment of harvest and post-harvest losses were carried out in fourteen vegetables (cabbage, cauliflower, green peas, mushroom, onion, potato, tomato, tapioca, bottle gourd, brinjal, beans, raddish, capsicum and okra). The extent of post-harvest losses at district level, state level and ACZ level are presented, respectively in **Appendix 5, Appendix 6** and **Appendix 7** and national level are reported in **Table 6.5.**

The survey for estimating the post-harvest losses of ***cabbage*** was conducted in 10 districts in 5 states covering 6 agro-climatic zones (ACZ2, ACZ3, ACZ4, ACZ7, ACZ11, and ACZ13). The overall post-harvest losses at district level were the highest in Muzaffarpur district (10.40%) and the lowest in Nagaon district (7.49%). In the district such as Nagaon, care was taken while harvesting of cabbage and operations like transportation, storage and handling were carried out properly at market level. Whereas in Kandhamal district the care was taken at farm operations such as harvesting, storage and transportation. The overall post-harvest losses at state level were the highest in West Bengal (8.71%), and the lowest in Assam (7.38%). It was observed that the majority of the cabbage were being harvested by adopting selection method, wherein only matured and marketable cabbage were being harvested very carefully by the farmers/labourers. The reasons for high post-harvest losses in Vaishali district of Bihar, Rajkot district of Gujarat and Nadia and Murshidabad districts of West Bengal were due to high losses during harvesting i.e, damage caused due to harvesting equipment. The overall post-harvest losses at ACZ level were the highest in ACZ3 (8.63%) and the lowest in ACZ2 (7.31%). Practice of loose packaging, delay in transport from farm to Mandi due to lack of transportation facilities in case of small and marginal farmers were other reasons for high losses. The post-harvest handling practices in cabbage such as proper harvesting and handling resulted in less overall post-harvest losses in ACZ2. Post-harvest losses in farm operations at national level were found to be 5.83 percent, while for market level, the post-harvest losses were 2.32 percent with overall total losses of 8.15 percent. Sorting grading operation (2.77%) and harvesting (1.39%) in farm operations and retail level (1.42%) in market operations majorly contributed to losses. The major causes of losses were improper harvesting methods such as damage of cabbage by using knife and sickle and rough handling by labourers. The present study reported lower post-harvest losses (8.15%) in comparison to the 9.37 percent losses reported in the previous study conducted by CIPHET (2015). The difference in losses was significant at 5 percent level of significance **(Fig. 6.5(a))**. This decrease in overall post-harvest losses may be due to overall improvements in various farm operations such as careful harvesting and collection of cabbage, use of plastic crates in some states and involvement of FPO and cooperatives in post-harvest handling.

The survey for estimating the post-harvest losses of ***cauliflower*** was conducted in 10 districts in 5 states covering 6 agro-climatic zones (ACZ3, ACZ4, ACZ7, ACZ9, ACZ11, and ACZ13). The overall post-harvest losses at district level were the highest in Sabar Kantha district (7.86%) and the lowest in Chhindwara district (6.04%). In the districts such as Chhindwara and Banas Kantha, care was taken while harvesting of cauliflower and operations like sorting and grading, storage and handling. It was observed that the majority of the cauliflower were being harvested by adopting a selection method, wherein only matured and marketable cauliflower were being harvested very carefully by the farmers/labourers. The reasons for high post-harvest losses in Sabar Kantha district of Gujarat, Mayurbhanj and Kandhamal districts of Odisha were improper harvesting, lack of packing materials during handling, improper packaging i.e. loose & bulk packs etc. The practice of loose and bulk packaging along with delay in transport from farm to Mandi led to losses during transportation. The overall post-harvest losses at state level were the highest in Gujarat (7.99%), and the lowest in Madhya Pradesh (6.46%). The overall post-harvest losses at ACZ level were the highest in ACZ11 (8.58%) and the lowest in ACZ7 (7.33%). The post-harvest handling practices such as proper harvesting and handling, storage resulted in less overall post-harvest losses in ACZ7. Post-harvest losses in farm operations at national level were found to be 5.76 percent, while for market level, the post-harvest losses were 2.13 percent. The overall total losses of cauliflower were 7.89 percent. The highest post-harvest losses were observed at sorting grading operation (2.51%) in farm operations and at transport level (0.73%) in market operations. At the market level, post-harvest losses were mainly observed due to rough handling of cauliflower and lack of storage facilities. As depicted in **Fig. 6.5(a),** ICAR-CIPHET (2015) reported an overall loss of 9.56 percent for cauliflower, while the present study reported lower post-harvest losses (7.89%). The difference was significant at 5 percent level of significance w.r.t the post-harvest losses reported by ICAR-CIPHET 2015. The improvements in various farm operations-harvesting (selection method) and collection, use of plastic crates in some states, involvement of FPO and cooperatives in post-harvest handling, varietal development by SAUs and ICAR, extension services of KVKs-agri-input outlets established at KVKs campus has positively contributed in reduction of post-harvest losses.

Table 6.5 Post-harvest losses of vegetables in percent at national level

Crop	Farm operations									Market operations						Overall total loss
	Harvest-ing	Collec-tion	Sorting/ Grading	Clean-ing	Trimm-ing	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retailers	Processing unit	Trans-port	Total loss in market level	
Cabbage	1.39 ± 0.0312	0.34 ± 0.0067	2.77 ± 0.0641	-	-	0.34 ± 0.0089	0.15 ± 0.0048	0.84 ± 0.0031	5.83 ± 0.0724	0.07 ± 0.0112	0.7 ± 0.1599	1.42 ± 0.4201	0.01 ± 0.0096	0.12 ± 0.0436	2.32 ± 0.4519	8.15± 0.4576 (9.37)
Cauliflower	1.79 ±0.0317	0.21 ±0.0036	2.51 ±0.0496	-	-	0.33 ±0.0129	0.07 ±0	0.85 ±0.0208	5.76 ±0.0638	0.05 ±0.0006	0.64 ±0.0388	0.71 ±0.0378	0.01 ±0.0006	0.73 ±0.0845	2.13 ±0.1004	7.89 ±0.1189 (9.56)
Green pea	1.93 ± 0.0513	0.51 ± 0.0311	1.70 ± 0.0747	-	-	0.13 ± 0.0042	0.005 ± 0.0004	0.40 ± 0.0145	4.67 ± 0.097	-	0.5 ± 0.1013	0.68 ± 0.1452	0.04 ± 0.0084	0.54 ± 0.0954	1.76 ± 1.0827	6.43 ± 1.0871 (7.45)
Mushroom	0.64 ±0.0189	0.27 ±0.0164	3.88 ±0.1539	0.18 ±0.008	0.46 ±0.0243	0.15 ±0.0082	0.28± 0.0107	0.35 ±0.0102	6.21 ±0.1589	-	0.36 ±0.0854	0.3 ±0.0654		0.32 ±0.0616	0.99 ±0.1239	7.2±0.6621 (9.51)
Onion	2.48 ± 0.0452	0.39 ± 0.0115	1.01 ± 0.0168	-	-	0.12 ± 0.0068	0.91 ± 0.028	0.4 ± 0.0149	5.31 ± 0.0593	0.34 ± 0.0059	0.69 ± 0.1485	0.57 ± 0.06	0.01 ± 0.0028	0.35 ± 0.0897	1.96 ± 0.1836	7.26 ± 0.193 (8.20)
Potato	2.13 ± 0.002	0.22 ± 0.0036	2.09 ± 0.0014	-	-	0.04 ± 0.0004	0.1 ± 0.0003	0.52 ± 0.0008	5.1 ± 0.0045	0.13 ± 0.0001	0.28 ± 0.0022	0.1 ± 0.0003	0.03 ± 0.0001	0.32 ± 0.0001	0.86 ± 0.0022	5.96 ± 0.005 (7.32)
Tomato	2.92 ±0.9325	0.61 ±0.3001	3.1 ±0.932	-	-	0.33 ±0.2111	0.59 ±0.2825	0.82 ±0.3624	8.37 ±1.4436	-	0.93 ±0.1971	1.1 ±0.2446	0.13 ±0.0449	1.09 ±0.2282	3.25 ±0.3909	11.61 ±1.4956 (12.44)
Tapioca	1.19 ±0.033	0.25 ±0.0089	1.05 ±0.021	-	-	0.08 ±0.0032	0.26 ±0.0053	0.55 ±0.0093	3.39 ±0.0417	-	0.4 ±0.1403	0.47 ±0.0389	0.16 ±0.0159	0.45 ±0.0768	1.48 ±0.1654	4.87 ±0.1705 (4.58)
Bottle gourd	1.95 ±0.1444	0.45 ±0.0202	1.12 ±0.0759	-	-	0.27 ±0.0107	0.21 ±0.0206	0.68 ±0.0432	4.69 ±0.1715	-	0.69 ±0.1338	0.82 ±0.2293	-	0.81 ±0.1386	2.32 ±0.2995	7.01 ±0.3451

Crop	Farm operations									Market operations						Overall total loss
	Harvest-ing	Collec-tion	Sorting/ Grading	Clean-ing	Trimm-ing	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retailers	Processing unit	Trans-port	Total loss in market level	
Brinjal	2.25 ±0.0321	0.48 ±0.023	0.9 ±0.0148	-	-	0.26 ±0.0061	0.32 ±0.0065	0.54 ±0.0153	4.75 ±0.0458	-	0.92 ±0.2218	1.19 ±0.1699	-	0.54 ±0.0985	2.66 ±0.2962	7.41 ±0.2998
Beans	1.11 ± 0.015	0.47 ±0.0062	1.23 ±0.0135	-	-	0.3 ±0.004	0.26 ±0.0028	0.31 ±0.0058	3.68 ±0.0224	0.12 ±0.0178	1.01 ±0.1615	0.73 ±0.173	-	1.57 ±0.2387	3.43 ±0.3366	7.11 ±0.3373
Radish	2.24 ±0.0274	0.1 ±0.0036	0.93 ±0.0107	-	-	0.47 ±0.0068	0.11 ±0.0024	0.11 ±0.0022	3.96 ±0.0306	-	0.71 ±0.1025	0.72 ±0.1604		1.07 ±0.1544	2.5 ±0.2451	6.46 ±0.247
Capsicum	1.41 ± 0.0976	0.13 ± 0.0172	0.55 ± 0.045	-	-	0.17 ± 0.0136	0.005 ± 0.001	0.33 ± 0.0303	2.60 ± 0.1138	-	1.05 ± 0.1896	0.73 ± 0.0746	-	0.77 ± 0.2306	2.55 ± 0.3077	5.15 ± 0.3281
Okra	1.5 ±0.0316	0.31 ±0.0065	1.16 ±0.0131	-	-	0.14 ±0.0024	0.17 ±0.0035	0.49 ±0.0092	3.76 ±0.0363	-	0.82 ±0.0966	0.87 ±0.0833	-	0.56 ±0.0553	2.25 ±0.1392	6.01 ±0.1437

Note: *Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*

Figures in parentheses represent the losses of previous study by ICAR-CIPHET (2015)

The survey for estimating post-harvest losses of ***green peas*** was conducted in 8 districts in 4 states covering 6 agro-climatic zones (ACZ1, ACZ5, ACZ6, ACZ7, ACZ8, and ACZ9). The overall post-harvest losses were the highest in Amritsar district (6.67%) and the lowest in Ujjain district (5.56%). Higher losses in Amritsar may be attributed to wrong harvesting practices adopted by the farmers like picking all pods (mature and immature) from the bush, overfilling of pods in the hands followed by rolling and squeezing them and harshly trashing pods in the collection container leading to opening of pods. In addition to this, overfilling of gunny bags during packaging and wrong staking in transport vehicles also contributed to post-harvest losses. In Ujjain, retention with wholesaler and retailer was found less due to availability of processing units in the vicinity which reduced the losses at these levels. The overall post-harvest losses at state level were the highest in Punjab (6.67%) and the lowest in Uttar Pradesh (6.27%). The overall post-harvest losses at ACZ level were the highest in ACZ6 (6.67%) and the lowest in ACZ8 (5.82%). Higher post-harvest losses in ACZ6 may be attributed to mishandling pods during harvesting and collection operations. Harvesting of green peas at the wrong stage (under-ripe and over-ripe) increased the losses at harvesting level. Further, higher post-harvest losses were observed during loading and unloading while transport as staking was not done properly in the vehicles. Poor road infrastructure and uncontrolled atmospheric conditions during long distance transport further increased the post-harvest losses. Skilled labourers adopting proper harvesting practices in ACZ8 lowered losses in ACZ8. Some large farmers were also found using pod picking machines and mechanical sorter grader. Post-harvest losses in farm operations at national level were found to be 4.67 percent, while for market level, the post-harvest losses were 1.76 percent. The overall post-harvest losses for green peas at the national level were observed as 6.43 percent. The post-harvest losses at harvesting (1.93%) and sorting-grading (1.70%) majorly contributed to farm level post-harvest losses, while losses at retail level (0.68%) were the highest in green peas. At farm level uneven ripening, poor handling of produce and dropping and splitting of pods during collection led to increased losses. Delay in harvesting due to low prices in market led ot over maturity of pods leading to post-harvest losses in some cases. At market level lack of immediate demand in the market and poor storage facilities leading to moisture loss being the major cause behind the higher post-harvest losses and it also lowered the pea pod quality, shelf life, market value and perceived freshness of pods. Continuous watering of the lot kept for sale was done to keep the pods fresh but it led to growth of black mould and white fungus which further aggravated the post-harvest loss by reducing the shelf life. ICAR-CIPHET (2015) reported an overall loss of 7.45 percent for green peas while the present study reported lower post-harvest losses (6.43%). The estimated post-harvest losses obtained from the present study were non-significantly lower than the earlier conducted study (ICAR-CIPHET 2015) at 5 percent level of significance **(Fig. 6.5(a))**. The decline in post-harvest losses may be attributed to use of varieties with synchronous maturity leading to even maturity, better handling practices at farm level and increase in green peas processing units.

The survey for estimating post-harvest losses of ***mushroom*** was conducted in 15 districts in 8 states covering 7 agro-climatic zones (ACZ1, ACZ2, ACZ5, ACZ6, ACZ7, ACZ9 and ACZ10). The highest overall post-harvest losses (17.41%) were estimated in Pune district and the lowest (2.01%) in Sonipat district. The losses at bigger units in Pune district could be attributed to the large requiring longer time to market. At wholesaler shops, mushrooms were kept for longer duration leading to post-harvest losses. Mushrooms were generally transported at normal temperature leading to spoilage. The mushroom production in Sonipat district was mainly carried out by marginal farmers with small quantities. The handling was easy and efficient in this district resulting in reduced post-harvest losses. The highest overall post-harvest losses (13.6%) were reported in Odisha and the lowest (2.74%) in Uttar Pradesh. The higher post-harvest losses in Odisha were due to unfavourable weather conditions. The higher temperature during day time led to spoilage of mushroom as it was highly perishable and temperature sensitive. The lowest post-harvest losses were obtained, where growers were aware about best practices to handle the harvested mushroom through the entire supply chain. The highest overall post-harvest losses (13.04%) were observed in ACZ10 and the lowest (2.74%) in ACZ2. The overall post-harvest losses at national level were 7.20 percent. Total farm level post-harvest losses reported at national level were 6.21 percent, whereas total market level losses reported were 0.99 percent in mushroom. At national level, the highest post-harvest losses in mushroom (3.88%) were observed at the time of trimming during farm level operation, whereas the highest post-harvest losses (0.73%) were observed at wholesaler's level during market operations. The main causes identified for post-harvest losses of mushroom were high level of perishability of the mushroom, lack of awareness and non-availability of cold chain structures for mushroom. ICAR-CIPHET (2015) reported an overall loss of 9.51 percent for mushroom, while the present study reported lower post-harvest losses (7.20%) and reduction in the post-harvest losses was significant at 5 percent level of significance. This significant reduction in losses might be due to extension support by KVK for small mushroom growers and increased number of small scale growers. Modern technologies developed by ICAR-DMR for Hi-tech mushroom production and processing also resulted in reduced post-harvest losses.

The survey for estimating the post-harvest losses of ***onion*** was conducted in 7 districts in 4 states covering 4 agro-climatic zones (ACZ7, ACZ9, ACZ10 and ACZ13). The overall post-harvest losses at district level were the highest in Bijapur district (9.11%) and the lowest in Ahmednagar district (6.71%). Higher losses in Bijapur district could be attributed to poor management at farm level. At farm level improper harvesting, incomplete drying and poorly ventilated storage structures led to increased losses. In Ahmednagar, lower post-harvest losses were due to careful harvesting, proper curing before storage and well ventilated storage conditions along with regular turning of bulbs. Some farmers were having access to cold stores as well and for prevention of sprouting in cold stores, farmers were found doing spray of Malic Hydrazide in the standing crop beforehand. The overall post-harvest losses at state level were the highest in Karnataka (8.70%) and the lowest in Maharashtra (6.78%). The overall post-harvest losses at ACZ level were the highest in ACZ10 (8.70%) and the lowest in ACZ 9 (6.78%). Post-harvest losses in farm operations at national level were found to be 5.31 percent, while for market level, the post-harvest losses were 1.96 percent. The overall post-harvest losses for onion at the national level were observed as 7.26 percent. At farm level, post-harvest losses at harvesting (2.48%) and sorting-grading (1.01%) and at market level, post-harvest losses at wholesaler (0.69%) and retailer level (0.57%) majorly contributed to overall post-harvest losses in onion. Unskilled labourers involved in harvesting adopted wrong practices like causal forking of soil before harvesting causing bruises to the bulbs and harsh pulling out of bulbs causing tops to break off from the neck portion. After harvesting, bulbs were found placed in bundles leading to improper drying as well. Even before collection, wrong cuts were made below the neck of the bulbs, while de-topping increasing susceptibility of bulbs to pathogens. In addition to this, during storage bulbs were placed directly on the floor and proper ventilation was not maintained leading to increased post-harvest losses. At market level lack of poor storage facilities leading to moisture loss, rotting of bulbs and sprouting of bulbs were the major causes behind the higher post-harvest losses. Transportation losses were low due to proper curing of bulbs. Few large farmers were using mechanical harvesters and graders as well. ICAR-CIPHET (2015) reported an overall loss of 8.20 percent for onion, while the present study reported lower post-harvest losses (7.26%). The estimated post-harvest losses obtained from the present study were significantly lower than the earlier conducted study (ICAR-CIPHET 2015) at 5 percent level of significance **(Fig. 6.5(b))**. This reduction in overall post-harvest losses for onion is obviously due to various schemes of GoI targeted for this crop, especially promotion of FPOs by GoI, NABARD and SFAC. Regular training by ICAR-DOGR (Directorate of Onion and Garlic Research), Pune, Maharashtra to onion growers and traders on post- harvest management and value addition has helped in reducing post-harvest losses to a great extent. There has been an increase in the number of dehydration units of onion especially in Gujarat and Maharashtra (Gujarat: 75 units in 2014 and 120 units in 2022). These units have helped farmers to sell their produce in time.

The survey for estimating the post-harvest losses of ***potato*** was conducted in 11 districts in 6 states covering 6 agro-climatic zones (ACZ3, ACZ4, ACZ5, ACZ6, ACZ9 and ACZ13). The overall post-harvest losses were the highest in Indore district (7.05%) and the lowest in Sabar Kantha district (5.36%). In the district such as Sabar Kantha, care was taken while harvesting of potato and operations like sorting and grading, storage and handling were carried out properly at farm level. Whereas, in Hooghly district the care was taken at farm operations such as harvesting, collection, sorting grading, and transportation. The overall post-harvest losses at state level were the highest in Madhya Pradesh (7.00%), and the lowest in Punjab (5.57%). The reasons for high post-harvest losses in Indore district of Madhya Pradesh, Nalanda and Firozabad districts of Bihar were due to loss in harvesting i.e, damage caused due to harvesting equipment, sorting grading (removal of over cut, rotten or small size potato). The non-availability of packing materials timely such as gunny bags led to post-harvest losses. Further, delay in transportation from farm to cold storage units due to lack of transportation facilities in case of small and marginal farmers led to losses. The overall post-harvest losses at ACZ level were the highest in ACZ9 (7.02%) and the lowest in ACZ4 (5.62%). The post-harvest handling practices such as proper harvesting and handling, storage, transportation resulted in less overall post-harvest losses in ACZ4. Post-harvest losses in farm operations at national level were found to be 5.10 percent, while for market level, the post-harvest losses were 0.86 percent. The overall total losses of potato were 5.96 percent. Harvesting (2.13%) and sorting grading operation (2.09%) in farm operations and transport level (0.32%) in market operations contributed majorly to post-harvest losses. ICAR-CIPHET (2015) reported an overall loss of 7.32 percent for potato, while the present study reported lower post-harvest losses (5.96%). The difference was significant at the 5 percent level of significance as depicted in **Fig. 6.5(b).** The reduction in overall post-harvest losses with respect to potato was mainly due to increased creation of cold storage capacity for potato in the country. Around 75% of the total cold storages in India were single commodity (mainly potatoes) and 25% were multi commodity cold storages. Accordingly, there were 5168 cold storages for potatoes as of March 2014, which has gone up to 6134 for 23.87 million metric MT capacity as of September 2020. Further, there has been continuous efforts by ICAR – Central Potato Research Institute, Shimla HP towards development of new varieties and technologies of production and processing potato right since 1949.

The survey for estimating the post-harvest losses of ***tomato*** was conducted in 8 districts in 5 states covering 4 agro-climatic zones (ACZ7, ACZ10, ACZ11 and ACZ13). The overall post-harvest losses at district level were the highest in Kedujhar district (13.02%), and the lowest in Kolar district (9.46%). The reason for high post-harvest losses in Kendujhar & Kordha district could be attributed to high post-harvest losses at harvesting, farm storage and farm transport. The overall post-harvest losses in tomato at state level were the highest in Odisha (13.19%), and the lowest in Karnataka (10.97%). The overall post-harvest losses at ACZ level were the highest in ACZ11 (12.94%), and the lowest in ACZ10 (10.87%). Post-harvest losses in farm operations at national level were found to be 8.37 percent, while for market level, the post-harvest losses were 3.25 percent. The overall total losses of tomato were 11.61 percent. The highest post-harvest losses were observed at sorting and grading (3.1%) in farm operations and at retailer (1.10%) level in market operations. The non-availability of the good packaging materials such as plastic crates and poor condition of roads in production centres were observed as major causes of the post-harvest losses in tomato. Improper harvesting methods such as hand pressing and improper twisting during plucking was a major reason for loss during harvesting. At the market level, losses were mainly observed due to improper handling of tomato at ambient conditions resulting in rotting and spoilage. Overloading and faulty stacking of boxes/ crates also resulted in damage of fruit leading to high post-harvest losses. Earlier ICAR-CIPHET (2015) reported an overall loss of 9.69 percent for tomatoes, while the present study reported lower post-harvest losses (7.71%), which were significantly lower at 5 percent level of significance **(Fig. 6.5(b))**. This decrease in overall post-harvest losses may be attributed to overall improvements in various farm operations such as careful harvesting and collection of tomato after 2015.

The survey for estimating the post-harvest losses of ***tapioca*** was conducted in 6 districts in 4 states covering 3 agro-climatic zones (ACZ2, ACZ8, and ACZ12). The highest post-harvest losses at district level in tapioca were reported from East Godavari district (5.36%) and the lowest in Idukki district (4.61%). The overall post-harvest losses were the highest in Nagaland state (5.42%), and the lowest in Kerala state (4.80%). The overall highest losses in tapioca were reported from ACZ2 (6.43%), while the lowest post-harvest losses (4.8%) were reported from ACZ12. The post-harvest handling practices such as proper harvesting and handling, storage, transportation resulted in less overall post-harvest losses in ACZ12. Post-harvest losses in farm operations in tapioca at national level were found to be 3.39 percent, while for market level, the post-harvest losses were 1.48 percent with overall total post-harvest losses of 4.87 percent. The highest post-harvest losses were observed at harvesting (1.19%) under farm operations and at retailer (0.47%) level under market level. Improper harvesting, tapioca left in soil, lack of packing material, rotting and attack of insect pests due to delay in transport to processing units majorly contributed to the losses in farm level operations. At the market level, post-harvest losses were mainly observed due to rough handling of tapioca and lack of storage materials. As depicted in **Fig. 6.5(b)**, ICAR-CIPHET (2015) reported an overall loss of 4.58 percent for tapioca, while the present study reported post-harvest losses (4.87%). The difference was insignificant at 5 percent level of significance.

The survey for estimating the post-harvest losses of ***bottle gourd*** was conducted in 7 districts in 5 states covering 3 agro-climatic zones (ACZ4, ACZ5, and ACZ7). The overall post-harvest losses at district level were the highest in Sundargarh district (5.44%) and the lowest in Agra district (6.17%). Improper harvest and post-harvest handling practices such as damage to bottle gourd due to sickle/knife during harvesting and piling of bottle gourd during collection by farmers in the districts of Jabalpur and Sundargarh resulted in high post-harvest losses. The overall post-harvest losses at state level were the highest in Odisha (7.57%), and the lowest in Uttar Pradesh (6.43%). The overall post-harvest losses in bottle gourd were the highest in ACZ4 (7.35%), and the lowest in ACZ5 (6.43%). Proper harvest and post-harvest handling practices followed in bottle gourd by farmers resulted in the lowest post-harvest losses in ACZ5 (Uttar Pradesh). Post-harvest losses in farm operations at national level were found to be 4.69 percent, while for market level, the post-harvest losses were 2.32 percent. The overall total post-harvest losses of bottle gourd were 7.01 percent. The highest post-harvest losses were observed at harvesting operation (1.95%) in farm operations and at retailer level (0.82%) in market operations. Among different operations, harvesting and sorting and grading majorly contributed to the losses in bottle gourd in farm level operations. At the market level, post-harvest losses were mainly observed due to handling of bottle gourd at ambient conditions and unavailability of storage systems in the value chain.

Fig. 6.5(a) Comparison of post-harvest losses of vegetables

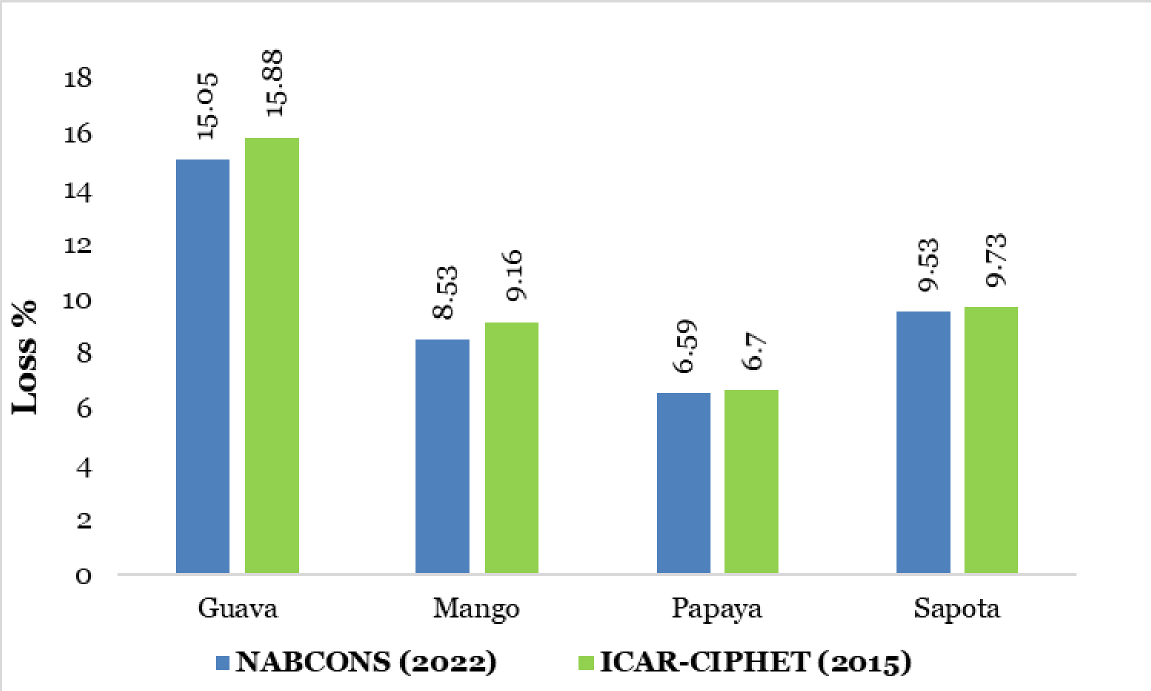
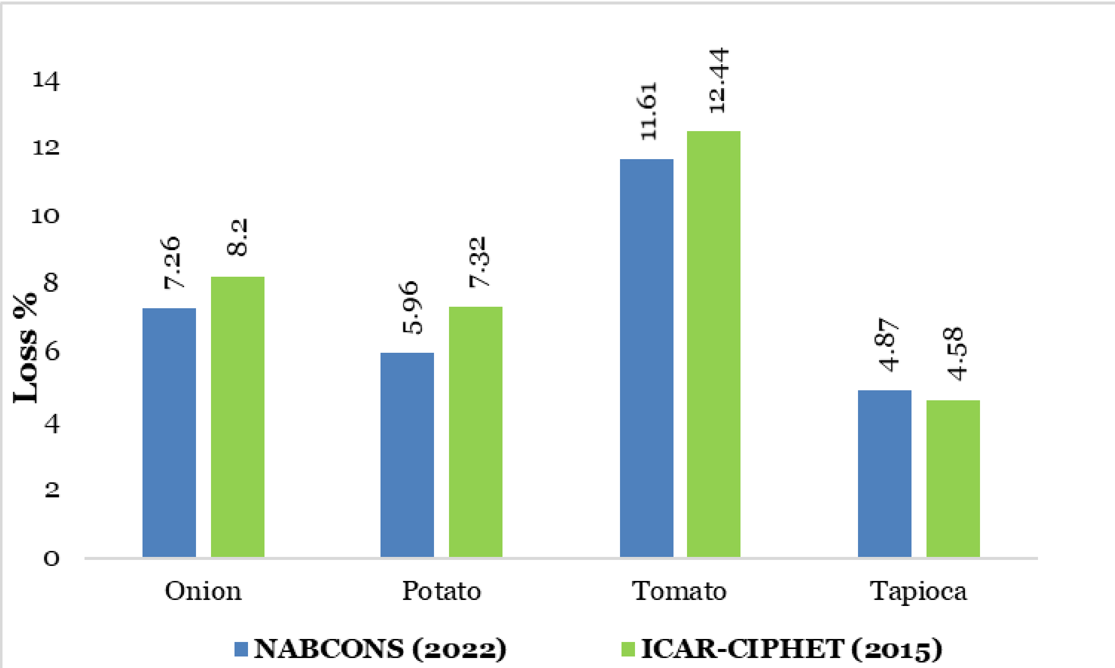


Fig. 6.5(b) Comparison of post-harvest losses of vegetables



The survey for estimating the post-harvest losses of ***brinjal*** was conducted in 7 districts in 4 states covering 4 agro-climatic zones (ACZ3, ACZ4, ACZ7, and ACZ13). The overall post-harvest losses at district level were the highest in Murshidabad district (8.32%) and the lowest in Anand district (6.88%). Better harvest and post-harvest handling practices followed by farmers in the districts of Anand and Vadodara contributed to lower post-harvest losses. The overall post-harvest losses at state level were the highest in West Bengal (7.85%), and the lowest in Gujarat (6.99%) in brinjal. The overall post-harvest losses were the highest in ACZ3 (7.89%), and the lowest in ACZ13 (6.95%) in brinjal. Post-harvest losses in farm operations at national level were found to be 4.75 percent in brinjal, while for market level, the post-harvest losses were 2.66 percent. The overall total post-harvest losses of brinjal were 7.41 percent. The highest post-harvest losses were observed at harvesting operation (2.25%) in farm operations and at retailer (1.19%) level in market operations. Among different operations, harvesting and sorting and grading majorly contributed to the losses in farm level operations. At the market level, post-harvest losses were mainly observed due to handling of brinjal at ambient conditions and unavailability of storage systems in the value chain.

The survey for estimating the post-harvest losses of ***beans*** was conducted in 6 districts in 4 states covering 4 agro-climatic zones (ACZ3, ACZ10, ACZ11, and ACZ13). The overall post-harvest losses at district level were the highest in Krishnagiri district (8.27%) and the lowest in Paraganas North district (5.48%). Use of family labour for harvesting and small farm size easy to manage in North Paraganas district helped in better harvest and post-harvest management. The overall post-harvest losses at state level were the highest in Tamil Nadu (7.82%), and the lowest in West Bengal (5.48%). The reason for high post-harvest losses of beans in Tamil Nadu and Karnataka was due to the improper harvesting method, which resulted in breakage of beans while harvesting. The overall post-harvest losses at ACZ level were the highest in ACZ10 (8.02%), and the lowest in ACZ3 (5.48%). Careful harvesting and handling of beans by farmers resulted in lower post-harvest losses in ACZ3 (West Bengal). Post-harvest losses in farm operations at national level were found to be 3.68 percent, while for market level, the post-harvest losses were 3.43 percent. The overall total post-harvest losses of beans were 7.11 percent. Among different operations, harvesting (1.11%) and sorting and grading (1.23%) majorly contributed to the losses in farm level operations. Improper harvesting methods resulting in splitting of beans was majorly observed during harvesting. Sudden decline in the market prices below the minimum level forced some farmers to delay the harvesting. This led to post-harvest losses as the tenderness of beans was lost. At all the market channels, beans were handled at ambient temperatures which resulted in higher post-harvest losses.

The survey for estimating the post-harvest losses of ***radish*** was conducted in 10 districts in 5 states covering 5 agro-climatic zones (ACZ2, ACZ3, ACZ4, ACZ6, and ACZ7). Sonipat district had the highest post-harvest losses (8.00%), while Kamrup district (4.97%) had the lowest post-harvest losses. This may be due to the reason that in districts such as Nalanda Vaishali and Kamrup, farmers dug out the radish carefully resulting in less harvesting losses, whereas split portion of radish were left in soil in various other districts such as Murshidabad, Nadia, Sonipat and Karnal. The highest overall post-harvest losses were observed in Haryana (7.51%), and the lowest in Assam (5.14%). The highest overall post-harvest losses were observed in ACZ6 (7.51%), and the lowest in ACZ2 (5.14%). The high losses in ACZ6, ACZ7 and ACZ3 could be attributed to improper harvesting methods resulting in the splitting of radish and thus an edible portion of radish remained in the field accounted as harvesting loss. Post-harvest losses in farm operations at national level observed were 3.96 percent, while for market level, the post-harvest losses were 2.50 percent. The overall total post-harvest losses of radish were 6.46 percent. Among different operations, harvesting (2.24%) and sorting and grading (0.93%) majorly contributed to the losses in farm level operations. Improper harvesting methods resulting in splitting of radish was majorly observed during harvesting. At the market level, losses were mainly observed due to spoilage as a result of delay in sales, unavailability of closed space for selling of radish at the wholesaler and retailer level and unavailability of storage system in the value chain.

The survey for estimating the post-harvest losses of ***capsicum*** was conducted in 6 districts in 4 states covering 4 agro-climatic zones (ACZ1, ACZ6, ACZ9, and ACZ10). The overall post-harvest losses at district level were the highest in Bangalore rural district (5.86%) and the lowest in Sangrur district (4.31%). Carelessness while harvesting caused cuts in attached capsicum making it prone to disease and pest attack, placing the harvested produce directly on the ground, packaging in gunny bags and stacking them improperly in the transport vehicles were the main reasons behind higher losses in Bangalore rural district. Lower losses in Sangrur district may be attributed to better post-harvest management practices at farm level. The overall post-harvest losses at state level were the highest in Karnataka (5.75%) and the lowest in Punjab (4.31%). The overall post-harvest losses at ACZ level were the highest in ACZ10 (5.75%) and the lowest in ACZ6 (4.31%). Poor handling of harvested produce by unskilled labours, poor packaging of produce before transport, lack of immediate demand in the market and poor storage facilities leading to moisture loss being the major cause behind the higher post-harvest losses in ACZ10. Post-harvest losses in farm operations at national level were found to be 2.69 percent while for market level, the post-harvest losses were 2.55 percent. The overall post-harvest losses for capsicum at the national level were observed as 5.15 percent. At farm level, post-harvest losses at harvesting (1.41%) and sorting-grading (0.55%), and at market level, post-harvest losses at wholesale level (1.05%) majorly contributed to overall post-harvest losses in capsicum. Relatively high amounts of water loss during storage and transport resulted in reduced turgor and firmness that lowered fruit quality, shelf life, market value and perceived freshness and increased post-harvest losses.

The survey for estimating the post-harvest losses of ***okra*** was conducted in 10 districts in 5 states covering 5 agro-climatic zones (ACZ3, ACZ4, ACZ7, ACZ11 and ACZ13). The overall post-harvest losses at district level were the highest in Nalanda district (6.47%) and the lowest in Surat district (5.48%). The reasons for lower losses in Surat district of Gujarat were mainly due to the fact that the okra was harvested carefully by the famers and the harvesting operation was conducted at the right stage avoiding losses due to harvesting of over ripe okra. The overall post-harvest losses at state level were the highest in Bihar (6.27%), and the lowest in Gujarat (5.79%). The overall post-harvest losses at ACZ level were the highest in ACZ4 (6.27%), and the lowest in ACZ13 (5.79%). The higher losses in ACZ4 and ACZ11 covering the states of Bihar and Odisha respectively, were due to improper harvesting and mishandling of okra during various operations leading to damage to okra. Post-harvest losses in farm operations at national level were found to be 3.76 percent, while for market level, the post-harvest losses were 2.25 percent. The overall total losses of okra were 6.01 percent. Among different operations at the farm level, harvesting (1.50%) and sorting and grading (1.16%) and at the market level, retailer (0.87%) and wholesaler (0.81%) were the major contributors to the losses. At the farm level, the losses were mainly due to harvesting of over ripe okra, poor harvesting and handling practices leading to damage to okra. The losses at the market level were because of unavailability of proper storage infrastructures and cold storage facilities leading to accelerated wilting and spoilage of okra. Also, the wholesaler and retailers continuously sprayed water on okra in order to maintain the freshness, however it led to increase in spoilage and fungal attack.

Among vegetables covered under the study, the estimated post-harvest losses in cabbage, cauliflower, mushroom, onion, potato and tomato were significantly lower than the earlier conducted study (ICAR-CIPHET 2015) which can be attributed to the overall improvements such as improved varieties, increased use of crates for handling & transport, and involvement of FPO and cooperatives in post-harvest handling. Out of around 4100 FPOs registered under the schemes promoted by NABARD, more than 1200 FPOs are working in fruits and vegetable sector.

Box 6.6 Good practices to be followed for vegetables

- *Use of improved varieties of tomato with better shelf-life developed by ICAR- Indian Institute of Horticulture Research, Bangalore.*
- *Use of mechanical digger/ harvesters in all root crops*
- *Use of equipment like nippers, pluckers, mechanical sorter-grader, and automatic packers at farm level.*
- *Collection of vegetables in plastic crates / bamboo baskets as is being done for vegetables in Himachal Pradesh*
- *Packaging vegetables in CFB cartons and plastic crates*
- *Crop specific packing materials e.g. packing of mushrooms in punnets*
- *Pre-cooling of vegetables in Zero Energy Cool Chambers (ZECC)*
- *Automated flow wrap packaging machines for french beans, okra, etc. which are being used by advanced countries like Japan, China etc.*

Box 6.6 Good practices to be followed for vegetables

- *Hot water rinsing (55°C for 12 seconds) followed by chemical washing and individual shrink wrap packaging in capsicum.*
- *Packaging of radish in a breathable poly bag with zipper locking system*
- *Use of cold chain (precooling, reefer transport, cold storage & air conditioned retail outlets) for domestic markets, as is being done by organized retail.*
- *Use of IQF for production of frozen peas, cauliflower, carrots etc.*
- *Use of irradiation in potato, onion and garlic*
- *Organized marketing of vegetables through FPOs/ FPC*
- *Minimum processing of vegetables (washing - peeling - trimming - curing) at retail level as is being followed in supermarkets.*
- *End-to-end agriculture supply chain of fresh agricultural produce supply chain through agri-startups.*

6.6 Plantation crops

Plantation crops are high value commercial crops with a total production of 15.85 million MT (DAC&FW, 2021-22) in the country. Sugarcane is included in the category of plantation crops. Though plantation crops are very hardy, there are still considerable post-harvest losses, leading to quantity and value loss. The extent and type of post-harvest losses occurring under the plantation sector is discussed in this section.

6.6.1 Farm level operations

Harvesting- Harvesting of areca nut, cashew nut and coconut was done manually by farmers or labourers. Tree climbers were used for harvesting areca nut and coconut in some regions. Cashew nuts were harvested by collection of dropped fruit from the ground. Plucking of cashew fruit by hand, when the plants were young was also observed during the study. The major causes of losses observed during harvesting include, spilling of nuts, breakage due to improper and rough handling of nuts by the labourers. Birds and animals also damaged considerable quantities of nuts. Harvesting of sugarcane was done manually by cutting the sugarcane from the bottom with the help of axes. The use of combine harvesters for sugarcane was not common. Harvesting operation was carried out by skilled male labourers in most parts of the country selected for study. Post-harvest stubble losses were observed at the time of harvesting due to improper harvesting performed by labourers. It was observed that sugarcane was not cut from the ground level and hence the sugar rich internodes were left over the field, which were considered as a post-harvest loss.

Cleaning- In this operation, areca nuts were separated from the branches. This was done on a flat surface in the threshing yard / processing shed and simultaneously in the garden, when the plucking process was going on. The post-harvest losses observed during this stage include improper and rough handling of nuts by the labourers resulting in wastage of nuts along with tree branches. The cleaning of the coconut was done by removing the husk. The operation usually referred to as dehusking was done manually with the help of an iron rod driven to the ground by skilled labour. Following dehusking, cleaning of coconut was carried out to remove the adhering dirt, if present on the surface of coconut. The main cause of loss at this level was due to breaking of nuts at the time of dehusking and due to improper handling. Cleaning of sugarcane was done in the field itself by skilled labourers to strip the green and dry leaves from the harvested sugarcane and detopping to remove immature top internodes. The operation was carried out with the help of a detrasher (a small hand tool). Post-harvest losses were reported at the time of cleaning and detopping, due to inappropriate height of sugarcane cutting.

Collection- Collection of plantation crops was done manually by farmers. It was done by using baskets or gunny bags and carrying them on head or by directly loading in the vehicles. The nuts were transported using a head load, bullock cart, power tiller and tractor. In case, the quantity was more in the range of 30-40 gunny bags tempos/ trucks were used for transporting nuts. In the collection operation cleaned and graded bundles of sugarcane were collected from the entire field at one place. Further, they were loaded to tractors/trucks and lifted to the collection centre/factory. Post-harvest losses observed during collection were mostly due to late lifting of canes by factory owners and long waiting time after harvesting. The extended waiting period also led to stalling loss during other in-field operations i.e. cleaning, bundling and loading to trucks. Weather conditions such as heavy sunshine and unusual rainfall also contributed to rotting and stalling losses of sugarcane in the fields, before and during collection.

Mechanical threshing of areca nut

Threshing/ dehusking- Threshing/dehusking of the areca nut was done manually as well as mechanically using mechanical threshers. The raw nuts were peeled in order to get the kernel. This was done within a day or two after harvesting. It involved heavy and intensive manual work and 7-8 people were engaged to accomplish this task. Areca nut dehusking was also carried out using dehusking machines, wherein 2-3 people were needed to work with the dehusking machines. During this operation, the hard shell of cashew nut was separated from the fruit. The process was mainly done by labourers, wherein shells and nuts were separated using a knife. The major reason for post-harvest loss during this operation was damage of nuts during mechanical threshing and mishandling losses during manual threshing.

Drying- The sun drying was done in case of areca nut, cashew nut and coconut. The areca nuts were uniformly spread on plastic sheets. The harvested nuts were sun dried for 45 days. During the drying process, the nuts were turned over periodically to facilitate uniform drying. Artificial drying using fabricated areca nut dryers were also used for drying of areca nut and coconut when adequate sunlight was not available. The major causes of losses recorded at the time of drying were improper handling, damages by rodents, unfavourable weather conditions, and insect infestation. At some places lack of appropriate storage facility, non-availability of fumigation facility, improper stacking practices and exposure to sunlight were also observed to cause the spoilage in coconut while drying operation.

Use of heated air dryer for drying of coconut

Sorting & grading - Sorting and grading of areca nuts was carried out by skilled graders manually to get higher market rates. After sorting & grading, nuts were packed in clean gunny bags holding about 60 to 70 kg of processed nuts. There was no post-harvest loss observed during this operation as sorting and grading was done only to get a better market price.

Packaging- Farm level packaging of cashew nut, coconut and areca nut was done to make it ready for transportation. The packaging was mainly done in gunny bags. The main reason for loss during packaging was due to improper packaging practices and improper stacking. It was also observed that many farmers were not packaging the fruit and were transporting it in loose form from the field. Improper handling was observed as a cause of loss at this level of operation.

Storage- The areca nuts, coconuts and cashew nuts were stored by the farmers at their home. Weather conditions, inadequate storage facilities, rotting and damage due to rodents were observed as major causes of losses during the storage.

Transport- The plantation crops were transported by tempos, trucks, etc. to various markets in the same district and different districts/ states. Losses were observed during the transportation owing to physical damage due to faulty packaging/ stacking, impact during the journey and spillage of the nuts. Breaking of nuts during loading and unloading as a result of improper and rough handling occurred in coconut. Physical loss due to rough handling during transportation, plucking by strangers and waiting time due to mismanagement of harvesting and transportation operation were the main reasons for post-harvest losses in sugarcane.

6.6.2 Market level operations

Wholesalers and Retailers - All operations in markets were carried out manually at both the wholesaler and retailer stages. In coconut and areca nuts, post-harvest losses that occurred at the wholesaler and retailer level were due to the removal of unfit nuts caused by improper handling. Water loss and sprouting due to disease infestation were also detected at this stage. In cashew nuts, there was spoilage due to high moisture content and improper drying during storage.

Storage - Storage of coconut, cashew nut and areca nut was done at normal warehouses. Rotting and fungal infection was observed in stored coconut due to humidity, lack of proper storage facilities, rodent attack and due to lack of quality control measures.

Transporters- In this operation, coconut, cashew nut and areca nut were transported through various routes i.e. within the same district, from one district to another and from one state to another state to processing units or other places. Nuts were packed in gunny bags, sacs, and boxes for transportation. Post-harvest losses at the transportation level were observed due to injury during loading and unloading or spillage resulting from improper handling. In the case of sugarcane, transportation from collection centres in the villages to sugar mills took place in trucks/ tractors / bullock carts. Huge stalling loss of sugarcane was reported during transportation to the mill from the field/collection centres. Delay in bringing the sugarcane to mills, waiting period at mill gates and poor road conditions were the main causes of stalling loss. Pulling sugarcane stubbles from trucks by strangers was also one of the causes identified for post-harvest losses.

Processing unit – The main cause of post-harvest loss during this operation was mainly due to spillage, breakage, mishandling during loading and unloading and improper storage measures at processing units. In sugarcane, losses were studied at the sugar mills from unloading after transportation to mills till crushing of sugarcane. Due to mismanagement of harvesting and milling schedules, it was observed that huge quantities of canes arrived at the mills at a particular point of time and hence disparagingly long waiting time was reported at mills before crushing. In common practice, crushing of the sugarcane should be carried out within 24 hours from the harvesting time. Due to glut conditions and long duration of waiting time, heavy stalling and weight loss was observed before crushing at mill level. In some places hi-tech sugar mills with all the advanced and fully automated processing lines were operating. Therefore, proper scheduling was also maintained with coordination of harvesting and milling. Negligible post-harvest losses were reported at such places in case of sugarcane.

Transportation of sugarcane

6.6.3 Post-harvest losses at farm and market level

Four plantation crops (areca nut, cashew nut, coconut and sugarcane) were taken for assessment of post-harvest losses in this study. The extent of post-harvest losses at district level, state level and ACZ level respectively, are presented in **Appendix 5**, **Appendix 6** and **Appendix 7** and national level are reported in **Table 6.6.**

The survey for estimating the post-harvest losses of ***areca nut*** was conducted in 8 districts in 3 states covering 3 agro-climatic zones (ACZ10, ACZ12 and ACZ15). The highest overall losses in areca nuts were found in Nicobar district (7.25%) and the lowest in Kasargod district (4.21%), which could be due to high moisture content and humid tropical climate leading to fast deterioration. The highest market level post-harvest losses at North Kannada district of Karnataka occurred due to improper storage and unfavourable weather conditions. At state level, the highest overall post-harvest losses in areca nut was observed in Nicobar district (6.44%) and the lowest in Kerala (4.01%). The main cause of the higher post-harvest losses in Andaman & Nicobar Islands was due to adverse weather conditions and non-availability of technology support for small scale farmers, while at market level post-harvest losses were high because of the long transportation gap of cargo ship docks, the frequency of which was three to four months. The procurement by CAMPCO and interventions of FPO/FPCs resulted in reduced post-harvest losses in Kerala. The highest overall post-harvest losses at ACZ level in areca nut were reported in ACZ15 (7.25%) and the lowest in ACZ10 (4.29%). At national level, the post-harvest losses in areca nut observed at farm level operations were 3.58 percent, whereas, for market level operations post-harvest losses were 0.83 percent. The overall post-harvest losses in the areca nut at the national level were reported as 4.41 percent. The highest post-harvest losses were reported at harvesting operation at farm level (1.18%) and the lowest (0.01%) in godown at market level. The major cause of post-harvest losses at harvesting and threshing was non-awareness about modern farm machinery among the farmers. Though adequate technologies were available for harvesting and processing of areca nuts, operations were being carried out manually in most of the districts surveyed. These manual operations led to the losses during harvesting and threshing of areca nuts. Scientific storages which can store areca nuts for long were not available in districts surveyed. The Andaman & Nicobar was the most neglected area and required attention. Soon after harvest, this crop was liable to accelerated physiological, chemical, and microbial processes due to high moisture content and humid tropical climate, which invariably led to deterioration and post-harvest loss. ICAR-CIPHET (2015) reported an overall loss of 4.91 percent for areca nut, while the present study reported significantly lower post-harvest losses (4.41%) at 5 percent level of significance **(Fig. 6.6)**. The reduction in post-harvest losses at harvesting and threshing level compared to the previous study was observed, which may be due to improvement in handling practices at farm level. Organised post-harvest management and marketing of nuts by formation of Farmer Producer Organisations (FPOs) and introduction of carbon fibre telescopic harvesting technology for harvesting the produce has positively contributed to reduction in post-harvest losses. The introduction of mechanical threshing and artificial drying using areca nut dryers have positively contributed to reduction in post-harvest losses. Use of tree climbers machine/bike not only ensured safety of pluckers, but also helped in reduction of losses.

The survey for estimating the post-harvest losses of ***cashew nuts*** was conducted in 5 districts in 4 states covering 3 agro-climatic zones (ACZ2, ACZ10 and ACZ12). The highest overall losses at district level in cashew nuts were estimated in Vishakhapatnam district (4.85%) and the lowest in Kannur district (3.17%). The lowest post-harvest losses in Kannur district of Kerala was because of pioneering work done by The Kerala Agriculture university and Cashew Research Station, Madakkathara, KAU, Thrissur, Kerala. Various trainings were conducted at CRS, Madakkathara on cashew apple utilisation including off campus training. The constant transfer of technology initiatives by the Madakkathara Centre has resulted in the establishment of several units by private entrepreneurs and Self Help Groups. The highest overall post-harvest losses at state level in cashew nuts were observed in Andhra Pradesh (4.82%) and the lowest in Kerala (3.07%). At ACZ level, the highest overall post-harvest losses in cashew nuts were observed in ACZ10 (4.82%) and the lowest in ACZ12 (3.12%). The highest post-harvest loss at ACZ10 was apparently due to lack of awareness among farmers about pre and post-harvest management practices. At the market level, post-harvest losses were mainly observed due to rough handling, rodent attack, spillage, spoilage due to weather conditions and impact during the transportation. The post-harvest losses observed at farm level operations in cashew nuts were 3.11 percent, whereas, for market level operations post-harvest losses were 0.61 percent. The overall post-harvest losses in the cashew nut at the national level were found as 3.72 percent. The highest post-harvest losses were found at harvesting operation (1.36%) followed by threshing operation (1.08%) at farm level and processing unit (0.26%) at market level. As depicted in **Fig. 6.6,** ICAR-CIPHET (2015) reported an overall loss of 4.17 percent for cashew nut, while the present study reported lower post-harvest losses (3.72%), however, the difference was insignificant at 5 percent level of significance. Reduction in post-harvest losses as compared to the previous study may be due to overall improvement in farm operations such as careful harvesting of cashew nuts.

The survey for estimating the post-harvest losses of ***coconut*** was conducted in 6 districts in 4 states covering 3 agro-climatic zones (ACZ10, ACZ12 and ACZ15). The highest overall post-harvest losses in coconut at district level were observed in East Godavari district (4.58%) and the lowest in Kozhikode district (2.86%). Coconuts were harvested manually by climbing trees in Nicobar and breakage of nuts occurred while dropping them down from the tree by the climber. Unexpected weather conditions, sudden rainfall and higher humidity in Nicobar led to spoilage of immature/partially mature nuts during the drying process. In Kozhikode district, harvesting was mostly carried out with the help of mechanical tree climbers, owing to reduced harvest losses. At market level assessment the highest market level post-harvest losses in East Godavari occurred due to improper storage conditions and unusual rainfall during storage. The highest post-harvest losses (4.32%) were observed in Andhra Pradesh and the lowest in Kerala (3.60%). The coconut farmers were not aware of the latest technologies on harvest and post-harvest management of coconut in the Union territory of Andaman and Nicobar Islands. There was a lack of proper storage and transportation facilities in UT. Hence, the highest post-harvest losses were observed in farm level and market level post-harvest management. The highest overall post-harvest losses were reported in ACZ15 (4.25%) and the lowest in ACZ12 (3.61%). This occurred due to the non-availability of a proper transportation facility and lack of connectivity between Port Blair and Nicobar. Lack of modern storage facilities was also the main cause identified during the survey. The use of mechanical harvesters for coconut and heated air dryers in ACZ 12 contributed towards efficient harvesting and collection of coconut. Post-harvest losses in farm operations at national level in coconut were estimated as 2.85 percent, while the post-harvest losses were 1.005 percent at market level. The overall total losses of coconut were 3.86 percent. The highest post-harvest losses during harvesting (0.90%) were reported at farm level operations, whereas the lowest post-harvest losses (0.13%) were reported at wholesaler level during market level assessment. ICAR-CIPHET (2015) reported an overall loss of 4.77 percent for coconut, while the present study reported lower post-harvest losses (3.86%), and reduction in the post-harvest losses was significant at 5 percent level of significance **(Fig. 6.6)**. This reduction in post-harvest losses of coconut may be attributed to the positive impact of different schemes being implemented by the Coconut Development Board, especially the scheme on “Mission for Integrated Development of Horticulture”. Under this scheme various services and training programmes to individuals, farmers’ collective groups and women groups were provided to create awareness about best post-harvest management practices of coconut. Marketing, market intelligence services, statistics & strengthening of export promotion council (EPC) helped coconut growers in providing access to international markets. Technology Mission on Coconut (TMOC) scheme was implemented for development, demonstration and adoption of technologies for post-harvest management of coconut resulting in reduced post-harvest losses at processing channels.

The survey for estimating the post-harvest losses of ***sugarcane*** was conducted in 13 districts in 8 states covering 7 agro-climatic zones (ACZ1, ACZ4, ACZ5, ACZ6, ACZ9, ACZ10 and ACZ11). The highest overall losses in sugarcane were reported in Cuddalore district (9.84%) and the lowest in Haridwar district (3.12%). The highest post-harvest losses in Cuddalore district were mainly due to inefficient manual harvesting by labourers and heavy sunshine causing moisture loss in sugarcane at farm level, while at market level post-harvest losses were due to waiting time for the trucks, which caused stalling loss in sugarcane after harvesting. Covering of harvested sugarcane with the help of straw and stripped leaves in Haridwar district were found helpful in reducing the staling loss in sugarcane. The highest overall post-harvest losses in sugarcane at state level were observed in Bihar (8.99%) and the lowest in Punjab (4.38%). The highest overall post-harvest losses at ACZ level in sugarcane were reported in ACZ4 (10.49%) and the lowest in ACZ10 (4.83%). The main cause of the higher post-harvest losses in

Bihar in sugarcane was due to delay in lifting of the harvested canes causing the stalling loss, whereas adverse weather conditions i.e. heavy sunshine and unconditional rainfall at some places were main reasons for post-harvest losses in Tamil Nadu. The immediate disposal of sugarcane from farms to mills in Punjab by better coordination between factories and farmers resulted in reduced post- harvest losses. During market level survey higher stalling losses were reported at sugar mills, when surveyed during unloading and before crushing attributed to extended waiting period for unloading and crushing due to low capacity of sugar mills. The post-harvest losses observed at farm level operations in sugarcane were 5.62 percent, whereas, for market level operations post-harvest losses were 1.70 percent. The overall post-harvest losses in the sugarcane at the national level were reported as 7.33 percent. The highest post-harvest losses were reported at harvesting operation at farm level (1.62%) and transport channel (1.01%) at market level. The major cause of post-harvest losses at harvesting and collection was non-awareness about modern farm machinery among the farmers. Manual operations led to the stubble losses. The post-harvest losses at transport and mill level occurred due to stalling of stubbles because of a long gap between the harvesting and crushing time resulting from a mismanaged harvesting schedule set by the mill owner for farmers. The sugarcane should be processed within 24 hours of harvesting, but it was not being followed at the factories, which caused the post-harvest losses in the sugarcane. Earlier, ICAR-CIPHET (2015) reported an overall loss of 7.89 percent for sugarcane, while the present study reported lower post-harvest losses (7.33%), however, the difference was non-significant at 5 percent level of significance as depicted in **Fig. 6.6**. Reduction in post-harvest losses at harvesting level compared to the previous study was observed, which may be due to adoption of mechanical detrasher-cum-harvester by farmers for harvesting, which made the operation less labour intensive and efficient.

Very significant advancement in the sugar industry was GPS based sugarcane crop area survey initiated by the Indian Sugar Mills Association (ISMA), which made planning and monitoring of harvesting along with all post-harvest operations effortless resulting in reduced post-harvest losses. Implementation of ICT tools like Sugarcane Information System (SIS) mobile application developed by Government of Uttar Pradesh also benefited the post-harvest activities like sale procurement and marketing of sugarcane easier. A high level of modernization and capacity expansion has taken place in the sugarcane processing industries, enabling these units to perform better, and resulted in reduced stalling losses due to higher waiting time. Sugarcane was brought directly to mills with minimum handling and excluding the other market level channels viz. wholesaler and retailer, which resulted in minimum post-harvest losses during transportation.

Box 6.7 Good practices to be followed for plantation crops

- *Use of tree climbers machine / scooters for harvesting*
- *Use of machines e.g. threshers in areca nuts, dehusking machines in coconut, cashew nut shelling machines etc.*
- *Use of electric dryers/ sundryer in coconut, cashew nuts, etc.*
- *System for bulk loading and unloading in the vehicles during transportation.*
- *Modernization of sugar mills and use of ICT tools for survey of sugarcane fields.*
- *Organized marketing of plantation crops through FPOs/ FPC*

Fig. 6.6 Comparison of post-harvest losses of plantation crops

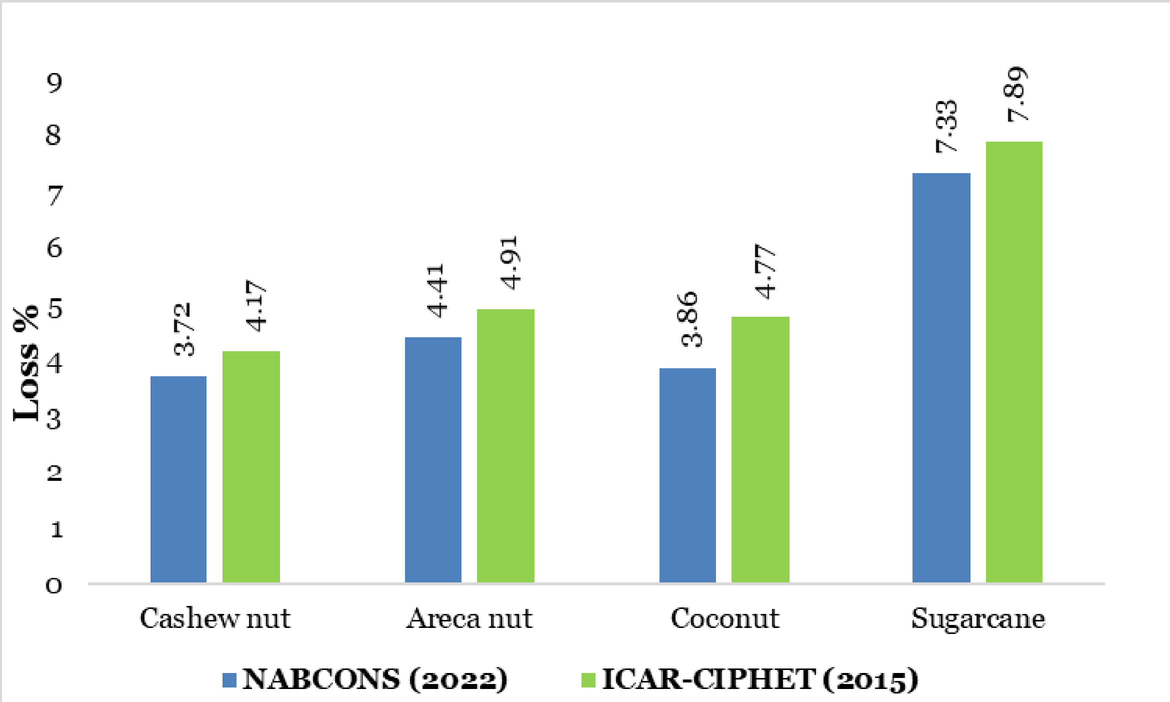


Table 6.6 Post-harvest losses of plantation crops and spices in percent at national level

Crop	Farm operations										Market operations						Overall total loss
	Harves-ting	Collec-tion	Sorting/ Grading	Thresh-ing	Winnow-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operat-ions	Godown	Whole-salers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	
Areca nut	1.18 ±0.0457	0.28 ±0.011	-	0.44 ±0.0173	0.91 ±0.039	0.14 ±0.0043	0.12 ±0.0046	0.25 ±0.0098	0.27 ±0.0101	3.58 ±0.1419	0.01 ±0.0037	0.4 ±0.121	0.03 ±0.0198	0.16 ±0.0429	0.23 ± 0.0469	0.83 ±0.1382	4.41 ±0.1981 (4.91)
Cashew nut	1.36 ±0.0329	0.35 ±0.0053	-	1.08 ±0.025	0.23 ±0.0054	0.03 ±0.0007	-	0.02 ±0.0005	0.04 ±0.001	3.11 ±0.042	0.00 ±0.000	0.14 ±0.03	0.02 ±0.0048	0.26 ±0.0359	0.19 ±0.0319	0.61 ±0.0568	3.72 ±0.0707 (4.17)
Coconut	0.9 ± 0.0163	0.32 ± 0.0059	-	-	0.34 ± 0.0063	0.34 ± 0.0055	0.21 ± 0.0041	0.55 ± 0.011	0.19 ± 0.0064	2.85 ± 0.0234	0.15 ± 0.0197	0.13± 0.0344	0.21± 0.0749	0.23 ± 0.0267	0.3± 0.0627	1.00 ± 0.1087	3.86 ± 0.1112 (4.77)
Sugarcane	1.62 ±0.0156	1.26 ±0.0133	1.44 ±0.015	-	-	-	-	-	1.31 ±0.0207	5.63 ±0.0328	-	-	-	0.7± 0.0814	1.01 ±0.1535	1.71 ±0.1737	7.33 ±0.1768 (7.89)
Black pepper	0.53 ± 0.0296	0.2 ± 0.012	-	0.22 ± 0.0094	0.03 ± 0.0022	0.07 ± 0.0033	0.02 ± 0.0016	0.01 ± 0.0012	0.02 ± 0.0012	1.11 ± 0.0337	0.01 ± 0.0037	0.07 ± 0.0067	0.07 ± 0.0088	0.02 ± 0.0035	0.01 ± 0.0029	0.19 ± 0.009	1.29 ± 0.0349 (1.18)
Chilli	1.07 ±0.0096	0.78 ±0.0081	2.14 ±0.023	-	-	0.08 ±0.001	0.03 ±0.0008	0.93 ±0.0123	0.17 ±0.0023	5.2 ±0.029	0.08 ±0.0197	0.35 ±0.0633	0.41 ±0.082	-	0.06 ±0.0375	0.91 ±0.1119	6.11 ±0.1255 (6.51)
Coriander	1.97 ± 0.0429	0.69 ± 0.0163	-	1.15 ± 0.0325	0.53 ± 0.0163	0.06 ± 0.0016	0.13 ± 0.0037	0.06 ± 0.002	0.12 ± 0.0029	4.72 ± 0.0588	0 ± 0.00	0.16 ± 0.0147	0.1 ± 0.0043	0.00 ± 0.00	0.33 ± 0.037	0.6 ± 0.0398	5.32 ± 0.071 (5.87)
Turmeric	1.74 ±0.0683	0.26 ±0.0002	0.48 ±0.001	-	0.27 ±0.0196	0.47 ±0.057	0.17 ±0.0001	0.06 ±0.0003	0.33 ±0.0001	3.78 ±0.089	0.49 ±0.1406	1.09 ±0.4746	0.64 ±0.3438	-	0.89 ±0.6736	1.57 ±0.7307	5.36 ±0.7361 (4.44)

Note: *Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*
Figures in parentheses represent the losses of previous study by ICAR-CIPHET (2015)

6.7 Spices

India is known for its spices in the world and is one of the largest consumers and exporters of spices. Currently, India produces 10.82 million MT of spices (DAC&FW, 2021-22), but a considerable quantity of spices go waste due to lack of adequate Post-harvest management infrastructure for these crops. This section deals in the type and extent of post-harvest losses in spices in India.

6.7.1 Farm level operations

Harvesting- Harvesting of spices viz. black paper, coriander, chilli and turmeric was done manually by the farmers, either by simply plucking by hands or using sickle. Black pepper was also harvested by climbing trees with vines using ladders. The major causes of losses observed during harvesting of black pepper included improper harvesting, harvesting of over ripe berries, shattering or dropping of berries while harvesting and damage to berries (crushing/ bruising) due to rough handling by farmers. The post-harvest losses during harvesting of coriander were mainly due to scattering of seed, insect infestation, improper harvesting and spillage due to mishandling. Post-harvest losses observed during harvesting turmeric were small fingers of turmeric left underground and also physical damage to rhizomes.

Collection- Harvested spices were collected manually by farmers. The losses observed at this operation were low and were mainly due to improper handling, shattering of berries, damage by rain/bird and delay in collection

Threshing- Threshing operation of black pepper was done to separate the berries from the stalk. Majority of the farmers were using mechanical threshers for threshing of coriander and black paper and a few farmers were still doing manual threshing. During this operation, the losses observed were mainly due to spilling of berries and coriander seeds, physical damage and splitting.

Threshing of black pepper

Sorting and Grading- This operation was one of the most important operations after harvesting of turmeric and chilly and was done manually on the field. During this operation, rotten and mouldy chillies and damaged rhizomes, which were unfit for consumption, were removed.

Sorting grading of chilli

Winnowing/ cleaning- The threshed black pepper berries and coriander were cleaned by winnowing or picking of extraneous matter. The operation was done manually by the farmers. The losses during the operation were mainly due to spilling and physical damage to berries and swaying away of coriander seed. Usually the undamaged spilled berries were collected back by the farmers.

Drying- The spices were dried after threshing and cleaning. The black pepper, coriander seeds, turmeric and chillies were sun dried by spreading on the floor. During drying, the losses in black paper and coriander were observed mainly due to spoilage and rotting, physical damage and damage due to bad weather like unpredicted rain and hailstorms.

Packaging- spices were packaged in gunny bags, jute bags and plastic bags. The losses during the operation and were mainly due to spilling or physical damage due to manual handling.

Transport- The transportation was done through various types of fleet such as tempo, jeep and trucks to various destinations such as markets in the same district or different districts/states. The post-harvest losses during transportation observed were due to spilling as a result of defective packaging.

Storage- Storage of spices was done at farmer level for a period ranging from few days to few months before the sale to traders. Black pepper was stored after proper packaging in gunny bags/ plastic bags. The losses observed in black pepper at farmer level were mainly due to spilling of berries as a result of improper packaging, improper stacking, crushing & physical damage. Rotting of chilli and rhizomes in turmeric was observed during storage.

6.7.2 Market level operations

Wholesalers and retailers- The operations at the wholesalers and retailers level were done manually. The post-harvest losses were found to be very less and were mainly due to spilling and physical damage/ crushing of spices (chilli, black pepper, turmeric and coriander) due to improper stacking and storage.

Transporters- Transportation of black pepper was done using different types of fleet such as tempo and trucks. Losses observed during the transportation were very less as the spices were properly packed before loading. The minute losses were mainly due to spilling and physical damage/ crushing of seeds due to improper stacking.

Storage- Black pepper and chilli were stored in godowns/warehouses without any temperature controlled conditions. Coriander and turmeric were stored in cold stores as well. The losses observed during storage of spices were mainly due to physical damage/ crushing of black pepper due to improper stacking and spilling.

Processing- Post-harvest losses at processing units were estimated for black pepper and coriander. The losses observed at the processing units were very low and were mainly due to spilling and physical damage/ crushing of black pepper and coriander seeds as a result of mishandling.

6.7.3 Post-harvest losses at farm and market level

Survey for assessment of harvest and post-harvest losses was conducted for 4 spices (black pepper, chilli, coriander and turmeric). The extent of post-harvest losses at district level, state level and ACZ level respectively, are presented in **Appendix 5, Appendix 6** and **Appendix 7** and national level are reported in **Table 6.6.**

The survey for estimating the post-harvest losses of ***black pepper*** was conducted in 6 districts in 2 states covering 2 agro-climatic zones (ACZ10 and ACZ12). The highest overall post-harvest losses at district level in black pepper were observed in South Kannada district (1.3%) and the lowest in Chikmangalur district (1.26%). The lower losses in Chikmangalur and Idukki were due to careful harvesting of black pepper spikes from the vines with nipper by many farmers. However, most of the farmers in all other districts manually plucked spikes of pepper leading to losses. The variation in post-harvest losses amongst different districts was not high due to the fact that crop was handled very carefully owing to its importance as a high value crop. The highest overall post-harvest losses at state level in black pepper were observed in Karnataka (1.31%) followed by Kerala (1.26%). At ACZ level, the overall post-harvest losses were observed almost similar in ACZ12 (1.3%) and ACZ10 (1.29%) in black pepper. At national level, the post-harvest losses in black pepper observed at farm level operations were 1.11 percent, whereas, for market level operations post-harvest losses were 0.19 percent with overall post-harvest losses of 1.29 percent. Among different operations at farm level, the highest post-harvest losses were observed at harvesting (0.53%) followed by threshing (0.22%), collection (0.2%) and drying (0.07%) and the lowest were observed at storage (0.01%). During harvesting and threshing, the losses were mainly due to spilling and physical damage/ crushing. At the farmer level, lower losses during storage were also due to the fact that black pepper was not stored for a longer period of time and was sold to the traders to fetch immediate cash and also due to the fact that many farmers did not have the facilities required to store their harvested pepper. The major causes of losses in black pepper at the market level were due to spilling and physical damages or crushing of berries occurring during loading and unloading.

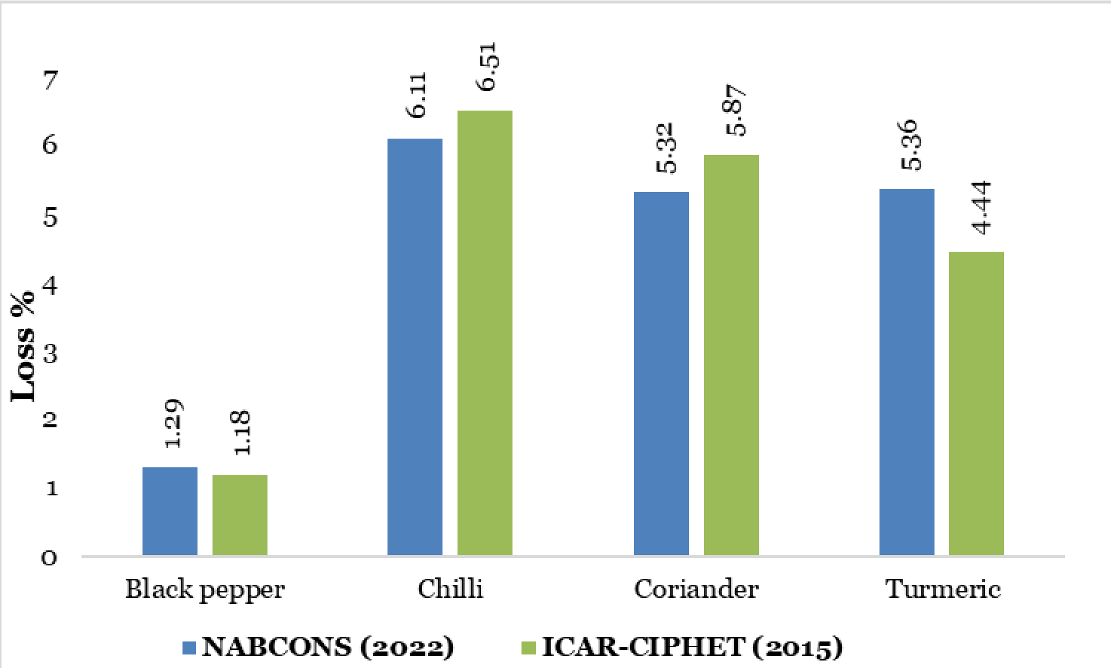
The overall losses for black pepper reported in ICAR-CIPHET (2015) were 1.18 percent. The present study estimated slightly higher overall post-harvest losses (1.29%) in comparison to previous study conducted by ICAR-CIPHET (2015), however, the difference was non-significant at 5 percent level of significance **(Fig. 6.7)**. The slight non-significant increase in black pepper post-harvest losses were mainly due to slightly increased losses during harvesting and drying operations. This could be attributed to the seasonal variations affecting sun drying operation in black pepper leading to losses.

The survey for estimating the post-harvest losses of ***chilli*** was conducted in 5 districts in 3 states covering 2 agro-climatic zones (ACZ9 and ACZ10). Overall the highest post-harvest losses for chilli were observed in Barwani district (7.43%) and the lowest in Warangal Rural district (5.92%). The highest overall post-harvest losses for chilli at state level were observed in Andhra Pradesh (7.06%) and the lowest in Telangana (6.06%). At ACZ level, the highest overall post-harvest losses for chill were observed in ACZ10 (6.15%) followed by ACZ9 (6.09%). Post-harvest losses in farm operations at national level for chilli were found to be 5.2 percent. Among different operations, sorting & grading (2.14%) followed by harvesting (1.07%) and storage (0.93%) majorly contributed to the losses in farm level operations in chilli. At market level, the highest post-harvest losses were reported at retailer (0.41%) level followed by wholesaler (0.35%) and at storage level (0.08%) and overall post-harvest losses at market were 0.91 percent. It can be observed that the overall losses at national level were 6.87 percent. The losses estimated were slightly higher than the losses reported in the earlier study (6.51%) by ICAR-CIPHET (2015). There was no significant difference between the estimates of post-harvest losses obtained from the present study and the earlier conducted study by ICAR-CIPHET (2015) at 5 percent level of significance as depicted in **Fig. 6.6**.

The survey for estimating the post-harvest losses of ***coriander*** was conducted in 5 districts in 2 states covering 3 agro-climatic zones (ACZ7, ACZ8 and ACZ9). The overall post-harvest losses in coriander at district level were the highest in Rajgarh district (6.04%) and the lowest in Guna district (5.5%). The reason for high post-harvest losses in coriander in Rajgarh district were due to high losses observed during harvesting and threshing operations as a result of seed shattering and spillage of coriander seed. The overall post-harvest losses at state level in coriander were the highest in Madhya Pradesh (6.01%) and the lowest was reported in Rajasthan (5.7%). The overall post-harvest losses at ACZ level in coriander were the highest in ACZ9 (7.60%) and the lowest in ACZ13 (4.87%). Improper harvesting method and mishandling resulted in high losses in ACZ9 (Maharashtra). Post-harvest losses in farm operations at national level in coriander were found to be 4.72 percent, while for market level, the post-harvest losses were 0.6 percent. The overall total losses of coriander were 5.32 percent. Among different operations, harvesting (1.44%) and threshing (1.17%) majorly contributed to the post-harvest losses in farm level operations of coriander and at wholesaler (0.24%) and retailer (0.23%) level in market operations. Spillage during loading and unloading was observed during the transportation and improper storage conditions and mishandling resulted in losses at various market stakeholders.

Earlier, ICAR-CIPHET (2015) reported an overall loss of 5.87 percent for coriander, while the present study reported lower post-harvest losses (5.32%), which were significantly lower at 5 percent level of significance. This decrease in overall post-harvest losses may be attributed to release of improved varieties of coriander for cultivation, improved crop protection and production technologies and training conducted on post-harvest management practices.

Fig. 6.7 Comparison of post-harvest losses of spices



The survey for estimating the post-harvest losses of ***turmeric*** was conducted in 5 districts in 2 states covering 2 agro-climatic zones (ACZ2 and ACZ10). The overall highest losses at district level in turmeric were observed in West Jaintia Hills district (5.48%) and the lowest was reported in Chamarajanagar district (3.76%). The highest overall post-harvest losses at state level in turmeric were observed in Meghalaya (5.64%) followed by Andhra Pradesh (4.97%) and the lowest was observed from Karnataka (4.63%). At ACZ level, the highest overall post-harvest losses in turmeric were reported in ACZ2 (5.84%) and the lowest was observed in ACZ10 (4.88%). At national level, the post-harvest losses observed at farm level operations were 3.78 percent, whereas, for market level operations post-harvest losses were 1.57 percent. The overall post-harvest losses in turmeric at the national level were reported as 5.36 percent. The highest post-harvest losses were reported at harvesting operation at farm level (1.74%) followed by sorting & grading (0.48%). At market level the highest post-harvest losses were observed at wholesale level (1.09%) followed by transport (0.89%). The major cause of post-harvest losses at harvesting was due to the small fingers left under soil. It has been observed that the overall post-harvest losses at national level were

5.36 percent. The post-harvest losses estimated were slightly higher than the post-harvest losses reported in the earlier study (4.44%) by ICAR-CIPHET (2015), however, the difference was non-significant at 5 percent level of significance as depicted in **Fig. 6.7**.

Among spices covered under the study, the estimated post-harvest losses in coriander were significantly lower than the earlier conducted study, which can be attributed to the overall improvements done during the recent years. The results of other spices were in conformity with the earlier study by ICAR-CIPHET 2015.

Box 6.8 Good practices to be followed for spices

- *Mechanical harvesters and use of plucking tools and implements e.g. plucking tools developed at National Innovation Foundation.*
- *Pre-treatment such as blanching before drying followed by use of solar dryers for drying of black pepper.*
- *Irradiation of spices to increase shelf life*
- *Cold storage of spices, as is being done for chilli at Guntur, coriander and other spices at Kundli, Haryana etc.*
- *Production of spices in organic way of cultivation*
- *Branding and packaging of spices for marketing.*
- *Organized marketing of spices through FPOs/ FPC.*

6.8 Eggs

Eggs are considered as functional food that provides health benefits along with their basic nutrient content. Egg with a production of 122.05 billion numbers (DAHD, 2021-22) could meet the nutritional requirement of households but considerable post-harvest losses in India result in quantity and value loss. The extent and type of losses occurring in eggs are discussed in this section.

6.8.1 Farm level operations

Handling- Handling and collection of eggs was done manually by the majority of the farmers. In the sample surveyed, it was observed that some of the bigger poultry farms (commercial farms) were having automatic eggs collection facilities. The losses ranged from the eggs being cracked, leaked, having holes in their shells or being smashed. On discussion with poultry farmers, it was observed that in most cases, the retailer/ wholesaler came directly to the farm and collected all eggs they have, which resulted in lower losses at the farm gate level.

Sorting/ grading- Some farmers sorted/ graded their eggs to fetch a better price for their produce. Except in some big commercial farms, sorting/ grading was done manually. Only the eggs that were smashed, leaked or cracked were considered as loss. Only losses were when the eggs were discarded and were not sold further.

Packaging- The eggs were packaged in a variety of materials such as bamboo baskets, plastic crates/boxes/bags and cardboard/plastic eggs trays. Post-harvest losses included eggs getting cracked, smashed, leaked either due to holes in the shell, having thin shells or by mishandling of the eggs during packaging. When the packaging was done by automated machines, the losses observed were low in comparison to the packaging done manually.

Storage- The farmers stored the eggs at normal temperature in godowns/ at home. The eggs were stored for a short duration of time ranging 1-2 days before they were picked up by the aggregator/ retailer/ wholesaler. Few eggs ended up cracked or leaked during transfer from the godown. In a few cases, eggs ended up getting spoiled when in storage for a longer time as there was no provision for temperature and humidity control.

Commercial poultry farms

Transportation- Due to the delicate nature of eggs, when transported from one place to another, losses were observed. The eggs were transported over short distances by the farmer and often resulted in leaked and smashed eggs. Improper stacking of eggs trays as well as inadequate packaging were observed as the main reason for losses. For the eggs that were being sold at the farm gate, the retailers/ wholesalers were responsible for the transportation thus no losses were observed in the case of farmers.

6.8.2 Market level operations

Wholesalers and retailers- The losses during cleaning, sorting and grading, marketing and storage were observed. Losses occurred mainly due to improper handling, which resulted in the eggs getting smashed and leaked. Certain eggs that were stored for longer times ended up getting spoilt as little to no cold stores were available in the market. There was lack of proper knowledge amongst the stakeholders regarding cold stores, which resulted in poor handling and ultimately led to losses.

Egg storage at wholesaler level

Transporters- Eggs were transported in tempos and trucks by stacking trays. Mostly the trays were made of cardboard or plastic. In most cases, the eggs were transported to nearby areas. The far-off distances were covered by trucks and the losses in such cases were either due to bad roads, which caused breakage or due to the improper stacking done by the transporter. Very small losses due to breakage while loading and unloading were observed when packed properly. The major reason for losses observed however was use of improper infrastructure (trucks and roads) which resulted in higher losses.

Storage- Out of the sample covered, only a few districts were having cold storage facilities (maintained temperature and humidity). The temperature was not regulated but ranged between 2-5 °C. In most of the districts, eggs were kept in godowns with no proper ventilation. Major losses included spillages and breakages of eggs during transfers. Some places had poor ventilation that resulted in spoilage of the eggs. The reasons ranged from low demand for the commodity or the lack of a nearby market which resulted in the eggs being stored for longer duration. During discussions it was also observed that the eggs which were cracked or had small holes were sold to nearby bakeries.

6.8.3 Post-harvest losses at farm and market level

The survey for estimating the post-harvest losses of ***eggs*** was conducted in 19 districts in 14 states covering 10 agro-climatic zones (ACZ1, ACZ2, ACZ3, ACZ5, ACZ6, ACZ7, ACZ9, ACZ10, ACZ11 and ACZ12). The overall post-harvest losses at district level in eggs were the highest in West Bardhaman district (7.22%) and the lowest in Pune district (5.04%). Higher post-harvest losses in West Bardhaman district may be attributed to poor handling of eggs, improper packaging, cracking and smashing of eggs during loading and unloading, improper stacking of eggs during transport, long travel distance due to lack of immediate demand and poor availability of nearby markets. The overall post-harvest losses at state level in eggs were the highest in West Bengal (7.07%) and the lowest in Telangana (5.52%). The overall post-harvest losses at ACZ level in eggs were the highest in ACZ3 (7.07%) and the lowest in ACZ11 (5.11%). The reasons behind higher losses in West Bengal state of ACZ3 were lack of automatic eggs collection and packaging facility, mechanical sorter and grader and uncovered space for carrying out sorting and grading, which led to more cracking and leakage of eggs. Farmers in ACZ11 properly handled the eggs and were found using mechanical sorters and graders, advanced packaging material like plastic cartons and bubble earth wraps which may have contributed to lower losses. Post-harvest losses in farm operations at national level in eggs were found to be 4.40 percent while for market level, the post-harvest losses were 1.63 percent. The overall total losses of eggs were 6.03 percent. Among different operations, collection (1.52%) and sorting and grading (1.23%) majorly contributed to the losses in farm level operations and wholesaler level (0.76%) in market operations. Physical damage (cracking, smashing and leakage) due to mis-handling of eggs was the major cause behind the higher post-harvest losses at each level.

Earlier, CIPHET (2015) reported an overall loss of 7.19 percent for eggs while the present study reported significantly lower post-harvest losses (6.03%) at 5 percent level of significance as depicted in **Fig. 6.8**. This decrease in overall post-harvest losses may be due to following reasons:

- Significant shift of backyard poultry farms to organised poultry farms under Rural Backyard Poultry Development (RBPD) component of National Livestock Mission (NLM) (80% organised poultry farms and 20% backyard farms).
- Increase in number of eggs processing units (Processing capacity: 12,000 MT annually in 2015 and 15,500 MT annually in 2022).
- Increase in exports of eggs products: ₹474 crores in 2015, ₹1,500 crores in 2022)

Box 6.9 Good practices to be followed for eggs

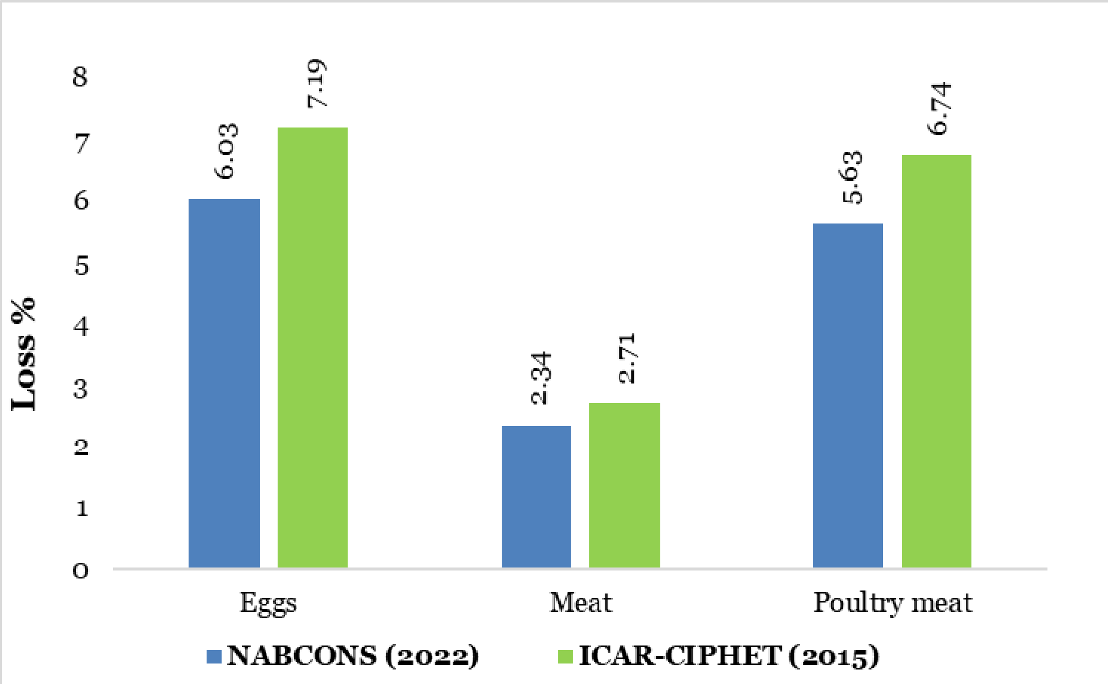
- *Use of mechanical sorters and graders.*
- *Use of automated packing machines.*
- *Cage free and ventilated barns for birds along with vaccination of birds against diseases like bird flu.*
- *Use of plastic cartons, bubble earth wraps and wooden eggs trays for packaging.*
- *Processing of eggs into dehydrated/frozen products, for export purpose and use in bakeries and other food and non-food industries.*

Table 6.7 Post-harvest losses of livestock produce in percent at national level

	Farm Operations								Market operation						
Crop	**Harvest-ing**	**Collection**	**Sorting/ grading**	**Drying**	**Packaging**	**Farm level storage**	**Transport**	**Total loss in farm operations**	**Godown**	**Wholesalers**	**Retailers**	**Processing unit**	**Transport**	**Total loss in storage**	**Overall total loss**
Egg	-	1.52 ± 0.0504	1.23 ± 0.0786	-	1.10 ± 0.0423	0.24 ± 0.0058	0.30 ± 0.0152	4.40 ± 0.1038	0.05 ± 0.0049	0.39 ± 0	0.44 ± 0.0093	-	0.39 ± 0	1.63 ± 0.0545	6.03 ± 0.1173 (7.19)
Inland fish	1.63 ± 0.0118	0.37 ± 0.002	1.42 ± 0.0073	-	0.16 ± 0.0008	0.08 ± 0.001	0.14 ± 0.0008	3.8 ± 0.0139	-	0.22 ± 0.011	0.7 ± 0.0592	-	0.14 ± 0.0159	1.06 ± 0.0612	4.86 ± 0.0627 (5.23)
Marine fish	5.49 ± 0.2142	0.56 ± 0.0163	0.38 ± 0.0088	0.14 ± 0.0057	0.01 ± 0.0004	-	0.42 ± 0.0139	7 ± 0.2156	0.19 ± 0.0335	0.57 ± 0.08	0.21 ± 0.0255	0.27 ± 0.0388	0.52 ± 0.06	1.76 ± 0.0845	8.76 ± 0.2315 (10.52)
Meat	1.43 ± 0.0479	-	-	-	-	-	-	1.43 ± 0.0479	-	0.38 ± 0	0.25 ± 0	0.24 ± 0	0.02 ± 0	0.91 ± 0	2.34 ± 0.0479 (2.71)
Poultry meat	2.51 ± 0.0155	-	-	-	-	-	-	2.51 ± 0.0155	-	1.83 ± 0	0.92 ± 0	0.36 ± 0	0.02 ± 0	3.12 ± 0	5.63 ± 0.0155 (6.74)
Milk	0.25 ± 0.0078	0.08 ± 0.0046	-	-	-	-	0.21 ± 0.0153	0.54 ± 0.0178	-	-	0.05 ± 0.0088	0.17 ± 0.0275	0.12 ± 0.0229	0.33 ± 0.0368	0.87 ± 0.0409 (0.92)

Note: Figures in the table represent "Loss ± Standard Error (SE) ×1.96"
Figures in parentheses represent the losses of previous study by ICAR-CIPHET (2015)

Fig. 6.8 Comparison of post-harvest losses of eggs and meat



6.9 Fish

Fisheries is a major sector in India, which provides employment to millions of people, especially the marginalised and vulnerable communities, and contributes to the food security of the country. Being one of the major items of agri export, this sector also earns foreign exchange for the country. The fish production in India reached an all-time high of 14.16 million metric tons during 2019-20 (Handbook of Fisheries Statistics, FAH&D, 2020), but a considerable quantity goes waste due to lack of adequate post-harvest management infrastructure. This section deals in the type and extent of losses in fisheries in India.

6.9.1 Farm level operations

Harvesting- Harvesting or capturing of marine fish was done in deep sea and fishermen took the boats or ships to sea with ice and ice boxes on board. The average fishing trip took about two to ten days. Both mechanised and non- mechanised crafts were used for harvesting or catching of marine fish. The non-mechanised crafts were used to operate within 5 km of the coast and mechanised ones went farther in the sea. Also, different types of fishing gear were used by fishermen, such as gillnets, boatswains, drag nets, cast nets and traps, etc. The fishermen usually used ice or sometimes salt to preserve the catch on board. The losses during harvesting or catching of marine fish included damage to fish during unloading from the net and dumping on the deck, delayed hauling of nets or gears for a long period of time resulting in damage to caught fish and use of damaged and unhygienic craft and gears.

Harvesting of inland fish covered capture of fish in natural water bodies such as rivers, lakes and large reservoirs including both freshwater and brackish water, and culture fisheries (aquaculture) raised in tanks or ponds, etc. In case of capture fisheries, wild untended fish were harvested while in the culture fishery (aquaculture), the production of the fish was based on control over water quality and other variables. The harvesting of the inland fish was done by using various types of nets, crafts and gears. The various types of losses occurring at harvesting of inland fish included delayed hauling of nets or fishing craft or gears for a long period of time in the water bodies resulting in spoilage of caught fish, use of damaged and unhygienic craft and gears and physical damage while to fish during unloading the fish from the net. Also, losses due to death of fish as a result of poor quality of water in case of culture fish contributed to the losses.

Harvesting of inland fishes

Collection- Harvested fish were unloaded from the net and were collected at one place. The collected fish were collected in baskets or crates with ice. The unloading of inland fish from the nets took place at the landing centres or on the ground near the water bodies, where the collected fish were transferred from the net to the vessel or collected for sorting and grading or for further packaging. The post-harvest losses during unloading and collection included losses due to falling from nets or baskets, physical damage while handling and transferring to the vessels or crates. Losses due to exposure to high temperatures while holding on boards and during unloading and delayed collection also contributed to the losses and spoilage of fish.

Sorting and grading- Collected marine and inland fish were sorted and graded based on the fish varieties and size. The damaged or spoiled fish were removed during this operation. Also, the unwanted fish known as bycatch were also removed from the collected fish. These removed fish contributed to the post-harvest losses during sorting and grading. The damage due to manual handling during sorting and grading also added to the post-harvest losses.

Sorting & grading of fishes on boat and landing centre

Drying- Drying of marine fish was commonly practised by many of the fishermen, wherein various varieties of marine fish were sun dried by spreading the fish on surface and dried for 3-4 days. The various types of post-harvest losses during drying of marine fish include losses due to spoilage of fish due to improper drying and poor weather conditions, damage by birds, insects and rodents, damage to fish during drying rotation. Also quality deterioration was common in case of improper drying and unhygienic handling practices during drying.

Drying of marine fishes

Packaging- Sorted and graded fish were packaged in various types of containers such as ice boxes, plastic crates with or without ice, baskets, utensils or vessels and small plastic bags. The post-harvest losses during packaging operation mainly were due to physical damage to fish due to improper handling, spoilage in fish due to delayed packaging of the fish and improper packaging materials.

Packaging of marine fishes

Storage- Packaged inland fish in ice boxes or crates with ice were stored for very short periods of time at the fishermen level extending up to only for a few hours. Due to lack of cold storage facilities at fishermen level, the fishermen could not store the fish for long. The post-harvest losses were mainly due to spoilage of fish due to improper storage conditions, biochemical deterioration of the fish due to temperature fluctuations and unavailability of ice at the fishermen level. Storage under unhygienic conditions lead to contamination of fish resulting in rapid spoilage at the fishermen level due to lack of required infrastructure.

Transportation- Marine fish packaged in ice boxes or crates were transported in tempo, trucks, cycle, refrigerated vans etc. The losses in refrigerated vans were low due to low temperature control during the transit reducing the decay rate. During transportation without temperature control, the post-harvest losses mainly were due to spoilage due to exposure to adverse temperatures, physical damage due to impact in transit due to improper packaging or ruptured package and improper handling practices during loading and unloading from the fleet.

6.9.2 Market level operations

Wholesalers and retailers- Marine and inland fish were handled manually at the wholesale and retail levels. The major losses at operations conducted by wholesalers and retailers were due to physical damage as a result of improper handling by the stakeholders and due to lack of proper storage facilities like cold storage or deep freezers. Further, sometimes fish supplied to the market at the wrong time also led to huge losses at the wholesale and retail level.

Fishes at retailer level

Transporters- Fish, both marine and inland, were transported after packaging in ice boxes or ice crates or vessels with water or plastic crates without any ice, through tempo, vans, rickshaw, auto, trucks, refrigerated vans or reefer vans. Reefer vans were mostly used for long distance transportation. Damage to fish due to impact in transit or due to improper packaging, spoilage of fish as a result of uncontrolled temperature in transit or due to vehicle breakdown were the major contributors to post-harvest losses during transportation. Also, mishandling of fish by labourers during loading and unloading of fish was found prominent and led to post-harvest losses.

Processing- Post-harvest losses due to storage and unloading from the fleet at the processing units were observed and it was found that the major causes of losses at the processing units were physical damage to marine fish due to improper and unhygienic handling by labourers, spoilage of marine fish during storage before and after processing due to temperature fluctuations and insufficient facilities for storage of all the raw materials.

Processing of marine fish

Storage- Marine fish were stored in cold storages and under freezing conditions. Usually the storage facilities were integrated with the processing units and standalone cold storage facilities were limited. The post-harvest losses during storage of marine fish included spoilage to fish due to temperature fluctuation and damage due to improper manual handling of fish. Very limited storage facilities were available for inland fish.

Storage of marine fish

6.9.3 Post-harvest losses at farm and market level

The survey for estimating the post-harvest losses of ***inland fish*** was conducted in 11 districts in 6 states covering 6 agro-climatic zones (ACZ2, ACZ3, ACZ4, ACZ5, ACZ7 and ACZ10). The extent of post-harvest losses at district level, state level and ACZ level, respectively are presented in **Appendix 5, Appendix 6** and **Appendix 7** and national level are reported in **Table 6.8**. The highest overall post-harvest losses at district level in inland fish were reported in Darbhanga district (6.00%) and the lowest were found in Nadia district (3.74%), which may be due to inefficient manual harvesting, use of damaged nets, improper handling while collection and sorting and grading in Darbhanga district. Also, biological losses due to spoilage of fish as a result of unhygienic practices by fishermen contributed to the increased post-harvest losses in the district. The highest overall post-harvest losses at state level in inland fish were estimated in Bihar (6.11%) and the lowest in West Bengal (3.82%). At ACZ level, the highest overall post-harvest losses in inland fish were observed in ACZ7 (5.83%) and the lowest overall in ACZ3 (3.74%). The higher losses in ACZ7 and ACZ4 covering the states of Odisha and Bihar, respectively, were mainly due to improper harvesting and unhygienic handling of the fish by the fishermen and other stakeholders at market level. Also, the use of gears for fishing was limited in the states and there was a lack of cold storage facilities and shortage of ice for storage of fish. At national level, the post-harvest losses observed at farm level operations in inland fish were 3.8 percent, whereas, for market level operations post-harvest losses were 1.06 percent. The overall post-harvest losses for inland fish at the national level were observed as 4.86 percent. Among different operations at farm level, the highest post-harvest losses were observed at harvesting (1.63%), sorting and grading (1.42%) and collection (0.37%). The highest post-harvest losses at the market level were reported at retailer level (0.7%) followed by wholesaler level (0.22%) and transportation level (0.14%). The overall losses for inland fish reported in ICAR-CIPHET (2015) were 5.23 percent. The estimated post-harvest losses obtained from the present study (4.86%) were lower than the earlier conducted study (ICAR-CIPHET 2015), however, the difference was non-significant at 5 percent level of significance.

The survey for estimating the post-harvest losses of ***marine fish*** was conducted in 23 districts in 8 states covering 5 agro-climatic zones (ACZ3, ACZ10, ACZ11, ACZ12 and ACZ13). The highest overall post-harvest losses at district level in marine fish were reported in Jagatsinghpur district (9.86%) and the lowest were found in Gir Somnath district (8.23%). The highest post-harvest losses in Jagatsinghpur were mainly due to poor handling practices at the fishermen level, long hauling of nets in the sea and due to low availability of ice boxes on the boat. The lower losses in Gir Somnath district can be attributed to the good handling practices followed by the fishermen and availability of infrastructure to the fishermen such as fishing vessels, good availability of ice and reefer vans in the districts. The highest overall post-harvest losses at state level in marine fish were estimated in Odisha (9.89%) and the lowest in Gujarat (8.48%). At ACZ level, the highest overall post-harvest losses in marine fish were observed in ACZ11 (9.37%) and the lowest overall in ACZ13 (8.41%). The higher losses in ACZ11 covering the states of Odisha and Tamil Nadu, were because of the unhygienic handling of the fish by the fishermen, insufficient availability of cold storage facilities and shortage of ice for storage of fish and unorganised marketing. The lower losses in ACZ3 and ACZ10 covering the states of Gujarat and Andhra Pradesh was due to careful and hygienic handling of the marine fish by the farmers and other market stakeholders, availability of ice and cold storage facilities near the landing centres/harbours and more number of reefer vans and processing units. Also, at the market level, majority of wholesalers and retailers in the state of Andhra Pradesh used facilities such as ice boxes and hygienic practices such as cleaning and sorting before selling, while in most of the other states used unhygienic practices while marketing and storage of marine fish at market level such as storage of fish at room temperature without any ice or cold storage facilities and many retailers were selling the fish on ground or in open and unhygienic spaces. At national level, the post-harvest losses observed at farm level operations in marine fish were 7.00 percent, whereas, for market level operations post-harvest losses were 1.76 percent. The overall post-harvest losses for marine fish at the national level were observed as 8.76 percent. Among different operations at farm level, the highest post-harvest losses were observed at harvesting (5.49%), collection (0.56%) and sorting and grading (0.38%). During harvesting, the losses were high mainly due to the fact that the catch of marine fish was performed in deep sea wherein the fishermen used to go for fishing for long periods of time, such as for 2-10 days along, with ice boxes and load of ice and the prolonged duration of fishing resulted in higher incidences of spoilage of fish on board due to shortage of ice, fluctuation in temperature because of improper storage on board and higher amounts of bycatch (unwanted and uneconomical fish) thrown back in the sea. Also, damage to fish due to improper handling on board, during sorting & grading and during unloading at harbours or landing centres contributed to the post-harvest losses. As depicted in **Fig. 6.9,** the overall losses for marine fish reported in ICAR-CIPHET (2015) were 10.52 percent. The estimated post-harvest losses obtained from the present study (8.76%) were significantly lower than the earlier conducted study (ICAR-CIPHET 2015) at 5 percent level of significance.

Among inland fish and marine fish, the estimated post-harvest losses in marine fish were significantly lower than the earlier conducted study, which can be attributed to the overall improvements done during the recent years. The reduction of harvest and post-harvest losses for fish can be attributed to various improvements in the supply chain as result of multiple initiatives undertaken by GoI and governmental and intergovernmental bodies. Support and development in fisheries supply under Blue Revolution launched by Department of Fisheries, GoI during 2014-15 to 2017-18, included sanction of 17,499 units of fish transportation facilities viz., refrigerated & insulated trucks, auto rickshaws, motorcycles & bicycles with icebox, skill training provided to 63,290 fish farmers & other stakeholders, assistance provided for bringing 29,127.73 ha area under aquaculture and sanction of 6,812 units of fish markets & fish mobile markets.

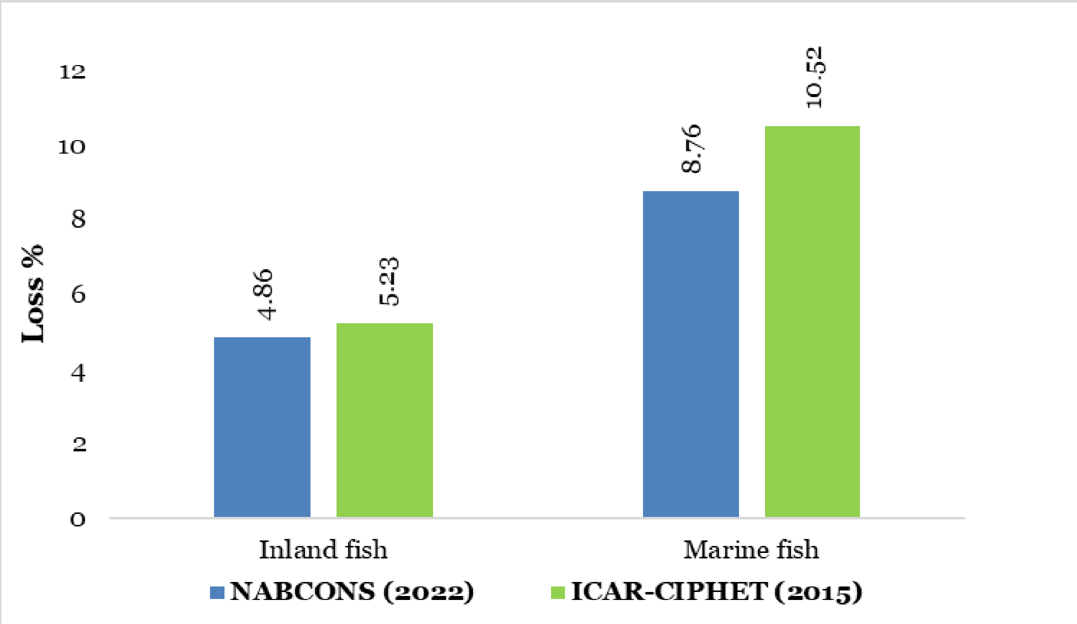
Also, since 2015, 20 cold chain projects have been completed under the Integrated Cold Chain scheme of MoFPI. Further, launch of Pradhan Mantri Matsya Sampada Yojana (PMMSY) for the development of marine & inland fisheries by GoI in the year 2020 with a highest ever total investment of 20,050 crore gave a boost to inland fisheries and aquaculture. The extension of facilities of Kisan Credit Cards (KCC) to fishers & fish farmers has helped fishermen in meeting their working capital needs resulting in better management of fisheries at the farm level. As on December 2021, a sum of 1038.60 Crore has been disbursed to various KCC card holder beneficiaries against the 70,818 KCCs issued. The various initiatives of the National Fisheries Development Board (NFDB) have been extensively playing a role in fish production and fishery development including conducting various training programs in collaboration with the States/UTs, ICAR fisheries Institutes.

During FY 2020-21, 1016 programmes/events were organised by NFDB. The outreach activities of NFDB have covered more than 7 lakh stakeholders in collaboration with Central Institutes, ICAR institutes, fisheries organisations, State Fisheries Departments, KVKs, relevant organisations. Similarly, the National Institute of Fisheries Post-Harvest Technology and Training (NIFPHATT) envisages the best post-harvest fish utilisation and consumption with the least post-harvest losses and delivery of the best quality fish and fish products. Likewise, the contribution from Fishery Survey of India (FSI), Indian Council of Agricultural Research (ICAR), Central Inland Fisheries Research Institute (CIFE), Central Institute of Freshwater Aquaculture (CIFA), Central Institute of Fisheries Technology (CIFT) and Department of Fisheries, Ministry of Fisheries, GoI has impacted the post-harvest handling practices positively. The reduction of harvest and post-harvest losses for marine fish can be attributed to various reasons such as increased availability of ice and ice plants near the harbours/ landing centres, establishment and renovation of harbours and landing centres and increased availability of insulated vans for transportation of fish.

Box 6.10 Good practices to be followed for fish

- *Cluster based approach for aquaculture*
- *Setting up of cold chain infrastructure including cold storages, ice plants and freezing plants*
- *Modern wholesale and retail infrastructure*
- *Increased use of refrigerated or insulated vans for fish transportation*
- *Replacement of damaged fishing vessels and crafts*
- *Implementation of hygienic practices and HACCP plans at processing units*
- *GPS based tracking systems and technologies for fish catching in sea*
- *Solar dryers and fish drying platforms for drying of fish*
- *Organized marketing of fisheries through FPOs/ FPC*

Fig. 6.9 Comparison of post-harvest losses of fish



6.10 Meat

The Government of India has recognized meat and poultry meat as an important sector for the socio-economic development of the country. The total meat production in India (including poultry meat) was 8.80 million metric tons during 2020-21 (DAHD, 2021-22). Improvement of production and post-harvest management will promote consumption of healthy, safe and hygienic meat products among the Indian population, however significant losses occur in the value chain of meat. This section deals in the type and extent of losses in meat and poultry meat in India.

6.10.1 Farm level operations

Slaughtering and dressing – Butcher shops and slaughter houses were surveyed under this operation. It involved slaughtering and dressing of meat including separation of edible and inedible offals. Rejection of the whole carcass or a part of it because of contamination during slaughter and bruises was the main reason for losses during harvesting. Losses of edible offals were observed due to physical damage, while cleaning and further due to lack of immediate offal collection mechanism leading to spoilage of offals.

6.10.2 Market level operations

Wholesalers and retailers - Wholesalers received slaughtered meat from slaughterhouses in bulk and supplied to retailers in smaller quantities as per the market demand. The slaughtered meat was kept hanged for display to customers. Lack of adequate market infrastructure including open market system, lack of general hygiene and sanitation, lack of proper disposal of waste material and lack of automation in markets majorly contributed to post-harvest losses. At many places, unavailability of markets and customers nearby forced the wholesalers and retailers to store the slaughtered meat for long duration, which led to further increase in post-harvest losses as no cold storage facility was available for slaughtered poultry meat and cost of renting storage area was very high.

Transporters – Transport of slaughtered meat and poultry meat took place through bikes and tempos. Rough handling during loading and unloading, use of improper packaging material highly prone to tear while transit (plastic bags) and lack of refrigerated vans were responsible for post-harvest losses during transport.

Processing units – Primary objective of poultry meat processing was to inhibit microbial growth and to stop deteriorative quality changes in the meat. Meat processing utilised the principles of refrigeration, freezing, smoking, curing, dehydration, freeze drying, canning and ionising. Post-harvest losses during storage before processing were observed, which was minimal due to availability of cold store chambers in the processing units.

6.10.3 Post-harvest losses at farm and market level

The survey for estimating the post-harvest losses of ***meat*** was conducted in 11 districts in 8 states covering 8 agro-climatic zones (ACZ2, ACZ3, ACZ5, ACZ6, ACZ9, ACZ10, ACZ11 and ACZ12). The extent of post-harvest losses at district level, state level and ACZ level, respectively for meat are presented in **Appendix 5, Appendix 6** and **Appendix 7** and national level are reported in **Table 6.8.** The overall harvest and post-harvest losses at district level in meat were the highest in Meerut district (5.69%) and the lowest in Hyderabad district (1.61%). Lack of hygienic slaughter houses, slaughtering of meat by unskilled labourers causing cuts and bruises in edible parts and lack of offal collection facility were the major causes behind higher losses in Meerut. The lower post-harvest losses in Hyderabad may be due to the presence of post-harvest infrastructure such as cold storage facilities and modern retail stores with covered display units. The overall harvest and post-harvest losses at state level in meat were the highest in Assam (3.44%) and the lowest in Telangana (1.61%). The overall harvest and post-harvest losses at ACZ level in meat were the highest in ACZ2 (3.44%) and the lowest in ACZ 10 (1.61%). Low post-harvest losses in ACZ10 can be attributed to better handling, storage, packaging and transport of meat. At national level, post-harvest losses in meat were found to be 1.43 percent at farm level and 0.91 percent at market level. Overall losses at national level in meat were 2.34 percent. Among different operations at market level, the highest post-harvest losses in meat were observed at wholesale level (0.38%) and retailer level (0.25%). Earlier study by ICAR-CIPHET (2015) reported an overall loss of 2.71 percent for meat, while the present study reported lower overall losses (2.34%). The estimated post-harvest losses obtained from the present study were significantly lower than the earlier study at 5 percent level of significance as depicted in **Fig. 6.8**.

The survey for estimating the post-harvest losses *of* ***poultry meat*** was conducted in 11 districts in 9 states covering 7 agro-climatic zones (ACZ2, ACZ3, ACZ5, ACZ6, ACZ9, ACZ10 and ACZ12). The overall post-harvest losses at district level in poultry meat were the highest in Rangareddy district (7.93%) and the lowest in Jind district (4.50%). Birds suffered from bird flu due to lack of general hygiene and sanitation in a farm in Rangareddy. Lack of integrated facilities along with poor storage, packaging and processing at farm level caused higher post-harvest losses in Rangareddy. Further, the open market system, poor storage facility in the market aggravated the losses at retail level. The overall post-harvest losses at state level in poultry meat were the highest in Telangana (7.93%) and the lowest in Haryana (4.50%). The overall post-harvest losses in poultry meat at ACZ level were the highest in ACZ10 (7.12%) and the lowest in ACZ6 (4.63%). The reasons behind higher losses in Andhra Pradesh and Telangana states of ACZ10 were poor handling of live birds and slaughtered meat at farm level along with lack of cold storage facilities, poor refrigerated and reefer vans for transport and improper packaging material. Post-harvest losses in farm/harvest operations in poultry meat at national level were found to be 2.51 percent, while for market level, the post-harvest losses were 3.12 percent. The overall total losses of poultry meat were 5.63 percent. Among different market level operations, losses at wholesale level (1.83%) and retailer level (0.92%) majorly contributed to the post-harvest losses. ICAR-CIPHET (2015) reported an overall loss of 6.74 percent for poultry meat while the present study reported lower post-harvest losses (5.63%). The estimated post-harvest losses obtained from the present study were significantly lower at 5 percent level of significance. In case of poultry meat and meat, slaughtering by unskilled labourers, physical damage during transport, inadequate storage facilities and poor availability of nearby markets were the main reasons behind higher losses at farm level. At market level, losses were mainly observed due to poor storage facilities and uncovered display units causing rapid moisture loss and lack of modern retail units. Poor handling of slaughtered meat at farm by unskilled labourers, lack of storage facilities, and unavailability of offal collection centres led to higher post -harvest losses at farm level. At market level, lack of adequate marketing infrastructure and lack of cold storage facilities were the main reasons behind higher post-harvest losses.

Slaughtering of meat

Box 6.11 Good practices to be followed for meat and poultry meat

- *Fly proof poultry dressing units (integrated units) at retail level with facilities for modified atmosphere packaging, vacuum packaging and category based packaging to increase shelf life of products.*
- *Modern retail shops, inspired by western countries, have two sections, first section with live butchery, where the customers can choose the product and get it cut and packed freshly and the second section with pre-cut meats, marinated meat products, cold cuts, sausages etc.*
- *Organized marketing of meat and poultry meat through FPOs/ FPC.*
- *Development of platforms like TraceX's blockchain enabled proprietary software to digitize the end to end process of the supply chain in livestock, poultry, dairy, sea food and agri produce.*

The estimated losses for both poultry meat and meat were significantly lower than the losses reported by ICAR-CIPHET (2015). This decrease in overall post-harvest losses may be due to following reasons:

- MoFPI's scheme of Backward and Forward Linkages has supported creation of modern meat units in different States.
- There were 20 cold chain projects for meat supply chain (as on 01.10.2021) in various states like Punjab, West Bengal, Nagaland, Karnataka, Tamil Nadu, Andaman and Nicobar Islands etc. under implementation under MoFPI's scheme for cold chain, value addition and preservation infrastructure.
- Significant shift of standalone facilities (slaughter only) to integrated ones (slaughter, storage and processing) with improvement in farm level animal husbandry practices.

- Poultry processing capacity has increased from 86,500 B.P.H (birds per hour) in 2015 to 1,40,000 B.P.H. in 2022. Significant increase in exports of poultry products from ₹ 30 crore in 2015 to ₹ 1360 crore in 2022 has improved market infrastructure and processing standards of poultry meat.

6.11 Milk

India is the world's top milk producer. Milk production in India did not show any sign of slowdown even during the COVID-19 pandemic. Milk production was 209.96 million MT in 2020-21 (Economic survey, 2021-22). Post-harvest losses across all stages of the value chain were minimum. This section deals in the type and extent of losses in milk in India.

6.11.1 Farm level operations

Milking- Milking was done manually by the majority of the farmers. Some of the bigger farmers had portable milking machines in the sample surveyed. The losses ranged from small spillages due to the milking activity, which included striping of milk outside the container (bucket), the milk falling due to mishandling while milking as well as spoilage of milk due to unclean udder or mastitic milk. On some rare occasions, it was also observed that while milking, the cow kicked the bucket in which the milk was stored leading to losses to the farmers. Most of the farmers interviewed were small-scale farmers and had old buckets/ containers, which were distorted or broken, thus leading to more losses during milking.

Collection – After milking, milk was collected in large steel/ aluminium cans either for delivery to the neighbouring areas or for transport to the cooperative units. Losses in such cases were mainly due to spillage of the milk while transferring it to the cans.

Milking by farmer

Collection of milk at collection centre

Transport - Due to the short shelf life of milk, it was transported on the same day. Some farmers transported their own produce while in other cases, the processor/ aggregator/ cooperatives collected milk from their farms. Milk was transported by farmers on bikes in steel cans. The milk was lost due to mishandling of milk, which led to spillage. Certain quantity of milk was also wasted due to leakages.

6.11.2 Market level operations

Retailers- At retailer level, post-harvest losses during handling, marketing and storage were observed. Post-harvest losses during the storage and marketing for milk were caused by spillage during loading/ unloading or improper packaging. At the retailer's level, packaged milk was spoiled and loose milk collected from various small farmers/ dairies got curdled due to improper storage/ higher temperature.

Milk packets for sale at retailer shop

Transporters- Milk was transported in tempos, trucks, insulated containers and tankers. In most cases, the milk was transported to nearby areas. The duration of the journey ranged from a few minutes (45-50 minutes) to a few hours (8-10 hours). The far off distances were covered by insulated tankers and containers. Very small losses due to spillage while loading/ unloading or leakage in the milk container were observed.

Processing units- At this level, the storage carried out in the processing units was captured. The post-harvest losses observed were mostly attributed to mishandling and spillages/ leakages.

Milk containers in the processing unit

6.11.3 Post-harvest losses at farm and market level

The survey for estimating the post-harvest losses of ***milk*** was conducted in 13 districts in 9 states covering 8 agro-climatic zones (ACZ1, ACZ3, ACZ5, ACZ6, ACZ7, ACZ9, ACZ10 and ACZ13). The overall post-harvest losses at district level in milk were the highest in Saharanpur district (1.64%) and the lowest in Guntur district (0.52%) and Navsari district (0.52%). The higher post-harvest losses in Saharanpur at farm level were due to higher spillage of milk, while milking and poor handling during loading-unloading at farm level. At market level, poor storage facilities increased losses at retailer level in the form of curdling of milk. The overall post-harvest losses at state level were the highest in Rajasthan (1.35%) and the lowest in Gujarat (0.50%). The overall post-harvest losses at ACZ level in milk were the highest in ACZ7 (1.35%) and the lowest in ACZ13 (0.50%). Adequate post-harvest management practices like cleaning of udders and teats of animals before milking, full hand method of milking, stripping at the end of milking, milking in separate dry and clean areas were being followed in Gujarat state of ACZ13 which resulted in lower losses. Sale of milk to home was minimal and milk procurement in Gujarat was majorly done by cooperatives which further reduced the losses during farm level transport. Poor milking and handling practices led to comparatively high post-harvest losses in ACZ7.

At national level, post-harvest losses in milk were found to be 0.54 percent in farm operations and 0.33 percent in market channels. The overall total losses of milk were 0.87 percent. Among different operations, milking (0.25%) and transport (0.21%) majorly contributed to the losses in farm level operations and processing units (0.33%) in market operations. Poor handling of milk at farm level, improper packaging prone to tear during transit, no temperature control during short distance transport, tank leakage and poor conditions of roads were the main reasons behind post-harvest losses.

As depicted in **Fig. 6.10,** ICAR-CIPHET (2015) reported an overall loss of 0.92 percent for milk while the present study reported lower post-harvest losses (0.87%). The estimated post-harvest losses obtained from the present study were non-significantly lower than the earlier conducted study (ICAR-CIPHET 2015) at 5 percent level of significance.

This decrease in overall post-harvest losses may be due to following reasons:

- Milk processing capacity has increased to 151.16 lakh litres per day (as on May 2022) attributed to MoFPI's Scheme of Cold Chain, Value Addition and Preservation Infrastructure.
- MoFPI's scheme of Forward and Backward Linkages has supported creation of 6 units in the dairy sector in Gujarat, Maharashtra, Rajasthan, Kerala and Telangana.
- Significant increase in cooperative milk unions, increasing milk procurement per day from 42.56 million litre in 2015 to 51.80 million litre in 2022 under the Village Based Milk Procurement System (VBMPS) of National Dairy Plan.
- Dairy cooperative societies have increased from 52911 in 2011 to 68508 in 2021. Cold chain infrastructure capacity has increased to 89262 thousand litres per day by March 2021.
- Out of around 4100 FPOs registered under the schemes promoted by NABARD, about 400 FPOs are related to milk/ dairy sector.
- During the period from 2014-15 to 2021-22, under National Programme for Dairy Development (NPDD) of Department of Animal Husbandry and Dairying (DAHD), 5.10 lakh new farmers were benefitted, 21.20 lakh litres per day new milk processing capacity were established, 2110 bulk milk coolers with 36.46 Lakh litres chilling capacity, 15882 automatic milk collection unit, data processing and milk collection unit were installed at village level dairy cooperative societies.
- Since 2018-19, 45 LLPD (Lakh Litres Per Day) Milk Processing Capacity, 113 BMCs (Bulk Milk Coolers) with 3.4 LLPD capacity, 165 MTPD (Metric tonnes per day) drying capacity and 6.76 LLPD VAP (Value Added Products) capacity has been established under Dairy Processing & Infrastructure Development Fund (DIDF) scheme of DAHD.
- With the objective of increasing of milk and meat processing capacity and product diversification, 38 projects have been strengthened through Animal Husbandry Infrastructure Development Fund (AHIDF) of DAHD.

An attempt has been made for cross country comparison of post-harvest losses for major crops/ commodities and is presented in **Table 6.8.**

Box 6.12 Good practices to be followed for milk

- *Cleaning of udders and teats of animals before milking, full hand method of milking, stripping at the end of milking, milking in separate dry and clean areas.*
- *Genetic improvement of cattle population through artificial insemination and cross breeding with exotic breeds.*
- *Cool chamber for pre-cooling during farm storage before transportation.*
- *Establishment of automatic milk collection units in milk processing plants which is an integrated unit comprising of automatic milk weighing system and automatic milk analyzer. The current number stands at 10217 in India.*
- *Processing of milk into functional dairy value added products like cheese, yogurt, butter, ice-cream, bottled flavored milk, probiotics, milk powder, bread spread etc.*

Box 6.12 Good practices to be followed for milk

- *Increase in bulk milk cooler (BMC) capacity which at present is 3646 KL.*
- *Production of high value clean milk, A2 milk etc.*
- *Branding and promotion of milk in the lines of AMUL, which is the largest milk product marketing organization and largest exporter of dairy products in India, with daily milk procurement of approximately 26.3 liters.*
- *Organized marketing of milk through FPOs/ FPCs*
- *Agri-startups to support dairy farmers and cooperatives to maximize their profits by digitizing and optimizing milk procurement & cold chain management as done by Stellapps through its IoT-based SmartMoo platform.*
- *Use of robotic milking for less human handling.*

Fig. 6.10 Comparison of post-harvest losses of milk

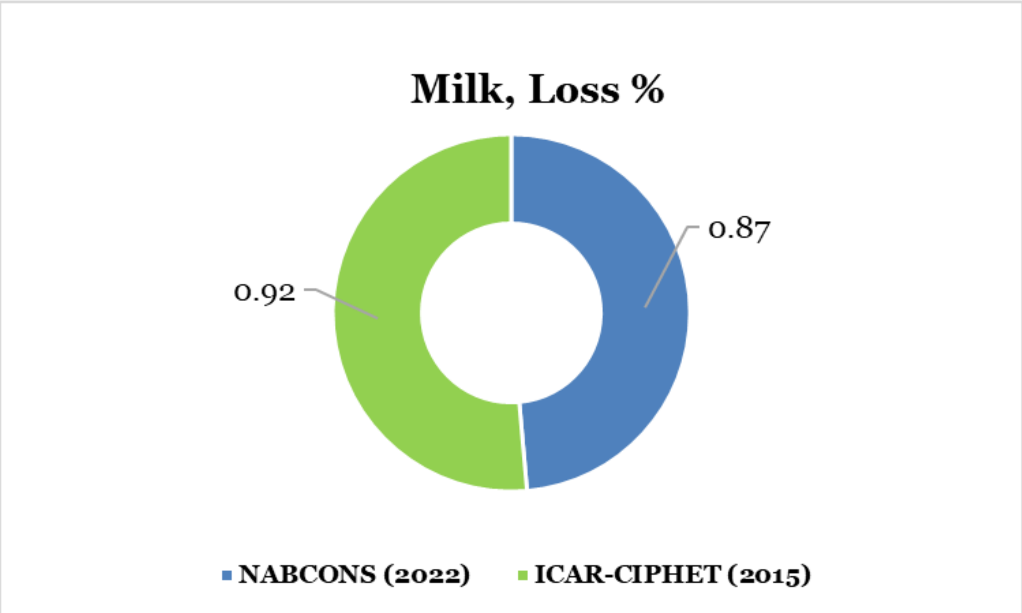


Table 6.8 Cross country comparison of post-harvest losses

S.No	Name of crop	Results of ICAR-CIPHET study 2015 (loss %)	Estimate of NABCONS study 2022 (loss %)	Comparison with international studies		
				Country name	Loss %	References
1	Apple	10.39	9.51	Bhutan	12.78	Rinchenz et al. (2019)
2	Banana	7.76	7.57	Sri Lanka	28.8	FAO (2018(a))
3	Citrus	9.69	7.71	Nepal	20.29	FAO (2018(a))
4	Grapes	8.63	7.15	Egypt	31 percent (2016) 45.76 percent (2017)	FAO (2021)
				South Africa	Farm level- 5.97 Retail Marketing- 3.65 Export- 4.36	Blanckenberg et al. (2021)
5	Guava	15.88	15.05	North-Western Ethiopia	Producer level - 16	Muluken et al. (2017)
				Chapadinha, Brazil	Groceries Stores - 17 Supermarkets - 21	Mendes et al. (2019)
6	Mango	9.16	8.53	Ghana	farm level - 3.6 wholesale level- 5.26 retail level - 4.5	Kitinoja and Cantwell (2010)
				Republic of Guyana.	28	FAO (2018 (b))
				Santa Lucia, Eastern Caribbean island	23	FAO (2018 (g))
7	Muskmelon	Not covered in the CIPHET study	6.83	Ethiopia	16.7	Kitinoja and Kader (2015)
8	Papaya	6.7	6.59	South Wollo, Ethiopia	Handling at - Farmers level - 1.5 Transportation - 1 Storage - 3.3	Seid et al. (2013)
				Maranhão, Brazil	Marketing and post-harvest losses in retail markets - 17.9	Ferreira et al. (2020)
9	Pineapple	Not covered in the CIPHET study	6.02	Bangladesh	43	Hassan et al. (2010)
10	Pomegranate	Not covered in the CIPHET study	6.82	South Africa	Packhouse- 6.74-7.69	Opara et al.(2021)
11	Sapota	9.73	9.53	Tropical countries	25–30	Siddiqui et al.(2014)
12	Beans	Not covered in the CIPHET study	7.11	Sri Lanka	27.91	FAO (2018(a))
13	Bottlegourd	Not covered in the CIPHET study	7.01	Not available		
14	Brinjal	Not covered in the CIPHET study	7.41	Bangladesh	23.38	Kaysar et al. (2016)
				Bangladesh	Farm Level- 13.90	Khatun and Rahman (2019)
15	Cabbage	9.37	8.15	Philippines	26.5	Gonzales and Acedo (2016)
16	Capsicum	Not covered in the CIPHET study	5.17	Brazil	33.33	Santos et al. (2020)
17	Cauliflower	9.56	7.89	Eastern Hills of Nepal	46	Udas et al. (2005)
				Nepal	18.3	FAO. (2018(a))

S.No	Name of crop	Results of ICAR-CIPHET study 2015 (loss %)	Estimate of NABCONS study 2022 (loss %)	Comparison with international studies		
				Country name	Loss %	References
18	Green Pea	7.45	6.43	Asia Pacific Region	Refrigerated transport - 10, Non-refrigerted transport - 11.6	APO (2006)
19	Mushroom	9.51	7.202	Not available		
20	Okra	Not covered in the CIPHET study	6.01	Bangladesh	32.3	Molla et al. (2011)
				Ghana	24.2	Ridolfi et al. (2018)
21	Onion	8.2	7.26	United Kingdom	Farm level - 3-5, Sorting-grading - 9-20, Storage - 3-10, packaging - 2-3	Porat et al. (2018)
				Ghana	13.5	FAO. (2010)
22	Potato	7.32	5.96	Ecuador and Peru	17.8	Luciana et al. (2017)
23	Radish	Not covered in the CIPHET study	6.46	South Korea	Packaged Radishes- 10 Non Packaged Radishes- 18	Chandra et al. (2018)
24	Tapioca	4.58	4.87	Nigeria	13	Bamikole et al. (2022)
				Nigeria	Farmers- 14.59	Oguntade (2013)
				Developing countries	12.4	Naziri et al. (2014)
25	Tomato	12.44	11.61	Republic of Trinidad and Tobago	28	FAO (2018(c))
				Republic of Guyana	1700%	FAO (2018(d))
26	Milk	0.92	0.87	Ethiopia	Milking- 9.4 Postharvest handling & storage- 0.5 Processing and packaging- 6	Minten et al. (2019)
				Europe	Milking- 3.5 Postharvest handling & storage- 0.5 Processing and packaging- 1.2	FAO (2011)
				North America	Milking- 3.5 Postharvest handling & storage- 0.5 Processing and packaging- 1.2	
				Latin America	Milking- 3.5 Postharvest handling & storage- 6 Processing and packaging- 2	
				Asia	Milking- 3.5 Postharvest handling & storage- 1 Processing and packaging- 1.2	
27	Egg	7.19	6.03	North Africa, West and Central Asia	8	Gustafsson et al. (2013)
				Ghana	12	FAO. (2010)
28	Meat	2.71	2.34	Europe	Postharvest handling & storage- 0.7 Processing and packaging- 5	FAO (2011)
				North America	Postharvest handling & storage- 1 Processing and packaging- 5	

S.No	Name of crop	Results of ICAR-CIPHET study 2015 (loss %)	Estimate of NABCONS study 2022 (loss %)	Comparison with international studies		
				Country name	Loss %	References
				Latin America	Postharvest handling & storage- 1.1 Processing and packaging- 5	
				Asia	Postharvest handling & storage- 0.6 Processing and packaging- 5	
				Ghana	14	FAO. (2010)
29	Poultry Meat	6.74	5.63	USA	4	Buzby et al. (2014)
				Ghana	4	FAO. (2010)
30	Inland fish	4.88	5.23	Bangladesh	Culture fish- 4.47 Capture fish- 7.01	Rashid and Sarkar (2020)
				Ghana	21.93	FAO. (2010)
31	Marine fish	10.52	8.77	Bangladesh	11.67	Rashid and Sarkar (2020)
				Ghana	21.08	FAO. (2010)
				Kenya	Drying- 1-2	FAO. (2014)
	Fish and seafood			Europe	Fishing- 9.4 Postharvest handling & storage- 0.5 Processing and packaging- 6	FAO (2011)
				North America	Fishing- 12 Postharvest handling & storage- 0.5 Processing and packaging- 6	
				Latin America	Fishing- 5.7 Postharvest handling & storage- 5 Processing and packaging- 9	
				Asia	Fishing- 15 Postharvest handling & storage- 2 Processing and packaging- 6	
				USA	Retail- 8	Buzby et al. (2014)
32	Bajra	5.23	4.37	Bangladesh	3.81	Molla et al. (2011)
33	Paddy	5.53	4.77	Democratic Republic of Timor-Leste	23.5	FAO (2018(h))
				Democratic Republic of the Congo	Harvesting- 4, Drying- 4 Farm storage- 5 Manual hulling- 11 Mechanised hulling- 22	FAO, IFAD and WFP (2021)
				Ghana	13.5	Ridolfi et al. (2018)
34	Maize	4.65	3.89	Zimbabwe	Harvesting-4.4, shelling- 2.9, cleaning - 1.7, drying-1.6, storage - 3.5	FAO (2020)
35	Wheat	4.93	4.17	Korea	16.35	FAO (2014)
36	Sorghum	5.99	5.92	Ethiopia	11.6	U.S. Department of State (2013).
37	Black gram	7.07	5.83	Bangladesh	Infestation- 7.7 to 13.7	Mannan and Tarannum (2011)
38	Chickpea	8.41	6.74	Tigray, Amhara, Oromia & SNNP of Ethiopia	Harvesting : 2% Storage loss : 2.5 % Cleaning Loss : 0.01 %	Mekasha Chichaybelu (2014)
39	Green gram	6.6	6.19	Not Available		
40	Pigeon pea	6.36	5.65	Tanzania	10.2	Abass et al. (2014)

S.No	Name of crop	Results of ICAR-CIPHET study 2015 (loss %)	Estimate of NABCONS study 2022 (loss %)	Comparison with international studies		
				Country name	Loss %	References
41	Groundnut	6.03	5.73	Zimbabwe	Harvesting -7.2, threshing/shelling -3.2, winnowing/cleaning-2.9, drying-3.6, storage -4.9	FAO (2020)
42	Mustard	5.54	4.46	Not Available		
43	Soybean	9.96	7.51	Europe	6	Fine et al. (2015)
44	Sunflower	5.26	4.38	Uganda	harvesting - 2.50, threshing-2.40, drying - 3, storage -2.00, milling - 1.5	FAO, WFP and IFAD. (2019)
45	Cottonseed	3.08	2.87	America	Not Available	Not Available
46	Safflower	3.24	3.06	Australia	Harvesting - 3 to4	GRDC (2017)
47	Black pepper	1.18	1.29	Not Available		
48	Coriander	5.87	5.32	Not Available		
49	Chili	6.51	6.11	Not Available		
50	Turmeric	4.44	5.36	Not Available		
51	Areca nut	4.91	4.41	Not Available		
52	Cashew nut	4.17	3.72	Srilanka	15	Priyashantha et al. (2020)
53	Coconut	4.77	3.86	Tanzania	Harvesting - 10, Storage - 5 to 10	Punchihewa and Arancon (1999)
54	Sugarcane	7.89	7.33	Tandojam, Pakistan	Storage-Kept Open (10 days) -27.8, Storage-Covered with Trash (10 days) -15.8	Hussain et al. (2018)

CHAPTER 7 ECONOMIC VALUE OF POST-HARVEST LOSSES

Information of estimated post-harvest losses for crops and commodities help in understanding and identifying various operations contributing to losses. This also helps in identification of measures/ strategies to be taken up to avoid such losses. The reduction in losses in turn helps make more food available for consumption. The creation of post-harvest infrastructure and implementation of measures for reducing post-harvest losses requires huge investments. Therefore, it is essential to estimate the extent of post-harvest losses in absolute physical terms and monetary terms.

Based on the estimates of the study on post-harvest losses for 54 crops, the extent of loss in physical quantity and monetary value were estimated at the national level. The estimation was based on the production of all crops during 2020-21 (latest available) and average wholesale price for 3 years (2019-20, 2020-21 and 2021-22). The wholesale prices of 2019-20 and 2020-21, were brought to 2021-22 level by adjusting inflation rates for years 2019-20 and 2020-21. Using production data and the estimated percent loss, quantity lost was estimated. Based on the estimated quantity lost and average wholesale price, monetary losses were estimated and presented in **Table 7.1.**

7.1 Losses in absolute terms

As indicated in **Table 7.1**, in terms of physical quantity, 12.49 million MT of cereals, 1.37 million MT of pulses, 2.11 million MT of oilseeds, 7.36 million MT of fruits, 11.97 million MT of vegetables, 30.59 million MT of plantation crops (including sugarcane) and spices, 3.01 million MT of livestock excluding eggs and 7363 million eggs were lost due to post-harvest losses. Therefore, if effective post-harvest facilities are put in place, there is a possibility of making additional quantities of food available for consumption by reducing the post-harvest loss to the tune of 39.19 million MT of food, 29.72 million MT of sugarcane and 7363 million eggs.

7.2 Losses in monetary terms

Agriculture sector contributes 19.9 percent to the national GDP (2020-21). The study has estimated a monetary loss of ₹1,52,790.42 crore for 54 crops and commodities, based on the average wholesale annual prices for 3 years which was higher than previous assessment by ICAR-CIPHET (2015). Increase in monetary loss observed was primarily contributed by increase in volume of production and inflation in prices of 45 commodities which was to the tune of ₹33,849.92 crores and ₹39,024.16 crores, respectively.

The highest contribution in the economic loss was by livestock produce (21.70%), followed by fruits (19.34%), vegetables (17.97%), cereals (17.02%), plantation crops and spices (10.74%), oilseeds (7.15%) and pulses (6.08%). Crop-wise details of physical and monetary loss are presented in **Appendix 8(a)** and crop-wise details of production and prices are presented in **Appendix 8(b)**. The perishable nature of livestock produce, fruits and vegetables resulted in the higher post-harvest losses, thereby resulting in high contribution in the economic losses. Thus, perishable commodities required special attention for creation of better post-harvest technology, policy interventions and infrastructure.

Table 7.1 Category wise absolute losses in physical quantity and monetary value

S. No.	Category	Production (million MT)	Quantity lost (million MT) NABCONS (2020-21)	Monetary loss NABCONS (₹ in Crore)	% Contribution of the total loss (in monetary value)
1	Cereals	281.28	12.49	26000.79	17.02
2	Pulses	21.55	1.37	9289.21	6.08
3	Oilseeds	37.27	2.11	10924.97	7.15
4	Fruits	90.82	7.36	29545.07	19.34
5	Vegetables	164.74	11.97	27459.08	17.97
6	Plantation crops (including sugarcane) and spices	426.13	30.59	16412.56	10.74
7	Livestock produce (milk, meat and fish)	232.86	3.01	29871.41	21.70
Total		**1254.65**	**68.90**	**149503.10**	
8	Eggs*	122110	7363	3287.32	
	Grand Total			**152790.42**	

**For eggs, production in million numbers and price per eggs were taken*

Among all the 54 crops/ commodities, the major contributors to the economic value of losses in the country includes paddy, wheat and chickpea (cereals and pulses), mango, banana, tomato, onion and potato (fruits and vegetables), soybean (oilseeds), sugarcane (plantation crops), poultry meat, milk, marine fish and inland fish (livestock produce). These crops and commodities are responsible for 67.69 percent of the total loss of ₹ 1,52,790.42 crore. The total monetary loss is 2.35 percent of national GDP (at current prices for Q1 of 2022-23).

7.3 Saving in losses

GoI, State Government and private sector has taken up various initiatives to reduce post-harvest losses. As a result, out of 45 commodities, a decline in post-harvest losses since 2015 was observed in 42 commodities, of which there was a significant decline in losses in 25 commodities.

The previous study of 2015 reported monetary loss of ₹92,651 crore. With the premise that if the extent of losses recorded in 2015 would have continued, the value loss estimated would have been ₹1,66,593.72 crore. However, the current study estimated the monetary loss of ₹1,46,153.15 crore for 45 crops covered under previous study. The difference of monetary losses of 45 commodites resulted in savings of **₹20,440.58 crore**. This clearly indicates that the creation of post-harvest infrastructure, introduction of improved technology and policy changes have resulted in reduction of losses, thereby increasing the available food for meeting the demands of the growing population. Better post-harvest management and technological advancements are needed to further reduce the losses to the maximum extent possible in order to keep up with the pace of changing patterns of production of crops/ commodities and climate conditions.

Various strategies/ measures to be taken up for better post-harvest management across the categories of cereals, pulses, fruits, vegetables, plantation crops, spices, egg, milk, poultry meat, meat and fish are discussed in detail in **Chapter 8** of this report.

CHAPTER 8 STRATEGY TO REDUCE POST-HARVEST LOSSES

8.1 Fruit and vegetable sector

8.1.1 Post-harvest infrastructure

(i) Farm level measures (Harvesting, preparation, storage & transport)

- Support for harvesting equipment such as pluckers, nippers, sapota/mango harvester to farmers.
- Support for plastic/ wooden/ bamboo baskets or crates to small and medium farmers for collection of harvested crops.
- Promotion of mechanical harvesting for fruits and vegetables especially root crops may be promoted.
- Support for setting up farm level covered collection centres / pre cooling units with washing arrangement.
- Support for equipment like mechanical sorter-grader, and automatic packers through custom hiring centres may be provided to the farmers. Automated grading and packing machines need to be promoted near to production centres for fruits such as apples, citrus, etc.
- Hot water treatment using a hot water tank and a net cover as practised in other papaya producing countries like Taiwan, Hawaii, Thailand etc. can be introduced for effective post-harvest treatment against post-harvest diseases.
- Waxing and packaging in corrugated boxes with maximum 2-3 layers need to be introduced.
- Facilities for vapour heat treatment for fruits like mango, litchi etc.
- Usage of telescopic Corrugated Fibre Board (CFB) cartons (for e.g. 20 kg boxes) wherein packed papaya can be placed vertically.
- Replacement of gunny bags with plastic crates and HDPE bags for vegetables such as capsicum, bean, radish, green peas etc.

- Use of shrink wrap packaging for high value fruits and vegetables like apple, capsicum, beans, lettuce, etc.
- Individual cushioning of fruits to be encouraged and introduction of modern polystyrene wrapping to be promoted.
- Precooling units / pack house facilities for fruits and vegetables at farm level for domestic marketing of produce. As per report published by NCCD 2015, there is a requirement of 70,080 number of pack house.
- Support and promotion of Zero Energy Cool Chamber (ZECC) for precooling at farms before transportation.
- Minimum covered storage facility for temporary storage and handling of produce at farm level.
- Controlled Atmospheric (CA) storage for high value produce like apple, pear, cherry etc.
- Bulk cold storage for storage of crops like potato, onion, red chillies, etc. As per study conducted by NCCD in 2015, the requirement of cold storage in the country was 35 million MT, while capacity of such storage is around 32 million MT.
- Setting up of Individual Quick Freezing (IQF) units near catchment areas for green peas.
- Promotion of setting of dehydration units for onions.
- Support for farm pick up vans in group mode to Farmers' club, SHGs, FPOs etc.
- Support for setting up ropeways (span used to carry only farm produce) may be provided in hilly areas in a group mode.

(ii) Market level measures

- Cold storage hub for holding during transit time for all kinds of fruits and vegetables.
- Use of technology in marketing e.g. e-commerce, artificial intelligence, block chain and GPS for traceability.
- Supporting cold chain infrastructure may be provided.
- Modern retail infrastructure with displays needs to be promoted and encouraged.
- Promoting electronic auction & linking farmers to e-NAM.
- Setting up irradiation processing facility to irradiate products like onion, potato to extend the shelf life.
- Application of Ozone for increasing the shelf life of fruits and vegetables.
- Promoting Agri Startup for e marketing of perishable produce.
- Supporting FPOs / FPCs for aggregation, transport and forwarding of produce to markets.
- Increasing the availability of reefer vans as there was a gap of 52,826 number of reefer vans in the country (NCCD, 2015), which has further increased with increase in the agriculture production in the recent years.
- Increasing the availability of closed vehicles instead of open vehicles to reduce the losses related to external environment.
- Improvement in road conditions for long haul transportation.
- Promote use of "Kisan Rail" to increase transport through railways.
- Setting up perishable cargo complexes at the select airports.

8.1.2 Training and capacity building

- Establishment of training units for the vendors at retailer level on handling of perishable commodities.
- Training on better post-harvest management practices may be provided to all relevant stakeholders.
- Awareness to be created among various stakeholders regarding storage practices of perishables.
- Field demonstrations of new technologies to increase the farmer's acceptance.

8.1.3 Policy advocacy

- Organised marketing through creation of FPOs/FPCs.
- Promotion of high-density plantation of fruit crops like apple, papaya, guava, etc.
- Promotion of production of processable varieties.

- Improvement in crop protection and production technology.
- Organised marketing in crops like apple and citrus to be introduced in crops like sapota, papaya etc
- Making legal provision for use of crop specific "Standard Size Boxes".
- Creation of an institutional framework and policies for utilisation of the food waste into value added products
- Government support for research and development and technology transfer in production and post production for agri-produce may be increased.

8.2. Milk and milk products

8.2.1 Post-harvest infrastructure

(i) Farm level measures (Harvesting, preparation, storage & transport)

- Promoting setting up of modern dairy farms (hygienic feed to animals and vaccination) for production of clean milk.
- Promoting use of milking machines and automated collection of milk.
- Increasing automatic milkers and vacuum cups at farm level to reduce handling losses.
- Support and promotion of Zero Energy Cool Chamber (ZECC) for precooling at farms before transportation.
- Expanding coverage of insulated/ refrigerated milk vans to remote villages.

(ii) Market level measures

- Maintaining refrigeration during storage at the market level.
- Milk parlours, milk vans and token milk to be promoted.
- Maintaining refrigeration during supply to the markets.
- Increased use of technology such as GPS control and tracking transport vehicles.

8.2.2 Training and capacity building

- Adequate focus on farmers training on quality feed production, milking practices and handling of milk post milking.

8.2.3 Policy advocacy

- Expansion of co-operatives and processing units to states including Uttar Pradesh, Madhya Pradesh, Odisha, West Bengal, Chhattisgarh and Jharkhand for meeting the procurement target, as milk production has increased.
- Establishment of new units and strengthening of existing manufacturing units for value addition and innovation for milk products.
- Promoting genetic improvement of cattle population through artificial insemination and cross breeding with exotic breeds.
- Need to bring structural changes in the unorganised sector so that initial storage and processing of milk can be taken up at the village level itself.
- Creation of an institutional framework and policies for utilisation of the food waste into value added products

8.3. Eggs and poultry meat sector

8.3.1 Post-harvest infrastructure

(i) Farm level measures (Harvesting, preparation, storage & transport)

- Promotion of 'grow out houses' (cage free and ventilated barns) along with vaccination of birds against diseases like bird flu.
- Need to shift from backyard poultry farming to mechanised and organised eggs production farms.
- Mechanical grader and sorters need to be promoted at farm level for eggs.
- Promoting use of plastic cartons, bubble earth wraps and wooden eggs trays for packaging of eggs.
- Promotion of integrated units (slaughter, storage and processing) rather than standalone facilities (slaughter only) at

farm level so that safe storage can take place in case when immediate disposal is not possible.

- Need of cold storage units at farm level for eggs.
- Increasing the availability of reefer vans.

(ii) Market level measures

- Promotion of cold storage facilities at retailer level.
- Country wide expansion of modern retail stores like FIPOLA retail stores prevalent in southern states of India with well-managed backward integration.
- Promotion of retail shops inspired by western countries with two sections, first section with live butchery, where the customers can choose the product and get it cut and packed freshly. Second section with pre-cut meats, marinated meat products, cold cuts, sausages and imported meat.
- Country wide expansion of modern and automatic processing units like those present in Venky's along with installation of modern equipment.
- Promotion of modified atmosphere packaging, vacuum packaging and category based packaging to increase shelf life of products and to prevent loss during transport.
- Setting up fly proof poultry dressing units at retail level with facilities for packing.
- Better conditions of roads and increasing the availability of reefer vans for long haul transport.

8.3.2 Training and capacity building

- Acquaintance with equipment, demonstrations of slaughtering of birds and handling of eggs through extension activities supported by farmers' cooperatives to impart skill among labourers.

8.3.3 Policy advocacy

- Capacity expansion for abattoirs-cum-meat processing plants all over India with support of local municipal authorities.
- Need to increase eggs processing capacity.
- Supporting collectivization of smallholders through co-operatives and producer companies to ensure the supply of inputs and to facilitate and enable market linkages.
- Creation of an institutional framework and policies for utilisation of the food waste into value added products

8.4 Meat Sector

8.4.1 Post-harvest infrastructure

(i) Farm level measures (Harvesting, preparation, storage & transport)

- Development of ventilated barns with proper animal husbandry practices supported by timely vaccination against pathogenic diseases.
- Need to establish modern slaughter-houses to bring improvements in meat-handling practices.
- Need to increase availability of category and need based packaging material at farm (and market level) like shrink films for wrapping large and uneven cuts of fresh meat and vacuum packaging for long term storage.
- Promotion of integrated units (slaughter, storage and processing) rather than standalone facilities (slaughter only) at farm level so that safe storage can take place in case when immediate disposal is not possible.
- Increasing the availability of reefer vans.

(ii) Market level measures

- Promotion of cold storage facilities at retailer level.
- Country wide expansion of modern retail stores like FIPOLA retail stores prevalent in southern states of India with well-managed backward integration.

- Need to shift from wet markets to frozen/chilled meat delivery systems that essentially deliver fresh meat within 2-3 hours and guarantee safety and hygiene.
- Country wide expansion of modern and automatic processing units like those present in Hyderabad and promotion of retail shops inspired by western countries.
- Increasing the availability of reefer vans for long haul transport.

8.4.2 Training and capacity building

- Promoting farmers' cooperatives which can play a major role in the field of production and marketing of quality animals, extension education and encouragement of backward integration / contract farming.

8.4.3 Policy advocacy

- Capacity expansion for abattoirs-cum-meat processing plants all over India with support of local municipal authorities.
- Creation of an institutional framework and policies for utilisation of the food waste into value added products

8.5 Fisheries sector (Inland and marine fisheries)

8.5.1 Post-harvest infrastructure

(i) Farm level measures

- Improvisation and modernization of fishing vessels in order to reduce losses to be promoted.
- Modernization and up gradation of existing fishing harbours and establishment of new harbours/ landing centres.
- Promoting use of Geo-informatics and GPS based applications and technologies for catching fish such as application for catch reports which gives data analytics that can be used by fishermen to target regions for fishing.
- Increasing availability of ice boxes and ice at fishermen.
- Support for solar dryers (economical) may be given to the fishermen for drying of fish.
- Cold storage facilities need to be promoted.
- Increasing availability with fleets with provision for ice boxes such as rickshaws with ice box units.
- Increasing the number of insulated/ refrigerated vans and trucks for transportation of fishes.

(ii) Market level measures

- Strengthening the cold chain infrastructure including cold storages, ice plants, freezing plants etc.
- Cold storage facilities need to be promoted
- Increasing availability of ice boxes and holds during fish auctioning and sale.
- Modern wholesale and retail infrastructure needs to be promoted and encouraged.
- Popularization of low cost hygienic fish vending kiosk in roadside markets
- Promotion of e-marketing of fish and fish products.
- Promotion and setting up of freezing units and processing units for fisheries.
- Implementation of traceability system.

8.5.2 Training and capacity building

- Training on hygienic post-harvest handling of fish may be provided.
- Awareness and skill development training for all market stakeholders.

8.5.3 Policy advocacy

- Promotion of cluster based approach for aquaculture for better inland fish post-harvest management.
- Marketing strategisation for an effective sale and marketing of fisheries.
- Rigorous licensing and registration of fish markets and traders.
- Preparation and distribution of training module in regional languages for fishermen and important stakeholders for better post-harvest management practices.
- Credit facilitation to be made easier to the fishermen and small-scale processors from the financial institutions.
- To promote Kisan Credit Cards (KCC) under scheme of GoI.

- Creation of an institutional framework and policies for utilisation of the food waste into value added products

8.6. Cereals, pulses and oilseed sector

8.6.1 Post-harvest infrastructure

(i) Farm level measures

- Promotion and support for combine harvester and reaper binder wherever applicable may be done.
- Timely harvesting, proper collection and stacking may be practised.
- Support for equipment like mechanical thresher, sheller, hullers and strippers through custom hiring centres may be provided to farmers.
- Use of good quality and infestation free packaging bags to avoid pest attack may be practised.
- Technologies like hermetic packaging and solar bubble dryers may be promoted.
- Use of CAP storage for farm level storage of grains may be promoted.
- Support for development of community godown near to farms may be provided.
- Use of organic insecticides such as Neem leaves and bio insecticides may be promoted at farm level storage.
- Use of dunnage for stacking the grain bags on floor may be promoted for farm level storage.
- Support for cleaning, grading and winnowing machinery at farm level to farmers may be provided.
- Support for tractors and trolleys for farm level transportation may be provided to farmers.
- Interventions such as systems for bulk transport for direct loading into the vehicle will help reduce the losses in transportation and handling during loading and unloading.

(ii) Market level measures

- Support for establishment of scientific storage and construction of grain silos may be provided.
- System for proper fumigation and drying at all the storage levels for control of pest infestation may be created.
- Use of technology in marketing e.g. e-commerce, artificial intelligence and block chain for traceability
- Modern retail infrastructure with displays needs to be promoted and encouraged.
- Promoting electronic auction & linking farmers to e-NAM.
- Supporting FPOs / FPCs.
- Interventions such as systems for bulk transport for direct loading into the vehicle will help reduce the losses in transportation and handling during loading and unloading.
- Promote use of “Kisan Rail” to increase transport through railways.

8.6.2 Training and capacity building

- Training on better post-harvest management practices may be provided to all relevant stakeholders.
- Training may be provided to farmers to control the pre-harvest factors such as weed and insect pests responsible for huge post-harvest losses in cereals and pulses.
- Training and capacity building of farmers for scientific handling of grains need to be provided.
- Centre of excellence may be established to facilitate transfer of latest agricultural technologies.
- Awareness about drying and optimum storage moisture content may be created.

8.6.3 Policy advocacy

- Organised marketing through creation of FPOs/FPCs.
- Promotion of better storage practices of cereals, pulses and oilseeds.
- Improvement in crop protection and production technology.

- Direct linkage of farmers with grain processing units, dal mills and oil mills may reduce the handling losses.
- Standard practices for reusing of packaging materials.
- Quality certification, assaying and testing.
- Creation of an institutional framework and policies for utilisation of the food waste into value added products.
- Government support for research and development and technology transfer in production and post-production for agri-produce may be increased.

8.7 Plantation and spices sector

8.7.1 Post-harvest infrastructure

(i) Farm level measures

- Promotion of sugarcane harvester, tree climbers for coconut and areca nut, coconut dehusker may be done wherever applicable.
- Support for equipment and tools like nippers, pluckers to farmers may be provided.
- Timely harvesting, proper collection practices may be promoted.
- Plucking tools especially developed for black pepper and turmeric harvester may be promoted.
- Synchronised harvesting schedule of sugarcane with sugar mills functioning may be adopted to reside waiting time and staling losses in sugarcane.
- Use of mechanical dryers/solar tunnel dryers may be promoted for farm level drying.
- Support for sorting, grading and cleaning equipment may be provided to farmers.
- Best post-harvest management practices may be promoted for plantation crops and spices.
- In case of rains or high humidity, the drying of black pepper in the sun was severely hampered. Use of mechanical dryer could effectively help to dry the black pepper with reduced losses.
- In countries like Sri Lanka, a combination of wood-burning and solar dryers were used.
- Practices like covering harvested sugarcane with stray leaves before collection and during transportation may be promoted for reducing post-harvest losses in sugarcane.
- Minimal processing of plantation crops and spices may be promoted.
- A proper management and cooperation must be provided from factories to farmers for scheduling of harvesting and for quick disposal of harvested sugarcane to factories to reduce the post-harvest losses of sugarcane.
- Innovative coconut products have been launched in European countries like coconut bread and chocolate coconut bars in the Netherlands and organic coconut fruit bars in Germany; which are getting wide popularity, similar products can be processed and marketed in India.
- Use of pre-treatment such as blanching before drying of black pepper.
- Recent technologies available for extraction, packaging and preservation of fresh sugarcane juice may be promoted among the growers to generate an alternate source of income and reduce the dependency upon the sugar mills in case of emergency
- Technologies like irradiation and electron beam technology must be adopted to pasteurise and sterilise the products as well as packaging materials of coconut products to develop the export quality products.
- Support for tractors and trolleys for farm level transportation may be provided to farmers.
- Plastic crates may be promoted for packaging of spices for transportation of spices.
- Timely loading and unloading of sugarcane and its quick disposal to factories after harvesting may be managed to reduce the staling losses

(ii) Market level measures

- Support for establishment of scientific godowns may be provided.
- Temperature and moisture conditions of the warehouse may be maintained and monitored regularly to reduce the post-harvest losses due to weather conditions.
- Support of storage facilities with humidity control for black pepper storage can help reduce the losses at storage level and will aid in improving the quality of the pepper for exports.
- Cost of utilising the services of cold storage for chilli may be made affordable to the farmers.
- Use of technology in marketing e.g. e-commerce, artificial intelligence and block chain for traceability.
- Modern retail infrastructure with displays needs to be promoted and encouraged.
- Promoting electronic auction & linking farmers to e-NAM.
- Promote use of "Kisan Rail" to increase transport through railways.
- Covering of sugarcane during transportation may be practised.
- Use of contamination free vehicles and clean collecting station may be practised.

8.7.2 Training and capacity building

- Training on better post-harvest management practices for plantation crops and spices may be provided to all relevant stakeholders.
- Training and capacity building of farmers for scientific handling of grains need to be provided.

8.7.3 Policy advocacy

- Organised marketing through creation of FPOs/FPCs.Promotion of better storage practices of plantation crops and spices may be done.
- Promotion of improved and spoilage resistant varieties may be done.
- Policies may be intervened for establishment of processing units for value added products from plantation crops and spices to get the potential foreign markets.
- Support may be provided to farmers for organised aggregation of coconut by forming FPC/ FPOs so that farm level costs can be minimised. The organised marketing by forming cooperatives or farmer producer companies may be introduced.
- Creation of an institutional framework and policies for utilisation of the food waste into value added products.

CHAPTER 9 SUMMARY AND CONCLUSION

India with 1.3 billion populations was only the second to China in the World (Census 2011). As per UN World Population Report 2022, India's population stood at 1.412 billion in 2022, compared to China's 1.426 billion and is likely to surpass China by 2023. Therefore, the demand for food in India is likely to go up in the coming years. In order to provide food to an additional population, India needs not only to increase its production but also focus on reducing food loss and waste effectively. Therefore, an effective post-harvest management is very important for planning effective utilization of the agricultural and livestock produce to meet the additional food requirements of the growing Indian population. It becomes important to check different channels through which crop and livestock produce reaches consumers. Each channel and operation while handling the crop and livestock produce at farm level as well as market level causes some post-harvest losses of the agricultural commodities. Study of these channels and post-harvest losses therein will help in making strategies for improving these farm practices and thus reducing post-harvest losses. Therefore, the present study was conducted to assess the post-harvest losses of 54 crops and livestock produce. Data for estimating post-harvest losses were collected from 292 districts covering 15 agro-climatic zones. Stratified multistage random sampling method as described in chapter 3 was used to select the respondents.

The data were collected through enquiry and by observations visiting the fields by trained manpower/ enumerators. CAPI based application was used for data collection from all the respondents. The collected data was cross checked, scrutinized and analysed for estimation of post-harvest losses at district level, state level, agro-climatic zone level and national level.

The salient findings of the study are summarized below: -

In the ***cereals*** category, the post-harvest losses for paddy, wheat, maize, Bajra and sorghum were estimated. The post-harvest losses ranged from **3.89** percent in maize to **5.92** percent in sorghum. Use of manual methods for operations such as sickle and sticks for harvesting and threshing majorly contributed to the post-harvest losses.

The estimated post-harvest losses in ***pulses*** (pigeon pea, chickpea, black gram and green gram) ranged from **5.65** percent in pigeon pea to **6.74** percent in chickpea. Shattering of grains during harvesting, spillage during various operations, and mishandling resulted in losses.

In case of ***oilseeds,*** post-harvest losses of crops such as mustard, cottonseed, soybean, safflower, sunflower and groundnut were estimated. The losses were ranging from **2.87** percent in cottonseed to **7.51** percent in soybean. Harvesting and threshing at farm operations and improper storage conditions at market channels resulted in losses.

Among 11 ***fruits*** (apple, banana, citrus, grapes, guava, mango, papaya, sapota, pineapple, pomegranate and muskmelon) selected for study, losses were in the range of **6.02** percent in pineapple to **15.05** percent in guava. Harvesting and sorting/ grading in farm operations and improper handling and storage conditions majorly contributed towards losses. This indicated that better harvesting tools, proper handling and effective cold chain infrastructure was essential for further reduction in post-harvest losses.

Among 14 ***vegetables*** (cabbage, cauliflower, green peas, mushroom, onion, potato, tomato, tapioca, bottle gourd, brinjal, beans, radish, capsicum and okra) the estimated losses ranged from **4.82** percent in tapioca to **11.61** percent in tomato. Farm operations such as harvesting and sorting/grading resulted in post-harvest losses. Lack of proper storage and improper handling practices by various stakeholders in market channels were the major contributing factors towards losses. The cold chain infrastructure for vegetables in the country was still at a very nascent stage and needed special focus for reduction in post-harvest losses as well as for retention of good quality as desirable by the consumers.

The losses of ***plantation crops*** (areca nut, cashew nut, coconut and sugarcane) ranged from **3.72** percent in cashew nut to **7.33** percent in sugarcane. Operations such as harvesting and threshing contributed towards losses.

In the case of ***spices*** (black pepper, coriander, chilli and turmeric), the estimated post-harvest losses were in the range of **1.29** percent in black pepper to **6.11** percent in chilli. Shattering during harvesting and spillage at various stages contributed towards post-harvest losses.

The post-harvest loss of ***eggs*** was **6.03** percent attributed to mishandling of eggs at farm level and lack of cold storage facilities. Increase in the number of eggs processing units and significant shift to organized poultry farms resulted in a significant decline in the post-harvest losses in eggs.

The post-harvest loss of ***inland fish*** was **4.86** percent, whereas post-harvest loss of marine fish was **8.76** percent. Post-harvest loss in marine fish was majorly attributed by throwing the uneconomical fish after harvesting along with post-harvest losses at wholesaler and retailer levels, which can be overcome by strengthening of cold chain infrastructure including cold storages, ice plants, freezing plants.

The post-harvest loss of ***meat*** was **2.34** percent, whereas post-harvest loss of poultry meat was **5.63** percent. The overall improvement in farm level operations and significant shift of standalone slaughter houses to integrated facilities led to reduced post-harvest losses.

The post-harvest loss of ***milk*** was observed to be **0.87** percent. Improvement in handling of milk at farm level and in processing units may help in further reduction of post-harvest losses.

In comparison to post-harvest losses during 2014-15, the post-harvest losses during 2020-22 reduced significantly for 25 crops and non-significant reduction was observed in 17 crops. However, in 3 crops viz. black paper, turmeric and tapioca, estimated post-harvest losses were slightly higher than previous assessment in 2015.

As can be seen from **Table 9.1**, the average range of losses for all crops and commodities was found to be 15.05 percent in guava to 0.87 percent in milk.

Table 9.1 Category wise range of estimated overall national average post-harvest losses

Categories	Overall national average post-harvest losses (%)
Fruits	6.02 (pineapple) - 15.05 (guava)
Vegetables	4.87 (tapioca) - 11.61 (tomato)
Cereals	3.89 (maize) - 5.92 (sorghum)
Pulses	5.65 (pigeon pea) - 6.74 (chickpea)
Oilseeds	2.87 (cottonseed) - 7.51 (soybean)
Plantation crops & sugarcane	3.72 (cashew nut) – 4.41 (areca nut) 7.33 (sugarcane)
Spices	1.29 (black pepper) - 6.11 (chilli)
Milk	0.87
Egg	6.03
Meat	2.34 (meat) – 5.63 (poultry meat)
Inland fish	4.86
Marine fish	8.76

This study provided the estimates of post-harvest losses in various farm level operations and market channels including storage and processing units. It also presented the changes in the scenario of harvest and post-harvest losses over the past 7 years. The study has clearly brought out gaps identified in the post-harvest management and strategies to reduce the post-harvest losses. It can be pinpointed that the losses in farm level operations viz. harvesting and threshing need to be tackled at priority by standardization/ improvements in harvesting and threshing machinery.

Good farm level practices and existing modern machinery available for post-harvest practices may be promoted and made accessible to farmers. Processing of agricultural produce to suitable value added products and appropriate storage of excess production during glut can help to reduce the post-harvest losses and also may give relief from price fluctuation. Cold chain for perishable products and cold storage for cereal, pulses, oilseed, plantation crops and spices need to be promoted in a big way. Training and demonstration of modern technologies and good handling practices may be carried out through extension activities to reduce post-harvest losses. Interventions of favourable policies and development in infrastructure for processing and storage is needed for further reduction of post-harvest losses.

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APPENDICES

Appendix 1 Questionnaires/ schedules for collection of data for the estimation of post-harvest losses

Questionnaire for Cereals, Pulses, Oilseeds and Spices (Farm Level)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Crop Category Type	
3.	Crop/ commodity selected	
4.	State	
5.	District	
6.	Tehsil/Taluk	
7.	Block/Mandal	
8.	Village	

Schedule I (b): Brief Detail of Farmers

S. No	Particulars	Response
1.	Name of the respondent (Farmer)	
2.	Father's name	
3.	Name of head of household	
4.	Address	
5.	Contact	
6.	Total Operational Holding (in Ha)	
7.	Irrigated Land (in Ha)	
8.	Unirrigated Land (in Ha)	
9.	Category (Marginal, Small, Medium & Large)	
10.	Ownership type (ha)	
11.	Major crops grown	
12.	Total area under Cereals/Pulses/Oilseeds/Spices cultivation (Ha.)	
13.	Area under selected crop (Ha)	
14.	Cultivation variety of the selected Cereals/Pulses/Oilseeds/Spices	
15.	Method of Harvesting	
16.	Equipment used for Harvesting	
17.	Method of Threshing	
18.	Type of Extension Services	
19.	Source of planting material/seed	

FORM II: Harvest and Post-Harvest Losses at Farm Level (at producer level) – Interview Mode

Schedule II (a) Production Details

S. No.	Particular	Details
1.	Season of Crop	
2.	Month of Sowing	
3.	Month of Harvesting	
4.	Total Quantity Produced (MT)	

S. No.	Particular	Details
5.	Quantity Consumed (MT)	
6.	Marketable Surplus (MT)	
7.	Total Quantity Sold (MT)	

Schedule II (b): Harvest and Post-Harvest Loss at each level

Operation	Method of Operation	Equipment used	Gross weight of one lot (Kgs)	Quantity lost (Kgs)	Type of loss	Cause of loss
Harvesting/Picking						
Collection						
Threshing/dehusking/ Sorting & grading						
Winnowing/Sieving/ Cleaning						
Drying						
Packaging						
Storage						
Transport (From threshing floor to store and mandi)						

FORM III: Gap Assessment

S. No.	Operations	Details
1	Infrastructure gaps in harvesting	
2	Infrastructure gaps in threshing, sorting & grading and cleaning	
3	Infrastructure gaps in drying	
4	Infrastructure gaps in storage and marketing	
5	Infrastructure gaps in transportation	
6	Any other gaps?	

FORM IV: Harvest and Post-Harvest Losses at Farm Level (at Producer level) – Observation Mode

Schedule IV (a): Losses during harvesting (in the field)

S. No.	Particulars	Observation
1.	Method of harvesting	
2.	Equipment used for harvesting	
3.	Size of the sampled plot	
4.	Weight of the selected lot harvested (kg)	
5.	Losses observed during harvesting of the selected lot (kg)	
6.	Type of losses observed	
7.	Reason for the loss	

Schedule IV (b): Loss during threshing

S.No.	Particulars	Observation
1.	Type of threshing floor	
2.	Method of threshing	
3.	Equipment used for threshing	
4.	Weight of bundles/pods from the selected lot before threshing (kg)	
5.	Weight of grain obtained after threshing of the selected lot of bundles/pods (Kg)	
6.	Weight of straw obtained from the selected lot (Kg)	
7.	Loss of grains (in Kg)	

S.No.	Particulars	Observation
8.	Weight of grains lost with straw (250/500 g sample)	
9.	Type of losses	
10.	Causes of losses	

Schedule IV (c): Loss during cleaning/winnowing/sieving

S. No.	Particulars	Observation
1.	Method of cleaning/winnowing/sieving	
2.	Equipment used for cleaning/winnowing/sieving	
3.	Weight of sample lot before cleaning/winnowing/sieving (Kg)	
4.	Weight of grain after cleaning/winnowing/sieving of the lot (Kg)	
5.	Weight of straw and other material obtained during cleaning/winnowing/sieving (Kg)	
6.	Loss of grains (in Kg)	
7.	Weight of grains lost with straw (250/500 g sample)	
8.	Type of losses	
9.	Cause of losses	

Schedule IV (d) Loss during drying at the farm level (only in oilseeds)

S. No.	Particulars	Observation
1.	Weight of the lot observed during drying (in Kg)	
2.	Duration of drying (no. of days/ hrs)	
3.	Weight of grains spoiled/ damaged during drying (250 g sample)	
4.	Type of Loss	
5.	Cause of loss	

Schedule IV (e): Loss during Storage at the farm level

S. No.	Particulars	Observation
1.	Weight of the lot observed during storage (Kg)	
2.	Duration of storage (no. of days)	
3.	Weight of grains spoiled/damaged during storage (250 g sample)	
4.	Type of Loss	
5.	Cause of loss	

QUESTIONNAIRE FOR CEREALS, PULSES, OILSEEDS AND SPICES (WHOLESALER/RETAILER)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Crop Category Type	
3.	Crop/ commodity selected	
4.	State	
5.	District	
6.	Block	
7.	Village name	
8.	Tehsil/Taluk	
9.	Pin code	

Schedule I (b): Brief detail of the Wholesaler/Retailer

S. No.	Particulars	Response
1.	Whether Wholesaler or Retailer	
2.	Name of Concern	
3.	Address	
4.	Name of Contact Person	
5.	Contact No.	
6.	Type of Business	
	If Wholesaler,	
7.	Location	
8.	Crop / Commodity	
9.	Licence No.	
10.	Volume of Trade	
11.	Crop / Commodity	
12.	Volume of Trade	

Schedule II: Losses at Market Level (Wholesaler /Retailer) – Personal Interview Method

S. No	Particulars	Response
1.	Total quantity handled during one month (Kg)	
2.	Loss during cleaning/sorting/ grading/curing/blanching/sun drying/ packaging (Kg)	
3.	Loss during storage/ marketing (Kg)	
4.	Any other loss (please specify) (Kg)	
5.	Type of loss	
6.	Cause of losses	

Form III: Gaps in Infrastructure – Personal Interview Method

S. No.	Operations	Details
1.	Infrastructure gap in sorting/ grading	
2.	Infrastructure gap in storage and marketing	
3.	Any other gaps	

Form IV: Losses at Market Level (Wholesaler /Retailer) Observation method

Schedule IV (a) - Loss during cleaning/sorting/ grading/ packaging (Kg)

S. No.	Particulars	Response
1.	Quantity of lot handled (Kgs)	
2.	Quantity lost during cleaning(Kg)	
3.	Quantity lost during sorting, grading, packing etc. (Kg)	
4.	Any other loss (Kg)	
5.	Type of loss	
6.	Causes of losses	

Schedule IV (b) Loss during Storage / Marketing

S. No.	Particulars	Response
1.	Quantity of sample (Kg)	
2.	Quantity lost during storage (Kg)	
3.	Quantity lost during marketing/handling (Kg)	
4.	Any other loss (Kg)	
5.	Type of loss	

S. No.	Particulars	Response
6.	Causes of losses	

QUESTIONNAIRE FOR CEREALS, PULSES, OILSEEDS AND SPICES (STORAGE)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No	Particulars	Response
1.	Agro-climatic zone	
2.	Category type	
3.	Crop/Commodity selected	
4.	State	
5.	District	
6.	Block	
7.	Village name	
8.	Tehsil/Taluk	
9.	Pin code	

Schedule I (b): Brief detail of the Warehouse/Silo

S. No	Particulars	Response
1.	Name of concern	
2.	Name of contact person	
3.	Address	
4.	Contact no.	
5.	Installed capacity (MT)	
6.	Type of warehouse	
7.	Type of registration	
8.	Registration no.	

FORM II: Post-harvest losses at Warehouse/Silo level – Personal Interview Method

Schedule II (a) Warehouse/ Silo

S. No.	Particulars	Response
1.	Unit of measurement	
2.	Previous year balance	
3.	Addition during enquiry period	
4.	Quantity withdrawal during enquiry period	
5.	Quantity lost	
6.	Type of storage	
7.	Period of storage (months)	
8.	Type of losses	
9.	Causes of losses	

FORM III: Gap Assessment – Personal Interview Method

S. No.	Operations	Details
1.	Infrastructure gap in storage	

FORM IV: Post-harvest losses at storage level – Observation Method

S. No	Particulars	Response
1.	Bag/ lot number	
2.	Variety of crop	
3.	Date of visit	
4.	Type of storage	
5.	Storage temperature (degree Celsius)	
6.	Relative Humidity (%)	
7.	Weight of sampled bags (in Kg)	
8.	Loss in 1 Kg sample	
9.	Date of next visit	
10.	Type of losses	
11.	Causes of losses	

QUESTIONNAIRE FOR CEREALS, PULSES, OILSEEDS AND SPICES (PROCESSING)

FORM I: Basic information **Date of survey**

Schedule I (a): Crop survey details

S. No	Particulars	Response
1.	Agro-climatic zone	
2.	Category type	
3.	Crop/Commodity selected	
4.	State	
5.	District	
6.	Block	
7.	Village name	
8.	Tehsil/Taluka	
9.	Pin code	

Schedule I (b): Brief detail of the Primary Processing Unit

S. No	Particulars	Response
1.	Name of concern	
2.	Name of contact person	
3.	Address	
4.	Contact number	
5.	Installed capacity	
6.	Type of storage at the unit	
7.	Type of registration	
8.	Registration number	

FORM II: Post-harvest losses at Primary Processing Level – Personal interview method

S. No	Particulars	Response
1.	Type of processing unit	
2.	Installed capacity per annum	
3.	Total quantity processed	
4.	Size of the lot (Kg)	
5.	Loss during transportation and handling (kg)	

S. No	Particulars	Response
6.	Loss during cleaning (Kg)	
7.	Loss during drying and cooling (Kg)	
8.	Processing losses (Kg)	
9.	Loss during sorting, grading and packaging (Kg)	
10.	Loss during storage (Kg)	
11.	Any other losses (please specify) (Kg)	
12.	Type of losses	
13.	Caused of losses	

FORM III: Gap Assessment – Personal Interview Method

S. No.	Operations	Details
1.	Infrastructure gap in storage	
2.	Infrastructure gap in processing facility	

FORM IV: Post-harvest losses at Primary Processing Level– Observation Method

S. No.	Particulars	Response
1.	Quantity of sample (Kg)	
2.	Quantity lost during storage (Kg)	
3.	Quantity lost during primary processing (Kg)	
4.	Any other loss (please specify) (Kg)	
5.	Type of losses	
6.	Cause of losses	

QUESTIONNAIRE FOR CEREALS, PULSES, OILSEEDS AND SPICES (TRANSPORTATION)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Category type	
3.	Crop/Commodity selected	
4.	State	
5.	District	
6.	Block/Mandal	
7.	Village name	
8.	Tehsil/Taluka	
9.	Pin code	

Schedule I (b): Brief detail of the transporter

S. No.	Particulars	Response
1.	Name of transporter	
2.	Address	
3.	Contact number	
4.	Type of fleet	

FORM II: Post-harvest losses at transport level – Personal interview method

S. No.	Particulars	Response
1.	Source site & address	
2.	Destination site & address	
3.	Date of loading	
4.	Distance travelled in km	
5.	Date of unloading	
6.	Measuring unit	
7.	Gross weight before loading	
8.	Gross weight after unloading	
9.	Weight of loss	
10.	Transit time	
11.	Type of losses	
12.	Cause of losses	

FORM III: Gap Assessment – Personal interview method

S. No.	Operations	Details
1.	Infrastructure gap in transportation	
2.	Any other gaps	

FORM IV: Post-harvest losses at transport level – Observation method

S. No.	Particulars	Response
1.	Name of the transporter/driver	
2.	Type of vehicle	
3.	Registration number of the vehicle	
4.	Transporter's / driver's mobile number	
5.	Source location (city)	
6.	Source location (pin code)	
7.	Destination	
8.	Destination location (pin code)	
9.	Distance between source and destination (Km)	
10.	Measuring unit	
11.	Total number of bags / vehicle	
12.	Total weight of grains	
13.	Date of loading	
14.	Date of arrival	
15.	Weight of quantity unloaded	
16.	Quantity of grains lost	
17.	Type of losses	
18.	Causes of losses	

QUESTIONNAIRE FOR FRUITS AND VEGETABLES (FARM LEVEL)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Crop Category Type	

S. No.	Particulars	Response
3.	Crop/ Commodity selected	
4.	State	
5.	District	
6.	Block	
7.	Village Name	
8.	Pin code	

Schedule I (b): Brief Detail of Farmers

S. No	Particulars	Response
1.	Name of head of household	
2.	Father's name	
3.	Address	
4.	Sub village	
5.	Contact	
6.	Total operational holding (in Ha)	
7.	Irrigated Land (in Ha)	
8.	Unirrigated Land (in Ha)	
9.	Category of farmer (Marginal, Small, Medium & Large)	
10.	Area under Fruits/Vegetables (Ha)	
11.	Ownership type	
12.	Major crops grown	
13.	Cultivation variety of the selected Fruit/Vegetable	
14.	Number of trees of selected fruit	
15.	Area under selected fruit/vegetable (Ha)	
16.	Method of Harvesting	
17.	Type of Extension Services	
18.	Source of planting material/seed	

FORM II: Harvest and Post-harvest Losses at Farm Level (at producer level) – Interview Mode

Schedule II (a) Production Details

S. No.	Particular	Details
1.	Month of harvesting	
2.	Measuring unit	
3.	Total quantity produced	
4.	Quantity consumed	
5.	Total quantity sold	

Schedule II (b): Harvest and Post-harvest Loss at each level

Operation	Methods of Operation	Equipment used	Gross weight of one lot (Kgs)	Quantity lost (Kgs)	Type of loss	Cause of loss
Harvesting						
Collection						
Precooling						
Sorting & Grading						
Trimming						
Cleaning						

Operation	Methods of Operation	Equipment used	Gross weight of one lot (Kgs)	Quantity lost (Kgs)	Type of loss	Cause of loss
Packaging						
Storage						
Transport						
Any other (specify)						

FORM III: Gaps in infrastructure

S. No.	Operations	Details
1	Harvesting	
2	Farm level infrastructure	
3	Transport level	
4	Storage level	
5	Market level	

FORM IV: Harvest and Post-Harvest Losses at Farm Level (at Producer level) – Observation Mode

Schedule IV (a): Losses during harvesting (in the field)

S.No.	Particulars	Observation
1.	Method of harvesting	
2.	Sample weight of one lot (Kg)	
3.	Weight of fruits/vegetables damaged during harvesting (Kg)	
4.	Number of fruits damaged during harvesting	
5.	Type of loss	
6.	Reason for the loss	

Schedule IV (b): Losses during sorting, grading and cleaning

S.No.	Particulars	Observation
1.	Method of operation	
2.	Sample weight of one lot (Kg)	
3.	Sample weight of fruits/vegetables damaged during the operation (Kg)	
4.	Number of fruits damaged during the operation	
5.	Type of loss	
6.	Reason for the loss	

Schedule IV (c): Loss during on farm storage

S.No.	Particulars	Observation
1.	Method of operation	
2.	Sample weight of one lot (Kg)	
3.	Sample weight of fruits/vegetables damaged during storage (Kg)	
4.	Number of fruits rejected during the operation	
5.	Type of loss	
6.	Reason for the loss	

Schedule IV (d) Loss during transportation

S.No.	Particulars	Observation
1.	Source name (loading point)	
2.	Destination name (unloading point)	
3.	Transit distance-source to destination (Km)	
4.	Mode of farm level transportation	
5.	Type of packaging	
6.	Sample weight of one lot transported (Kg)	
7.	Sample weight of fruits/vegetables spoiled during transportation (Kg)	
8.	Type of loss	
9.	Reason for the loss	

QUESTIONNAIRE FOR FRUITS AND VEGETABLES (WHOLESALER/RETAILER)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Category Type	
3.	Crop/ commodity selected	
4.	State	
5.	District	
6.	Block	
7.	Village name	
8.	Pin code	
9.	Type of respondent - Wholesaler or Retailer	

Schedule I (b): Brief detail of the Wholesaler/Retailer

S. No.	Particulars	Response
1.	Name of Concern	
2.	Type of Concern	
3.	Address	
4.	Name of contact person	
5.	Contact number	
6.	Location	
7.	Licence No.	
8.	Volume of trade	
9.	Crops/commodities traded	

Schedule II (a): Losses at Market Level (Wholesaler /Retailer) – Personal Interview Method

S. No	Particulars	Response
1.	Measuring unit	
2.	Previous stock available	
3.	Stock added during the current period	
4.	Total quantity traded during the current period	
5.	Stock available for trading	
6.	Loss during transportation and handling	
7.	Loss during storage/marketing	

S. No	Particulars	Response
8.	Loss due to ripening/rotting	
9.	Any other loss (please specify)	

Schedule II (b): Post-harvest losses in each operation at Market level

Operation	Method of operation	Gross weight of one lot (Kg)	Quantity lost (Kgs)	Type of loss	Cause of losses
Precooling					
Sorting and Grading					
Primary processing					
Ripening					
Waxing					
Bulk Packaging					
Retail Packaging					
Transport					
Storage					
Any other (please specify)					

Form III: Gaps in Infrastructure – Personal Interview Method

S. No.	Operations	Details
1.	Transport level	
2.	Storage level	
3.	Marketing level	
4.	Any other gaps	

Form IV: Losses at Market Level (Wholesaler /Retailer) Observation method

S.No.	Particulars	Response
1.	Weight of sample lot (Kg)	
2.	Number of commodity in the sample lot	
3.	Weight of spoiled fruit/vegetable in the selected lot (Kg)	
4.	Total number of sample commodities in which any post-harvest loss is observed	
5.	Type of loss	
6.	Causes of losses	

QUESTIONNAIRE FOR FRUITS AND VEGETABLES (STORAGE)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No	Particulars	Response
1.	Agro-climatic zone	
2.	Category type	
3.	Crop/Commodity selected	
4.	State	
5.	District	
6.	Block	
7.	Village name	

Schedule I (b): Basic information - Storage facility

S. No	Particulars	Response
1.	Name of concern	
2.	Type of concern	
3.	Address	
4.	Name of contact person	
5.	Contact number	
6.	Type of storage	
7.	Installed capacity	
8.	Type of registration	

FORM II: Post-harvest losses at Storage level – Personal Interview Method

Schedule II (a) Storage level

S. No.	Particulars	Response
1.	Measuring unit	
2.	Previous stock available	
3.	Addition during enquiry period	
4.	Quantity withdrawal during enquiry period	
5.	Quantity lost	
6.	Period of storage	
7.	Type of loss	
8.	Causes of loss	

FORM III: Gaps in Infrastructure – Personal Interview Method

S. No.	Operations	Details
1.	Transport level	
2.	Storage level	
3.	Other gaps	

FORM IV: Post-harvest losses at storage level – Observation Method

S. No	Particulars	Response
1.	Variety of the commodity	
2.	Date of visit	
3.	Type of storage	
4.	Storage temperature (degree Celsius)	
5.	Relative Humidity (%)	
6.	Gross weight of sample	
7.	Net weight of sample	
8.	No. of commodity in a sample	
9.	Physical appearance	
10.	Total number of fruits/vegetables in which any post-harvest loss is observed	
11.	Weight of fruits/vegetables in which any post-harvest loss is observed	
12.	Date of next visit	
13.	Type of loss	
14.	Cause of loss	

QUESTIONNAIRE FOR FRUITS AND VEGETABLES (PROCESSING)

FORM I: Basic information **Date of survey**

Schedule I (a): Crop survey details

S. No	Particulars	Response
1.	Agro-climatic zone	
2.	Category type	
3.	Crop/Commodity selected	
4.	State	
5.	District	
6.	Block	
7.	Village name	
8.	Pin code	

Schedule I (b): Brief detail of the Processing Unit

S. No	Particulars	Response
1.	Name of concern	
2.	Type of concern	
3.	Address	
4.	Name of contact person	
5.	Contact number	
6.	Type of activity	
7.	Installed capacity	

FORM II: Post-harvest losses at Processing Level – Personal interview method

S. No	Particulars	Response
1.	Measuring unit	
2.	Previous stock balance	
3.	Addition during the enquiry period	
4.	Quantity withdrawal during the enquiry period	
5.	Total quantity processed	
6.	Type of activity	
7.	Quantity lost	
8.	Period of storage	
9.	Type of loss	
10.	Causes of loss	

FORM III: Gaps in Infrastructure – Personal Interview Method

S. No.	Operations	Details
1.	Transport level	
2.	Storage level	
3.	Other gaps	

FORM IV: Post-harvest losses at Processing Level– Observation Method

S. No.	Particulars	Response
1.	Name of the process/activity	
2.	Variety of the commodity	
3.	Temperature in the processing unit (degree Celsius)	

S. No.	Particulars	Response
4.	Relative Humidity (%)	
5.	Initial gross weight of the sample lot (Kg)	
6.	Initial net weight of the sample lot (Kg)	
7.	Final gross weight of the sample lot (Kg)	
8.	Final net weight of the sample lot (Kg)	
9.	Total number of commodity in which any post-harvest loss is observed	
10.	Weight of commodity in which any post-harvest loss is observed (Kg)	
11.	Type of loss	
12.	Causes of loss	

QUESTIONNAIRE FOR FRUITS AND VEGETABLES (TRANSPORTATION)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No	Particulars	Response
1.	Agro-climatic zone	
2.	Category type	
3.	Crop/Commodity selected	
4.	State	
5.	District	
6.	Block	
7.	Village name	

Schedule I (b): Brief detail of the transporter

S. No.	Particulars	Response
1.	Name of transporter	
2.	Address	
3.	Contact number	
4.	Type of fleet	

FORM II: Post-harvest losses at transport level – Personal interview method

S. No.	Particulars	Response
1.	Source site name & address	
2.	Source district	
3.	Destination site name & address	
4.	Destination district	
5.	Date of loading	
6.	Total distance travelled (Km)	
7.	Gross weight before loading	
8.	Gross weight after unloading	
9.	Date of unloading	
10.	Transit time unit	
11.	Transit time	
12.	Type of loss	
13.	Cause of loss	

FORM III: Gaps in Infrastructure – Personal Interview Method

S. No.	Operations	Details
1.	Transport level	
2.	Other gaps	

FORM IV: Post-harvest losses at transport level – Observation method
Schedule IV (a): General details of the loading point

S. No.	Particulars	Response
1.	Source district	
2.	Source location (city)	
3.	Source location (pin code)	
4.	Destination district	
5.	Destination location (city)	
6.	Destination location (pin code)	
7.	Distance between source and destination (Km)	
8.	Total number of samples for loading	
9.	Total quantity at loading (Kg)	
10.	Date of loading	
11.	Registration number of the vehicle	
12.	Transporter mobile number	
13.	Estimated transit time unit	
14.	Estimated transit time	
15.	Date of reaching destination	

Schedule IV (b): Sample information at the loading point

S. No.	Particulars	Response
1.	Variety of commodity	
2.	Type of packaging of a sample	
3.	Gross weight of sample (Kg)	
4.	Net weight of sample (Kg)	
5.	No. of commodity in a sample	
6.	Physical appearance	
7.	Total number of fruits/vegetables in which any post-harvest loss is observed	
8.	Weight of fruits/vegetables in which any post-harvest loss is observed (Kg)	
9.	Type of loss	
10.	Cause of losses	

Schedule IV (c): General details of the unloading point

S. No.	Particulars	Response
1.	Source district	
2.	Source location (city)	
3.	Source location (pin code)	
4.	Destination district	
5.	Destination location (city)	
6.	Destination location (pin code)	
7.	Distance between source and destination (Km)	
8.	Total number of samples for unloading	
9.	Total quantity at unloading (Kg)	

S. No.	Particulars	Response
10.	Date of unloading	
11.	Registration number of the vehicle	
12.	Transporter mobile number	
13.	Estimated transit time unit	
14.	Estimated transit time	
15.	Date of reaching destination	

Schedule IV (d): Sample information at the unloading point

S. No.	Particulars	Response
1.	Variety of commodity	
2.	Type of packaging of a sample	
3.	Gross weight of sample (Kg)	
4.	Net weight of sample (Kg)	
5.	No. of commodity in a sample	
6.	Physical appearance	
7.	Total number of fruits/vegetables in which any post-harvest loss is observed	
8.	Weight of fruits/vegetables in which any post-harvest loss is observed (Kg)	
9.	Type of loss	
10.	Cause of losses	

QUESTIONNAIRE FOR PLANTATION (FARM LEVEL)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Crop Category Type	
3.	Crop/ commodity selected	
4.	State	
5.	District	
6.	Block/Mandal	
7.	Village	
8.	Pin code	

Schedule I (b): Brief Detail of Farmers

S. No	Particulars	Response
1.	Name of head of household	
2.	Address	
3.	Sub village	
4.	Mobile	
5.	Total area information: Irrigated Land (in ha.)	
6.	Total area information: Unirrigated Land (in ha.)	
7.	Ownership type (ha.)	
8.	Major crops grown	
9.	Cultivation variety of the selected plantation	
10.	Number of trees of selected crop for survey	
11.	Area under selected crop (ha)	

S. No	Particulars	Response
12.	Method of Harvesting	
13.	Source of planting material/seed	

FORM II: Harvest and Post-Harvest Losses at Farm Level (at producer level) – Interview Mode

Schedule II (a) Production Details

S. No.	Particular	Details
1.	Month of Harvesting	
2.	Measuring unit	
3.	Total Quantity Produced	
4.	Quantity Consumed	
5.	Total Quantity Sold (MT)	

Schedule II (b): Harvest and Post-Harvest Loss in each operation at farm level (Farm level by enquiry)

Operation	Methods of Operation	Equipment used	Gross weight of one lot (Kgs)	Quantity lost (Kgs)	Type of loss	Cause of loss
Infield handling/Harvesting applicability						
Collection applicability						
Sorting & grading applicability						
Cleaning applicability						
Drying applicability						
Packaging applicability						
Transport applicability						
Storage applicability						
Any other applicability						
Rate per unit quantity (Rs/kg)						

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1	Harvesting	
2	Farm level infrastructure	
3	Transport level	
4	Storage level	
5	Processing level	
6	Market level	

FORM IV: Post-Harvest Losses in each operation at farm level (Farm level by observation method)

Operation	Methods of Operation	Equipment used	Gross weight of one lot (Kgs)	Quantity lost (Kgs)	Type of loss	Cause of loss
Harvesting applicability						
Sorting & grading applicability						
Collection applicability						
Cleaning applicability						
Packaging applicability						
Storage applicability						
Transport applicability						

Operation	Methods of Operation	Equipment used	Gross weight of one lot (Kgs)	Quantity lost (Kgs)	Type of loss	Cause of loss
Any other applicability						
Rate per unit quantity (Rs/Kg)						

Schedule IV (a): Losses during harvesting (in the field)

S. No.	Particulars	Observation
1.	Method of harvesting	
2.	Sample weight of one lot (kg)	
3.	Weight of plantation lost, left on tree, field during harvesting (kg)	
4.	Type of loss	
5.	Reason for the loss	

Schedule IV (b): Loss during sorting grading

S. No.	Particulars	Observation
1.	Method of operation	
2.	Sample weight of one lot (kg)	
3.	Weight of plantation lost, left on tree, field during sorting and grading (kg)	
4.	Type of loss	
5.	Reason for the loss	

Schedule IV (c): Loss during collection

S. No.	Particulars	Observation
1.	Method of operation	
2.	Sample weight of one lot (kg)	
3.	Weight of plantation lost, left on tree, field during collection (kg)	
4.	Type of loss	
5.	Reason for the loss	

Schedule IV (d) Loss during cleaning

S. No.	Particulars	Observation
1.	Method of operation	
2.	Sample weight of one lot (kg)	
3.	Weight of plantation lost, left on tree, field during cleaning (kg)	
4.	Type of loss	
5.	Reason for the loss	

Schedule IV (e): Loss during packaging at the farm level

S. No.	Particulars	Observation
1.	Method of operation	
2.	Sample weight of one lot (kg)	
3.	Weight of plantation lost, left on tree, field during packaging (kg)	
4.	Type of loss	
5.	Reason for the loss	

Schedule IV (f): Loss during storage at the farm level

S. No.	Particulars	Observation
1.	Method of operation	
2.	Sample weight of one lot (kg)	

S. No.	Particulars	Observation
3.	Weight of plantation lost, left on tree, field during storage (kg)	
4.	Type of loss	
5.	Reason for the loss	

Schedule IV (e): Loss during transportation at the farm level

S. No.	Particulars	Observation
1.	Sample trader name	
2.	Source name	
3.	Destination name	
4.	Distance-source to destination (km)	
5.	Method of operation	
6.	Type of packaging	
7.	Layers of staking in vehicle	
8.	Total weight of produce transported (kg)	
9.	Total number of produce transported	
10.	Total weight of sample produce transported (kg)	
11.	Total number of sample produce transported	
12.	Weight of sample spoiled or rejected after unloading (kg)	
13.	Number of sample spoiled or rejected after unloading	
14.	Type of loss	
15.	Reason for the loss	

QUESTIONNAIRE FOR PLANTATION (WHOLESALER/RETAILER)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Crop Category Type	
3.	Crop/ commodity selected	
4.	State	
5.	District	
6.	Block/Mandal	
7.	Village name	
8.	Tehsil/Taluk	

Schedule I (b): Brief Detail of Wholesaler/Retailer

S. No	Particulars	Response
1.	Type of business	
2.	Name of concern	
3.	Type of concern	
4.	Address	
5.	Name of contact person	
6.	Contact number	
7.	Location	
8.	Licence number	
9.	Volume of trade	
10.	Crop/Commodity traded	

Schedule II (a): Post-harvest losses at market level (Personal Interview method)

S. No	Particulars	Response
1.	Measuring unit	
2.	Stock during last month	
3.	Stock added during the month	
4.	Stock withdrawn	
5.	Loss during transportation and handling	
6.	Loss during storage/marketing	
7.	Loss due to ripening/ rotting	

Schedule II (b): Post-harvest losses at market level (Personal Interview Method)

Operation	Method of Operation	Equipment used	Gross weight of one lot (kgs)	Quantity lost per lot during operation (Kgs)	Type of loss	Cause of losses
Bulk packaging applicability						
Retail packaging applicability						
Transport applicability						
Storage applicability						

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Transport level	
2.	Storage level	
3.	Market level	

Schedule II (b): Post-harvest losses at market level Observation method (Wholesaler/Trader/Retailer)

S. No	Particulars	Response
1.	Weight of fruit lot (kg)	
2.	Weight of spoiled/Damaged fruit in the selected lot (kg)	
3.	Type of losses	
4.	Cause of losses	

QUESTIONNAIRE FOR PLANTATION (TRANSPORTATION)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No	Particulars	Response
1.	Name of zonal coordinator	
2.	Agro-climatic zone	
3.	Category type	
4.	Crop/Commodity selected	
5.	State	
6.	District	
7.	Block/Mandal	
8.	Village name	
9.	Tehsil/Taluk	

Schedule I (c): Brief Detail of Transporter

S. No	Particulars	Response
1.	Name of transporter	
2.	Address	
3.	Contact number	
4.	Type of fleet	

Schedule II (a): Post-harvest losses at transport level (By enquiry)

S. No	Particulars	Response
1.	Source site and address	
2.	Destination site and address	
3.	Date of loading	
4.	Distance travelled (km)	
5.	Please indicate the measuring unit?	
6.	Gross weight before loading	
7.	Gross weight after unloading	
8.	Date of unloading	
9.	Transit time	
10.	Types of losses	
11.	Reasons of losses	

FORM III: Gaps in Infrastructure

Operations	Details
Transport level	

Schedule II (b): Post-harvest losses at market level in each operation at transport level (Observation method)

S. No	Particulars	Response
1.	Source location (City)	
2.	Source location (Pin code)	
3.	Destination district	
4.	Destination location (City)	
5.	Destination location (Pin code)	
6.	Distance travelled between source and destination (km)	
7.	Total number of samples for loading	
8.	Total quantity at loading (kg)	
9.	Date of loading	
10.	Transit time	
11.	Registration number of vehicles	
12.	Transporter mobile number	
13.	Estimated transit time unit	
14.	Estimated transit time	
15.	Date of reaching destination	

Schedule III (a): Post-harvest losses at Transport level loading point sample information

S. No	Particulars	Response
1.	Box/ Lot number	
2.	Type of packaging of a sample	
3.	Please indicate the measuring unit?	
4.	Sample unit	
5.	Number of commodity in a sample	
6.	Physical appearance e.g. colour, size	
7.	No. of the rotten commodity	
8.	No. of the healthy commodity	
9.	Weight of the rotten commodity	
10.	Weight. of the healthy commodity	
11.	Type of loss	
12.	Cause of loss	

Schedule IV (a): Post-harvest losses at Transport level unloading point sample information

S. No	Particulars	Response
1.	Box/ Lot number	
2.	Type of packaging of a sample	
3.	Please indicate the measuring unit?	
4.	Sample unit	
5.	Number of commodity in a sample	
6.	Physical appearance e.g. colour, size	
7.	No. of the rotten commodity	
8.	No. of the healthy commodity	
9.	Weight of the rotten commodity	
10.	Weight. of the healthy commodity	
11.	Type of loss	
12.	Cause of loss	

QUESTIONNAIRE FOR PLANTATION (STORAGE)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No	Particulars	Response
1.	Name of zonal coordinator	
2.	Agro-climatic zone	
3.	Category type	
4.	Crop/Commodity selected	
5.	State	
6.	District	
7.	Block/Mandal	
8.	Village name	
9.	Tehsil/Taluk	

Schedule I (c): Brief Detail of warehouse

S. No	Particulars	Response
1.	Name of the concern	
2.	Type of concern	
3.	Address	
4.	Name of contact person	
5.	Contact number	
6.	Type of storage	
7.	Installed capacity (MT)	
8.	Type of registration	

Schedule II (a): Post-harvest losses at cold storage/warehouse level (By enquiry)

S. No	Particulars	Response
1.	Measuring unit	
2.	Previous month balance	
3.	Addition during current month	
4.	Quantity withdrawal during current month	
5.	Quantity lost	
6.	Type of activity	
7.	Period of storage (months)	
8.	Type of losses	
9.	Causes of losses	

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Transport level	
2.	Storage level	

Schedule II (b): Post-harvest losses at market level in each operation at storage level (Observation method)

S. No	Particulars	Response
1.	Box/ lot number	
2.	Sample unit	
3.	Name of commodity	
4.	Type of storage	
5.	Storage temperature (Degree Celsius)	
6.	Relative humidity (%)	
7.	Gross weight of commodity (kg)	
8.	Total weight of healthy commodity (kg)	
9.	Total weight of rotten commodity (kg)	
10.	Date of next visit	

QUESTIONNAIRE FOR PLANTATION (PROCESSING)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No	Particulars	Response
1.	Name of zonal coordinator	
2.	Agro-climatic zone	
3.	Category type	
4.	Crop/Commodity selected	
5.	State	
6.	District	
7.	Block/Mandal	
8.	Village name	
9.	Tehsil/Taluk	

Schedule I (c): Brief Detail of Processing level

S. No	Particulars	Response
1.	Unit name	
2.	Address of unit	
3.	Year of establishment	
4.	Name of contact person	
5.	Contact number	
6.	Installed capacity	

Schedule II (a): Post-harvest losses at processing level (Personal interview)

S. No	Particulars	Response
1.	Measuring unit	
2.	Previous month balance	
3.	Addition during enquiry period	
4.	Quantity withdrawal during enquiry period	
5.	Type of storage	
6.	Quantity lost (kg)	
7.	Period of storage (Months)	
8.	Causes of losses	
9.	Revisit required	

Schedule II (b): Post-harvest losses at processing level (Personal interview)

Operation	Type of activities	Method of operation	Gross weight of one lot (kgs)	Quantity lost per lot during operation (Kgs)	Value of quantity (Rs/Kg)	Type of loss	Cause of losses
Collection applicability							
Dehusking/Deshelling applicability							
Steam roasting/dry roasting applicability							
Sorting/grading applicability							
Packaging applicability							

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Transport level	
2.	Storage level	
3.	Processing level	

Schedule II (b): Post-harvest losses at market level in each operation at processing level (Observation method)- Collection/Dehusking/Sorting and grading and Packaging operation

S. No	Particulars	Response
1.	Box/Lot number	
2.	Sample unit	
3.	Commodity name	
4.	Gross weight of the commodity (kg)	
5.	Weight of the spoiled commodity (kg)	
6.	Type of losses	
7.	Causes of losses	
8.	Date of next visit	

QUESTIONNAIRE EGG (FARM LEVEL)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Category Type	
3.	Crop/ commodity selected	
4.	State	
5.	District	
6.	Block/Mandal	
7.	Village	
8.	Pin code	

Schedule I (b): Brief Details of the Poultry Farm

S. No	Particulars	Response
1.	Name of the respondent/Farmer/Firm	
2.	Address	
3.	Sub village	
4.	Mobile	
5.	Type of birds	
6.	System of poultry rearing	
7.	Category	

S. No	Particulars	Response
8.	Type of farm	
9.	Ownership of farm	
10.	Method of eggs collection	
11.	Method of transportation to packaging yard	
12.	Type of extension service, Please specify	
13.	Fees source	

Schedule I (c): Losses at producer level (Farm level by Enquiry)

S. No	Particulars	Response
1.	Month of assessment	
2.	Total quantity produced (Numbers)	
3.	Quantity consumed (Numbers)	
4.	Total quantity sold (Numbers)	

FORM II: Post-harvest losses in each operation at Farm level (Farm level by Enquiry)

Operation	Methods of Operation	Equipment used	Gross weight of one lot (Kgs)	Quantity lost (Kgs)	Type of loss	Cause of loss
Handling applicability						
Cleaning, Sorting and grading						
Packaging applicability						
Storage applicability						
Transport applicability						
Rate per unit quantity (Rate/eggs in Rs)						

Form III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Poultry house level	
2.	Sorting grading	
3.	Packaging/Logistics level	

FORM IV: Post-Harvest Losses in each operation at eggs farm level (Farm level by observation method)

Schedule IV (a): Losses during collection

S. No.	Particulars	Observation
1.	Sample number	
2.	Sample farmers name	
3.	Date of survey	
4.	Sample weight of one lot (Measurement)	
5.	Sample weight of one lot (unit)	
6.	Total number of eggs collected	
7.	No. of eggs damaged/spoiled	
8.	Type of loss	
9.	Cause of loss	

Schedule IV (b): Losses during Packaging of eggs

S. No.	Particulars	Observation
1.	Total number of eggs packed in one lot	
2.	No. of eggs damaged/spoiled	
3.	Type of loss	
4.	Cause of loss	

Schedule IV (b): Losses during storage applicability

S. No.	Particulars	Observation
1.	Total number of eggs in the sample lot (marked) before storage (Number)	
2.	Duration of storage (No. of days)	
3.	Number of eggs in sample lot (Marked) after storage period (Number)	
4.	Type of loss	
5.	Cause of loss	

Schedule IV (c): Losses during transportation applicability (Poultry farm to Mandi/Retailer/Wholesaler

S. No.	Particulars	Observation
1.	**Date of visit**	
2.	Method of loading & Unloading	
3.	Mode of transport	
4.	Type of packaging	
5.	Source name (Village and district)	
6.	Destination name (Village/town and district)	
7.	Distance travelled (km)	
8.	Staking of vehicle	
9.	Total weight of one crate (kg)	
10.	Total number of crates transported (Number)	
11.	Number of the sample lot (Marked) at the time of loading (Number)	
12.	Number of the sample lot (Marked) at the time of unloading (Number)	
13.	Type of loss	
14.	Cause of loss	

QUESTIONNAIRE FOR EGG (WHOLESALER/ RETAILER LEVEL)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S.No.	Particulars	Response
1.	Name of Zonal Coordinator	
2.	Agro-climatic zone	
3.	Type of respondent	
4.	Type of survey	
5.	Category type	
6.	Commodity selected	
7.	State	
8.	District	
9.	Block	
10.	Village Name	
11.	Tehsil/ Taluka	
12.	Pin code	

Schedule I (b): Brief Detail of the Wholesaler/ Retailer

S. No	Particulars	Response
1.	Name of Concern	
2.	Address	
3.	Sub village	
4.	Name of contact person	
5.	Mobile	
6.	Type of Business	
7.	Commodity	
8.	Whether wholesaler or Retailer	
9.	Location	

FORM II (a): Losses at Market level (Wholesaler/ Retailer)- Personal interview method

S. No.	Particular	Details
1.	Total quantity handled during one week (Number)	
2.	Loss during unloading/sorting/grading/packaging (Number)	
3.	Loss during temporary storage at market (Number)	
4.	Any other loss, Please specify (Number)	
5.	Type of loss	
6.	Cause of loss	

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Infrastructure gap I cleaning/sorting/grading	
2.	Infrastructure gaps in storage	

Any other observations - ____________________________

FORM IV: Post-Harvest Losses Wholesaler/ Retailer/Trader level operational losses- Observation method
Form IV (a) Losses at Market level (Wholesaler/Retailer/Trader) Observation method – Sorting/ grading/Packaging

S. No.	Particulars	Observation
1.	Quantity of one lot handled (Number)	
2.	Quantity lost during sorting, grading, packing (Number)	
3.	Type of loss	
4.	Cause of loss	

Form IV (b) Losses at Market level (Wholesaler/Retailer/Trader) Observation method – Storage/marketing applicability

S. No.	Particulars	Observation
	Date of visit	
1.	Quantity of sample (Number) with the wholesaler/Retailer	
2.	Quantity lost during storage in godown of wholesale trader/Wholesale market (Number)	
3.	Quantity lost during marketing at wholesale/retail trader establishment (Number)	
4.	Type of loss	
5.	Cause of loss	

QUESTIONNAIRE FOR EGG (TRANSPORTER LEVEL)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Type of respondent	
3.	Type of survey	
4.	Category Type	
5.	Crop/ commodity selected	
6.	State	
7.	District	
8.	Block	
9.	Village Name	
10.	Tehsil/ Taluka	
11.	Pin code	

Schedule I (b): Brief Detail of the Transporter

S. No	Particulars	Response
1.	Name of Transporter	
2.	Address	
3.	Mobile	
4.	Type of Fleet	

FORM II: Post-harvest Losses of Egg at Transport level – Enquiry method

S. No.	Particular	Details
1.	Source site & address	
2.	Destination site & address	
3.	Date of loading	
4.	Distance travelled in Km	
5.	Gross weight of eggs before loading (MT)	
6.	Gross weight of eggs after unloading (MT) at destination	
7.	Date of unloading	
8.	Transit time	
9.	% of losses (By enquiry)	
10.	Type of loss	
11.	Cause of loss	

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Infrastructure gaps in transportation	

Schedule II (b): Losses at transport level – operational losses- observation method

S. No	Particulars	Response
1.	Name of the transporter/ driver	
2.	Type of vehicle	
3.	Registration number of the vehicle	
4.	Transporter's/ Driver's Mobile number	
5.	Source location (City)	
6.	Source location (Pin code)	
7.	Destination (Wholesale market/Processing)	
8.	Destination location (city)	
9.	Destination location (Pin code)	
10.	Distance between source and destination (km)	
11.	Total number of eggs	
12.	Date of loading	
13.	Date of Arrival	
14.	Quantity of eggs lost	
15.	Type of loss	
16.	Cause of loss	

QUESTIONNAIRE FOR EGG (STORAGE)

FORM I: Basic Information **Date of survey**

Schedule I (a): Crop Survey Details

S. No	Particulars	Response
1.	Name of zonal coordinator	
2.	Agro-climatic zone	
3.	Category type	
4.	Crop/Commodity selected	
5.	State	

S. No	Particulars	Response
6.	District	
7.	Block	
8.	Village name	
9.	Pin Code	
10.	Type of respondent	
11.	Type of survey	

Schedule I (b): Brief Detail of Warehouse/Cold store

S. No	Particulars	Response
1.	Name of concern	
2.	Name of contact person	
3.	Address	
4.	Sub village	
5.	Mobile	
6.	Type of concern	
7.	Installed capacity (MT)	
8.	Type of registration	

Schedule I (c): Post-harvest losses at Warehouse/Cold store – Personal interview method

S. No	Particulars	Response
1.	Previous week balance (Number)	
2.	Addition during enquiry period (Number)	
3.	Quantity withdrawal during enquiry period (number)	
4.	Total quantity stored during enquiry period (Number)	
5.	Type of storage (Normal/cold)	
6.	Quantity lost during enquiry period (Number)	
7.	Period of storage (Days)	
8.	Loss %	
9.	Type of loss	
10.	Causes of losses	

FORM II: Gaps in Infrastructure

S. No.	Operations	Details
1.	Storage	
2.	Processing facility	

Any other observations - ______________________________

Schedule III (a): Post-harvest losses at storage level operational losses – observation method

S. No	Particulars	Response
1.	Crate No.	
2.	Date of Visit	
3.	Type of storage	
4.	Storage temperature (degree Celsius)	
5.	Relative humidity (%)	
6.	Number of eggs in the crate	
7.	Observation of 5 eggs drawn from the sample crate	
8.	Quantity of eggs in the crate	
9.	Quantity lost during storage	
10.	Type of loss	
11.	Cause of losses	

QUESTIONNAIRE FOR MILK (FARM LEVEL)

FORM I: Basic Information **Date of surve**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Crop Category Type	
3.	Crop/ commodity selected	
4.	State	
5.	District	
6.	Block/Mandal	
7.	Village	
8.	Pin code	

Schedule I (b): Details at producer level: Basic details of the producer/ farmer in the village

S. No	Particulars	Response
1.	Name of the respondent	
2.	Name of head of household	
3.	Fathers name	
4.	Address	
5.	Sub village	
6.	Mobile	
7.	Total area information: Irrigated Land (in ha.)	
8.	Total area information: Unirrigated Land (in ha.)	
9.	Types of Animals	
10.	Total Animals	
11.	Ownership type (ha)	
12.	Method of milking of Animals	
13.	Method of collection	
14.	Type of extension services	
15.	Feed source	
16.	Breeding service	
17.	Veterinary service	

FORM II (a): Losses at producer level (Farm level by enquiry)

S. No.	Particular	Details
1.	Month of assessment	
2.	Total Quantity Produced (Litre)	
3.	Quantity Consumed (Litre)	
4.	Total Quantity Sold (Litre)	

Schedule II (b): Post milking losses in each operation at farm level (Farm level by enquiry)

Operation	Methods of Operation	Equipment used	Gross volume of one lot (Ltr)	Quantity lost (Ltr)	Type of loss	Cause of loss
Milking of animal applicability						
Direct sale to home applicability						
Transport to chilling centre applicability						

Operation	Methods of Operation	Equipment used	Gross volume of one lot (Ltr)	Quantity lost (Ltr)	Type of loss	Cause of loss
Processor level applicability						
Storage at farm level applicability						
Rate per unit quantity (Rs/ltr)						

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Producer level	
2.	Packaging/logistic level	
3.	Storage level	

Any other observations - ______________________________

FORM IV: Post-Harvest Losses in each operation at farm level (Farm level by observation method)

Operation	Percent loss	Type of loss	Cause of loss
Supply chain applicability			
Storage applicability			
Transportation applicability			
Any other applicability			
Rate per unit quantity (Rs/kg)			

Schedule IV (a): Losses during milking applicability

S. No.	Particulars	Observation
1.	Percent loss	
2.	Type of loss	
3.	Cause of loss	

Schedule IV (b): Loss during handling at producer level applicability

S. No.	Particulars	Observation
1.	Percent loss	
2.	Type of loss	
3.	Cause of loss	

Schedule IV (c): Loss during chilling centre applicability

S. No.	Particulars	Observation
1.	Percent loss	
2.	Type of loss	
3.	Cause of loss	

Schedule IV (d) Loss at processing unit applicability

S. No.	Particulars	Observation
1.	Percent loss	
2.	Type of loss	
3.	Cause of loss	

Schedule IV (e): Losses at storage applicability

S. No.	Particulars	Observation
1.	Weight of the sample lot before storage (Ltr)	
2.	Duration of storage (Number of days)	
3.	Weight of the sample lot after storage (Ltr)	

S. No.	Particulars	Observation
4.	Type of loss	
5.	Cause of loss	

Schedule IV (f): Loss during transportation

S. No.	Particulars	Observation
1.	Date of visit	
2.	Method of loading and unloading	
3.	Mode of transport	
4.	Type of packaging	
5.	Source name	
6.	Destination name	
7.	Distance travelled (km)	
8.	Staking in vehicle	
9.	Total weight of produce transported (Ltr)	
10.	Total weight of sample lot at the time of loading (Ltr)	
11.	Total weight of sample lot at the time of unloading (Ltr)	
12.	Type of loss	
13.	Cause of loss	

Schedule IV (g): Transportation (Dairy Farm to Collection centre) Applicability

S. No.	Particulars	Observation
1.	Date of visit	
2.	Method of loading and unloading	
3.	Mode of Transport	
4.	Type of packaging	
5.	Source name	
6.	Destination name	
7.	Distance-source to destination (km)	
8.	Stacking in vehicle	
9.	Total weight of sample produce transported (kg)	
10.	Weight of sample lot (Marked) at the time of loading (Ltr)	
11.	Weight of sample lot (Marked) at the time of unloading (Ltr)	
12.	Type of loss	
13.	Cause of loss	

QUESTIONNAIRE FOR MILK (WHOLESALER/ RETAILER LEVEL)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Name of Zonal Coordinator	
2.	Agro-climatic zone	
3.	Crop Category Type	
4.	Crop/ commodity selected	
5.	State	
6.	District	

S. No.	Particulars	Response
7.	Block	
8.	Village Name	
9.	Pin code	
10.	Tehsil/ Taluka	
11.	Type of respondent	
12.	Type of Survey	

Schedule I (b): Brief Detail of the Wholesaler/ Retailer

S. No	Particulars	Response
1.	Name of Concern	
2.	Address	
3.	Name of contact person	
4.	Mobile	
5.	Type of Business	

FORM II (a): Losses at Market level (Wholesaler/ Retailer/ Trader)

S. No.	Particular	Details
1.	Total quantity handled daily (Litres)	
2.	Loss during packaging (Litres)	
3.	Loss during Storage/ marketing (Litres)	
4.	Type of loss	
5.	Cause of loss	

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Storage level	

Any other observations - ______________________________

FORM IV: Post-Harvest Losses Retailer level operational losses- Observation method

S. No.	Particulars	Observation
1.	Quantity of sample (Litres)	
2.	Quantity lost during storage (Litres)	
3.	Quantity lost in Marketing (Litres)	
4.	Type of loss	
5.	Cause of loss	

QUESTIONNAIRE FOR MILK (TRANSPORTER LEVEL)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Name of Zonal Coordinator	
2.	Agro-climatic zone	
3.	Crop Category Type	
4.	Crop/ commodity selected	

S. No.	Particulars	Response
5.	State	
6.	District	
7.	Block	
8.	Village Name	
9.	Pin code	
10.	Tehsil/ Taluka	
11.	Type of respondent	
12.	Type of Survey	

Schedule I (b): Brief Detail of the Transporter

S. No	Particulars	Response
1.	Name of Transporter	
2.	Address	
3.	Mobile	
4.	Type of Fleet	

FORM II (a): Post-harvest Losses at Transport level – Personal interview method

S. No.	Particular	Details
1.	Source site & address	
2.	Destination site & address	
3.	Date of loading	
4.	Distance travelled in Km	
5.	Gross weight before loading (MT)	
6.	Gross weight after unloading (MT)	
7.	Date of unloading	
8.	Transit time	
9.	Type of loss	
10.	Cause of loss	

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Transportation level	

Schedule II (b): Losses at transport level by observation

S. No	Particulars	Response
1.	Name of the transporter/ driver	
2.	Type of vehicle	
3.	Registration number of the vehicle	
4.	Transporter's/ Driver's Mobile number	
5.	Source location (City)	
6.	Source location (Pin code)	
7.	Destination (Mandi/ unit)	
8.	Destination location (Pin code)	
9.	Distance travelled between source and destination (km)	
10.	Total weight of empty vehicle at source	
11.	Total weight of filled vehicle at source	

S. No	Particulars	Response
12.	Date of loading	
13.	Date of Arrival	
14.	Weight of empty vehicle after transporting	
15.	Weight of filled vehicle after transporting	

QUESTIONNAIRE FOR PLANTATION (PROCESSING)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No	Particulars	Response
1.	Name of zonal coordinator	
2.	Agro-climatic zone	
3.	Category type	
4.	Crop/Commodity selected	
5.	State	
6.	District	
7.	Block/Mandal	
8.	Village name	
9.	Pin Code	
10.	Tehsil/Taluk	
11.	Type of respondent	

Schedule I (b): Brief Detail of Processing unit

S. No	Particulars	Response
1.	Name of concern	
2.	Address	
3.	Name of contact person	
4.	Mobile	
5.	Type of concern	
6.	Installed capacity (MT)	
7.	Type of registration	

Schedule I (c): Personal level- Daily details- enquiry

S. No	Particulars	Response
1.	Month of assessment	
2.	Total quantity received (Litres)	
3.	Total quantity sold (Litres)	
4.	Type of unit	
5.	Quantity lost (Litres)	
6.	Type of loss	
7.	Causes of losses	

Schedule II (a): Post-harvest losses at processor level operational losses- Enquiry

Operation	Method of operation	Equipment used	Gross weight of one lot (Litres)	Quantity lost per lot during operation (Litres)	Type of loss	Cause of losses
Unloading applicability						
Processing applicability						
Packaging of milk applicability						
Storage applicability						

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Processing facility	

Any other observations - ______________________________

Schedule II (b): Post-harvest losses at processor level operational losses – observation

S. No	Particulars	Response
1.	Bag/Lot/ Crate number	
2.	Type of milk	
3.	Type of storage	
4.	Storage temperature (degree Celsius)	
5.	Relative humidity (%)	
6.	Number of Samples products	
7.	Observation of 5 pouches from sampled pooches	
8.	Quantity of sample (number)	
9.	Quantity lost during storage	
10.	Quantity lost in marketing (number)	

QUESTIONNAIRE FOR MEAT/ POULTRY MEAT (SLAUGHTER HOUSE)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Category Type	
3.	Crop/ commodity selected	
4.	State	
5.	District	
6.	Block/Mandal	
7.	Village	
8.	Tehsil/ Taluka	
9.	Pin code	
10.	Type of respondent	
11.	Type of survey	

Schedule I (b): Brief Details of slaughter facility

S. No	Particulars	Response
1.	Name of the slaughter facility	
2.	Address	
3.	Name of contact person	
4.	Contact number	
5.	Type of concern	
6.	Slaughter facility type	
7.	Installed capacity	

FORM II: Gaps in Infrastructure

S. No.	Operations	Details
1.	Live animal transport	
2.	Slaughtering & Packaging	
3.	Transport level	
4.	Storage	

QUESTIONNAIRE FOR MEAT/ POULTRY MEAT (WHOLESALER/ RETAILER LEVEL)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Name of Zonal Coordinator	
2.	Agro-climatic zone	
3.	Category Type	
4.	Crop/ commodity selected	
5.	State	
6.	District	
7.	Block	
8.	Village Name	
9.	Tehsil/ Taluka	
10.	Pin code	
11.	Type of respondent	
12.	Type of Survey	

Schedule I (b): Brief Detail of the Wholesaler/ Retailer

S. No	Particulars	Response
1.	Name of Concern	
2.	Address	
3.	Name of contact person	
4.	Contact number	
5.	Type of Business	
6.	Commodity	
7.	Volume of trade (average kg per month)	

FORM II (a): Losses at Market level (Wholesaler/ Retailer)

S. No.	Particular	Details
1.	Total quantity handled during one week (kg)	
2.	Loss during marketing (kg)	
3.	Loss during storage (kg)	
4.	Total quantity lost (kg)	
5.	Type of loss	
6.	Cause of loss	

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Marketing	
2.	Storage	

Any other observations - ______________________________

FORM IV: Post-Harvest Losses Wholesaler/ Retailer level operational losses- Observation method

S. No.	Particulars	Observation
1.	Quantity of sample (kg)	
2.	Quantity lost during storage (kg)	
3.	Quantity lost in Marketing (kg)	
4.	Type of loss	
5.	Cause of loss	

QUESTIONNAIRE FOR MEAT/ POULTRY MEAT (TRANSPORTER LEVEL)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Name of Zonal Coordinator	
2.	Agro-climatic zone	
3.	Category Type	
4.	Crop/ commodity selected	
5.	State	
6.	District	
7.	Block	
8.	Village Name	
9.	Tehsil/ Taluka	
10.	Pin code	
11.	Type of respondent	
12.	Type of Survey	

Schedule I (b): Brief Detail of the Transporter

S. No	Particulars	Response
1.	Name of Transporter	
2.	Address	
3.	Mobile	
4.	Type of Fleet	

FORM II : Post-harvest Losses of meat at Transport level – Enquiry method

S. No.	Particular	Details
1.	Source site & address	
2.	Destination site & address	
3.	Date of loading	
4.	Distance travelled in Km	
5.	Gross weight of material before loading (MT)	
6.	Gross weight of material after unloading (MT)	
7.	Date of unloading	
8.	Transit time	
9.	% of losses	
10.	Type of loss	
11.	Cause of loss	

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Infrastructure gaps in transportation	

Schedule II (b): Losses at transport level – operational losses- observation method

S. No	Particulars	Response
1.	Name of the transporter/ driver	
2.	Type of vehicle	
3.	Registration number of the vehicle	
4.	Transporter's/ Driver's Mobile number	
5.	Source location (City)	
6.	Source location (Pin code)	
7.	Destination (Chiller/ Meat processing unit)	
8.	Destination location (Pin code)	
9.	Distance between source and destination (km)	
10.	Total number of carcasses	
11.	Total weight of carcasses on loading (kg)	
12.	Date of loading	
13.	Date of Arrival	
14.	Total weight of carcasses on unloading (kg)	
15.	Quantity of carcass meat lost	
16.	Type of loss	
17.	Cause of loss	

QUESTIONNAIRE FOR MEAT/ POULTRY MEAT (PROCESSING)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No	Particulars	Response
1.	Name of zonal coordinator	
2.	Agro-climatic zone	
3.	Category type	
4.	Crop/Commodity selected	
5.	State	
6.	District	

S. No	Particulars	Response
7.	Block	
8.	Village name	
9.	Pin Code	
10.	Tehsil/Taluk	
11.	Type of respondent	
12.	Type of survey	

Schedule I (b): Brief Detail of Processing unit

S. No	Particulars	Response
1.	Name of concern	
2.	Address	
3.	Name of contact person	
4.	Mobile	
5.	Type of concern	
6.	Installed capacity (MT)	
7.	Type of registration	

Schedule I (c): Post-harvest losses at meat processing level enquiry

S. No	Particulars	Response
1.	Date of survey	
2.	Number of wholesome and hygienic carcasses received in the survey day for processing	
3.	Weight of carcasses received in the survey day for processing (kg)	
4.	Quantity lost during processing- physical/ chemical/ biological/ others (kg)	
5.	Type of loss	
6.	Causes of losses	

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Transport level	
2.	Processing unit	
3.	Storage	

Any other observations - ______________________________

Schedule II (b): Post-harvest losses at processor level operational losses – observation

S. No	Particulars	Response
1.	Lot number	
2.	Total number of carcasses in a lot	
3.	Sample unit (number of wholesome and hygienic carcasses)	
4.	Type of meat	
5.	Date of Visit	
6.	Process hall temperature (degree Celsius)	
7.	Process hall relative humidity (%)	
8.	Gross weight of sample carcasses (kg)	
9.	Quantity lost bones/ tendons/ cartilage in the carcass(kg)	
10.	Spoiled/ contaminated meat	
11.	Type of loss	
12.	Causes of losses	

QUESTIONNAIRE FOR MEAT/ POULTRY MEAT (STORAGE)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No	Particulars	Response
1.	Name of zonal coordinator	
2.	Agro-climatic zone	
3.	Category type	
4.	Crop/Commodity selected	
5.	State	
6.	District	
7.	Block	
8.	Village name	
9.	Tehsil/Taluk	
10.	Pin Code	
11.	Type of respondent	
12.	Type of survey	

Schedule I (b): Brief Detail of frozen cold storage/ Chiller

S. No	Particulars	Response
1.	Type of storage	
2.	Name of storage facility	
3.	Address	
4.	Name of contact person	
5.	Mobile	
6.	Type of concern	

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Storage level	

Any other observations - ____________________________

QUESTIONNAIRE FOR INLAND FISH (FARM LEVEL)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Crop Category Type	
3.	Crop/ commodity selected	
4.	State	
5.	District	
6.	Tehsil/Taluk	
7.	Block/Mandal	
8.	Village	
9.	Pin code	

Schedule I (b): Brief Detail of Fisherman

S. No	Particulars	Response
1.	Name of the beneficiary	
2.	Father's name	
3.	Address	
4.	Sub village	
5.	Contact	
6.	Total area of the land (in Ha)	
7.	Area of fish pond (in Ha/Acre)	
8.	Category (Small, Medium & Large)	
9.	Ownership of land	
10.	Type of culture	
11.	Species cultivated	
12.	Source of water body	
13.	Method of Harvesting	
14.	Equipment used for Harvesting	
15.	Type of Extension Services	
16.	Whether FFDA is functioning in the district	
17.	Services available from FFDA	
18.	Services available from fisheries department	
19.	Seed source	
20.	Subsidy available/Subsidy percentage	
21.	Kisan credit card availability	
22.	Interest subvention available	
23.	Name of the bank	
24.	Bank loan availed (Rs)	
25.	Interest rate of bank loan availed	
26.	Whether data maintained at farm level	

FORM II: Harvest and Post-Harvest Losses at Farm Level (at producer level) – Interview Mode

Schedule II (a) Production Details

S. No.	Particular	Details
1.	Month of Harvesting	
2.	Total Quantity Produced (MT/per year)	
3.	Quantity Consumed (MT/Per year)	
4.	Total Quantity Sold (MT/per year)	
5.	Spoilage (MT/Per year)	

Schedule II (b): Post-Harvest Losses in each operation (Farm level by enquiry)

S.No.	Operation	Methods of Operation	Equipment used	Gross weight of one lot (Kgs)	Quantity lost (Kgs)	Type of loss	Cause of loss
1	Harvesting/catching						
2	Collection						
3	Sorting & grading						
4	Packaging						
5	Storage						
6	Transport						
7	Any other (specify)						

FORM III: Gap Assessment

S. No.	Operations	Details
1	Infrastructure gaps in harvesting	

S. No.	Operations	Details
2	Infrastructure gaps in various operations at farm level	
4	Infrastructure gaps in storage	
5	Infrastructure gaps in transportation	
6	Infrastructure gaps in market	
7	Any other gaps?	

FORM IV: Harvest and Post-Harvest Losses at Farm Level (at Producer level) – Observation Mode

Schedule IV (a): Losses during harvesting (in the field)

S. No.	Particulars	Observation
1.	Method of harvesting	
2.	Equipment used for harvesting of fish	
3.	Total catch of fish (kg) on the date of visit	
4.	Weight of damaged fish during harvesting (Kg) on the date of visit	
5.	Type of losses observed	
6.	Reason for the loss	

Schedule IV (b): Loss during sorting

S. No.	Particulars	Observation
1.	Sample weight of one lot (kg)	
2.	Weight of spoiled fish (kg)	
3.	Causes of losses	

Schedule IV (b): Loss during drying

S. No.	Particulars	Observation
1.	Method of drying	
2.	Equipment used for drying	
3.	Time taken for drying (hrs/days)	
4.	Weight of the sample taken (kg)	
5.	Weight of damaged/spoilt fishes in the sample (kg)	
6.	Type of loss	
7.	Cause of loss	

Schedule IV (c): Loss during storage at fisherman

S. No.	Particulars	Observation
1.	Type of storage	
2.	Duration of storage (Days)	
3.	Weight of sample taken (kg)	
4.	Weight of the damaged/spoiled fishes in the sample (kg)	
5.	Type of loss	
6.	Cause of loss	

Schedule IV (c): Loss during transportation

S. No.	Particulars	Observation
1.	Source name (loading point)	
2.	Destination name (unloading point)	
3.	Distance – Source to destination (km)	

S. No.	Particulars	Observation
4.	Method of operation	
5.	Type of packaging	
6.	Total weight of produce transported (kg)	
7.	Number of weight of sample spoiled or rejected after unloading (kg)	
8.	Cause of loss	

QUESTIONNAIRE FOR MARINE FISH (FARM LEVEL)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Crop Category Type	
3.	Crop/ commodity selected	
4.	State	
5.	District	
6.	Tehsil/Taluk	
7.	Block/Mandal	
8.	Village	
9.	Pin code	

Schedule I (b): Brief Detail of Fisherman

S. No	Particulars	Response
1.	Name of the boat owner	
2.	Registration number of boat	
3.	Father’s name	
4.	Address	
5.	Sub village	
6.	Contact	
7.	Size of the boat	
8.	Storage facility available in the boat	
9.	Duration of trip (hrs)	
10.	Time taken to reach the landing centre after catch	
11.	Name of the landing centre	
12.	Varieties of fish captured	
13.	Type of fishing craft (boat) used	
14.	Gear (net) for catching fish	
15.	Extension service	
16.	Subsidy available	
17.	Subsidy percentage	
18.	Additional facility required from fisheries department	
19.	Kisan credit card availability	
20.	Interest subvention available	
21.	Name of the bank	
22.	Bank loan availed (Rs)	
23.	Interest rate of bank loan availed	

FORM II: Harvest and Post-Harvest Losses at Farm Level (during fishing/capturing- fish catch) – Interview Mode
Schedule II Production Details

S. No.	Particular	Details
1.	Total fish catch (Quantity, MT)	
2.	Total sales (MT)	
3.	Total surplus (MT)	
4.	Total Bycatch (MT)	
5.	Total wastage	
6.	Type of storage for surplus catch	
7.	Whether cold storage is available in nearby area	
8.	Availability of ice in nearby areas	
9.	In there any shortage for ice?	

Schedule III (a): Post-Harvest Losses in each operation (Farm level by enquiry)

Operation	Methods of Operation	Equipment used	Gross weight of one lot (Kgs)	Quantity lost (Kgs)	Type of loss	Cause of loss
Harvesting/catching						
Sorting						
Unloading						
Drying						
Storage						
Transportation						
Any other (specify)						

FORM III: Gap Assessment

S. No.	Operations	Details
1	Infrastructure gaps in fishing	
2	Infrastructure gaps in landing centre	
4	Sorting	
5	Storage	
6	Transport	
7	Marketing	
8	Any other gaps	

FORM IV: Harvest and Post-Harvest Losses at fisherman level marine fish – Observation Mode

Schedule IV (a): Losses during fishing at the fisherman

S. No.	Particulars	Observation
1.	Method of fishing	
2.	Type of fishing craft used	
3.	Type of fishing equipment used	
4.	Facilities available in the boat & subsidy available for the boat, percentage	
5.	Subsidy received	
6.	Subsidy percentage	
7.	Varieties by catch	
8.	Total weight of fish catch before landing (kg)	
9.	Loss of fish in the craft before landing (kg)	
10.	Type of losses observed	
11.	Reason for the loss	

Schedule IV (b): Loss during Landing Centre

S. No.	Particulars	Observation
1.	Total weight of fish catch from the boat at the time of landing (kg)	
2.	Bycatch (kg)	
3.	Loss during transferring (kg)	
4.	Loss of fish at landing centre (kg)	
5.	Loss of fish during sorting at landing centre (kg)	
6.	Type of loss	
7.	Causes of losses	

Schedule IV (b): Loss during drying

S. No.	Particulars	Observation
1.	Method of drying	
2.	Equipment used for drying	
3.	Time taken for drying (hrs/days)	
4.	Weight of the sample taken (kg)	
5.	Weight of damaged/spoilt fishes in the sample (kg)	
6.	Type of loss	
7.	Cause of loss	

Schedule IV (c): Loss during storage

S. No.	Particulars	Observation
1.	Type of storage	
2.	Variety of fish stored	
3.	Duration of storage (Days)	
4.	Weight of sample taken (kg)	
5.	Weight of the damaged/spoiled fishes in the sample (kg)	
6.	Type of loss	
7.	Cause of loss	

Schedule IV (d): Loss during transportation

S. No.	Particulars	Observation
1.	Source name (loading point)	
2.	Destination name (unloading point)	
3.	Distance – Source to destination (km)	
4.	Method of operation	
5.	Type of packaging	
6.	Total weight of produce transported (kg)	
7.	Total weight of sample produced transported (kg)	
8.	Number or weight of sample spoiled or rejected after unloading (kg)	
9.	Cause of loss	

QUESTIONNAIRE FOR FISH (WHOLESALER/ RETAILER LEVEL)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Name of Zonal Coordinator	
2.	Agro-climatic zone	
3.	Category type	
4.	Crop/ Commodity selected	
5.	State	
6.	District	
7.	Block	
8.	Village Name	
9.	Tehsil/ Taluka	
10.	Pin code	
11.	Name of concern	
12.	Address	
13.	Name of contact person	
14.	Mobile	
15.	Type of Business	
16.	Location	
17.	Name of fish market	

FORM II (a): Losses at Market level (Wholesaler/ Retailer) - Personal interview method

S. No.	Particular	Details
1.	**Measuring unit**	
2.	Total quantity handled weight of fish purchased	
3.	Loss during sorting	
4.	Loss during transportation and handling	
5.	Loss during storage/ marketing	

FORM II (b): Operational losses at Market level (Wholesaler/ Retailer) - Personal interview method

Operation	Methods of Operation	Equipment used	Average period of operation	Unit period of operation	Gross weight of one lot (Kgs)	Quantity lost (Kgs)	Type of loss	Cause of loss
Sorting applicability								
Storage applicability								
Marketing applicability								
Any other applicability								

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Transport	
2.	Storage	
3.	Market	

Any other observations - ______________________________

FORM IV: Post-Harvest Losses Wholesaler/ Retailer level operational losses- Observation method

S. No.	Particulars	Observation
1.	Please indicate measuring unit	
2.	Weight of fish purchased- lot	
3.	Quantity lost during cleaning in the lot	
4.	Quantity lost during sorting in the lot	
5.	Weight of fish spoiled	
6.	Any other loss	
7.	Type of loss	
8.	Cause of loss	

QUESTIONNAIRE FOR FISH (TRANSPORTER LEVEL)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No.	Particulars	Response
1.	Agro-climatic zone	
2.	Category Type	
3.	Crop/ commodity selected	
4.	State	
5.	District	
6.	Block	
7.	Village Name	
8.	Tehsil/ Taluka	

Schedule I (b): Brief Detail of the Transporter

S. No	Particulars	Response
1.	Name of Transporter	
2.	Address	
3.	Mobile	
4.	Type of Fleet	

FORM II: Post-harvest Losses of fish at Transport level – Enquiry method

S. No.	Particular	Details
1.	Source site & address	
2.	Destination site & address	
3.	Date of loading	
4.	Distance travelled in Km	
5.	Measuring unit	
6.	Total weight of fishes at the time of loading (Kg/MT)	
7.	Gross weight of at unloading	
8.	Date of unloading	
9.	Transit time	
10.	Type of packaging used for transportation	
11.	Whether ice is used for packing	
12.	Type of loss	
13.	Cause of loss	

FORM III: Gaps in Infrastructure

S. No.	Operations	Details
1.	Transportation level	

Schedule II (b): Losses at transport level – operational losses- observation method

S. No	Particulars	Response
1.	Source location (City)	
2.	Source location (Pin code)	
3.	Destination district	
4.	Destination location (city)	
5.	Destination location (Pin code)	
6.	Distance between source and destination (km)	
7.	Total number of samples for loading	
8.	Total quantity at loading (kg)	
9.	Date of loading	
10.	Transit time	
11.	Date of reaching destination	
12.	Registration number of the vehicle	
13.	Mobile number of transporter	
14.	Lot number/ box	
15.	Type of Packaging	
16.	Sample unit	
17.	Variety fish	
18.	Type of loss	
19.	Cause of loss	

QUESTIONNAIRE FOR FISH (STORAGE PROCESSING LEVEL)

FORM I: Basic Information **Date of survey:**

Schedule I (a): Crop Survey Details

S. No	Particulars	Response
1.	Name of zonal coordinator	
2.	Agro-climatic zone	
3.	Category type	
4.	Crop/Commodity selected	
5.	State	
6.	District	
7.	Block	
8.	Village name	
9.	Tehsil/ Taluka	

Schedule I (b): Brief Detail of Cold storage/ primary processing

S. No	Particulars	Response
1.	Name of cold storage	
2.	Type	
3.	Address	
4.	Name of contact person	
5.	Contact number	

S. No	Particulars	Response
6.	Type of storage	
7.	Registration number	
8.	Installed capacity (MT)	
9.	Maximum utilisation (MT)	
10.	Rental facility of storage of fish in cold storage	

Schedule II: Post-harvest losses at storage and processing level- Operational losses

Schedule II (a): Losses at pre-processing cold storage level

S. No	Particulars	Response
1.	Measuring unit	
2.	Weight of fish at pre-processing unit	
3.	Spoiled fish/ wastage	
4.	Time taken for cleaning one basket	
5.	Causes of losses	

Schedule II (b): Losses at processing level

S. No	Particulars	Response
1.	Type of processing unit	
2.	Installed capacity (MT) per annum	
3.	Total quantity processed (MT) per annum	
4.	Time taken for cleaning one basket	
5.	Please indicate the measuring unit	
6.	Size of the lot	
7.	Loss during transportation and handling	
8.	Loss during storage	
9.	Any other losses during processing activities	
10.	Type of loss	
11.	Cause of loss	

FORM II: Gaps in Infrastructure

S. No.	Operations	Details
1.	Transport	
2.	Storage	

Any other observations - ______________________________

Schedule III (a): Post-harvest losses at pre-processing/cold storage level operational losses – observation method

S. No	Particulars	Response
1.	Box/ Lot number	
2.	Sample unit	
3.	Variety of fish	
4.	Date of visit	
5.	Type of storage	
6.	Storage temperature (degree Celsius)	
7.	Relative humidity (%)	
8.	Please indicate the measuring unit	
9.	Total weight of fish in a lot	

S. No	Particulars	Response
10.	Total weight of spoiled fish	
11.	Type of loss	
12.	Cause of losses	

Appendix 2 Sample size (no. of respondents) for estimation of loss in farm operations at the national level

S.No	Crops	Harvesting	Collection	Sorting/ Grading	Trimming	Threshing	Winnowing/ cleaning	Drying	Packaging	Transportation	Storage
Grains (Cereals, Millets, Pulses, Oilseeds)											
1	Paddy	2400	2045	-	-	2261	2267	2256	2268	2400	2233
2	Wheat	1900	1402	-	-	1462	1646	1493	1847	1783	1778
3	Maize	2300	2300	-	-	2165	2165	2160	2125	2300	2300
4	Bajra	1800	1800	-	-	1800	1800	1800	1800	1800	1800
5	Sorghum	2400	2400	-	-	2400	2400	2400	2400	2400	2400
6	Pigeon pea	1400	1400	-	-	1400	1400	1400	1400	1400	1400
7	Chick pea	1900	1900	-	-	1900	1900	1900	1900	1900	1900
8	Black gram	1200	1165	-	-	1165	1165	1035	1123	1200	1200
9	Green gram	1800	1800	-	-	1800	1800	1800	1800	1800	1800
10	Mustard	1500	1500	-	-	1500	1500	1500	1500	1500	1500
11	Cottonseed	500	500	-	-			500	500	500	500
12	Soybean	700	551	-	-	432	700	700	700	700	700
13	Safflower	500	500	500	-	500	500	500	500	500	500
14	Sunflower	900	900	900	-	900	900	900	900	900	900
15	Groundnut	1300	1300	1300	-	1300	1300	1300	1300	1300	1300
Fruit & Vegetables											
16	Apple	500	500	500	-	-	-	-	500	500	500
17	Banana	800	800	800	-	-	-	-	800	800	800
18	Citrus	900	900	559	-	-	-	-	448	716	900
19	Grapes	500	500	500	-	-	-	-	500	500	500
20	Guava	1003	1003	1003	-	-	-	-	1003	1003	1003
21	Mango	600	600	600	-	-	-	-	600	600	600
22	Papaya	500	500	442	-	-	-	-	496	500	191
23	Sapota	900	900	900	-	-	-	-	900	900	900
24	Pineapple	600	600	600	-	-	-	-	600	600	600
25	Pomegranate	600	600	600	-	-	-	-	600	600	600
26	Muskmelon	700	700	700	-	-	-	-	384	700	700
27	Cabbage	1200	1200	1200	-	-	-	-	1200	1200	1200
28	Cauliflower	1200	1200	1200	-	-	-	-	1200	1200	1200
29	Green pea	800	800	800	-	-	-	-	800	800	800
30	Mushroom	1500	1500	1500	1500	-	1315	-	1488	1444	1482

S.No	Crops	Harvesting	Collection	Sorting/ Grading	Trimming	Threshing	Winnowing/ cleaning	Drying	Packaging	Transportation	Storage
31	Onion	700	700	700	-	-	-	-	700	700	700
32	Potato	1300	1300	1300	-	-	-	-	1300	1300	1300
33	Tomato	800	800	800	-	-	-	-	800	800	800
34	Tapioca	600	600	600	-	-	-	-	600	600	600
35	Bottle gourd	700	700	700	-	-	-	-	700	700	700
36	Brinjal	800	800	800	-	-	-	-	800	800	800
37	Beans	600	600	600	-	-	-	-	600	600	600
38	Radish	1000	1000	1000	-	-	-	-	1000	1000	1000
39	Capsicum	600	600	600	-	-	-	-	600	600	600
40	Okra	1000	1000	1000	-	-	-	-	1000	1000	1000
Plantation crops & Spices											
41	Areca nut	700	700	700	-	700	700	700	700	700	700
42	Black pepper	500	500	-	-	500	500	500	500	500	500
43	Cashew	600	600	-	-	600	600	600	600	600	12
44	Chilli	500	500	500	-	-	-	500	500	500	500
45	Coconut	602	602	-	-	-	582	560	601	570	553
46	Coriander	500	500	-	-	500	500	500	500	500	500
47	Sugarcane	1316	1316	-	-	-	1316	-	-	1317	-
48	Turmeric	500	500	500	-	-	500	500	500	500	10
Livestock Produce											
49	Egg	-	1064	906	-	-	-	-	743	229	1019
50	Inland Fish	1400	1400	1400	-	-	-	-	1400	1400	1400
51	Marine Fish	2400	2400	2400	-	-	-	2400	2400	2400	-
52	Meat	275	-	-	-	-	-	-	-	-	-
53	Poultry Meat	550	-	-	-	-	-	-	-	-	-
54	Milk	1300	-	-	-	-	-	-	-	632	1077
Total		**54046**	**51948**	**29110**	**1500**	**23285**	**27456**	**27904**	**50126**	**52394**	**48558**

Appendix 3 Sample size (no. of respondents) for estimation of loss at market operations at the national level

S.No.	Crops	Wholesaler	Retailer	Processing	Transport	Storage
Grains (Cereals, Millets, Pulses, Oilseeds)						
1	Paddy	48	48	48	48	48
2	Wheat	38	38	38	38	38
3	Maize	46	46	46	46	46
4	Bajra	36	36	36	36	36
5	Sorghum	48	48	48	48	48
6	Pigeon pea	28	28	28	28	28
7	Chick pea	38	38	38	38	38
8	Black gram	24	24	24	24	24
9	Green gram	36	36	36	36	36
10	Mustard	30	30	30	30	30
11	Cottonseed	30	30	30	30	30
12	Soybean	14	14	14	14	14
13	Safflower	10	10	10	10	10
14	Sunflower	18	18	18	18	18
15	Groundnut	26	26	26	26	26
Fruit & Vegetables						
16	Apple	10	10	10	10	10
17	Banana	16	16	16	16	16
18	Citrus	18	18	18	18	18
19	Grapes	10	10	10	10	10
20	Guava	18	18	18	18	-
21	Mango	12	12	12	12	12
22	Papaya	10	10	10	10	10
23	Sapota	18	18	18	18	18
24	Pineapple	12	12	12	12	12
25	Pomegranate	12	12	12	12	12
26	Muskmelon	14	14	-	14	14
27	Cabbage	20	20	20	20	20
28	Cauliflower	20	20	20	20	20
29	Green pea	16	16	9	16	-
30	Mushroom	30	30	-	30	-
31	Onion	14	14	14	14	14

S.No.	Crops	Wholesaler	Retailer	Processing	Transport	Storage
32	Potato	22	22	22	22	22
33	Tomato	16	16	16	16	-
34	Tapioca	10	10	10	10	0
35	Bottle gourd	14	14	-	14	-
36	Brinjal	16	16	-	16	-
37	Beans	12	12	-	12	12
38	Radish	20	20	-	20	-
39	Capsicum	12	12	-	12	-
40	Okra	20	20	-	20	20
Plantation crops & Spices						
41	Arecanut	14	14	12	14	12
42	Black pepper	10	10	10	10	10
43	Cashew	12	12	12	12	12
44	Chilli	10	10	0	10	10
45	Coconut	12	12	10	12	12
46	Coriander	10	10	10	10	10
47	Sugarcane	-	-	26	26	-
48	Turmeric	10	10	0	10	10
Livestock Produce						
49	Egg	19	19	-	19	19
50	Inland Fish	70	70	-	70	-
51	Marine Fish	120	120	72	120	72
52	Meat	11	11	11	11	-
53	Poultry Meat	11	11	11	11	-
54	Milk	-	26	26	26	-
Total		**1173**	**1199**	**907**	**1225**	**877**

Appendix 4 Extent of national coverage of crops/ livestock produce by sampling

S.No.	Crops	ACZ covered (numbers)	States covered (numbers)	Districts surveyed (numbers)	Production in surveyed districts (,000 tonnes) 2018-19	All India production (,000 tonnes) 2018-19	% of National production
Grains (Cereals, Millets, Pulses, Oilseeds)							
1	Paddy	13	13	24	21,005.98	136,306.45	15.41
2	Wheat	10	9	19	12,732.72	103,600.00	12.29
3	Maize	13	14	23	7,086.78	28,804.80	24.60
4	Bajra	9	8	18	3,544.53	8,660.00	40.93
5	Sorghum	6	6	24	1,398.49	3,833.28	36.48
6	Pigeon pea	5	5	14	1,246.55	3,312.16	37.64
7	Chick pea	6	6	19	862.67	9,940.00	8.68
8	Black gram	5	7	12	466.09	3,220.92	14.47
9	Green gram	7	8	18	427.71	2,546.13	16.80
10	Mustard	7	6	15	3,352.46	9,255.70	36.22
11	Cottonseed	6	6	15	682.40	3,121.88	21.86
12	Soybean	4	4	7	2,056.53	12,415.42	16.56
13	Safflower	2	2	5	11.41	24.60	46.40
14	Sunflower	5	5	9	47.09	203.53	23.14
15	Groundnut	7	8	13	1,981.68	7,015.21	28.25
Fruit & Vegetables							
16	Apple	1	3	5	854.88	2,316.00	36.91
17	Banana	5	4	7	6,214.33	31,514.52	19.72
18	Citrus	4	5	9	4,012.00	13,183.71	30.43
19	Grapes	2	2	5	2,550.98	3,023.32	84.38
20	Guava	6	5	9	363.29	4,253.00	8.54
21	Mango	3	4	6	3,858.21	21,378.09	18.05
22	Papaya	3	3	5	1,375.49	5,856.61	23.49
23	Sapota	5	4	9	300.53	1,040.73	28.88

S.No.	Crops	ACZ covered (numbers)	States covered (numbers)	Districts surveyed (numbers)	Production in surveyed districts (,000 tonnes) 2018-19	All India production (,000 tonnes) 2018-19	% of National production
24	Pineapple	4	4	6	49.32	1,723.04	2.86
25	Pomegranate	2	3	5	1,430.96	3,114.45	45.95
26	Muskmelon	5	4	7	369.34	1,245.97	29.64
27	Cabbage	6	5	10	1,042.44	8,944.05	11.66
28	Cauliflower	6	5	10	1,318.71	8,974.74	14.69
29	Green pea	6	4	8	1,624.37	5,370.25	30.25
30	Mushroom*	7	8	15	-	182.00	-
31	Onion	4	4	7	6,605.62	21,636.53	30.53
32	Potato	6	6	11	12,089.70	50,085.30	24.14
33	Tomato	4	5	8	18,181.34	19,007.24	95.65
34	Tapioca	3	4	6	149.39	5,359.44	2.79
35	Bottle gourd	3	5	7	276.84	2,973.92	9.31
36	Brinjal	4	4	7	1,443.68	12,717.84	11.35
37	Beans	4	4	6	219.42	2,057.38	10.67
38	Radish	5	5	10	432.78	2,795.57	15.48
39	Capsicum	4	4	6	61.63	479.27	12.86
40	Okra	5	5	10	741.83	6,176.14	12.01
Plantation crops & Spices							
41	Arecanut	3	3	7	795.67	1,108.00	71.81
42	Black pepper	2	2	5	78.93	134.00	58.90
43	Cashew nut	3	4	5	70.11	743.00	9.44
44	Chilli	2	3	5	671.42	2,185.90	30.72
45	Coconut	3	4	6	2,880.77	13,648.97	21.11
46	Coriander	3	2	5	274.74	592.00	46.41
47	Sugarcane	7	3	13	116,428.36	401,114.45	29.03
48	Turmeric	2	2	5	192.47	1,977.91	9.73

S.No.	Crops	ACZ covered (numbers)	States covered (numbers)	Districts surveyed (numbers)	Production in surveyed districts (,000 tonnes) 2018-19	All India production (,000 tonnes) 2018-19	% of National production
Livestock Produce							
49	Egg (in lakh numbers)	10	14	19	237,637.00	1,143,827.00	20.78
50	Inland Fish	6	6	14	959.28	9,740.27	9.85
51	Marine Fish	5	8	24	1,930.29	3,855.29	50.07
52	Meat	8	8	11	301.75	6,908.92	4.37
53	Poultry Meat	7	9	11	794.56	4,061.78	19.56
54	Milk	8	9	13	12,181.12	187,749.00	6.49

* District wise production for mushroom was not available

Appendix 5 Post-harvest losses of crops/ commodities at district level

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
1	**Apple**	Kinnaur	7.57 ±0.2041	1.57 ±2.1773	9.14 ±2.1868
		Shimla	7.53 ±0.1361	1.45 ±0.9648	8.98 ±0.9744
		Nainital	9.54 ±0.9624	2.28 ±1.0298	11.83 ±1.4095
		Anantanag	7.53 ±0.1162	2.46 ±1.1134	9.99 ±1.1194
		Baramulla	7.5 ±0.4079	1.82 ±1.8477	9.32 ±1.8922
2	**Areca nut**	Davanagere	3.61 ±0.2575	0.76 ±0.0391	4.37 ±0.2604
		Tumkur	3.54 ±0.0462	0.75 ±0.0484	4.29 ±0.0669
		Kasargod	3.41 ±0.3366	0.81 ±0.136	4.21 ±0.363
		North Kannada	3.6 ±0.1466	0.83 ±0.4892	4.43 ±0.5107
		Shimoga	3.62 ±0.0277	0.7 ±0.0424	4.31 ±0.0506
		South Kannada	3.5 ±0.127	0.75 ±0.0771	4.25 ±0.1485
		Nicobars	6.46 ±0.5003	0.79 ±0.1971	7.25 ±0.5378
3	**Bajra**	Firozabad	3.45 ±0.0595	0.7 ±0.5732	4.15 ±0.5762
		Ghazipur	3.4 ±0.0664	0.79 ±0.5391	4.19 ±0.5432
		Agra	3.29 ±0.0651	0.73 ±0.536	4.02 ±0.5399
		Aligarh	3.48 ±0.0204	0.78 ±0.7581	4.26 ±0.7583
		Bhiwani	2.93 ±0.0831	0.8 ±0.6755	3.74 ±0.6806
		Mahendergarh	2.98 ±0.086	0.77 ±0.5538	3.75 ±0.5604
		Alwar	3.55 ±0.0073	0.75 ±0.565	4.3 ±0.5651
		Jaipur	3.77 ±0.1732	0.71 ±0.5363	4.48 ±0.5635
		Morena	3.81 ±0.0757	0.73 ±0.7138	4.55 ±0.7178
		Dausa	3.82 ±0.1964	0.83 ±0.627	4.65 ±0.657
		Dhule	3.38 ±0.1425	0.79 ±0.6084	4.17 ±0.6249
		Nashik	3.52 ±0.0437	0.85 ±0.7741	4.37 ±0.7753
		Koppal	4.23 ±0.2653	0.8 ±0.638	5.03 ±0.6909
		Prakasam	4.36 ±0.0338	0.75 ±0.5945	5.11 ±0.5955
		Banas Kantha	3.89 ±0.0858	0.89 ±0.7849	4.77 ±0.7896

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Kheda	4.12 ±0.036	0.87 ±0.8593	4.99 ±0.86
		Jalore	3.87 ±0.0425	0.64 ±0.5028	4.51 ±0.5046
		Sikar	3.81 ±0.0105	0.61 ±0.4422	4.42 ±0.4423
4	**Banana**	Jalgaon	3.96 ±0.0499	1.58 ±0.3339	5.54 ±0.3376
		Nanded	4.78 ±0.0394	2.25 ±0.5628	7.03 ±0.5642
		East Godavari	6.24 ±0.0469	3.09 ±0.7942	9.33 ±0.7955
		Tuticorin	5.82 ±0.3113	3 ±1.3554	8.81 ±1.3906
		Theni	3.98 ±0.1903	1.86 ±0.77	5.84 ±0.7932
		Anand	4.37 ±0.0437	1.91 ±0.6473	6.28 ±0.6488
		Bharuch	4.65 ±0.2672	2.31 ±0.5866	6.95 ±0.6446
5	**Beans**	Paraganas_North	2.61 ±0.0209	2.87 ±0.3834	5.48 ±0.384
		Kolar	4.42 ±0.0893	2.91 ±0.8676	7.33 ±0.8721
		Krishnagiri	4.36 ±0.0037	3.91 ±1.0067	8.27 ±1.0068
		Vellore	3.37 ±0.0225	3.73 ±0.745	7.1 ±0.7453
		Anand	2.96 ±0.0464	3.89 ±1.1196	6.85 ±1.1206
		Banas_Kantha	3.23 ±0.0582	3.46 ±0.8756	6.69 ±0.8776
6	**Black gram**	Budaun	4.46 ±0.0863	0.72 ±0.2866	5.18 ±0.2867
		Kaushambi	4.52 ±0.0374	0.68 ±0.2223	5.2 ±0.224
		Jhansi	6.13 ±0.0935	0.84 ±0.3088	6.97 ±0.3102
		Lalitpur	5.2 ±0.1238	0.94 ±0.2148	6.14 ±0.2151
		Bundi	4.99 ±0.0856	0.8 ±0.1804	5.8 ±0.1806
		Alirajpur	5.94 ±0.05	0.89 ±0.2824	6.83 ±0.2825
		Nanded	6.95 ±0.0252	0.85 ±0.2611	7.8 ±0.2653
		Washim	6.45 ±0.1201	0.73 ±0.1638	7.18 ±0.164
		Guntur	4.8 ±0.0141	0.91 ±0.2984	5.71 ±0.2992
		Krishna	4.67 ±0.0652	0.93 ±0.2164	5.59 ±0.2188
		Cuddalore	5.12 ±0.083	0.91 ±0.1868	6.03 ±0.1869
		Chhota Udaipur	4.05 ±0.0432	0.82 ±0.1561	4.87 ±0.1561
7	**Black Pepper**	Chikmagalur	1.08 ± 0.086	0.18 ± 0.0209	1.26 ± 0.0885

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Hassan	1.1 ± 0.0296	0.19 ± 0.0131	1.29 ± 0.0324
		Idukki	1.08 ± 0.0175	0.19 ± 0.0148	1.27 ± 0.0229
		Kodagu	1.1 ± 0.095	0.19 ± 0.0218	1.29 ± 0.0975
		South Kannada	1.12 ± 0.0428	0.19 ± 0.034	1.3 ± 0.0547
8	**Bottlegourd**	Muzaffarpur	4.46 ±0.0862	2.62 ±0.5932	7.07 ±0.5994
		Vaishali	4.74 ±0.0687	2.63 ±0.4602	7.37 ±0.4653
		Agra	4.22 ±0.1765	1.95 ±0.3186	6.17 ±0.3642
		Mainpuri	4.58 ±0.3133	2.02 ±0.5862	6.6 ±0.6646
		Durg	4.28 ±0.6967	2.49 ±1.6881	6.77 ±1.8262
		Jabalpur	5.2 ±0.7115	2.26 ±0.9288	7.46 ±1.17
		Sundargarh	5.44 ±0.0886	2.13 ±0.4749	7.57 ±0.4831
9	**Brinjal**	Murshidabad	5.26 ±0.0714	3.06 ±0.6126	8.32 ±0.6167
		Nadia	5.3 ±0.1125	2.36 ±1.0923	7.66 ±1.0981
		Nalanda	5.12 ±0.08	2.4 ±0.6285	7.52 ±0.6336
		Kalahandi	4.1 ±0.0341	3.01 ±0.794	7.11 ±0.7948
		Kendujhar	4.32 ±0.0676	2.94 ±0.8705	7.26 ±0.8731
		Anand	4.31 ±0.3949	2.57 ±0.6793	6.88 ±0.7858
		Vadodara	4.55 ±0.0832	2.42 ±0.5097	6.97 ±0.5165
10	**Cabbage**	Barpeta	5.38 ±0.1698	2.25 ± 0.3915	7.62 ± 0.4267
		Nagaon	5.68 ± 0.0419	1.81 ± 0.2322	7.49 ± 0.2359
		Murshidabad	5.64 ± 0.0386	3.27 ± 0.108	8.92 ± 0.1147
		Nadia	5.33 ± 0.0499	3.09 ± 0.5459	8.42 ±0.5482
		Muzaffarpur	8.12 ± 0.0053	2.28 ± 0.7734	10.4 ± 0.7734
		Vaishali	6.7 ± 0.062	1.72 ± 0.1634	8.42 ± 0.1748
		Kendujhar	6.1 ± 0.0845	2.03 ± 0.2131	8.13 ± 0.2293
		Kandhamal	5.04 ± 0.0556	2.47 ± 0.6072	7.51 ± 0.6097
		Rajkot	6.16 ± 0.0092	2.38 ± 0.5805	8.54 ± 0.5805
		Sabar Kantha	6.09 ± 0.053	2.29 ± 0.1778	8.39 ± 0.1856
11	**Capsicum**	Sirmaur	2.59 ± 0.3357	2.64 ± 0.3545	5.24 ±0.4882

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Solan	3.21 ± 0.4621	2.35 ± 1.0904	5.57 ± 1.1843
		Sangrur	1.49 ± 0.0937	2.82 ± 0.5496	4.31 ± 0.5575
		Jhabua	2.24 ± 0.0419	2.09 ± 0.5818	4.33 ±0.5833
		Banglore (Rural)	3.7 ± 0.0182	2.16 ± 0.370	5.86 ±0.3704
		Kolar	2.98 ±0.0306	2.85 ± 0.5368	5.83 ± 0.5377
12	Cashew nut	West Garo Hills	3.82 ±0.202	0.63 ±0.0874	4.45 ±0.2201
		South Garo Hills	3.83 ±0.1906	0.62 ±0.1821	4.46 ±0.2636
		Vishakhapatnam	4.2 ±0.0254	0.65 ±0.117	4.85 ±0.1197
		Srikakulam	4.19 ±0.0715	0.64 ±0.1581	4.82 ±0.1735
		South Kannada	2.61 ±0.1552	0.57 ±0.0988	3.17 ±0.184
		Kannur	2.52 ±0.03	0.55 ±0.11	3.07 ±0.114
13	Cauliflower	Murshidabad	5.34 ±0.0033	1.25 ±0.3471	6.59 ±0.3471
		Nadia	5.66 ±0.0209	1.18 ±0.3443	6.84 ±0.3449
		Muzaffarpur	5.66 ±0.0212	0.86 ±0.2172	6.52 ±0.2182
		Vaishali	5.47 ±0.0068	1.1 ±0.229	6.57 ±0.2291
		Chhindwara	5.16 ±0.0807	0.88 ±0.141	6.04 ±0.1625
		Indore	5.51 ±0.1025	1.14 ±0.2623	6.65 ±0.2817
		Mayurbhanj	5.89 ±0.4316	1.39 ±0.2703	7.28 ±0.5093
		Kandhamal	5.65 ±0.0854	1.5 ±0.2505	7.15 ±0.2647
		Banas Kantha	5.23 ±0.1016	1.23 ±0.2688	6.46 ±0.2874
		Sabar Kantha	6.11 ±0.0696	1.75 ±0.3877	7.86 ±0.3939
14	Chickpea	East Godavari	4.58 ±0.0113	1.3 ±0.2821	5.88 ±0.2824
		Guntur	4.68 ±0.0137	1.58 ±0.1909	6.25 ±0.1913
		Krishna	4.55 ±0.0329	1.08 ±0.1367	5.63 ±0.1407
		Prakasam	4.75 ±0.0179	1.25 ±0.3478	5.99 ±0.3482
		Bhiwani	5.04 ±0.1385	1.94 ±0.296	6.98 ±0.3268
		Hissar	5.92 ±0.1375	1.46 ±0.288	7.38 ±0.3192
		Chhindwara	4.98 ±0.1368	2.26 ±0.2447	7.24 ±0.2804
		Dewas	4.61 ±6.0803	1.9 ±0.2953	6.51 ±6.3756

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Neemuch	5.78 ±5.2771	1.93 ±0.213	7.71 ±5.4901
		Amravati	4.5 ±6.0008	1.13 ±0.2958	5.63 ±6.2966
		Nashik	5.21 ±4.4919	1.87 ±0.3171	7.07 ±4.809
		Sangli	4.96 ±4.6216	2.04 ±0.2502	7 ±4.8719
		Bhandara	5.02 ±5.165	1.94 ±0.4208	6.96 ±5.5858
		Cuttack	5 ±4.4899	1.83 ±0.3999	6.83 ±4.8897
		Ganjam	4.8 ±4.6105	2.24 ±0.4845	7.04 ±5.095
		Alwar	5.89 ±4.6159	2.91 ±1.0856	8.8 ±5.7015
		Udaipur	5.71 ±4.6079	1.86 ±0.4287	7.57 ±5.0366
		Baran	5.05 ±4.6031	3.34 ±0.655	8.38 ±5.2581
		Sikar	5.92 ±4.6042	2.41 ±0.6334	8.33 ±5.2376
15	Citrus	Muktsar	4.23 ±0.4142	1.96 ±0.5151	6.19 ±0.6609
		Fazilka	3.47 ±0.0965	1.56 ±0.2253	5.03 ±0.2451
		Chhindwara	4.83 ±0.1242	2.57 ±1.0369	7.4 ±1.0443
		Amravati	6.95 ±0.0388	2.68 ±0.5272	9.63 ±0.5287
		Aurangabad	6.17 ±0.3904	1.75 ±0.7363	7.92 ±0.8334
		Jalna	5.95 ±0.207	1.72 ±0.7001	7.67 ±0.7301
		Nagpur	4.47 ±0.1089	2.15 ±0.5148	6.63 ±0.5262
		Anantapur	5.84 ±0.0061	2.12 ±0.7306	7.97 ±0.7306
		Nalgonda	5.37 ±0.0652	2.35 ±0.2594	7.73 ±0.2675
16	Coconut	Nicobars	3.4± 0.2557	0.85± 0.8499	4.25± 0.8875
		Malappuram	3.22± 0.0279	0.95± 0.7044	4.17± 0.705
		Kozhikode	2.25± 0.0625	0.61± 0.4351	2.86± 0.4396
		East Godavari	3.23± 0.0204	1.36± 0.7502	4.58± 0.7505
		Hassan	3.05± 0.2007	0.79± 0.4776	3.84± 0.518
		Tumkur	3.06± 0.0183	0.64± 0.1782	3.69± 0.1791
17	Coriander	Baran	4.95 ± 0.0799	1.03 ± 0.0936	5.98 ± 0.1231
		Jhallawar	4.86 ± 0.1199	1.01 ± 0.053	5.87 ± 0.131
		Guna	4.51 ± 0.0864	0.99 ± 0.0741	5.5 ± 0.1139

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Mandsaur	4.92 ± 0.2497	1.02 ± 0.0622	5.93 ± 0.2574
		Rajgarh	5.03 ± 0.1154	1.01 ± 0.215	6.04 ± 0.244
18	**Cottonseed**	Guntur	2.57 ± 0.0075	0.46 ± 0.2646	3.03 ± 0.2647
		Khammam	2.63 ± 0.0297	0.41 ± 0.1318	3.03 ± 0.1351
		Nalgonda	2.61 ± 0.0082	0.42 ± 0.1711	3.03 ± 0.1713
		Bhavnagar	2.57 ± 0.0358	0.41 ± 0.2685	2.97 ± 0.2708
		Rajkot	2.53 ± 0.1047	0.38 ± 0.2701	2.91 ± 0.2897
		Vadodara	2.46 ± 0.0704	0.32 ± 0.2302	2.78 ± 0.2407
		Bhiwani	1.8 ± 0.1013	0.35 ± 0.1112	2.14 ± 0.1505
		Mahender-Garh	2.07 ± 0.1242	0.2 ± 0.1027	2.28 ± 0.1611
		Hanumang-Arh	1.92 ± 0.0133	0.29 ± 0.194	2.21 ± 0.1945
		Jodhpur	2.36 ± 0.034	0.41 ± 0.1684	2.76 ± 0.1718
		Alwar	1.97 ± 0.0224	0.26 ± 0.1843	2.23 ± 0.1857
		Bhilwara	2.78 ± 0.0512	0.41 ± 0.233	3.19 ± 0.2386
		Amravati	2.49 ± 0.0337	0.35 ± 0.1185	2.84 ± 0.1232
		Aurangabad	2.35 ± 0.136	0.34 ± 0.2336	2.69 ± 0.2703
		Jalgaon	2.67 ± 0.0156	0.66 ± 0.6496	3.32 ± 0.6498
19	**Dry Chilli**	Khammam	5.87 ±0.0016	0.7 ±0.1788	6.58 ±0.0908
		Warangal Rural	5.28 ±0.0148	0.64 ±0.4224	5.92 ±0.1914
		Guntur	5.2 ±0.0438	1.86 ±0.2353	7.06 ±0.1119
		Khargone	5.17 ±0.0204	0.79 ±0.5821	5.96 ±0.2444
		Barwani	6.54 ±0.0398	0.89 ±0.4748	7.43 ±0.1861
20	**Eggs**	Jammu	4.57 ± 0.2446	1.52 ± 0.2602	6.08 ± 0.3571
		Nagaon	4.29 ± 0.1114	1.99 ± 0.0447	6.28 ± 0.1201
		North 24 Pargana	5.29 ± 0.1311	1.8 ± 0	7.09 ± 0.1311
		West Bardhaman	5.23 ± 0.3245	1.99 ± 0	7.22 ± 0.3245
		Kanpur Nagar	4.68 ± 0.1603	1.69 ± 0	6.36 ± 0.1603
		Mirzapur	4.14 ± 0.2573	1.42 ± 0	5.56 ± 0.2573
		Jind	5.16 ± 0.375	1.58 ± 0.4747	6.74 ± 0.605

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Jabalpur	4.66 ± 0.4657	1.66 ± 0	6.33 ± 0.4657
		Pune	3.66 ± 0.0555	1.38 ± 0	5.04 ±0.0555
		Nashik	4.16 ± 0.0475	1.72 ± 0	5.87 ± 0.0475
		Namakkal	4.54 ± 0.1418	1.1 ± 0	5.64 ± 0.1418
		East Godavari	4.12 ± 0.0073	1.34 ± 0	5.46 ±0.0073
		Krishna	4.14 ± 0.0464	2.11 ± 0	6.25 ±0.0464
		Rangareddy	4.42 ± 0.0075	1.18 ± 0	5.61 ± 0.0075
		Siddipet	4.86 ±0.0708	2 ± 0	6.86 ±0.0708
		Bellary	4.53 ± 0.0092	1.38 ± 0	5.91 ±0.0092
		Ganjam	4.29 ± 0.066	1.7 ± 0	5.99 ± 0.066
		Vellore	3.41 ± 0.0438	1.7 ± 0	5.11 ± 0.0438
		Ernakulum	4.15 ± 0.0328	1.38 ± 0.0494	5.53 ±0.0594
21	Grapes	Nashik	4.6 ±0.0627	2.08 ±0.9964	6.69 ±0.9984
		Sangli	4.89 ±0.0782	2.28 ±0.1922	7.17 ±0.2075
		Solapur	5.65 ±0.0561	2.17 ±0.2132	7.82 ±0.2204
		Bijapur	5.75 ±0.2859	2.17 ±0.4057	7.92 ±0.4963
		Chikkaball-Apur	5.52 ±0.176	2.08 ±0.6177	7.6 ±0.6423
22	Meat	Kamrup	2.45 ± 0	0.99 ± 0	3.44 ± 0
		Nadia	1.46 ± 0	0.96 ± 0	2.41 ± 0
		Aligarh	1.69 ± 0	1.08 ± 0	2.77 ± 0
		Meerut	4.65 ± 0	1.05 ± 0	5.69 ± 0
		Unnao	4.52 ± 0	1.08 ± 0	5.6 ± 0
		Mewat	2.12 ± 0.2986	0.76 ± 0	2.88 ±0.2986
		Nashik	2.47 ± 0.0217	0.43 ± 0	2.91 ± 0.0217
		Aurangabad	1.94 ± 0.231	0.94 ± 0	2.87 ± 0.231
		Hyderabad	0.69 ± 0	0.93 ± 0	1.61 ± 0
		Kancheepuram	1.89 ± 0	1.16 ± 0	3.04 ± 0
		Ernakulum	1.49 ± 0	0.71 ± 0	2.2 ± 0
23	Groundnut	Anantapur	5.51 ± 0.005	1.07 ± 0.5064	6.58 ± 0.5064

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Junagadh	4.29 ± 0.0744	1.09 ± 0.6544	5.38 ± 0.6586
		Sabar Kantha	3.84 ± 0.0217	1 ± 0.4546	4.84 ± 0.4551
		Shivpuri	5 ± 0.0415	1.11 ± 0.4194	6.11 ± 0.4215
		Kolhapur	5.1 ± 0.0463	1.08 ± 0.4059	6.18 ± 0.4085
		Satara	4.92 ± 0.0144	1.06 ± 0.3924	5.98 ± 0.3927
		Bikaner	4.07 ± 0.0808	1.11 ± 0.3855	5.17 ± 0.3939
		Jaipur	3.98 ± 0.0442	0.91 ± 0.3311	4.88 ± 0.334
		Jodhpur	3.99 ± 0.0696	0.95 ± 0.406	4.94 ± 0.4119
		Vellore	4.38 ± 0.0414	0.92 ± 0.4579	5.3 ± 0.4598
		Villupuram	4.62 ± 0.0115	1.01 ± 0.5186	5.63 ± 0.5188
		Mahabubnagar	5.57 ± 0.0063	1.1 ± 0.2873	6.67 ± 0.2874
		Purba Medinipur	6.15 ± 0.0359	1.09 ± 0.3642	7.24 ± 0.3659
24	**Guava**	South 24 Paraganas	12.6 ±0.6929	4.01 ±2.8708	16.61 ±2.9532
		Rohtas	11.12 ±0.4949	3.4 ±5.0895	14.52 ±5.1135
		Nalanda	12.23 ±0.4047	3.26 ±2.6702	15.49 ±2.7007
		Badaun	11.3 ±0.193	3.67 ±4.1673	14.98 ±4.1718
		Prayagraj	11.78 ±0.1018	3.14 ±5.8673	14.91 ±5.8682
		Korba	10.66 ±4.4733	2.97 ±2.4503	13.63 ±5.1004
		Sehore	11.57 ±0.5763	2.55 ±3.2144	14.13 ±3.2656
		Rewa	11.55 ±0.8346	2.49 ±4.0296	14.04 ±4.1151
		Indore	11.02 ±2.5977	3.19 ±2.3357	14.21 ±3.4933
25	**Inland fish**	East Godavari	3.82 ± 0.0125	1.02 ± 0.0798	4.84 ± 0.0807
		Nellore	3.47 ± 0.0089	1.07 ± 0.0764	4.54 ± 0.0769
		Bhabua	4.46 ± 0.2803	1.04 ± 0.1227	5.51 ± 0.306
		Darbhanga	4.75 ± 0.0834	1.15 ± 0.6063	5.9 ± 0.612
		Belgaum	3.87 ± 0.0108	1.12 ± 0.0439	4.99 ± 0.0452
		Bijapur	3.7 ± 0.1662	1.02 ± 0.1046	4.72 ± 0.1964
		Cuttack	4.45 ± 0.2182	1.08 ± 0.1054	5.54 ± 0.2423
		Ganjam	4.58 ± 0.0734	1.03 ± 0.115	5.61 ± 0.1364

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Kalahandi	4.86 ± 0.0437	0.99 ± 0.1831	5.85 ± 0.1883
		Barabanki	3.86 ± 0.1989	1.13 ± 0.0621	4.99 ± 0.2084
		Basti	3.73 ± 0.0439	0.83 ± 0.0544	4.56 ± 0.0699
		Mirzapur	4.09 ± 0.0288	1.16 ± 0.1229	5.24 ± 0.1262
		Jalpaiguri	3.13 ± 0.045	1.01 ± 0.0685	4.13 ± 0.082
		Nadia	2.69 ± 0.0109	1 ± 0.0629	3.7 ± 0.0638
26	**Maize**	Kangra	1.94± 0.03	0.33±0.22	2.38±0.222
		Rajouri	2.11±0.044	0.42±0.859	2.53±0.86
		Cooch Behar	2.74±0.096	0.22±0.326	2.97±0.339
		North Dinajpur	3.51± 0.034	0.67±0.229	4.17±0.232
		Aurangabad	2.47± 0.089	0.6±0.196	3.06±0.215
		Begusarai	2.46± 0.067	0.98±0.499	3.44±0.504
		Bhagalpur	2.79±0.074	0.83±0.319	3.63±0.328
		Katihar	3.16±0.142	0.45±0.186	3.61±0.234
		Bulandshaher	3.88± 0.105	0.43±0.288	4.3±0.307
		Mainpuri	4.63±0.104	0.48±0.331	5.11±0.347
		Rupnagar	3.36±0.233	0.95±0.261	4.31±0.351
		Hoshiarpur	4.07±0.135	0.39±0.198	4.46±0.239
		Bhilwara	4.15±0.105	1.19±0.34	5.34±0.356
		Udaipur	2.94±0.06	0.61±0.64	3.55±0.642
		Chhindwara	3.16±0.111	1.14±0.574	4.3±0.584
		Chittorgarh	3.09±0.036	0.46±0.247	3.54±0.25
		Nashik	4.1±0.033	0.38±0.086	4.48±0.092
		Guntur	3.98±0.028	0.48±0.094	4.46±0.098
		Vikarabad	2.01±0.03	1.53±0.725	3.54±0.726
		Perambalur	3.14±0.145	0.53±0.157	3.67±0.213
		Shimoga	1.64±0.155	0.87±0.509	2.51±0.532
		Dohad	3.56±0.105	0.44±0.378	3.99±0.393
		Rajsamand	4.12±0.224	0.47±0.19	4.6±0.294

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
27	**Mango**	Darbhanga	6.98 ±0.0799	3.42 ±0.5343	10.4 ±0.5402
		Lucknow	6.11 ±0.2825	2.94 ±0.3472	9.05 ±0.4476
		Saharanpur	6.96 ±0.238	3.36 ±0.5397	10.32±0.589
		Chittoor	5.25 ±0.0865	2.62 ±0.1742	7.87 ±0.1945
		Krishna	5.47 ±0.0587	2.87 ±0.4691	8.34 ±0.4727
		Ratnagiri	6.06 ±0.279	2.77 ±0.4322	8.82 ±0.5145
28	**Marine fish**	East_Godavari	6.87 ± 0.0122	1.47 ± 0.2828	8.34 ± 0.2831
		Guntur	6.94 ± 0.0295	1.54 ± 0.2954	8.49 ± 0.2968
		Krishna	6.94 ± 0.1381	1.6 ± 0.3178	8.54 ± 0.3465
		West_Godavari	7.37 ± 0.0796	1.61 ± 0.2332	8.98 ± 0.2464
		Amreli	6.77 ± 0.2914	1.86 ± 0.4299	8.63 ± 0.5193
		Gir_Somnath	6.52 ± 0	1.71 ± 0.4003	8.23 ± 0.4003
		Kutch	6.68 ± 0.0382	1.85 ± 0.2131	8.53 ± 0.2165
		Porbandar	6.7 ± 0.4127	1.61 ± 0.2681	8.31 ± 0.4921
		Dakshina_Kannada	7.02 ± 0	1.96 ± 0.3099	8.98 ± 0.3099
		Kannur	7.2 ± 0.4057	1.55 ± 0.241	8.74 ± 0.4719
		Kasaragod	7.38 ± 0.3415	1.85 ± 0.3028	9.23 ± 0.4564
		Kollam	6.75 ± 0.0435	1.75 ± 0.5012	8.5 ± 0.5031
		Thiruvananthapuram	7.3 ± 1.2031	1.83 ± 0.3689	9.13 ± 1.2583
		Mumbai	7.16 ± 1.5074	1.82 ± 0.3082	8.97 ± 1.5386
		Raigarh_Raigad	7.38 ± 0.1124	1.82 ± 0.4164	9.2 ± 0.4313
		Ratnagiri	7.09 ± 0.3095	1.91 ± 0.3852	9 ± 0.4941
		Balasore	7.86 ± 0.0268	1.87 ± 0.3245	9.73 ± 0.3256
		Jagatsinghapur	7.91 ± 0.3541	1.96 ± 0.3608	9.86 ± 0.5055
		Kancheepuram	6.86 ± 0.0981	1.92 ± 0.2616	8.77 ± 0.2793
		Kanyakumari	7.53 ± 0.1185	1.76 ± 0.3482	9.29 ± 0.3679
		Nagapattinam	6.78 ± 0.6882	1.86 ± 0.3287	8.64 ± 0.7627
		Ramanathapuram	7.27 ± 0.4974	2.01 ± 0.3551	9.28 ± 0.6112
		Tuticorin	7.29 ± 0.669	1.72 ± 0.3051	9.01 ± 0.7353

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		East_Medinipur	7.26 ± 0.0949	1.68 ± 0.2941	8.94 ± 0.3091
29	Milk	Udham Singh Nagar	0.89 ± 0.0507	0.26 ±0.0853	1.16 ± 0.0992
		Bankura	0.61 ± 0.0211	0.48 ± 0.0415	1.09 ±0.0466
		Meerut	0.52 ± 0.0194	0.41 ± 0.1441	0.93 ± 0.1454
		Saharanpur	1.07 ± 0.0359	0.57 ± 0.0001	1.64 ± 0.0359
		Karnal	0.55 ± 0.039	0.20 ± 0.1031	0.74 ± 0.1102
		Moga	0.51 ± 0.0326	0.12 ± 0.0285	0.63 ±0.0433
		Jaipur	0.98 ± 0.0563	0.37 ± 0.1344	1.35 ± 0.1458
		Indore	0.33 ± 0.0561	0.21 ± 0.0496	0.53 ±0.0749
		Pune	0.32 ± 0.0129	0.33 ± 0.0735	0.65 ±0.0746
		Chittoor	0.28 ±0.0029	0.26 ± 0.1546	0.54 ± 0.1546
		Guntur	0.35 ± 0.0079	0.15 ± 0.0539	0.52 ±0.0545
		Mehsana	0.16 ±0.0148	0.37 ±0.0946	0.53 ±0.0958
		Navsari	0.24 ±0.0249	0.27 ± 0.0841	0.52 ±0.0877
30	Green peas	Shimla	4.36 ± 0.2164	1.99 ± 0.3701	6.35 ±0.4287
		Kasganj	4.65 ± 0.1844	1.71 ± 0.4344	6.36 ± 0.4719
		Amritsar	4.77 ± 0.1197	1.89 ± 0.2684	6.67 ±0.2938
		Jalaun	4.68 ± 0.1776	1.75 ± 0.562	6.43 ±0.5893
		Lalitpur	4.75 ± 0.1294	1.51 ± 0.454	6.26 ± 0.472
		Mahoba	4.13 ± 0.3079	1.68 ± 0.518	5.82 ±0.6026
		Ratlam	4.29 ± 0.178	1.99 ± 0.6813	6.28 ±0.7042
		Ujjain	4.57 ± 0.124	1.00 ± 0.7435	5.56 ± 0.7538
31	Green gram	Bankura	5.38 ±0.0026	0.95 ±0.3541	6.33 ±0.3541
		Murshidabad	5.19 ±0.2976	0.78 ±0.2005	5.98 ±0.3589
		South 24 Parganas	5.23 ±0.0528	1.27 ±0.9422	6.5 ±0.9437
		Muzaffarpur	5.14 ±0.1732	0.91 ±0.4931	6.05 ±0.5227
		Hanumangarh	5.14 ±0.006	0.78 ±0.1998	5.92 ±0.1999
		Sri Ganganagr	5 ±0.0188	0.8 ±0.3292	5.8 ±0.3297
		Ajmer	4.75 ±0.0738	0.79 ±0.1288	5.54 ±0.1485

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Pali	5.65 ±0.1807	0.78 ±0.2476	6.43 ±0.3065
		Tonk	5.26 ±0.2093	0.77 ±0.0789	6.03 ±0.2237
		Buldana	5.03 ±0.0674	0.89 ±0.2737	5.92 ±0.2819
		Washim	5.12 ±0.0416	0.97 ±0.306	6.09 ±0.3088
		Prakasham	5.17 ±0.0272	0.88 ±0.2477	6.05 ±0.2492
		Cuttack	5.25 ±0.2535	0.94 ±0.2939	6.19 ±0.3881
		Puri	4.91 ±0.0078	0.91 ±0.3762	5.82 ±0.3763
		Khammam	5.37 ±0.0998	0.95 ±0.4494	6.32 ±0.4604
		Vikarabad	5.41 ±0.0348	0.91 ±0.4844	6.32 ±0.4856
		Jagatsinghpur	5.19 ±0.0572	0.89 ±0.315	6.08 ±0.3201
		Kutch	5.08 ±0.0988	0.88 ±0.282	5.97 ±0.2988
32	Mushroom	Kangra	3.61±0.5929	1.11±0.6302	4.73±0.8653
		Solan	5.62±0.241	0.98±0.169	6.6±0.2943
		Golaghat	3.89±0	0.59±0.1587	4.48±0.1587
		Bareilly	1.82±0.0571	0.51±0.5304	2.33±0.5335
		Bara Banki	1.73±0.2228	1.47±0.4387	3.2±0.4921
		Hoshiarpur	6.45±0.2342	0.75±0.5561	7.21±0.6034
		Ludhiana	5.05±0.8643	1.72±0.5509	6.77±1.0249
		Rohtak	7.83±0.1662	1.3±0.3193	9.13±0.36
		Sonipat	1.15±0.0475	0.85±0.1568	2.01±0.1639
		Bhandara	7.57±0.2532	0.85±0.2219	8.42±0.3367
		Kolhapur	7.29±0.5876	0.41±0.2809	7.7±0.6513
		Pune	16.21±0.3373	1.2±0.7549	17.41±0.8269
		Ganjam	13.11±0.8636	1.37±0.4508	14.48±0.9742
		Puri	10.44±0.5117	1.68±0.4001	12.12±0.6496
		Dharmapuri	10.12±0.3634	0.91±0.2192	11.03±0.4244
33	Muskmelon	Firozabad	3.68 ±0.0493	2.96 ±0.4323	6.64 ±0.4351
		Aligarh	3.91 ±0.0303	2.55 ±0.6927	6.46 ±0.6933
		Farrukhabad	4.78 ±0.0839	2.97 ±0.827	7.75 ±0.8313

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Kapurthala	3.25 ±0.0526	2.25 ±0.3668	5.5 ±0.3705
		Mandsaur	3.48 ±0.0866	2.28 ±0.4477	5.76 ±0.456
		Anantapur	4.16 ±0.0079	3.41 ±0.4997	7.58 ±0.4998
		Chittoor	4.35 ±0.0098	3.31 ±0.9019	7.66 ±0.902
34	**Okra**	Nalanda	3.89 ± 0.0265	2.57 ± 0.3343	6.47 ± 0.3354
		Vaishali	3.95 ± 0.0137	2.05 ± 0.296	6 ± 0.2963
		Surat	3.53 ± 0.1099	2.08 ± 0.1778	5.61 ± 0.209
		Tapi	3.55 ± 0.1398	2.25 ± 0.4894	5.81 ± 0.509
		Chhindwara	3.75 ± 0.1051	2.07 ± 0.3071	5.82 ± 0.3246
		Mandla	3.74 ± 0.061	2.09 ± 0.6623	5.83 ± 0.6651
		Baleshwar	3.92 ± 0.0816	2.29 ± 0.2185	6.21 ± 0.2333
		Bhadrak	3.96 ± 0.1385	2.38 ± 0.326	6.33 ± 0.3542
		Nadia	3.86 ± 0.0043	2.11 ± 0.2188	5.97 ± 0.2188
		South 24 Paragnas	3.85 ± 0.0592	2.38 ± 0.4506	6.23 ± 0.4545
35	**Onion**	Khandwa	5.87 ± 0.3216	2.24 ± 0.442	8.11 ± 0.5466
		Nashik	4.76 ± 0.0188	2.18 ± 0.1786	6.94 ± 0.1796
		Ahmednagar	4.65 ± 0.079	2.06 ±0.3482	6.71 ± 0.3571
		Bijapur	7.2 ± 0.3306	1.92 ± 0.4527	9.11 ± 0.5605
		Chitradurga	6.95 ± 0.0584	1.71 ± 0.4773	8.66 ±0.4808
		Gadag	7.03 ± 0.352	1.74 ± 0.7453	8.76 ±0.8242
		Bhavnagar	5.82 ± 0.0367	1.75 ± 0.4116	7.57 ± 0.4133
36	**Paddy**	Udham Singh Nagar	2.33 ±0.0335	0.57 ±0.3741	2.91 ±0.3741
		Nagaon	5.62 ±0.0709	0.53 ±0.2047	6.15 ±0.2049
		Sonitpur	5.43 ±0.0797	0.57 ±0.2829	6 ±0.283
		Barddhaman	4.54 ±0.0196	0.57 ±0.1078	5.11 ±0.1078
		Paschim Medinipur	4.68 ±0.0155	0.5 ±0.2046	5.19 ±0.205
		Aurangabad	4 ±0.0469	0.66 ±0.3149	4.67 ±0.3151
		Rohtas	4.03 ±0.1038	0.61 ±0.0905	4.64 ±0.0912
		Bara Banki	3.79 ±0.0483	0.57 ±0.2731	4.36 ±0.2732

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Shahjahanpur	3.27 ±0.0521	0.77 ±0.173	4.04 ±0.1749
		Ludhiana	2.54 ±0.0665	0.53 ±0.0753	3.07 ±0.0909
		Sangrur	2.79 ±0.113	0.54 ±0.1236	3.32 ±0.1238
		Bargarh	3.43 ±0.0157	0.59 ±0.2494	4.02 ±0.2516
		Gondiya	3.45 ±0.147	0.5 ±0.1111	3.95 ±0.1114
		Janjgir Champa	3.53 ±0.041	0.48 ±0.0834	4.02 ±0.097
		Shahdol	3.77 ±0.0647	0.51 ±0.3104	4.28 ±0.3119
		Kolhapur	2.82 ±0.0644	0.63 ±0.0803	3.45 ±0.0807
		East Godavari	4.96 ±0.056	0.78 ±0.1436	5.74 ±0.1438
		West Godavari	4.87 ±0.0348	0.74 ±0.1862	5.61 ±0.1864
		Ganjam	3.59 ±0.0379	0.71 ±0.2338	4.3 ±0.2342
		Thanjavur	5.36 ±0.0899	0.62 ±0.1044	5.98 ±0.1074
		Raigarh	5.06 ±0.0592	0.56 ±0.1071	5.61 ±0.1112
		Thane	3.94 ±0.0605	0.67 ±0.0988	4.61 ±0.099
		Ahmedabad	4.11 ±0.0352	0.59 ±0.167	4.7 ±0.1671
		Anand	3.98 ±0.0217	0.45 ±0.2964	4.43 ±0.2964
37	**Papaya**	Anantapur	4.41 ± 0.0032	2.19 ± 0.5493	6.6 ± 0.5493
		Chittoor	3.83 ± 0.0121	2.3 ± 0.5446	6.13 ± 0.5447
		Kadapa	4.09 ± 0.0353	2.36 ± 0.6052	6.45 ±0.6063
		Kachchh	2.96 ± 0.0781	2.85 ± 0.6773	5.81 ± 0.6818
		Sahebganj	3.45 ± 0.1553	3.28 ± 0.8971	6.73 ± 0.9105
38	**Pigeon pea**	Chandrapur	5.1 ± 4.1002	1.13 ± 0.5341	6.23 ± 4.1349
		Chhindwara	5.18 ± 4.3682	1.25 ± 0.61	6.43 ± 4.4106
		Narsinghpur	4.78 ± 4.077	1.01 ± 0.5503	5.78 ± 4.1139
		Singrauli	5.72 ± 4.3572	1.35 ± 0.7022	7.07 ± 4.4134
		Latur	5.97 ± 4.6585	1.42 ± 0.7569	7.4 ± 4.7196
		Amravati	3.68 ± 3.5877	1.13 ± 0.5844	4.8 ± 3.635
		Yavatmal	6.1 ± 4.8359	1.53 ± 0.7512	7.63 ± 4.8939
		Akola	4.82 ± 3.6164	1.18 ± 0.5799	6 ± 3.6626

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Gulbarga	3.76 ± 3.1752	0.67 ± 0.4189	4.43 ± 3.2027
		Bidar	3.92 ± 3.3219	0.85 ± 0.518	4.77 ± 3.3621
		Mahabubnagar	4.11 ± 3.4274	0.9 ± 0.4401	5.01 ± 3.4556
		Vadodara	3.69 ± 2.9845	0.58 ± 0.4461	4.27 ± 3.0176
		Bharuch	3.38 ± 2.8876	0.5 ± 0.2452	3.89 ± 2.898
		Panch Mahals	3.72 ± 3.2164	0.91 ± 0.468	4.63 ±3.2503
39	**Pineapple**	Darjeeling	5.67 ±0.1957	2.34 ±0.246	8.01 ±0.3143
		Dima Hasao	5.43 ±0.2769	1.75 ±0.4193	7.18 ±0.5025
		Dinajpur Uttar	3.11 ±0.0998	1.37 ±0.1591	4.47 ±0.1878
		Kishanganj	3.87 ±0.2278	2.05 ±0.5289	5.92 ±0.5759
		Shimoga	2.09 ±0.0528	1.25 ±0.4506	3.34 ±0.4537
		Uttar Kannada	1.9 ±0.0232	2.25 ±0.3785	4.15 ±0.3793
40	**Pomegranate**	Ahmednagar	3.02 ±0.0328	2.45 ±0.413	5.47 ±0.4143
		Nashik	4.43 ±0.108	2.4 ±0.5172	6.83 ±0.5284
		Solapur	3.67 ±0.0223	1.99 ±0.2229	5.65 ±0.224
		Anantapur	5.48 ±0.0367	3.21 ±0.5356	8.69 ±0.5368
		Chitradurga	5.53 ±0.0607	2.66 ±0.4499	8.19 ±0.4539
41	**Potato**	Bardhaman	5 ± 0.0084	1.03 ± 0.4822	6.03 ±0.4823
		Hooghly	4.69 ± 0.006	1.00 ± 0.4649	5.69 ±0.4649
		Nalanda	4.91 ± 0.008	0.89 ±0.3695	5.8 ± 0.3696
		Firozabad	4.71 ± 0.0053	0.91 ± 0.5726	5.61 ± 0.5726
		Agra	5.25 ± 0.0065	0.84 ± 0.6101	6.09 ± 0.6101
		Hathras	5.08 ±0.0054	1.01 ± 0.5151	6.09 ± 0.5151
		Hoshiarpur	5.11 ± 0.0034	0.89 ±0.6099	6 ± 0.6099
		Jalandhar	5.59 ± 0.0039	0.79 ± 0.6313	6.37 ± 0.6313
		Indore	5.92 ± 0.0069	1.13 ± 0.63	7.05 ± 0.6301
		Banas Kantha	5.17 ± 0.0054	1.02 ± 0.5988	6.19 ± 0.5988
		Sabar Kantha	4.58 ±0.0058	0.78 ± 0.546	5.36 ± 0.5461
42	**Poultry Meat**	Nagaon	1.91 ± 0.0321	4.7 ± 0	6.61 ± 0.0321

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		North 24 Parganas	2.38 ± 0.0431	3.03 ± 0	5.41 ± 0.0431
		Kanpur Nagar	2.88 ± 0.0544	4.36 ± 0	7.24 ±0.0544
		Jind	2.25 ± 0	2.38 ± 0	4.63 ± 0
		Pune	2.34 ± 0.078	2.54 ± 0	4.88 ± 0.078
		Nashik	2.35 ± 0.1495	2.31 ± 0	4.66 ± 0.1495
		East Godavari	3.17 ± 0	4.31 ± 0	7.48 ± 0
		Krishna	3.21 ± 0	4.09 ± 0	7.3 ± 0
		Rangareddy	3.39 ± 0	4.74 ± 0	8.12 ± 0
		Namakkal	2.3 ± 0	3.78 ± 0	6.08 ± 0
		Ernakulum	1.58 ± 0	3.75 ± 0	5.32 ± 0
43	**Radish**	Kamrup	3.31 ±0.1475	1.66 ±0.8183	4.97 ±0.8315
		Barpeta	3.96 ±0.3405	1.41 ±1.0497	5.38 ±1.1035
		Murshidabad	4.29 ±0.069	2.57 ±0.742	6.85 ±0.7452
		Nadia	4.3 ±0.0513	1.9 ±0.9618	6.2 ±0.9632
		Nalanda	2.73 ±0.0635	2.5 ±1.034	5.23 ±1.0359
		Vaishali	2.66 ±0.0657	2.55 ±0.3228	5.21 ±0.3294
		Sonipat	5.75 ±0.0342	2.26 ±0.3457	8 ±0.3474
		Karnal	4.26 ±0.0789	2.42 ±0.9576	6.68 ±0.9608
		Kondagaon	2.49 ±0.0468	3.85 ±0.4298	6.34 ±0.4323
		Korba	2.13 ±0.0755	4.64 ±0.4691	6.76 ±0.4752
44	**Rapeseed/ Mustard**	Murshidabad	6.03 ±0.0178	0.87 ±0.0778	6.9 ±0.0798
		Nadia	6.26 ±0.0695	1.08 ±0.0534	7.34 ±0.0876
		Mathura	5.76 ±0.6238	1.11 ±0.0557	6.87 ±0.6263
		Agra	5.65 ±0.2716	0.83 ±0.0781	6.48 ±0.2826
		Bhiwani	4.16 ±0.3398	0.81 ±0.0176	4.97 ±0.3403
		Sri Ganganagar	4.69 ±0.1141	0.95 ±0.0733	5.64 ±0.1357
		Bharatpur	3.8 ±0.0412	0.92 ±0.0539	4.72 ±0.0678
		Sawai_Madhopur	4.48 ±0.2007	0.86 ±0.0928	5.34 ±0.2211
		Alwar	3.98 ±0.041	0.8 ±0.0741	4.78 ±0.0847

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Tonk	4.83 ±0.1934	0.9 ±0.0536	5.73 ±0.2007
		Mandsaur	4.51 ±0.0494	0.71 ±0.04	5.22 ±0.0636
		Banas_Kantha	4.05 ±0.0142	0.81 ±0.1356	4.86 ±0.1363
		Mehsana	3.88 ±0.0551	0.84 ±0.0715	4.72 ±0.0902
		Jhunjhunu	4.14 ±0.0268	0.86 ±0.0789	5 ±0.0833
		Nagor	4.48 ±0.0264	0.8 ±0.0279	5.28 ±0.0383
45	**Safflower**	Hingoli	2.62 ± 0.0393	0.38 ± 0.0711	3 ± 0.0813
		Latur	2.65 ± 0.0213	0.39 ± 0.0932	3.04 ± 0.0956
		Parbhani	2.64 ± 0.0893	0.38 ± 0.0622	3.01 ± 0.1088
		Bidar	2.67 ± 0.0624	0.37 ± 0.0188	3.05 ± 0.0652
		Gulbarga	2.74 ± 0.0398	0.44 ± 0.1245	3.19 ± 0.1307
46	**Sapota**	Solapur	6.46 ±0.015	3.34 ±0.9096	9.81 ±0.9097
		Pune	5.74 ±0.1383	3.25 ±0.7004	8.98 ±0.7139
		Dharwad	6.5 ±0.2411	3.39 ±0.7438	9.89 ±0.7819
		Dindigul	7.3 ±0.0252	3.38 ±0.6126	10.68±0.6132
		Tirupathur	7.54 ±0.2105	4.09 ±0.8395	11.63±0.8655
		Virudhunagar	7.43 ±0.0337	3.89 ±0.4579	11.33±0.4592
		Palghar	4.96 ±0.4871	3.05 ±0.3175	8.01 ±0.5814
		Junagadh	4.6 ±0.0145	2.76 ±0.4264	7.36 ±0.4267
		Navsari	4.72 ±0.1219	2.91 ±0.3213	7.63 ±0.3436
47	**Sorghum**	Ajmer	4.611±0.178	1.145±0.131	5.756±0.221
		Bhilwara	4.657±0.143	1.679±0.144	6.335±0.203
		Bundi	4.855±0.15	0.97±0.049	5.825±0.158
		Tonk	4.751±0.068	1.113±0.176	5.863±0.189
		Chittorgarh	4.918±0.084	1.134±0.317	6.052 ±0.327
		Solapur	4.984±0.097	1.168±0.336	6.151±0.35
		Sangli	5.193±0.128	1.319±0.173	6.512±0.215
		Osmanabad	4.943±0.09	1.101±0.25	6.044±0.266
		Khargone	5.496±0.11	1.272±0.716	6.768±0.725

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Dewas	5.74±0.244	1.324±0.822	7.064±0.857
		Neemuch	6.025±0.048	1.391±0.772	7.416±0.774
		Bagalkot	4.517±0.005	1.023±0.327	5.539±0.327
		Bijapur	4.577±0.133	1.354±0.145	5.93±0.196
		Belgaum	4.619±0.231	1.425±0.48	6.044±0.533
		Gulbarga	4.611±0.39	1.043±0.236	5.655±0.456
		Kurnool	4.639±0.01	1.202±0.524	5.841±0.524
		Guntur	4.54±0.052	0.816±0.242	5.356±0.247
		Prakasam	4.355±0.004	1.005±0.318	5.36±0.318
		Kadapa	4.58±0.043	1.299±0.729	5.878±0.731
		Dharmapuri	4.955±0.178	1.123±1.023	6.078±1.039
		Tuticorin	4.933±0.073	0.999±0.352	5.932±0.36
		Virudhunagar	5.037±0.179	1.287±0.361	6.323±0.403
		Ramanathapuram	4.928±0.145	1.015±0.46	5.944±0.483
		Nagaur	4.902±0.041	0.921±0.101	5.823±0.109
48	**Soybean**	Betul	5.35 ± 0.056	1.35 ± 1.008	6.7 ± 1.01
		Sagar	6.12 ± 0.136	1.22 ± 0.105	7.34 ± 0.172
		Baran	5.52 ± 0.068	0.67 ± 0.092	6.19 ± 0.114
		Jhalawar	5.41 ± 0.076	0.95 ± 0.152	6.36 ± 0.17
		Ujjain	6.57 ± 0.367	0.32 ± 0.069	6.89 ± 0.374
		Latur	7.59 ± 0.162	1.26 ± 0.213	8.84 ± 0.267
		Bidar	5.91 ± 0.212	0.72 ± 0.238	6.63 ± 0.319
49	**Sugarcane**	Udham Singh Nagar	3.76±0.1157	0.44±0.4904	4.2±0.5039
		Haridwar	2.83±0.0115	0.29±0.8388	3.12±0.8388
		West Champaran	7.73±0.1804	0.61±0.145	8.34±0.2315
		Bijnor	3.81±0.0732	0.05±0.432	3.86±0.4382
		Muzaffarnagar	7.57±0.0462	0.18±0.2805	7.75±0.2843
		Gurdaspur	3.15±0.0967	0.24±0.7344	3.39±0.7408
		Yamunanagar	5.37±0.0125	0.49±0.3947	5.86±0.3949

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
		Kolhapur	5.44±0.0101	0.89±0.2363	6.33±0.2365
		Pune	6.91±0.0467	0.27±0.2578	7.17±0.262
		Bagalkot	3.82±0.1253	0.81±0.8749	4.63±0.8838
		Belgaum	3.73±0.045	0.18±0.2639	3.91±0.2678
		Cuddalore	9.78±0.0335	0.07±0.3757	9.85±0.3772
		Villupuram	7.62±0.1228	0.13±0.2203	7.75±0.2522
50	**Sunflower**	Madhepura	4.12 ± 0.0207	1.08 ± 0.2067	5.2 ± 0.2077
		Supaul	4.1 ± 0.082	1.03 ± 0.2322	5.13 ± 0.2462
		Kurukshetra	3.31 ± 0.2544	1.13 ± 0.5228	4.44 ± 0.5814
		Koppal	3.12 ± 0.1227	1.08 ± 0.2093	4.2 ± 0.2426
		Raichur	3.32 ± 0.066	1.08 ± 0.2808	4.41 ± 0.2884
		Latur	3.95 ± 0.075	1.01 ± 0.2855	4.96 ± 0.2952
		Osmanabad	3.62 ± 0.128	1.14 ± 0.31	4.76 ± 0.3354
		Solapur	3.88 ± 0.0575	1.15 ± 0.2348	5.04 ± 0.2417
		South 24 Parganas	4.17 ± 0.004	1.04 ± 0.258	5.21 ± 0.2581
51	**Tapioca**	East Godavari	3.92 ±0.0305	1.44 ±0.7395	5.36 ±0.7401
		Idukki	3.5 ±0.1702	1.12 ±0.2249	4.61 ±0.282
		Thiruvananthapuram	3.44 ±0.0123	1.41 ±0.5156	4.85 ±0.5157
		Kollam	3.57 ±0.0293	1.45 ±0.1562	5.02 ±0.1589
		East Garo Hills	3.62 ±0.0474	1.56 ±0.8614	5.17 ±0.8627
		Mokokchung	3.42 ±0.1063	1.5 ±0.1884	4.92 ±0.2163
52	**Tomato**	Anantapur	9.27 ±0.1855	3.03 ±0.5121	12.3 ±0.5447
		Banas_Kantha	7.9 ±0.067	3.35 ±1.493	11.25 ±1.4945
		Ahmadabad	7.79 ±0.1353	3.36 ±0.8721	11.15 ±0.8825
		Vadodara	7.22 ±0.0902	3 ±0.7998	10.23±0.809
		Kolar	7.03 ±0.1619	2.43 ±0.9978	9.46 ±1.0108
		Chhindwara	8.7 ±0.0838	3.22 ±1.3962	11.92 ±1.3987
		Khordha	8.83 ±0.0979	3.5 ±0.5509	12.33±0.5595
		Kendujhar	9.46 ±0.0396	3.56 ±1.1832	13.02±1.1838

S.No.	Crop	District	Total loss in farm operations	Total loss in market level	Over all total loss
53	Turmeric	West Jaintia Hills	3.92 ±0.206	1.56 ±0.8692	5.48 ±0.8932
		Guntur	4.19 ±0.007	0.89 ±0.3904	5.08 ±0.3905
		Kurnool	4.11 ±0.0188	1.16 ±0.1495	5.27 ±0.1507
		Krishna	3.75 ±0.0048	0.66 ±0.1098	4.4 ±0.1099
		Chamarajanagar	2.26 ±0.0848	1.5 ±0.912	3.76 ±0.9159
54	Wheat	Kangra	2.84±0.0645	0.48±0.19	3.32±0.2006
		Udham Singh Nagar	0.79±0.0353	0.6±0.1946	1.39±0.1978
		Murshidabad	5.36±0.0565	0.27±0.4952	5.63±0.4984
		Azamgarh	2.74±0.0379	1±0.1681	3.74±0.1723
		Jaunpur	5.36±0.0193	1.2±0.2462	6.56±0.247
		Hardoi	5.87±0.0373	0.59±0.3063	6.46±0.3086
		Shahjahanpur	2.77±0.0235	0.58±0.2398	3.35±0.241
		Sirsa	2.84±0.054	0.22±0.3351	3.06±0.3395
		Sangrur	3.48±0.0492	0.47±0.1184	3.95±0.1282
		Bundi	2.84±0.0645	0.48±0.19	3.32±0.2006
		Jalaun	4.73±0.4034	0.83±0.3815	5.55±0.5552
		Jhansi	4.16±0.1629	0.41±0.6128	4.57±0.6341
		Baran	3.31±0.0763	0.53±0.4726	3.85±0.4787
		Indore	4.69±0.1355	0.47±0.3806	5.17±0.404
		Ujjain	3.33±0.1944	0.49±0.7795	3.82±0.8034
		Sabar Kantha	3.08±0.186	0.62±0.6474	3.69±0.6736
		Ahmadabad	2.96±0.1134	0.38±0.5923	3.35±0.6031
		Jhunjhunun	4.56±0.5089	0.49±0.1274	5.04±0.5246
		Sikar	5.31±0.1823	0.57±0.4826	5.88±0.5159

Appendix 6 Post-harvest losses of crops/ commodities at state level

Andhra Pradesh

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collect-ion	Sorting/ Grading	Threshin g	Winno-wing/ cleaning	Drying	Packag -ing	Farm level storage	Trans -port	Total loss in farm operations	Godown	Wholes-alers	Retail -ers	Process -ing unit	Trans -port	Total loss in market level	Overall total loss
Bajra	1.17 ±0.0187	0.41 ±0.0069	-	1.69 ±0.0263	0.15 ±0	0.21 ±0.0037	0.26 ±0.0044	0.28 ±0.00	0.18 ±0.0032	4.36 ±0.0337	0.15 ±0.06	0.01 ±0.0252	0.12 ±0.0029	0.13 ±0.0035	0.3 ±0.044	0.72 ±0.0787	5.07 ±0.0856
Banana	1.67 ±0.0238	0.39 ±0.0052	2.07 ±0.0297	-	-	-	0.21 ±0.0028	0.06 ±0.0009	1.83 ±0.0267	6.24 ±0.0469	0.1 ±0.025	1.32 ±0.582	0.5 ±0.224	0.01 ±0.0063	1.16 ±0.4911	3.09 ±0.7942	9.33 ±0.7955
Black gram	1.06 ±0.0309	0.6 ±0.0168	-	0.99 ±0.0316	0.67 ±0.0196	0.29 ±0.0083	0.26 ±0.0075	0.69 ±0.0189	0.13 ±0.0032	4.69 ±0.0558	0.04 ±0.0351	0.22 ±0.0427	0.23 ±0.0338	0.11 ±0.0459	0.27 ±0.1772	0.87 ±0.1942	5.56 ±0.2021
Cashew nut	1.77 ±0.0274	0.56 ±0.0086	-	1.36 ±0.0211	0.33 ±0.0051	0.03 ±0.0004	-	0.07 ±0.0012	0.07 ±0.0012	4.181 ±0.0361	0.001 ±0.0008	0.13 ±0.0649	0.02 ±0.0101	0.3 ±0.073	0.17 ±0.0201	0.63 ±0.1003	4.82 ±0.1066
Chickpea	1.27 ±0.0076	0.84 ±0.0042	-	1.87 ±0.0105	0.2 ±0.001	0.27 ±0.0009	0.16 ±0.0006	0.16 ±0.0008	0.24 ±0.0017	5.01 ±0.0139	0.01 ±0.011	0.16 ±0.0264	0.11 ±0.0215	0.07 ±0.0147	0.35 ±0.0734	0.69 ±0.0809	5.71 ±0.0821
Chilli	1.02 ±0.0146	0.72 ±0.0123	2.13 ±0.0351	-	-	0.11 ±0.0015	0.02 ±0.0003	1.05 ±0.0175	0.15 ±0.0027	5.2 ±0.0438	0.39 ±0.0628	1.72 ±0.0014	3.31 ±0.2246	-	0.04 ±0.0313	1.86 ±0.2353	7.06 ±0.1162
Citrus	0.74 ±0.0022	0.39 ±0.0009	2.67 ±0.0025	-	-	-	0.56 ±0.0023	0.62 ±0.0017	0.86 ±0.0041	5.84 ±0.0061	0.08 ±0.0204	0.68 ±0.2159	0.62 ±0.1427	0.03 ±0.0059	0.71 ±0.6829	2.12 ±0.7306	7.97 ±0.7306
Coconut	1.3 ± 0.0171	0.23 ± 0.0031	-	-	0.36 ± 0.0047	0.36 ± 0.000	0.2 ± 0.0027	0.68 ± 0.009	0.09± 0.0014	3.23 ± 0.0204	0.02 ± 0.0239	0.08 ± 0.0619	0.38± 0.0156	0.26 ± 0.0191	0.36 ± 0.1177	1.1 ± 0.1374	4.32 ± 0.1389
Cottonseed	1.98 ±0.0073	0.34 ± .0013	-	-	-	0.02 ± 0.0001	0.05 ± 0.0002	0.05 ± 0.00	0.14 ± 0.0006	2.57 ± 0.0075	0.02 ± 0.0034	0.25 ±0.0443	0.03 ± 0.0239	0.03 ± 0.024	0.13 ± 0.0362	0.46 ± 0.0665	3.03 ± 0.0669
Egg	1.00 ±0.008	-	1.09 ± 0.0089	-	-	-	1.39 ± 0.0108	0.47 ± 0.003	0.25 ± 0.0047	4.21 ± 0.0170	0.03 ± 0.00	0.48 ± 0	0.43 ± 0	-	0.32 ± 000	1.26 ± 0.00	5.47 ± 0.017
Green gram	1.47 ±0.00	0.73 ±0.00	-	1.5 ±0.00	0.37 ±0.00	0.32 ±0.00	0.24 ±0.00	0.39 ±0.00	0.15 ±0	5.17 ±0.0364	0.33 ±0.018	0.31 ±0.0166	0.02 ±0.0033	0.14 ±0.0334	0.16 ±0.1142	0.97 ±0.1215	6.13 ±0.1268
Groundnut	2.13 ± 0.0037	0.58 ±0.0007	-	1.81 ± 0.0031	0.41 ± 0.0011	0.19 ± .0003	0.13 ± 0.0001	0.09 ± 0.0001	0.17 ± 0.0002	5.51 ± 0.005	0.04 ± 0.0624	0.4 ± 0.2257	0.08 ± 0.0236	0.38 ± 0.3118	0.17 ± 0.1934	1.07 ± 0.4315	6.58 ± 0.4315
Inland fish	1.5 ± 0.0063	0.33 ±0.0011	1.43 ± .0045	-	-	-	0.17 ± 0.0006	0.08 ± 0.0004	0.13 ± 0.0007	3.64 ± 0.0078	-	0.36 ± 0.0712	0.57 ± 0.0537	-	0.11 ± 0.0194	1.04 ± 0.0587	4.68 ± 0.0592
Maize	1.2 ± 0.014	0.38 ± 0.016	-	1.26 ± 0.014	0.57± 0.006	0.17 ± 0.007	0.12 ± 0.005	0.26 ± 0.005	0.01± 0.001	3.98 ± 0.028	0.01 ±0.003	0.46± 0.057	0.25 ±0.058	1.27 ± 0.037	0.01 ± 0.006	2.00 ± 0.09	5.98 ± 0.094
Mango	1.66 ±0.0488	0.26 ±0.0041	2.56 ±0.0785	-	-	-	0.05 ±0.0027	5.59 ±0.0954	0.81 ±0.0224	5.59 ±0.0954	-	0.22 ±0.1122	1.1 ±0.2417	0.16 ±0.0826	0.82 ±0.285	2.3 ±0.3902	7.9 ±0.4017
Marine fish	5.42 ±0.0328	0.58 ± 0.0032	0.35 ± 0.0019	-	-	0.13 ± 0.0009	0.01 ± 0.00	-	0.41 ± 0.0021	6.89 ± 0.0331	0.19 ± 0.0491	0.47 ± 0.1521	0.21 ± 0.0499	0.29 ± 0.0611	0.43 ± 0.1086	1.6 ± 0.1491	8.5 ± 0.1527

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collect-ion	Sorting/ Grading	Threshing	Winno-wing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Milk	0.16 ± 0.0033	0.10 ± 0.0021	-	-	-	-	-	-	0.08 ± 0.0014	0.34 ±0.0041	-	-	0.06 ± 0.0084	0.14 ± 0.1033	0.13 ± 0.0346	0.24 ± 0.1092	0.58 ± 0.1093
Muskmelon	1.5 ±0.003	0.34 ±0.0043	1.25 ±0.0014	-	-	-	0.31 ±0.0062	0.11 ±0.0005	0.76 ±0.001	4.26 ±0.0083	0.03 ±0.0104	0.59 ±0.0906	1.45 ±0.5366	-	1.09 ±0.1786	3.16 ±0.5728	7.42 ±0.5729
Paddy	2.04 ±0.0267	0.39 ±0.0057	-	1.07 ±0.015	0.65 ±0.0082	0.13 ±0.0018	0.1 ±0.0014	0.43 ±0.0072	0.17 ±0.0026	4.97 ±0.0332	0.02 ±0.0056	0.33 ±0.077	0.03 ±0.0089	0.14 ±0.0826	0.24 ±0.1478	0.76 ±0.1863	5.73 ±0.1892
Papaya	1.43 ± 0.0099	0.43 ± 0.0032	0.86 ± 0.0065	-	-	-	0.32 ± 0.0023	0.26 ± 0.0022	0.96 ± 0.0066	4.27 ± 0.0142	0.01 ± 0.0059	0.41 ± 0.1183	1.2 ± 0.313	0.05 ± 0.0208	0.64 ± 0.1861	2.32 ± 0.3835	6.58 ± 0.3837
Pomegranate	1.64 ±0.0186	0.16 ±0.0017	2.97 ±0.0312	-	-	-	0.21 ±0.003	0.25 ±0.0028	0.25 ±0.0027	5.48 ±0.0367	0.21 ±0.0608	0.84 ±0.2162	1.04 ±0.1754	0.11 ±0.0158	1.02 ±0.4532	3.21 ±0.5356	8.69 ±0.5368
Poultry Meat	3.19 ± 0.00	-	-	-	-	-	-	-	-	3.19 ± 0.00	-	2.28 ± 0.00	1.45 ± 0.00	0.01 ± 0.00	0.46 ± 0.00	4.2 ± 0.00	7.39 ± 0.00
Sorghum	1.38 ±0.021	0.32 ±0.0048	-	1.78 ±0.0267	0.35± 0.0053	0.08± 0.0012	0.3 ±0.0044	0.2 ±0.0035	0.08 ±0.0012	4.5 ±0.0352	0.09 ±0.0394	0.72 ±0.2401	0.14 ±0.0589	0.02 ±0.0102	0.2 ±0.0895	1.17 ±0.266	5.67 ±0.2683
Tapioca	1.2 ±0.0062	0.26 ±0.0033	1.03 ±0.0082	-	-	-	0.07 ±0.0009	0.26 ±0.0014	0.62 ±0.0056	3.44 ±0.0123	0.47 ±0.3255	0.5 ±0.0866	0.18 ±0.0308	0.45 ±0.1941	1.59 ±0.39	5.04 ±0.3902	-
Tomato	2.99 ±0.2352	0.68 ±0.0478	3.21 ±0.2488	-	-	-	0.31 ±0.0217	1.22 ±0.0879	0.86 ±0.0672	9.27 ±0.3637	-	0.79 ±0.2874	1.46 ±0.4071	0.13 ±0.024	0.71 ±0.1174	3.1 ±0.5119	12.36 ±0.628
Turmeric	2.32 ±0.0066	0.23 ±0.0008	0.69 ±0.0034	-	0.27 ±0.0013	0.16 ±0.0005	0.1 ±0.0003	0.08 ±0.0003	0.17 ±0.0017	4.02 ±0.0078	0.49 ±0.1134	0.55 ±0.3232	0.61 ±0.2878	-	0.41 ±0.2429	0.95 ±0.3254	4.97 ±0.3254

Assam

Crop	Farm Operations											Market Level Assessment						
	Harvest-ing	Collec-tion	Sorting/ Grading	Trimming	Threshing	Winno-wing/ clean-ing	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm opera-tions	Godown	Wholesalers	Retail-ers	Processing unit	Trans-port	Total loss in market level	Overall total loss
Cabbage	1.21 ±0.1119	0.11 ± 0.0103	2.62 ± 0.2432	-	-	-	-	0.35 ± 0.0395	0.11 ± 0.011	1.06 ± 0.016	5.46 ± 0.2715	0.05 ± 0.0107	0.7 ± 0.2217	1.08 ± 0.2462	0.01 ± 0.0056	0.07 ± 0.035	1.92 ± 0.3334	7.38 ± 0.43
Egg	1.56 ± 0.0954	-	0.88 ± 0.0511	-	-	-	-	1.08 ± 0.0135	0.25 ± 0.0228	0.29 ± 0.0167	4.06 ± 0.1126	0.05 ± 0	0.87 ± 0.00	0.50 ± 0.0447	-	0.57 ± 0.00	1.99 ± 0.0447	6.05 ± 0.1212
Meat	2.45 ± 0.00	-	-	-	-	-	-	-	-	-	2.45 ± 0	-	0.5 ± 0.00	0.27 ± 0	0.19 ± 0	0.03 ± 0	0.99 ± 0	3.44 ± 0
Mushroom	0.49 ±0.00	0.2 ±0.00	1.51 ±0.00	0.51± 0	-	0.31±0	-	0.1 ±0	0.1 ±0	0.68 ±0	3.89 ±0	-	0.27 ± 0.0905	0.14 ± 0.0923	-	0.18 ± 0.092	0.59± 0.1587	4.48 ±0.1587
Paddy	2.02 ±0.0409	0.62 ±0.0257	-	-	1.55 ±0.0357	0.57 ±0.013	0.4 ±0.0113	0.05 ±0.0046	0.18 ±0.0048	0.09 ±0.0037	5.48 ±0.0629	0.06 ±0.023	0.18 ±0.087	0.03 ±0.0151	0.12 ±0.1237	0.21 ±0.2401	0.59 ±0.285	6.07 ±0.2919

Crop	Farm Operations											Market Level Assessment						Overall total loss
	Harvest-ing	Collec-tion	Sorting/ Grading	Trim ming	Threshi ng	Winno-wing/ clean-ing	Drying	Packag -ing	Farm level storage	Trans-port	Total loss in farm opera -tions	Godown	Wholesal ers	Retail-ers	Processi ng unit	Trans-port	Total loss in market level	
Pineapple	2.17 ±0.2099	0.09 ±0.0059	2.18 ±0.167	-	-	-	-	0.03 ±0.0017	0.05 ±0.0062	0.9 ±0.0681	5.43 ±0.2769	0.01 ±0.0137	0.49 ±0.2948	0.41 ±0.1148	0.13 ±0.0478	0.64 ±0.2405	1.68 ±0.4005	7.11 ±0.4869
Poultry Meat	1.91 ± 0.034	-	-	-	-	-	-	-	-	-	1.91 ± 0.034	-	2.79 ± 0.00	1.38 ± 0	0.03 ± 0	0.49 ± 0	4.7 ± 0	6.61 ± 0.034
Radish	2.5 ±0.1381	0.17 ±0.0179	0.27 ±0.0126	-	-	-	-	0.44 ±0.0379	0.07 ±0.005	0.15 ±0.0127	3.6 ±0.1455	-	0.37 ±0.2363	0.35 ±0.3531	-	0.82 ±0.5295	1.54 ±0.6789	5.14 ±0.6943

Andaman and Nicobar Islands

Crop	Farm Operations									Market Level Assessment					Overall total loss
	Harvest ing	Collec-tion	Thresh-ing	Winnowin g/ cleaning	Drying	Packag -ing	Farm level storage	Trans-port	Total loss in farm opera-tions	Godown	Whole-salers	Retail -ers	Processi -ng unit	Trans-port	Total loss in market level	
Areca nut	1.95 ±0.3268	0.36 ±0.00	1.36 ±0.2305	1.26 ±0.2884	0.23 ±0.0306	0.11 ±0.0324	0.19 ±0.0594	0.51 ±0.0418	5.96 ±0.5003	-	0.45 ±0.1936	0.03 ±0.0199	-	0.3 ±0.0313	0.49 ±0.1971	6.44 ±0.5378
Coconut	0.82 ± 0.1289	0.27 ± 0.0228	-	0.52 ± 0.0138	0.77 ± 0.2061	0.42 ± 0.0558	0.29 ± 0.0215	0.32 ± 0.0451	3.4 ± 0.2557	0.37 ± 0.0453	0.16 ± 0.061	0.21 ± 0.1722	-	0.11 ± 0.1109	0.85 ± 0.2184	4.25 ± 0.3363

Bihar

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnow ing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	
Bottlegourd	1.84 ±0.0554	0.45 ±0.0035	1.31 ±0.0144	-	-	-	0.37 ±0.0049	0.2 ±0.0043	0.62 ±0.0095	4.79 ±0.0585	-	0.71 ±0.0999	1.02 ±0.3606	-	0.82 ±0.1998	2.56 ±0.4242	7.35 ±0.4282
Brinjal	2.66 ±0.0672	0.28 ±0.0054	1.16 ±0.0401	-	-	-	0.25 ±0.0043	0.37 ±0.0073	0.4 ±0.0133	5.12 ±0.08	-	0.72 ±0.5139	1.12 ±0.2711	-	0.57 ±0.2398	2.4 ±0.6285	7.52 ±0.6336
Cabbage	1.61 ± 0.0469	0.5 ± 0.0134	3.11 ± 0.0858	-	-	-	0.35 ± 0.0085	0.11 ± 0.0043	1.01 ± 0.0008	6.7 ± 0.0991	0.06 ± 0.0065	0.39 ± 0.0742	1.39 ± 0.5527	0.01 ± 0.0043	0.15 ± 0.0381	2 ± 0.559	8.7 ± 0.5677
Cauliflower	2.00 ±0.0062	0.22 ±0.0017	2.42 ±0.0097	-	-	-	0.23 ±0.0009	0.04 ±0	0.72 ±0.0038	5.64 ±0.0122	0.03 ±0.0011	0.19 ±0.1039	0.29 ±0.0558	0.01 ± 0.0004	0.61 ±0.1871	1.12 ±0.2212	6.76 ±0.2215
Green gram	1.55 ±0.1119	0.64 ±0.0432	-	1.51 ±0.1129	0.31 ±0.023	0.36 ±0.0219	0.18 ±0.0113	0.47 ±0.0409	0.13 ±0.0093	5.14 ±0.0183	0.32 ±0.418	0.29 ±0.0336	0.02 ±0.0131	0.16 ±0.0071	0.13 ±0.0141	0.93 ±0.4198	6.07 ±0.4202
Guava	5.02 ±0.0474	0.31 ±0.0078	4.84 ±0.0706	-	-	-	0.35 ±0.0047	0.44 ±0.0117	0.67 ±0.0101	11.62 ±0.0869	-	2.09 ±0.1464	2.92 ±0.3399	1.44 ±0.2298	0.96 ±0.3046	3.45 ±0.5315	15.07 ±0.5386

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnow ing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Inland fish	2.32 ± 0.0764	0.34 ± .0095	1.62 ± 0.0445	-	-	-	0.15 ± 0.0042	0.1 ± 0.004	0.18 ± 0.0046	4.72 ± 0.0892	-	0.36 ± 0.0559	0.8 ± 0.37	-	0.12 ± 0.0398	1.28 ± 0.3723	5.99 ± 0.3828
Maize	0.54 ± 0.022	0.11 ± 0.007	-	2.2± 0.119	0.2± 0.009	0.27± 0.009	0.2 ± 0.026	0.16 ± 0.008	0.04 ± 0.002	3.72 ± 0.125	0.03 ±0.0004	1.02 ± 0.148	0.4 ± 0.054	0.48 ± 0.023	0.19 ± 0.023	2.11 ± 0.161	5.83 ± 0.204
Mango	2.08 ±0.0729	0.47 ±0.0106	3.1 ±0.0598	-	-	-	0.03 ±0.0018	0.39 ±0.0041	0.91 ±0.0227	6.98 ±0.0977	-	0.57 ±0.4595	1.12 ±0.1585	0.3 ±0.0644	1.13 ±0.2219	3.13 ±0.5343	10.1 ±0.5431
Okra	1.57 ± 0.0127	0.27 ± 0.0023	1.17 ± 0.0073	-	-	-	0.18 ± 0.0011	0.16 ± 0.0011	0.57 ± 0.0075	3.93 ± 0.0167	-	0.82 ± 0.1742	0.96 ±0.0459	-	0.56 ± 0.1621	2.34 ± 0.2423	6.27 ± 0.2429
Paddy	1.32 ±0.046	0.35 ±0.0116	-	0.99 ±0.0373	0.32 ±0.0118	0.42 ±0.0144	0.24 ±0.0096	0.32 ±0.0094	0.09 ±0.0036	4.04 ±0.0647	0.07 ±0.0238	0.23 ±0.1563	0.04 ±0.0161	0.15 ±0.1412	0.11 ±0.0315	0.6 ±0.2149	4.65 ±0.2244
Pineapple	1.72 ±0.2019	0.1 ±0.007	1.22 ±0.0866	-	-	-	0.03 ±0.0053	0.12 ±0.0124	0.69 ±0.0582	3.87 ±0.2278	0.06 ±0.0467	0.57 ±0.028	0.56 ±0.3805	0.1 ±0.0795	0.74 ±0.3586	2.02 ±0.5316	5.89 ±0.5784
Potato	2.11 ± 0.0021	0.29 ± 0.0051	2.19 ± 0.0022	-	-	-	0.03 ± 0.0002	0.12 ± 0.0004	0.65 ± 0.0011	5.4 ± 0.0061	0.07 ± 0.0453	0.25 ± 0.0421	0.11 ± 0.2191	0.04 ± 0.0249	0.25 ± 0.0042	0.71 ± 0.2291	6.11 ± 0.2292
Radish	1.34 ±0.0408	0.26 ±0.0099	0.35 ±0.011	-	-	-	0.52 ±0.0163	0.17 ±0.0038	0.1 ±0.0027	2.74 ±0.0466	-	0.47 ±0.1248	0.83 ±0.345	-	1.08 ±0.4092	2.37 ±0.5496	5.11 ±0.5516
Sugarcane	1.31 ±0.0712	1.86 ±0.0837	-	-	2.24 ±0.1008	-	-	-	2.31 ±0.1016	7.73 ±0.1804	-	-	-	0.23 ±0.1016	1.02 ±0.8207	1.26 ±0.8269	8.99 ±0.8464
Sunflower	1.16 ± 0.0317	0.46 ± 0.0111	-	1.79 ± 0.0541	0.27 ± 0.008	0.1 ± 0.003	0.12 ± 0.0023	0.06 ± 0.0018	0.13 ± 0.0018	4.1 ± 0.0643	0.02 ± 0.0027	0.17 ± .0968	0.04 ± 0.0106	0.74 ± 0.0482	0.12 ± 0.097	1.08 ± 0.1456	5.18 ± 0.1592

Chhattisgarh

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnow ing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Bottlegourd	1.93 ±0.6299	0.28 ±0.0883	1.18 ±0.1754	-	-	-	0.14 ±0.0418	0.22 ±0.0897	0.53 ±0.201	4.28 ±0.6967	-	0.67 ±0.3045	0.95 ±1.421	-	0.88 ±0.8589	2.49 ±1.6881	6.77 ±1.8262
Guava	4.62 ±0.6936	0.38 ±0.0809	4.45 ±0.5971	-	-	-	0.28 ±0.0586	0.5 ±0.1098	0.59 ±0.0426	10.83 ±0.9281	-	1.93 ±0.3341	2.85 ±0.2309	1.12 ±0.0469	0.6 ±0.1565	2.97 ±0.4377	13.79 ±1.0261
Paddy	1.04 ±0.0253	0.35 ±0.0114	-	0.83 ±0.0218	0.32 ±0.0141	0.5 ±0.0122	0.16 ±0.0054	0.29 ±0.0073	0.05 ±0.0018	3.53 ±0.041	0.04 ±0.0473	0.18 ±0.0475	0.03 ±0.0157	0.06 ±0.0477	0.07 ±0.0325	0.38 ±0.0898	3.92 ±0.0987
Radish	1.48 ±0.0355	0.04 ±0.0031	0.19 ±0.0166	-	-	-	0.46 ±0.0069	0.13 ±0.0109	0.11 ±0.0014	2.4 ±0.0414	-	1.34 ±0.2132	0.84 ±0.1909	-	2.07 ±0.1428	4.26 ±0.3199	6.66 ±0.3225

Gujarat

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	
Green gram	1.76 ±0.0704	0.65 ±0.0255	-	1.5 ±0.0593	0.29 ±0.0116	0.36 ±0.0141	0.13 ±0.0052	0.28 ±0.0158	0.1 ±0.0041	5.08 ±0.0382	0.32 ±0.0314	0.28 ±0.071	0.02 ±0.0029	0.15 ±0.0057	0.01 ±0.0005	0.77 ±0.0779	5.85 ±0.0868
Bajra	1.52 ±0.0593	0.3 ±0.0087	-	0.96 ±0.0361	0.15 ±0.0029	0.22 ±0.0054	0.24 ±0.007	0.24 ±0.0076	0.27 ±0.0074	3.91 ±0.0714	0.12 ±0.0215	0.08 ±0.0512	0.13 ±0.0118	0.11 ±0.0186	0.06 ±0.8437	0.5 ±0.8458	4.41 ±0.8488
Tomato	2.69 ±0.2327	0.44 ±0.0947	3.24 ±0.2835	-	-	-	0.26 ±0.076	0.24 ±0.0915	0.77 ±0.1188	7.64 ±0.4145	-	0.82 ±0.3361	1.22 ±0.4735	0.19 ±0.0546	1.11 ±0.2668	3.34 ±0.6391	10.97 ±0.7617
Cottonseed	1.73 ± 0.0483	0.25 ± 0.0053	-	-	-	0.17 ± 0.0047	0.1 ± 0.0026	0.1 ± 0	0.17 ± 0.0039	2.52 ± 0.049	0.01 ± 0.0044	0.23 ± 0.1026	0.02 ± 0.0098	0.02 ± 0.0107	0.12 ± 0.0367	0.41 ± 0.11	2.93 ± 0.1204
Groundnut	1.85 ± 0.0478	0.32 ± 0.0086	-	1.24 ± 0.0297	0.39 ± 0.0086	0.14 ± 0.0033	0.16 ± 0.0039	0.06 ± 0.0021	0.11 ± 0.0037	4.27 ± 0.0579	0.05 ± 0.0455	0.43 ± 0.3333	0.09 ± 0.0342	0.29 ± 0.1161	0.19 ± 0.1471	1.04 ± 0.3839	5.31 ± 0.3883
Okra	1.46 ± 0.0737	0.25 ± 0.0139	1.06 ± 0.0503	-	-	-	0.15 ± 0.0084	0.12 ± 0.006	0.47 ± 0.0244	3.51 ± 0.0941		0.77 ± 0.25	0.89 ± 0.2672		0.61 ± 0.1303	2.27 ± 0.3884	5.79 ± 0.3996
Gujarat	5.22 ± 0.1238	0.49 ± 0.0082	0.45 ± 0.0059	-	-	0.12 ± 0.0066	0.01 ± 0.0003	-	0.38 ± 0.0074	6.67 ± 0.1246	0.19 ± 0.0419	0.62 ± 0.2086	0.2 ± 0.0384	0.26 ± 0.0802	0.54 ± 0.1615	1.81 ± 0.201	8.48 ± 0.2365
Cabbage	1.36 ± 0.0361	0.4 ± 0.0146	3.19 ± 0.1005	-	-	-	0.3 ± 0.0044	0.12 ± 0.0029	0.64 ± 0.0003	6.02 ± 0.1079	0.1 ± 0.0183	0.77 ± 0.1883	1.35 ± 0.3828	0.03 ± 0.0212	0.13 ± 0.0551	2.37 ± 0.4311	8.39 ± 0.4444
Cauliflower	1.85 ±0.0243	0.16 ±0.0036	2.42 ±0.0466	-	-	-	0.66 ±0.0061	0.14 ±0	1.31 ±0.0291	6.52 ±0.0605	0.02±0.0013	0.47 ±0.1323	0.17 ±0.0453	0.01 ±0.0004	0.8 ±0.1925	1.47 ±0.2379	7.99 ±0.2455
Milk	0.06 ± 0.0026	0.07 ± 0.0106	-	-	-	-	-	-	0.06 ± 0.0066	0.18 ± 0.0128	-	-	0.13 ± 0.0191	0.04 ± 0.0281	0.35 ± 0.0555	0.32 ± 0.0651	0.50 ± 0.0663
Onion	2.94 ± 0.0343	0.49 ± 0.0043	1.23 ± 0.005	-	-	-	0.15 ± 0.0009	0.77 ± 0.0105	0.24 ± 0.0033	5.82 ± 0.0367	0.22 ± 0.037	0.49 ± 0.137	0.57 ± 0.1976	0.01 ± 0.0057	0.46 ± 0.3321	1.75 ± 0.4116	7.57 ± 0.4133
Mustard	1.68 ±0.0076	0.39 ±0.0013	-	1.52 ±0.0119	0.36 ±0.0014	0.02 ±0.0000	0.02 ±0.0002	0.01 ±0.00001	0.05 ±0.0002	4.04 ± 0.0142	0.01 ±0.0078	0.04 ±0.0181	0.03 ±0.0156	0.02 ±0.031	0.23 ±0.0776	0.34 ±0.0812	4.38 ±0.0824
Potato	2.26 ± 0.0039	0.27 ± 0.0025	1.85 ± 0.0027	-	-	-	0.04 ± 0.0007	0.12 ± 0.0006	0.57 ± 0.0018	5.11 ± 0.0058	0.14 ± 0.0149	0.23 ± 0.0197	0.11 ± 0.2111	0.02 ± 0.0506	0.29 ± 0.0085	0.79 ± 0.2186	5.9 ± 0.2187
Wheat	0.72± 0.037	0.44± 0.0322	-	0.22± 0.0168	0.15± 0.0102	0.03± 0.0028	0.003 ±0.0003	1.93± 0.0414	0.32± 0.0198	3.82± 0.07	0.052± 0.0034	0.28 ±0.0164	0.31 ±0.0111	0.7 ±0.1886	0.03 ±0.0262	1.37 ±0.1915	5.18 ±0.2039
Maize	0.48± 0.041	0.65± 0.044	-	0.52± 0.065	0.17± 0.011	0.61± 0.035	0.51 ± 0.023	0.11± 0.008	0.5± 0.033	3.56± 0.105	0.002± 0.001	0.35± 0.164	0.92± 0.07	0.31± 0.027	0.04± 0.009	1.62± 0.18	5.18± 0.208
Beans	0.44 ±0.0149	0.43 ±0.0143	1.08 ±0.0297	-	-	-	0.34 ±0.0083	0.38 ±0.0071	0.47 ±0.016	3.14 ± 0.0411	0.15 ±0.0588	1.15 ±0.4563	0.56 ±0.1555	-	1.65 ±0.552	3.51 ±0.7352	6.65 ±0.7364
Brinjal	1.84 ±0.1337	0.95 ±0.136	0.46 ±0.047	-	-	-	0.27 ±0.035	0.32 ±0.0214	0.59 ±0.0486	4.43 ± 0.2064	-	0.98 ±0.3808	1.22 ±0.3116	-	0.37 ±0.1456	2.56 ±0.5131	6.99 ±0.5531
Banana	1.28 ±0.0792	0.25 ±0.0025	1.48 ±0.0149	-	-	-	0.15 ±0.0017	0.03 ±0.0001	2.25 ±0.006	5.44 ±0.0808	0.05 ±0.0101	0.88 ±0.3341	0.37 ±0.2638	0.01 ±0.0037	0.85 ±0.1395	2.15 ±0.4481	7.59 ±0.4553
Black gram	1.08 ±0.0222	0.65 ±0.0189	-	0.95 ±0.0249	0.52 ±0.0165	0.26 ±0.0064	0.13 ±0.0047	0.3 ±0.0066	0.17 ±0.0035	4.05 ± 0.0432	0.07 ±0.0058	0.2 ±0.0533	0.24 ±0.0156	0.05 ±0.0293	0.19 ±0.1368	0.75 ±0.1506	4.8 ±0.1567
Papaya	0.92 ± 0.045	0.55 ± 0.028	0.29 ± 0.0214	-	-	-	0.32 ± 0.0155	0.15 ± 0.0305	0.73 ± 0.0408	2.96 ± 0.0781	0.02 ± 0.0059	0.5 ± 0.2092	1.96 ± 0.6409	0.04 ± 0.0132	0.33 ± 0.064	2.85 ± 0.6773	5.81 ± 0.6818

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Paddy	1.01 ±0.0066	0.45 ±0.0076	-	0.72 ±0.0123	0.56 ±0.0085	0.29 ±0.0029	0.28 ±0.003	0.32 ±0.004	0.41 ±0.0046	4.02 ± 0.0195	0.05 ±0.043	0.27 ±0.0881	0.05 ±0.0103	0.09 ±0.0891	0.06 ±0.0769	0.52 ±0.1535	4.54 ±0.1548
Pigeon Pea	0.79 ± 0.0201	0.24 ± 0.0069	-	1.54 ± 0.0575	0.17 ± 0.0065	0.16 ± 0.0052	0.13 ± 0.0037	0.42 ± 0.0117	0.14 ± 0.0044	3.59 ± 0.0632	0.08 ± 0.047	0.08 ± 0.028	0.15 ± 0.0761	0.19 ± 0.0713	0.19 ± 0.0617	0.69 ± 0.1246	4.27 ± 0.1397

Haryana

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collecti-on	Sorting/ Grading	Thres h-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Bajra	0.69 ±0.0189	0.29 ±0.0081	-	-	0.83 ±0.0236	0.15 ±0.0046	0.22 ±0.0065	0.24 ±0.009	0.31 ±0.0095	0.24 ±0.0082	2.96 ±0.0358	0.14 ±0.0772	0.05 ±0.0213	0.16 ±0.0161	0.14 ±0.0497	0.32 ±0.0363	0.81 ±0.1023
Bajra	0.69 ±0.0189	0.29 ±0.0081	-	-	0.83 ±0.0236	0.15 ±0.0046	0.22 ±0.0065	0.24 ±0.009	0.31 ±0.0095	0.24 ±0.0082	2.96 ±0.0358	0.14 ±0.0772	0.05 ±0.0213	0.16 ±0.0161	0.14 ±0.0497	0.32 ±0.0363	0.81 ±0.1023
Chickpea	1.17 ±0.0677	0.8 ±0.0352	-	-	1.59 ±0.0306	0.26 ±0.0157	0.39 ±0.0107	0.16 ±0.0058	0.17 ±0.0096	0.19 ±0.0036	4.73 ±0.0852	0.01 ±0.0182	0.19 ±0.0332	0.13 ±0.0379	0.13 ±0.0243	0.47 ±0.1216	0.93 ±0.1316
Cottonseed	1.7 ± 0.159	0.2 ± 0.0255	-	-	-	-	0.05 ± 0.0049	0.03 ± 0.0062	0.03 ± 0.0001	0.28 ± 0.0235	2.28 ± 0.1629	0.01 ± 0.0015	0.16 ± 0.0468	0.02 ± 0.0065	0.03 ± 0.0106	0.05 ± 0.0257	0.27 ± 0.0548
Egg	2.26 ± 0.3016	-	1.27 ± 0.1184	-	-	-	-	1.14 ± 0.1813	0.08 ± 0.0128	0.41 ± 0.0513	5.16 ± 0.375	0.06 ± 0	0.69 ± 0.4747	0.41 ± 0	-	0.42 ± 0	1.58 ± 0.4747
Meat	2.12± 0.2986	-	-	-	-	-	-	-	-	2.12 ±0.2986	-	0.15 ± 0	0.20 ± 0	0.39 ± 0	0.03 ± 0	0.76 ± 0	2.88± 0.2986
Milk	0.12 ± 0.0027	0.09 ± 0.0222	-	-	-	-	-	-	-	0.34 ± 0.032	0.55 ±0.039	-	-	0.03 ± 0.0049	0.02 ± 0.0329	0.24 ± 0.0976	0.20 ± 0.1031
Mushroom	0.1± 0.0018	0.11± 0.0124	4.24± 0.0877	0.04± 0.005	-	0.04 ±0.0093	-	0.24± 0.0245	0.06± 0.0023	0.004 ±0.0007	4.83± 0.0926	-	0.32± 0.0515	0.22± 0.1044	-	0.54± 0.1345	1.08 ±0.1779
Mustard	1.58 ±0.2609	0.53 ±0.0985	-	-	1.33 ±0.2205	0.44 ±0.0716	0.14 ±0.0245	0.12 ±0.0218	0.01 ±0.0017	0.05 ±0.0092	4.2 ±0.3643	0.02 ±0.0009	0.03 ±0.0087	0.03 ±0.0122	0.04 ±0.0037	0.2 ±0.0093	0.32 ±0.0176
Poultry Meat	2.25 ± 0	-	-	-	-	-	-	-	-	-	2.25 ± 0	-	1.31 ± 0	0.52 ± 0	0.01 ± 0	0.54 ± 0	2.38 ± 0
Radish	2.36 ±0.0295	0.06 ±0.0049	2.26 ±0.0307	-	-	-	-	0.48 ±0.0044	0.08 ±0.001	0.11 ±0.0015	5.35 ±0.0431	-	0.69 ±0.2094	0.36 ±0.3152	-	1.11 ±0.3476	2.16 ±0.5138
Sugarcane	0.96±0.0 051	0.47 ±0.0014	-	-	-	1.08 ±0.0051	-	-	-	2.86 ±0.0101	5.37 ±0.0125	-	-	-	0.32± 0.0569	0.65 ±0.7322	0.97 ±0.7344
Sunflower	0.99 ± 0.143	0.5 ± 0.0744	-	-	1.36 ± 0.1921	0.18 ± 0.0334	0.05 ± 0.0072	0.08 ± 0.0107	0.06 ± 0.0091	0.11 ± 0.0217	3.31 ± 0.2544	0.02 ± 0.0077	0.18 ± 0.0841	0.04 ± 0.0121	0.78 ± 0.4634	0.12 ± 0.0064	1.13 ± 0.4712
Wheat	1.62±0.0 205	0.56 ±0.0064	-	-	1± 0.0134	0.03 ±0.0009	0.03 ±0.0004	0.005 ±0.0001	0.26 ±0.0045	0.13 ±0.0017	3.64 ±0.0258	0.001 ±0.0007	0.21 ±0.0246	0.14± 0.0153	0.38± 0.0463	0.004 ±0.0071	0.73 ±0.0551

Himachal Pradesh

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harve st-ing	Collecti -on	Sorting/ Grading	Thresh-ing	Winnowi ng/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operation s	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	
Apple	3.42 ±0.026 3	0.01 ±0.0009	3.09 ±0.0223	-	-	-	-	0.03 ±0.0004	0.21 ±0.0049	0.76 ±0.009	7.53 ±0.036	1.06 ±0.1734	0.02 ±0.0069	1.6 ±0.2656	0.64 ±0.1102	1.01 ±0.2647	1.67 ±0.4276
Capsicum	1.64 ±0.2631	0.07 ±0.010	0.74 ± 0.129	-	-	-	-	0.11 ± 0.042	0.005 ± 0.0017	0.58 ± 0.0854	3.14 ± 0.3083	-	1.00 ±0.363	0.61 ± 0.1442	-	1.00 ± 0.7604	2.61 ± 0.8549
Green Pea	1.90 ±0.1456	0.32 ± 0.0305	1.71 ± 0.1559	-	-	-	-	0.09 ± 0.0053	0.004 ± 0.0006	0.34 ± 0.0194	4.36 ± 0.2164	-	0.71 ± 0.2649	0.94 ± 0.2549	0.03 ± 0.0421	0.31 ± 0.1028	1.99 ± 0.3841
Mushroom	0.82 ±0.056	0.09± 0.014	3.36 ± 0.3013	0.39 ±0.0988	-	0.08 ±0.0198	-	0.03 ± 0.0104	0.1± 0.0243	0.11± 0.0247	4.98± 0.3246	-	0.36± 0.1945	0.26 ± 0.1506	-	0.4 ± 0.217	1.02± 0.3281
Wheat	1.17±0. 0437	0.03± 0.0018	-	-	1.32 ±0.0461	0.18 ±0.0067	0.003 ±0.0001	0.00004 ±0.000002	0.11 ±0.0059	0.03 ±0.0015	2.84 ±0.0642	0.005 ±0.0108	0.13 ±0.091	0.05 ±0.0473	0.28 ±0.1595	0.004 ±0.0021	0.48 ±0.19
Maize	0.53± 0.018	0.07± 0.004	-	-	1.05 ±0.023	0.12 ±0.004	0.04± 0.003	0.05± 0.003	0.04± 0.002	0.04± 0.003	1.94± 0.03	0.001± 0.0001	0.05± 0.026	0.05± 0.026	0.06± 0.005	0.18± 0.217	0.33 ±0.22

Jharkhand

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh -ing	Winnowing / clean-ing	Dryin g	Packagi -ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	
Papaya	1.63 ± 0.1222	0.32 ± 0.0436	0.59 ± 0.0421	-	-	-	-	0.22 ± 0.0316	0.1 ± 0.0077	0.59 ± 0.0667	3.45 ± 0.1553	0.02 ± 0.0101	0.71 ± 0.2029	1.58 ± 0.8358	0.04 ± 0.0241	0.92 ± 0.2537	3.28 ± 0.8971

Jammu & Kashmir

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti -on	Sorting/ Grading	Thresh -ing	Winnowing / clean-ing	Dryin g	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process -ing unit	Trans-port	Total loss in market level	
Apple	3.47 ±0.054	0.01 ±0.0002	3.79 ±0.0793	-	-	-	0.02 ±0.0005	0.01 ±0.0004	0.2 ±0.0046	7.5 ±0.0961	1.04 ±0.2067	0.01 ±0.0019	1.94 ±0.2381	1.00 ±0.2123	1.21 ±0.3555	2.01 ±0.5204	9.51 ±0.5292
Egg	1.92 ± 0.0615	-	1.44 ± 0.1922	-	-	-	0.8 ± 0.1288	0.14 ± 0.0289	0.26 ± 0.0414	4.57 ± 0.2446	0.03 ± 0	0.70 ± 0.2471	0.45 ± 0.0814	-	0.34 ± 0	1.52 ± 0.2602	6.08 ± 0.3571
Maize	0.9± 0.029	0.17± 0.003	-	0.86± 0.032	0.01± 0.001	0.03± 0.001	0.02± 0.002	0.08± 0.005	0.04± 0.008	2.11± 0.044	0.03± 0.012	0.24± 0.152	0.5± 0.08	0.24± 0.025	0.16± 0.28	1.18± 0.329	3.29± 0.332

Karnataka

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	
Areca nut	1.18 ±0.0386	0.27 ±0.0086	0.27 ±0.0086	0.41 ±0.014	0.94 ±0.034	0.12 ±0.0041	0.24 ±0.0088	0.28 ±0.0107	3.57 ±0.056	3.57 ±0.056	0.01 ±0.0078	0.37 ±0.113	0.04 ±0.0289	0.16 ±0.0498	0.19 ±0.0492	0.58 ±0.1363	4.15 ±0.1473
Bajra	1.16 ±0.1515	0.34 ±0.0197	-	1.67 ±0.215	0.15 ±0	0.23 ±0.0064	0.24 ±0.0133	0.21 ±0	0.24 ±0.0144	4.23 ±0.2645	0.12 ±0.135	0.1 ±0.0499	0.12 ±0.0163	0.15 ±0.0066	0.32 ±0.1507	0.8 ±0.2092	5.03 ±0.3373
Beans	1.78 ±0.0739	0.58 ±0.0189	1.34 ±0.0445	-	-	-	0.35 ±0.0145	0.14 ±0.006	0.24 ±0.0105	4.42 ±0.0903	0.13 ±0.0552	0.4 ±0.4134	0.91 ±0.1312	-	1.48 ±0.7493	2.91 ±0.8676	7.33 ±0.8722
Black pepper	0.54 ± 0.0452	0.2 ± 0.0175	-	0.22 ± 0.0127	0.03 ± 0.003	0.07 ± 0.0047	0.02 ± 0.0023	0.02 ± 0.0018	0.02 ± 0.0016	1.12 ± 0.0505	0.01 ± 0.0032	0.07 ± 0.0096	0.07 ± 0.0125	0.02 ± 0.0042	0.01 ± 0.0051	0.19 ± 0.0121	1.31 ± 0.0519
Capsicum	1.24 ± 0.0779	0.30 ± 0.0482	0.87 ± 0.0486	-	-	-	0.37 ± 0.004	0.006 ± 0.002	0.28 ± 0.0303	3.06 ± 0.1082	-	1.15 ± 0.2674	0.73 ± 0.2155	-	0.8 ± 0.1608	2.68 ± 0.3793	5.75 ± 0.3944
Cashew nut	1.18 ±0.1294	0.22 ±0.0081	-	0.96 ±0.0831	0.18 ±0.0188	0.02 ±0.0026	-	0.01 ±0	0.03 ±0.0034	2.606 ±0.1552	0.002 ±0.0006	0.12 ±0.0744	0.02 ±0.0047	0.22 ±0.0561	0.2 ±0.0326	0.57 ±0.0988	3.17 ±0.184
Coconut	1.29± 0.0392	0.21± 0.0065	-	-	0.36± 0.0105	0.36± 0.0004	0.18± 0.0055	0.6± 0.0184	0.08± 0.0028	3.07± 0.0455	0.04± 0.0104	0.01± 0.0019	0.07± 0.1333	0.13± 0.0212	0.41± 0.0345	0.65± 0.1397	3.72± 0.1469
Egg	1.31 ± 0.085	-	1.25 ± 0.067	-	-	-	0.99 ± 0.0601	0.59 ± 0.0336	0.39 ± 0.0604	4.54 ± 0.1418	0.05 ± 0	0.5 ± 0	0.27 ± 0	-	0.28 ± 0	1.1 ± 0	5.64 ± 0.1418
Grapes	1.8 ±0.1455	0.34 ±0.0251	2.45 ±0.1874	-	-	-	0.12 ±0.0091	0.38 ±0.0318	0.65 ±0.053	5.75 ±0.2467	0.01 ±0.0046	0.55 ±0.3086	0.99 ±0.1809	0.09 ±0.0512	0.63 ±0.1287	2.28 ±0.3836	8.02 ±0.4561
Inland fish	1.48 ± 0.0864	0.35 ± 0.0116	1.49 ± 0.0509	-	-	-	0.19 ± 0.0116	0.1 ± 0.0055	0.15 ± 0.0098	3.76 ± 0.1023	-	0.38 ± 0.0307	0.55 ± 0.0543	-	0.14 ± 0.0159	1.07 ± 0.0567	4.82 ± 0.117
Maize	0.47± 0.091	0.06± 0.009	-	0.72± 0.118	0.07± 0.014	0.1± 0.015	0.02± 0.003	0.18± 0.035	0.03± 0.004	1.64± 0.155	0.003± 0.001	0.23± 0.071	1.88± 0.378	0.32± 0.001	0.32± 0.071	2.76± 0.391	4.4± 0.42
Marine fish	5.31 ± 0	0.68 ± 0.0078	0.35 ± 0.0081	-	-	0.22 ± 0.0024	0.01 ± 0.0003		0.44 ± 0.008	7.01 ± 0.014	0.18 ± 0.0662	0.68 ± 0.2466	0.23 ± 0.0655	0.28 ± 0.0221	0.59 ± 0.2815	1.96 ± 0.3099	8.97 ± 0.3102
Onion	2.85 ± 0.1223	0.51 ± 0.0541	1.29 ± 0.0485	-	-	-	0.29 ± 0.0332	1.37 ± 0.1344	0.68 ± 0.0749	7.00 ± 0.2122	0.3 ± 0.1053	0.53 ± 0.2563	0.59 ± 0.2491	0.02 ± 0.0046	0.25 ± 0.073	1.69 ± 0.3797	8.70 ± 0.435
Pigeon Ppea	0.85 ± 0.1029	0.18 ± 0.0203		1.62 ± 0.1967	0.12 ± 0.0164	0.12 ± 0.0135	0.14 ± 0.0162	0.5 ± 0.0604	0.23 ± 0.029	3.77 ± 0.2343	0.09 ± 0.0175	0.09 ± 0.0225	0.14 ± 0.0664	0.27 ± 0.1788	0.21 ± 0.0448	0.8 ± 0.1972	4.57 ± 0.3063
Pineapple	1.19 ±0.0371	0.04 ±0.0022	0.71 ±0.0252	-	-	-	0.04 ±0.0022	0.04 ±0.0014	0.02 ±0.0007	2.04 ±0.0449	0.03 ±0.033	0.63 ±0.1679	0.45 ±0.1847	0.11 ±0.0533	0.47 ±0.1667	1.69 ±0.3066	3.73 ±0.3099
Pome-granate	1.66 ±0.0389	0.28 ±0.0058	2.88 ±0.0453	-	-	-	0.22 ±0.004	0.25 ±0.0056	0.24 ±0.0058	5.53 ±0.0607	0.15 ±0.0634	0.62 ±0.1637	0.72 ±0.1671	0.17 ±0.0278	1 ±0.1966	2.66 ±0.3133	8.19 ±0.3191
Safflower	1.04 ± 0.0532	0.41 ± 0.0095	-	0.49 ± 0.0139	0.26 ± 0.0084	0.14 ± 0.0065	0.17 ± 0.0044	0.01 ± 0.0004	0.15 ± 0.0041	2.68 ± 0.0571	0.02 ± 0.0051	0.15 ± 0.0139	0.04 ± 0.0161	0.08 ± 0.0116	0.13 ± 0.0897	0.42 ± 0.0929	3.1 ± 0.1091
Sorghum	1.38 ±0.0729	0.33 ±0.0191	-	1.85 ±0.1241	0.36 ±0.0218	0.08 ±0.0045	0.3 ±0.0176	0.18 ±0.0096	0.08 ±0.0045	4.57 ±0.1484	0.04 ±0.0088	0.65 ±0.1359	0.14 ±0.0254	0.01 ±0.0104	0.27 ±0.0992	1.12 ±0.1707	5.69 ±0.2262
Soybean	2.39 ± 0.1728	0.88 ± 0.0552	-	0.89 ± 0.0663	0.34 ± 0.0275	0.13 ± 0.0093	0.24 ± 0.0162	0.87 ± 0.0811	0.16 ± 0.012	5.91 ± 0.2124	0.21 ± 0.0012	0.01 ± 0.1	0.22 ± 0.2115	0.11 ± 0.0126	0.16 ± 0.0049	0.72 ± 0.2343	6.63 ± 0.3163
Sugarcane	0.83 ±0.0335	1.41 ±0.0359	-	-	1.32 ±0.0237	-	-	-	0.02 ±0.0014	3.57 ±0.0545	-	-	-	0.96 ±0.5119	0.3 ±0.1307	1.26 ±0.5283	4.84 ±0.5311

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Sunflower	0.85 ± 0.0413	0.35 ± 0.0217	-	1.41 ± 0.0884	0.2 ± 0.0076	0.09 ± 0.0034	0.13 ± 0.0067	0.04 ± 0.0032	0.1 ± 0.0059	3.16 ± 0.1008	0.02 ± 0.01	0.17 ± 0.0954	0.03 ± 0.0172	0.75 ± 0.1294	0.08 ± 0.0416	1.06 ± 0.167	4.22 ± 0.1951
Tomato	2.19 ±0.1634	0.48 ±0.0269	3.1 ±0.263	-	-	-	0.24 ±0.017	0.16 ±0.0203	0.85 ±0.0583	7.03 ±0.3173	-	0.77 ±0.8454	1.11 ±0.3319	0.15 ±0.0845	0.9 ±0.413	2.94 ±0.9977	9.97 ±1.0469
Turmeric	0.74 ±0.0658	0.25 ±0.026	0.1 ±0.0079	-	0.25 ±0.0265	0.16 ±0.0104	0.22 ±0.0228	0 ±0	0.54 ±0.0279	2.26 ±0.0848	1.12 ±0.1442	0.23 ±0.0242	0.2 ±0.0514	-	1.74 ±0.9113	2.37 ±0.9155	4.63 ±0.9194

Kerala

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Areca nut	1.15 ±0.2549	0.27 ±0.0561	-	0.44 ±0.0924	0.87 ±0.1749	0.13 ±0.0254	0.12 ±0.0235	0.19 ±0.0405	0.25 ±0.0565	3.41 ±0.3366	0.01 ±0	0.42 ±0.0119	0.01 ±0.0037	0.16 ±0.0194	0.2 ±0.02	0.6 ±0.0305	4.01 ±0.3379
Black pepper	0.51 ± 0.01	0.2 ± 0.0073	-	0.21 ± 0.0116	0.03 ± 0.0022	0.06 ± 0.003	0.03 ± 0.0011	0.02 ± 0.0008	0.01 ± 0.001	1.07 ± 0.0175	0.01 ± 0.0025	0.08 ± 0.011	0.07 ± 0.024	0.02 ± 0.0075	0.01 ± 0.0054	0.19 ± 0.0148	1.26 ± 0.0229
Cashew nut	0.95 ±0.0089	0.3 ±0.0076	-	1.03 ±0.0271	0.17 ±0.005	0.03 ±0.0005	-	0.01 ±0	0.03 ±0.0009	2.519 ±0.03	0.002 ±0.0002	0.16 ±0.0491	0.02 ±0.0099	0.17 ±0.03	0.19 ±0.0932	0.55 ±0.11	3.07 ±0.114
Coconut	0.43± 0.0151	0.44± 0.0118	-	-	0.33± 0.0109	0.32± 0.0118	0.24± 0.0078	0.46± 0.0187	0.33± 0.0139	2.55± 0.0351	0.03± 0.0361	0.08± 0.0804	0.27± 0.1108	0.28± 0.0509	0.39± 0.1449	1.06± 0.2089	3.61± 0.2118
Egg	1.71 ± 0.006	-	1.06 ± 0.013	-	-	-	1.06 ± 0.0288	0.02 ± 0.0002	0.3 ± 0.0064	4.15 ± 0.0328	0.05 ± 0.0494	0.65 ± 0	0.26 ± 0	-	0.42 ± 0	1.38 ± 0.0494	5.53 ± 0.0594
Marine fish	5.46 ± 0.1458	0.55 ± 0.0112	0.36 ± 0.0058	-	-	0.16 ± 0.0046	0.01 ± 0.0003	-	0.44 ± 0.0114	6.98 ± 0.1468	0.24 ± 0.0714	0.53 ± 0.1833	0.21 ± 0.0513	0.3 ± 0.0676	0.49 ± 0.1494	1.77 ± 0.1943	8.75 ± 0.2436
Meat	1.49 ± 0	-	-	-	-	-	-	-	-	1.49 ± 0	-	0.26 ± 0	0.25 ± 0	0.18 ± 0	0.02 ± 0	0.71 ± 0	2.20 ± 0
Poultry Meat	1.58 ± 0	-	-	-	-	-	-	-	-	1.58 ± 0	-	2.08 ± 0	1.22 ± 0	0.01 ± 0	0.44 ± 0	3.75 ± 0	5.32 ± 0
Tapioca	1.2 ±0.0375	0.26 ±0.0101	1.05 ±0.0238	-	-	-	0.09 ±0.0036	0.27 ±0.006	0.54 ±0.0054	3.41 ±0.0464	-	0.35 ±0.1275	0.45 ±0.0594	0.16 ±0.0136	0.44 ±0.0865	1.4 ±0.1657	4.8 ±0.1721

Madhya Pradesh

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing / cleaning	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godow n	Wholes-alers	Retail-ers	Process -ing unit	Trans-port	Total loss in market level	Overall total loss
Bajra	1.38 ±0.0573	0.35 ±0.014	-	0.9 ±0.0406	0.15 ±0.0061	0.21 ±0.01	0.26 ±0.0044	0.36 ±0.0158	0.19 ±0.0096	3.8 ±0.075	0.11 ±0.0008	0 ±0.0043	0.12 ±0.0037	0.14 ±0.0162	0.36 ±0.014	0.73 ±0.0221	4.53 ±0.0782
Black gram	1.41 ±0.0226	0.94 ±0.0208	-	1.73 ±0.0303	0.66 ±0.0225	0.45 ±0.0001	0.38 ±0.0081	0.24 ±0.0071	0.12 ±0.0042	5.94 ±0.05	0.01 ±0.003	0.27 ±0.0386	0.23 ±0.1461	0.05 ±0.0498	0.24 ±0.1784	0.79 ±0.239	6.73 ±0.2442
Bottlegourd	2.13 ±0.5535	0.46 ±0.0825	1.44 ±0.4025	-	-	-	0.28 ±0.0495	0.36 ±0.0911	0.53 ±0.1427	5.2 ±0.7115	-	0.87 ±0.818	0.64 ±0.3787	-	0.75 ±0.2239	2.26 ±0.9288	7.46 ±1.17
Capsicum	1.92 ± 0.2568	0.04 ± 0.0116	0.02 ± 0.0055	-	-	-	0.02 ± 0.0042	0.004 ± 0.0013	0.24 ± 0.0405	2.24 ± 0.2603	-	1.03 ± 0.5435	0.6 ± 0.1046	-	0.46 ± 0.1791	2.09 ± 0.5818	4.33 ± 0.6374
Cauliflower	1.6 ±0.0618	0.18 ±0.0098	2.42 ±0.0994	-	-	-	0.31 ±0.0155	0.06 ±0	0.74 ±0.0303	5.32 ±0.1222	0.01 ±0.0013	0.21 ±0.0386	0.26 ±0.0733	0.01 ±0.0004	0.65 ±0.201	1.14 ±0.2174	6.46 ±0.2494
Chickpea	1.21 ±0.0807	0.83 ±0.0332	-	1.61 ±0.1027	0.24 ±0.014	0.26 ±0.0108	0.16 ±0.0069	0.17 ±0.0098	0.29 ±0.0201	4.77 ±0.138	0.02 ±0.0202	0.31 ±0.0415	0.13 ±0.0809	0.13 ±0.0269	0.49 ±0.1345	1.08 ±0.1623	5.85 ±0.213
Chilli	1.23 ±0.0029	0.93 ±0.0023	2.17 ±0.0044	-	-	0.02 ±0.0003	0.07 ±0.003	0.61 ±0.0171	0.2 ±0.0056	5.23 ±0.0191	0.24 ±0.1065	1.2 ±0.0322	1.01 ±0.558	-	0.07 ±0.091	0.86 ±0.5762	6.09 ±0.2475
Citrus	1.47 ±0.0843	0.56 ±0.0263	0.96 ±0.0688	-	-	-	0.35 ±0.019	0.64 ±0.0296	0.86 ±0.0405	4.83 ±0.1242	0.02 ±0.0125	0.66 ±0.3197	0.69 ±0.9372	0.03 ±0.0058	1.18 ±0.3073	2.57 ±1.0369	7.4 ±1.0443
Coriander	2.07 ± 0.0617	0.86 ± 0.0205	-	1.11 ± 0.0354	0.51 ± 0.0176	0.08 ± 0.0021	0.15 ± 0.0042	0.06 ± 0.0022	0.13 ± 0.0039	4.97 ± 0.0763	0.02 ± 0.0063	0.3 ± 0.0558	0.3 ± 0.0165	0.07 ± 0.0174	0.33 ± 0.0556	1.03 ± 0.0809	6.01 ± 0.1112
Egg	1.34 ± 0.070	-	1.7 ± 0.4326	-	-	-	1.31 ± 0.1418	0.04 ± 0.0125	0.28 ± 0.0678	4.66 ± 0.4657	0.07 ± 0	0.72 ± 0	0.34 ± 0	-	0.54 ± 0	1.66 ± 0	6.33 ± 0.4657
Green Pea	1.68 ± 0.0452	0.79 ± 0.0782	1.43 ± 0.0434	-	-	-	0.08 ± 0.0129	0.005 ± 0.0008	0.62 ± 0.0392	4.60 ± 0.1084	-	0.57 ± 0.3046	0.66 ± 0.4775	0.06 ± 0	0.38 ± 0.1828	1.67 ± 0.5952	6.27 ± 0.605
Groundnut	1.99 ± 0.0383	0.65 ± 0.0032	-	1.43 ± 0.0141	0.43 ± 0.0066	0.14 ± 0.0003	0.15 ± 0.0022	0.09 ± 0.0009	0.14 ± 0.0011	5 ± 0.0415	0.07 ± 0.0268	0.35 ± 0.1378	0.07 ± 0.0357	0.41 ± 0.0724	0.2 ± 0.1825	1.11 ± 0.2425	6.11 ± 0.2461
Guava	4.93 ±0.1202	0.66 ±0.0228	5.12 ±0.1449	-	-	-	0.35 ±0.0101	0.44 ±0.0169	0.83 ±0.0195	12.33 ±0.1917	-	1.5 ±0.2682	2.4 ±0.2268	1.32 ±0.0891	0.63 ±0.0685	2.6 ±0.3687	14.93 ±0.4156
Maize	1± 0.019	0.09± 0.023	-	1.82± 0.104	0.04± 0.003	0.01± 0.002	0.04± 0.005	0.07± 0.002	0.09± 0.024	3.16± 0.111	0.08± 0.008	1.81± 0.126	0.29±0.081	0.81± 0.079	0.14± 0.0001	3.14± 0.17	6.3± 0.203
Milk	0.18 ± 0.047	0.06 ± 0.0299	-	-	-	-	-	-	0.09 ± 0.0063	0.33 ±0.0561	-	-	0.22 ± 0.0103	0.12 ± 0.0458	0.03 ±0.016	0.21 ± 0.0496	0.53 ± 0.0749
Muskmelon	1.32 ±0.0138	0.26 ±0.1121	0.95 ±0.011	-	-	-	0.36 ±0.1418	0.05 ±0.0025	0.55 ±0.0097	3.48 ±0.1819	0.02 ±0.0088	0.54 ±0.1195	1.04 ±0.3702	-	0.68 ±0.2215	2.28 ±0.4477	5.76 ±0.4833
Mustard	1.46 ±0.02	0.55 ±0.0099		1.6 ±0.0429	0.56 ±0.0095	0.14 ±0.0027	0.13 ±0.0024	0.01 ±0.0001	0.06 ±0.0006	4.51 ±0.0494	0.004 ±0.0036	0.04 ±0.0166	0.04 ±0.0067	0.02 ±0.0143	0.24 ±0.0358	0.34 ±0.04	4.86 ±0.0636
Okra	1.42 ± 0.093	0.33 ± 0.0194	1.23 ± 0.0138	-	-	-	0.1 ± 0.0046	0.2 ± 0.0117	0.47 ± 0.0244	3.75 ± 0.0999	-	0.76 ± 0.306	0.77 ± 0.1596	-	0.55 ± 0.1261	2.08 ± 0.3674	5.84 ± 0.3808
Onion	2.67 ± 0.2962	0.42 ± 0.0327	1.19 ± 0.1032	-	-	-	0.08 ± 0.0202	1.02 ± 0.0565	0.49 ± 0.0196	5.87 ± 0.3216	0.36 ± 0.243	0.99 ± 0.2712	0.59 ± 0.2312	0.01 ± 0.0064	0.28 ± 0.0963	2.24 ± 0.442	8.11 ± 0.5466
Paddy	1.03 ±0.0246	0.61 ±0.0057	-	0.74 ±0.0267	0.24 ±0.0047	0.35 ±0.0178	0.32 ±0.0054	0.28 ±0.0495	0.2 ±0.0044	3.77 ±0.0647	0.04 ±0.0368	0.18 ±0.2362	0.02 ±0.0423	0.07 ±0.0858	0.12 ±0.1711	0.43 ±0.3091	4.2 ±0.3158

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing / cleaning	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godow n	Wholes-alers	Retail-ers	Process -ing unit	Trans-port	Total loss in market level	
Pigeon Pea	1.26 ± 0.1324	0.4 ± 0.0552	-	2.08 ± 0.2402	0.55 ± 0.0281	0.24 ± 0.0219	0.16 ± 0.0138	0.4 ± 0.0455	0.16 ± 0.0139	5.24 ± 0.2863	0.16 ± 0.0539	0.12 ± 0.0415	0.18 ± 0.0682	0.45 ± 0.1623	0.28 ± 0.0964	1.19 ± 0.2052	6.43 ± 0.3522
Potato	2.54 ± 0.0031	0.2 ± 0.0156	2.38 ± 0.0031	-	-	-	0.04 ± 0.0042	0.12 ± 0.0004	0.64 ± 0.0011	5.92 ± 0.0168	0.27 ± 0.0748	0.32 ± 0.0639	0.1 ± 0.1883	0.02 ± 0.0969	0.37 ± 0.0207	1.07 ± 0.2344	7.00 ± 0.235
Sorghum	1.95 ±0.0924	0.36 ±0.0319	-	1.96 ±0.0271	0.45 ±0.0203	-	-	-	0.14± 0.0069	5.37 ±0.1044	0.08± 0.0203	0.94± 0.6056	0.15± 0.0351	0.02± 0.0133	0.21± 0.0642	1.39± 0.6105	6.76± 0.6194
Soybean	2.62 ± 0.1959	0.83 ± 0.0577	-	1.09 ± 0.0962	0.25 ± 0.0288	0.09 ± 0.0057	0.43 ± 0.0288	0.71 ± 0.0697	0.36 ± 0.0253	6.37 ± 0.2411	0.05 ± 0.0053	0.52 ± 0.3399	0.18 ± 0.0375	0.18 ± 0.0837	0.06 ± 0.0085	0.99 ± 0.3522	7.36 ± 0.4269
Tomato	3.14 ±0.0801	0.68 ±0.0259	3.01 ±0.1345	-	-	-	0.58 ±0.011	0.34 ±0.0282	0.95 ±0.0296	8.7 ±0.1643	-	0.74 ±0.2984	1 ±0.2324	0.13 ±0.275	1.28 ±1.3439	3.15 ±1.3961	11.85 ±1.4058
Wheat	0.86 ±0.0876	0.27 ±0.0187	-	1.03 ±0.0957	0.86 ±0.1668	0.05 ±0.0031	0.045 ±0.01	0.36 ±0.0157	0.1 ±0.0107	3.58 ±0.2133	0.024 ±0.0022	0.6 ±0.0382	0.07 ±0.012	0.41 ±0.119	0.26 ±0.3149	1.37 ±0.339	4.94 ±0.400

Maharashtra

Crop	Farm Operations											Market Level Assessment						Overall total loss
	Harves ting	Collectio n	Sorting/ Grading	Trim mmin g	Thres h-ing	Winnowi ng/ clean-ing	Drying	Packag-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes alers	Retailers	Process-ing unit	Trans-port	Total loss in market level	
Bajra	0.97 ±0.0455	0.3 ±0.019	-	-	0.97 ±0.04 32	0.17 ±0.0086	0.22 ±0.0097	0.24 ±0.0108	0.26 ±0.0132	0.26 ±0.0123	3.4 ±0.0701	0.11 ±0.015	0.07 ±0.0105	0.13 ±0.0173	0.16 ±0.0105	0.34 ±0.0757	0.81 ±0.0804	4.21 ±0.1067
Banana	1.24 ±0.0226	0.19 ±0.0117	1.55 ±0.0152	-	-	-	-	0.12 ±0.0065	0.02 ±0.0004	1.04 ±0.0335	4.15 ±0.0453	0.05 ±0.0156	0.76 ±0.2146	0.27 ±0.1252	0.01 ±0.003	0.82 ±0.2816	1.91 ±0.3758	6.06 ±0.3785
Black gram	1.88 ±0.016	1.09 ±0.0106	-	-	1.55 ±0.019 3	0.72 ±0.0128	0.46 ±0.0001	0.32 ±0.0055	0.51 ±0.0104	0.14 ±0.0022	6.66 ±0.0324	0.03 ±0.0214	0.34 ±0.2007	0.18 ±0.0379	0.05 ±0.033	0.22 ±0.1471	0.82 ±0.2547	7.48 ±0.2568
Chickpea	1.16 ±0.0234	0.96 ±0.0123	-	-	1.54 ±0.031 1	0.2 ±0.0041	0.24 ±0.0032	0.16 ±0.0022	0.17 ±0.003	0.17 ±0.0038	4.59 ±0.0415	0.01 ±0.0186	0.1 ±0.0228	0.12 ±0.0225	0.1 ±0.0248	0.48 ±0.1241	0.8 ±0.1281	5.40 ±0.1347
Citrus	1.59 ±0.0366	0.07 ±0.0041	2.53 ±0.0285	-	-	-	-	0.3 ±0.0056	1.15 ±0.0153	1.06 ±0.0955	6.71 ±0.1075	0.05 ±0.0105	0.62 ±0.2135	0.63 ±0.2087	0.02 ±0.0048	0.77 ±0.2719	2.08 ±0.404	8.79 ±0.4180
Cottonseed	1.83 ± 0.0307	0.28 ± 0.0061	-		-	-	0.08 ± 0.0021	0.1 ± 0.0025	0.1 ± 0	0.16 ± 0.0045	2.55 ± 0.0318	0.01 ± 0.0022	0.23 ± 0.041	0.03 ± 0.0077	0.03 ± 0.0119	0.13 ± 0.0325	0.44 ± 0.0542	2.99 ± 0.0628
Eggs	1.14 ± 0.0107		0.93 ± 0.0257	-	-	-	-	1.18 ± 0.0144	0.42 ± 0.0065	0.33 ± 0.0224	4.00 ± 0.0391	0.04 ± 0	0.67 ± 0	0.61 ± 0		0.24 ± 0	1.55 ± 0	5.55 ± 0.0391
Grapes	1.45 ±0.033	0.1 ±0.0025	2.42 ±0.0406	-	-	-	-	0.04 ±0.001	0.54 ±0.0122	0.35 ±0.0068	4.9 ±0.0542	0.02 ±0.0081	0.44 ±0.0728	0.8 ±0.1974	0.06 ±0.0282	0.51 ±0.3086	1.83 ±0.3746	6.73 ±0.3785

Crop	Farm Operations											Market Level Assessment						Overall total loss
	Harvesting	Collection	Sorting/ Grading	Trimming	Thresh-ing	Winnowing/ clean-ing	Drying	Packag-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesalers	Retailers	Process-ing unit	Trans-port	Total loss in market level	
Green gram	1.7 ±0.0322	0.59 ±0.0137	-	-	1.4 ±0.03	0.33 ±0.0073	0.32 ±0.0077	0.17 ±0.0038	0.37 ±0.0121	0.14 ±0.003	5.03 ±0.0357	0.33 ±0.0072	0.28 ±0.1109	0.02 ±0.0029	0.12 ±0.0643	0.12 ±0.0172	0.87 ±0.1296	5.90 ±0.1344
Groundnut	2.11 ± 0.0169	0.48 ± 0.0077	-	-	1.55 ± 0.0241	0.42 ± 0.0057	0.12 ± 0.0015	0.18 ± 0.0022	0.08 ± 0.001	0.16 ± 0.0027	5.09 ± 0.0312	0.04 ± 0.0438	0.39 ± 0.1612	0.08 ± 0.0364	0.39 ± 0.0644	0.17 ± 0.1455	1.07 ± 0.2295	6.16 ± 0.2316
Maize	1.47± 0.018	0.41± 0.018	-	-	1.32± 0.006	0.23± 0.002	0.34± 0.02	0.04± 0.002	0.27± 0.001	0.03± 0.006	4.1± 0.033	0.001± 0	0.52± 0.049	0.1± 0.007	0.55 ±0.022	0.03 ±0.002	1.2 ±0.054	5.30 ±0.064
Mango	1.99 ±0.2562	0.11 ±0.0146	2.91 ±0.3021	-	-	-	-	0.02 ±0.0026	0.13 ±0.0092	0.9 ±0.0776	6.06 ±0.404	-	0.34 ±0.1615	0.93 ±0.2541	0.14 ±0.0899	1.13 ±0.3101	2.54 0.4322	8.6± 0.5917
Marine fish	5.3 ± 0.5241	0.6 ± 0.0434	0.33 ± 0.0225	-	-	-	0.19 ± 0.022	0.01 ± 0.0008	-	0.48 ± 0.0519	6.9 ± 0.5294	0.2 ± 0.0677	0.68 ± 0.2779	0.19 ± 0.0507	0.32 ± 0.0945	0.51 ± 0.1165	1.9 ± 0.2016	8.8 ± 0.5665
Meat	2.10±0. 0925	-	-	-	-	-	-	-	-	-	2.10 ±0.0925	-	0.21 ± 0	0.27 ± 0	0.19 ± 0	0.02 ± 0	0.68 ± 0	2.78 ± 0.0925
Milk	0.16 ± 0.0056	0.10 ± 0.0111	-	-	-	-	-	-	-	0.06 ± 0.0033	0.32 ± 0.0129	-	-	0.18 ± 0.0279	0.22 ± 0.0001	0.07 ± 0.068	0.33 ± 0.0735	0.65 ± 0.0746
Mushroom	1.36± 0.0768	0.55± 0.0713	4.66± 0.5191	0.77 ±0.093	-	0.19± 0.0257	-	0.12± 0.0161	0.3± 0.0313	0.35± 0.0261	8.3 ±0.54	-	0.43 ±0.3366	0.32± 0.1219	-	0.2± 0.0548	0.95 ±0.3622	9.24 ±0.6502
Onion	2.28 ± 0.0316	0.34 ± 0.0055	0.87 ± 0.0124	-	-	-	-	0.08 ± 0.0011	0.76 ± 0.0142	0.31 ± 0.0049	4.63 ± 0.0375	0.49 ± 0.0796	0.75 ± 0.1162	0.51 ± 0.132	0.01 ± 0.0052	0.39 ± 0.0612	2.15 ± 0.2026	6.78± 0.2060
Paddy	1 ±0.0281	0.49 ±0.0226	-	-	1.04 ±0.0366	0.35 ±0.0169	0.28 ±0.0133	0.13 ±0.006	0.43 ±0.0209	0.14 ±0.0062	3.87 ±0.0602	0.09 ±0.0186	0.29 ±0.0394	0.03 ±0.0031	0.11 ±0.0345	0.05 ±0.0191	0.57 ±0.0589	4.44 ±0.0842
Pigeon Pea	1.56 ± 0.0226	0.37 ± 0.0048	-	-	2.16 ± 0.0233	0.24 ± 0.0033	0.3 ± 0.0112	0.25 ± 0.0034	0.53 ± 0.0071	0.24 ± 0.0041	5.64 ± 0.036	0.16 ± 0.0591	0.14 ± 0.0342	0.2 ± 0.061	0.47 ± 0.1063	0.3 ± 0.0856	1.26 ± 0.1537	6.90 ± 0.1579
Pomegranate	1.31 ±0.0335	0.18 ±0.0111	1.42 ±0.0476	-	-	-	-	0.22 ±0.0111	0.28 ±0.0132	0.31 ±0.0181	3.73 ±0.0643	0.13 ±0.0288	0.61 ±0.1164	0.7 ±0.1454	0.09 ±0.0378	0.76 ±0.1566	2.29 ±0.2479	6.02 ±0.2561
Poultry Meat	2.36 ± 0.0857	-	-	-	-	-	-	-	-	-	2.36 ± 0.0857	-	1.2 ± 0	0.8 ± 0	0.02 ± 0	0.41 ± 0	2.42 ± 0	4.78 ± 0.0857
Safflower	1.04 ± 0.0272	0.4 ± 0.0101	-	-	0.47 ± 0.0103	0.25 ± 0.0056	0.12 ± 0.0028	0.17 ± 0.003	0.02 ± 0.0005	0.17 ± 0.0042	2.64 ± 0.0319	0.02 ± 0.0038	0.15 ± 0.0652	0.04 ± 0.0174	0.06 ± 0.0276	0.11 ± 0.0315	0.38 ± 0.0795	3.02 ± 0.0856
Sorghum	1.71 ±0.0483	0.34 ±0.0084	-	-	1.78 ±0.0552	0.38 ±0.0197	0.1 ±0.0033	0.22 ±0.0112	0.26 ±0.0093	0.13 ±0.0051	4.93 ±0.078	0.04 ±0.007	0.76 ±0.1485	0.21 ±0.041	0.02 ±0.0084	0.2 ±0.0488	1.22 ±0.1619	6.16 ±0.1798
Soybean	3.24 ± 0.1421	0.93 ± 0.035	-	-	1.44 ± 0.0525	0.13 ± 0.0052	0.15 ± 0.0034	0.55 ± 0.0194	0.86 ± 0.0371	0.28 ± 0.0116	7.59 ± 0.1616	0.18 ± 0.0055	0.27 ± 0.0515	0.16 ± 0.0253	0.51 ± 0.0317	0.13 ± 0.0319	1.26 ± 0.073	8.84 ± 0.1773
Sugarcane	2.27 ±0.0181	1.71 ±0.0093	-	-	-	2.27 ±0.0152	-	-	-	0.02 ±0.0004	6.26 ±0.0254	-	-	-	0.48 ±0.1287	1.87 ±0.2141	2.35 ±0.2498	8.61 ±0.2511
Sunflower	1.11 ± 0.0299	0.39 ± 0.0117	-	-	1.68 ± 0.043	0.23 ± 0.0071	0.07 ± 0.001	0.13 ± 0.0051	0.07 ± 0.0019	0.13 ± 0.0049	3.81 ± 0.0547	0.02 ± 0.0073	0.18 ± 0.0506	0.04 ± 0.0207	0.74 ± 0.0747	0.13 ± 0.0406	1.12 ± 0.1011	4.93 ± 0.1149

Meghalaya

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Tapioca	1.15 ±0.115	0.33 ±0.0255	1.13 ±0.1072	-	-	-	0.08 ±0.0064	0.28 ±0.0172	0.53 ±0.0548	3.5 ±0.1694	-	0.35 ±0.4298	0.46 ±0.0827	0.17 ±0.0661	0.51 ±0.1086	1.49 ±0.4558	4.99 ±0.4862
Turmeric	1.57 ±0.1418	0.28 ±0.0536	0.44 ±0.0565	-	0.28 ±0.0392	0.8 ±0.0772	0.21 ±0.0207	0 ±0	0.34 ±0.0913	3.92 ±0.206	0.59 ±0.2807	1.02 ±0.5879	0.41 ±0.3771	-	1.07 ±0.8062	1.72 ±0.8936	5.64 ±0.9171
Cashew nut	1.79 ±0.1031	0.3 ±0.0193	-	1.35 ±0.1189	0.34 ±0.0285	0.02 ±0.0017	-	0.07 ±0.0071	0.02 ±0.0029	3.888 ±0.1613	0.002 ±0.0004	0.13 ±0.0352	0.02 ±0.0086	0.28 ±0.0719	0.2 ±0.0794	0.64 ±0.1131	4.53 ±0.197

Nagaland

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing / clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Tapioca	1.54 ±0.0232	0.29 ±0.0037	1.19 ±0.0171	-	-	-	0.11 ±0.0015	0.27 ±0.0034	0.52 ±0.0079	3.92 ±0.0303	-	0.39 ±0.1313	0.44 ±0.0601	0.22 ±0.0004	0.45 ±0.0887	1.5 ±0.1695	5.42 ±0.1722

Odisha

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing / cleaning	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Bottle gourd	2.38 ±0.0557	0.69 ±0.0249	1.66 ±0.0566	-	-	-	-	0.12 ±0.0045	0.1 ±0.0048	0.49 ±0.0297	5.44 ±0.0886	-	0.72 ±0.2492	0.57 ±0.2543	-	0.84 ±0.3143	2.13 ±0.4749
Brinjal	1.99 ±0.0161	0.33 ±0.0076	0.47 ±0.0065	-	-	-	-	0.25 ±0.0018	0.44 ±0.0149	0.82 ±0.0366	4.31 ±0.0439	-	0.94 ±0.5203	1.21 ±0.2309	-	0.85 ±0.2463	3 ±0.6203
Cabbage	1.35 ± 0.0506	0.35 ± 0.0188	2.59 ± 0.0921	-	-	-	-	0.28 ± 0.0141	0.12 ± 0.0061	0.74 ± 0.0023	5.44 ± 0.1079	0.08 ± 0.0116	0.78 ± 0.1453	1.54 ± 0.4477	0.01 ± 0.0022	0.24 ± 0.1311	2.65 ± 0.4887
Cauliflower	2.05 ±0.105	0.18 ±0.0087	2.38 ±0.1264	-	-	-	-	0.37 ±0.0187	0.07 ±0	1.23 ±0.0668	6.28 ±0.1786	0.03 ±0.0011	0.38 ±0.071	0.28 ±0.0876	0.01 ±0.0006	0.67 ±0.1925	1.37 ±0.2231
Chickpea	1.17 ±0.0285	0.8 ±0.017	-	-	1.59 ±0.047	0.26 ±0.0076	0.55 ±0.0093	0.16 ±0.0034	0.17 ±0.0048	0.56 ±0.0071	5.25 ±0.0595	0.02 ±0.0181	0.28 ±0.0493	0.12 ±0.049	0.13 ±0.0242	0.28 ±0.1209	0.84 ±0.1394
Egg	1.14 ± 0.0038	-	1.11 ± 0.0032	-	-	-	-	1.41 ± 0.0036	0.5 ± 0.0013	0.37 ± 0.0041	4.52 ± 0.0075	0.05 ± 0	1.20 ± 0	0.46 ± 0	-	0.41 ± 0	2.11 ± 0
Greengram	1.74 ±0.0663	0.67 ±0.0301		-	1.45 ±0.0543	0.34 ±0.0124	0.31 ±0.0141	0.2 ±0.0118	0.32 ±0	0.13 ±0.0044	5.16 ±0.0172	0.33 ±0.3259	0.31 ±0.0536	0.02 ±0.0105	0.13 ±0.0792	0.16 ±0.1029	0.94 ±0.355

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing / cleaning	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	
Inland fish	1.99 ± 0.0468	0.44 ± 0.0082	1.71 ± 0.039	-	-	-	-	0.19 ± 0.004	0.11 ± 0.0031	0.19 ± 0.0047	4.63 ± 0.0619	-	0.37 ± 0.0302	0.54 ± 0.1108	-	0.13 ± 0.0229	1.04 ± 0.1132
Marine fish	6.47 ± 0.2114	0.5 ± 0.0307	0.43 ± 0.0324	-	-	-	0.12 ± 0.0094	0.01 ± 0.0004	-	0.41 ± 0.0372	7.94 ± 0.2195	0.24 ± 0.1789	0.61 ± 0.2602	0.22 ± 0.0913	0.37 ± 0.0441	0.5 ± 0.2723	1.94 ± 0.3541
Mushroom	1.04± 0.0662	0.04± 0.0043	8.31± 0.5551	0.71± 0.0555	-	0.23± 0.0343	-	0.22± 0.0217	0.73± 0.047	0.81± 0.0518	12.1 ±0.5676	-	0.7±0.2291	0.43± 0.1933	-	0.38± 0.0695	1.5 ±0.3077
Okra	1.78 ± 0.0401	0.34 ± 0.019	1.08 ± 0.0553	-	-	-	-	0.15 ± 0.0064	0.16 ± 0.0071	0.46 ± 0.019	3.96 ± 0.074	-	0.89 ± 0.1378	0.83 ± 0.201	-	0.57 ± 0.1203	2.29 ± 0.2717
Paddy	1.08 ±0.0126	0.37 ±0.0048	-	-	0.89 ±0.0108	0.13 ±0.0013	0.35 ±0.0047	0.18 ±0.002	0.4 ±0.0066	0.06 ±0.0007	3.46 ±0.0193	0.08 ±0.0454	0.25 ±0.1251	0.04 ±0.0083	0.12 ±0.0879	0.23 ±0.1651	0.71 ±0.2297
Tomato	3.99 ±0.1012	0.76 ±0.0269	2.25 ±0.0532	-	-	-	-	0.24 ±0.006	1.2 ±0.0356	1.11 ±0.0315	9.56 ±0.1268	-	1.03 ±0.2108	1.08 ±0.5762	0.25 ±0.048	1.26 ±0.2345	3.63 ±0.6569

Punjab

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retailers	Process-ing unit	Trans-port	Total loss in market level	
Capsicum	1.30 ±0.133	0.02 ± 0.0009	0.03 ± 0.0021	-	-	-	-	0.03 ± 0.0008	0.006 ± 0.0012	0.11 ± 0.0233	1.49 ± 0.1351	-	1.02 ± 0.2763	1.00 ± 0.1041	-	0.8 ± 0.4636	2.82 ± 0.5496
Citrus	0.9 ±0.0347	0.01 ±0.0003	1 ±0.0812	-	-	-	-	0.03 ±0.0053	0.8 ±0.0447	0.89 ±0.0369	3.62 ±0.1058	0.03 ±0.0224	0.31 ±0.2277	0.53 ±0.1118	0.02 ±0.0063	0.89 ±0.1371	1.78 ±0.2892
Green Pea	2.10 ± 0.0924	0.41 ± 0.0214	1.73 ± 0.071	-	-	-	-	0.20 ± 0.0081	0.007 ± 0.0011	0.33 ± 0.0148	4.77 ± 0.1197	-	0.37 ± 0.1559	0.64 ± 0.2184	0.03 ± 0	0.85 ± 0.2961	1.89 ± 0.3996
Maize	0.9± 0.043	1.24± 0.084	-	-	1.01± 0.05	0.2± 0.011	0.18± 0.037	0.11± 0.02	0.2± 0.011	0.2± 0.022	4.04± 0.118	0.01± 0.002	1±0.116	1.41± 0.085	0.38± 0.081	0.06± 0.023	2.87± 0.167
Milk	0.23 ± 0.0113	0.08 ± 0.0211	-	-	-	-	-	-	-	0.2 ± 0.0222	0.51 ±0.0326	-	-	0.13 ± 0.0161	0.05 ± 0.0398	0.03 ± 0.0234	0.12 ± 0.0489
Mushroom	0.21± 0.0212	0.18± 0.0194	4.44± 0.461	0.11± 0.0159	-	0.11± 0.0155	-	0.16± 0.0232	0.11± 0.0106	0.12± 0.0155	5.45± 0.4634		0.31± 0.0964	0.64± 0.4163	-	0.3± 0.2666	1.25± 0.5036
Muskmelon	1.2 ±0.0036	0.24 ±0.0056	0.78 ±0.0037	-	-	-	-	0.34 ±0.1222	0.04 ±0.0006	0.61 ±0.0031	3.23 ±0.1225	0.01 ±0.0053	0.51 ±0.1614	1.14 ±0.2516	-	0.6 ±0.2124	2.25 ±0.3668
Paddy	1 ±0.0562	0.39 ±0.0073	-	-	0.36 ±0.0049	0.23 ±0.0165	0.27 ±0.0129	0.12 ±0.0092	0.26 ±0.03	0.07 ±0.0045	2.7 ±0.0684	0.08 ±0.0423	0.17 ±0.043	0.04 ±0.0052	0.06 ±0.0149	0.1 ±0.0219	0.45 ±0.0661
Potato	2.13 ± 0.0073	0.11 ± 0.002	2.1 ± 0.0041	-	-	-	-	0.03 ± 0.0005	0.09 ± 0.0009	0.44 ± 0.0022	4.91 ± 0.0089	0.1 ± 0.0206	0.25 ± 0.0402	0.05 ± 0.0914	0.03 ± 0.0366	0.24 ± 0.028	0.66 ± 0.1119
Sugarcane	0.3 ±0.0214	0.22 ±0.0129	-	-	-	0.99±0.0514	-	-	-	1.64 ±0.078	3.15 ±0.0967	-	-	-	0.22 ±0.0578	1 ±0.2238	1.23 ±0.2311
Wheat	1.63 ±0.0362	0.41 ±0.0261	-	-	0.57 ±0.0302	0.06 ±0.0034	0.04 ±0.0005	0.002 ±0.0004	0.04 ±0.0014	0.08 ±0.0013	2.84 ±0.054	0.00004 ±0	0.13 ±0.0044	0.7 ±0.0158	0.86 ±0.0635	0.02 ±0.0027	1.71 ±0.0656

Rajasthan

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retailers	Process-ing unit	Trans-port	Total loss in market level	
Bajra	1.76 ±0.0704	0.65 ±0.0255	-	1.25 ±0.0473	0.16 ±0.0061	0.22 ±0.008	0.25 ±0.0094	0.3 ±0.0129	0.28 ±0.0108	3.72 ±0.0638	0.12 ±0.135	0.1 ±0.0499	0.12 ±0.0163	0.15 ±0.0066	0.32 ±0.1507	0.8 ±0.2092	5.03 ±0.3373
Black gram	1.68 ±0.0468	0.89 ±0.0233	-	1.43 ±0.0643	0.23 ±0.0184	0.3 ±0	0.02 ±0.0042	0.33 ±0.0097	0.13 ±0.0027	4.99 ±0.0856	0.07 ±0.0499	0.26 ±0.1472	0.21 ±0.0656	0.04 ±0.0139	0.17 ±0.064	0.75 ±0.181	5.74 ±0.2002
Chickpea	1.43 ±0.0161	0.88 ±0.0063	-	1.2 ±0.0162	0.51 ±0.0044	0.46 ±0.0031	0.17 ±0.0011	0.17 ±0.0015	0.08 ±0.0006	4.88 ±0.0244	0.02 ±0.0157	0.4 ±0.1223	0.1 ±0.0285	0.2 ±0.021	0.24 ±0.1048	0.95 ±0.1636	5.83 ±0.1654
Coriander	1.91 ± 0.0656	0.7 ± 0.0274	-	1.25 ± 0.0415	0.5 ± 0.0217	0.09 ± 0.0043	0.11 ± 0.0046	0.02 ± 0.0022	0.12 ± 0.0041	4.7 ± 0.0855	0.02 ± 0.0056	0.32 ± 0.021	0.26 ± 0.0206	0.07 ± 0.0129	0.33 ± 0.0346	1 ± 0.0429	5.7 ± 0.0957
Cottonseed	1.62 ± 0.0145	0.23 ± 0.0037	-	-	-	0.05 ± 0.0005	0.05 ± 0.0013	0.05 ± 0	0.15 ± 0.0019	2.14 ± 0.0152	0.01 ± 0.0034	0.2 ± 0.0212	0.03 ± 0.005	0.04 ± 0.0093	0.1 ± 0.0276	0.39 ± 0.0364	2.53 ± 0.0394
Greengram	1.82 ±0.0852	0.69 ±0.0229	-	1.55 ±0.0677	0.31 ±0.0125	0.3 ±0.0095	0.19 ±0.0057	0.36 ±0.0162	0.1 ±0.0044	5.33 ±0.0325	0.33 ±0.0448	0.3 ±0.0835	0.02 ±0.0018	0.12 ±0.0181	0.12 ±0.0135	0.88 ±0.0975	6.21 ±0.1028
Groundnut	1.6 ± 0.0409	0.52 ± 0.0171	-	1.17 ± 0.0334	0.38 ± 0.0102	0.09 ± 0.002	0.1 ± 0.0033	0.1 ± 0.0035	0.1 ± 0.0014	4.06 ± 0.0567	0.05 ± 0.0204	0.39 ± 0.0732	0.07 ± 0.0219	0.28 ± 0.0724	0.18 ± 0.1235	0.98 ± 0.1623	5.03 ± 0.1719
Maize	0.78± 0.02	0.27± 0.007	-	1.53± 0.039	0.17± 0.004	0.16± 0.004	0.13± 0.005	0.21± 0.005	0.18± 0.022	3.44± 0.05	0.01± 0.001	0.9± 0.081	0.51± 0.011	0.83± 0.107	0.13± 0.033	2.37± 0.139	5.81± 0.147
Milk	0.41 ± 0.0257	0.06 ± 0.0071	-	-	-	-	-	-	0.52 ± 0.0496	0.98 ±0.0563	-	-	0.26 ± 0.0345	0.22 ± 0.0882	0.10 ± 0.0954	0.37 ± 0.1344	1.35 ± 0.1458
Mustard	1.9 ±0.04	0.44 ±0.0198	-	1.24 ±0.0264	0.38 ±0.0091	0.14 ±0.0052	0.10 ±0.0045	0.08 ±0.0045	0.05 ±0.0023	4.33 ±0.0534	0.01 ±0.0041	0.06 ±0.0134	0.05 ±0.0206	0.04 ±0.0164	0.26 ±0.041	0.42 ±0.0478	4.75 ±0.0716
Sorghum	1.55 ±0.0556	0.29 ±0.0104	-	1.67 ±0.0650	0.38 ±0.0093	0.14 ±0.0043	0.22 ±0.0073	0.28 ±0.0074	0.2 ±0.0069	4.72 ±0.0877	0.04 ±0.0055	0.74 ±0.0864	0.16 ±0.0191	0.02 ±0.007	0.17 ±0.0272	1.12 ±0.093	5.85 ±0.1278
Soybean	1.85 ± 0.039	1.06 ± 0.0216	-	1.37 ± 0.0255	0.16 ± 0.004	0.03 ± 0.0008	0.37 ± 0.007	0.4 ± 0.0087	0.3 ± 0.004	5.54 ± 0.0529	0.08 ± 0.0108	0.48 ± 0.0712	0.15 ± 0.0022	0.21 ± 0.05	0.09 ± 0.0254	1 ± 0.0913	6.54 ± 0.1055
Wheat	1.83 ±0.1189	0.38 ±0.0248	-	1.22 ±0.0701	0.29 ±0.0327	0.12 ±0.0056	0.024 ±0.001	0.51 ±0.0099	0.09 ±0.0042	4.46 ±0.1445	0.002 ±0.0003	0.34 ±0.0418	0.55 ±0.0373	0.81 ±0.065	0.08 ±0.0135	1.78 ±0.0869	6.25 ±0.1686

Tamil Nadu

Crop	Farm Operations											Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Trimmi-ng	Thresh-ing	Winnowing/ cleaning	Drying	Packaging	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retailers	Process-ing unit	Trans-port	Total loss in market level	
Banana	1.41 ±0.0972	0.27 ±0.0184	1.77 ±0.1298	-	-	-	-	0.16 ±0.0105	0.04 ±0.0034	1.38 ±0.1009	5.03 ±0.1922	0.06 ±0.0412	1.03 ±0.6645	0.44 ±0.2603	0.01 ±0.006	0.91 ±0.4059	2.45 ±0.822	7.47 ±0.8442
Beans	1.47 ±0.003	0.5 ±0.0007	1.27 ±0.0083	-	-	-	-	0.34 ±0.0019	0.2 ±0.0022	0.32 ±0.0028	4.09 ±0.0098	0.12 ±0.0245	1.21 ±0.2846	0.76 ±0.4135	-	1.64 ±0.4651	3.73 ±0.6847	7.82 ±0.6848

Crop	Farm Operations											Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Trimmi-ng	Thresh-ing	Winnowing/ cleaning	Drying	Packaging	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retailers	Process-ing unit	Trans-port	Total loss in market level	
Black gram	0.88 ±0.0286	0.97 ±0.0447	-	-	1.04 ±0.0454	0.43 ±0.0201	0.54 ±0.0213	0.13 ±0.0062	0.98 ±0.0325	0.15 ±0.0077	5.12 ±0.083	0.02 ±0.0041	0.24 ±0.0137	0.22 ±0.1351	0.06 ±0.0101	0.27 ±0.048	0.81 ±0.1445	5.93 ±0.1666
Egg	1.19 ± 0.0255		0.98 ± 0.0234	-	-	-	-	1.54 ± 0.0033	0.51 ± 0.0011	0.51 ± 0.0021	4.75 ± 0.0348	0.03 ± 0	0.91 ± 0	0.50 ± 0	-	0.41 ± 0	1.85 ± 0	6.60 ± 0.0348
Groundnut	1.66 ± 0.0131	0.6 ± 0.0055	-	-	1.25 ± 0.0117	0.37 ± 0.0033	0.15 ± 0.0012	0.12 ± 0.0012	0.05 ± 0.0009	0.15 ± 0.002	4.35 ± 0.0189	0.05 ± 0.0162	0.35 ± 0.1033	0.07 ± 0.0173	0.32 ± 0.2272	0.17 ± 0.1647	0.96 ± 0.2995	5.31 ± 0.3001
Maize	0.85± 0.121	0.2± 0.013	-	-	0.73± 0.042	0.17± 0.036	0.23± 0.019	0.27± 0.023	0.24± 0.034	0.45± 0.03	3.14± 0.145	0.001± 0	0.33± 0.145	0.22± 0.002	0.43± 0.016	0.27 ±0.057	1.25 ±0.157	4.39 ±0.213
Marine fish	5.69 ± 0.2843	0.53 ± 0.0212	0.38 ± 0.016	-	-	-	0.14 ± 0.006	0.02 ± 0.0007	-	0.5 ± 0.0272	7.25 ± 0.2869	0.2 ± 0.0622	0.63 ± 0.1911	0.22 ± 0.0546	0.28 ± 0.0575	0.55 ± 0.0954	1.89 ± 0.1511	9.14 ± 0.3242
Meat	1.89±0	-	-	-	-	-	-	-	-	-	1.89±0	-	0.56 ± 0	0.26 ± 0	0.32 ± 0	0.02 ± 0	1.16 ± 0	3.04 ± 0
Mushroom	1.04± 0.0547	0.04± 0.002	6.4± 0.3527	0.67± 0.0361		0.25± 0.0117		0.22± 0.0102	0.74± 0.0392	0.77± 0.0407	10.12 ±0.3634	-	0.49± 0.0557	0.33± 0.2089	-	0.09 ±0.0358	0.91 ±0.2192	11.03 ±0.4244
Paddy	2.03 ±0.0648	0.39 ±0.0136	-	-	1.55 ±0.0532	0.52 ±0.0244	0.13 ±0.0044	0.1 ±0.0033	0.45 ±0.0144	0.19 ±0.0066	5.36 ±0.0899	0.06 ±0.0177	0.21 ±0.093	0.04 ±0.0282	0.17 ±0.0154	0.14 ±0.0301	0.62 ±0.1044	5.98 ±0.1378
Poultry Meat	2.3 ± 0	-	-	-	-	-	-	-	-	-	2.3 ± 0	-	2.18 ± 0	1.26 ± 0	0 ± 0	0.34 ± 0	3.78 ± 0	6.08 ± 0
Sorghum	1.79 ±0.0581	0.33 ±0.0118	-	-	1.78 ±0.0842	0.42 ±0.0198	0.1 ±0.0034	0.23 ±0.0082	0.31 ±0.0138	0.2 ±0.007	5.17 ±0.1064	-	-	0.18 ±0.0467	0.01 ±0.0052	0.17 ±0.0725	1.13 ±0.3502	6.29 ±0.366
Sugarcane	2.88 ±0.0621	1.6 ±0.039	-	-	-	1.01 ±0.0183	-	-	-	2.38± 0.0468	7.87 ±0.0889	-	-	-	0.59 ±0.1854	0.19 ±0.1386	0.78 ±0.2315	8.65 ±0.248

Telangana

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winno-wing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	
Chilli	1.29 ±0.0048	0.77 ±0.0024	2.1 ±0.0063	-	-	0.04 ±0.0001	0.02 ±0.0001	1.04 ±0.0035	0.18 ±0.0006	5.44 ±0.009	0.09 ±0.0378	1.02 ±0.02	0.63 ±0.3286	-	0.06 ±0.0782	0.61 ±0.3405	6.06 ±0.1579
Citrus	0.85 ±0.0215	0.37 ±0.0085	1.4 ±0.0363	-	-	-	0.41 ±0.0092	1.53 ±0.0442	0.8 ±0.0189	5.37 ±0.0652	0.08 ±0.0108	0.7 ±0.1589	0.69 ±0.1036	0.02 ±0.0085	0.87 ±0.1764	2.35 ±0.2594	7.73 ±0.2675
Cotton seed	2.03 ± 0.0109	0.34 ± 0.0045	-	-	-	0.02 ± 0.0001	0.05 ± 0.0006	0.05 ± 0	0.14 ± 0.0008	2.62 ± 0.0118	0.01 ± 0.0042	0.28 ± 0.0387	0.03 ± 0.0128	0.03 ± 0.0095	0.07 ± 0.0169	0.42 ± 0.0452	3.04 ± 0.0467
Egg	1.01 ± 0.0176		0.85 ± 0.009	-	-	-	1.11 ± 0.0314	0.53 ± 0.0107	0.33 ± 0.0085	3.82 ± 0.0396	0.04 ± 0	0.70 ± 0	0.46 ± 0	-	0.34 ± 0	1.54 ± 0	5.36 ± 0.0396
Green gram	1.8 ±0.0328	0.78 ±0.0129	-	1.46 ±0.0217	0.34 ±0.0046	0.25 ±0.0045	0.16 ±0.0029	0.43 ±0	0.13 ±0.0021	5.33 ±0.135	0.33 ±0.1408	0.29 ±0.2145	0.02 ±0.0061	0.14 ±0.0334	0.12 ±0.0175	0.9 ±0.2594	6.24 ±0.2924

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winno-wing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	
Groundnut	2.09 ± 0.0039	0.71 ± 0.0012	-	1.74 ± 0.0047	0.48 ± 0.0009	0.15 ± 0.0004	0.17 ± 0.0005	0.1 ± 0.0003	0.14 ± 0.0003	5.57 ± 0.0063	0.05 ± 0.0436	0.39 ± 0.0932	0.08 ± 0.0213	0.45 ± 0.0806	0.13 ± 0.1598	1.1 ± 0.2031	6.67 ± 0.2032
Maize	0.41± 0.014	0.28± 0.008	-	0.67± 0.023	0.26± 0.009	0.12± 0.003	0.09± 0.002	0.15± 0.006	0.03± 0.001	2.01± 0.03	0.17±0.002	1.91± 0.688	1.62±0.041	0.56±0.033	0.24± 0.115	4.5± 0.699	6.51± 0.7
Meat	0.69±0		-							0.69±0		0.46 ± 0	0.25 ± 0	0.19 ± 0	0.03 ± 0	0.93 ± 0	1.61 ± 0
Pigeon Pea	0.98 ± 0.0206	0.21 ± 0.0037	-	1.75 ± 0.0241	0.14 ± 0.0019	0.15 ± 0.0026	0.17 ± 0.003	0.62 ± 0.009	0.09 ± 0.0014	4.11 ± 0.0335	0.1 ± 0.0369	0.1 ± 0.0636	0.2 ± 0.1207	0.3 ± 0.1244	0.2 ± 0.1535	0.9 ± 0.2402	5.01 ± 0.2425
Poultry Meat	3.39 ± 0	-	-	-	-	-	-	-	-	3.39 ± 0	-	3.05 ± 0	1.4 ± 0	0.02 ± 0	0.26 ± 0	4.74 ± 0	8.12 ± 0

Uttarakhand

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	
Apple	3.99 ±0.1796	0.24 ±0.015	4.75 ±0.2781	-	-	-	0.04 ±0.0031	0.18 ±0.0069	0.34 ±0.0104	9.54 ±0.3316	0.23 ±0.0477	0.01 ±0.0106	1.83 ±0.4739	0.63 ±0.1947	1.39 ±0.099	1.96 ±0.5241	11.5 ±0.6202
Milk	0.54 ± 0.0019	0.08 ± 0.0187	-	-	-	-	-	-	0.27 ± 0.0471	0.89 ±0.0507	-	-	0.12 ± 0.024	0.16 ± 0.0791	0.09 ± 0.0206	0.26 ± 0.0853	1.16 ± 0.0992
Paddy	1.72 ±0.0262	0.09 ±0.0076	-	0.08 ±0.008	0.11 ±0.0101	0.12 ±0.0074	0.09 ±0.0052	0.08 ±0.0114	0.04 ±0.0008	2.33 ±0.0335	0.07 ±0.021	0.19 ±0.3022	0.02 ±0.011	0.15 ±0.1071	0.12 ±0.1843	0.54 ±0.3706	2.87 ±0.3721
Sugar cane	0.77 ±0.014	0.48±0.0115	-		1.1 ±0.0143				0.52 ±0.0168	2.87 ±0.0286	-	-	-	0.83 ±0.2956	1.75 ±0.4298	2.58 ±0.5216	5.46 ±0.5224
Wheat	0.25 ±0.030	0.04 ±0.004	-	0.07 ±0.0123	0.11 ±0.0081	0.002 ±0.0005	0.00047± 0.00009	0.21 ±0.0016	0.11 ±0.0024	0.79 ±0.0343	0.013±0.0011	0.28±0.0023	0.11±0.0021	1.3 ±0.0473	0.02±0.0103	1.73±0.0485	2.51±0.0594

Uttar Pradesh

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	
Bajra	0.72 ±0.0159	0.25 ±0.0048	-	-	1.24 ±0.0253	0.15 ±0.0032	0.22 ±0.0053	0.25 ±0.0057	0.28 ±0.0058	0.24 ±0.0069	3.35 ±0.0327	2.49 ± 0	1.45 ± 0	0.02 ± 0	0.4 ± 0	4.36 ± 0	7.24 ± 0.0577

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Black gram	1.91 ±0.0596	0.87 ±0.0315	-	-	1.4 ±0.0412	0.2 ±0.0081	0.25 ±0.0006	0.2 ±0.0095	0.39 ±0.0159	0.13 ±0.007	5.36 ±0.0818	0.03 ±0.0111	0.16 ±0.061	0.28 ±0.1141	0.06 ±0.0217	0.22 ±0.0887	0.75 ±0.1588
Bottle gourd	2.03 ±0.1377	0.55 ±0.0289	0.53 ±0.0377	-	-	-	-	0.19 ±0.0091	0.14 ±0.0089	1.09 ±0.0853	4.52 ±0.1692	-	0.48 ±0.1146	0.67 ±0.2901	-	0.76 ±0.1509	1.91 ±0.3466
Egg	2.18 ± 0.087	-	0.92 ± 0.089	-	-	-	-	1.15 ± 0.0797	0.01 ± 0.0011	0.24 ± 0.0133	4.50 ± 0.1484	0.03 ± 0	0.73 ± 0	0.46 ± 0	-	0.34 ± 0	1.56 ± 0
Green Pea	2.03 ± 0.0779	0.50 ± 0.0387	1.68 ± 0.1064	-	-	-	-	0.10 ± 0.0047	0.005 ± 0.0006	0.36 ± 0.0192	4.68 ± 0.1389	-	0.43 ± 0.1571	0.56 ± 0.223	0.04 ± 0	0.62 ± 0.1835	1.66 ± 0.3288
Inland fish	1.82 ± 0.0638	0.34 ± 0.0096	1.38 ± 0.0497	-	-	-	-	0.19 ± 0.005	0.08 ± 0.0032	0.15 ± 0.0044	3.96 ± 0.0818	-	0.38 ± 0.0662	0.55 ± 0.0587	-	0.09 ± 0.0263	1.03 ± 0.0655
Maize	0.99± 0.031	0.54± 0.042	-	-	0.54± 0.016	0.19± 0.002	0.42± 0.018	0.56± 0.026	0.18± 0.001	0.45± 0.083	3.88± 0.104	0.02± 0.001	0.35± 0.06	1.09± 0.049	0.27± 0.061	0.16± 0.167	1.88± 0.194
Meat	1.73 ± 0	-	-	-	-	-	-	-	-	-	1.73 ± 0	-	0.48 ± 0	0.29 ± 0	0.27 ± 0	0.03 ± 0	1.07 ± 0
Milk	0.29 ± 0.0044	0.09 ± 0.0087	-	-	-	-	-	-	-	0.24 ± 0.0176	0.62 ±0.0201	-	-	0.17 ± 0.0051	0.14 ± 0.0605	0.46 ± 0.0432	0.50± 0.0746
Mushroom	0.11± 0.0187	0.35± 0.0537	0.69± 0.0859	0.12± 0.0175		0.13 ±0.0139	-	0.24± 0.0376	0.14 ± 0.0263	0.05 ± 0.0073	1.84± 0.1152	-	0.21± 0.1115	0.27± 0.2655	-	0.42 ±0.2295	0.9± 0.3683
Muskmelon	1.6 ±0.0026	0.28 ±0.0043	1.2 ±0.0012	-	-	-	-	0.34 ±0.0151	0.06 ±0.0003	0.74 ±0.0011	4.22 ±0.016	0.01 ±0.0032	0.62 ±0.1379	1.49 ±0.4756	-	0.91 ±0.2426	3.03 ±0.5514
Mustard	1.9 ±0.1367	0.49 ±0.1002	-	-	1.2 ±0.1184	0.37 ±0.0521	0.14 ±0.0298	0.12 ±0.0235	0.14 ±0.0285	1.29 ±0.1604	5.65 ±0.271	0.01 ±0.0046	0.02 ±0.0135	0.02 ±0.0066	0.02 ±0.0182	0.25 ±0.0456	0.32 ±0.048
Paddy	1.24 ±0.031	0.24 ±0.0042	-	-	0.56 ±0.0094	0.87 ±0.0173	0.3 ±0.0051	0.04 ±0.001	0.16 ±0.0088	0.05 ±0.0014	3.46 ±0.0384	0.06 ±0.0656	0.27 ±0.1659	0.04 ±0.0095	0.07 ±0.0835	0.1 ±0.0989	0.54 ±0.2206
Potato	1.93 ± 0.0028	0.29 ± 0.0097	2.18 ± 0.0029	-	-	-	-	0.04 ± 0.0009	0.1 ± 0.0005	0.58 ± 0.0015	5.11 ± 0.0107	0.09 ± 0.0479	0.27 ± 0.0657	0.1 ± 0.1918	0.03 ± 0.0329	0.22 ± 0.0145	0.71 ± 0.2114
Poultry Meat	2.88 ± 0.0577	-	-	-	-	-	-	-	-	-	2.88 ± 0.0577	-	-	-	-	-	-
Sugarcane	1.68± 0.0268	0.87±0.0146	-	-	-	0.95 ±0.0171	-	-	-	2.16 ±0.0412	5.65 ±0.0541	-	-	-	0.17 ±0.1009	1.09 ±0.2275	1.26 ±0.2489
Wheat	1.15± 0.0262	0.39± 0.006	-	-	1.58 ±0.04	0.15 ±0.0079	0.04± 0.0022	0.008 ±0.001	0.05 ±0.0019	0.12 ±0.0039	3.48 ±0.0491	0.433 ±0.0227	0.83 ±0.0491	0.66 ±0.0214	0.9 ±0.1017	0.06 ±0.0125	2.88 ±0.1178

West Bengal

Crop	Farm Operations										Market Level Assessment						
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnow ing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Beans	0.33 ±0.0038	0.19 ±0.01	1.29 ±0.0157	-	-	-	0.33 ±0.0049	0.39 ±0.006	0.07 ±0.0024	2.61 ±0.0206	0.09 ±0.0149	0.96 ±0.19	0.67 ±0.318	-	1.15 ±0.0977	2.87 ±0.3834	5.48 ±0.3839
Brinjal	2.53 ±0.0596	0.5 ±0.0082	1.45 ±0.032	-	-	-	0.25 ±0.0046	0.18 ±0.004	0.3 ±0.0077	5.21 ±0.0688	-	1.06 ±0.3272	1.25 ±0.5631	-	0.33 ±0.205	2.64 ±0.6828	7.85 ±0.6862
Cabbage	1.35 ± 0.0284	0.37 ± 0.009	2.54 ± 0.0531	-	-	-	0.36 ± 0.0078	0.16 ± 0.0028	0.76 ± 0.0004	5.54 ± 0.0614	0.06 ± 0.0077	0.73 ± 0.1465	2.28 ± 0.4216	0.01 ± 0.0066	0.09 ± 0.047	3.17 ± 0.4489	8.71 ± 0.4531
Cauliflower	1.85 ±0.0081	0.27 ±0.0013	2.87 ±0.0156	-	-	-	0.22 ±0.0015	0.04 ±0	0.75 ±0.0043	6 ±0.0182	0.02 ±0.0012	0.32 ±0.1041	0.47 ±0.0912	0.01 ±0.001	0.44 ±0.2347	1.26 ±0.2725	7.26 ±0.2731
Eggs	1.78 ± 0.1146	-	1.63 ± 0.1076	-	-	-	1.04 ± 0.0726	0.41 ± 0.0401	0.38 ± 0.0206	5.26 ± 0.1789	0.05 ± 0	0.91 ± 0	0.61 ± 0		0.24 ± 0	1.81 ± 0	7.07 ± 0.1789
Green gram	1.74 ±0.0449	0.64 ±0.017	-	1.48 ±0.0368	0.33 ±0.0086	0.33 ±0.0082	0.19 ±0.0043	0.43 ±0.0142	0.09 ±0.0023	5.23 ±0.0167	0.33 ±0.0603	0.29 ±0.0553	0.02 ±0.0031	0.13 ±0.0278	0.23 ±0.0428	1.01 ±0.0965	6.24 ±0.098
Groundnut	2.35 ± 0.0278	0.73 ± 0.0082	-	2.01 ± 0.0203	0.52 ± 0.0047	0.16 ± 0.0025	0.14 ± 0.001	0.1 ± 0.0013	0.13 ± 0.0019	6.15 ± 0.0359	0.05 ± 0.0191	0.42 ± 0.0885	0.08 ± 0.0394	0.39 ± 0.2179	0.14 ± 0.2131	1.09 ± 0.3198	7.24 ± 0.3218
Guava	4.76 ±0.0831	1.06 ±0.047	4.6 ±0.1008	-	-	-	0.28 ±0.0209	0.46 ±0.0272	0.68 ±0.0146	11.85 ±0.1438	-	1.98 ±0.2112	3.78 ±0.4771	1.3 ±0.0709	1.12 ±0.1599	4.01 ±0.5503	15.86 ±0.5688
Inland fish	1.24 ± 0.0102	0.31 ± 0.0016	0.92 ± 0.008	-	-	-	0.13 ± 0.0007	0.06 ± 0.001	0.12 ± 0.0007	2.77 ± 0.0132	-	0.35 ± 0.0308	0.54 ± 0.0496		0.12 ± 0.0243	1.02 ± 0.0553	3.79 ± 0.0569
Maize	0.64± 0.01	0.4± 0.019	-	1.48± 0.042	0.12± 0.003	0.35± 0.017	0.27± 0.009	0.18± 0.002	0.18± 0.009	3.62± 0.052	0.01± 0.0004	0.69± 0.174	0.55± 0.074	0.52± 0.038	0.02± 0.027	1.8± 0.195	5.42± 0.202
Marine fish	5.77 ± 0.0944	0.51 ± 0.0065	0.42 ± 0.005	-	-	0.17 ± 0.0022	0.03 ± 0.0004	-	0.37 ± 0.0031	7.26 ± 0.0949	0.15 ± 0.1317	0.57 ± 0.2153	0.2 ± 0.096	0.2 ± 0.1516	0.56 ± 0.1785	1.68 ± 0.2941	8.94 ± 0.3091
Meat	1.46±0	-	-	-	-	-	-	-	-	1.46 ±0	-	0.46 ± 0	0.23 ± 0	0.24 ± 0	0.03 ± 0	0.96 ± 0	2.41 ± 0
Milk	0.27 ± 0.0095	0.05 ± 0.0044	-	-	-	-	-	-	0.30 ± 0.0183	0.61 ±0.0211	-	-	0.14 ± 0.0204	0.40 ± 0.0001	0.04 ± 0.0361	0.48 ± 0.0415	1.09 ± 0.0466
Mustard	2.07 ±0.022	0.53 ±0.0038	-	1.38 ±0.0187	0.48 ±0.007	0.14 ±0.0011	0.11 ±0.0009	0.14 ±0.0011	1.01 ±0.0104	5.86 ±0.0318	0.02 ±0.0079	0.04 ±0.0501	0.03 ±0.011	0.04 ±0.0317	0.21 ±0.0793	0.35 ±0.0944	6.21 ±0.0996
Okra	1.48 ± 0.0188	0.39 ± 0.0046	1.16 ± 0.0157	-	-	-	0.16 ± 0.0022	0.19 ± 0.0025	0.44 ± 0.0072	3.81 ± 0.0262	-	0.84 ± 0.167	0.9 ± 0.185		0.49 ± 0.0545	2.24 ± 0.2551	6.05 ± 0.2565
Paddy	1.53 ±0.0086	0.55 ±0.0029	-	1.18 ±0.0074	0.18 ±0.0011	0.63 ±0.0034	0.07 ±0.0022	0.4 ±0.0021	0.09 ±0.0005	4.62 ±0.0127	0.08 ±0.0101	0.17 ±0.057	0.03 ±0.0345	0.09 ±0.0612	0.16 ±0.106	0.52 ±0.1397	5.14 ±0.1403
Pineapple	2.45 ±0.0941	0.09 ±0.0049	0.96 ±0.0479	-	-	-	0.04 ±0.0017	0.08 ±0.0042	0.2 ±0.0137	3.82 ±0.1067	0.15 ±0.0356	0.71 ±0.1028	0.54 ±0.0894	0.1 ±0.0344	0.35 ±0.1336	1.84 ±0.1972	5.66 ±0.2242
Potato	2.28 ± 0.0054	0.15 ± 0.0019	1.89 ± 0.0025	-	-	-	0.04 ± 0.0005	0.11 ± 0.0007	0.42 ± 0.0014	4.89 ± 0.0065	0.1 ± 0.0125	0.3 ± 0.0597	0.08 ± 0.1486	0.04 ± 0.0344	0.31 ± 0.0068	0.83 ± 0.1644	5.71 ± 0.1646
Poultry Meat	2.38 ± 0.0456	-	-	-	-	-	-	-	-	2.38 ± 0.0456	-	2.29 ± 0	0.53 ± 0	0.02 ± 0	0.2 ± 0	3.03 ± 0	5.41 ± 0.0456

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Thresh-ing	Winnowing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retail-ers	Process-ing unit	Trans-port	Total loss in market level	
Radish	3.09 ±0.0421	0.08 ±0.0031	0.4 ±0.0047	-	-	-	0.47 ±0.0054	0.14 ±0.0019	0.09 ±0.0012	4.27 ±0.0428	-	0.68 ±0.3191	1.2 ±0.5139	-	0.29 ±0.0841	2.17 ±0.6107	6.45 ±0.6122
Sunflower	1.15 ± 0.0035	0.49 ± 0.0015	-	1.92 ± 0.0009	0.23 ± 0.0004	0.1 ± 0.0003	0.09 ± 0.0002	0.05 ± 0.0005	0.13 ± 0.0005	4.17 ± 0.004	0.02 ± 0.0108	0.17 ± 0.0295	0.04 ± 0.0044	0.71 ± 0.1508	0.11 ± 0.0455	1.04 ± 0.1603	5.21 ± 0.1603
Wheat	1.52± 0.0314	0.53± 0.0121	-	1.46± 0.0326	0.63± 0.0213	0.01± 0.0001	0.025± 0.001	0.49± 0.0104	0.7± 0.0173	5.36 ±0.0553	0.018± 0.0036	0.14± 0.0171	0.1± 0.0079	0.47± 0.1001	0.04± 0.0265	0.77± 0.1053	6.13 ±0.119

Appendix 7 Post-harvest losses of crops/ commodities at ACZ level

AGRO CLIMATIC ZONE 1

Crop	Farm Operations											Market Operations						Overall total loss
	Harvest-ing	Collec-tion	Sort-ing/ Grad-ing	Trimm-ing	Threshi-ng	Winnow ing/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operati-ons	Godo-wn	Wholesa lers	Retail-ers	Process-ing unit	Transp ort	Total loss in market level	
Apple	3.55 ±0.0885	0.01 ±0.0006	3.61 ±0.1126	-	-	-	-	0.02 ±0.0008	0.05 ±0.0033	0.62 ±0.0241	7.87 ±0.1452	0.93 ±0.2178	0.01 ±0.007	1.76 ±0.3579	0.91 ±0.409	0.93 ±0.2178	2.03 ±0.6251	9.91 ±0.6417
Capsicum	1.64 ± 0.2631	0.07 ± 0.0100	0.74 ± 0.129	-	-	-	-	0.11 ± 0.042	0.005 ± 0.0017	0.58 ±0.0854	3.14 ± 0.3083	-	1.00 ± 0.363	0.61 ±0.1442	-	1.00 ±0.7604	2.61 ± 0.8549	5.74 ±0.908
Egg	1.92 ± 0.0615		1.44 ±0.1922	-	-	-	-	0.80 ± 0.1288	0.14 ± 0.0289	0.26 ± 0.0414	4.57 ± 0.2446	0.03 ± 0	0.7 ±0.0597	0.45 ±0.0325	-	0.34 ± 0	1.52 ± 0.0679	6.08 ±0.2539
Green pea	1.90 ± 0.1456	0.32 ± 0.0305	1.71 ±0.1559	-	-	-	-	0.09 ± 0.0053	0.004 ± 0.0006	0.34 ± 0.0194	4.36 ± 0.2164	-	0.71 ±0.2649	0.94 ±0.2549	0.03 ±0.0421	0.31 ±0.1028	1.99 ± 0.3841	6.35 ±0.440
Maize	0.82 ±0.017	0.12 ±0.002	-	-	1.06 ±0.023	0.05 ±0.002	0.03 ±0.002	0.03 ±0.002	0.07 ± 0.003	0.04 ±0.004	2.23 ±0.03	0.004 ±0.005	0.08 ±0.037	0.08 ±0.041	0.04 ±0.013	0.17 ±0.334	0.37 ±0.34	2.60 ±0.34
Milk	0.54 ± 0.0019	0.08 ±0.0187	-	-	-	-	-	-	-	0.27 ± 0.0471	0.89 ±0.0507	-	-	0.12 ±0.0243	0.16 ±0.0791	0.09 ±0.020	0.26 ± 0.0853	1.16 ±0.099
Mushroom	0.82 ±0.0546	0.09 ±0.014	3.36 ±0.3013	0.39 ±0.0988		0.08± 0.0198		0.03 ±0.0104	0.1 ±0.0243	0.11 ±0.0247	4.98 ±0.3246	-	0.72 ±0.1945	0.52 ±0.1506	-	0.40 ±0.217	1.02 ±0.3281	6.01 ±0.4615
Paddy	1.72 ±0.0262	0.09 ±0.0076	-	-	0.08 ±0.008	0.11 ±0.0101	0.12 ±0.004	0.09 ±0.0052	0.08 ±0.0114	0.04 ±0.0008	2.33 ±0.0335	0.07 ±0.021	0.19 ±0.3022	0.02 ±0.0112	0.18 ±0.1187	0.12 ±0.1843	0.57 ±0.3741	2.91 ±0.3756
Sugarcane	0.77 ±0.014	0.48 ±0.0115	-	-		1.1 ±0.0143	-	-	-	0.52 ±0.0168	2.87 ±0.0286	-	-	-	0.65±0	1.75 ±0.429	2.4 ±0.4298	5.28 ±0.4307
Wheat	0.45 ±0.0277	0.04 ±0.0027	-	-	0.56 ±0.0312	0.18 ±0.006	0.003±0. 0003	0.0003 ±0.0001	0.24 ±0.0093	0.08 ±0.0018	1.55 ±0.0434	0.004 ±0.009	0.16 ±0.0212	0.04 ±0.048	0.34 ±0.084	0.01 ±0.0212	0.55 ±0.0722	2.1 ±0.0722

AGRO CLIMATIC ZONE 2

Crop	Farm Operations											Market Operations						
	Harvest-ing	Collec-tion	Sorting/ Grading	Trimm-ing	Threshi-ng	Winnow-ing/ cleaning	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retailers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Cabbage	1.21 ± 0.1119	0.11 ± 0.0103	2.62 ± 0.2432	-	-	-	-	0.35 ± 0.0395	0.11 ± 0.0155	1.06 ± 0.016	5.46 ± 0.2718	0.05 ± 0.0107	0.7 ± 0.2217	1.08 ± 0.157	0.01 ± 0.0056	0.07 ± 0.035	1.84 ± 0.2742	7.31 ± 0.386
Cashew nut	1.79 ±0.1031	0.3 ±0.0193	-	-	0.79 ±0.0938	0.34 ±0.0285	0.02 ±0.0017		0.07 ±0.0071	0.02 ±0.0029	3.331 ±0.1438	0.002 ±0.0004	0.13 ±0.0352	0.02 ±0.0086	0.28 ±0.0719	0.2 ±0.0794	0.64 ±0.1131	3.97 ±0.1829
Egg	1.56 ± 0.0954	-	0.88 ± 0.0511	-	-	-	-	1.08 ± 0.0135	0.25 ± 0.0228	0.29 ± 0.0167	4.06 ± 0.1126	0.05 ± 0	0.87 ± 0	0.5 ± 0	-	0.57 ± 0	1.99 ± 0	6.05 ± 0.1126
Inland fish	1.26 ± 0.0395	0.35 ± 0.0076	1.15 ± 0.0191	-	-	-	-	0.16 ± 0.0051	0.07 ± 0.0004	0.15 ± 0.0035	3.13 ± 0.045	-	0.37 ± 0.0431	0.52 ± 0.0555	-	0.11 ± 0.0398	1.01 ± 0.0685	4.13 ± 0.082
Meat	2.45 ± 0	-	-	-	-	-	-	-	-	-	2.45 ± 0	-	0.50 ± 0	0.27 ± 0	0.19 ± 0	0.03 ± 0	0.99 ± 0	3.44 ± 0
Mushroom	0.49 ±0	0.2 ±0	1.51 ±0	0.51 ±0	-	0.31 ±0		0.1 ±0	0.1 ±0	0.68 ±0	3.89 ±0	-	0.54± 0.0905	0.28± 0.0923	-	0.18± 0.092	0.59± 0.1587	4.48± 0.1587
Paddy	2.02 ±0.0409	0.62 ±0.0257	-	-	1.55 ±0.0357	0.57 ±0.013	0.4 ±0.0113	0.05 ±0.0046	0.18 ±0.0048	0.09 ±0.0037	5.48 ±0.0629	0.06 ±0.023	0.18 ±0.087	0.03 ±0.0151	0.14 ±0.0555	0.21 ±0.2401	0.61 ±0.2627	6.1 ±0.2702
Pineapple	2.39 ±0.1503	0.09 ±0.0053	2.22 ±0.1116	-	-	-	-	0.03 ±0.0021	0.09 ±0.0049	0.53 ±0.0285	5.35 ±0.1895	0.07 ±0.024	0.69 ±0.1673	0.41 ±0.0915	0.14 ±0.0702	0.74 ±0.1498	2.06 ±0.2536	7.41 ±0.3166
Poultry Meat	1.91 ± 0.0321	-	-	-	-	-	-	-	-	-	1.91 ± 0.0321	-	2.79 ± 0	1.38 ± 0	0.03 ± 0	0.49 ± 0	4.7 ± 0	6.61 ± 0.0321
Radish	2.5 ±0.1381	0.17 ±0.0179	0.27 ±0.0126	-	-	-	-	0.44 ±0.0379	0.07 ±0.005	0.15 ±0.0127	3.6 ±0.1455	-	0.37 ±0.2363	0.35 ±0.3531		0.82 ±0.5295	1.54 ±0.6789	5.14 ±0.6943
Tapioca	1.6 ±0.0602	0.36 ±0.0138	2.06 ±0.0153	-	-	-	-	0.12 ±0.0036	0.31 ±0.0088	0.54 ±0.0256	4.99 ±0.0693	-	0.37 ±0.2343	0.45 ±0.0511	0.15 ±0.034	0.48 ±0.0893	1.45 ±0.2581	6.43 ±0.2673
Turmeric	1.57 ±0.1418	0.28 ±0.0536	0.44 ±0.0565	-	-	0.28 ±0.0392	0.8 ±0.0772	0.21 ±0.0207	0 ±0	0.34 ±0.0913	3.92 ±0.206	0.59 ±0.1487	1.47 ±0.5879	0.72 ±0.3771	-	1.07 ±0.8062	1.92 ±0.8692	5.84 ±0.8932

AGRO CLIMATIC ZONE 3

Crop	Farm Operations										Market Operations						
	Harves-ting	Collec-tion	Sorting/ Grading	Thresh-ing	Winnow-ing/ cleaning	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesa-lers	Retail-ers	Processi-ng unit	Trans-port	Total loss in market level	Overall total loss
Beans	0.33 ±0.0038	0.19 ±0.01	1.29 ±0.0157	-	-	-	0.33 ±0.0049	0.39 ±0.006	0.07 ±0.0024	2.61 ±0.0206	0.09 ±0.0149	0.96 ±0.19	0.67 ±0.318	-	1.15 ±0.0977	2.87 ±0.3834	5.48 ±0.3839
Brinjal	2.53 ±0.0596	0.5 ±0.0082	1.45 ±0.032	-	-	-	0.25 ±0.0046	0.18 ±0.004	0.3 ±0.0077	5.21 ±0.0688	-	1.06 ±0.3272	1.29 ±0.5203	-	0.33 ±0.1995	2.68 ±0.6462	7.89 ±0.6499
Cabbage	1.35 ± 0.0284	0.37 ± 0.009	2.54 ± 0.0531	-	-	-	0.36 ± 0.0078	0.18 ± 0.004	0.76 ± 0.0004	5.55 ± 0.0615	0.06 ± 0.0077	0.73 ± 0.1465	2.28 ± 0.2689	0.01 ± 0.0066	0.09 ± 0.047	3.08 ± 0.31	8.63 ± 0.316
Cauliflower	1.85 ±0.0081	0.27 ±0.0013	2.87 ±0.0156	-	-	-	0.22 ±0.0015	0.04 ±0	0.74 ±0.0034	5.99 ±0.018	0.05 ±0.0012	0.7 ±0.1041	0.45 ±0.0912	0.01 ±0.0006	0.44 ±0.2347	1.65 ±0.2725	7.64 ±0.2731
Egg	1.78 ± 0.0001	-	1.63 ± 0.1076	-	-	-	1.04 ± 0.0726	0.41 ± 0.0401	0.38 ± 0.0206	5.26 ± 0.1374	0.05 ± 0	0.91 ± 0	0.61 ± 0	-	0.24 ± 0	1.81 ± 0	7.07 ± 0.1374
Greengram	1.74 ±0.0252	0.64 ±0.0096	-	1.48 ±0.04	0.33 ±0.0052	0.33 ±0.0077	0.19 ±0.0077	0.43 ±0.0114	0.1 ±0.0083	5.24 ±0.0517	0.02 ±0.0031	0.33 ±0.0603	0.29 ±0.0553	0.13 ±0.0278	0.23 ±0.0428	1.01 ±0.0965	6.24 ±0.1095
Groundnut	2.35 ± 0.0278	0.73 ± 0.0082	-	2.01 ± 0.0203	0.52 ± 0.0047	0.16 ± 0.0025	0.14 ± 0.001	0.1 ± 0.0013	0.13 ± 0.0019	6.15 ± 0.0359	0.05 ± 0.0191	0.42 ± 0.0885	0.08 ± 0.0394	0.39 ± 0.2179	0.14 ± 0.2131	1.09 ± 0.3198	7.24 ± 0.3218
Guava	4.76 ±0.1629	1.06 ±0.092	5.12 ±0.1975	-	-	-	0.28 ±0.0409	0.46 ±0.0533	0.91 ±0.0286	12.6 ±0.2818	-	1.98 ±0.4139	4.66 ±0.9352	1.3 ±0.5456	1.12 ±0.3135	4.48 ±1.2007	17.08 ±1.2334
Inland fish	1.27 ± 0.0095	0.31 ± 0.0015	0.83 ± 0.0049	-	-	-	0.12 ± 0.0004	0.05 ± 0.001	0.12 ± 0.0007	2.69 ± 0.0109	-	0.33 ± 0.0409	0.54 ± 0.0566		0.13 ± 0.0268	1 ± 0.0629	3.7 ± 0.0638
Maize	0.75± 0.012	0.47± 0.022	-	1.73± 0.049	0.14± 0.004	0.41± 0.02	0.32± 0.011	0.21± 0.002	0.21± 0.01	4.23± 0.061	0.002± 0.038	0.3± 0.074	0.11± 0.0004	0.09± 0.038	0.02± 0.048	0.53± 0.114	4.76± 0.114
Marine fish	5.77 ± 0.0944	0.51 ± 0.0065	0.42 ± 0.005	-	-	0.17 ± 0.0022	0.03 ± 0.0004	-	0.37 ± 0.0031	7.26 ± 0.0949	0.15 ± 0.1317	0.57 ± 0.2153	0.2 ± 0.096	0.2 ± 0.1516	0.56 ± 0.1785	1.68 ± 0.2941	8.94 ± 0.3091
Meat	1.46 ± 0	-	-	-	-	-	-	-	-	1.46 ± 0	-	0.46 ± 0	0.23 ± 0	0.24 ± 0	0.03 ± 0	0.96 ± 0	2.41 ± 0
Milk	0.27 ± 0.0095	0.05 ± 0.0044	-	-	-	-	-	-	0.3 ± 0.0183	0.61 ± 0.0211	-	-	0.14 ± 0.0204	0.40 ± 0.0001	0.04 ± 0.0361	0.48 ± 0.0415	1.09 ± 0.0466

Crop	Farm Operations										Market Operations						
	Harves-ting	Collec-tion	Sorting/ Grading	Thresh-ing	Winnow-ing/ cleaning	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesa-lers	Retail-ers	Processi-ng unit	Trans-port	Total loss in market level	Overall total loss
Mustard	2.07 ±0.0431	0.53 ±0.0075	-	1.38 ±0.0367	0.48 ±0.0137	0.14 ±0.0021	0.11 ±0.0018	0.14 ±0.0021	1.01 ±0.0204	5.86 ±0.0588	0.06 ±0.0217	0.04 ±0.025	0.03 ±0.0055	0.04 ±0.018	0.21 ±0.0396	0.39 ±0.0472	6.25 ±0.0754
Okra	1.48 ± 0.0188	0.39 ± 0.0046	1.16 ± 0.0157	-	-	-	0.16 ± 0.0022	0.19 ± 0.0025	0.44 ± 0.0072	3.81 ± 0.0262	-	0.84 ± 0.167	0.9 ± 0.185	-	0.49 ± 0.0545	2.24 ± 0.2551	6.05 ± 0.2565
Paddy	1.53 ±0.0086	0.55 ±0.002		1.18 ±0.007	0.18 ±0.0011	0.63 ±0.0034	0.07 ±0.0022	0.4 ±0.0021	0.09 ±0.0005	4.62 ±0.0127	0.08 ±0.0101	0.17 ±0.057	0.03 ±0.0345	0.14 ±0.062	0.16 ±0.106	0.57 ±0.1401	5.19 ±0.1407
Pineapple	1.78 ±0.0747	0.09 ±0.005	1.11 ±0.0655	-	-	-	0.04 ±0.0031	0.04 ±0.004	0.05 ±0.0045	3.11 ±0.0998	0.13 ±0.0388	0.41 ±0.0422	0.65 ±0.1322	0.11 ±0.0539	0.06 ±0.040	1.37 ±0.1591	4.47 ±0.1878
Potato	2.28 ± 0.0054	0.15 ± 0.0019	1.89 ± 0.0025	-	-	-	0.04 ± 0.0005	0.11 ± 0.0007	0.42 ± 0.0014	4.89 ± 0.0065	0.10 ± 0.0002	0.3 ± 0.0365	0.08 ± 0.0024	0.05 ± 0.0004	0.36 ± 0.0061	0.89 ± 0.0371	5.77 ± 0.0376
Poultry Meat	2.38 ± 0.0431	-	-	-	-	-	-	-	-	2.38 ± 0.0431		2.29 ± 0	0.53 ± 0	0.02 ± 0	0.2 ± 0	3.03 ± 0	5.41 ± 0.0431
Radish	3.09 ±0.0421	0.08 ±0.0031	0.4 ±0.004	-	-	-	0.47 ±0.0054	0.14 ±0.0019	0.09 ±0.0012	4.27 ±0.0428		0.68 ±0.3191	1.2 ±0.5139	-	0.29 ±0.0841	2.17 ±0.6107	6.45 ±0.6122
Sunflower	1.15 ± 0.0035	0.49 ± 0.0015	-	1.92 ± 0.0009	0.23 ± 0.0004	0.1 ± 0.0003	0.09 ± 0.0002	0.05 ± 0.0005	0.13 ± 0.0005	4.17 ± 0.004	0.02 ± 0.0108	0.17 ± 0.0295	0.04 ± 0.0044	0.71 ± 0.1508	0.11 ± 0.0455	1.04 ± 0.1603	5.21 ± 0.1603
Wheat	1.52± 0.0314	0.53± 0.0121	-	1.46 ±0.0326	0.63 ±0.0213	0.01 ±0.0001	0.025 ±0.001	0.49 ±0.0154	0.7 ±0.0173	5.36 ±0.0565	0.004 ±0.1001	0.04 ±0.0079	0.01 ±0.003	0.18 ±0.1001	0.04 ±0.0751	0.27 ±0.1376	5.63 ±0.1387

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Crop	Farm Operations										Market Operations					
	Harvesti-ng	Collec-tion	Sorting/ Grading	Threshi-ng	Winnowing/ cleaning	Drying	Packaging	Farm level storage	Trans-port	Total loss in farm operatio-ns	Godown	Wholesa-lers	Retailers	Processi-ng unit	Trans-port	Total loss in market level	Overall total loss
Bajra	0.74 ±0.0252	0.28 ±0.009	-	1.25 ±0.04	0.15 ±0.0052	0.22 ±0.0077	0.26 ±0.0077	0.3 ±0.0114	0.25 ±0.0083	3.44 ±0.0517	0.13 ±0.0095	0.11 ±0.0846	0.07 ±0.0151	0.15 ±0.006	0.3 ±0.1664	0.76 ±0.1876	4.2 ±0.1946
Bottlegourd	1.84 ±0.0554	0.45 ±0.003	1.31 ±0.0144	-	-	-	0.37 ±0.0049	0.2 ±0.004	0.62 ±0.0095	4.79 ±0.0585	-	0.71 ±0.0999	1.02 ±0.360	-	0.82 ±0.1998	2.56 ±0.4242	7.35 ±0.4282
Brinjal	2.66 ±0.0672	0.28 ±0.005	1.16 ±0.0401	-	-	-	0.25 ±0.0043	0.37 ±0.0073	0.4 ±0.0133	5.12 ±0.08	-	0.72 ±0.5139	1.12 ±0.2711	-	0.57 ±0.2398	2.4 ±0.6285	7.52 ±0.6336
Cabbage	1.61 ± 0.0469	0.5 ± 0.0134	3.11 ± 0.0858	-	-	-	0.35 ± 0.0085	0.11 ± 0.0047	1.01 ± 0.0008	6.7 ± 0.0991	0.06 ± 0.0065	0.39 ± 0.0742	1.39 ± 0.3525	0.01 ± 0.0043	0.15 ± 0.0381	1.86 ± 0.3623	8.55 ± 0.3756
Cauliflower	2 ±0.0062	0.22 ±0.0017	2.42 ±0.0097	-	-	-	0.23 ±0.0009	0.04 ±0	0.72 ±0.0038	5.64 ±0.0122	0.07 ±0.0011	1.14 ±0.1039	0.71 ±0.0558	0.01 ±0.001	0.54 ±0.1871	2.46 ±0.2212	8.1 ±0.2215
Greengram	1.55 ±0.02	0.64 ±0.0052	-	1.51 ±0.0311	0.31 ±0.0041	0.36 ±0.0069	0.18 ±0.0074	0.47 ±0.0067	0.13 ±0.0068	5.14 ±0.0401	0.02 ±0.0131	0.32 ±0.418	0.29 ±0.0336	0.16 ±0.0071	0.13 ±0.0141	0.93 ±0.4198	6.07 ±0.4217
Guava	5.02 ±0.0929	0.31 ±0.0154	5.08 ±0.1383	-	-	-	0.35 ±0.0091	0.44 ±0.0229	0.55 ±0.0198	11.74 ±0.1703	-	2.09 ±0.287	3.12 ±0.6661	1.44 ±0.8544	0.96 ±0.597	3.56 ±1.2699	15.3 ±1.2812
Inland fish	2.21 ± 0.0585	0.34 ± 0.0073	1.6 ± 0.0342	-	-	-	0.15 ± 0.0032	0.09 ± 0.0031	0.17 ± 0.0035	4.57 ± 0.0684	-	0.37 ± 0.046	0.7 ± 0.193		0.1 ± 0.0253	1.17 ± 0.1948	5.74 ± 0.2065
Maize	0.54± 0.022	0.11± 0.007		2.2± 0.119	0.2± 0.009	0.27± 0.009	0.2± 0.026	0.16± 0.008	0.04± 0.002	3.72± 0.125	0.005± 0.023	0.44± 0.054	0.08± 0.0004	0.08± 0.023	0.19± 0.368	0.8± 0.393	4.52± 0.393
Mango	2.08 ±0.0729	0.47 ±0.0106	3.1 ±0.0598	-	-	-	0.03 ±0.0018	0.39 ±0.0041	0.91 ±0.0227	6.98 ±0.0977	0.01 ±0.0023	0.57 ±0.4595	1.12 ±0.1585	0.15 ±0.1063	1.13 ±0.2219	2.97 ±0.5343	9.95 ±0.5431
Muskmelon	1.46 ±0.0087	0.31 ±0.0071	1 ±0.0039	-	-	-	0.31 ±0.0477	0.05 ±0.0008	0.57 ±0.0035	3.68 ±0.0493	0.01 ±0.0032	0.64 ±0.2347	1.43 ±0.2962	-	0.87 ±0.21	2.96 ±0.4323	6.64 ±0.4351
Okra	1.57 ± 0.0127	0.27 ± 0.0023	1.17 ± 0.0073	-	-	-	0.18 ± 0.0011	0.16 ± 0.0011	0.57 ± 0.0075	3.93 ± 0.0167	-	0.82 ± 0.1742	0.96 ± 0.0459	-	0.56 ± 0.1621	2.34 ± 0.2423	6.27 ± 0.2429
Paddy	1.32 ±0.046	0.35 ±0.0116		0.99 ±0.0373	0.32 ±0.0118	0.42 ±0.0144	0.24 ±0.0096	0.32 ±0.0094	0.09 ±0.0036	4.04 ±0.0647	0.07 ±0.0238	0.23 ±0.1563	0.04 ±0.0161	0.17 ±0.0359	0.11 ±0.0315	0.62 ±0.1659	4.67 ±0.1781
Pineapple	1.72 ±0.2019	0.1 ±0.007	1.22 ±0.0866	-	-	-	0.03 ±0.0053	0.12 ±0.0124	0.69 ±0.0582	3.87 ±0.2278	0.06 ±0.0467	0.57 ±0.028	0.56 ±0.3805	0.13 ±0.0583	0.74 ±0.3586	2.05 ±0.5289	5.92 ±0.5759

Crop	Farm Operations										Market Operations						Overall total loss
	Harvesti-ng	Collec-tion	Sorting/ Grading	Threshi-ng	Winnowing/ cleaning	Drying	Packaging	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesa-lers	Retailers	Processi-ng unit	Trans-port	Total loss in market level	
Potato	2.12 ± 0.0051	0.12 ± 0.0015	2.07 ± 0.0028	-	-	-	0.04 ± 0.0004	0.08 ± 0.0006	0.43 ± 0.0015	4.85 ± 0.0062	0.11 ± 0.0005	0.26 ± 0.0303	0.08 ± 0.0086	0.02 ± 0.0001	0.3 ± 0.0243	0.77 ± 0.0398	5.62 ± 0.0403
Radish	1.34 ±0.0408	0.26 ±0.0099	0.35 ±0.011	-	-	-	0.52 ±0.0129	0.17 ±0.0038	0.1 ±0.0027	2.74 ±0.0455	-	0.47 ±0.1248	0.83 ±0.345	-	1.08 ±0.4092	2.37 ±0.5496	5.11 ±0.5515
Sugarcane	1.3 1±0.0712	1.86 ±0.0837	-	-	2.24 ±0.1008	-	-	-	2.31 ±0.1016	7.73 ±0.1804	-	-	-	1.74 ±0	1.02 ±0.8207	2.77 ±0.8207	10.49 ±0.8403
Sunflower	1.16 ± 0.0317	0.46 ± 0.0111	-	1.79 ± 0.0541	0.27 ± 0.008	0.1 ± 0.003	0.12 ± 0.0023	0.06 ± 0.0018	0.13 ± 0.0018	4.1 ± 0.0643	0.02 ± 0.0027	0.17 ± 0.0968	0.04 ± 0.0106	0.74 ± 0.0482	0.12 ± 0.097	1.08 ± 0.1456	5.18 ± 0.1592
Wheat	1.12± 0.0123	0.64± 0.0137	-	1.7 ±0.0103	0.06 ±0.0014	0.01 ±0.0002	0.001 ±0.00003	0.21 ±0.0043	0.12 ±0.0037	3.87 ±0.0219	0.196 ±0.025	0.28 ±0.0262	0.12 ±0.059	0.47 ±0.025	0.08 ±0.1477	1.14 ±0.1646	5.01 ±0.1646

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	Farm Operations											Market Operations						
Crop	**Harvest-ing**	**Collec-tion**	**Sorting/ Grading**	**Trim m-ing**	**Thresh-ing**	**Winnow-ing/ cleaning**	**Drying**	**Packag-ing**	**Farm level storage**	**Trans-port**	**Total loss in farm operations**	**Godown**	**Whole-salers**	**Retai-lers**	**Process-ing unit**	**Trans-port**	**Total loss in market level**	**Overall total loss**
Bajra	0.71 ±0.02	0.24 ±0.0052	-	-	1.23 ±0.0311	0.15 ±0.0041	0.22 ±0.0069	0.25 ±0.0074	0.27 ±0.0067	0.24 ±0.0068	3.3 ±0.0401	0.14 ±0.0249	0.07 ±0.0385	0.08 ±0.0113	0.14 ±0.0079	0.34 ±0.0149	0.78 ±0.0501	4.08 ±0.0641
Black gram	1.39 ±0.0456	1.25 ±0.0579	-	-	1 ±0.0391	0.13 ±0.0077	0.12 ±0	0.14 ±0.0062	0.3 ±0.0132	0.14 ±0.0057	4.46 ±0.0852	0.03 ±0.0154	0.14 ±0.0685	0.23 ±0.1962	0.14 ±0.05	0.23 ±0.1159	0.77 ±0.2436	5.23 ±0.2581
Bottlegourd	2.03 ±0.1377	0.55 ±0.0289	0.53 ±0.0377	-	-	-	-	0.19 ±0.0091	0.14 ±0.0853	1.09 ±0.0853	4.52 ±0.1893	-	0.48 ±0.1146	0.67 ±0.2901	-	0.76 ±0.1509	1.91 ±0.3466	6.43 ±0.3949
Egg	2.18 ± 0.087	-	0.92 ± 0.089	-	-	-	-	1.15 ± 0.0797	0.01 ± 0.0011	0.24 ± 0.0133	4.5 ± 0.1484	0.03 ± 0	0.73 ± 0	0.46 ± 0	-	0.34 ± 0	1.56 ± 0	6.06 ± 0.1484
Green Pea	1.41 ± 0.0931	0.56 ± 0.1117	1.91 ± 0.1071	-	-	-	-	0.27 ± 0.017	0.002 ± 0.0004	0.5 ± 0.033	4.65 ± 0.1844	-	0.41 ± 0.2471	0.75 ± 0.3573	0.03 ± 0	0.52 ± 0.2471	1.71 ± 0.4998	6.36 ± 0.5327
Guava	5.36 ±0.0625	0.19 ±0.0019	4.25 ±0.0406	-	-	-	-	0.31 ±0.0048	0.47 ±0.0071	0.74 ±0.0066	11.32 ±0.0753	-	1.23 ±0.6306	2.27 ±0.6854	1.31 ±0.9271	1.65 ±0.6252	3.44 ±1.4552	14.76 ±1.4572
Inland fish	1.81 ± 0.1564	0.36 ± 0.0238	1.24 ± 0.1193	-	-	-	-	0.24 ± 0.0124	0.07 ± 0.0077	0.13 ± 0.0102	3.86 ± 0.1989	-	0.36 ± 0.0536	0.63 ± 0.0374	-	0.14 ± 0.0488	1.13 ± 0.0621	4.99 ± 0.2084
Maize	0.99± 0.031	0.54± 0.042	-	-	0.54± 0.016	0.19± 0.002	0.42± 0.018	0.56± 0.026	0.18± 0.001	0.45± 0.083	3.88± 0.104	0.004± 0.061	0.15± 0.049	0.22± 0.001	0.05± 0.061	0.16± 0.309	0.58 ±0.335	4.46± 0.345
Mango	1.87 ±0.1464	0.29 ±0.0461	2.87 ±0.2356	-	-	-	-	0.04 ±0.0076	0.4 ±0.0535	0.91 ±0.0655	6.38 ±0.2937	0.01 ±0.002	0.63 ±0.2319	1.18 ±0.2258	0.13 ±0.0246	0.86 ±0.0667	2.81 ±0.3305	9.19 ±0.4421
Meat	1.73 ± 0	-	-	-	-	-	-	-	-	-	1.73 ± 0	-	0.48 ± 0	0.29 ± 0	0.27 ± 0	0.03 ± 0	1.07 ± 0	2.80 ± 0
Milk	0.29 ± 0.0044	0.09 ± 0.0087	-	-	-	-	-	-	-	0.24 ± 0.0176	0.62 ± 0.0201	-	-	0.17 ± 0.0051	0.14 ± 0.0605	0.46 ± 0.0432	0.50 ± 0.0746	1.12 ± 0.0772
Mushroom	0.11± 0.0187	0.35± 0.0537	0.69± 0.0859	0.12± 0.0175	-	0.13± 0.0139	-	0.24± 0.0376	0.14± 0.0263	0.05± 0.0073	1.84± 0.1152	-	0.43± 0.1115	0.55± 0.2655	-	0.42± 0.2295	0.9± 0.3683	2.74± 0.3859
Muskmelon	1.65 ±0.002	0.27 ±0.0058	1.28 ±0.0019	-	-	-	-	0.35 ±0.0528	0.06 ±0.0004	0.81 ±0.0015	4.43 ±0.0532	0.01 ±0.0042	0.63 ±0.1532	1.4 ±0.4873	-	0.88 ±0.3385	2.93 ±0.6128	7.35 ±0.6151
Mustard	1.9 ±0.268	0.49 ±0.1963	-	-	1.2 ±0.232	0.37 ±0.102	0.14 ±0.0584	0.12 ±0.0461	0.14 ±0.0559	1.29 ±0.3144	5.65 ±0.4244	0.06 ±0.0138	0.02 ±0.0135	0.02 ±0.0066	0.02 ±0.0086	0.25 ±0.0456	0.37 ±0.048	6.03 ±0.4271

Crop	Farm Operations											Market Operations						
	Harvest-ing	Collec-tion	Sorting/ Grading	Trim m-ing	Thresh-ing	Winnow-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retai-lers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Paddy	1.24 ±0.031	0.24 ±0.0042	-	-	0.56 ±0.0094	0.87 ±0.0173	0.3 ±0.0051	0.04 ±0.001	0.16 ±0.0088	0.05 ±0.0014	3.46 ±0.0384	0.06 ±0.0656	0.27 ±0.1659	0.04 ±0.0095	0.17 ±0.0421	0.1 ±0.0989	0.64 ±0.2085	4.09 ±0.212
Potato	1.9 ± 0.0033	0.32 ± 0.012	2.22 ± 0.0036	-	-	-	-	0.04 ± 0.0011	0.11 ± 0.0006	0.62 ± 0.0019	5.21 ± 0.0132	0.1 ± 0.0006	0.29 ± 0.0464	0.1 ± 0.0059	0.02 ± 0.0004	0.25 ± 0.0066	0.75 ± 0.0473	5.96 ± 0.0491
Poultry Meat	2.88 ± 0.0544	-	-	-	-	-	-	-	-	-	2.88 ± 0.0544	-	2.49 ± 0	1.45 ± 0	0.02 ± 0	0.4 ± 0	4.36 ± 0	7.24 ± 0.0544
Sugarcane	1.68 ±0.0268	0.87 ±0.0146	-	-	0.95 ±0.0171	-	-	2.16 ±0.0412	5.65 ±0.0541	-	-	-	0.17 ±0.1009	1.09 ±0.2275	1.26 ±0.2489	6.91 ±0.2547	-	-
Wheat	1.41 ±0.0172	0.61 ±0.0066	-	-	1.05 ±0.0227	0.09 ±0.0009	0.04 ±0.0007	0.003 ±0.0002	0.06 ±0.0015	0.15 ±0.0022	3.41 ±0.0294	0.001 ±0.1267	0.15 ±0.0289	0.1 ±0.0019	0.37 ±0.1267	0.03 ±0.059	0.64 ±0.1457	4.06 ±0.1457

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Crop	Farm Operations											Market Operations						
	Harvest-ing	Collec-tion	Sorting / Grading	Trimming	Thresh-ing	Winnow-ing/ cleaning	Drying	Packing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesalers	Retailers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Bajra	0.69 ±0.0357	0.29 ±0.0147	-	-	0.82 ±0.044	0.15 ±0.0086	0.22 ±0.0117	0.24 ±0.0165	0.31 ±0.0174	0.24 ±0.0123	2.96 ±0.066	0.16 ±0.0161	0.14 ±0.0772	0.05 ±0.0213	0.14 ±0.0497	0.32 ±0.0363	0.81 ±0.1023	3.76 ±0.1217
Capsicum	1.30 ± 0.133	0.02 ± 0.0009	0.03 ± 0.0021	-	-	-	-	0.03 ± 0.0008	0.006 ± 0.0012	0.11 ± 0.0233	1.49 ± 0.1351	-	1.02 ± 0.2763	1.00 ± 0.1041	-	0.8 ± 0.4636	2.82 ± 0.5496	4.31 ± 0.566
Chickpea	1.3 ±0.1466	0.85 ±0.1089	-	-	1.62 ±0.0609	0.42 ±0.031	0.38 ±0.0494	0.17 ±0.0211	0.56 ±0.0198	0.69 ±0.0484	5.02 ±0.1991	0.06 ±0.0204	0.19 ±0.0332	0.1 ±0.0379	0.07 ±0.0205	0.28 ±0.1216	0.7 ±0.1316	5.72 ±0.2387
Citrus	0.9 ±0.0347	0.01 ±0.0003	1 ±0.0812	-	-	-	-	0.03 ±0.0053	0.8 ±0.0447	0.89 ±0.0369	3.62 ±0.1058	0.03 ±0.0224	0.31 ±0.2277	0.53 ±0.1118	0.02 ±0.0063	0.89 ±0.1371	1.78 ±0.2892	5.4 ±0.308
Cottonseed	1.59 ± 0.0502	0.16 ± 0.008	-	-	-	-	0.04 ± 0.0016	0.02 ± 0.0022	0.02 ± 0	0.22 ± 0.0078	2.05 ± 0.0514	0.01 ± 0.0022	0.15 ± 0.0353	0.02 ± 0.0047	0.03 ± 0.0076	0.07 ± 0.0228	0.28 ± 0.0429	2.34 ± 0.067
Egg	2.26 ± 0.3016	-	1.27 ± 0.1184	-	-	-	-	1.14 ± 0.1813	0.08 ± 0.0128	0.41 ± 0.0513	5.16 ± 0.375	0.06 ± 0	0.69 ± 0.0723	0.41 ± 0	-	0.42 ± 0	1.58 ± 0.0723	6.74 ± 0.3819
Green Pea	2.10 ± 0.0924	0.41 ± 0.0214	1.73 ± 0.071	-	-	-	-	0.20 ± 0.0081	0.007 ± 0.0011	0.33 ± 0.0148	4.77 ± 0.1197	-	0.37 ± 0.1559	0.64 ± 0.2184	0.03 ± 0	0.85 ± 0.2961	1.89 ± 0.3996	6.67 ± 0.4172
Greengram	1.51 ±0.0357	0.61 ±0.0147	-	-	1.54 ±0.044	0.33 ±0.0086	0.29 ±0.0117	0.18 ±0.0165	0.4 ±0.0174	0.1 ±0.0123	4.96 ±0.066	0.02 ±0.0026	0.33 ±0.0909	0.31 ±0.1657	0.13 ±0.0395	0.01 ±0.0029	0.79 ±0.1931	5.75 ±0.2041
Maize	0.9± 0.043	1.24± 0.084	-	-	1.01± 0.05	0.2± 0.011	0.18± 0.037	0.11± 0.02	0.2± 0.011	0.2± 0.022	4.04± 0.118	0.002± 0.039	0.46± 0.025	0.27± 0.0001	0.13± 0.039	0.06± 0.117	0.93± 0.172	4.97± 0.174
Meat	2.12± 0.2986	-	-	-	-	-	-	-	-	-	2.12 ±0.2986	-	0.15 ± 0	0.2 ± 0	0.39 ± 0	0.03 ± 0	0.76 ± 0	2.88 ±0.2986
Milk	0.16 ± 0.005	0.08 ± 0.0157	-	-	-	-	-	-	-	0.01 ± 0.0254	0.26 ± 0.0303	-	-	0.11 ± 0.0453	0.07 ± 0.1325	0.17 ± 0.1242	0.22 ± 0.1872	0.48 ± 0.1896
Mushroom	0.16± 0.0107	0.14± 0.0115	4.27± 0.2571	0.08± 0.0083	-	0.07± 0.009	-	0.2± 0.0169	0.08± 0.0056	0.08± 0.0086	5.09± 0.2586	-	0.7± 0.0564	0.86± 0.2147	-	0.48± 0.2294	1.26± 0.3193	6.35± 0.4109
Muskmelon	1.22 ±0.004	0.24 ±0.0324	0.79 ±0.003	-	-	-	-	0.34 ±0.041	0.04 ±0.0007	0.61 ±0.0028	3.25 ±0.0526	0.01 ±0.0053	0.51 ±0.1614	1.14 ±0.2516	-	0.6 ±0.2124	2.25 ±0.3668	5.5 ±0.3705

Crop	Farm Operations											Market Operations						
	Harvest-ing	Collec-tion	Sorting / Grading	Trimming	Thresh-ing	Winnow-ing/ cleaning	Drying	Packing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesalers	Retailers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Mustard	1.13 ±0.3732	0.19 ±0.1043	-	-	1.55 ±0.4472	0.36 ±0.121	0.05 ±0.0286	0.04 ±0.0239	0.04 ±0.0229	0.2 ±0.0089	3.57 ±0.6051	0.04 ±0.0344	0.03 ±0.0087	0.03 ±0.0122	0.04 ±0.0124	0.2 ±0.009	0.35 ±0.0176	3.92 ±0.6053
Paddy	1 ±0.0562	0.39 ±0.0073	-	-	0.36 ±0.0049	0.23 ±0.0165	0.27 ±0.0129	0.12 ±0.0092	0.26 ±0.03	0.07 ±0.0045	2.7 ±0.0684	0.08 ±0.0423	0.17 ±0.043	0.04 ±0.0052	0.16 ±0.0486	0.1 ±0.0219	0.55 ±0.0807	3.25 ±0.1057
Potato	2.11 ± 0.0021	0.29 ± 0.0051	2.19 ± 0.0022	-	-	-	-	0.03 ± 0.0002	0.12 ± 0.0004	0.65 ± 0.0011	5.4 ± 0.0061	0.07 ± 0.0004	0.25 ± 0.0248	0.11 ± 0.008	0.04 ± 0.0002	0.26 ± 0.0041	0.73 ± 0.0264	6.13 ± 0.0271
Poultry Meat	2.25 ± 0	-	-	-	-	-	-	-	-	-	2.25 ± 0	-	1.31 ± 0	0.52 ± 0	0.01 ± 0	0.54 ± 0	2.38 ± 0	4.63 ± 0
Radish	2.36 ±0.0295	0.06 ±0.0049	2.26 ±0.0307	-	-	-	-	0.48 ±0.0044	0.08 ±0.001	0.11 ±0.0015	5.35 ±0.0431	-	0.69 ±0.2094	0.36 ±0.3152	-	1.11 ±0.3476	2.16 ±0.5138	7.51 ±0.5156
Sugarcane	0.86 ±0.0459	0.39 ±0.0094	-	-	-	1.04 ±0.0268	-	-	-	2.04 ±0.0447	4.33 ±0.0701	-	-	-	0.27 ±0.0406	0.85 ±0.402	1.12 ±0.4048	5.45±0.4108
Sunflower	0.99 ± 0.143	0.5 ± 0.0744	-	-	1.36 ± 0.1921	0.18 ± 0.0334	0.05 ± 0.0072	0.08 ± 0.0107	0.06 ± 0.0091	0.11 ± 0.0217	3.31 ± 0.2544	0.02 ± 0.0077	0.18 ± 0.0841	0.04 ± 0.0121	0.78 ± 0.4634	0.12 ± 0.0064	1.13 ± 0.4712	4.44 ± 0.5355
Wheat	1.38 ±0.022	0.4 ±0.0126	-	-	1.52 ±0.0371	0.3 ±0.0091	0.04 ±0.0012	0.005 ±0.001	0.04 ±0.002	0.1 ±0.0022	3.79 ±0.0459	0.0002 ±0.0437	0.05 ±0.0262	0.03 ±0.0001	0.26 ±0.0437	0.01 ±0.027	0.36 ±0.0738	4.15 ±0.0738

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Crop	Farm Operations											Market Operations						
	Harvesting	Collec-tion	Sorting / Grading	Trimm-ing	Thresh-ing	Winno-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retai-lers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Bajra	1.06 ±0.0207	0.33 ±0.0075	-	-	1.12 ±0.0287	0.16 ±0.0037	0.22 ±0.0048	0.25 ±0.0054	0.28 ±0.0084	0.27 ±0.0054	3.68 ±0.0384	0.12 ±0.0093	0.12 ±0.0034	0.08 ±0.0286	0.11 ±0.0206	0.31 ±0.0318	0.75 ±0.0485	4.44 ±0.0619
Black gram	1.63 ±0.0551	0.66 ±0.0236	-	-	1.15 ±0.0598	0.2 ±0.0112	0.34 ±0.0007	0.18 ±0.0113	0.44 ±0.0186	0.13 ±0.0107	4.73 ±0.0888	0.04 ±0.0197	0.27 ±0.0722	0.32 ±0.0805	0.12 ±0.0296	0.2 ±0.0942	0.95 ±0.1478	5.68 ±0.1724
Bottlegourd	2.04 ±0.3894	0.4 ±0.0539	1.22 ±0.2099	-	-	-	-	0.2 ±0.0288	0.26 ±0.1106	0.53 ±0.1106	4.65 ±0.4732	-	0.88 ±0.3715	0.77 ±0.5091	-	0.84 ±0.3318	2.5 ±0.7122	7.14 ±0.8551
Brinjal	1.99 ±0.0161	0.33 ±0.0076	0.47 ±0.0065	-	-	-	-	0.25 ±0.0018	0.44 ±0.0149	0.82 ±0.0366	4.31 ±0.0439	-	0.94 ±0.5203	1.21 ±0.2217	-	0.86 ±0.1926	3.01 ±0.5975	7.32 ±0.5991
Cabbage	1.52 ± 0.0811	0.38 ± 0.0209	2.95 ± 0.1581	-	-	-	-	0.32 ± 0.0153	0.16 ± 0.0144	0.79 ± 0.0014	6.12 ± 0.1801	0.08 ± 0.0128	0.82 ± 0.1368	0.95 ± 0.0985	0.01 ± 0.0031	0.17 ± 0.0517	1.86 ± 0.1768	7.98 ± 0.2524
Cauliflower	1.61 ±0.0575	0.2 ±0.0107	2.29 ±0.0839	-	-	-	-	0.3 ±0.0107	0.06 ±0	0.89 ±0.0318	5.35 ±0.1076	0.04 ±0.0008	0.54 ±0.0536	0.76 ±0.069	0.01± 0.0004	0.62 ±0.1549	1.97 ±0.1778	7.33 ±0.2079
Chickpea	1.43 ±0.067	0.89 ±0.0295	1.59 ±0.0592	-	-	0.32 ±0.0139	0.43 ±0.0143	0.17 ±0.0056	0.18 ±0.0078	0.6 ±0.0236	4.71 ±0.096	0.12 ±0.048	0.18 ±0.0188	0.11 ±0.0345	0.13 ±0.0561	0.2 ±0.1248	0.76 ±0.1308	5.47 ±0.1623
Citrus	1.47 ±0.0843	0.56 ±0.0263	0.96 ±0.0688	-	-	-	-	0.35 ±0.019	0.64 ±0.0296	0.86 ±0.0405	4.83 ±0.1242	0.02 ±0.0125	0.66 ±0.3197	0.69 ±0.9372	0.03 ±0.0058	1.18 ±0.3073	2.57 ±1.0369	7.4 ±1.0443
Coriander	1.87 ± 0.0583	0.57 ± 0.0228	-	-	1.19 ± 0.0537	0.55 ± 0.0246	0.04 ± 0.002	0.11 ± 0.0054	0.06 ± 0.0032	0.12 ± 0.0044	4.51 ± 0.0864	0.02 ± 0.0082	0.32 ± 0.0289	0.25 ± 0.0238	0.07 ± 0.0285	0.33 ± 0.0725	0.99 ± 0.0836	5.5 ± 0.1202
Cottonseed	1.7 ± 0.0285	0.32 ± 0.0051	-	-	-	-	0.07 ± 0.001	0.12 ± 0.0031	0.12 ± 0	0.11 ± 0.0028	2.43 ± 0.0292	0.01 ± 0.0047	0.21 ± 0.0361	0.04 ± 0.0086	0.04 ± 0.0131	0.11 ± 0.0493	0.41 ± 0.0631	2.84 ± 0.0695
Egg	1.34 ± 0.07	-	1.70 ± 0.4326	-	-	-	-	1.31 ± 0.1418	0.04 ± 0.0125	0.28 ± 0.0678	4.66 ± 0.4657	0.07 ± 0	0.72 ± 0	0.34 ± 0		0.54 ± 0	1.66 ± 0	6.33 ± 0.4657
Green Pea	2.06 ± 0.0806	0.5 ± 0.0432	1.68 ± 0.12	-	-	-	-	0.09 ± 0.0053	0.005 ± 0.0007	0.38 ± 0.0224	4.72 ± 0.1527	-	0.37 ± 0.2659	0.47 ± 0.2691	0.04 ± 0	0.75 ± 0.2466	1.64 ± 0.4515	6.36 ± 0.4767
Greengram	1.9 ±0.0207	0.71 ±0.0075	-	-	1.55 ±0.0287	0.3 ±0.0037	0.3 ±0.0048	0.19 ±0.0054	0.35 ±0.0084	0.1 ±0.0054	5.42 ±0.0384	0.02 ±0.0017	0.32 ±0.0394	0.29 ±0.0879	0.15 ±0.0031	0.01 ±0.0003	0.78 ±0.0964	6.2 ±0.1037
Groundnut	1.91 ± 0.0311	0.62 ± 0.0028	-	-	1.35 ± 0.0123	0.41 ± 0.0052	0.13 ± 0.0006	0.14 ± 0.0018	0.09 ± 0.0008	0.13 ± 0.0011	4.79 ± 0.0341	0.06 ± 0.0369	0.39 ± 0.0833	0.07 ± 0.0306	0.3 ± 0.0673	0.17 ± 0.1336	1 ± 0.174	5.78 ± 0.1773

	Farm Operations											Market Operations						
Crop	Harvesting	Collec-tion	Sorting / Grading	Trimm-ing	Thresh-ing	Winno-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retai-lers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Guava	4.89 ±0.3201	0.63 ±0.0446	4.76 ±0.344	-	-	-	-	0.38 ±0.0303	0.46 ±0.0517	0.77 ±0.0287	11.89 ±0.4766	-	1.74 ±0.5171	2.44 ±0.2328	1.27 ±0.5597	0.62 ±0.3066	2.7 ±0.8538	14.59 ±0.9778
Inland fish	2.09 ± 0.042	0.57 ± 0.008	1.76 ± 0.0091	-	-	-	-	0.17 ± 0.0005	0.08 ± 0.001	0.19 ± 0.0009	4.86 ± 0.0437	-	0.39 ± 0.0401	0.5 ± 0.1789	-	0.11 ± 0.0386	0.99 ± 0.1831	5.85 ± 0.1883
Maize	1.22± 0.022	0.11± 0.016	-	-	1.95± 0.074	0.08± 0.003	0.05± 0.002	0.07± 0.004	0.1± 0.003	0.14± 0.02	3.73± 0.081	0.005± 0.132	0.68± 0.031	0.13± 0.003	0.18± 0.132	0.09± 0.176	1.09± 0.236	4.81± 0.237
Milk	0.41 ± 0.0257	0.06 ± 0.0071	-	-	-	-	-	-	-	0.52 ± 0.0496	0.98 ±0.0563	-	-	0.26 ± 0.0345	0.22 ± 0.0882	0.10 ± 0.0954	0.37 ± 0.1344	1.35 ± 0.1458
Mushroom	0.51± 0.0263	0.52± 0.0396	4.45± 0.2385	0.58± 0.031	-	0.31± 0.032	-	0.11± 0.0119	0.62± 0.043	0.45± 0.0317	7.57± 0.2532	-	0.35± 0.128	0.69± 0.107	-	0.33± 0.1464	0.85± 0.2219	8.42± 0.3367
Mustard	1.48 ±0.0804	0.37 ±0.0427	-	-	0.99 ±0.0504	0.29 ±0.0164	0.12 ±0.0109	0.09 ±0.0096	0.05 ±0.0089	0.05 ±0.0055	3.43 ±0.1063	0.05 ±0.0074	0.07 ±0.0132	0.04 ±0.0081	0.04 ±0.0057	0.22 ±0.0216	0.41 ±0.0266	3.84 ±0.1096
Okra	1.42 ± 0.093	0.33 ± 0.0194	1.23 ± 0.0138	-	-	-	-	0.1 ± 0.0046	0.2 ± 0.0117	0.47 ± 0.0244	3.75 ± 0.0999	-	0.76 ± 0.306	0.77 ± 0.1596	-	0.55 ± 0.1261	2.08 ± 0.3674	5.84 ± 0.3808
Onion	2.67 ± 0.2962	0.42 ± 0.0327	1.19 ± 0.1032	-	-	-	-	0.08 ± 0.0202	1.02 ± 0.0565	0.49 ± 0.0196	5.87 ± 0.3216	0.36 ± 0.243	0.99 ± 0.2712	0.59 ± 0.2312	0.01 ± 0.0064	0.28 ± 0.0963	2.24 ± 0.442	8.11 ± 0.5466
Paddy	1.08 ±0.0231	0.4 ±0.0228	-	-	0.84 ±0.0313	0.33 ±0.0149	0.34 ±0.0133	0.16 ±0.0033	0.27 ±0.0127	0.09 ±0.0051	3.52 ±0.0513	0.06 ±0.0391	0.2 ±0.0844	0.04 ±0.0062	0.16 ±0.027	0.08 ±0.0331	0.54 ±0.1026	4.05 ±0.1147
Papaya	1.63 ± 0.1222	0.32 ± 0.0436	0.59 ± 0.0421	-	-	-	-	0.22 ± 0.0316	0.1 ± 0.0077	0.59 ± 0.0667	3.45 ± 0.1553	0.02 ± 0.0101	0.71 ± 0.2029	1.58 ± 0.8358	0.04 ± 0.0241	0.92 ± 0.2537	3.28 ± 0.8971	6.73 ± 0.9105
Pigeon Pea	1.04 ± 0.0876	0.39 ± 0.0334	-	-	1.99 ± 0.1899	0.94 ± 0.0271	0.27 ± 0.0172	0.19 ± 0.0109	0.57 ± 0.0636	0.15 ± 0.0129	5.56 ± 0.2241	0.13 ± 0.0374	0.11 ± 0.0454	0.16 ± 0.0577	0.41 ± 0.1202	0.27 ± 0.0417	1.08 ± 0.147	6.63 ± 0.268
Radish	1.48 ±0.0355	0.04 ±0.0031	0.19 ±0.0166	-	-	-	-	0.46 ±0.0069	0.13 ±0.0109	0.11 ±0.0014	2.4 ±0.0414	-	1.34 ±0.2132	0.84 ±0.1909	-	2.07 ±0.1428	4.26 ±0.3199	6.66 ±0.3225
Sorghum	1.56 ±0.0654	0.3 ±0.0123	-	-	1.65 ±0.0759	0.29 ±0.0109	0.13 ±0.005	0.22 ±0.0086	0.25 ±0.0085	0.19 ±0.008	4.59 ±0.1027	0.04 ±0.0078	0.85 ±0.0631	0.18 ±0.026	0.02 ±0.0095	0.17 ±0.3293	1.25 ±0.3365	5.85 ±0.3519
Soybean	2.31 ± 0.0739	0.93 ± 0.0271	-	-	0.98 ± 0.0579	0.29 ± 0.0134	0.08 ± 0.0035	0.24 ± 0.0147	0.91 ± 0.0257	0.25 ± 0.0083	5.99 ± 0.1033	0.06 ± 0.0066	0.7 ± 0.509	0.25 ± 0.0514	0.2 ± 0.1166	0.05 ± 0.01	1.25 ± 0.5248	7.24 ± 0.5349
Tomato	3.37 ±6.7688	0.74 ±1.4711	2.95 ±6.1949	-	-	-	-	0.47 ±1.0848	0.58 ±1.1467	0.83 ±1.7278	8.95 ±9.5831	-	0.85 ±0.1862	1.11 ±0.586	0.18 ±0.0738	1.27 ±0.687	3.41 ±0.922	12.36 ±9.6273

Crop	Farm Operations											Market Operations						
	Harvesting	Collection	Sorting / Grading	Trimming	Threshing	Winnowing/ cleaning	Drying	Packaging	Farm level storage	Transport	Total loss in farm operations	Godown	Wholesalers	Retailers	Processing unit	Transport	Total loss in market level	Overall total loss
Wheat	2.41 ±0.142	0.35 ±0.0254	-	-	0.71 ±0.0493	0.02 ±0.0021	0.03 ±0.0033	0.011 ±0.001	0.73 ±0.046	0.07 ±0.005	4.33 ±0.1594	0.004 ±0.1832	0.08 ±0.0097	0.03 ±0.0021	0.36 ±0.1832	0.09 ±0.1725	0.56 ±0.298	4.89 ±0.2981

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Crop	Farm Operations										Market Operations						
	Harvesti-ng	Collec-tion	Sorting/ Grading	Threshi-ng	Winnow-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesal-ers	Retaile-rs	Process-ing unit	Transport	Total loss in market level	Overall total loss
Bajra	0.95 ±0.2234	0.31 ±0.07	-	1.27 ±0.2707	0.16 ±0.0353	0.22 ±0.0486	0.25 ±0.057	0.38 ±0.0774	0.27 ±0.0657	3.82 ±0.3811	0.14 ±0.0069	0.13 ±0.1442	0.11 ±0.0362	0.13 ±0.0187	0.31 ±0.1289	0.83 ±0.1978	4.64 ±0.4294
Black gram	1.41 ±0.0226	0.94 ±0.0208	-	1.73 ±0.0303	0.66 ±0.0225	0.45 ±0.0001	0.38 ±0.0081	0.24 ±0.0071	0.12 ±0.0042	5.94 ±0.05	0.01 ±0.003	0.27 ±0.0386	0.23 ±0.1461	0.15 ±0.1585	0.24 ±0.1784	0.89 ±0.2824	6.83 ±0.2868
Chickpea	1.48 ±0.0131	0.81 ±0.0132	-	1.52 ±0.033	0.46 ±0.0085	0.64 ±0.0074	0.16 ±0.0008	0.16 ±0.0011	0.27 ±0.0054	4.88 ±0.0392	0.12 ±0.061	0.25 ±0.086	0.08 ±0.0567	0.18 ±0.0959	0.29 ±0.1531	0.92 ±0.1845	5.81 ±0.1887
Coriander	2.01 ± 0.0462	0.85 ± 0.0261	-	1.17 ± 0.0503	0.49 ± 0.0246	0.11 ± 0.0037	0.11 ± 0.0038	0.06 ± 0.0032	0.12 ± 0.0058	4.92 ± 0.0776	0.02 ± 0.007	0.33 ± 0.0109	0.28 ± 0.0099	0.07 ± 0.0028	0.33 ± 0.0516	1.03 ± 0.0533	5.95 ± 0.0941
Green Pea	1.81 ± 0.0269	0.41 ± 0.1082	1.68 ± 0.2869	-	-	-	0.11 ± 0.0065	0.004 ± 0.001	0.11 ± 0.0048	4.13 ± 0.3079	-	0.57 ± 0.2248	0.61 ± 0.4667	0.05 ± 0	0.46 ± 0.2717	1.68 ± 0.585	5.82 ± 0.661
Maize	1.01± 0.025	0.28± 0.008	-	0.64 ±0.016	0.36 ±0.011	0.24 ±0.007	0.15 ±0.008	0.3 ±0.008	0.1± 0.004	3.09± 0.036	0.0001 ±0.085	0.18± 0.033	0.07± 0.0001	0.06 ±0.085	0.15± 0.3	0.46± 0.316	3.54± 0.316
Paddy	1.03 ±0.0246	0.61 ±0.0057	-	0.74 ±0.0267	0.24 ±0.0047	0.35 ±0.0178	0.32 ±0.0054	0.28 ±0.0495	0.2 ±0.0044	3.77 ±0.0647	0.04 ±0.0368	0.18 ±0.2362	0.02 ±0.0423	0.14 ±0.0903	0.12 ±0.1711	0.51 ±0.3104	4.28 ±0.3171
Pigeon Pea	1.71 ± 0.0156	0.38 ± 0.003	-	2.45 ± 0.0124	0.15 ± 0.0012	0.31 ± 0.0064	0.23 ± 0.0021	0.4 ± 0.0037	0.19 ± 0.0038	5.81 ± 0.0219	0.15 ± 0.1043	0.12 ± 0.0755	0.18 ± 0.1387	0.45 ± 0.2825	0.35 ± 0.2591	1.26 ± 0.4153	7.07 ± 0.4159
Sorghum	1.68 ±0.0462	0.21 ±0.012	-	1.99 ±0.0621	0.2 ±0.0122	0.16 ±0.012	0.25 ±0.0146	0.21 ±0.0097	0.22 ±0.0167	4.92 ±0.0837	0.03 ±0.0062	0.73 ±0.2926	0.12 ±0.0434	0.01 ±0.0153	0.23 ±0.46	1.13 ±0.5471	6.05 ±0.5535
Soybean	1.85 ± 0.039	1.06 ± 0.0216	-	1.37 ± 0.0255	0.16 ± 0.004	0.03 ± 0.0008	0.37 ± 0.007	0.4 ± 0.0087	0.3 ± 0.004	5.54 ± 0.0529	0.08 ± 0.0108	0.48 ± 0.0712	0.15 ± 0.0022	0.21 ± 0.05	0.09 ± 0.0254	1 ± 0.0913	6.54 ± 0.1055

Crop	Farm Operations										Market Operations						
	Harvesti-ng	Collec-tion	Sorting/ Grading	Threshi-ng	Winnow-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesal-ers	Retaile-rs	Process-ing unit	Transp ort	Total loss in market level	Overall total loss
Tapioca	1.17 ±0.006	0.25 ±0.0032	1.01 ±0.0086	-	-	-	0.07 ±0.0009	0.25 ±0.0014	0.61 ±0.0055	3.36 ±0.0124	-	0.47 ±0.3255	0.5 ±0.0866	0.18 ±0.0308	0.45 ±0.1941	1.59 ±0.39	4.96 ±0.3902
Wheat	2.06 ±0.0484	0.27 ±0.0375	-	0.97 ±0.1187	0.91 ±0.0175	0.15 ±0.0134	0.013 ±0.00029	0.22±0.0042	0.1 ±0.0028	4.69 ±0.1355	0.003 ±0.3596	0.13 ±0.0727	0.07 ±0.0159	0.18 ±0.3596	0.14 ±0.274	0.53 ±0.4778	5.22 ±0.4778

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Crop	Farm Operations										Market Operations						Overall total loss
	Harvest-ing	Collec-tion	Sorting/ Grading	Trimming	Thresh-ing	Winnow-ing/ cleaning	Drying	Packaging	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesalers	Retailers	Processing unit	Transport	Total loss in market level	
Bajra	0.97 ±0.0455	0.3 ±0.019	-	-	0.97 ±0.0432	0.17 ±0.0086	0.22 ±0.0097	0.24 ±0.0108	0.26 ±0.0132	0.26 ±0.0123	3.4 ±0.0701	0.13 ±0.0173	0.11 ±0.015	0.07 ±0.0105	0.16 ±0.0105	0.34 ±0.0757	0.81 ±0.0804	4.21 ±0.1067
Banana	1.24 ±0.0226	0.19 ±0.0117	1.55 ±0.0152	-	-	-	-	0.12 ±0.0065	0.02 ±0.0004	1.04 ±0.0335	4.15 ±0.0453	0.05 ±0.0156	0.76 ±0.2146	0.27 ±0.1252	0.01 ±0.003	0.82 ±0.2816	1.91 ±0.3758	6.06 ±0.3785
Black gram	1.88 ±0.016	1.09 ±0.0106		-	1.55 ±0.0193	0.72 ±0.0128	0.46 ±0.0001	0.32 ±0.0055	0.51 ±0.0104	0.14 ±0.0022	6.66 ±0.0324	0.03 ±0.0214	0.34 ±0.2007	0.18 ±0.0379	0.17 ±0.0585	0.22 ±0.1471	0.94 ±0.2593	7.6 ±0.2613
Capsicum	1.92 ± 0.2568	0.04 ± 0.0116	0.02 ± 0.0055	-	-	-	-	0.02 ± 0.0042	0.004 ± 0.0013	0.24 ± 0.0405	2.24 ± 0.2603		1.03 ± 0.5435	0.6 ± 0.1046		0.46 ± 0.1791	2.09 ± 0.5818	4.33 ± 0.6374
Cauliflower	1.7 ±0.3078	0.1 ±0.0164	2.45 ±0.4943	-	-	-	-	0.7 ±0.1469	0.14 ±0.0006	0.92 ±0.2134	6.02 ±0.6376	0.05 ±0.0023	0.56 ±0.0532	0.63 ±0.0678	0.01 ±0.0004	0.67 ±0.2849	1.91 ±0.2977	7.92 ±0.7037
Chickpea	1.15 ±0.1366	0.86 ±0.0563		-	1.71 ±0.1747	0.22 ±0.0237	0.36 ±0.0257	0.16 ±0.0118	0.17 ±0.0166	0.57 ±0.0554	4.36 ±0.2314	0.07 ±0.0146	0.17 ±0.0196	0.15 ±0.0283	0.13 ±0.0249	0.42 ±0.087	0.94 ±0.0936	5.3 ±0.2496
Chilli	1.23 ±0.0029	0.93 ±0.0023	2.17 ±0.0044	-	-	-	0.02 ±0.0003	0.07 ±0.003	0.61 ±0.0171	0.2 ±0.0056	5.23 ±0.0191	0.24 ±0.1065	1.2 ±0.0322	1.01 ±0.558	-	0.07 ±0.091	0.86 ±0.5762	6.09 ±0.2637
Citrus	1.59 ±0.0366	0.07 ±0.0041	2.53 ±0.0285	-	-	-	-	0.3 ±0.0056	1.15 ±0.0153	1.06 ±0.0955	6.71 ±0.1075	0.05 ±0.0105	0.62 ±0.2135	0.63 ±0.2087	0.02 ±0.0048	0.77 ±0.2719	2.08 ±0.404	8.79 ±0.418
Coriander	2.13 ± 0.1028	0.8 ± 0.0344		-	1.09 ± 0.0486	0.52 ± 0.0313	0.06 ± 0.0027	0.19 ± 0.0083	0.06 ± 0.0031	0.12 ± 0.0057	4.98 ± 0.1233	0.02 ± 0.0064	0.29 ± 0.0831	0.31 ± 0.0246	0.07 ± 0.026	0.33 ± 0.0661	1.03 ± 0.1095	6 ± 0.1649
Cottonseed	1.83 ± 0.0307	0.28 ± 0.0061	-	-	-	-	0.08 ± 0.0021	0.1 ± 0.0025	0.1 ± 0	0.16 ± 0.0045	2.55 ± 0.0318	0.01 ± 0.0022	0.23 ± 0.041	0.03 ± 0.0077	0.03 ± 0.0119	0.13 ± 0.0325	0.44 ± 0.0542	2.99 ± 0.0628
Egg	1.14 ± 0.0107		0.93 ± 0.0257	-	-	-	-	1.18 ± 0.0144	0.42 ± 0.0065	0.33 ± 0.0224	4.00 ± 0.0391	0.04 ± 0	0.67 ± 0	0.61 ± 0	-	0.24 ± 0	1.55 ± 0	5.55 ± 0.0391
Grapes	1.45 ±0.033	0.1 ±0.0025	2.42 ±0.0406	-	-	-	-	0.04 ±0.001	0.54 ±0.0122	0.35 ±0.0068	4.9 ±0.0542	0.02 ±0.0081	0.44 ±0.0483	0.8 ±0.1974	0.06 ±0.0282	0.51 ±0.3086	1.83 ±0.3707	6.73 ±0.3746
Green Pea	1.68 ± 0.0452	0.79 ± 0.0782	1.43 ± 0.0434	-	-	-	-	0.08 ± 0.0129	0.005 ± 0.0008	0.62 ± 0.0392	4.60 ± 0.1084	-	0.57 ± 0.3046	0.66 ± 0.4775	0.06 ± 0	0.38 ± 0.1828	1.67 ± 0.5952	6.27 ± 0.605

Crop	Farm Operations										Market Operations							
	Harvest-ing	Collec-tion	Sorting/ Grading	Trimming	Thresh-ing	Winnow-ing/ cleaning	Drying	Packaging	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesalers	Retailers	Processing unit	Transport	Total loss in market level	Overall total loss
Greengram	1.7 ±0.2234	0.59 ±0.07		-	1.4 ±0.2707	0.33 ±0.0353	0.32 ±0.0486	0.17 ±0.057	0.37 ±0.0774	0.14 ±0.0657	5.03 ±0.3811	0.02 ±0.0032	0.33 ±0.0332	0.31 ±0.0084	0.13 ±0.0238	0.15 ±0.0446	0.93 ±0.0611	5.96 ±0.386
Groundnut	2.11 ± 0.0169	0.48 ± 0.0077	-	-	1.55 ± 0.0241	0.42 ± 0.0057	0.12 ± 0.0015	0.18 ± 0.0022	0.08 ± 0.001	0.16 ± 0.0027	5.09 ± 0.0312	0.04 ± 0.0438	0.39 ± 0.1612	0.08 ± 0.0364	0.39 ± 0.0644	0.17 ± 0.1455	1.07 ± 0.2295	6.16 ± 0.2316
Guava	4.82 ±0.7922	0.56 ±0.1363	4.26 ±0.6636	-	-	-	-	0.19 ±0.045	0.41 ±0.1009	0.77 ±0.1308	11.02 ±1.0563		1.39 ±0.6531	3.1 ±0.5573	1.24 ±0.3681	0.84 ±0.1342	3.13 ±0.9437	14.14 ±1.4165
Maize	1.47± 0.018	0.41± 0.018	-	-	1.32± 0.006	0.23± 0.002	0.34± 0.02	0.04± 0.002	0.27± 0.001	0.03± 0.006	4.1± 0.033	0.0002 ±0.022	0.22± 0.007	0.02± 0.00002	0.1± 0.022	0.03± 0.068	0.38± 0.079	4.48± 0.079
Meat	2.10 ±0.0925	-	-	-	-	-	-	-	-	-	2.10± 0.0925	-	0.21 ± 0	0.27 ± 0	0.19 ± 0	0.02 ± 0	0.68 ± 0	2.78± 0.0925
Milk	0.13 ± 0.0058	0.08 ± 0.0121	-	-	-	-	-	-	-	0.07 ± 0.003	0.28 ± 0.0137	-	-	0.20 ± 0.015	0.16 ± 0.0395	0.05 ± 0.0349	0.26 ± 0.0548	0.55 ± 0.0565
Mushroom	1.35± 0.1022	0.56± 0.1051	4.91± 0.7678	0.86± 0.1386	-	0.13± 0.035	-	0.12± 0.0234	0.2± 0.0408	0.28± 0.0342	8.41± 0.7968	-	1.05± 0.5	0.56± 0.1697	-	0.14± 0.037	0.95± 0.5293	9.36± 0.9566
Muskmelon	1.32 ±0.0037	0.26 ±0.0366	0.95 ±0.0025	-	-	-	-	0.36 ±0.0783	0.05 ±0.0006	0.55 ±0.0018	3.48 ±0.0866	0.02 ±0.0088	0.54 ±0.1195	1.04 ±0.3702	-	0.68 ±0.2215	2.28 ±0.4477	5.76 ±0.456
Mustard	1.46 ±0.0392	0.55 ±0.0193	-	-	1.6 ±0.0842	0.56 ±0.0186	0.14 ±0.0053	0.13 ±0.0047	0.01 ±0.0002	0.06 ±0.0012	4.51 ±0.0969	0.03 ±0.0084	0.04 ±0.0166	0.04 ±0.0067	0.02 ±0.0023	0.24 ±0.0358	0.37 ±0.04	4.89 ±0.1048
Onion	2.28 ± 0.0316	0.34 ± 0.0055	0.87 ± 0.0124	-	-	-	-	0.08 ± 0.0011	0.76 ± 0.0142	0.31 ± 0.0049	4.63 ± 0.0375	0.49 ± 0.0796	0.75 ± 0.1162	0.51 ± 0.132	0.01 ± 0.0052	0.39 ± 0.0612	2.15 ± 0.2026	6.78 ± 0.206
Paddy	0.76 ±0.0321	0.32 ±0.0127	-	-	0.73 ±0.0429	0.12 ±0.007	0.19 ±0.0078	0.02 ±0.0012	0.56 ±0.0304	0.11 ±0.0089	2.82 ±0.0644	0.07 ±0.0265	0.31 ±0.0601	0.03 ±0.0051	0.17 ±0.0391	0.05 ±0.024	0.63 ±0.0803	3.45 ±0.1029
Pigeon Pea	1.58 ± 0.0256	0.36 ± 0.0055	-	-	2.16 ± 0.0265	0.24 ± 0.0037	0.3 ± 0.0127	0.25 ± 0.0038	0.54 ± 0.0081	0.25 ± 0.0047	5.68 ± 0.0408	0.15 ± 0.0845	0.12 ± 0.0575	0.17 ± 0.0823	0.46 ± 0.1388	0.31 ± 0.1313	1.22 ± 0.2165	6.9 ± 0.2203
Pomegranate	1.31 ±0.0335	0.18 ±0.0111	1.42 ±0.0476	-	-	-	-	0.22 ±0.0111	0.28 ±0.0132	0.31 ±0.0181	3.73 ±0.0643	0.13 ±0.0288	0.61 ±0.1164	0.7 ±0.1454	0.09 ±0.0378	0.76 ±0.1566	2.29 ±0.2479	6.02 ±0.2561
Potato	2.54 ± 0.0031	0.2 ± 0.0156	2.38 ± 0.0031	-	-	-	-	0.04 ± 0.0042	0.12 ± 0.0004	0.64 ± 0.0011	5.92 ± 0.0168	0.27 ± 0.0009	0.32 ± 0.0396	0.11 ± 0.0193	0.02 ± 0.0002	0.39 ± 0.0136	1.1 ± 0.0461	7.02 ± 0.049

Crop	Farm Operations											Market Operations						
	Harvest-ing	Collec-tion	Sorting/ Grading	Trimming	Thresh-ing	Winnow-ing/ cleaning	Drying	Packaging	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesalers	Retailers	Processing unit	Transport	Total loss in market level	Overall total loss
Poultry Meat	2.36 ± 0.0808	-	-	-	-	-	-	-	-	-	2.36 ± 0.0808	-	1.2 ± 0	0.83 ± 0	0.02 ± 0	0.42 ± 0	2.47 ± 0	4.82 ± 0.0808
Safflower	1.04 ± 0.0272	0.4 ± 0.0101	-	-	0.47 ± 0.0103	0.25 ± 0.0056	0.12 ± 0.0028	0.17 ± 0.003	0.02 ± 0.0005	0.17 ± 0.0042	2.64 ± 0.0319	0.02 ± 0.0038	0.15 ± 0.0652	0.04 ± 0.0174	0.06 ± 0.0276	0.11 ± 0.0315	0.38 ± 0.0795	3.02 ± 0.0856
Sapota	1.9 ±0.075	0.38 ±0.0093	1.7 ±0.0411	-	-	-	-	0.3 ±0.0095	0.05 ±0.0014	1.47 ±0.0502	5.8 ± 0.1	0.18 ±0.0348	0.83 ±0.3044	0.98 ±0.0995	0.05 ±0.0148	1.26 ±0.4897	3.3 ±0.5863	9.1 ±0.5948
Sorghum	1.71± 0.0473	0.34± 0.0083	-	-	1.78± 0.054	0.38± 0.0192	0.1± 0.0032	0.22± 0.011	0.26± 0.0091	0.13± 0.005	4.93± 0.0763	0.06± 0.0108	0.9± 0.3146	0.17± 0.028	0.02± 0.008	0.21± 0.4033	1.36± 0.5124	6.29± 0.5181
Soybean	2.95 ± 0.187	0.84 ± 0.0543	-	-	1.28 ± 0.0902	0.19 ± 0.0262	0.13 ± 0.0065	0.54 ± 0.0272	0.79 ± 0.0649	0.25 ± 0.019	6.98 ± 0.2282	0.09 ± 0.0052	0.11 ± 0.0283	0.08 ± 0.0186	0.34 ± 0.0314	0.11 ± 0.0174	0.74 ± 0.0496	7.72 ± 0.2336
Sugarcane	2.27± 0.0181	1.71± 0.0093	-	-		2.27± 0.0152	-	-	-	0.02± 0.0004	6.26± 0.0254	-	-	-	0.48± 0.1287	1.87± 0.2141	2.35± 0.2498	8.61± 0.2511
Sunflower	1.11 ± 0.0299	0.39 ± 0.0117	-	-	1.68 ± 0.043	0.23 ± 0.0071	0.07 ± 0.001	0.13 ± 0.0051	0.07 ± 0.0019	0.13 ± 0.0049	3.81 ± 0.0547	0.02 ± 0.0073	0.18 ± 0.0506	0.04 ± 0.0207	0.74 ± 0.0747	0.13 ± 0.0406	1.12 ± 0.1011	4.93 ± 0.1149
Wheat	0.86± 0.0876	0.27± 0.0187	-	-	1.03± 0.0957	0.86± 0.1668	0.05± 0.0031	0.045± 0.0103	0.36± 0.0482	0.1± 0.0107	3.58± 0.2181	0.006± 0.119	0.17± 0.012	0.01± 0.0022	0.15± 0.119	0.26± 0.5112	0.6± 0.5685	4.17± 0.5686

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Crop	Farm Operations											Market Operations						
	Harvesting	Collection	Sorting/ Grading	Trimming	Thresh-ing	Winnowing/ cleaning	Drying	Packing	Farm level storage	Transport	Total loss in farm operations	Godown	Wholesalers	Retailers	Processing unit	Transport	Total loss in market level	Overall total loss
Areca nut	1.12 ±0.0857	0.28 ±0.0198	-	-	0.45 ±0.0339	0.91 ±0.0751	0.13 ±0.0082	0.12 ±0.009	0.23 ±0.0175	0.32 ±0.0243	3.55 ±0.1248	0.01 ±0.0033	0.37 ±0.0651	0.01 ±0.0084	0.15 ±0.0305	0.2 ±0.0095	0.74 ±0.0731	4.29 ±0.1447
Bajra	1.16 ±0.1356	0.34 ±0.0176	-	-	1.67 ±0.1924	0.15 ±0	0.22 ±0.0057	0.24 ±0.0119	0.21 ±0	0.24 ±0.0129	4.24 ±0.2368	0.12 ±0.0083	0.14 ±0.0742	0.07 ±0.0339	0.14 ±0.0054	0.34 ±0.1055	0.8 ±0.1337	5.04 ±0.2719
Banana	1.67 ±0.0238	0.39 ±0.0052	2.07 ±0.0297	-	-	-	-	0.21 ±0.0028	0.06 ±0.0009	1.83 ±0.0267	6.24 ±0.0469	0.1 ±0.025	1.32 ±0.582	0.5 ±0.224	0.01 ±0.0063	1.16 ±0.4911	3.09 ±0.7942	9.33 ±0.7955
Beans	1.76 ±0.0279	0.59 ±0.0072	1.31 ±0.0174	-	-	-	-	0.34 ±0.0056	0.15 ±0.0025	0.24 ±0.004	4.39 ±0.0344	0.13 ±0.0362	0.92 ±0.3158	0.92 ±0.1678	-	1.65 ±0.664	3.62 ±0.7551	8.02 ±0.7559
Black gram	1.06 ±0.0309	0.6 ±0.0168	-	-	0.99 ±0.0316	0.67 ±0.0196	0.29 ±0.0083	0.26 ±0.0075	0.69 ±0.0189	0.13 ±0.0032	4.69 ±0.0558	0.04 ±0.0351	0.22 ±0.0427	0.23 ±0.0338	0.15 ±0.0313	0.27 ±0.1772	0.91 ±0.1913	5.6 ±0.1993
Black pepper	0.54 ± 0.021	0.22 ± 0.0156	-	-	0.2 ± 0.0126	0.03 ± 0.0023	0.07 ± 0.0051	0.01 ± 0.0008	0.01 ± 0.0007	0.02 ± 0.0014	1.1 ± 0.0296	0.01 ± 0.0069	0.07 ± 0.0089	0.07 ± 0.0115	0.02 ± 0.0059	0.01 ± 0.0029	0.19 ± 0.0131	1.29 ± 0.0324
Capsicum	1.23 ± 0.0779	0.30 ± 0.0482	0.87 ± 0.0486	-	-	-	-	0.37 ± 0.004	0.006 ± 0.002	0.28 ± 0.0303	3.06 ± 0.1082		1.15 ± 0.2674	0.73 ± 0.2155	-	0.8 ± 0.1608	2.68 ± 0.3793	5.74 ± 0.3944
Cashew nut	1.77 ±0.0274	0.56 ±0.0086	-	-	1.36 ±0.0211	0.33 ±0.0051	0.03 ±0.0004		0.07 ±0.0012	0.07 ±0.0012	4.181 ±0.0361	0.001 ±0.0008	0.13 ±0.0649	0.02 ±0.0101	0.3 ±0.073	0.17 ±0.0201	0.63 ±0.1003	4.82 ±0.1066
Chickpea	1.27 ±0.0148	0.84 ±0.0082	-	-	1.87 ±0.0206	0.2 ±0.002	0.38 ±0.0026	0.16 ±0.0011	0.16 ±0.0016	0.55 ±0.0055	4.61 ±0.0268	0.06 ±0.017	0.17 ±0.0247	0.12 ±0.0218	0.08 ±0.0195	0.27 ±0.071	0.7 ±0.0782	5.3 ±0.0827
Chilli	1.01 ±0.013	0.73 ±0.0109	2.13 ±0.031	-	-	-	0.11 ±0.0013	0.02 ±0.0003	1.05 ±0.0155	0.15 ±0.0024	5.19 ±0.0387	0.22 ±0.0579	1.25 ±0.0165	1.3 ±0.4029	-	0.05 ±0.0665	0.96 ±0.4128	6.15 ±0.1861
Citrus	0.75 ±0.0024	0.39 ±0.001	2.11 ±0.0038	-	-	-	-	0.55 ±0.0022	1.01 ±0.0055	0.85 ±0.004	5.66 ±0.0085	0.08 ±0.0118	0.73 ±0.137	0.65 ±0.0883	0.02 ±0.0057	0.79 ±0.3527	2.27 ±0.3888	7.93 ±0.3889
Coconut	1.29± 0.0273	0.22± 0.0045	-	-	-	0.36± 0.0073	0.36± 0.0002	0.19± 0.0038	0.63± 0.0129	0.06 ± 0.0021	3.1±0.0317	0.03 ± 0.0112	0.15 ± 0.0222	0.13 ± 0.2612	0.18 ±0.0162	0.39± 0.0455	0.88± 0.2668	3.98± 0.2687
Cotton-seed	2.02 ± 0.0071	0.34 ± 0.0021	-	-	-	-	0.02 ± 0.0001	0.05 ± 0.0003	0.05 ± 0	0.14 ± 0.0005	2.61 ± 0.0074	0.01 ± 0.0035	0.27 ± 0.0305	0.03 ± 0.0117	0.03 ± 0.0104	0.11 ± 0.0266	0.45 ± 0.0434	3.06 ± 0.044
Egg	1.14 ± 0.008	-	1.00 ± 0.0048	-	-	-	-	1.16 ± 0.0096	0.58 ± 0.0041	0.34 ± 0.0036	4.22 ± 0.0145	0.03 ± 0	2.47 ± 0	1.4 ± 0	0.01 ± 0	0.38 ± 0	4.26 ± 0	7.12 ± 0

Crop	Farm Operations											Market Operations						
	Harvesting	Collection	Sorting/ Grading	Trimming	Thresh-ing	Winnowing/ cleaning	Drying	Packing	Farm level storage	Transport	Total loss in farm operations	Godown	Wholesalers	Retailers	Processing unit	Transport	Total loss in market level	Overall total loss
Grapes	1.8 ±0.1455	0.34 ±0.0251	2.45 ±0.1874	-	-	-	-	0.12 ±0.0091	0.38 ±0.0318	0.65 ±0.053	5.75 ±0.2467	0.01 ±0.0046	0.55 ±0.3086	0.99 ±0.1809	0.09 ±0.0512	0.63 ±0.1287	2.28 ±0.3836	8.02 ±0.4561
Green gram	1.76 ±0.0455	0.69 ±0.019	-	-	1.45 ±0.0432	0.34 ±0.0086	0.3 ±0.0097	0.19 ±0.0108	0.34 ±0.0132	0.13 ±0.0123	5.21 ±0.0701	0.02 ±0.0028	0.33 ±0.0718	0.3 ±0.0779	0.13 ±0.0184	0.14 ±0.0225	0.92 ±0.1099	6.13 ±0.1303
Ground-nut	2.12 ± 0.0036	0.59 ± 0.0007	-	-	1.81 ± 0.003	0.41 ± 0.0011	0.19 ± 0.0003	0.13 ± 0.0001	0.09 ± 0.0001	0.17 ± 0.0002	5.51 ± 0.0049	0.05 ± 0.0532	0.4 ± 0.1224	0.08 ± 0.016	0.37 ± 0.1807	0.15 ± 0.1259	1.04 ± 0.2527	6.55 ± 0.2527
Inland fish	1.54 ± 0.0105	0.34 ± 0.0019	1.57 ± 0.0086	-	-	-	-	0.18 ± 0.0011	0.1 ± 0.0019	0.14 ± 0.0013	3.86 ± 0.0139	-	0.36 ± 0.0299	0.56 ± 0.0564	-	0.14 ± 0.0123	1.05 ± 0.0578	4.92 ± 0.0595
Maize	0.95± 0.017	0.36± 0.013	-	-	1.19± 0.02	0.51± 0.009	0.16± 0.006	0.12± 0.004	0.14± 0.005	0.02± 0.001	3.44± 0.032	0.018±0.025	0.78± 0.093	0.14± 0.002	0.16 ±0.025	0.13 ±0.246	1.22 ±0.266	4.66 ±0.266
Mango	1.66 ±0.0488	0.26 ±0.0041	2.56 ±0.0785	-	-	-	-	0.05 ±0.0027	0.27 ±0.0058	0.81 ±0.0224	5.59 ±0.0954	0.01 ±0.0039	0.19 ±0.07	1.09 ±0.1865	0.15 ±0.0332	0.81 ±0.1688	2.25 ±0.2611	7.84 ±0.278
Marine fish	5.42 ± 0.0328	0.58 ± 0.0032	0.35 ± 0.0019	-	-	-	0.13 ± 0.0009	0.01 ± 0	-	0.41 ± 0.0021	6.89 ± 0.0331	0.19 ± 0.0491	0.47 ± 0.1521	0.21 ± 0.0499	0.29 ± 0.0611	0.43 ± 0.1086	1.6 ± 0.1491	8.5 ± 0.1527
Meat	0.69 ± 0	-	-	-	-	-	-	-	-	-	0.69 ± 0	-	0.46 ± 0	0.25 ± 0	0.19 ± 0	0.03 ± 0	0.93 ± 0	1.61 ± 0
Milk	0.16 ± 0.0033	0.10 ± 0.0021	-	-	-	-	-	-	-	0.08 ± 0.0014	0.34 ± 0.0041	-	-	0.06 ± 0.0084	0.14 ± 0.1033	0.13 ± 0.0346	0.24 ± 0.1092	0.58 ± 0.1093
Mushroom	1.04± 0.0503	0.04± 0.0029	7.94± 0.4033	0.7± 0.0389	-	0.24± 0.0232		0.22± 0.0149	0.74± 0.0357	0.8± 0.0391	11.72± 0.4127	-	1.3± 0.171	0.8± 0.1466	-	0.28± 0.0479	1.33± 0.2302	13.04 ±0.4725
Muskmelon	1.5 ±0.003	0.34 ±0.004	1.25 ±0.0014	-	-	-	-	0.31 ±0.005	0.1 ±0.0005	0.76 ±0.0009	4.26 ±0.0073	0.03 ±0.0104	0.59 ±0.0906	1.45 ±0.5366	-	1.09 ±0.1786	3.16 ±0.5728	7.42 ±0.5729
Onion	2.85 ± 0.1223	0.51 ± 0.0541	1.29 ± 0.0485	-	-	-	-	0.29 ± 0.0332	1.37 ± 0.1344	0.68 ± 0.0749	7.00 ± 0.2122	0.3 ± 0.1053	0.53 ± 0.2563	0.59 ± 0.2491	0.02 ± 0.0046	0.25 ± 0.073	1.69 ± 0.3797	8.70 ± 0.435
Paddy	1.83 ±0.0193	0.4 ±0.0048	-	-	0.99 ±0.0113	0.55 ±0.0062	0.21 ±0.003	0.11 ±0.0015	0.49 ±0.0073	0.14 ±0.0017	4.7 ±0.0251	0.03 ±0.0258	0.32 ±0.0598	0.03 ±0.0085	0.15 ±0.0609	0.25 ±0.1519	0.77 ±0.1764	5.47 ±0.1781
Papaya	1.43 ± 0.0099	0.43 ± 0.0032	0.86 ± 0.0065	-	-	-	-	0.32 ± 0.0023	0.26 ± 0.0022	0.96 ± 0.0066	4.27 ± 0.0142	0.01 ± 0.0059	0.41 ± 0.1183	1.2 ± 0.313	0.05 ± 0.0208	0.64 ± 0.1861	2.32 ± 0.3835	6.58 ± 0.3837

Crop	Farm Operations										Market Operations						
	Harvesting	Collection	Sorting/ Grading	Trimming	Thresh-ing	Winnowing/ cleaning	Drying	Packing	Farm level storage	Transport	Total loss in farm operations	Godown	Wholesalers	Retailers	Processing unit	Transport	Total loss in market level	Overall total loss
Pigeon Pea	0.89 ± 0.1015	0.18 ± 0.02	-	-	1.63 ± 0.194	0.12 ± 0.0162	0.12 ± 0.0133	0.14 ± 0.0159	0.5 ± 0.0595	0.23 ± 0.0287	3.82 ± 0.2311	0.08 ± 0.0216	0.09 ± 0.0201	0.13 ± 0.0507	0.26 ± 0.1355	0.22 ± 0.049	0.77 ± 0.1541	4.59 ± 0.2778
Pomegranate	1.66 ±0.0216	0.2 ±0.0022	2.84 ±0.0345	-	-	-	-	0.22 ±0.0024	0.24 ±0.0032	0.25 ±0.0032	5.4 ±0.0411	0.18 ±0.0465	0.73 ±0.1363	0.9 ±0.1255	0.14 ±0.0163	1.23 ±0.2859	3.18 ±0.3442	8.58 ±0.3467
Poultry Meat	2.86 ± 0	-	-	-	-	-	-	-	-	-	2.86 ± 0	-	2.2 ± 0	1.22 ± 0	0.35 ± 0	0.01 ± 0	3.79 ± 0	6.87 ± 0
Safflower	1.04 ± 0.0532	0.41 ± 0.0095	-	-	0.49 ± 0.0139	0.26 ± 0.0084	0.14 ± 0.0065	0.17 ± 0.0044	0.01 ± 0.0004	0.15 ± 0.0041	2.68 ± 0.0571	0.02 ± 0.0051	0.15 ± 0.0139	0.04 ± 0.0161	0.08 ± 0.0116	0.13 ± 0.0897	0.42 ± 0.0929	3.1 ± 0.1091
Sapota	2.84 ±0.0385	0.37 ±0.0055	2.43 ±0.0427	-	-	-	-	0.41 ±0.0065	0.05 ±0.0007	1.49 ±0.0209	7.59 ±0.0618	0.18 ±0.0332	0.86 ±0.2883	0.95 ±0.3879	0.05 ±0.0099	1.35 ±0.1794	3.38 ±0.5167	10.97 ±0.5204
Sorghum	1.4 ±0.0469	0.33 ±0.0121	-	-	1.87 ±0.0789	0.36 ±0.0138	0.08 ±0.0029	0.3 ±0.0111	0.21 ±0.0063	0.09 ±0.0029	4.63 ±0.0946	0.07 ±0.07	0.69 ±0.5006	0.2 ±0.1038	0.02 ±0.0211	0.23 ±0.4491	1.21 ±0.6843	5.84 ±0.6908
Soybean	2.39 ± 0.1728	0.88 ± 0.0552	-	-	0.89 ± 0.0663	0.34 ± 0.0275	0.13 ± 0.0093	0.24 ± 0.0162	0.87 ± 0.0811	0.16 ± 0.012	5.91 ± 0.2124	0.21 ± 0.0012	0.01 ± 0.1	0.22 ± 0.2115	0.11 ± 0.0126	0.16 ± 0.0049	0.72 ± 0.2343	6.63 ± 0.3163
Sugarcane	0.83 ±0.0335	1.41 ±0.0359	-	-		1.32 ±0.0237	-	-	-	0.02 ±0.0014	3.57 ±0.0545	-	-	-	0.96 ±0.5119	0.3 ±0.1307	1.26 ±0.5283	4.84 ±0.5311
Sunflower	0.85 ± 0.0413	0.35 ± 0.0217	-	-	1.41 ± 0.0884	0.2 ± 0.0076	0.09 ± 0.0034	0.13 ± 0.0067	0.04 ± 0.0032	0.1 ± 0.0059	3.16 ± 0.1008	0.02 ± 0.01	0.17 ± 0.0954	0.03 ± 0.0172	0.75 ± 0.1294	0.08 ± 0.0416	1.06 ± 0.167	4.22 ± 0.1951
Tomato	2.68 ±3.1543	0.57 ±1.4434	3.18 ±3.4616	-	-	-	-	0.27 ±1.0003	0.61 ±1.4188	0.82 ±1.7498	8.12 ±5.4855	-	0.85 ±0.4775	1.16 ±0.2994	0.1 ±0.0505	0.64 ±0.2147	2.75 ±0.6031	10.87 ±5.5186
Turmeric	1.9 ±0.0208	0.23 ±0.008	0.51 ±0.0034	-	-	0.26 ±0.0082	0.16 ±0.0032	0.14 ±0.007	0.12 ±0.0005	0.32 ±0.0088	3.65 ±0.0267	0.39 ±0.1319	0.7 ±0.3238	0.55 ±0.307	-	0.72 ±0.5075	1.23 ±0.5575	4.88 ±0.5581

AGRO CLIMATIC ZONE 11

Crop	Farm Operations										Market Operations						
	Harvest-ing	Collec-tion	Sorting/ Grading	Thresh-ing	Winnow-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retai-lers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Banana	1.56 ±0.1573	0.33 ±0.032	1.97 ±0.2012	-	-	-	0.18 ±0.0174	0.05 ±0.0057	1.73 ±0.1741	5.82 ±0.3113	0.08 ±0.0767	1.26 ±0.9815	0.56 ±0.4852	0.01 ±0.0058	1.09 ±0.7952	3 ±1.3554	8.81 ±1.3906
Beans	0.73 ±0.0113	0.21 ±0.0025	1.16 ±0.0119	-	-	-	0.35 ±0.0059	0.39 ±0.0074	0.54 ±0.011	3.37 ±0.022	0.11 ±0.0082	1 ± 0.2705	0.77 ±0.5704	-	1.85 ±0.3955	3.73 ±0.745	7.1 ±0.7453
Black gram	0.88 ±0.0286	0.97 ±0.0447	-	1.04 ±0.0454	0.43 ±0.0201	0.54 ±0.0213	0.13 ±0.0062	0.98 ±0.0325	0.15 ±0.0077	5.12 ±0.083	0.02 ±0.0041	0.24 ±0.0137	0.22 ±0.1351	0.16 ±0.1189	0.27 ±0.048	0.91 ±0.1868	6.03 ±0.2044
Cabbage	1.26 ± 0.0435	0.33 ± 0.0309	2.42 ± 0.0675	-	-	-	0.24 ± 0.0236	0.12 ± 0.0087	0.7 ± 0.0043	5.07 ± 0.0897	0.07 ± 0.0057	0.76 ± 0.1518	1.53 ± 0.3745	0.01 ± 0.0005	0.1 ± 0.028	2.37 ± 0.4051	7.44 ± 0.4149
Cauliflower	1.9 ±0.0691	0.2 ±0.0076	2.43 ±0.0977	-	-	-	0.41 ±0.0147	0.08 ±0	0.81 ±0.0419	5.82 ±0.1279	0.07 ±0.0018	0.43 ±0.097	0.99 ±0.1689	0.01 ±0.0006	1.26 ±0.1576	2.75 ±0.2505	8.58 ±0.2813
Egg	0.73 ± 0.0076	-	0.97 ± 0.0024	-	-	-	1.10 ± 0.0415	0.42 ± 0.0056	0.19 ± 0.0102	3.41 ± 0.0438	0.05 ± 0	0.95 ± 0	0.30 ± 0	-	0.40 ± 0	1.70 ± 0	5.11 ± 0.0438
Groundnut	1.66 ± 0.0131	0.6 ± 0.0055	-	1.25 ± 0.0117	0.37 ± 0.0033	0.15 ± 0.0012	0.12 ± 0.0012	0.05 ± 0.0009	0.15 ± 0.002	4.35 ± 0.0189	0.05 ± 0.0162	0.35 ± 0.1033	0.07 ± 0.0173	0.32 ± 0.2272	0.17 ± 0.1647	0.96 ± 0.2995	5.31 ± 0.3001
Maize	0.85± 0.121	0.2± 0.013	-	0.73± 0.042	0.17± 0.036	0.23± 0.019	0.27± 0.023	0.24± 0.034	0.45± 0.03	3.14± 0.145	0.0002 ±0.016	0.14± 0.002	0.04 ±0.00001	0.08 ±0.016	0.27 ±0.529	0.53± 0.548	3.67± 0.549
Marine fish	5.86 ± 0.2492	0.53 ± 0.0179	0.39 ± 0.0144	-	-	0.13 ± 0.0053	0.02 ± 0.0005	-	0.49 ± 0.0237	7.41 ± 0.2515	0.21 ± 0.0672	0.65 ± 0.16	0.22 ± 0.0433	0.31 ± 0.044	0.56 ± 0.1057	1.96 ± 0.147	9.37 ± 0.2913
Meat	1.89 ± 0	-	-	-	-	-	-	-	-	1.89 ± 0	-	0.56 ± 0	0.26 ± 0	0.32 ± 0	0.02 ± 0	1.16 ± 0	3.04 ± 0
Okra	1.78 ± 0.0401	0.34 ± 0.019	1.08 ± 0.0553	-	-	-	0.15 ± 0.0064	0.16 ± 0.0071	0.46 ± 0.019	3.96 ± 0.074	-	0.89 ± 0.1378	0.83 ± 0.201		0.57 ± 0.1203	2.29 ± 0.2717	6.25 ± 0.2816
Paddy	2.03 ±0.0648	0.39 ±0.0136	-	1.55 ±0.0532	0.52 ±0.0244	0.13 ±0.0044	0.1 ±0.0033	0.45 ±0.0144	0.19 ±0.0066	5.36 ±0.0899	0.06 ±0.0177	0.21 ±0.093	0.04 ±0.0282	0.18 ±0.0156	0.14 ±0.0301	0.62 ±0.1044	5.98 ±0.1378
Poultry Meat	2.17	-	-	-	-	-	-	-	-	2.17		1.89	1.07	0.31	0	3.27	5.44

Crop	Farm Operations										Market Operations						
	Harvest-ing	Collec-tion	Sorting/ Grading	Thresh-ing	Winnow-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retai-lers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
	± 0									± 0		± 0	± 0	± 0	± 0	± 0	± 0
Sapota	2.76 ±0.1047	0.48 ±0.0128	2.19 ±0.0793	-	-	-	0.35 ±0.0087	0.03 ±0.0008	1.6 ±0.0511	7.42 ±0.1418	0.23 ±0.1323	0.94 ±0.2799	1.07 ±0.3173	0.06 ±0.0105	1.76 ±0.2179	4.05 ±0.4941	11.48 ±0.514
Sorghum	1.78 ±0.0358	0.33 ±0.0145	-	1.72 ±0.0934	0.28 ±0.0159	0.09 ±0.0042	0.24 ±0.0109	0.23 ±0.017	0.23 ±0.0105	4.91 ±0.1049	0.03 ±0.0138	0.76 ±0.2869	0.15 ±0.0561	0.02 ±0.0063	0.17 ±0.3273	1.13 ±0.4391	6.04 ±0.4515
Sugarcane	2.88± 0.0621	1.6 ±0.039	-	-	1.01 ±0.0183	-	-	-	2.38 ±0.0468	7.87 ±0.0889	-	-	-	0.59 ±0.1854	0.19 ±0.1386	0.78 ±0.2315	8.65 ±0.248
Tomato	3.73 ±5.2454	0.59 ±2.094	2.73 ±4.4917	-	-	-	0.24 ±1.3199	1.36 ±3.1714	0.87 ±2.5329	9.53 ±8.3839	-	1.05 ±0.3747	0.89 ±0.1399	0.11 ±0.1275	1.36 ±0.3937	3.42 ±0.5612	12.94 ±8.4027

AGRO CLIMATIC ZONE 12

Crop	Farm Operations										Market Operations						
	Harvest-ing	Collec-tion	Sorting/ Grading	Thresh-ing	Winnow-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retai-lers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Areca nut	1.2 ±0.0533	0.28 ±0.0133	-	-	0.9 ±0.0444	0.14 ±0.0049	0.11 ±0.0053	0.25 ±0.012	0.23 ±0.0081	3.53 ±0.0749	0.01 ±0.004	0.39 ±0.0474	0.04 ±0.0266	0.16 ±0.0524	0.18 ±0.0744	0.79 ±0.1061	4.32 ±0.1298
Banana	1.22 ±0.1051	0.2 ±0.0155	1.42 ±0.1313	-	-	-	0.14 ±0.0109	0.03 ±0.0032	0.97 ±0.0868	3.98 ±0.1903	0.05 ±0.007	0.76 ±0.7302	0.3 ±0.1877	0.01 ±0.01	0.74 ±0.1563	1.86 ±0.77	5.84 ±0.7932
Black pepper	0.53 ± 0.0409	0.2 ± 0.0157	-	0.22 ± 0.0124	0.03 ± 0.0028	0.06 ± 0.0043	0.02 ± 0.0021	0.01 ± 0.0016	0.02 ± 0.0015	1.11 ± 0.0459	0.01 ± 0.0027	0.08 ± 0.0101	0.07 ± 0.0136	0.02 ± 0.0042	0.01 ± 0.0051	0.19 ± 0.0124	1.3 ± 0.0476
Cashew nut	1.12 ±0.0494	0.25 ±0.007	-	0.97 ±0.0372	0.18 ±0.0078	0.02 ±0.001	-	0 ±0	0.03 ±0.0014	2.567 ±0.0628	0.002 ±0.0004	0.14 ±0.0513	0.02 ±0.0057	0.19 ±0.0331	0.2 ±0.0494	0.55 ±0.0788	3.12 ±0.1007
Coconut	0.43± 0.0151	0.44± 0.0118	-	-	0.33± 0.0109	0.32± 0.0118	0.24± 0.0078	0.46± 0.0187	0.33 ± 0.0139	2.55± 0.0351	0.03 ± 0.0361	0.08 ± 0.0804	0.27 ± 0.5388	0.28 ±0.0509	0.39± 0.1449	1.06± 0.5671	3.61 ± 0.5682
Egg	1.71 ± 0.006	-	1.06 ± 0.013	-	-	-	1.06 ± 0.0288	0.02 ± 0.0002	0.30 ± 0.0064	4.15 ± 0.0328	0.05 ± 0.0357	0.65 ± 0	0.26 ± 0		0.42 ± 0	1.38 ± 0.0357	5.53 ± 0.0485
Maize	0.47± 0.091	0.06± 0.009	-	0.72± 0.118	0.07± 0.014	0.1± 0.015	0.02± 0.003	0.18± 0.035	0.03± 0.004	1.64± 0.155	0.001± 0.001	0.1± 0.378	0.39± 0.001	0.06± 0.001	0.32± 0.636	0.87± 0.756	2.51± 0.756
Mango	1.99 ±0.2562	0.11 ±0.0146	2.91 ±0.3021	-	-	-	0.02 ±0.0026	0.13 ±0.0092	0.9 ±0.0776	6.06 ±0.404	0.002± 0.0011	0.37 ±0.1591	0.93 ±0.2541	0.1 ±0.0514	1.13 ±0.3101	2.53 ±0.4313	8.58 ±0.591
Marine fish	5.5 ± 0.5002	0.59 ± 0.0379	0.35 ± 0.0204	-	-	0.16 ± 0.0129	0.01 ± 0.0007		0.43 ± 0.0322	7.04 ± 0.5032	0.22 ± 0.0447	0.61 ± 0.1454	0.21 ± 0.0341	0.3 ± 0.0493	0.51 ± 0.0941	1.85 ± 0.1266	8.89 ± 0.5189
Meat	1.49 ± 0	-	-	-	-	-	-	-	-	1.49 ± 0	-	0.26 ± 0	0.25 ± 0	0.18 ± 0	0.02 ± 0	0.71 ± 0	2.20 ± 0
Paddy	1.43 ±0.0322	0.6 ±0.0079	-	1.23 ±0.023	0.4 ±0.0135	0.37 ±0.0101	0.25 ±0.0151	0.49 ±0.0211	0.16 ±0.0077	4.94 ±0.0514	0.1 ±0.0271	0.31 ±0.0555	0.03 ±0.005	0.17 ±0.0296	0.06 ±0.0362	0.67 ±0.0776	5.61 ±0.0931
Pineapple	1.19 ±0.0371	0.04 ±0.0022	0.71 ±0.0252	-	-	-	0.04 ±0.0022	0.04 ±0.0014	0.02 ±0.0007	2.04 ±0.0449	0.03 ±0.0307	0.63 ±0.1679	0.45 ±0.1847	0.13 ±0.0559	0.47 ±0.1667	1.71 ±0.3069	3.75 ±0.3101

Crop	Farm Operations										Market Operations						
	Harvest-ing	Collec-tion	Sorting/ Grading	Thresh-ing	Winnow-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retai-lers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Poultry Meat	1.58 ± 0	-	-	-	-	-	-	-	-	1.58 ± 0	-	2.08 ± 0	1.22 ± 0	0.01 ± 0	0.44 ± 0	3.75 ± 0	5.32 ± 0
Sapota	1.86 ±0.3433	0.26 ±0.0414	1.62 ±0.3002	-	-	-	0.21 ±0.0334	0.02 ±0.0038	0.99 ±0.1625	4.96 ±0.4871	0.12 ±0.0675	0.78 ±0.1832	0.97 ±0.2441	0.03 ±0.0113	1.15 ±0.0547	3.05 ±0.3175	8.01 ± 0.5814
Tapioca	1.2 ±0.0375	0.26 ±0.0101	1.05 ±0.0238	-	-	-	0.09 ±0.0036	0.27 ±0.006	0.54 ±0.0105	3.41 ±0.0473	-	0.35 ±0.1275	0.45 ±0.0594	0.16 ±0.0136	0.44 ±0.0865	1.4 ±0.1657	4.8 ± 0.1723

AGRO CLIMATIC ZONE 13

Crop	Farm Operations										Market Operations						
	Harvest-ing	Collec-tion	Sorting/ Grading	Thresh-ing	Winnow-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retai-lers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Bajra	1.52 ±0.0593	0.3 ±0.0087		0.96 ±0.0361	0.16 ±1.96	0.23 ±0.0069	0.24 ±0.007	0.24 ±0.0076	0.27 ±0.0074	3.91 ±1.9613	0.13 ±0.0118	0.12 ±0.0215	0.08 ±0.0512	0.11 ±0.0186	0.43 ±0.0163	0.87 ±0.062	4.78 ±1.9623
Banana	1.28 ±0.0792	0.25 ±0.0143	1.58 ±0.0888	-	-	-	0.15 ±0.0089	0.03 ±0.002	1 ±0.0777	4.29 ±0.1431	0.05 ±0.0101	0.88 ±0.3341	0.37 ±0.2638	0.01 ±0.0037	0.85 ±0.1395	2.15 ±0.4481	6.44 ±0.4704
Beans	0.44 ±0.0149	0.43 ±0.0143	1.08 ±0.0297	-	-	-	0.34 ±0.0083	0.38 ±0.0071	0.47 ±0.016	3.14 ±0.0411	0.15 ±0.0588	1.15 ±0.4563	0.56 ±0.1555	-	1.65 ±0.552	3.51 ±0.7352	6.65 ±0.7364
Black gram	1.08 ±0.0222	0.65 ±0.0189	-	0.95 ±0.0249	0.52 ±0.0165	0.26 ±0.0064	0.13 ±0.0047	0.3 ±0.0066	0.17 ±0.0035	4.05 ±0.0432	0.07 ±0.0058	0.2 ±0.0533	0.24 ±0.0156	0.11 ±0.0503	0.19 ±0.1368	0.82 ±0.1561	4.87 ±0.162
Brinjal	1.84 ±0.1337	0.95 ±0.136	0.46 ±0.047	-	-	-	0.27 ±0.035	0.32 ±0.0214	0.59 ±0.0486	4.43 ±0.2064	-	0.98 ±0.3808	1.15 ±0.2622	-	0.4 ±0.1439	2.53 ±0.4842	6.95 ±0.5264
Cabbage	1.36 ± 0.0361	0.4 ± 0.0146	3.19 ± 0.1005	-	-	-	0.3 ± 0.0044	0.13 ± 0.0031	0.64 ± 0.0003	6.02 ± 0.1079	0.1 ± 0.0183	0.77 ± 0.1883	1.35 ± 0.2441	0.03 ± 0.0212	0.13 ± 0.0551	2.24 ± 0.3145	8.26 ± 0.3325
Cauliflower	1.85 ±0.0243	0.16 ±0.0036	2.42 ±0.0466	-	-	-	0.66 ±0.0061	0.14 ±0	1.31 ±0.0291	6.52 ±0.0605	0.05 ±0.0013	0.46 ±0.1323	0.71 ±0.0453	0.01 ±0.0004	0.84 ±0.1925	2.06 ±0.2379	8.58 ±0.2455
Cottonseed	1.73 ± 0.0483	0.25 ± 0.0053	-	-	-	0.17 ± 0.0047	0.1 ± 0.0026	0.1 ± 0	0.17 ± 0.0039	2.52 ± 0.049	0.01 ± 0.0044	0.23 ± 0.1026	0.02 ± 0.0098	0.02 ± 0.0107	0.12 ± 0.0367	0.41 ± 0.11	2.93 ± 0.1204
Greengram	1.76 ±0.1356	0.65 ±0.0176	-	1.5 ±0.1924	0.29 ±0	0.36 ±0.0057	0.13 ±0.0119	0.28 ±0	0.1 ±0.0129	5.08 ±0.2368	0.02 ±0.0045	0.33 ±0.0083	0.29 ±0.133	0.12 ±0.1231	0.11 ±0.0052	0.86 ±0.1815	5.95 ±0.2984
Groundnut	1.85 ± 0.0478	0.32 ± 0.0086	-	1.24 ± 0.0297	0.39 ± 0.0086	0.14 ± 0.0033	0.16 ± 0.0039	0.06 ± 0.0021	0.11 ± 0.0037	4.27 ± 0.0579	0.05 ± 0.0455	0.43 ± 0.3333	0.09 ± 0.0342	0.29 ± 0.1161	0.19 ± 0.1471	1.04 ± 0.3839	5.31 ± 0.3883
Maize	0.48± 0.041	0.65± 0.044	-	0.52± 0.065	0.17± 0.011	0.61± 0.035	0.51± 0.023	0.11± 0.008	0.5± 0.033	3.56± 0.105	0.0003 ±0.027	0.15± 0.07	0.19± 0.001	0.05 ±0.027	0.04 ±0.086	0.44± 0.155	3.99± 0.158
Marine fish	5.22 ± 0.1238	0.49 ± 0.0082	0.45 ± 0.0059	-	-	0.12 ± 0.0066	0.01 ± 0.0003	-	0.38 ± 0.0074	6.67 ± 0.1246	0.17 ± 0.0419	0.56 ± 0.2086	0.21 ± 0.0384	0.26 ± 0.0802	0.53 ± 0.1615	1.74 ± 0.201	8.41 ± 0.2365
Milk	0.06 ± 0.0749	0.06 ± 0.0795	-	-	-	-	-	-	0.06 ± 0.0744	0.18 ± 0.1322	-	-	0.13 ± 0.0191	0.04 ± 0.0281	0.35 ± 0.0555	0.32 ± 0.0651	0.50 ± 0.1473

Crop	Farm Operations										Market Operations						
	Harvest-ing	Collec-tion	Sorting/ Grading	Thresh-ing	Winnow-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retai-lers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Mustard	1.68 ±0.015	0.39 ±0.0025	-	1.52 ±0.0232	0.36 ±0.0027	0.02 ±0.0001	0.02 ±0.0004	0.01 ±0.0001	0.05 ±0.0004	4.04 ±0.0279	0.05 ±0.0262	0.04 ±0.0181	0.03 ±0.0156	0.02 ±0.0076	0.23 ±0.0776	0.38 ±0.0812	4.42 ±0.0858
Okra	1.46 ± 0.0737	0.25 ± 0.0139	1.06 ± 0.0503	-	-	-	0.15 ± 0.0084	0.12 ± 0.006	0.47 ± 0.0244	3.51 ± 0.0941	-	0.77 ± 0.25	0.89 ± 0.2672	-	0.61 ± 0.1303	2.27 ± 0.3884	5.79 ± 0.3996
Onion	2.94 ± 0.0343	0.49 ± 0.0043	1.23 ± 0.005	-	-	-	0.15 ± 0.0009	0.77 ± 0.0105	0.24 ± 0.0033	5.82 ± 0.0367	0.22 ± 0.037	0.49 ± 0.137	0.57 ± 0.1976	0.01 ± 0.0057	0.46 ± 0.3321	1.75 ± 0.4116	7.57 ± 0.4133
Paddy	1.01 ±0.0066	0.45 ±0.0076	-	0.72 ±0.0123	0.56 ±0.0085	0.29 ±0.0029	0.28 ±0.003	0.32 ±0.004	0.41 ±0.0046	4.02 ±0.0195	0.05 ±0.043	0.27 ±0.0881	0.05 ±0.0103	0.17 ±0.1447	0.06 ±0.0769	0.61 ±0.1913	4.63 ±0.1923
Papaya	0.92 ± 0.045	0.55 ± 0.028	0.29 ± 0.0214	-	-	-	0.32 ± 0.0155	0.15 ± 0.0305	0.73 ± 0.0408	2.96 ± 0.0781	0.02 ± 0.0059	0.5 ± 0.2092	1.96 ± 0.6409	0.05 ± 0.0132	0.33 ± 0.064	2.85 ± 0.6773	5.81 ± 0.6818
Pigeon Pea	0.79 ± 0.0201	0.24 ± 0.0069	-	1.54 ± 0.0575	0.17 ± 0.0065	0.16 ± 0.0052	0.13 ± 0.0037	0.42 ± 0.0117	0.14 ± 0.0044	3.59 ± 0.0632	0.08 ± 0.0592	0.06 ± 0.0282	0.11 ± 0.0673	0.19 ± 0.058	0.18 ± 0.0859	0.61 ± 0.1272	4.2 ± 0.142
Potato	2.26 ± 0.0039	0.27 ± 0.0025	1.85 ± 0.0027	-	-	-	0.04 ± 0.0007	0.12 ± 0.0006	0.57 ± 0.0018	5.11 ± 0.0058	0.15 ± 0.0005	0.25 ± 0.0162	0.11 ± 0.0026	0.02 ± 0.0003	0.34 ± 0.0076	0.88 ± 0.0181	5.98 ± 0.019
Sapota	1.79 ±0.059	0.23 ±0.0084	1.46 ±0.0588	-	-	-	0.21 ±0.0064	0.02 ±0.0006	0.96 ±0.0296	4.67 ±0.0891	0.13 ±0.0269	0.72 ±0.131	0.91 ±0.2091	0.03 ±0.0056	1.01 ±0.1523	2.8 ±0.2912	7.47 ±0.3046
Tomato	2.53 ±4.122	0.44 ±0.7619	3.23 ±5.2076	-	-	-	0.26 ±0.4767	0.24 ±0.46	0.78 ±1.275	7.47 ±6.8378	-	0.99 ±0.4677	1.21 ±0.7103	0.13 ±0.0892	1.08 ±0.4002	3.4 ±0.9399	10.88 ±6.902
Wheat	0.72± 0.037	0.44± 0.0322	-	0.22± 0.0168	0.15± 0.0102	0.03± 0.0028	0.003± 0.0003	1.93± 0.2481	0.32± 0.0198	3.82± 0.2544	0.012± 0.1886	0.08± 0.0111	0.04± 0.0034	0.26± 0.1886	0.03± 0.0577	0.42± 0.3222	4.23 ±0.3228

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Crop	Farm Operations										Market Operations						
	Harvest-ing	Collec-tion	Sorting/ Grading	Thresh-ing	Winnow-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retai-lers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Bajra	0.96 ±0.0064	0.28 ±0.0018	-	1.23 ± 0.014	0.15 ±0.0011	0.21 ±0.0014	0.28 ±0.0019	0.43 ±0.0028	0.27 ±0.0019	3.8 ± 0.0161	0.13 ±0.0367	0.09 ±0.0136	0.06 ±0.0123	0.09 ±0.0276	0.23 ±0.0328	0.6 ±0.0594	4.4 ±0.0615
Chickpea	1.48 ±0.0496	0.93 ±0.0146	-	1.68 ±0.0419	0.9 ±0.0074	0.65 ±0.0086	0.16 ±0.0023	0.17 ±0.0032	1.01 ±0.0071	5.62 ±0.0673	0.13 ±0.031	0.41 ±0.1714	0.09 ±0.0377	0.11 ±0.0458	0.33 ±0.114	1.08 ±0.2092	6.7 ±0.2198
Cottonseed	1.66 ± 0.0229	0.41 ± 0.0243	-	-	-	0.04 ± 0.0023	0.08 ± 0.0042	0.04 ± 0	0.12 ± 0.0044	2.36 ± 0.034	0.01 ± 0.0029	0.25 ± 0.0009	0.03 ± 0.0055	0.03 ± 0.0146	0.09 ± 0.0238	0.41 ± 0.0285	2.76 ± 0.0444
Groundnut	1.6 ± 0.0444	0.52 ± 0.0186	-	1.16 ± 0.0358	0.38 ± 0.0111	0.09 ± 0.0021	0.1 ± 0.0036	0.1 ± 0.0039	0.1 ± 0.0015	4.06 ± 0.0613	0.05 ± 0.0189	0.38 ± 0.0953	0.08 ± 0.0272	0.32 ± 0.0882	0.19 ± 0.1593	1.02 ± 0.2073	5.08 ± 0.2162
Maize	0.95± 0.08	0.77± 0.047	-	1.88± 0.202	0.14± 0.014	0.05± 0.008	0.09± 0.011	0.23± 0.019	0.01± 0.0001	4.12± 0.224	0.001± 0.07	0.1± 0.005	0.05± 0.001	0.1± 0.07	0.22± 0.435	0.47± 0.494	4.6± 0.494
Mustard	1.8 ±0.0346	0.47 ±0.0048	-	1.14 ±0.0185	0.4 ±0.0059	0.14 ±0.0016	0.1 ±0.0019	0.14 ±0.0014	0.05 ±0.0009	4.24 ±0.04	0.05 ±0.0086	0.03 ±0.006	0.02 ±0.0056	0.02 ±0.0099	0.23 ±0.0423	0.36 ±0.0431	4.59 ±0.0588
Sorghum	1.53± 0.0182	0.31± 0.003	-	1.89± 0.0356	0.33± 0.0072	0.21± 0.002	0.21± 0.002	0.22± 0.0029	0.21± 0.002	4.9 ±0.041	0.03± 0.008	0.64± 0.0956	0.12± 0.0178	0.01± 0.0047	0.11± 0.2215	0.92± 0.2421	5.82± 0.2455
Wheat	1.36± 0.0401	0.58± 0.0115	-	1.97± 0.0958	0.76± 0.0274	0.16± 0.003	0.052± 0.001	0.28± 0.0049	0.1± 0.0024	5.25± 0.1082	0.0001± 0.0169	0.13± 0.0473	0.09± 0.0001	0.22± 0.0169	0.07± 0.1346	0.52± 0.1798	5.77± 0.1799

AGRO CLIMATIC ZONE 15

Crop	Farm Operations										Market Operations						
	Harvest-ing	Collec-tion	Sorting / Gradin g	Thresh-ing	Winnow-ing/ cleaning	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operation s	Godown	Whole-salers	Retai-lers	Process-ing unit	Trans-port	Total loss in market level	Overall total loss
Areca nut	1.95 ±0.3268	0.36 ±0	-	1.36 ±0.2305	1.26 ±0.2884	0.46 ±0.0306	0.17 ±0.0324	0.4 ±0.0594	0.51 ±0.0418	6.46 ±0.5003	0 ±0	0.45 ±0.1936	0.03 ±0.0199	0 ±0	0.3 ±0.0313	0.79 ±0.1971	7.25 ±0.5378
Coconut	0.82± 0.1289	0.27± 0.0228	-	-	0.52± 0.0138	0.77± 0.2061	0.42± 0.0558	0.29± 0.0215	0.32 ± 0.0451	3.4± 0.2557	0.37 ± 0.0453	0.16 ± 0.061	0.21 ± 0.4206	-	0.11± 0.1109	0.85± 0.4416	4.25± 0.5103

Appendix 8 (a) Estimate of the monetary value of post-harvest losses in India

S.No.	Crops	Average price (Rs/ MT) (2019-20, 2020-21 and 2021-22)	Production in MT (2020-21 final estimates)	% Post-harvest losses NABCONS 2022	% Postharvest losses ICAR-CIPHET 2015	Quantity lost in MT NABCONS (2020-21)	Quantity lost in MT ICAR-CIPHET 2015	Monetary loss NABCONS (Rs. in Crore)		Monetary loss as per CIPHET (Rs. in Crore) (b)	Savings due to reduction in losses (b-a)
								54 crops	45 crops (a)		
Grains (Cereals, Millets, Pulses, Oilseeds)											
1	Paddy	20727.82	124370000	4.77	5.53	5932449	6877661	12296.67	12296.67	14255.89	1959.22
2	Wheat	21627.25	109590000	4.17	4.93	4569903	5402787	9883.45	9883.45	11684.74	1801.30
3	Maize	18372.35	31650000	3.88	4.65	1228020	1471725	2256.16	2256.16	2703.91	447.74
4	Bajra	18213.39	10860000	4.37	5.23	474582	567978	864.37	864.37	1034.48	170.11
5	Sorghum	24587.57	4810000	5.92	5.99	284752	288119	700.14	700.14	708.41	8.28
6	Pigeon pea	84688.16	4320000	5.65	6.36	244080	274752	2067.07	2067.07	2326.82	259.76
7	Chick pea	62105.85	11910000	6.74	8.41	802734	1001631	4985.45	4985.45	6220.71	1235.27
8	Black gram	66648.91	2230000	5.83	7.07	130009	157661	866.50	866.50	1050.79	184.30
9	Green gram	71636.25	3090000	6.19	6.6	191271	203940	1370.19	1370.19	1460.95	90.76
10	Mustard	55809.55	10210000	4.46	5.54	455366	565634	2541.38	2541.38	3156.78	615.40
11	Cottonseed	25584.40	3939845.2	2.87	3.08	113074	121347	289.29	289.29	310.46	21.17
12	Soybean	50011.48	12610000	7.51	9.96	947011	1255956	4736.14	4736.14	6281.22	1545.08
13	Safflower	41866.97	36000	3.06	3.24	1102	1166	4.61	4.61	4.88	0.27
14	Sunflower	47057.01	228000	4.38	5.26	9986	11993	46.99	46.99	56.43	9.44
15	Groundnut	56331.57	10244000	5.73	6.03	586981	617713	3306.56	3306.56	3479.68	173.12
Fruit & Vegetables											
16	Apple	89699.98	2276000	9.51	10.39	216448	236476	1941.53	1941.53	2121.19	179.66
17	Banana	23082.26	33062000	7.57	7.76	2502793	2565611	5777.01	5777.01	5922.01	145.00
18	Citrus	39580.89	14245000	7.71	9.69	1098290	1380341	4347.13	4347.13	5463.51	1116.38
19	Grapes	65060.19	3358000	7.15	8.63	240097	289795	1562.08	1562.08	1885.41	323.34
20	Guava	32150.18	4582000	15.05	15.88	689591	727622	2217.05	2217.05	2339.32	122.27
21	Mango	60853.22	20386000	8.53	9.16	1738926	1867358	10581.92	10581.92	11363.47	781.55

S.No.	Crops	Average price (Rs/ MT) (2019-20, 2020-21 and 2021-22)	Production in MT (2020-21 final estimates)	% Post-harvest losses NABCONS 2022	% Postharvest losses ICAR-CIPHET 2015	Quantity lost in MT NABCONS (2020-21)	Quantity lost in MT ICAR-CIPHET 2015	Monetary loss NABCONS (Rs. in Crore)		Monetary loss as per CIPHET (Rs. in Crore) (b)	Savings due to reduction in losses (b-a)
								54 crops	45 crops (a)		
22	Papaya	21531.83	5540000	6.59	6.7	365086	371180	786.10	786.10	799.22	13.12
23	Sapota	33589.81	822000	9.53	9.73	78337	79981	263.13	263.13	268.65	5.52
24	Pineapple	28046.17	1799000	6.02		108300		303.74			
25	Pomegranate	68722.72	3271000	6.82		223082		1533.08			
26	Muskmelon	23012.31	1478000	6.83		100947		232.30			
27	Cabbage	15787.29	9560000	8.15	9.37	779140	895772	1230.05	1230.05	1414.18	184.13
28	Cauliflower	23313.40	9225000	7.89	9.56	727853	881910	1696.87	1696.87	2056.03	359.16
29	Green pea	60875.64	5846000	6.43	7.45	375898	435527	2288.30	2288.30	2651.30	363.00
30	Mushroom	126830.43	243000	7.2	9.51	17496	23109	221.90	221.90	293.10	71.19
31	Onion	26659.55	26641000	7.26	8.2	1934137	2184562	5156.32	5156.32	5823.94	667.62
32	Potato	17126.63	56173000	5.96	7.32	3347911	4111864	5733.84	5733.84	7042.24	1308.39
33	Tomato	24078.38	21181000	11.61	12.44	2459114	2634916	5921.15	5921.15	6344.45	423.30
34	Tapioca	19007.33	6941000	4.87	4.58	338027	317898	642.50	642.50	604.24	-38.26
35	Bottle gourd	15235.73	3171000	7.01		222287		338.67			
36	Brinjal	20855.09	12874000	7.41		953963		1989.50			
37	Beans	41134.40	2595000	7.11		184505		758.95			
38	Radish	11961.13	3263000	6.46		210790		252.13			
39	Capsicum	36257.71	563000	5.15		28995		105.13			
40	Okra	28917.95	6466000	6.01		388607		1123.77			
Plantation crops & Spices											
41	Areca nut	306514.12	1563000	4.41	4.91	68928	76743	2112.75	2112.75	2352.29	239.54
42	Black pepper	351092.09	141000	1.29	1.18	1819	1664	63.86	63.86	58.41	-5.45
43	Cashew	297810.85	738000	3.72	4.17	27454	30775	817.60	817.60	916.50	98.90
44	Chilli	132736.85	2049000	6.11	6.51	125194	133390	1661.78	1661.78	1770.58	108.79
45	Coconut (1 year average)	42686.47	14301000	3.86	4.77	552019	682158	2356.37	2356.37	2911.89	555.52

S.No.	Crops	Average price (Rs/ MT) (2019-20, 2020-21 and 2021-22)	Production in MT (2020-21 final estimates)	% Post-harvest losses NABCONS 2022	% Postharvest losses ICAR-CIPHET 2015	Quantity lost in MT NABCONS (2020-21)	Quantity lost in MT ICAR-CIPHET 2015	Monetary loss NABCONS (Rs. in Crore)		Monetary loss as per CIPHET (Rs. in Crore) (b)	Savings due to reduction in losses (b-a)
								54 crops	45 crops (a)		
46	Coriander	83421.63	811000	5.32	5.87	43145	47606	359.92	359.92	397.13	37.21
47	Sugarcane	2993.36	405399000	7.33	7.89	29715747	31985981	8895.00	8895.00	9574.56	679.56
48	Turmeric	24113.88	1124000	5.36	4.44	60246	49906	145.28	145.28	120.34	-24.94
Livestock Produce											
49	Egg (production in numbers and price per egg)	4.46	122110000000	6.03	7.19	7363233000	8779709000	3287.32	3287.32	3919.70	632.39
50	Inland Fish (prod of 2019-20)	174004.32	10437000	4.86	5.23	507238	545855	8826.16	8826.16	9498.11	671.95
51	Marine Fish (prod of 2019-20)	174004.32	3727000	8.76	10.52	326485	392080	5680.98	5680.98	6822.37	1141.38
52	Meat	38888.89	4500000	2.34	2.71	105300	121950	409.50	409.50	474.25	64.75
53	Poultry Meat	246892.14	4300000	5.63	6.74	242090	289820	5977.01	5977.01	7155.43	1178.42
54	Milk	49162.69	209900000	0.87	0.92	1826130	1931080	8977.75	8977.75	9493.71	515.96
Grand Total								**152790.42**	**146153.15**	**166593.72**	**20440.58**

Appendix 8 (b) Production and average price details for the commodity

S.No.	Crops	% Post-harvest losses NABCONS 2022	Average price (Rs/ MT) (2019-20, 2020-21 and 2021-22)	Price (2014) (Rs/tonne)	Production in MT (2020-21 final estimates)	Production in MT (2012-13)
1	Paddy	4.77	20727.82	17918	124370000	104400000
2	Wheat	4.17	21627.25	17309	109590000	92460000
3	Maize	3.88	18372.35	12662	31650000	22230000
4	Bajra	4.37	18213.39	12666	10860000	8740000
5	Sorghum	5.92	24587.57	18456	4810000	5280000
6	Pigeon pea	5.65	84688.16	49028	4320000	3070000
7	Chick pea	6.74	62105.85	32838	11910000	8880000
8	Black gram	5.83	66648.91	48159	2230000	830000
9	Green gram	6.19	71636.25	60912	3090000	460000
10	Mustard	4.46	55809.55	34820	10210000	7820000
11	Cottonseed	2.87	25584.40	32275	3939845.2	3490000
12	Soybean	7.51	50011.48	36984	12610000	14680000
13	Safflower	3.06	41866.97	26260	36000	100000
14	Sunflower	4.38	47057.01	32576	228000	580000
15	Groundnut	5.73	56331.57	31769	10244000	4750000
16	Apple	9.51	89699.98	68078	2276000	1900000
17	Banana	7.57	23082.26	18601	33062000	27060000
18	Citrus	7.71	39580.89	14011	14245000	11470000
19	Grapes	7.15	65060.19	44564	3358000	2520000
20	Guava	15.05	32150.18	20628	4582000	2620000
21	Mango	8.53	60853.22	45355	20386000	17290000
22	Papaya	6.59	21531.83	16023	5540000	5190000
23	Sapota	9.53	33589.81	18770	822000	1500000
24	Cabbage	8.15	15787.29	10928	9560000	8530000
25	Cauliflower	7.89	23313.40	16321	9225000	7790000
26	Green pea	6.43	60875.64	33698	5846000	3870000
27	Mushroom	7.2	126830.43	119049	243000	40000
28	Onion	7.26	26659.55	16920	26641000	16660000
29	Potato	5.96	17126.63	16649	56173000	41090000
30	Tomato	11.61	24078.38	16510	21181000	17850000
31	Tapioca	4.87	19007.33	22436	6941000	7320000
32	Areca nut	4.41	306514.12	182865	1563000	530000
33	Black pepper	1.29	351092.09	570547	141000	50000
34	Cashew	3.72	297810.85	76026	738000	750000

S.No.	Crops	% Post-harvest losses NABCONS 2022	Average price (Rs/ MT) (2019-20, 2020-21 and 2021-22)	Price (2014) (Rs/tonne)	Production in MT (2020-21 final estimates)	Production in MT (2012-13)
35	Chilli	6.11	132736.85	64411	2049000	1310000
36	Coconut	3.86	42686.47	28587	14301000	15090000
37	Coriander	5.32	83421.63	80506	811000	530000
38	Sugarcane	7.33	2993.36	2100	405399000	338960000
39	Turmeric	5.36	24113.88	24845	1124000	980000
40	Egg (production in numbers and price per egg)	6.03	4.46	2.634	122110000000	69700000000
41	Inland Fish (production of 2019-20)	4.86	174004.32	125306	10437000	5740000
42	Marine Fish (production of 2019-20)	8.76	174004.32	125306	3727000	3280000
43	Meat	2.34	38888.89	350000	4500000	1300000
44	Poultry Meat	5.63	246892.14	150000	4300000	3900000
45	Milk	0.87	49162.69	36000	209900000	132400000
46	Pineapple	6.02	28046.17		1799000	
47	Pomegranate	6.82	68722.72		3271000	
48	Muskmelon	6.83	23012.31		1478000	
49	Bottle gourd	7.01	15235.73		3171000	
50	Brinjal	7.41	20855.09		12874000	
51	Beans	7.11	41134.40		2595000	
52	Radish	6.46	11961.13		3263000	
53	Capsicum	5.15	36257.71		563000	
54	Okra	6.01	28917.95		6466000	

Note: In the current study, wholesale prices for 3 years (2019-20, 2020-21 and 2021-22) as affected by inflation over years were considered for estimating average wholesale prices. Wholesale prices were taken from agmarket for agricultural crops, horticultural crops and fisheries, National Egg Co-Ordination Committee for egg and Department of Consumer Affairs for milk. In case of sugarcane, Fair and Remunerative Price (FRP) was considered. Production data were taken from Department of Agriculture and Farmers' Welfare for horticulture and agriculture commodities, annual report (2021-22) of Department of Animal Husbandry and Dairying for egg, meat and poultry meat and National Dairy Development Board (NDDB) for milk.

Appendix 9 Data analysis procedure for estimation of losses

Data analysis procedure

The data analysis procedure used (Ahmad et al. 2021, Jha et al., 2015) is explained at various operations in the subsequent subsections. The notations used in the procedure are given below:

$\widehat{Y_i}$	Estimate of quantity handled for a particular farm operation of the crop/commodity in i^{th} district (by enquiry)
B_i	Total number of blocks in i^{th} district
b_i	Number of selected blocks in i^{th} district
V_{ib}	Total number of villages in b^{th} selected block of i^{th} district
v_{ib}	Number of selected villages in b^{th} selected block of i^{th} district for a farm operation
F_{ibv}	Total number of farmers growing a particular crop/commodity in v^{th} selected village of b^{th} selected block from i^{th} district
f_{ibv}	Number of selected farmers growing a crop/commodity in v^{th} selected villages of b^{th} selected block of i^{th} district for a farm operation
y_{ibvf}	Quantity handled for a farm operation of a crop/commodity by the f^{th} selected farmer in v^{th} selected village of b^{th} selected block of i^{th} district (by enquiry)
$\widehat{\delta_i}$	Estimate of quantity lost for a farm operation of a crop/commodity in i^{th} district (by enquiry)
δ_{ibvf}	Quantity of crop/commodity lost at a particular farm operation by the f^{th} selected farmer in v^{th} selected village of b^{th} selected block for i^{th} district (by enquiry)
$\widehat{L_i}$	Estimate of loss (%) by enquiry for the i^{th} district
$\hat{V}(\widehat{L_i})$	Estimate of variance of percent loss by enquiry for i^{th} district
$\hat{V}(\widehat{Y_i})$	Estimate of variance of quantity handled (by enquiry) for an operation in the crop for i^{th} district
$\hat{V}(\widehat{\delta_i})$	Estimate of variance of quantity lost (by enquiry) for an operation in the crop for i^{th} district
$\widehat{Y_i'}$	Estimate of quantity handled at a particular farm operation of the crop/commodity in i^{th} district (by observation)
y'_{ibvf}	Quantity handled at a particular farm operation of the crop/ commodity of the f^{th} selected farmer in v^{th} selected village of b^{th} selected block of i^{th} district (by observation)
$\widehat{\delta_i'}$	Estimates of quantity lost for a particular farm operation of the crop/ commodity in i^{th} district (by observation)
δ'_{ibvf}	Quantity lost at particular farm operation of the crop/ commodity by the f^{th} selected farmer in v^{th} selected village of b^{th} selected block of i^{th} district (by observation)
$\widehat{L_i'}$	Estimate of percent loss by observation for i^{th} district
$\hat{V}(\widehat{L_i'})$	Estimate of variance of percent loss by observation for i^{th} district
$\hat{V}(\widehat{\delta_i'})$	Estimate of variance of quantity lost (by observation) for an operation in a crop / commodity of i^{th} district
$\hat{V}(\widehat{Y_i'})$	Estimate of variance of quantity handled (by observation) for an operation in a crop / commodity of i^{th} district
$\widehat{L_i}^{(c)}$	Estimate of percentage loss of c^{th} crop/commodity in a farm operation of i^{th} district
$\widehat{s_i}$	Estimate of standard error of percentage loss in a farm operation of i^{th} district obtained by enquiry

$\widehat{s_i'}$	Estimate of standard error of percentage loss in a farm operation of i^{th} district obtained by actual observation
$\widehat{S_i}$	Estimate of standard error of pooled estimate of percentage loss in a farm operation of i^{th} district
n_i	Number of data points obtained through method of enquiry in a particular farm operation for a particular crop/commodity in i^{th} district
n_i'	Number of data points obtained through method of actual observation in a particular farm operation for a particular crop/commodity in i^{th} district
$\widehat{P_{iz}}$	Production of crop/commodity for the i^{th} district falling in z^{th} ACZ in the agricultural year for which the percentage loss is being estimated.
s_s	Estimate of standard error of percentage loss for data collected by enquiry/observation in s^{th} state
L_{is}^*	Estimate of percentage loss obtained for data collected by enquiry/observation in i^{th} district falling in s^{th} state.
$\widehat{L_{is}}$	Estimate of percent loss (by enquiry/observation) of the crop/commodity during storage in the channel for the i^{th} district falling in s^{th} state
d	number of sampled districts in s^{th} state.
$\widehat{L_{iz}}$	Estimate of percent loss (by enquiry) of the crop/commodity in a farm operation for the i^{th} district falling in z^{th} ACZ
$\widehat{L_z}$	Estimated percent loss of the crop/commodity in a farm operation for z^{th} ACZ (by enquiry)
$\widehat{L_{iz}'}$	Estimate of percent loss (by observation) of the crop/commodity in a farm operation for the i^{th} district falling in z^{th} ACZ
$\widehat{L_z'}$	Estimated percent loss of the crop/commodity in an operation for z^{th} ACZ (by observation)
$\widehat{s_{iz}}$	Estimate of standard error of estimated of loss percent in a farm operation of i^{th} district in z^{th} ACZ by enquiry/observation
$\widehat{S_z}$	Estimate of standard error of estimated loss percent in a farm operation of z^{th} ACZ by enquiry/observation
$\widehat{L_{iz}^*}$	Estimate of percent loss of the crop/commodity in a farm operation for the i^{th} district falling in z^{th} ACZ by enquiry /observation
$\widehat{L_N^{(c)}}$	Estimate of percentage loss of c^{th} crop/commodity at national level.
$\widehat{L_{iN}}$	Estimated loss (%) of crop/commodity after pooling the enquiry and actual observation data of i^{th} ACZ
$\widehat{P_{iN}}$	Production of crop/commodity in i^{th} ACZ for which the percentage loss is being estimated.
$\widehat{S_N}$	Estimate of standard error of estimate of percentage loss of the crop in a farm operation at national level
$\hat{P}_i$	Estimate of total quantity of crop/commodity withdrawn from the store by the selected farmers of the i^{th} district during total enquiry period.
p_{ibvft}	Quantity of crop/commodity withdrawn from the storage between previous and t^{th} visit to f^{th} selected farmer in v^{th} selected village of b^{th} selected block of i^{th} district (by enquiry)
$\hat{\zeta}_i$	Estimate of total quantity loss of crop/commodity of selected farmers of the i^{th} district during total enquiry period.
ζ_{ibvft}	Quantity loss of crop/commodity between previous and t^{th} visit to f^{th} selected farmer in v^{th} selected village of b^{th} selected block of i^{th} district (by enquiry)

d_{ibvft}	Weight/number of crops/commodities damaged in the sample drawn at the time of t^{th} visit to f^{th} selected farmer in v^{th} selected village of b^{th} selected block of i^{th} district (by observation)
u_{ibvft}	Weight/number of crops/commodities undamaged in the sample drawn at the time of t^{th} visit to f^{th} selected farmer in v^{th} selected village of b^{th} selected block of i^{th} district (by observation)
d_{ibt}	Weight/number of crop/commodity damaged in the sample drawn at the time of t^{th} visit to b^{th} respondent (godown/wholesaler/retailer/ processing unit) of i^{th} district (by observation)
u_{ibt}	Weight/number of crop/commodity undamaged in the sample drawn at the time of t^{th} visit to b^{th} respondent (godown/ wholesaler/ retailer/ processing unit) of i^{th} district (by observation)
$\widehat{S_i'}(\widehat{d_i})$	Estimate of standard error of weight/number of crops/commodities damaged in stores of farmers of i^{th} district (by observation)
$\widehat{S_i'}(\widehat{TG_i})$	Estimate of standard error of total weight/number of crops/ commodities drawn from stores of farmers of i^{th} district (by observation)
$\widehat{L_{Zs}}$	Estimate of loss (%) of a crop/commodity during storage in a channel at ACZ level (enquiry/observation)
$\widehat{S_z}$	Estimate of standard error of estimate of storage loss (%) at ACZ level (enquiry/observation)
$\widehat{L_{SN}^{(c)}}$	Estimate of percentage loss of c^{th} crop/commodity during storage in a channel at national level
$\widehat{S_{SN}}$	Estimate of standard error of estimate of storage loss (%) at national level
SE	Standard Error of estimates
$\widehat{L_{TS}}$	Estimate of total percentage loss of a crop/commodity during storage in different marketing channels
$\widehat{L_F}$	Estimated loss of crops / commodity during storage at farm
$\widehat{R_F}$	Estimated percent retention of crops / commodity in storage at farm
$\widehat{L_G}$	Estimated loss of crops / commodity during storage at godown
$\widehat{R_G}$	Estimated percent retention of crops / commodity in storage at godown
$\widehat{L_W}$	Estimated loss of crops / commodity during storage at wholesaler level
$\widehat{R_W}$	Estimated percent retention of crops / commodity in storage at wholesaler level
$\widehat{L_R}$	Estimated loss of crops / commodity during storage at retailer level
$\widehat{R_R}$	Estimated percent retention of crops / commodity for storage at retailer level
$\widehat{L_p}$	Estimated loss of crops / commodity during storage at processing unit
$\widehat{R_p}$	Estimated percent retention of crops / commodity for storage at processing unit
d	Number of districts in z^{th} ACZ or s^{th} state
a	Number of ACZs for selected crops
C	Channels
Y	Years
$d.f.$	Degrees of freedom
CSS	Channel sum of squares
YSS	Year sum of squares
ESS	Error sum of squares
TSS	Total sum of squares

(i) Estimation of post-harvest loss in farm operations

Estimation of losses was carried out at district level for enquiry and actual measurements data before estimating at ACZ/State level. Thereafter, both data were merged to obtain final loss at ACZ, state and national level.

Estimation of post-harvest loss at district level

After maturity of crop, usually complete produce passes through a series of farm operations (harvesting, collection, sorting/grading, threshing, winnowing, drying, packaging, storage and transportation). Each operation was performed separately and hence the losses were also different. Therefore, the estimation procedures of farm operations and storage channels were different and were computed separately both for data obtained by enquiry and actual measurement methods.

Estimation procedure for data collected by enquiry:

An estimate of quantity of a crop/commodity handled for a particular farm operation in a district was obtained by using eqn. (1)

$$\widehat{Y_\iota} = \frac{B_i}{b_i}\sum_{b=1}^{b_i}\frac{V_{ib}}{v_{ib}}\sum_{v=1}^{v_{ib}}\frac{F_{ibv}}{f_{ibv}}\sum_{f=1}^{f_{ibv}} y_{ibvf} \qquad \textbf{(1)}$$

In eqn. (1), the quantity of produce handled in a given farm operation by a farmer was taken to the total quantity handled at the village level, then to the block level and finally to the district level. An estimate of quantity of the crop/commodity lost in the same farm operation in i^{th} district was computed using eqn. (2)

$$\widehat{\delta_\iota} = \frac{B_i}{b_i}\sum_{b=1}^{b_i}\frac{V_{ib}}{v_{ib}}\sum_{v=1}^{v_{ib}}\frac{F_{ibv}}{f_{ibv}}\sum_{f=1}^{f_{ibv}} \delta_{ibvf} \qquad \textbf{(2)}$$

In eqn. (2), the quantitative loss in a given farm operation was taken from farmer level through the village level, block level and finally to the district level.

The loss (%) obtained by enquiry for the crop/commodity in i^{th} district was estimated by dividing the total quantity lost by the total quantity handled, using the eqn. (3)

$$\widehat{L_\iota} = \frac{\widehat{\delta_\iota}}{\widehat{Y_\iota}} \times 100 \qquad \textbf{(3)}$$

Estimated variance of $\widehat{L_\iota}$ was calculated using eqn. (4), after ignoring higher order terms.

$$\hat{V}(\widehat{L_\iota}) = \left(\frac{\widehat{\delta_\iota}}{\widehat{Y_\iota}} \times 100\right)^2 \left(\frac{\hat{V}(\widehat{\delta_\iota})}{(\widehat{\delta_\iota})^2} + \frac{\hat{V}(\widehat{Y_\iota})}{(\widehat{Y_\iota})^2}\right) \qquad \textbf{(4)}$$

where the estimate of variance of $\widehat{\delta_\iota}$ and $\widehat{Y_\iota}$ was obtained using the expressions given in eqn. (5)

$$\hat{V}(\widehat{X_\iota}) = \frac{1}{b_i(b_i-1)}\sum_{b=1}^{b_i}\left(\hat{X}_{ib} - \hat{\bar{X}}_i\right)^2 \qquad \textbf{(5)}$$

where, $\hat{X}_{ib} = \frac{V_{ib}}{v_{ib}}\sum_{v=1}^{v_{ib}}\frac{F_{ibv}}{f_{ibv}}\sum_{f=1}^{f_{ibv}} x_{ibvf}$, $\hat{\bar{X}}_i = \frac{1}{b_i}\sum_{b=1}^{b_i}\hat{X}_{ib}$ *and* $\hat{\bar{X}}_i$ *is the mean of variable (quantity handled/lost) for* i^{th} *district and* $\hat{X}_{ib}$ *is estimate of quantity handled /lost for* b^{th} *block in* i^{th} *district.*

Estimation procedure for data collected by actual observation:

The estimates of quantity handled for an operation of a crop /commodity in the district were obtained in a manner similar to that of the data collected by enquiry, by using the estimator given by eqn. (6)

$$\widehat{Y_\iota'} = \frac{B_i}{b_i}\sum_{b=1}^{b_i}\frac{V_{ib}}{v_{ib}}\sum_{v=1}^{v_{ib}}\frac{F_{ibv}}{f_{ibv}}\sum_{f=1}^{f_{ibv}} y'_{ibvf} \qquad \textbf{(6)}$$

The estimate of quantity lost was calculated by eqn. (7)

$$\widehat{\delta_\iota'} = \frac{B_i}{b_i}\sum_{b=1}^{b_i}\frac{V_{ib}}{v_{ib}}\sum_{v=1}^{v_{ib}}\frac{F_{ibv}}{f_{ibv}}\sum_{f=1}^{f_{ibv}} \delta'_{ibvf} \qquad \textbf{(7)}$$

and the estimate of percentage loss for the district was calculated by eqn. (8)

$$\widehat{L_\iota'} = \frac{\widehat{\delta_\iota'}}{\widehat{Y_\iota'}} \times 100 \tag{8}$$

Estimate of variance of $\widehat{L_\iota'}$, after ignoring higher order terms was obtained by the eqn. (9)

$$\hat{V}(\widehat{L_\iota'}) = \left(\frac{\widehat{\delta_\iota'}}{\widehat{Y_\iota'}} \times 100\right)^2 \left(\frac{\hat{V}(\widehat{\delta_\iota'})}{\widehat{\delta_\iota'}^2} + \frac{\hat{V}(\widehat{Y_\iota'})}{(\widehat{Y_\iota'})^2}\right) \tag{9}$$

where, the estimate of variance of $\widehat{\delta_\iota'}$ and $\widehat{Y_\iota'}$ was obtained by employing eqn.(10)

$$\hat{V}(\widehat{X_\iota'}) = \frac{1}{b_i(b_i-1)} \sum_{b=1}^{b_i} \left(\hat{X}'_{ib} - \hat{\bar{X}}'_i\right)^2 \tag{10}$$

where, X was a variable for quantity handled/quantity lost in i^{th} district as expressed below:

$$\hat{X}'_{ib} = \frac{V_{ib}}{v_{ib}} \sum_{v=1}^{v_{ib}} \frac{F_{ibv}}{f_{ibv}} \sum_{f=1}^{f_{ibv}} x'_{ibvf}$$

$$\hat{\bar{X}}'_i = \frac{1}{b_i} \sum_{b=1}^{b_i} \hat{X}'_{ib}$$

Pooling of enquiry and actual observation based estimators:

In order to estimate the loss during farm operations at district level for different crops/commodities, the estimate of percentage loss of c^{th} crop / commodity in the i^{th} district was obtained by pooling an estimate of percentage loss by enquiry and actual observation using the weighted estimator (eqn.11).

$$\hat{L}_i^{(c)} = \frac{\widehat{s_\iota'}^2 \hat{L_\iota} + \hat{s_\iota}^2 \widehat{L_\iota'}}{\widehat{s_\iota'}^2 + \hat{s_\iota}^2} \tag{11}$$

The estimate of standard error of pooled estimate of percentage loss in a farm operation of i^{th} district was obtained by using eqn. (12)

$$\hat{S_\iota} = \sqrt{\frac{\hat{s_\iota}^2 \times \widehat{s_\iota'}^2}{\hat{s_\iota}^2 + \widehat{s_\iota'}^2}} \tag{12}$$

(ii) Estimation of post-harvest loss in farm operations at state level

Estimation procedure for data collected by enquiry:

The estimate of percentage loss of a crop/commodity handled in a farm operation at state level was given by the weighted average of production of the selected districts (eqn.13)

$$\hat{L_s} = \frac{\sum_{i=1}^{d} \hat{P_{\iota s}} \times \hat{L_{\iota s}}}{\sum_{i=1}^{d} \hat{P_{\iota s}}} \tag{13}$$

Estimation procedure for data collected by actual observation: The estimate of percentage loss of a crop/commodity in a farm operation at state level was obtained by using eqn. (14)

$$\hat{L'_s} = \frac{\sum_{i=1}^{d} \hat{P_{\iota s}} \times \widehat{L'_{\iota s}}}{\sum_{i=1}^{d} \hat{P_{\iota s}}} \tag{14}$$

The estimate of standard error of percentage loss for data collected by enquiry/observations was computed using eqn. (15)

$$\hat{S_s} = \sqrt{\frac{\sum_{i=1}^{d} P_{is}^2 \times \hat{V}(\widehat{L_{\iota s}^*})}{\left(\sum_{i=1}^{d} P_{is}\right)^2}} \tag{15}$$

where,

$\hat{s_s}$: Estimate of standard error of percentage loss for data collected by enquiry/observation in s^{th} state using the respective variance expression given in equations (4) and (9) respectively.

$\widehat{L_{is}^{*}}$: Estimate of percentage loss obtained for data collected by enquiry/observation in i^{th} district falling in s^{th} state.

The estimate of loss (%) and its standard error for data collected by enquiry and observation at state level was obtained using pooled estimator similar to equations (11) and (12) respectively.

(iii) Estimation of post-harvest loss in farm operations at ACZ level

Estimation procedure for data collected by enquiry:

The estimate of percentage loss of a crop/commodity handled in a farm operation at ACZ level was given by the weighted average of production of the selected districts (eqn.16)

$$\widehat{L_z} = \frac{\sum_{i=1}^{d} \widehat{P_{iz}} \times \widehat{L_{iz}}}{\sum_{i=1}^{d} \widehat{P_{iz}}} \qquad \textbf{(16)}$$

Estimation procedure for data collected by actual observation: The estimate of percentage loss of a crop/commodity in a farm operation at ACZ level was obtained by using eqn. (17)

$$\widehat{L_z'} = \frac{\sum_{i=1}^{d} \widehat{P_{iz}} \times \widehat{L_{iz}'}}{\sum_{i=1}^{d} \widehat{P_{iz}}} \qquad \textbf{(17)}$$

The estimate of standard error of percentage loss for data collected by enquiry/observations was computed using eqn. (18)

$$\widehat{S_z} = \sqrt{\frac{\sum_{i=1}^{d} P_{iz}^2 \times \hat{V}(\widehat{L_{iz}^{*}})}{\left(\sum_{i=1}^{d} P_{iz}\right)^2}} \qquad \textbf{(18)}$$

where,

$\widehat{s_z}$: Estimate of standard error of percentage loss for data collected by enquiry/observation in z^{th} ACZ using the respective variance expression given in equations (4) and (9) respectively.

$\widehat{L_{iz}^{*}}$: Estimate of percentage loss obtained for data collected by enquiry/observation in i^{th} district falling in z^{th} ACZ.

The estimate of loss (%) and its standard error for data collected by enquiry and observation at ACZ level was obtained using pooled estimator similar to equations (11) and (12) respectively.

(iv) Estimation of post-harvest loss in farm operations at national level

Estimation of losses at national level in different farm operations were obtained from pooled estimates of loss (enquiry and observation) at ACZ level. The estimate of percentage loss of c^{th} crop/commodity at national level was obtained by using weighted estimator given by eqn. (19)

$$\widehat{L_N^{(c)}} = \frac{\sum_{i=1}^{a} \widehat{P_{iN}} \times \widehat{L_{iN}}}{\sum_{i=1}^{a} \widehat{P_{iN}}} \qquad \textbf{(19)}$$

where,

$\hat{P}_{iN}$: Production of crop/commodity in i^{th} ACZ

$\widehat{L_{iN}}$: Estimate of loss (%) of crop/commodity after pooling the enquiry and observation data of i^{th} ACZ.

The estimate of standard error of estimate of percentage loss at national level was obtained by using eqn.(20)

$$\widehat{S_N} = \sqrt{\frac{\sum_{i=1}^{a} \widehat{P_{iN}}^2 \times \hat{V}(\widehat{L_{iN}})}{\left(\sum_{i=1}^{a} \widehat{P_{iN}}\right)^2}} \qquad \textbf{(20)}$$

Production data of some of the crops were not available at district and ACZ level. In such cases, the numbers of observations were taken in place of production to obtain standard error using eqn. (18) and simple averages were taken to estimate the loss.

(v) Estimation of post-harvest loss during storage

In order to estimate loss percent from the data collected through enquiry and observation, district -wise estimates are obtained separately and then pooled through an optimum pooling technique.

Estimation of farm level storage post-harvest loss at district level

Estimation procedure for data collected by enquiry: An estimate of total quantity of crop /commodity withdrawn from the store by the selected farmers of the i^{th} district during total enquiry period was computed by eqn. (21)

$$\hat{P}_i = \frac{B_i}{b_i}\sum_{b=1}^{b_i}\frac{V_{ib}}{v_{ib}}\sum_{v=1}^{v_{ib}}\frac{F_{ibv}}{f_{ibv}}\sum_{f=1}^{f_{ibv}}\left(\sum_{t=1}^{T} p_{ibvft}\right) \quad \textbf{(21)}$$

The estimated total quantity lost in i^{th} district was calculated using eqn. (22)

$$\hat{\zeta}_i = \frac{B_i}{b_i}\sum_{b=1}^{b_i}\frac{V_{ib}}{v_{ib}}\sum_{v=1}^{v_{ib}}\frac{F_{ibv}}{f_{ibv}}\sum_{f=1}^{f_{ibv}}\left(\sum_{t=1}^{T} \zeta_{ibvft}\right) \quad \textbf{(22)}$$

The estimate of farm level storage loss (%) of crop/commodity by enquiry in i^{th} district was given by eqn. (23)

$$\hat{L}_i = \frac{\hat{\zeta}_i}{\hat{P}_i} X100 \quad \textbf{(23)}$$

The estimated variance of $\hat{L}_i$ was obtained using eqn. (4) after ignoring the higher order terms.

Estimation procedure for data collected by actual observation: Estimate of farm level storage loss (%) of crop/commodity for data collected by actual observation used in i^{th} district was given by eqn. (24)

$$\widehat{L_i'} = \frac{\hat{d}_i}{\widehat{TG}_i} \times 100 = \frac{\frac{B_i}{b_i}\sum_{b=1}^{b_i}\frac{V_{ib}}{v_{ib}}\sum_{v=1}^{v_{ib}}\frac{F_{ibv}}{f_{ibv}}\sum_{f=1}^{f_{ibv}}\left(\sum_{t=1}^{T} d_{ibvft}\right)}{\frac{B_i}{b_i}\sum_{b=1}^{b_i}\frac{V_{ib}}{v_{ib}}\sum_{v=1}^{v_{ib}}\frac{F_{ibv}}{f_{ibv}}\sum_{f=1}^{f_{ibv}}\left(\sum_{t=1}^{T} d_{ibvft}+\sum_{t=1}^{T} u_{ibvft}\right)} \times 100 \quad \textbf{(24)}$$

and approximate estimate of variance of above estimators was given by eqn. (25)

$$\hat{V}\left(\widehat{L_i'}\right) = \left(\widehat{L_i'}\right)^2\left\{\frac{\left(\widehat{S_i'}(\hat{d}_i)\right)^2}{\left(\hat{d}_i\right)^2} + \frac{\left(\widehat{S_i'}(\widehat{TG}_i)\right)^2}{(\widehat{TG}_i)^2}\right\} \quad \textbf{(25)}$$

The estimate of variance of d_i and TG_i was obtained by eqn. (26)

$$\hat{V}(\hat{X}_i) = \frac{1}{b_i(b_i-1)}\sum_{b_i=1}^{b_i}\left(\hat{X}_{ib} - \hat{\bar{X}}_i\right)^2 \quad \textbf{(26)}$$

where,

$$\hat{X}_{ib} = \frac{V_{ib}}{v_{ib}}\sum_{v=1}^{v_{ib}}\frac{F_{ibv}}{f_{ibv}}\sum_{f=1}^{f_{ibv}}\sum_{t=1}^{T} x_{ibvft}$$

$$\hat{\bar{X}}_i = \frac{1}{b_i}\sum_{b=1}^{b_i}\hat{X}_{ib}$$

where X is the variable d_i orTG_i.

Pooled estimate of the estimates of loss percentage for data collected by enquiry and observation was obtained using equations (11) and (12).

***Note:** In the present study, the analysis for estimation of farm level storage was done similar to that of other farm operations as crop/commodity was stored temporarily by the farmer. Total quantity withdrawn/loss of crop/commodity was obtained by summing the quantity observed during various visits.*

(vi) Estimation of post-harvest loss in storage and marketing channels (wholesaler, retailer, godown and processing unit) at district level

Data for this purpose were collected from respondents of different marketing channels selected using stratified two stage random sampling in the lines of CIPHET (2015) as described in chapter-3.

Estimation procedure for data collected by enquiry: The estimate of loss (%) for different crops/commodity and its estimate of variance for data collected by enquiry were estimated using the equations similar to (18), (19), (20) and (4).

Estimation procedure for data collected by actual observation: Estimate of loss (%) during storage for data collected through actual observation was obtained by using eqn. (27)

$$\widehat{L_i'} = \frac{\hat{d}_i}{\widehat{TG}_i} \times 100 = \frac{\frac{B_i}{b_i}\sum_{b=1}^{b_i}\sum_{t=1}^{T} d_{ibt}}{\left(\frac{B_i}{b_i}\sum_{b=1}^{b_i}\sum_{t=1}^{T} d_{ibt} + \frac{B_i}{b_i}\sum_{b=1}^{b_i}\sum_{t=1}^{T} u_{ibt}\right)} \times 100 \qquad \mathbf{(27)}$$

The approximate estimate of variance of percentage loss during storage in i^{th} district was obtained as given in eqn. (28).

$$\hat{V}\left(\widehat{L_i'}\right) = \left(\widehat{L_i'}\right)^2 \left\{ \frac{\left(\widehat{S_i'}(\hat{d}_i)\right)^2}{\left(\hat{d}_i\right)^2} + \frac{\left(\widehat{S_i'}(\widehat{TG}_i)\right)^2}{(\widehat{TG}_i)^2} \right\} \qquad \mathbf{(28)}$$

The estimate of variance of d_i and TG_i was obtained as given by eqn. (26). Again, merging the estimate of loss (%) from data collected through enquiry and observation was carried out using eqns. (11) and (12) in order to get the pooled estimate of percentage loss.

(vii) Estimation of post-harvest loss at storage in different channels at state level

After production of the crop, the produce was distributed in different channels where it was stored or used for further processing and consumption. Production, therefore, may not be used as weights. The estimate of loss of a crop/commodity, therefore, during storage in a channel at state level was estimated separately for enquiry and observation data using eqn. (29).

$$\widehat{L_{Ss}} = \frac{\sum_{i=1}^{d} \widehat{L_{is}}}{d} \qquad \mathbf{(29)}$$

where,

$\widehat{L_{is}}$ is the estimate of percent loss (by enquiry/observation) of the crop/commodity during storage in the channel for the i^{th} district falling in s^{th} state for which the estimate is to be obtained.

d is number of sampled districts in s^{th} state

The estimate of standard error of estimate of storage loss (%) for data collected by enquiry/observation was obtained using eqn. (30).

$$\hat{S}_s = \sqrt{\frac{\sum_{i=1}^{d} \hat{V}(\widehat{L_{is}})}{d^2}} \qquad \mathbf{(30)}$$

The estimates of loss (%) and its standard error for pooled data collected by enquiry and observation at state level were obtained using estimator similar to equations (11) and (12).

(viii) Estimation of post-harvest loss at storage in different channels at ACZ level

After production of the crop, the produce was distributed in different channels where it was stored or used for further processing and consumption. Production, therefore, may not be used as weights. The estimate of loss of a crop/commodity, therefore, during storage in a channel at ACZ level was estimated separately for enquiry and observation data using eqn. (31).

$$\widehat{L_{Zs}} = \frac{\sum_{i=1}^{d} \widehat{L_{iz}}}{d} \qquad \mathbf{(31)}$$

where,

$\widehat{L_{iz}}$ is the estimate of percent loss (by enquiry/observation) of the crop/commodity during storage in the channel for the i^{th} district falling in z^{th} ACZ for which the estimate is to be obtained.

d is number of sampled districts in z^{th} ACZ.

The estimate of standard error of estimate of storage loss (%) for data collected by enquiry/observation was obtained using eqn. (32).

$$\widehat{S_z} = \sqrt{\frac{\sum_{i=1}^{d} \hat{V}(\widehat{L_{iz}})}{d^2}} \quad \textbf{(32)}$$

The estimates of loss (%) and its standard error for pooled data collected by enquiry and observation at ACZ level were obtained using estimator similar to equations (11) and (12).

(ix) Estimation of post-harvest loss at storage in different channels at national level

Estimate of percentage loss of c^{th} crop/commodity during storage in a channel at national level by pooling data of loss (%) at ACZ level was given by eqn. (33).

$$\widehat{L_{SN}^{(c)}} = \frac{\sum_{i=1}^{a} \widehat{L_{iS}}}{a} \quad \textbf{(33)}$$

where ,

$\widehat{L_{iS}}$ is the estimate of percentage loss of c^{th} crop/commodity during storage in a channel at ACZ level.

a is number of ACZs for selected crops

The estimate of standard error of estimate of storage loss (%) at national level for each crop/commodity was computed using eqn. (34).

$$\widehat{S_{SN}} = \sqrt{\frac{\sum_{i=1}^{a} \hat{V}(\widehat{L_{iS}})}{a^2}} \quad \textbf{(34)}$$

(x) Estimation of total post-harvest loss of crop/commodities at national level

In order to estimate the overall total loss of a crop/ commodity during storage, it is essential to know the quantity of crop/commodity retention/handling in each operation and channel during storage. Since, the total produce is handled in each of the farm operations, the total loss of a crop/commodity in all farm operations was taken as arithmetic sum of losses in individual operations. Estimate of total percentage loss of a crop/ commodity during storage in different channels was given by eqn. (35)

$$\widehat{L_{TS}} = \frac{\hat{L}_G \times \hat{R}_G + \hat{L}_W \times \hat{R}_W + \hat{L}_R \times \hat{R}_R + \hat{L}_P \times \hat{R}_P}{100} \quad \textbf{(35)}$$

To estimate the overall total loss of a crop/livestock produce, it is imperative to know the quantity of crop/ commodity retention/handling in each operation and channels during storage. Since, the total produce is handled in each of the farm operations, the total loss of a crop/ livestock produce in all farm operations was taken as an arithmetic sum of losses in individual operations. However, to estimate the total loss during storage/ holding in various marketing channels, data of percent retention in each market channel was required. Therefore, the percent retention was obtained from the field by the wide consultation with all the market stakeholders.

(xi) Estimation of moisture content and 1000 grain weight of cereals and spices

For 6 crops viz. paddy, wheat, maize, pigeon pea, green gram and coriander, tests such as moisture content, 1000 grain weight, details of number and weight of undamaged grains, infested/ damaged grains were conducted by institutes/ labs such as National Institute of Food Technology Entrepreneurship and Management (NIFTEM), Equinox Labs and AGSS Analytical and Research Lab Private Limited. For estimation, a sample of 100-150 g was withdrawn (subject to the availability with the respondent and willingness to provide the same) from the storage at farm level and different channels. The collected samples to be packed into polythene bags with the identity slips. The identity slip for the sample collected and results obtained are indicated below:

Estimates of percent storage of major crops and livestock produce in different channels at national level

S No	Crop/Commodity	Stored in godowns	Retained by wholesaler	Retailer level storage	Stored in processing unit
Grains (Cereals, Pulses, Oilseeds)					
1	Paddy	12.50	28.90	5.50	53.10
2	Wheat	22.80	27.80	11.70	37.70
3	Maize	18.70	43.20	20.50	17.60
4	Bajra	15.20	50.20	19.60	0
5	Sorghum	17.05	61.86	16.86	4.23
6	Pigeon Pea	7.10	29.90	28.40	34.60
7	Chick Pea	15.00	45.00	20.00	20.00
8	Black gram	18.01	35.49	25.83	20.67
9	Green Gram	50.20	19.60	15.20	15.00
10	Mustard	10.00	35.00	15.00	40.00
11	Cottonseed	9.00	54.00	12.00	25.00
12	Soybean	15.92	51.28	12.60	20.20
13	Safflower	9.00	25.00	5.50	60.50
14	Sunflower	3.30	21.20	4.50	71.00
15	Groundnut	7.00	43.20	11.70	38.10
Fruits					
16	Apple	18.02	42.05	22.21	17.73
17	Banana	6.00	78.00	15.00	1.00
18	Citrus	8.50	44.93	32.31	14.26
19	Grapes	19.10	30.51	34.10	16.29
20	Guava	0	41.23	53.09	5.68
21	Mango	5.10	37.03	40.17	17.70
22	Papaya	1.50	45.00	50.00	3.50
23	Sapota	10.00	43.00	42.00	5.00
24	Muskmelon	1.00	44.00	55.00	0
25	Pineapple	8.23	43.16	36.11	12.50
26	Pomegranate	12.24	39.16	37.70	10.90
Vegetables					
27	Tapioca	0	48.90	44.60	6.50
28	Tomato	0	46.22	33.22	20.46
29	Okra	0	57.90	42.10	0
30	Mushroom	0	50.06	49.94	0
31	Onion	28.09	34.58	24.15	13.18
32	Potato	50.00	27.00	14.00	9.00
33	Cabbage	5.00	50.00	35.00	10.00

S No	Crop/Commodity	Stored in godowns	Retained by wholesaler	Retailer level storage	Stored in processing unit
34	Cauliflower	5.00	45.00	40.00	10.00
35	Green pea	0	47.10	37.60	15.30
36	Capsicum	0	56.82	43.18	0
37	Beans	7.50	57.30	35.20	0
38	Bottle gourd	0	53.95	46.05	0
39	Brinjal	0	52.95	47.05	0
40	Radish	0	52.50	47.50	0
Plantation crops and spices					
41	Chilli	35.37	28.83	35.80	0
42	Turmeric	46.00	25.00	29.00	0
43	Areca nut	10.20	60.00	15.00	14.70
44	Coriander	6.00	50.00	37.00	7.00
45	Black Pepper	29.00	30.05	18.25	22.70
46	Coconut	14.00	37.00	22.00	27.00
47	Sugarcane	0	0	0	100.00
48	Cashew nut	6.00	21.89	10.10	62.01
Livestock Produce					
49	Egg	8.46	53.67	37.86	0
50	Inland Fish	0	39.00	61.00	0
51	Marine Fish	0	44.50	19.50	24.00
52	Meat	0	48.30	38.55	13.15
53	Poultry Meat	0	50.08	38.70	11.22
54	Milk	0	0	33.72	66.27

Identity slip for sample collection

S.No	Particulars	
1	Agro-Climatic Zone	
2	State	
3	District	
4	Tehsil/ Taluk	
5	Block/ Mandal	
6	Village/ Name of the Market	
7	Name of the farmer/ Trader/ Godown/ Processing Unit	
8	Type of Storage	
9	Name of the crop for which sample was taken	
10	Weight of the sample drawn (g)	
11	Date of the sample drawn	

Results of testing of 6 crops

Crops	Moisture Content of grains (%)	1000 Grain weight (g)	Infested/damaged grains (g/100g)	Infested/damaged grains (g/100g)
Paddy	11.70	21.35	2.62	89.17
Wheat	8.20	42.16	0.60	99.40
Maize	9.86	303.53	2.88	97.12
Green gram	10.04	24.87	0.10	99.90
Pigeon pea	11.13	21.49	2.42	89.25
Coriander	5.04	11.06	1.72	98.28

(xii) Testing Statistical significance of difference between losses of present and previous study (ICAR-CIPHET 2015)

Z test was applied for individual operational channels for testing the statistical difference between losses at $\alpha-$ level of significance. The Z statistics were given by eqn. (36).

$$\boldsymbol{Z} = \frac{\widehat{L_c}^{(1)} - \widehat{L_c}^{(2)}}{\sqrt{\frac{\widehat{V_c}^{(1)}}{n_1} + \frac{\widehat{V_c}^{(2)}}{n_2}}} \sim Z_\alpha \qquad \textbf{(36)}$$

where, $\widehat{L_c}^{(1)}$ and $\widehat{L_c}^{(2)}$ denotes the % loss at channel/operation C in the previous year and the current year respectively; $\widehat{V_c}^{(1)}$ and $\widehat{V_c}^{(2)}$ denotes the estimated variance of % loss at channel/operation C in the previous year and the current year respectively; n_1 and n_2 denotes the sample size taken to estimate the % loss at channel/operation C in the previous year and the current year respectively.

In the present study, the test was applied at 5% level of significance. Further 95% confidence intervals i.e. $[estimate \pm Z_{(5\%)}\, SE(estimate)]$ were computed for each channel and reported.

(xiii) Testing difference between overall total losses in studies

Two-way Analysis of Variance (ANOVA) was performed for testing overall difference between percent of losses between two studies (current and previous of Jha et al., 2015). In this, year was considered as treatment i.e. one source of variation and channel was considered as blocks i.e. second source of variation. The ANOVA has been presented below. The overall differences of pooled losses were tested at 5% level of significance.

Analysis of variance for comparing the estimates of two studies.

Sources of Variation	d.f.	Sum of Squares	Mean Sum of Squares	F- value
Years (Y)	$y-1$	YSS	$MYSS = YSS/\,(y-1)$	$MYSS/MESS$
Channels (C)	$c-1$	CSS	$MCSS = CSS/\,(c-1)$	$MCSS/MESS$
Error (E)	$(y-1)(c-1)$	ESS	$MESS = ESS/\,(y-1)(c-1)$	
Total (T)	$yc-1$	TSS	$MTSS$	

In case the variances of both studies were not statistically significant, t – test assuming equal variance was performed to check the significance of difference between overall total losses in two studies. When the variance of both studies were found statistically significant, t- test assuming unequal variance was performed.

In the present study, the Hansen-Hurwitz type estimators of PPS sampling were also attempted to estimate the loss (%) at state and ACZ level. The procedure is described as follows:

Estimation procedure for data collected by enquiry: An estimate of quantity of a crop/commodity handled for a particular farm operation at ACZ level was given by eqn.(37)

$$\widehat{Y_z} = \frac{1}{d}\sum_{i=1}^{d} \frac{\widehat{Y_\iota}}{p_{iz}} \qquad (37)$$

where $p_{iz} = \frac{\mathrm{P_{iz}}}{\mathrm{P_z}}$, denotes the probability of selecting the i^{th} district in the sample, $\mathrm{P_{iz}}$ is the production of the crop in the i^{th} district of z^{th} ACZ and $\mathrm{P_z}$ is the total production of the crop in the z^{th} ACZ.
An estimate of quantity of the crop/commodity lost in the same farm operation at ACZ level can be computed using eqn. (38)

$$\widehat{\delta_z} = \frac{1}{d}\sum_{i=1}^{d} \frac{\widehat{\delta_\iota}}{p_{iz}} \qquad (38)$$

The loss (%) obtained by enquiry for the crop/commodity at ACZ level was estimated by dividing the total quantity lost by the total quantity handled, using the eqn.(39)

$$\widehat{L_z} = \frac{\widehat{\delta_z}}{\widehat{Y_z}} \times 100 \qquad (39)$$

Estimated variance of $\widehat{L_z}$ is calculated using eqn. (40), after ignoring higher order terms.

$$\hat{V}(\widehat{L_z}) = \left(\frac{\widehat{\delta_z}}{\widehat{Y_z}} \times 100\right)^2 \left(\frac{\hat{V}(\widehat{\delta_z})}{(\widehat{\delta_z})^2} + \frac{\hat{V}(\widehat{Y_z})}{(\widehat{Y_z})^2}\right) \qquad (40)$$

where the estimates of variance of $\widehat{Y_z}$ and $\widehat{\delta_z}$ are given by equations (41) and (42), respectively.

$$\hat{V}(\widehat{Y_z}) = \frac{1}{d(d-1)}\sum_{i=1}^{d}\left(\frac{\widehat{Y_\iota}}{p_{iz}} - \widehat{Y_z}\right)^2 \qquad (41)$$

$$\hat{V}(\widehat{\delta_z}) = \frac{1}{d(d-1)}\sum_{i=1}^{d}\left(\frac{\widehat{\delta_\iota}}{p_{iz}} - \widehat{\delta_z}\right)^2 \qquad (42)$$

Estimation procedure for data collected by actual observation: An estimate of quantity of a crop/commodity handled, lost and loss percentage obtained by actual observation for a particular farm operation, at ACZ level was given by equations (43), (44) and (45)

$$\widehat{Y_z{}'} = \frac{1}{d}\sum_{i=1}^{d} \frac{\widehat{Y_\iota{}'}}{p_{iz}} \qquad (43)$$

$$\widehat{\delta_z{}'} = \frac{1}{d}\sum_{i=1}^{d} \frac{\widehat{\delta_\iota{}'}}{p_{iz}} \qquad (44)$$

$$\widehat{L'_z} = \frac{\widehat{\delta_z{}'}}{\widehat{Y_z{}'}} \times 100 \qquad (45)$$

Estimated variance of $\widehat{L'_z}$ was calculated using eqn.(46), after ignoring higher order terms.

$$\hat{V}(\widehat{L'_z}) = \left(\frac{\widehat{\delta_z{}'}}{\widehat{Y_z{}'}} \times 100\right)^2 \left(\frac{\hat{V}(\widehat{\delta_z{}'})}{(\widehat{\delta_z{}'})^2} + \frac{\hat{V}(\widehat{Y_z{}'})}{(\widehat{Y_z{}'})^2}\right) \qquad (46)$$

where the estimates of variance of $\widehat{Y_z{}'}$ and $\widehat{\delta_z{}'}$ are given by equations (47) and (48), respectively.

$$\hat{V}(\widehat{Y_z{}'}) = \frac{1}{d(d-1)}\sum_{i=1}^{d}\left(\frac{\widehat{Y_\iota{}'}}{p_{iz}} - \widehat{Y_z{}'}\right)^2 \qquad (47)$$

$$\hat{V}(\widehat{\delta_z{}'}) = \frac{1}{d(d-1)}\sum_{i=1}^{d}\left(\frac{\widehat{\delta_\iota{}'}}{p_{iz}} - \widehat{\delta_z{}'}\right)^2 \qquad (48)$$

The estimate of loss (%) and its standard error for data collected by enquiry and observation at ACZ level was obtained using pooled estimator similar to equations (11) and (12) respectively.

Note: *In similar to ACZ level estimations as given above, the state level estimates were obtained for the losses of farm and marketing operations.*

The estimates of losses using PPS methodology at national level are presented below:

Post-harvest losses of cereals in percent at national level

Crop	Farm Operations									Market Level Assessment						Overall total loss
	Harvest-ing	Collect-ion	Thresh-ing	Winnowi-ng/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesal-ers	Retailers	Process-ing unit	Trans-port	Total loss in market level	
Paddy	1.43 ±0.9058	0.41 ±0.1961	0.97 ±0.2707	0.5 ±0.3	0.31 ±0.2078	0.12 ±0.11	0.15 ±0.0589	0.36 ±0.1366	4.24 ±1.0486	0.06 ±0.0142	0.19 ±0.0408	0.03 ±0.0103	0.15 ±0.0598	0.12 ±0.0383	0.55 ±0.0837	4.79 ±1.052
Wheat	1.45 ±0.8159	0.32 ±0.1962	1.2 ±1.357	0.41 ±0.4909	0.06 ±0.0493	0.01 ±0.0159	0.24 ±0.1144	0.09 ±0.0332	3.78 ±1.6744	0.08 ±0.0076	0.12 ±0.0717	0.06 ±0.0298	0.27 ±0.1673	0.09 ±0.045	0.62 ±0.19	4.4 ±1.6851
Maize	1.12 ±0.1778	0.33 ±0.3042	1.15 ±0.627	0.43 ±0.0513	0.17 ±0.0887	0.11 ±0.0378	0.41 ±0.0673	0.11 ±0.0296	3.84 ±0.7312	0.007 ±0.0033	0.26 ±0.4286	0.18 ±0.0618	0.08 ±0.047	0.13 ±0.0746	0.66 ±0.4419	4.5 ±0.8543
Bajra	0.94 ±0.3477	0.3 ±0.0965	1.17 ±0.4692	0.16 ±0.0642	0.22 ±0.0811	0.25 ±0.0812	0.28 ±0.0927	0.26 ±0.0845	3.58 ±0.6192	0.85 ±0.3136	0.19 ±0.1207	0.36 ±0.2273	0.87 ±0.3501	0.31 ±0.2408	0.73 ±14.7643	4.32 ±0.6643
Sorghum	1.64 ±1.2693	0.93 ±1.1336	1.96 ±1.4995	0.36 ±0.2767	0.11 ±0.0818	0.25 ±0.204	0.2 ±0.1572	0.13 ±0.0991	5.58 ±2.303	0.05 ±0.0287	0.68 ±0.3641	0.15 ±0.0863	0.02 ±0.0101	0.17 ±0.0736	1.06 ±0.3826	6.64 ±2.3346

***Note:** Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*

Post-harvest losses of pulses in percent at national level

Crop	Farm Operations									Market Level Assessment						Overall total loss
	Harvest-ing	Collect-ion	Thresh-ing	Winnowi-ng/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesal-ers	Retailers	Process-ing unit	Trans-port	Total loss in market level	
Pigeon Pea	1.13 ±0.7125	0.3 ±0.2203	2.02 ±0.8395	0.2 ±0.1027	0.23 ±0.1343	0.2 ±0.1162	0.55 ±0.3552	0.18 ±0.0866	4.81 ±1.1985	0.13 ±0.0341	0.11 ±0.0372	0.17 ±0.1066	0.4 ±0.1711	0.25 ±0.0819	1.06 ±0.2209	5.87 ±1.2187
Chickpea	1.54 ±0.5854	1.05 ±0.4965	1.67 ±0.6521	0.31 ±0.1484	0.36 ±0.144	0.2 ±0.144	0.79 ±0.2909	0.04 ±0.0153	5.95 ±1.0784	0.67 ±0.2139	0.9 ±0.1218	0.56 ±0.13	0.81 ±0.213	0.4 ±0.0972	3.34 ±0.3637	9.29 ±1.1381
Greengram	1.67 ±1.2271	0.66 ±0.6319	1.45 ±1.1091	0.39 ±0.4385	0.28 ±0.2826	0.18 ±0.1814	0.33 ±0.2555	0.13 ±0.1005	5.11 ±1.875	0.02 ±0.0888	0.29 ±0.6646	0.29 ±0.8956	0.12 ±0.66	0.13 ±1.9235	0.86 ±2.321	5.97 ±2.9837
Blackgram	1.32 ±0.8293	0.76 ±0.3713	1.16 ±0.7334	0.44 ±0.1839	0.34 ±0.1881	0.2 ±0.1487	0.55 ±0.2913	0.14 ±0.0885	4.91 ±1.244	0.04 ±0.014	0.21 ±0.0691	0.22 ±0.0927	0.11 ±0.0446	0.23 ±0.071	0.82 ±0.1435	5.73 ±1.2523

***Note:** Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*

Post-harvest losses of oilseeds in percent at national level

Crop	Farm Operations									Market Level Assessment						Overall total loss
	Harvest-ing	Collect-ion	Thresh-ing	Winnowi-ng/ clean-ing	Drying	Packagi-ng	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesal-ers	Retailers	Process-ing unit	Trans-port	Total loss in market level	
Mustard	1.89 ±0	0.48 ±0	1.29 ±0	0.4 ±0	0.13 ±0	0.1 ±0	0.1 ±0	0.21 ±0	4.6 ±0	0.05 ±0.4542	1.07 ±0.3284	2.55 ±0.2526	0.45 ±0.1076	0.98 ±0.1056	5.16 ±0.4417	9.76 ±0.4417
Cottonseed	1.54 ±0.6744	0.27 ±0.0878	-	-	0.08 ±0.0214	0.07 ±0.0171	0.07 ±0.0357	0.16 ±0.0653	2.19 ±0.6846	0.01 ±0.0033	0.19 ±0.0482	0.03 ±0.008	0.03 ±0.0104	0.09 ±0.0376	0.35 ±0.0625	2.54 ±0.6876
Soybean	2.81 ± 1.5188	0.91 ± 0.5606	1.52 ± 0.6266	0.19 ± 0.0987	0.15 ± 0.0612	0.44 ± 0.306	0.88 ± 0.2807	0.3 ± 0.1331	7.2 ± 1.7937	0.13 ± 0.056	0.51 ± 0.5332	0.17 ± 0.1442	0.17 ± 0.0931	0.09 ± 0.0135	1.07 ± 0.5631	8.27 ± 1.88
Safflower	0.57 ±0.1794	0.22 ±0.0682	0.26 ±0.0802	0.14 ±0.0439	0.07 ±0.0212	0.09 ±0.0295	0.01 ±0.0034	0.09 ±0.0275	1.45 ±0.4533	0.15 ±0.222	0.04 ±0.0746	0.02 ±0.0257	0.08 ±0.1151	0.11 ±0.1732	0.4 ±0.3132	1.85 ±0.551
Sunflower	0.96 ±0.3524	0.39 ±0.3524	1.45 ±1.2756	0.21 ±0.2266	0.09 ±0.098	0.12 ±0.1231	0.05 ±0.0641	0.1 ±0.1043	3.37 ±1.4024	0.17 ±0.0635	0.04 ±0.0135	0.02 ±0.0073	0.77 ±0.2152	0.1 ±0.0316	1.1 ±0.227	4.47 ±1.4206
Groundnut	1.59 ±1.1401	0.56 ±0.429	1.4 ±1.0582	0.39 ±0.2756	0.13 ±0.0951	0.14 ±0.1045	0.06 ±0.0421	0.12 ±0.0882	4.39 ±1.6458	0.04 ±0.0796	0.38 ±0.1637	0.07 ±0.0351	0.38 ±0.6511	0.14 ±0.5841	1.01 ±0.8908	5.4 ±1.8716

Note: *Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*

Post-harvest losses of fruits in percent at national level

Crop	Farm Operations							Market Level Assessment						Overall total loss
	Harvest-ing	Collec-tion	Sorting/ Grading	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retailers	Process-ing unit	Trans-port	Total loss in market level	
Apple	3.55 ±0.0885	0.05 ±0.4542	3.61 ±0.1126	0.02 ±0.0008	0.05 ±0.0033	0.62 ±0.0241	7.87 ±0.0933	0.05 ±0.4542	0.01 ±0.026	0.38 ±2.6446	0.13 ±1.0444	0.73 ±0.9129	1.29 ±3.0208	9.17 ±3.0222
Banana	1.49 ±0.6084	0.32 ±1.1556	1.97 ±0.0216	0.18 ±2.2854	0.06 ±0.8853	1.48 ±0.6748	5.5 ±2.858	0.06 ±0.0209	1.14 ±0.3279	0.44 ±0.0917	0.01 ±0.0022	0.95 ±0.2949	2.6 ±0.4509	8.11 ±2.8934
Citrus	1.34 ±0.4238	0.22 ±0.1391	1.85 ±0.8733	0.26 ±0.1503	1.00 ±0.4271	0.97 ±0.9941	5.63 ±1.4679	0.04 ±0.0348	0.39 ±0.178	0.62 ±0.3333	0.02 ±0.013	0.84 ±0.3947	1.92 ±0.5477	7.55 ±1.5668
Grapes	1.65 ±0.9835	0.16 ±0.1399	2.43 ±1.4035	0.49 ±0.3229	0.06 ±0.0495	0.57 ±0.4498	5.34 ±1.8072	0.01 ±0.0068	0.49 ±0.2913	0.79 ±0.2578	0.07 ±0.0784	0.34 ±0.296	1.69 ±0.4951	7.04 ±1.8737
Guava	5.05 ±0.1527	0.45 ±0.0259	4.59 ±0.1396	0.31 ±0.0112	0.45 ±0.0216	0.73 ±0.0227	11.59 ±0.2112	-	1.57 ±0.6875	2.92 ±1.1304	1.31 ±0.4122	1.24 ±0.2443	3.51 ±1.4071	15.1 ±1.4229

Crop	Farm Operations							Market Level Assessment						Overall total loss
	Harvest-ing	Collec-tion	Sorting/ Grading	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Whole-salers	Retailers	Process-ing unit	Trans-port	Total loss in market level	
Mango	1.34 ±0.263	0.27 ±0.1652	2.16 ±0.4672	0.04 ±0.0339	0.29 ±0.1787	0.83 ±0.2051	4.94 ±0.6244	-	0.4 ±0.122	1.03 ±0.1001	0.08 ±0.0191	0.98 ±0.1056	2.5 ±0.1908	7.43 ±0.6529
Papaya	1.28 ± 0.8255	0.45 ± 0.265	0.66 ± 0.5001	0.3 ± 0.1879	0.2 ± 0.1573	0.85 ± 0.5588	3.76 ± 1.1722	0.02 ± 0.0011	0.72 ± 0.0422	1.64 ± 0.0506	0.04 0.0044	0.71 ± 0.0369	3.14 ± 0.0756	6.89 ± 1.1746
Sapota	0.74 ±0.6807	0.35 ±0.2635	0.49 ±0.4072	0.3 ±0.2034	0.01 ±0.0082	0.88 ±0.7324	2.77 ±1.1298	0.15 ±0.0667	0.82 ±0.2913	0.96 ±1.1498	0.04 ±0.0092	1.38 ±0.4252	3.34 ±1.2618	6.11 ±1.6937
Pineapple	3.09 ±1.8532	0.08 ±0.1037	1.86 ±0	0.03 ±0.0414	0.08 ±0.0425	0.29 ±0.3792	5.43 ±1.8954	0.07 ±0.0194	0.66 ±0.3561	0.53 ±0.2642	0.12 ±0.0391	0.6 ±0.1448	1.99 ±0.4685	7.42 ±1.9524
Pomegranate	1.44 ±0.6041	0.15 ±0.113	1.49 ±0.9683	0.19 ±0.0794	0.18 ±0.1302	0.16 ±0.087	3.61 ±1.1603	0.16 ±0.0511	0.64 ±0.621	0.73 ±0.6057	0.12 ±0.0511	1.00 ±0.6877	2.64 ±1.1093	6.25 ±1.6052
Muskmelon	1.63 ±1.0374	0.29 ±0.1811	1.04 ±0.5848	0.32 ±0.2259	0.08 ±0.0595	0.23 ±0.1047	3.59 ±1.2314	0.03 ±0.0045	0.54 ±0.1046	1.26 ±0.367	-	0.83 ±0.3108	2.66 ±0.4922	6.25 ±1.3262

Note: *Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*

Post-harvest losses of vegetables in percent at national level

Crop	Farm Operations							Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Packaging	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesale-rs	Retailers	Processing unit	Trans-port	Total loss in market level	
Cabbage	1.52 ±0.4106	0.33 ±0.1052	3.12 ±0.9504	0.32 ±0.1028	0.13 ±0.038	0.76 ±0.2403	6.18 ±1.0736	0 ±0.00	0.24 ±0.075	0.25 ±0.0949	0.01 ±0.0021	0.61 ±0.1908	1.11 ±0.2259	7.29 ±1.0971
Cauliflower	1.84 ±1.1073	0.23 ±0.1438	2.59 ±1.5292	0.3 ±0.1924	0.06 ±0.0388	0.79 ±0.5153	5.8 ±1.9722	0.04 ±0.0005	0.6 ±0.0272	0.78 ±0.0929	0 ±0.00	0.79 ±0.2245	2.2 ±0.2445	8.01 ±1.9873
Green pea	1.84 ± 1.2344	0.64 ± 0.7176	1.67 ± 0.9606	0.14 ± 0.1013	0.004 ± 0.0037	0.56 ± 0.6162	4.85 ± 1.8307	-	0.51 ± 0.7223	0.65 ± 0.5226	0.03 ± 0.0216	0.53 ± 0.4003	1.71 ± 1.3639	6.56 ± 2.2829
Onion	2.48 ± 0.6217	0.39 ± 0.1271	1.01 ± 0.2791	0.12 ± 0.0554	0.91 ± 0.3083	0.4 ± 0.1506	5.31 ± 0.7755	0.33 ± 0.0073	0.77 ± 0.0963	0.57 ± 0.0687	0.02 ± 0.0034	0.35 ± 0.049	2.05 ± 0.1283	7.31 ± 0.786
Potato	1.8 ± 0.182	0.21 ± 0.0163	2.08 ± 0.2098	0.04 ± 0.0036	0.11 ± 0.0133	0.49 ± 0.0483	4.73 ± 0.2827	0.14 ± 0.0337	0.26 ± 0.0822	0.1 ± 0.0306	0.03 ± 0.0024	0.34 ± 0.0749	0.87 ± 0.1202	5.6 ± 0.3072
Tomato	2.87 ±0.0000	0.61 ±0.0000	3.18 ±0.0000	0.37 ±0.0000	0.59 ±0.0000	0.89 ±0.0000	8.51 ±0.0000	-	1.84 ±1.7427	3.19 ±5.5304	0.95 ±1.6041	1.06 ±1.244	7.04 ±6.1436	15.55 ±6.1436

Crop	Farm Operations							Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting/ Grading	Packaging	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesale-rs	Retailers	Processing unit	Trans-port	Total loss in market level	
Tapioca	1.19 ±0.033	0.25 ±0.0089	1.05 ±0.021	0.08 ±0.0032	0.26 ±0.0053	0.55 ±0.0093	3.39 ±0.0417	-	0.42 ±0.1533	0.45 ±0.3354	0.13 ±0.0413	0.42 ±0.1595	1.42 ±0.4039	4.81 ±0.406
Bottlegourd	1.57 ±0.4953	0.93 ±0.2719	1.02 ±0.2962	0.29 ±0.1525	0.13 ±0.0828	0.2 ±0.0735	4.14 ±0.6652	-	0.55 ±0.5247	0.33 ±0.3077	-	0.72 ±0.6407	1.6 ±0.8834	5.74 ±1.1059
Brinjal	2.37 ±0.535	0.47 ±0.2116	0.97 ±0.326	0.26 ±0.1308	0.38 ±0.0821	0.61 ±0.139	5.07 ±0.6931	-	0.8 ±0.2589	1.14 ±0.3256	-	0.53 ±0.1563	2.47 ±0.4444	7.53 ±0.8234
Beans	1.12 ±1.6303	0.46 ±0.5771	1.21 ±1.2704	0.3 ±0.3258	0.27 ±0.2011	0.32 ±0.325	3.68 ±2.2039	0.11 ±0.0182	0.92 ±0.3089	0.8 ±0.3106	-	1.54 ±0.4521	3.37 ±0.6298	7.05 ±2.2921
Radish	2.13 ±0.7898	0.1 ±0.0313	0.82 ±0.2589	0.49 ±0.1542	0.1 ±0.0431	0.1 ±0.0277	3.74 ±0.8475	-	0.7 ±0.2921	0.7 ±0.2659	-	0.94 ±0.5149	2.35 ±0.649	6.08 ±1.0674
Capsicum	1.4 ± 0.5408	0.21 ± 0.1841	0.5 ± 0.2205	0.17 ± 0.1188	0.01 ± 0.002	0.34 ± 0.0535	2.63 ± 0.6261	-	0.82 ± 0.7842	0.72 ± 0.666	-	0.8 ± 1.4737	2.33 ± 1.7973	4.97 ± 1.9032
Okra	1.48 ±0.7144	0.32 ±0.1848	1.12 ±0.5913	0.15 ±0.0651	0.16 ±0.0976	0.45 ±0.2407	3.68 ± 0.9827	-	0.81 ±0.3397	0.88 ±0.2844	-	0.56 ±0.1997	2.25 ±0.4861	5.94 ±1.0964

Note: *Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*

Estimates for mushroom were not calculated using PPS methodology due to unavailability of district-wise data required for analysis by PPS.

Post-harvest losses of plantation crops and spices in percent at national level

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting and Grading	Threshi ng	Winnowi ng/ cleaning	Drying	Packagi ng	Farm level storage	Trans-port	Total loss in farm operatio-ns	Godown	Wholesal-ers	Retailers	Processi ng unit	Trans-port	Total loss in market level	
Are ca nut	1.18 ±0.0457	0.28 ±0.011	-	0.44 ±0.0173	0.91 ±0.039	0.14 ±0.0043	0.12 ±0.0046	0.25 ±0.0098	0.27 ±0.0101	3.58 ±0.1419	0.01 ±0.014	1.21 ±0.2191	0.02 ±0.0069	0.16 ±0.1585	0.22 ±0.1614	1.61 ±0.3153	5.19 ±0.3457
Cashew nut	1.36 ±0.0329	0.35 ±0.0053	-	1.08 ±0.025	0.23 ±0.0054	0.03 ±0.0007	-	0.02 ±0.0005	0.04 ±0.001	3.11 ±0.042	0.00 ±0.00	0.18 ±0.0858	0.02 ±0.016	0.27 ±0.1655	0.18 ±0.1256	0.65 ±0.2253	3.76 ±0.2292
Coconut	0.86 ± 0.3857	0.26 ± 0.2222	-	-	0.31 ± 0.2055	0.31 ± 0.2211	0.2 ± 0.2211	0.51 ± 0.3	0.2± 0.2104	2.65 ± 0.6873	0.05 ±0.0167	0.07 ± 0.0261	0.35± 0.1364	0.18± 0.1209	0.19 ± 0.049	1.00 ± 0.1913	3.86 ± 0.7134
Sugarcane	2.46	1.26	1.44	-	-	-	-	-	1.26	6.42	-	-	-	0.67	0.95	1.62	8.04

Crop	Farm Operations										Market Level Assessment						Overall total loss
	Harvest-ing	Collecti-on	Sorting and Grading	Threshing	Winnowing/ cleaning	Drying	Packaging	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholesal-ers	Retailers	Processing unit	Trans-port	Total loss in market level	
	±0.646	±0.2627	±0.6161						±0.1433	±0.9415				±0.1862	±0.386	±0.4286	±1.0345
Black pepper	0.55 ± 0.3728	0.2 ±0.1098	-	0.22 ± 0.1468	0.03 ± 0.0227	0.07 ± 0.0416	0.02 ± 0.0178	0.02 ±0.0149	0.01 ±0.0076	1.12 ± 0.4189	0.01 ± 0.0147	0.08 ± 0.1221	0.07 ± 0.1145	0.02 ± 0.0316	0.01 ± 0.0171	0.19 ± 0.1309	1.31 ± 0.4388
Chilli	1.07 ±0.0188	0.78 ±0.0158	2.14 ±0.045	-	-	0.08 ±0.0019	0.03 ±0.0016	0.93 ±0.0241	0.17 ±0.0045	5.2 ±0.0569	0.03 ±0.0504	0.09 ±0.0286	0.43 ±0.1279	-	0.08 ±0.0199	0.63 ±0.1418	5.84 ±0.1528
Coriander	1.97 ±0.0429	0.69 ±0.0163	-	1.15 ± 0.0325	0.53 ± 0.0163	0.06 ±0.0016	0.13 ± 0.0037	0.06 ±0.002	0.12 ±0.0029	4.72 ± 0.0588	0.00	0.15 ± 0.1107	0.09 ± 0.0478	0.01 ± 0.0018	0.33 ± 0.0721	0.58 ± 0.1325	5.3 ± 0.1448
Turmeric	1.74 ±0.0683	0.26 ±0.0002	0.48 ±0.001	-	0.27 ±0.0196	0.47 ±0.057	0.17 ±0.0001	0.06 ±0.0003	0.33 ±0.0001	3.78 ±0.089	0.14 ±0.1406	1.32 ±0.4746	0.35 ±0.3438	-	0.74 ±0.6736	1.23 ±0.7307	5.01 ±0.7361

Note: *Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*

Post-harvest losses of livestock produce in percent at national level

	Farm Operations								Market operation						Overall total loss
Crop	Harvest-ing	Collect-ion	Sorting/ grading	Drying	Packag-ing	Farm level storage	Trans-port	Total loss in farm operations	Godown	Wholes-alers	Retailers	Processing unit	Transport	Total loss in storage	
Egg	-	1.52 ± 0.0504	1.25 ± 0.2192	-	1.17 0.1962	0.29 ± 0.0784	0.29 ±0.0593	4.19 ± 0.3698	0.04 ± 0.0226	0.7 ± 0.2409	0.37 ± 0.2144	-	0.46 ± 0.132	1.57 ± 0.3492	5.76 ± 0.5087
Inland Fish	1.57 ± 0.7587	0.34 ± 0.1997	1.36 ± 0.7848	-	0.17 ±0.1107	0.09 ± 0.049	0.15 ±0.0857	3.68 ± 1.1196	-	0.23 ± 0.0439	0.64 ± 6.3151	-	0.15 ± 0.2046	1.02 ± 6.3185	4.7 ± 6.4168
Marine Fish	5.66 ± 3.7906	0.56 ± 3.4492	0.37 ± 0.3755	0.15 ± 3.0456	0.01 ± 0.218	-	0.45 ± 0.3916	7.2 ± 5.9902	0.19 ± 0.0721	0.54 ± 0.2446	0.2 ± 0.0907	0.28 ± 0.1478	0.51 ± 0.2154	1.72 ± 0.2985	8.92 ± 5.9976
Meat	1.43 ± 0.2864	-	-	-	-	-	-	1.43 ± 0.2864	-	0.41 ± 0.1182	0.24 ± 0.132	0.02 ± 0.0142	0.19 ± 0.0925	0.86 ± 0.2004	2.3 ± 0.3495
Poultry Meat	2.51 ± 0.0166	-	-	-	-	-	-	2.51 ± 0.0166	-	1.81 ± 0.3435	0.88 ± 0.3267	0.01 ± 0.0032	0.3 ± 0.0496	3 ± 0.4767	5.51 ± 0.477
Milk	0.12 ± 0.0827	0.07 ± 0.0197	-	-	-	-	0.14 ±0.1204	0.33 ± 0.1474	-	-	0.04 ± 0.0742	0.15 ± 0.0877	0.15 ± 0.4087	0.34 ± 0.4245	0.67 ± 0.4494

Note: *Figures in the table represent "Loss ± Standard Error (SE) ×1.96"*

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NABARD Consultancy Services

Corporate Office: 24 Rajendra Place, 7th Floor,
NABARD Building, New Delhi -110008,
website: www.nabcons.in